

Applied Quantum Computers

Learn about the Concept, Architecture, Tools, and Adoption Strategies
for Quantum Computing and Artificial Intelligence

DR. PATANJALI KASHYAP



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Dedicated to

My Mother

Late, Dr Meena Singh and

Maternal aunty – Late, Dr Vahedi Sharma

*They both believed in me against all the odds and even after
knowing
my weakness and limitations*

&

*My Father Late, Dr Damoder Parsad Singh and my maternal
uncle*

Late, Vibhuti Narayan Singh

Foreword



Technology evolves and even more rapidly. In my role as the Head of Corporate Functions IT, what I see today is remotely comparable to I witnessed during my college days at NIT Surathkal or even at IIM Calcutta. Whilst the basic tenets of ‘whats’ and ‘whys’ remain the same, its more of ‘how’ part we continue to address in our current technology days. The ‘current’ of today is the ‘past’ of tomorrow. Thus, is the speed and unpredictability of any emerging technology.

Technology begets technology; even after more than two decades of my stint in IT, there’s always anticipation of what’s next. When Patajanli approached me to write the foreword for this book, I enthusiastically said ‘yes’ as the topic and its relevance, is immense.

Whilst his first book “Machine Learning for Decision Makers” focused on the usage of an emerging tech, this book goes deeper into the underlying of data and its processing. The intricate topics, in this book, are woven seamlessly into an elegant fabric of Quantum computing, starting with the basics and giving the reader a detailed insight of its applications. The principles of Quantum physics are adeptly entrenched into computing space, evolving into various Quantum computing technologies.

This book is for anyone who is keen to pursue Quantum as a stream of choice. The language is basic for a wider reach and all the jargons one needs to dwell into, in its simplified format.

In the near future, I will hardly be surprised, if it becomes the obvious way-forward for all technology implementations. Especially, in an era of burgeoning data and its many avatars. So, get on to a world of Quantum computing.

-Yatish Shivaprasad

Director

Corporate Functions Technology

About the Author



Dr Patanjali Kashyap holds a degree in PhD (Physics) and MCA. He is working as a senior leader at a French multinational bank. He deals with high-impact mission-critical financial and innovative new generation technology projects on a day-to-day basis. He has worked with the technology giants like Infosys and Cognizant technology solutions. He is an expert on the agile process, machine learning, big data, Azure cloud, DevOps, Planisware, Azure quantum cloud, and other computing paradigm. He possesses a sound understanding of Microsoft Azure and cognitive computing platforms like Watson and Microsoft cognitive services.

He introduces .net technologies as his first love to his friends and colleagues.

He is vastly experienced in Software development methodologies, application support, and maintenance projects. His view of “everything is linked with the other” reflects in his work. For instance, he has filed a patent on improving and measuring the performance of an individual by using emotional, social and moral intelligence.

Patanjali had published multiple research and white papers on multiple topics including quantum, optical and DNA computers. He is a go-to person for incorporating “positivity and enthusiasm” into enterprises. His fresh way of synthesizing “Indian Vedic philosophies with the western practical management insight for building flawless organizational dynamics” is much appreciated in the corporate circle.

Apart from MCA, Patanjali holds masters in bioinformatics, physics (Radiophysics and electronics), computer science (M.Phil.) and a postgraduate diploma in computer applications.

About the Reviewer

Mr. Gourav Gupta is a Data specialist and having 4+ years of experience in Big Data, Artificial Intelligence, Internet of Things and Digital Twin. Mr. Gourav has worked on several interdisciplinary real time project which are the conglomerations of Digital Technologies. His expertise is on architectural optimization and technical solutioning on Big Data, AI, Computer Vision, and Internet of Things. Gourav loves to write research article and serving as a reviewer with Springer Journal.

Acknowledgements

In the summer of 2020, while returning from the hospital where I went for a checkup related to my liver disease with my wife Amrita, Manish Jain (BPB) called and asked me if I am interested in writing a book on quantum computing for BPB.

I took advice from my mother and wife Amrita on whether I should go ahead with the proposal or not. Stuck in this confusion, I called my friend Mukul Garg (SAP Delivery Center Head, Allstate), his response was positive. I thought for a few seconds and then accepted the proposal.

I want to take this opportunity to thank my wife, Late mother, and adorable son, Madhav (finally he accepted that I am a software engineer by heart and manager by profession) for their cooperation. I want to acknowledge that; due to this book and my occupation, I am not able to give them the quality time that they deserve. I felt guilt for missing out on spending time with my son. There were occasions when he and Simba (my dog) were standing by the closed door of my study room asking me to open and I did not open the door because I was busy writing. Numerous other incidents like this happened in the last few years.

When it comes to providing motivation “no one can beat my mother”, even now after her death, she encourages me. However, she is not with me as God called her near to him during the coronavirus pandemic because he also needs the motivation to run the show. My mother was a university professor by profession. Without her continuous moral, and emotional support, this book would have been impossible...so thanks Maa.

I also want to thank my brother, Sumukh, knowingly or unknowingly, he is always with me in my journey of writing this book.

I am thankful to the complete BPB team for giving me this opportunity to explore my hidden potential.

I would like to mention the names of some interns who worked for me, providing their intellectual insights. Thanks to Angela Ruohan Wang (Solution Architect at Amazon Web Services, Mount Holyoke College,

US), Kristin Emodi (Medical Student at NYU School of Medicine, US), Sabin Park (Mint.com, iOS Developer in San Francisco Bay Area), Simisola Otukoya (King's College London Structural Engineer, UK) and Neil Sharma (University of Pittsburgh School of Medicine and Oxford University, London).

Knowing the professional capabilities of my colleague Natrajan Gurumurthy, I asked for his help and suggestions to improve the content of the book and he without thinking a minute become ready to offer help. Thanks, Natarajan.... For this. The same happened with Yatish Shivaprasad (my mentor), during a casual meeting, I asked him to write the forward of the book and his spontaneous reaction was "send me the details I will do this for you". Thanks, Yatish for accepting my request.

Without Abhishek Bhargava, AB's (my Boss') encouragement, writing the final phase of this book would have been difficult. He always says "improve yourself individually Learn new things and delegate work so that you get time for productive work and provide opportunities to others to grow". This is a very motivating and guiding statement for me. Last but not least ... I am thankful to Praveen Krishnappa, Mary Fernandez and Mr Manjunath Channabasappa who were always available to support me and give suggestions for improvements.

I would also like to thank the authors of the Indian scriptures like Bhagavad Gita, Veda and other mythological literature for providing such intellectually insightful reading to the world.

Finally, I would like to thank my father, the late Dr Damador Parsad. Without his support, this book would not have come to fruition. He helped me in every situation, even if he wasn't in the real world. Especially when I am in a situation where there is no escape route.

Thanks, Dada...

Dr. Patanjali Kashyap
17.07.2022 (on the day of
my son Madhav's 7th birthday)

Preface

Technologies are mounting quicker than ever. Social media, the Internet of Things, metaverse, 5 G, Blockchain, edge computing, Big Data, smart mobile devices, and artificial intelligence are altering the way we live. The entire world is getting linked with the help of these technologies.

Quantum theory is one of the most successful theories that influenced the progress of science in the 20th century. It introduced new scientific thinking, predicted completely unimaginable situations, and influenced several areas of modern technology. More than 50 years after its founding, quantum theory merged with computer science, another major intellectual victory of the 20th century, and a new subject of quantum computation was born. Quantum computing is a fusion of two major scientific revolutions in the 20th century, quantum physics and computer science.

Quantum computing is the design of hardware and software that replaces Boolean logic with quantum laws at the algorithmic level. Quantum computing also applies to artificial intelligence and machine learning.

Quantum computers along with artificial intelligence can potentially incorporate intelligence everywhere. This will produce an all-inclusive system of smarter applications, products, and understandings for users, workforces, consumers, and customers. In this background, this book is inscribed to deliver holistic insights to the readers. The book will help them to grasp the concepts of quantum computers

Having said that, the recommendation is that the reader would treat this book as a journey and the journey must be enjoyed in a holistic fashion (from start to finish), not in bits and pieces... with these words, I request the readers to join me on the trip of the future...through this book.

[Chapter 1: Tools for Imaginations, Innovation, Technologies, and Creativity](#)

To describe ground-breaking ideas, thoughts, methodology, and inimitability-- some exceptional tools and techniques are required. It is a pre-requisite that these tools must be intuitive. Therefore, few state-of-the-

art, underutilized, casual tools, techniques, and methodologies are used in this book. These are helpful in the amplification of the concepts and ideas related to Artificial intelligence, QC, and associated technologies. This chapter consolidates and explains all the tools and techniques used in this book. This method will aid the person who reads to know more about them.

Chapter 2: Quantum Physics as an Enabler of a Quantum Computer

This chapter will provide a premier of Quantum physics and explain the fundamental concepts which are relevant from a quantum computer perspective. It goes into the details of entanglement, superposition, coherence, and decoherence and tries to explain them in simple words.

Also, we will examine the interrelation between quantum physics and quantum computers. We will figure out the what, how and why of quantum computers as well.

Chapter 3: Mathematics of Quantum Computers: The Fundamentals

This chapter will introduce some of the most fundamentals aspects of mathematics from a quantum computer perspective.

Chapter 4: From Bits to Qubits to Qubytes

This chapter contains the fundamentals of quantum computers, including their building blocks like qubit, logic gates, registers, and circuits. It gives readers a perspective on the complete ecosystems of QC. From hardware to the software stack. After going through this chapter, the readers have become comfortable with most of the quantum terminologies and their working at least at a superficial level.

Chapter 5: Artificial Intelligence and Associated Technologies: A Review

Companies are developing a circuit-based quantum computer that simulates associative memory using the execution of an algorithm which writes memory states into superpositions, and then it uses a Grover-like quantum search to retrieve the memory state closest to a given input with the ultimate goal being to simulate working of the human brain. Also, ideas have been proposed about modelling black holes as QNNs (Quantum Neural networks). Because theories say that black holes and brains may store memories in similar ways. Additionally, there is a lot of research happening

to create a network of AI-based quantum computers to improve functioning of humanity in the future. These technical directions are also fulling a type of social transformation. This chapter would present a perspective on these topics to you.

This chapter will discuss AI and its interrelation with quantum computers. And, present a synopsis of the future pathway. It explains how QC along with robotics, nanotechnology, big data, AI and IoT will shape the next few decades. This chapter also presents the case strongly that why QC is here to stay.

Chapter 6: Quantum Algorithms for Everyone ...!!!

This chapter focuses on quantum algorithms. As quantum computers turn out to be accessible to a wide-ranging sample of the public, the requirement has stood up to train a unit of quantum programmers.

This chapter becomes very important when one is looking for one-stop solutions for quantum algorithms and want to explore multiple dimensions of it.

Chapter 7: Quantum Machine Learning

Quantum machine learning is an interdisciplinary method that syndicates machine learning and the philosophies of Quantum Physics. Quantum chips can be used to chart out computer algorithms to resolve multifaceted problems.

This chapter will explore the combined potential of ML (Machine Learning) and QC. It sees ML in a holistic framework of big data and IoT. And, explains the technical, functional and process-based details of quantum machine learning.

Chapter 8: Quantum Cryptography: The Future of Security

This chapter will provide you with a basic understanding of the quantum internet. By giving a glimpse of the mesmerizing world of quantum technologies such as quantum cryptography, the internet, and protocols like BB84. The reader will be able to foresee the potential influence of quantum computing and the quantum internet on business. Also, this chapter would document cryptography in detail. Especially quantum cryptography.

The chapter is meant for a comprehensive and varied spectator including policy-makers, people with a scientific or personal interest, business executives, and students at all levels.

Chapter 9: The Architecture of a Quantum Computer

This chapter covers different types of proposed and working quantum hardware and their potential along with their applications.

Chapter 10: DNA, Quantum and Photonic Computers

This chapter will deal with the fundamentals of DNA, and optical / photonic computers. It also provides insights into the architecture of DNA and optical computers. However, before going into the details of DNA computers it would provide the foundation of DNA concepts. Finally, it showcases how DNA, optical and the quantum computer would work together.

Chapter 11: Let's Realize It: Quantum Start-Ups and Giants in Action

In recent years the field of QC has seen marvelous development and has moved out of the hypothetical monarchy to the tech world. Enterprises like Google, AT&T, Microsoft, Raytheon, Honeywell, and Intel, are creating a robust quantum portfolio.

Apart from all these corporates two of the world's top banks, Goldman Sachs and JP Morgan Chase are also having quantum motivations. IonQ, recently signed a long-term partnership with Microsoft. They are working on trapped-ion technology and hoping this will get up and go with novel advances in physical science, drug discovery and progressions in AI along with quantum computers. These startups are not alone, they are accompanied by Atom Computing, Artiste-qb.net, Aurora Quantum Technologies, Xanadu AppliedQubit and so on. This chapter will cover some stories and technologies these companies are offering.

This chapter would briefly cover the quantum patent landscape. Further, it would discuss the role of start-ups, tech giants and patents in the field of quantum innovation.

Chapter 12: The Quantum Strategies

This chapter will discuss how and what strategies are needed make to take maximum advantage of quantum technologies. What a corporation would

do to make it future-ready. It also discusses, what strategies will enable them to train their workforce to lead the quantum revolution.

Chapter 13: The Human Side of Quantum Computer

This chapter will provide a view on how to build a team for quantum computer-centric initiatives within an organization. Broadly discuss what are the skills, technologies, and tools required for the quantum computer team. It is through light on the behavioral and technical competencies requirements of a team member, manager, and leader. Also, discuss the new job role and mindsets emerging in the arena.

Annexure 1: Q# for quantum computation

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CHAPTER 1

Tools for Imaginations, Innovation, Technologies, and Creativity

Welcome to the first chapter of one of the most important books of your life's reading journey. This book is important because it will introduce you to the skills that will make you relevant in the job market, now and in the coming decades. You will not only be relevant in "the so-called job market" but become a change enabler for the future. Hence, without further ado, let's start with the structure of the chapter; thereafter, we will cover the objective and plan for the same, followed by the main content.

Structure

In this chapter, we will discuss the following topics:

- Mind Maps
- Quick tips and techniques
- Important question and answers
- Jargon busters
- Newest trends
- Latest research
- Audio and video links
- Industry bites
- Point of views
- Expert opinion
- Thought cartoons
- Before we wind up

Objectives

After studying this chapter, you will be able to understand the concept of productivity tools and use them for your benefits wherever required. Additionally, you will become a master in the following ways:

- Understanding the concepts of mind map and their uses in different scenarios.
- Understanding the concepts like jargon buster, expert opinion, and **Point of View (POV)**.
- Summarizing the concepts in a scientific and visual way.
- Using and depicting cartoons in a productive way.

Plan of the chapter

Before moving ahead, it is good to understand the overall plan of the chapter. [Figure 1.1](#) explains the entire flow of the chapter:

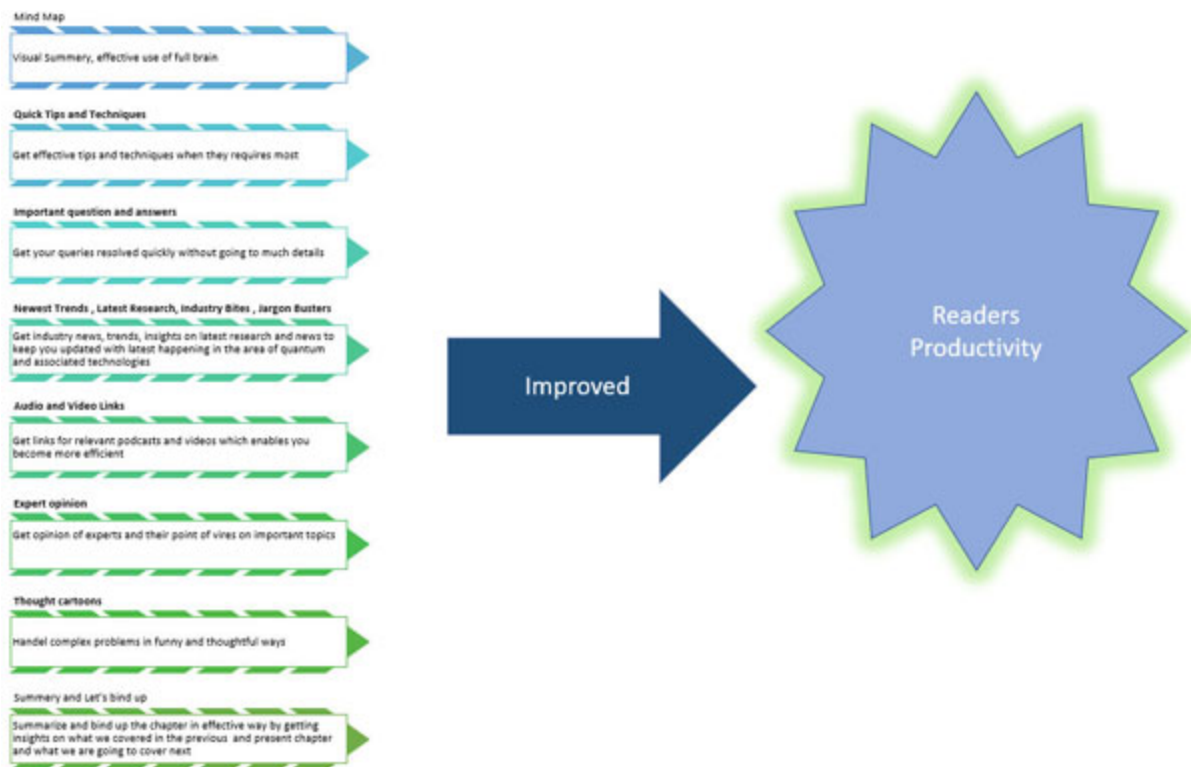


Figure 1.1: Plan of the chapter

This chapter provides you the blueprint of the book. I have taken a unique approach to elucidate the concepts which are associated with quantum

computers and artificial intelligence. In my previous book on machine learning, I took the same approach. Usually, this kind of method is not thought to be mainstream, particularly in the technology space. There might be many reasons for that. One bulbous reason, though, is that readers and publishers both are not eager and open for experimentation. They choose to go with the exasperated, verified, and well-known presentation methodologies. However, I ventured to experimentation since the board addressees of the book are, I assumed to be innovators and unconventional thinkers. Henceforth, their thought process is expected to be disruptive in nature.

To clarify ground-breaking trials, exceptionality, and disruptions, some unusual tools and techniques are essential, which must be innate in nature. Therefore, several state-of-the-art along with, unnoticed, underutilized, informal tools, methods, and practices are adopted in this book. They are certainly proved to be supportive in amplification and explanation of the thoughts and ideas connected to quantum computers, artificial intelligence, and related technologies.

As we progress in the journey of reading, this chapter introduces you to multiple tools and techniques. Also, it narrates a concise story of all the future chapters, which I will cover in the book. This overview of chapters will help the readers get a feel of the subject matter to come.

I liked writing this book and am confident that reading it will be an enjoyable and educational experience for you as well. This is because the ideas contained in the book are path-breaking and will help you uninterruptedly innovate and discover novel paths. I am willing to activate disruptive thought processes in your brain through my writing. The creative and intellectual coverage of the subject matter of quantum computing and artificial intelligence is the potential to bring that change to you.

After these initial thoughts, it is high time to go into the real subject matter. So, let's jump to the tools and techniques used in the book. Let's start with one of my favourite tools called Mind Maps.

Mind Maps: An overview

Mind mapping is a pictorial form of note-taking that provides a summary of a theme and its related information. Hence, it permits people to understand, produce novel concepts, and shape acquaintances, through the use of colors, pictures, and words. Mind mapping inspires people to start with a central idea and inflate outward to further in-depth sub-topics. In a nutshell, it is a graphical depiction of hierarchical information that comprises a chief idea enclosed by linked divisions of related topics.

The following are the ways in which Mind Maps are useful to you and me:

- To help in the brainstorm and see the sights of any idea, thought, or problem.
- Simplify the understanding of relationships and associations among ideas and concepts.
- It is easy to communicate new ideas and thought processes through a mind map.
- Allows people to effortlessly recall information.
- Helps to take notes and design tasks.
- Stay relaxed to establish ideas and concepts.
- A great Summarizing tool; helps to summarize areas effectively.
- Uses both parts of the brain (we will cover it shortly).

Mind Map and brain

Our brain is a processing unit and all the information which we gather is processed there. Now, think about how our brain reacts when we sense some fragrance of flowers or pay attention to music. The response is both simple and amazingly complex.

Every bit of information incoming to the brain, all sensations, memory, or thoughts, that includes the word, numeral, code, food, colour, picture, beat, note, and consistency can be signified as a central sphere from which emit tens, hundreds, thousands, even millions of knobs or hooks. Each hook or neuron characterizes a connotation, and each connotation has its individual unlimited array of relations and associates. From this enormous information, processing capability, learning, and dimensions, which derive

the idea of Radiant Thinking, are born and organized by our brain. Mind Map is a manifestation of the same working process.

Brain's Radiant Thinking pattern may, therefore, be seen as a massive dividing connotation machine. Just imagine it as a super bio-computer with outlines of thought radiating from an almost infinite number of data lumps, as shown in [Figure 1.2](#):

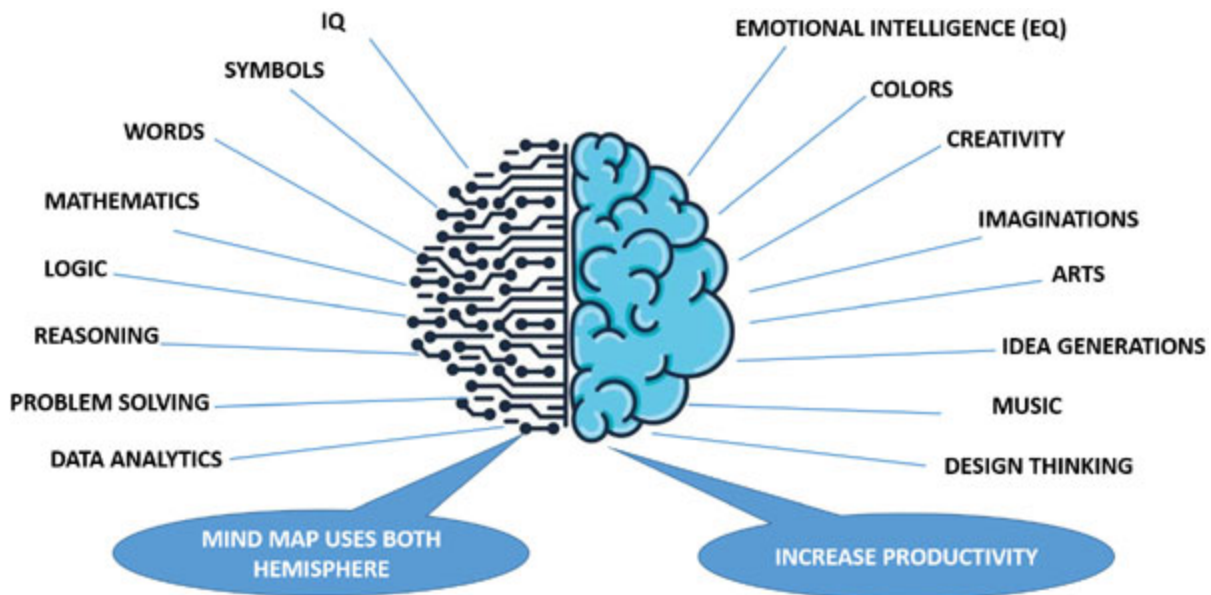


Figure 1.2: How our brain organizes information

Scientific research already established the mind map's rationality as a brain-compatible thinking technique. Scientists already confirmed that the evolutionarily newest portion of the brain, the thinking cap of the Cerebral Cortex (part of the brain), was separated into two main hemispheres, and those hemispheres did an all-inclusive variety of intellectual tasks, termed cortical skills. The tasks comprised: Reason, Beat, Lines, Colour, Lists, Daydreaming, Numbers, Imagination, Word, and Gestalt. The left hemisphere is responsible for all sorts of intellectual and reasoning tasks, whereas the right one is responsible for creativity ([Figure 1.2](#)).

The research established that the more these doings were combined, the more the brain's act turns out to be cooperative, with each knowledgeable skill enhancing the act of other rational areas. When you are Mind Mapping (a creative technique which we will learn in a while), you are not only

working out the essential memory powers and information dispensation, but you are also consuming your complete array of cortical skills.

The Mind Map is made even more influential by the practice of all the left and right brain-thinking tools, which improve the clarity, structure, and association of our thinking. And since the Mind Map beneficially uses the individual's imagination, connotation, and place identification capabilities, it could also think as a tool for combining the thought process of the left and right brain. We can contemplate the Mind Map as the eventual thinking tool that joins all the noteworthy and strong conducts of rationale into its individual structure and map it to creativity.

How to create a mind map

All mind maps commence with a chief concept or idea. And the remaining map rotates around it; so selecting that idea or topic is the first and crucial stage. We can start by making an image or writing a word that characterizes that first main idea or theme. From that main idea, divisions could be generated, which signifies a solitary word that relates to the chief topic. It's good to use dissimilar colors and images to distinguish the divisions and sub-topics. Then, one can draw sub-branches that shoot from the chief branches to additionally enlarge on ideas and concepts. These sub-branches will also comprise words that expands on the topic of the branch its trunks from. This helps advance and expand on the mind map, together with descriptions and drafts that can also be considered in brainstorming and creating the sub-branch topics. The process of mind map creation is depicted in [*Figure 1.3*](#):

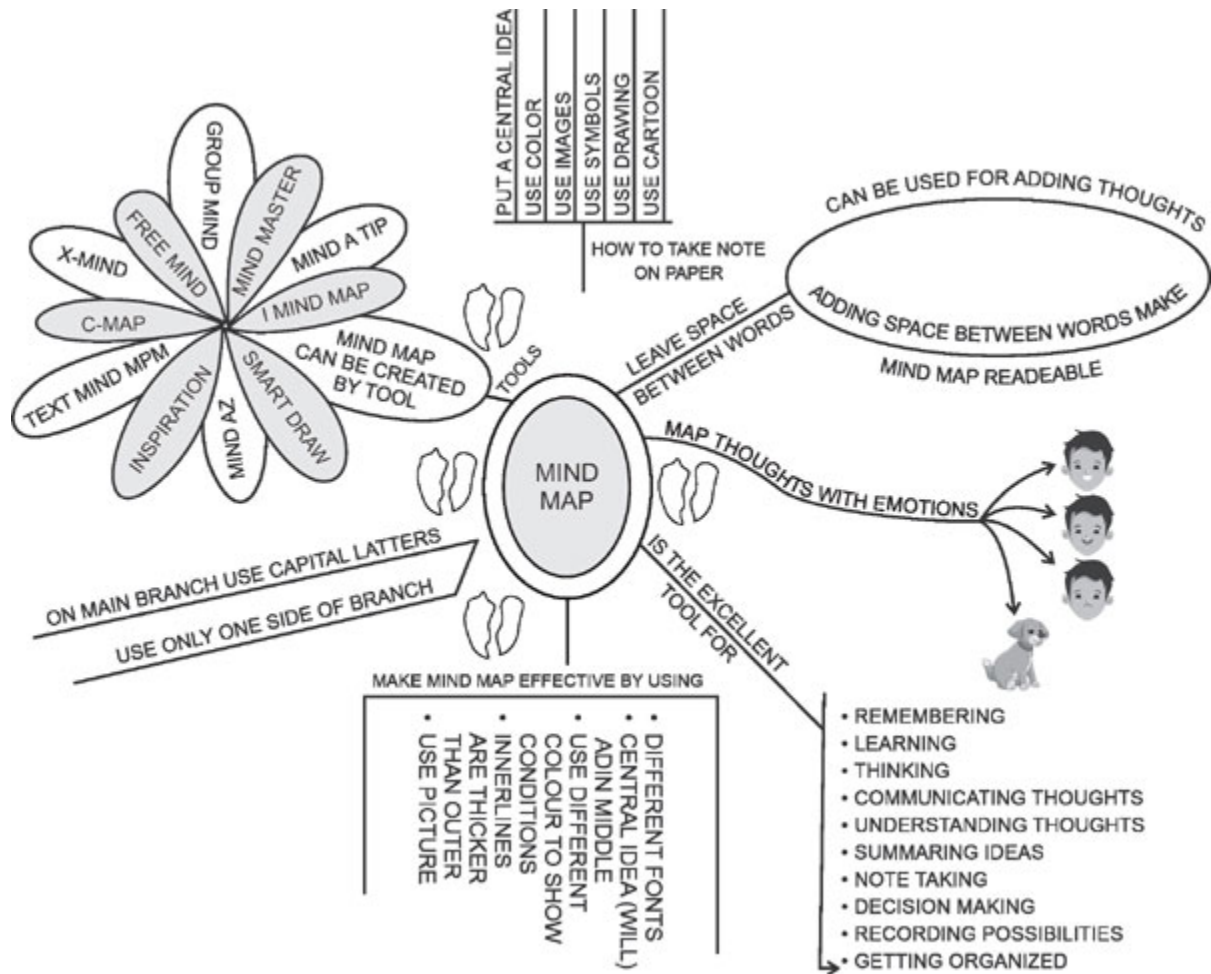


Figure 1.3: How to create an effective mind map

Mind maps can be shaped on paper but are more effortlessly and gracefully shaped on a computer with mind mapping software. Few mind mapping tools are Mind Master, X Map and so on.

Point of View (POV)

A **Point of View (POV)** is a meaningful and actionable problem statement, which will allow us to ideate in a goal-oriented manner. Someone's POV captures design vision by defining the RIGHT challenge to address in the ideation sessions. A POV also involves reframing a design challenge into an actionable problem statement. Good articulation of a POV can be done by combining knowledge about the subject and providing insights. In a nutshell, a POV is a combination of three elements – user, need, and insight.

In this book, the reader will find multiple POVs which are spread across the chapters. The POV presented in the chapters is sometimes deferred from its original definition. Here, they are used as a tool for capturing the vision of an individual or group of people related to a subject so that the reader becomes familiar with the concept of a product.

Note: POV: Tony Buzan on Mind Map

Tony Buzan is one of the strong proponents of a Mind map. He wrote multiple books on the subject and popularized mind mapping techniques to its core. The official website of Tony Buzan described Mind Map as:

An influential visuals method that offers a universal key to reveal the potential of the brain. It harnesses the full range of cortical skills including word, image, number, logic, rhythm, colors, and spatial awareness—in a single, uniquely powerful manner. In order to do so, it provides you the liberty to rove the immeasurable areas of your brain. The mind map can be applied to each facet of life where better-quality learning and vibrant thinking is required to improve human act.

The mind map about Leonardo da Vinci (he was one of the greatest innovators who does make important discoveries in mathematics, anatomy, engineering, cartography, sculpture, botany and geology) is depicted in [Figure 1.4](#):

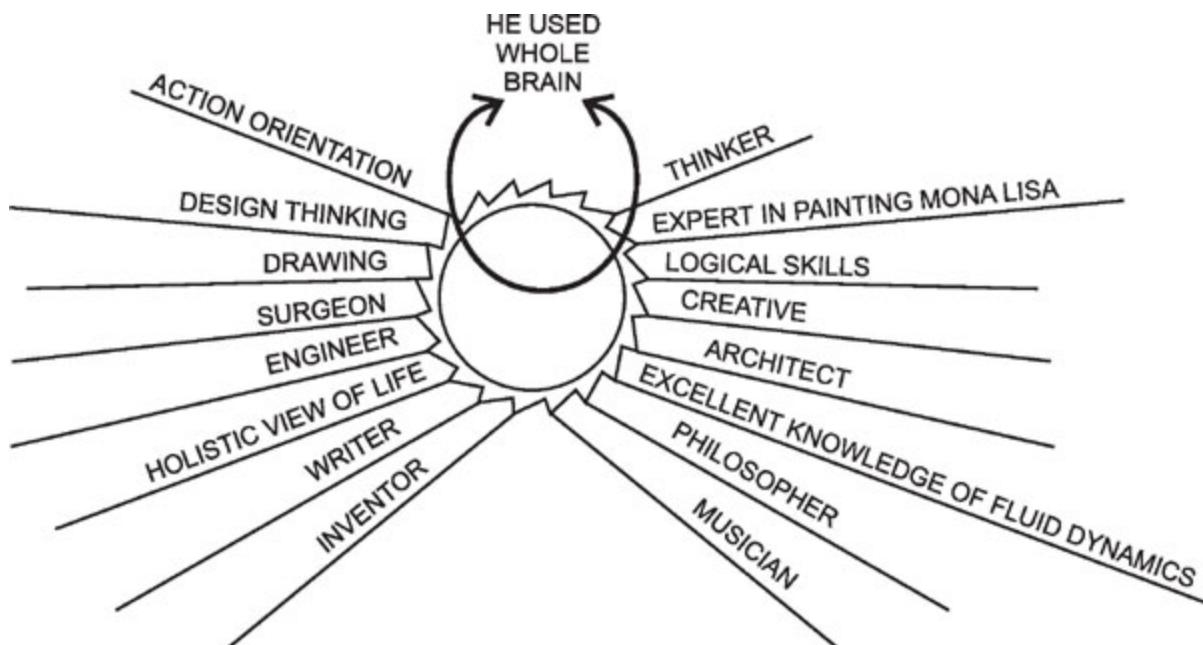


Figure 1.4: Mind map about Leonardo da Vinci

Expert opinion

Expert opinion is a comparatively casual method which can be used to serve a variety of purposes and may be used to support in problem identification, in expounding the problems applicable to a specific topic, and in the assessment of products. Expert opinion is often used to classify possible difficulties with products before they are released for a more comprehensive evaluation but can be used at any stage of design. However, it is significant to safeguard that those experts consulted have no prior involvement or interest in the design of the product to be evaluated; otherwise, it will be difficult to obtain impartial views.

The expert opinion presented in the chapters of this book is sometimes deferred from their original definition. Here, they are used as a tool for capturing the opinion of an individual or group of people who are expert in the subject so that the reader becomes familiar with the expert opinion about a product.

Jargon busters

Jargon-buster is an adjective, a word, such as weighty, beautiful, or strong, that is used to describe a noun, adverb, or a word. Also, terms that is used to provide additional information about an adjective, verb, or another adverb can fall under this category. Jargon Buster is intended to aid the reader to clean up through complications and directly provide the meaning, clarification, and content that he or she desires.

Jargon mentions to a gathering of domain-specific terminology with exact and dedicated meanings. Typically, this section in the book clarifies some frequently used jargon that's explicit to quantum computing, artificial intelligence, and associated technologies.

The jargon buster segments are very significant in a book like this, as there is a lot of jargon linked with these know-hows. This section is destined to aid the readers to comprehend and decipher the exact jargon in a rapid and precise way.

Customer stories, case studies, and use cases

Nowadays, corporates are giving a lot of focus on customer stories, case studies, and use cases. The corporates are using them as tools to capture and explain the real achievements and prospects of a company's product, vision, and roadmap.

Practically these can be used to analyse and display vendor competences. Their benefits are many if they are used correctly. I used customer stories, case studies, and use cases in the book to explain and provide real-time problem situations, hurdles, and issues along with solutions which created success for them.

Further, these can also be used for providing the actions taken on issues/problems by a respective corporation and showcasing the consequences of the action to the inner and outer world. It also explains why the product or idea is the best for a specific purpose. Let's see them in a bit of detail, as follows:

- Case studies explain the details of a customer's situation, their problems, and the process by which those problems were addressed. Case studies sometimes include use cases, but often they are more condensed than a use case, if available on its own. Case studies can be released independently, listed on a website, made available on the blog, or even presented in the form of videos.

This book provides some prodigious case studies. These case studies will enable the reader to know about a corporation's journey on quantum and associated technologies. Mostly, case studies are provided in the form of stories. Why? Because everyone wants to hear stories, not just mere facts. The stories include but are not limited to (or strictly revolve around) the following items:

- Who is the example client and what do they do?
- What were the client's goals?
- What were the client's requirements?
- How did corporates gratify requirements and aid their client to encounter their goals or product vision?

A customer story establishes the paybacks of products and services of a company to the wider world. It is not just a tool for presenting facts. This type of distinctive presentation can be the rotary point that aids a view to envisage in a content client's shoes for a corporation.

Typically, the emphasis of success stories of a company is on the accomplishment or outcome of a specific product or products or services. Sometimes it is about the company's achievement as well. Typically, customer accomplishment or success stories are quicker in comparison to case studies. These days, speaking about customer stories has to turn out to be a trend. Each corporation has a distinct section on their website for this. In this book, I kept my attention on actual customer stories and used them as and when needed. In this book, you will find some great customer/ client success/achievement stories, sometimes they are part of the case study, whereas on occasions, they are covered alone as well.

Typically, a use case describes exactly how the application is implemented and why the product or idea is the best for the job. Use cases are truthfully decent from marketing to technical recipients, chiefly for experts who may have a prodigious knowledge and understanding of the know-how, but cannot comprehend the precise product to know why it's the best fitting for a specific state or condition. By understanding use cases, decision-makers can learn more about how the product is precisely distinguished.

[Industry bites](#)

Quantum computers are developing and rising fields of study. The businesses and enterprises around the technology is maturing, disappearing, and growing at a very quick speed. Therefore, it is natural that anyone who is a core developer or leader or is playing any other role which includes some kind of informed decisions making related to the know-how must be based on the progress of the field (technology, coding standards, implementation strategies). Hence, the expectation is the decision must be as simple. It however ranges from -- which toolkit would be used for coding or as complex as, whether can we invest in the technology or not. This state needs alignment of ideas and thoughts with the quickly changing industry.

Hence, to pay attention to other voices, if envisaging peer plans become important, then only a competitive and informed decision would be made.

But the biggest challenge is that not much information is offered in the combined and central form on the easily available platforms like LinkedIn, web portals, and other mediums of knowledge, not even in the research literature and resources like books (including online). And if you find something, you have to pay an ample amount of cost for using them. This book has come as a rescuer and frees you from these headaches. The industry byte section of the book tries to fill those gaps which were mentioned earlier.

The industry byte delivers relevant, scientific, up-to-date and modern information related to industry happenings. Also, it's coverage is not only limited to technical stuff. It goes beyond technology boundaries and provides some quick statements apart from core areas of quantum computing and artificial intelligence such as leadership, organizational psychology, and behavioral sciences.

Thought cartoons / cartoons / TechToons

Quantum computers, Artificial intelligence, big data, cognitive, and cloud computing are technically complex subjects and sometimes become boring and difficult to digest and interpret. In a situation like this, TechToons, thought cartoons, and cartoons serve as a very important tool because of their unique way of presenting thoughts and concepts. They have the capability and potential to explain ideas in an easy and humorous way. Also, they are visual and graphical in nature, so their impact on our brain's positive chemicals is faster and long-lasting. Let's see what they are.

What are TechToons?

My definition of TechToons for the purpose of this book goes like this – if any technical, process or people-related ideas come from the world of science, technology, and business and are represented in the form of humorous drawings with or without comments, it is termed as TechToons.

What are cartoons?

The formal definition of cartoons comes as – a drawing in a newspaper, magazine, or book intended as a humorous comment on something.

What are thought cartoons?

Any technical, process, people or life-related ideas which come from the world of science, technology, business and any other field of human endeavour that have capabilities to triggers the thinking in the brain is termed as thought cartoons. It can be represented in the form of humorous drawings which may or may not have comments.

What are the differences between cartoons and TechToons?

TechToon is specific to technology, science, and business, whereas cartoon is generic and its scope can be anything varying from politics to the animal kingdom. To make it more technical, we could say that a cartoon is a superset and TechToon is a subset of this.

A few examples of these are depicted in [Figure 1.5](#) and [Figure 1.6](#):

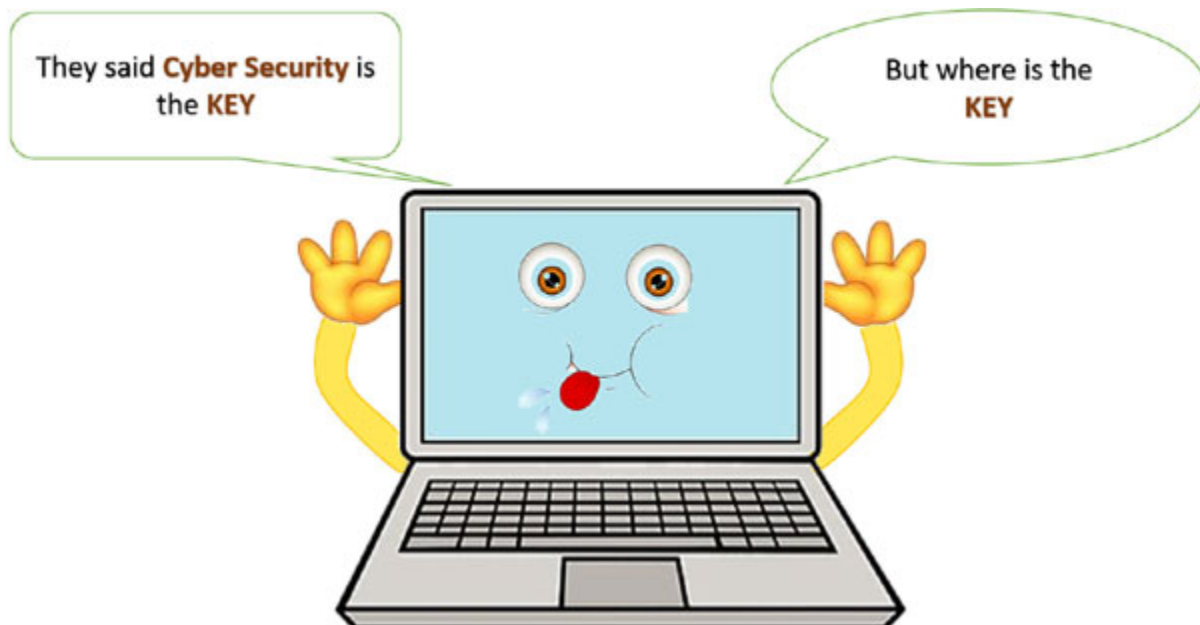


Figure 1.5: Tech Toons

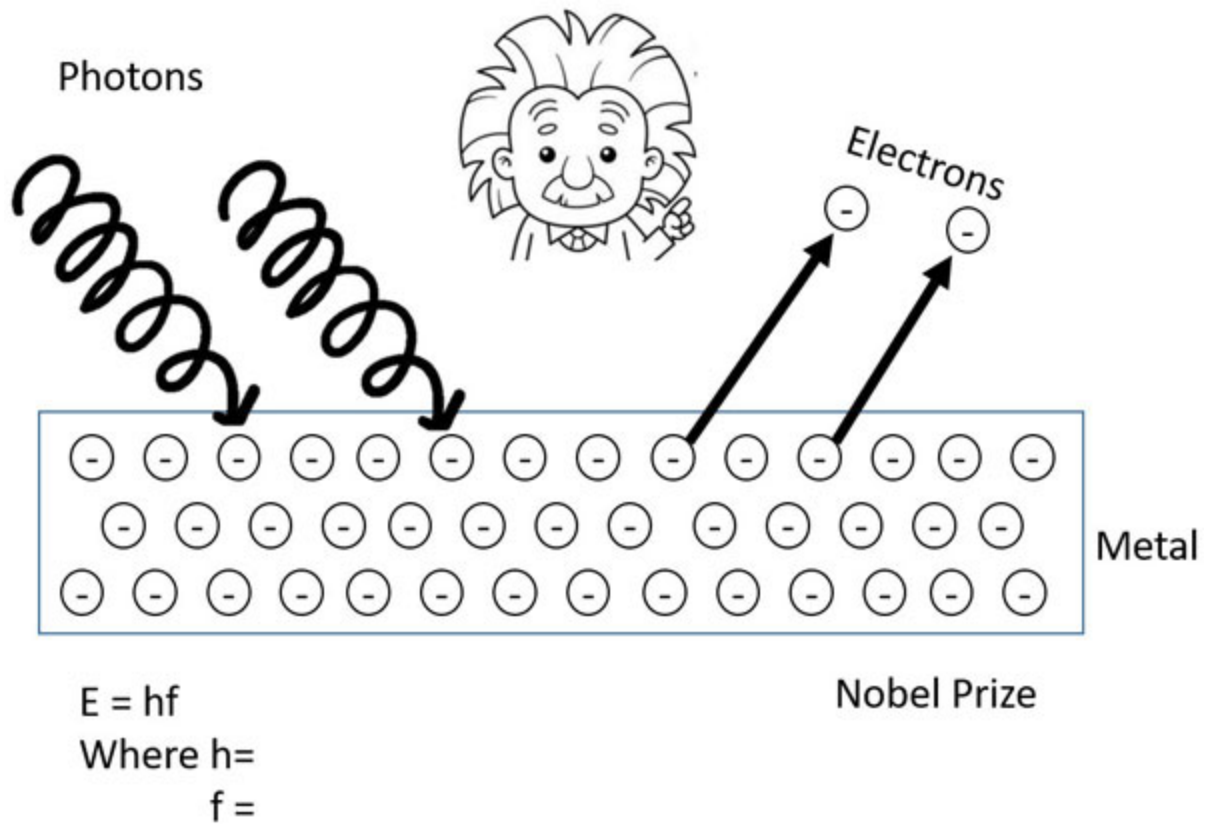


Figure 1.6: Thought cartoons

Mind Map of the chapter

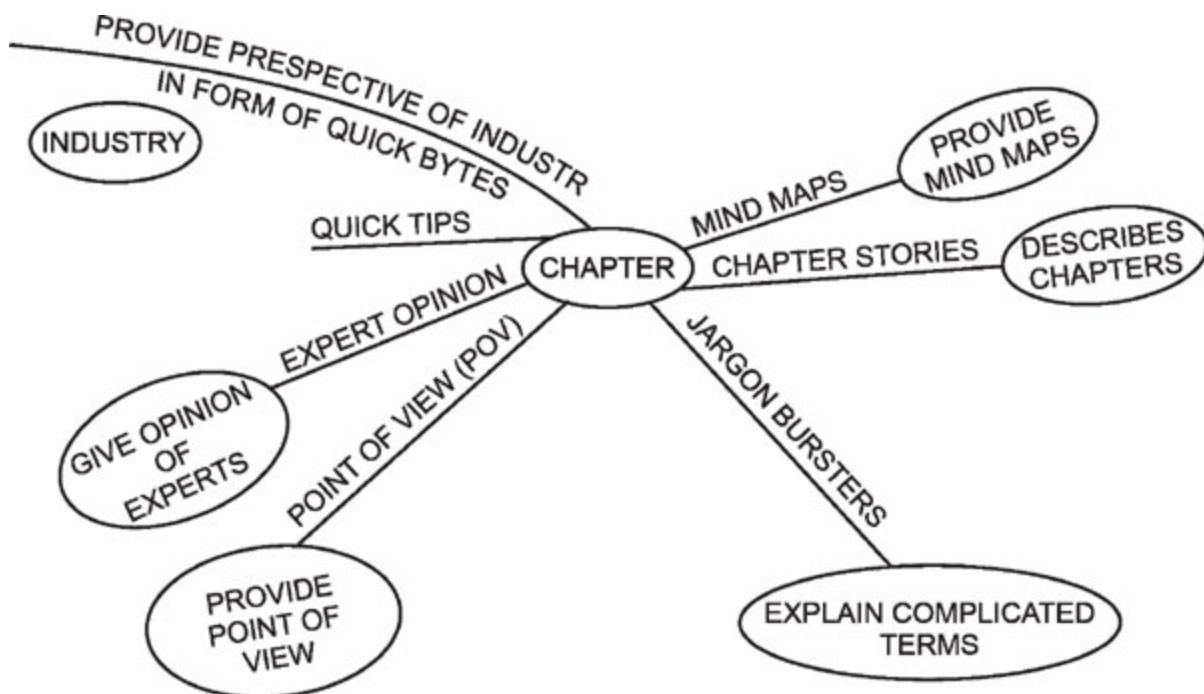


Figure 1.7: Mind map of the chapter

Conclusion

This chapter introduced multiple tools and techniques which are used in the book. Broadly, these tools are used for improving the productivity of the reader. Several tools are discussed in this chapter, including mind maps, jargon buster, and point of view.

CHAPTER 2

Quantum Physics as an Enabler of a Quantum Computer

“The world is running out of computing capacity. Moore’s law is running out of steam ... [we need quantum computing to] create all of these rich experiences we talk about, all of this artificial intelligence.”

—Satya Nadella, Microsoft CEO

“Quantum computation will be the first technology that allows useful tasks to be performed in collaboration between parallel universes.”

—David Deutsche, Physicist at the Centre for Quantum Computation, Oxford University

In the last four decades, there has been a melodramatic contraction in computer technology. We already have the elementary memory components of a computer to the extent of individual atoms. At such scales, the mathematical theory behind modern computer science will turn out to be useless. Hence, in the quest of finding solutions, computer scientists have seen hope in something called quantum computers. This is a new-fangled field of technology which is developing at a very fast rate at the moment. Computer scientists, mathematicians, and physicists are together looking for a breakthrough in the field, so that technology would become available for companies and individuals quickly and more easily.

Now, let’s discuss the quantum computing. The name of the area of study is related to – “quantum computing” technology. So, we can figure out intuitively that there must be some relation between the “quantum” and “computers”. Further, “quantum” is the study matter of “quantum physics”. Which is a branch of physics. Then how does this is associated with “computers”? Finding the answer of this question is the focus area of the chapter. We would investigate this million-dollar “curiosity” in this chapter.

Structure

Broadly, this chapter could be divided into three parts. The first part will set the context of the need and requirements of the chapter. Whereas, the second part will cover the fundamentals of quantum and modern physics. Finally, the third part will provide insights into the interrelations between quantum physics and computers. It gives some thought-provoking use cases and applications of quantum computers. Apart from this, it highlights some management and leadership practices that will be required for any organization that wants to implement quantum computing practices within its DNA.

In this chapter, we will discuss the following topics:

- Objectives
- Introduction
- From quantum information to quantum physics to the quantum computer
- Quantum Computers
- Why we need Quantum physics for quantum computers
- Quantum 1.0 and 2.0
- Quantum cryptography
- Quantum Computer and AI
- Classical Physics, Modern Physics, Quantum physics, and mechanics and their relationship
- The birth of quantum physics
- Photons and quantum computers
- What is Photoelectric Effect?... by the way
- Let's summarize: some fundamental concepts of modern physics and radiation
- Particles of light
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 - Coherence
 - Decoherence
 - Fault tolerance
 - Interference
 - Tunnelling
 - Randomness in behaviour
- Uncertainty principle and how it is useful in QC
- Young's double-slit experiment
- The interrelation between entanglement and coherence
- Before quantum physics, there is no law... right ...!!!
- Why quantum computers ... now
- Sample opportunity areas for quantum computing: explanations for few important ones
- The social and commercial effects of quantum technology
- Story 1: quantum organization and leadership
- Quantum computers: just a piece of a holistic system
- Facts from research, innovation, and industry
- Some questions and answers
- Human brain and quantum computers: The QuBrain project
- Mind Map
- Summery
- Key terminology
 - IT terminology
 - Qubits associated terminologies
 - Quantum Physics terminology

Objectives

After studying this chapter, you should be able to understand the concept of the quantum computer, and be able to discuss the need for it and why it is required. Also, you would be able to explain why and how quantum physics is the foundational block of it. At the same time, you become familiar with some of the use cases for it.

Introduction

Quantum information processing is an area of study that comprises quantum computing, quantum cryptography, and quantum communications. Further, this discovers the benefits of using quantum mechanics as an alternative to traditional mechanics to model information and its processing. Quantum information processing is not about altering the bodily substrate on which multiplication is done from traditional to quantum but somewhat changing the idea of computation itself. The alteration jolts at the utmost elementary level of information, that is, 0 or 1. In the quantum world, the essential unit of computation is no more the bit, but somewhat the quantum bit or 0, 1 or something in between of 0 and 1. Employing computation on a quantum mechanical underpinning is directed to the detection of quicker algorithms, novel cryptographic mechanisms, and better-quality communication protocols. This chapter uses a lot of technical terms therefore, it is good to have some idea of important concepts beforehand. In sections “jargon buster,” you will find terms which make your reading easy. Even if you are not getting the terminologies completely, do not worry... in upcoming sections of the book, you would find a detailed description of them.

Jargon Buster

- A Turing machine is a math-based model of computation that describes an intangible machine, which manipulates symbols on a ribbon of tape conferring to a table of rules. Despite the model’s uncomplicatedness, for any specified computer algorithm, a Turing machine is proficient in simulating that algorithm’s logic.
- A quantum Turing machine (TQM), to a universal quantum computer, is an intangible machine used to model the consequence of a quantum computer. It delivers a very simple model that imprisons all of the power

of quantum computation. Any quantum algorithm can be articulated properly as a specific quantum Turing machine.

- A quantum Turing machine (TQM) is also known as a universal quantum computer.
- A logic gate is a physical electronic device used for applying a Boolean function. A logical operation is done on one or more binary inputs that crop a sole binary output.
- An electronic circuit is made from distinct electronic components, such as resistors, transistors, capacitors, inductors, and diodes. These components are connected by conductive wires through which electric current can flow.
- An algorithm is a step-by-step process to do a calculation, or an order of instructions to resolve a problem. Technically, each step of an algorithm can be done on a computer. Consequently, an algorithm may be called a quantum algorithm when it can be done on a quantum computer. In principle, it is possible to run all classical algorithms on a quantum computer. Though, the terminology quantum algorithm is pragmatic to algorithms, of that at least one of the phases is definitely ‘quantum’ in nature. Meaning, it can use quantum property like superposition or entanglement.
- In the early 1990's, researchers developed the first truly quantum algorithms. Despite the probabilistic nature of quantum mechanics, the Shor algorithm was the first quantum algorithm, for which superiority over classical algorithms was proved. It had given the correct answer with certainty for the calculation of prime number factorization. They improve upon classical algorithms by solving in polynomial time with certainty – a problem that can be solved in polynomial (an expression of more than two algebraic terms, especially the sum of several terms that contain different powers of the same variables) time only with high probability (the extent to which an event is likely to occur, measured by the ratio of the favourable cases to the whole number of cases possible) using classical techniques. Such a result is of no direct practical interest since the impossibility of building a perfect machine reduces any practical machine running any algorithm to solving a problem only with high probability. But such results were of high theoretical interest since they showed for the first time that quantum computation is theoretically

more powerful than classical computation for certain computational problems.

- Traditional or contemporary or classical or old style computers --- these terms are used interchangeably throughout in this book.
- To represent "square". For instance, M square N -- M^N or M^N is used interchangeably throughout in the book. Also, in this chapter and in chapter four, concepts like Superposition, Interference, coherence, decoherence, fault tolerance and entanglement are used very frequently. Therefore, if you want to get a grip of the content from the beginning please refer to "Fundamental concepts of quantum physics". The same is looked at as a separate section in this chapter that deals with them in detail. For a shorter version and quick reference, you may refer to "Qubit associated terminologies", at the last of [chapter 2](#). However, going through it early is not necessary. Even if do not go through these concepts at the start you learn them progressively, provided you follow the natural flow of the chapter and the book.

[From quantum information to quantum physics to quantum computer](#)

The field of quantum information processing advanced in the decade of the eighties and early nineties as a minor group of researchers, scientists, and investigators worked out a theory of quantum information and quantum information dispensation. At the most fundamental level, it started when David Deutsch advanced a concept of a quantum mechanical Turing machine. Then the model was enhanced by several innovators and scientists. Further, it was recognized that a quantum Turing machine could put on a classical Turing machine. The standard quantum circuit model was then defined, which led to an understanding of quantum complexity in the footings of a set of elementary quantum revolutions named quantum gates. These gates are theoretical concepts that might or might not have a straight analogue in the physical components of an actual quantum computer.

Quantum physics provides an exact description of the electrical conductivity of materials, including semiconductors. There is something called a micro transistor which is nothing but a mixture of incapacitated semiconductor elements. Their style of operation is mainly determined by the movement of

electrons inside them. These conclusions are achieved with the help of the laws of quantum physics. Semiconductor machinery are the building chunks of all electronics, and certainly, the whole computer and information technologies that impacted our lives so deeply today are the by-product of this. In “integrated circuits”, transistors are wrapped in billions of minor chips so that extremely complex electronic circuits can be interrelated. At the moment, the separate fundamentals of these integrated circuits contain a few dozen atomic layers; whatever takes place in them follows the laws of quantum physics. The reason of recapitulate these highlights of the technologies is to remind us that quantum physics is already part of the technical developments. Now for building successful quantum computers the core concepts of quantum physics like interference, entanglement, and superposition along with something called “thermodynamics” are used with their true potential. So now it is clear that quantum physics is the blood that flows in the nerves of QC. But why so?

Quantum computers?

In this section, we will proceed in the way same is described in the [*Figure 2.1*](#) why? Because it is necessary to understand the concepts in a structured way, as shown in [*Figure 2.1*](#):

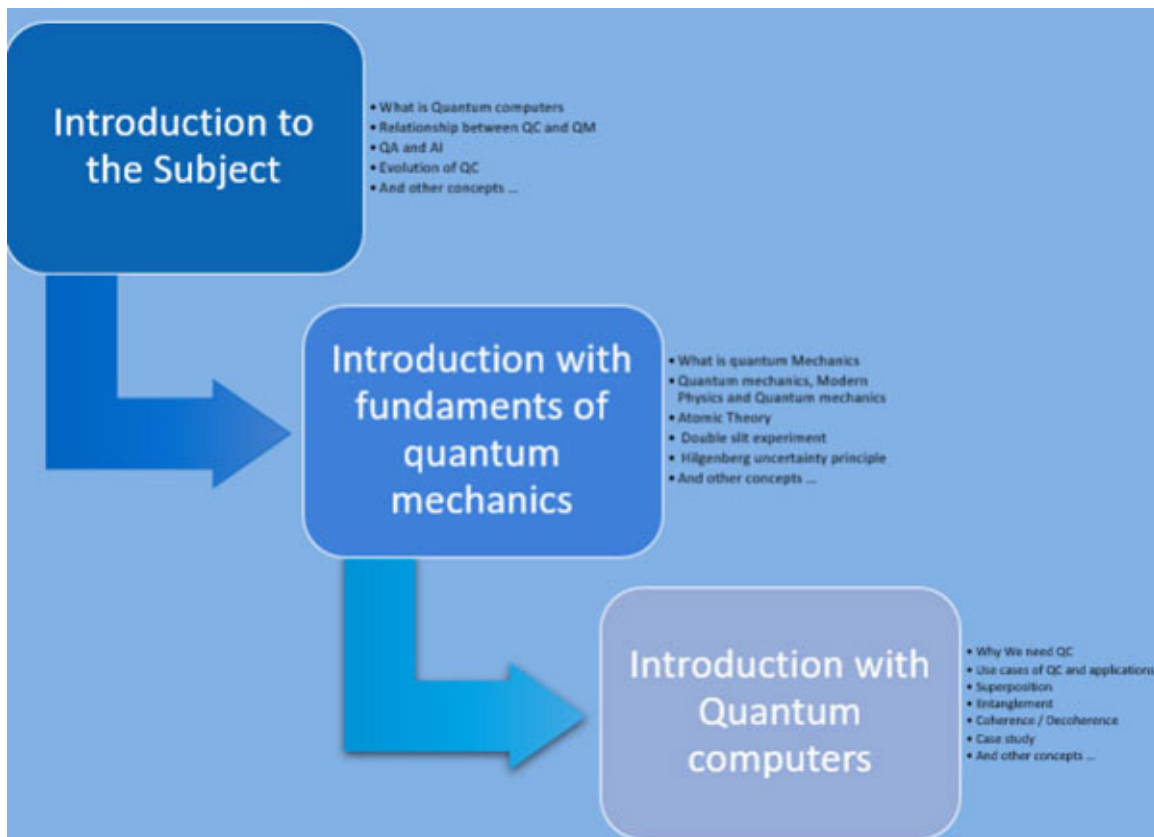


Figure 2.1: Brief plan of the chapter

Quantum Computer (QC) uses quantum mechanics to perform operations on data, and vary from the ‘classical’ binary computers that we use nowadays. A quantum computer is founded on the knowledge of the ‘quantum bit’ or qubit. In a traditional computer, the elementary constituent is the bit, which is either on or off (0 or 1). However, a qubit can occur in both states at once, hence, due to quantum superposition of 0 and 1 happen similar time ([Figure 2.2](#)). By entangling (can be done through Laser manipulation) several qubits together, we can attain the ultimate parallel processing, resounding out all calculations concurrently. We can also visualize quantum computing as an amalgamation of quantum physics, computer science, and Information Theory, as shown in [Figure 2.2](#):

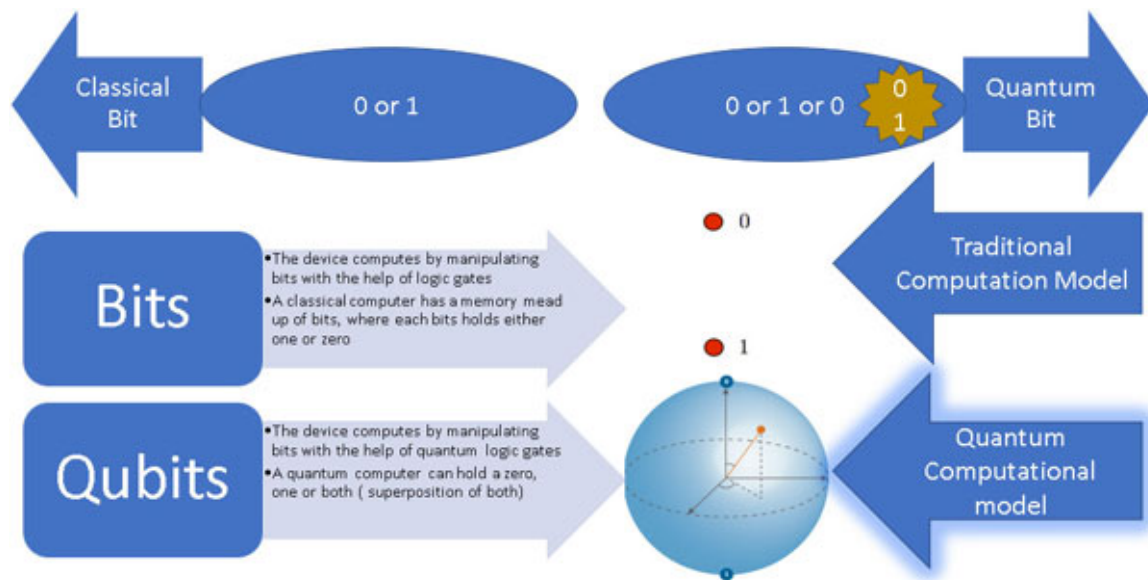


Figure 2.2: fundamentals of bits and qubits

As the quantum computer is one of the evolving fields, hence in the future, how real-world quantum computing and quantum information ultimately take shape is still unknown. No ultimate physical principles are identified that forbid the construction of large-scale and dependable quantum computers. Multiple engineering problems still are continuing in space. However, laboratory experiments have confirmed quantum computations with quite a few quantum bits carrying out lots of quantum operations. Innumerable hopeful methods are being discovered by logicians and experimentalists, but there is much doubt as to how and when a quantum computer proficient in resonant general quantum computations on hundreds of qubits will be built.

Quantum computers can be capable of performing certain tasks several times more rapidly than even the greatest powerful classical computer right now. They are consequently expected to serve various significant requests concurrently. There are presently several approaches to building a practical quantum computer that achieves concurrency with ease. For this, all trust in the impression of operating entangled superpositions of atoms, but all approaches eventually suffer from a similar problem that is, how to stop these subtle superpositions from dripping away and decohering (if you are not getting few terms here, no problem, we will cover them in detail in the upcoming sections and chapters).

The expression quantum computing is closer to analogue computing since the computational model for analogue computing varies from that of normal

digital computing. In analogue computing, we commonly have a range of values, rather than a disconnected set. While the expressions are matching, the two models diverge momentarily. As we will study in the forthcoming sections that analogue computation does not provision entanglement, an important resource for quantum computation, and measurements of a quantum computer's registers can harvest only a minor, isolated group of values. Additionally, while a qubit can take on a gamut of values, in several methods, a qubit bears a resemblance to a bit with its two distinct values, more than it does analogue computation.

Further, quantum computing can do far more things than just resolving commercial optimization glitches. Quantum computing can also be used to precisely model the natural world. For instance, in molecular chemistry, understanding molecules and being able to simulate their behaviour is vital to drug discovery. Yet, the finest supercomputers in the world can only simulate a scheme of about 43 electrons, which is appallingly insufficient for precisely designing the pharmaceuticals, vaccines, and antibiotics that will be vigorous. Quantum computers appear exclusively well-matched for resolving this kind of problem(s).

Hypothetically, the calculations of quantum theory can be used to compute any procedure in the world. Hence, the same expectations are from QCs as well because they are based on the same principles. Though even for simple molecules, the calculations are so multifaceted that they necessitate the wildest computers obtainable today, and physicists must yet gratify themselves with only estimated outcomes.

[Why do we need quantum physics for quantum computers?](#)

The compelling experts are working in area of development to integrate the basics of computer science with quantum physics. The reason for this holistic approach is simple – quantum computers are based on the fundamental laws of Quantum physics and computer science (but not limited to!). Also, we have to keep in mind that quantum physics is the utmost precise model of nature's realism that is presently known to us. Therefore, quantum computers may become one of the tools to realize nature.

To get a good understanding of quantum computing, there is a need for an elementary understanding of quantum mechanics, which includes the behaviour of light, photons, entanglement, superposition, interference, and multiple other concepts. We will go through these concepts in the latter part of the chapter. Quantum computers use quantum physics, which is essential to make them functional. The following concepts sit in the heart of quantum computers:

- **Linear algebra:** Quantum computers use linear algebra. A lot of Quantum Computing is linear algebra; hence, its knowledge would help in understanding Quantum Computers. The entire Quantum Mechanics works on Hilbert Space (allow generalizing the methods of linear algebra) and the math of (quantum Mechanics) is just linear operations on Hilbert space. Matrix, transpose, conjugate transpose, linear combination, basis, eigenvalue, eigenvector, inner product, matrix power series, and matrix exponential are all essentials of Quantum Computing (You will find details of these in [chapter 3](#)). Hilbert space outspreads the procedures of vector algebra and calculus from the two-dimensional Euclidean plane and three-dimensional space to spaces with any limited or unlimited numeral of dimensions.
- **The Fourier transform:** This happens in numerous dissimilar forms throughout classical computing; it includes areas oscillating from signal processing to data compression to complexity theory. The **Quantum Fourier Transforms (QFT)** is the quantum application of the discrete Fourier transform over the amplitudes of a wavefunction.
- **Many-body systems:** Are significant in QC because entanglement depends on it. It is also essential to understand the limitations of quantum computing.
- These concepts are above and beyond the core concepts like superposition, Entanglement, and other core quantum physics properties.

Before going into the details of quantum physics and computers, it is good to have a running look at the major areas of our life which will be affected by quantum computers.

[Quantum 1.0 and 2.0](#)

Knowingly or unknowing, we have encountered quantum technology in ordinary life, for instance, computers, LED, solar cells, lasers, and so on. Though, these technologies only use part of the properties of quantum mechanics and is communally known as quantum technology 1.0, as shown in [Figure 2.3](#):

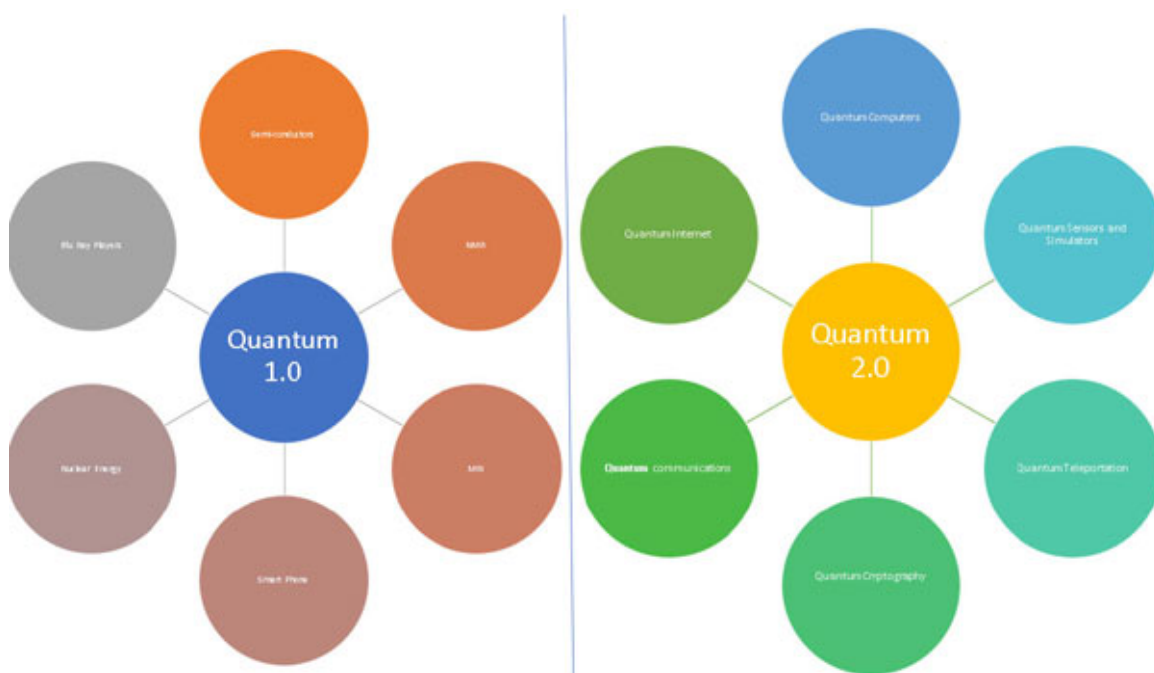


Figure 2.3: Quantum 1.0 and 2.0 – applications

In quantum technology 2.0, discrete atoms are operated to harness the complete capabilities of quantum mechanics. Hence, the new generation of devices that are based on quantum theory, its principles, and experiments that use the full potential of quantum physics are sometimes called Quantum 2.0. These types of devices are becoming famous nowadays because of their capability to bring changes in our lives. In general terms, Quantum 2.0 broadly signifies the new group of devices that are based on quantum technologies. Refer to [Figure 2.3](#) for the broader application areas of quantum 1.0 and 2.0.

The outcome of the first quantum revolution was LASER, microchips, and MRI scanners, whereas Quantum technologies, attached themselves with 2.0, include a class of devices that can operate striking states of matter through quantum superposition, tunnelling, or entanglement. Developments in areas such as quantum information theory, quantum electronics, quantum optics, and nanotechnology are serving to grow such devices. Quantum 2.0 gave us extremely precise sensors, atomic clocks, quantum processors, and safe

communication tools. Quantum 2.0 often has a very close companion, an emerging area called quantum cryptography. It would potentially affect the way we communicate now and keep our information safe. In a nutshell, quantum 2.0 applications can be divided into the following four major types:

- **Quantum computers:** A novel kind of computer that can enormously outdo traditional computers; it is based on qubits which can be “on and off” at the same time.
- **Quantum simulators:** Specialized quantum computers which are designed for material, drug design, and simulating the essential properties of elementary physics or broadly nature itself.
- **Quantum communication:** Quantum communication is a field of applied quantum physics. Quantum teleportation and quantum information processing are two areas of practical quantum physics that are strongly related to quantum communication.
- **Quantum sensors:** It is generally a super-high-performance sensor. This gives quality-new data about our world. Further, this data can be used to learn important things about our surroundings. This will support advancements in the Internet of Things, brain imaging, driverless vehicles, and navigation.

Within the last two decades, quantum technologies have made tremendous progress, moving from Nobel Prize award-winning experiments on quantum physics into a cross-disciplinary field of applied research. International Investment of more than 30 billion US dollars has been raised by governments and private companies such as Google, IBM, Intel, Microsoft, Alibaba, Huawei, Airbus, and many more.

[Quantum cryptography](#)

Sometimes, Quantum 2.0 uses quantum cryptography, which is again a very thrilling area of development. Quantum cryptography is the discipline that takes advantage of the quantum mechanical properties of quantum physics to do cryptographic tasks. The best-known example of quantum cryptography is the quantum key distribution that bids information hypothetically and provides a protected resolution of information by the key exchange. Quantum cryptography is the single known technique for communicating a secret key over a distance that is secure in principle and grounded on the rules of quantum

physics. It permits two parties to produce a common random secret key known only to them, which can then be used to encrypt and decrypt messages.

Quantum cryptography, for instance, trusts something named quantum key distribution which promises secure communication. Since the key in Quantum cryptography is used to encrypt and decrypt messages and information which depend on a couple of quantum-entangled particles, to get the key, any listener or hacker would have to interrupt and gauge one of the entangled particles. However, in order to do so, they unavoidably abolish the subtle quantum state, thus hovering the alarm. Thus, they would be unable to decrypt the information or message. To understand it better, let's take a small case. Before going into the details of the case, just keep in mind that in the following flow, Gopal represents the sender, whereas Madhav signifies the Receiver. OK, so here we go...

Assume Gopal wishes to transfer a 'secret' to Madhav. Gopal mops his top secret in a treasure trunk, an analogous treasure trunk is somehow formed in Madhav's room, Gopal expresses which kind of key he used to lock it and Madhav uses the same key to crack the treasure trunk in his room and gather the secret. Classically, it looks impossible. However, in the world of quantum physics, it is possible. Due to the properties of Quantum mechanics, it is likely to 'entangle' electrons. Therefore, if you have an electron in one state (accredited by its spin, a property that can be detected), one can produce an 'entangled electron' miles apart, and knowing the state of one's electron, he can automatically infer the properties of the other.

What occurs in quantum cryptography is, as Gopal entangles his electron which is taking the message with Madhav's electron, the message is attached or encoded by applying specific actions on the state of the electron. This tells the states, along with the message carried by the two electrons. Henceforth, even if there is no physical movement of data, information has moved diagonally to the distance. Now Gopal 'classically' connects the state of his electron to Madhav. And knowing the state of Madhav, he puts on operations on his electron state to decipher the message. After all, any listener can't perhaps chop any information since even if he has the key, there is no treasure trunk to open.

Broadly, quantum cryptography is a two-step communication between the dispatcher or sender and the receiver, as shown in [Figure 2.4](#):

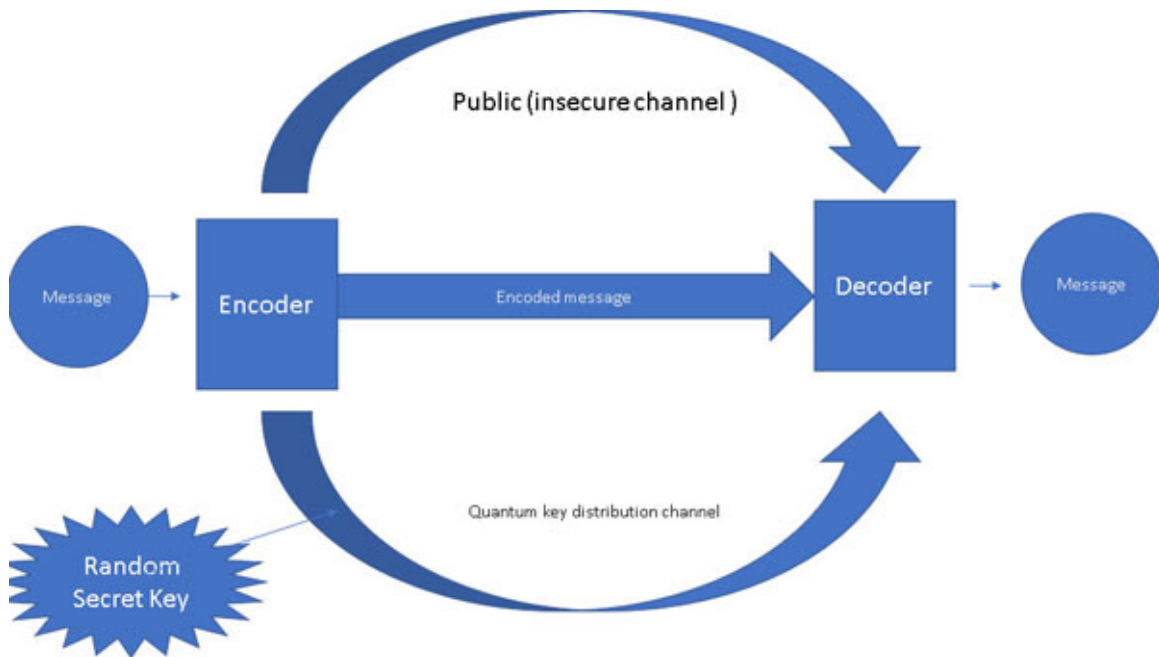


Figure 2.4: Quantum cryptography – process flow

The complete process is represented in the form of “steps” for the ease of understanding of the reader. Also, refer to [Figure 2.4](#) for the process flow.

The following steps contain two human entities, Gopal and Madhav. Gopal is the sender of the information, whereas Madhav is the receiver. Further, the following steps explain how the overall flow of information happens in the world of quantum cryptography. We have a separate chapter that would explain these concepts in detail. However, for kick-start, the following information will be helpful. Let’s start.

- Gopal directs the encrypted key in the method of entangled polarized photons, as follows:
- Gopal sends a message in the form of a key-value that exists in the form of a polarized photon.
- Madhav gets the photons and measures the key value by measuring the polarization.
- Madhav sends back the measured value of the key that is, the polarization values.
- Gopal transmits (sends) the basis on which each photon was polarized.
- Madhav sends the basis on which each photon’s polarization was measured.

- Gopal equates the values of only those photons for which both the sender in this case Gopal and receiver (Madhav) used the identical basis to generate and measure the polarization.
- Any inconsistency in these values means a conceivable change is attempted by an intruder. Therefore, if this situation occurs, the operation is terminated and is started again.

Quantum cryptography is a futuristic concept (well not really... few companies are already working in the same area and achieving it practically, for instance, QNu). For now, we can make do with 'non-quantum' public-key cryptosystems. The public key encryption, which is still difficult to crash, signifies we can securely use our debit card or net banking details online. However, when we create quantum computers, public key cryptosystems would be compromised and we would then need to move to quantum encryption.

Another area that will be heavily affected by a quantum computer is AI. We will discuss later in the chapter that how the complete landscape of AI itself is changing by the advent of QC. Multiple-use cases are already being defined which are revolving around the combo. So, it is exciting to see how they will change almost everything.

We will learn more about all this in the upcoming chapters.

[Quantum computer and AI](#)

After the advent of the quantum computer, Google defines AI in a slightly different way. According to Google, quantum computing is a novel archetype that will play a great role in quickening tasks for AI. Therefore, they want investigators and developers to access open-source frameworks and computing power that can function outside classical competencies.

We, humans, harvest 2.6 exabytes of data every day. This includes every minute of the day that 3.4 billion global internet users remain to feed the data banks with thousands of pins on Pinterest, lacks tweets, millions of Facebook likes. In addition to all the other data, we produce by taking pictures and videos, saving documents, opening accounts, and more.

The complexity and extent of data sets are rising sooner than our computing resources and hence placing substantial drain on our computing fabric. It's forecasted that artificial intelligence, and in specific, machine learning, can

take advantage of developments in quantum computing technology. Even before a full quantum computing solution is obtainable, Quantum computing algorithms permit us to improve what's already conceivable with machine learning.

Quantum computers cannot only operate on massive arrays of data in a solitary step but also plug into understated patterns that traditional computer cannot accomplish. This happens because of a natural linking between the numerical nature of computing and machine learning.

As QC is supposed to do massive data crunching concurrently, the human brain also has huge data processing capabilities in a parallel fashion. Therefore, this is one of the prime reasons why quantum computers are thought to stimulate the human brain by using artificial neural networks – because neural networks are designed to mimic human brains in pattern recognition.

Quantum computing is also helpful in organizing and analyzing big, unsorted data sets to discover patterns or anomalies very rapidly. Quantum computers would access all items in the datastore and at the same time, it recognizes resemblances among them in seconds. In the financial sector, the amalgamation of AI with quantum computing may help advance and battle fraud recognition. Models trained using a quantum computer could be proficient in noticing patterns that are tough to find using conservative gear. The quickening of algorithms would harvest prodigious compensations in terms of the capacity of information that the machines would be capable to handle for this purpose. Also, one of the newest drifts in banking right now is giving clients custom-made products and services using unconventional endorsement systems. For this, numerous quantum models have already been projected which are designed at enhancing these systems' act.

Additionally, quantum computers are probably excellent in the integration of very dissimilar data sets. The potential is that quantum computers will permit for rapid examination and incorporation of our huge data sets which will advance and transform our machine learning and artificial intelligence capabilities.

The usage of quantum algorithms in AI techniques will help in the improvement of machines' learning capabilities. This will lead to developments in the growth, amongst others, of forecasting systems, including

the financial and retail industry. Though, we'll have to wait to these developments being rolled out.

The aptitude to signify and handle numerous state types of quantum computing makes it passable for resolving problems in a diversity of fields. Intel has unlocked numerous lines of research on quantum algorithms. The first applications they are successful to get are in areas such as material sciences, where the modeling of minor molecules is a computing rigorous task. Bigger machines will permit scheming medicines or enhancing logistics to, for instance, find the utmost effectual route amongst any number of alternatives. Currently, most industrial applications of artificial intelligence come from the so-called supervised learning, used in tasks such as image recognition or consumption forecasting.

AI and QC would be the game changer... but the real beauty lies in the combination of QC and QP (quantum physics). Therefore, it is necessary to know their story at least in brief.

There is a lot of overlap happening in terms like modern physics, classical physics, quantum machines, and so on. The reason is obvious; they are related and sometimes dependent on each other. So, let's understand them for the sake of clarity.

[Classical physics, modern physics, and quantum physics their relationships](#)

Modern physics defines the world that we see when we go beyond the normal phenomenon of everyday workings. This generally cannot be clarified with the classical laws of physics. At this point, modern physics comes to the rescue. Quantum physics or quantum mechanics is the subdivision of modern physics that deals with light and associated gears that are very minor, for instance, molecules, single atoms, subatomic particles. Max Planck created the term quantum in 1900. Planck and Niels Bohr made the first quantum model of the hydrogen atom, thereafter futurists like Albert Einstein, Derick, and Richard Feynman took Quantum physics to a further advanced level.

Quantum mechanics need a whole reconsidering of the landscape of reality at the utmost basic level. Quantum mechanics defines a completely strange world, where nothing is sure and objects don't have fixed properties until you measure them. It's discovered and describes a world where aloof objects are

linked in bizarre ways, where there are whole cosmoses with dissimilar histories right next to our own, and where virtual particles are accessible in and out of being in otherwise unfilled spaces. Quantum physics may look like the paraphernalia of made-up fiction, but it's a discipline in itself. The flora and fauna defined in quantum theory is our world, at a tiny scale. The weird properties forecasted by quantum physics are actual, with real significances and bids. Quantum theory has been verified to be a far-fetched level of exactness. It is one of the furthestmost precisely confirmed scientific theories.

Some additional insights are given below on topics and related interests

We now know that quantum physics is one portion of modern physics because broadly, modern physics is the physics which is grounded on rules revealed after 1900; hence termed as modern. Then what is classical or conventional physics? Well, classical physics is the physics that is founded on the laws and principles of physics that were advanced before 1900. Likewise, classical physics is the physics of everyday stuff. So, for instance, it explains why something comes back to earth if we toss it in the air, or why something flows from top to bottom, not from bottom to top. Its biological extension is biological mechanics which explains why we become tied. On similar lines, classical thermodynamics describes the physics of heating and conserving objects and the process of engines and fridges. Whereas, classical electromagnetism clarifies the behaviour of tube lights, radio sets, and magnets, as depicted in [Figure 2.5](#):

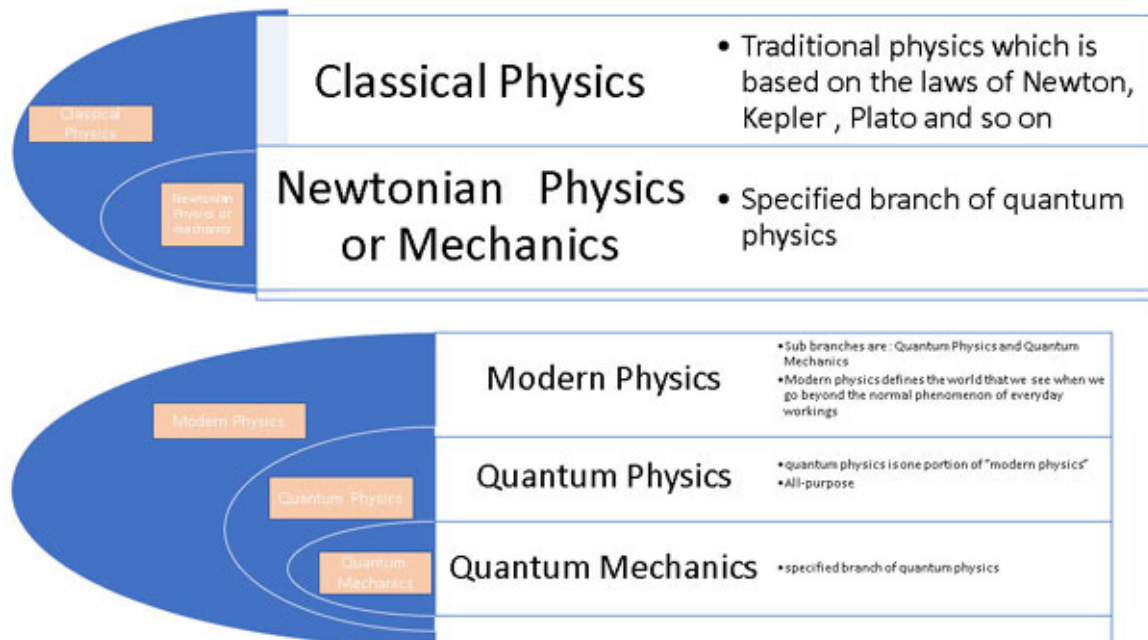


Figure 2.5: Classical and modern physics—branches and sub-branches

- Quantum physics is all-purpose, whereas quantum mechanics is a specified branch of quantum physics that pacts with the moving behavior of quantum variables of microscopic systems. It is similar to traditional physics, where general physics is all-inclusive while classical mechanics pacts with big scale systems unfolding their motion, collaboration like collision, angular momentum, and scattering.
- Quantum physics has been essential for our understanding of nature. It is pervasive, but it is also intensely cagey. Quantum mechanics clarifies why there are atoms. Likewise, quantum physics also describes how atoms syndicate into miscellaneous molecules. It provides the foundation for the understanding of the forms and connections of those molecules.
- The rulebooks of quantum mechanics clarify how electrons organize themselves in atoms and how atoms fit composed to style molecules and henceforth administrate the nature of all the materials we see everywhere. It throws light on their stability and establishes the reason behind their dissimilar chemical properties. For instance, without quantum tunnelling, we would not have understood how electricity is led in semiconductors. Likewise, without an understanding of the quantum nature of the electron, it would be impossible to design semiconductor chips. The electrons spit out photons of light that led to the discovery of the LASER. In turn, it is cast off in all kinds of medical and industrial bids, relaxation and entertainment industry, and applications. A few examples are several laser-based devices including blue-ray players.
- Quantum mechanics is used for the tunnelling diode, which is cast off as a very debauched switch in microprocessors. Quantum tunnelling also offers us nuclear power, electron microscopes, and MRI scanners, which brand the use of quantum spin. Without an understanding of the quantum landscape of light and atoms, it would be terrible to create the LASER which is used to send messages over fibre-optics communication channels.
- Even the burn sensors in the home trust quantum tunnelling of subatomic particles. Its applicability spread from a smartphone to HD TV displays to new innovative technologies like quantum dot-based displays. Even in biology, things are dependent on the principles of quantum physics.

- Quantum mechanics is the game-changer; it provided a new perspective to visualize the world. If quantum physics is not around, we will not be able to understand and decode life itself. Quantum mechanics elucidates the possessions of materials, such as what brands a metal, as a conductor of electrical energy, while another one is an insulator. It enlightens us about the working of light and radioactivity and serves as the foundational forces of nuclear physics. It explains the cosmos and tells us why stars and galaxies behave the way they do. We would not even be able to create computer chips without quantum physics. Its areas of applications are wide-ranging. And now it is fulling to one of the greatest technological revolutions of humankind called quantum computers. As mentioned earlier, Quantum computers would change almost everything in the coming time and it affects almost all areas of our life. Similarly, all things which are present are having quantum stuff – either it is elementary particles, or atoms and molecules. Also, all things possess both a particle and a wave nature because fundamentally they are quantum beings.
- A switch in a classical manner could be defined as a small button or something similar that you press up or down in order to turn on the electricity. Whereas the term "quantum switch" refers to a quantum operation in which a quantum system is affected by two or more quantum channels, the order of which is determined by the state of an order quantum system. Additionally, one can establish a quantum superposition of the various orders of application by carefully selecting the state of the order system, which enables one to do communication tasks that are not achievable using the normal quantum Shannon theory.

The narration of quantum physics is required for anyone who wants to appreciate the technicalities of QC. Hence, in the next few sections of the chapter, we will cover fundamentals, classical concepts of quantum physics, interrelation among quantum physics and quantum computers, and the History of QP. Hold on, covering fundamentals are OK...But why history? Because history predicts the future.

[The birth of quantum physics](#)

To understand the gamut of light produced by hot bodies, commonly understood as blackbody radiation, quantum physics was born. Understanding

blackbody radiation was the starting point of quantum mechanics. As per the concept of black body radiation, when hot matter is luminous, it becomes hotter and hotter. Slowly it turns out to be brighter and starts glowing. The spectrum of the light is extensive, with a top that changes from red to yellow and to end to blue as the temperature is elevated. It should have been feasible to comprehend the form of the spectrum by uniting thoughts from thermodynamics and electromagnetic theory, but all efforts are unsuccessful. Nevertheless, we will presume that the energies of the vibrant electrons that emit the light are quantized (when something is quantized, it means that it derives in distinct lumps, named “quanta”). Planck got a countenance that approved it nicely with experimentation. But as he documented all too well, the concept was physically seeming silly, an act of distraction, as he later labelled it.

Planck applied his understanding of quantum proposition to the energy of the vibrators in the walls of a glowing or radiating body (the temperature of the radiating body limits the strength and features of the radiation it produces). This set the ground for Einstein to pitch in 1905. He determined that if a vibrator’s energy is quantized, then the energy of the electromagnetic field that it emits commonly known as light must also be quantized. Einstein, therefore, permeated light with particle-like behaviour. This behaviour of light is one of the most important and foundational elements of quantum computers. How? that we will see in upcoming sections.

Photons and quantum computers

Photons were discovered by Einstein and he called them energy quanta. A photon is the least likely separable unit of light and is quantum-behaving. It would take at least a hundred photons, directed closely, for the average blinking human eye to record even a weak glimmer of light. Any beam of light or image we usually see includes millions to trillions of photons.

Quantum physicists ever so often run trials using single photons or other elementary particles, since by using minor numbers, the experts can eliminate other needless mess that would otherwise only obscure their research, the outcomes, and mathematical pieces of evidence. Initial proof of quantum properties was primarily revealed in trials using photons while examining radiation, electromagnetic waves, and the photoelectric effect. Einstein’s work

on photons was perilous to understanding quantum theory. Even his work to invalidate quantum mechanics only enhanced our understanding.

Experts have been able to produce single photons, direct them along many paths in trials, and measure what occurs using light-sensitive gear called photomultiplier tubes. A photomultiplier is able to take one visible photon and multiply it into sufficient other photons that an electrical current can be activated to register and confirm the early visible single photon.

Photons are gifted applicants for quantum computation since they can spread extended distances without transporting up the rear information. However, when they are kept in the matter, they turn out to be fragile and vulnerable to decoherence. Now investigators with the Photonics Initiative at the **Advanced Science Research Centre (ASRC)** have advanced a novel protocol for storing and freeing a single photon in an embedded eigenstate. The eigenstate is a quantum state that is virtually unpretentious by loss and decoherence. The new protocol brings major speed enhancements in goals to the development of quantum computers.

[What is photoelectric effect?... by the way](#)

The photoelectric effect was the start of the foundation for the creation of modern-day quantum mechanics. The observable light we see is just one kind and variety of electromagnetic radiation crossways. It is named the electromagnetic spectrum. The electromagnetic spectrum labels all types of electromagnetic radiation, including the visible light, which we can see, and all the kinds we can't (such as x-rays, microwaves, gamma waves, and radio waves). The dissimilar types of electromagnetic radiation vary chiefly by wavelength (noticeable light has a wavelength of 400 to 700 nanometres (nm) and x-rays have 0.10 to 10 nm), frequency, intensity, direction, and other properties. All types of electromagnetic radiation move in a relatively straight line, if unhindered (by an object, gravity, and so on), at the speed of light (which is 299,792,458 meters per second in a vacuum). Frequency and wavelength can be transformed into each other through the speed of light and are truly the same variable.

Light has momentum and energy (but no mass). Planck and Einstein understood that when light (or other forms of electromagnetic radiation) knockout other material, the material would often produce electrons (which are always negatively charged) from the subsequent transmission of energy from

the photon to the material, as signified in [Figure 2.5](#). The higher the intensity of light, the more electrons emitted.

The photoelectric effect ([Figure 2.6](#)) occurs when light hits most materials but is most readily observed when it hits metals and other extremely conductive materials. The photoelectric effect is how the sun's vigour is transformed into electricity by solar cells. The photoelectric effect is also in arrears the important way digital cameras work and record images, as shown in [Figure 2.6](#):

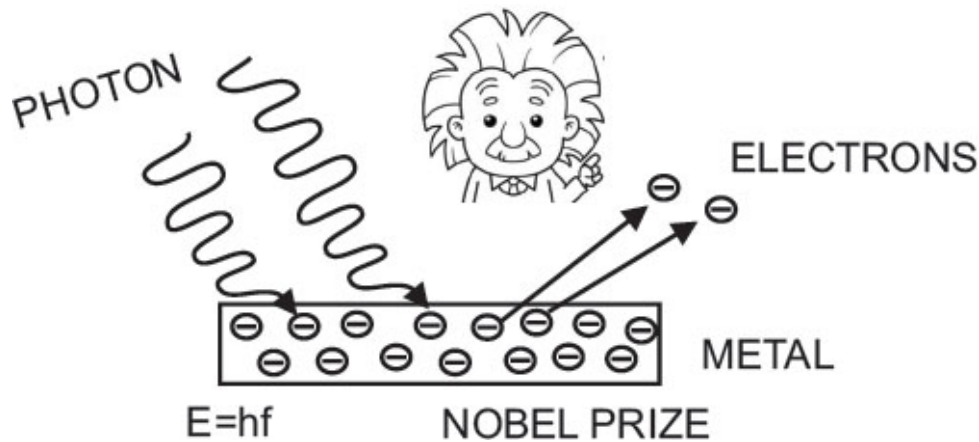


Figure 2.6: Photoelectric (TechToon created by Patanjali Kashyap)

In the photoelectric effect, light falls on an electrically exciting metal plate that can knock free electrons from its surface. But the energy of the free electrons would not hinge on the light's illumination or strength. In its place, their energy was found to hang on its frequency. This is an unforeseen outcome. If the light is a wave, growing a wave's intensity (its height) should upsurge its energy. Think of water waves roaring in contrast to the coastline; they have additional energy – the higher they are, the slower they disrupt against the shore. In the photoelectric effect, high-intensity light did not consequence in electrons being bashed free with more energy – just additional electrons being bashed free.

Einstein effectively described what was happening by suggesting that all electromagnetic radiation is eventually made up of minute protuberances of energy – particles that we today call photons. Each electron is thus bashed out when it is triumphed by a single photon, the energy of which rests on its frequency. The reason we do not typically see this particle-like nature of light

is due to the big number of photons involved, just as we cannot typically see separate pixels on a photo.

Let's summarize: some fundamental concepts of modern physics and radiation

[Figure 2.7](#) summarizes the fundamental concepts which we discussed in this chapter; apart from that, it also highlights some essential ingredients of quantum physics which would help understand the quantum computer and computation in a better way:

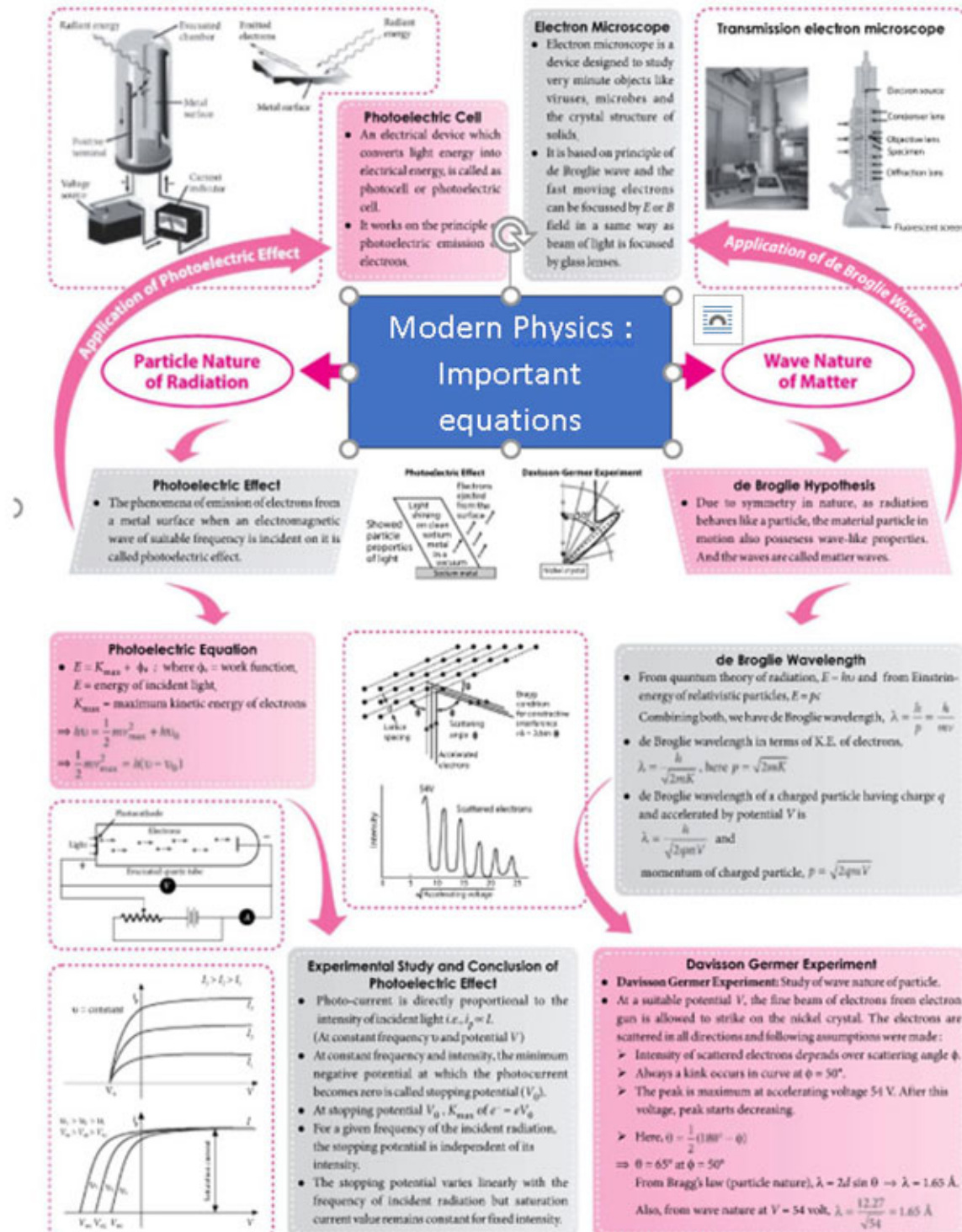


Figure 2.7: Fundamental of quantum physics

Particles of light

Trials on the photoelectric effect exposed that when light is engrossed, its energy essentially come in separate packages, which is a property of a particle. The twin nature of light, which is categorized as particle and wave contingent on what one looks for, is the first case of a puzzling theme that reappears all over quantum physics. The duality founded a theoretic mystery in the early years of the development of quantum physics for the next few decades.

The first step on the road to quantum theory had been caused by a predicament about radiation. The second step was triggered by a quandary about the matter. It was recognized that atoms comprise positively and negatively charged particles. But oppositely charged particles attract. According to electromagnetic philosophy, they should corkscrew into each other, radiating light in a comprehensive spectrum until they collapse.

Niels Bohr in 1913 proposed a fundamental theory, which says, electrons in an atom occur only in certain immobile states, together with a ground state. Electrons alter their energy by hopping among the motionless states, producing light whose wavelength hinges on the energy change. By uniting recognized laws with strange assumptions about quantum behaviour, Bohr brushed away the problem of atomic constancy. Bohr's theory was filled with inconsistencies, but it provided a measurable explanation of the gamut of the hydrogen atom. He recognized both the achievement and the inadequacies of his model. With strange anticipation, he united physicists to produce a new physics. His idea was ultimately contented, though it took more than a decade.

The hypothesis presented by Bohr was not adequate. Hence, multiple attempts occur to fine-tune this. Multiple of them failed measurably. Then a sequence of advances completely altered the progression of thinking. In 1923, Louis de Broglie, in his Ph.D. dissertation, anticipated that the particle behaviour of light should have its complement in the wave behaviour of particles. He connected a wavelength with the momentum of a particle.

He proposed, the bigger the momentum, the smaller the wavelength. The clue was fascinating, but no one recognized what a particle's wave nature might mean or how it connected to the atomic structure. However, de Broglie's theory was a significant forerunner for proceedings soon to take place. In 1924, there was yet additional pioneer, Satyendra N. Bose who gave a completely new method to clarify the Planck radiation law. He treated light as if it were a gas of massless particles that do not follow the traditional laws of Boltzmann

statistics but act according to a new kind of statistics founded on particles' hazy nature. This behaviour is now known as photons.

Einstein instantly applied Bose's perception of a real gas of huge particles and attained a new law. This law became famous as the Bose-Einstein distribution, which essentially explains how energy is shared by the particles in a gas. Then in the next few years, nothing major happened. But unexpectedly, an unrestrained sequence of proceedings occurred, concluding in a scientific revolution. In the three years beginning from January 1925 to January 1928, multiple events occurred and took shape.

The event which triggered the modern thoughts: A bit of more history

Technically, quantum mechanics started with Wolfgang Pauli when he proposed the exclusion principle, which provided a theoretical basis for the Periodic Table (a list of chemical elements that are ordered in atomic number order, typically in rows, with similar atomic structure elements (and thus, similar chemical characteristics) appearing in vertical columns). Just after that, Werner Heisenberg, along with Max Born and Pascual Jordan, came up with matrix mechanics. However, this was the first version of quantum mechanics, but it proved itself very useful. Thereafter, Schrödinger developed wave mechanics, a subsequent form of quantum mechanics in which the state of a system is defined by a wave function. Before this, matrix and wave mechanics, was seemingly discordant, but after this invention, they were shown to be equal.

While Bohr and Heisenberg were working on their mathematical depiction of atoms, Erwin Schrödinger was giving a dissimilar method, saying that the whole subatomic world is made up of waves. His philosophy turned out to be known as wave mechanics and the well-known equation that takes his name describes it.

Around this time, Dirac came into the picture. He exhibited the two dissimilar ways to the quantum world. He created a novel equation that now has his name. The Dirac equation well foretold the being of antimatter. Nowadays, we recognize that for each matter particle, there can be its antimatter complement, but if a particle and its antiparticle originate into interaction, they will overwhelm in a spurt of unadulterated energy. The procedure of matter-

antimatter obliteration can also occur in the opposite, whereby a quantum of energy, such as a photon of light, can be rehabilitated into a pair of particles, such as an electron and its antimatter partner, a positron.

During this period, it became standard that all particles follow either Fermi-Dirac statistics (A kind of quantum statistics known as Fermi-Dirac statistics deals with the physics of a system made up of several identical, non-interacting particles that adhere to the Pauli exclusion principle. The Fermi-Dirac distribution of particles over energy levels is one outcome.) or Bose Einstein statistics (A group of non-interacting, indistinguishable particles may occupy a set of available discrete energy states in thermodynamic equilibrium in one of two ways, according to Bose-Einstein (B-E) statistics.. Also, they had two classes that had dissimilar properties. This happen due to Fermi-Dirac statistics.

Heisenberg articulated the Uncertainty Principle during this era which became the foundation for multiple other concepts and theories in a later point of time followed by Paul A.M. Dirac who advanced a relativistic wave equation for the electron that described electron spin and forecasted antimatter. Dirac placed the basics of quantum field theory by giving a quantum explanation of the electromagnetic field. Well after that, Bohr proclaimed the complementarity principle, a metaphysical opinion that assisted in the resolution for ostensible paradoxes of quantum theory, mainly wave-particle duality.

Atomic theory

For more than 2,500 years, truth-seekers have contended with the existential crisis of humankind. In the west, Democritus questioned whether the material could be divided boundlessly into lesser and lesser portions, and had come to the conclusion that there must be tiny elements that are inseparable, the atoms. Whereas in the east, Kanad also reached the same conclusion long before Democritus. However, he called the smallest particle “kana”. On similar lines, Aristotle and Plato were attentive in what way we as spectators relate to the observed. The endeavour continued in the 20th century. In 1912, Ernest Rutherford revealed in a surprising trial that the atom contains a nucleus of protons with electrons circling it. The atomic nucleus contains elementary particles carrying a positive electric charge (protons). But then again, the question was like-charged particles deter each other.

Then how can atomic nuclei be steady? Why should not the numerous protons in the atomic nucleus fly separately? If this is not happening, then there must be some other force that had to act nicely and affectively at the very short distances inside the atomic nucleus, and powerfully sufficient to balance the electric force.

Bohr took this as a starting point of his invention and constructed upon Rutherford's model of the atom. In Rutherford's model, maximum of the atom's mass is focused on the center and electrons edging the positive mass to some degree like a cloud. Bohr's greatest input was the quantization of the model. He thought that electrons moved around the nucleus in circular orbits with quantized potential and kinetic energies. So, it was not likely for electrons to inhabit just any energy level.

In the Bohr model, he describes atom as a minor, positively charged nucleus enclosed by electrons that travel in round orbits around the nucleus. This is very similar to the structure of the Solar System. Each orbit resembles a dissimilar energy level. Variations of energy, such as the change of an electron from one orbit to another around the nucleus of an atom, is done in distinct quanta. The term quantum leap denotes the sudden movement from one discrete energy level to another, with no smooth transition. There is no in-between.

Quantum leap was special and since the movement of the electron was not liberal, it just vanished from one orbit and appear in the following orbit with no in-between state and produced (or engrossed) a precise amount of energy. Bohr clarified that the amount of energy at this level cannot be additionally subdivided. An exact minimum quantity of energy is quanta. First insights into this kind of energy level were provided by a physicist called Planck. In a nutshell, the story summarizes that the energy of an electron in an atom is quantized.

Foundation of quantum mechanics

So, the foundations of quantum mechanics were essentially laid down by 1930 on the basis of all these theories, concepts, ideas, and philosophy. The mad stride with which it happened was unbelievable. In 1925, the idea of electron spin had been anticipated by Samuel Goudsmit and George Uhlenbeck. Bohr was intensely sceptical but finally, it matured after having a constant, instant,

and passionate discussion among all the players involved in the development. The formation of quantum mechanics activated a technical gold rush.

On its principles, Heisenberg placed the fundamentals for atomic structure theory by gaining a rough explanation to Schrödinger's equation for the helium atom in 1927, and over-all techniques for manipulating the structures of atoms were developed rapidly after by John Slater, Douglas Rayner Hartree, and Vladimir Fock. The construction of the hydrogen molecule was resolved by Fritz London and Walter Heitler. Linus Pauling constructed on their outcomes to originate theoretic chemistry. Arnold Sommerfeld and Pauli put the nitty-gritty of the theory of electrons in metals, and Felix Bloch shaped band structure theory. The journey did not stop here; Heisenberg takes it further by explaining the cause of ferromagnetism. The paradox of the random nature of radioactive deterioration by alpha particle release was clarified in 1928 by George Gamow. He determined that it happens by quantum-mechanical tunnelling. In the subsequent years, Hans Bethe rested the basics for nuclear physics and enlightened the world about the energy basis of stars. So, in a nutshell, with these growths, atomic, molecular, solid-state, and nuclear physics entered the modern age.

Wave function

One of the amazing detections in quantum mechanics is that for electrons, the wave function at all times vicissitudes sign. The significances are theatrical, for if two electrons are in a similar quantum state, then the wave function has to be its adverse reverse. Thus, the wave function essentially vanishes. Therefore, the probability of the discovery of two electrons in the same state is zero. In technical terms, we know the Pauli exclusion principle. All particles with half-integer spin, including electrons, act this way are termed fermions (There are two types of particles: "fermions," which are the building blocks of matter, and "bosons," which carry forces. Fermions occupy space, whereas bosons can stack on top of one another, making this difference.). For particles with integer spin, including photons, the wave function does not alter the sign. This type of particle is named bosons.

Electrons in an atom organize themselves in shells since they are fermions, but the light from a laser arises in a single super-intense beam which is fundamentally a solitary quantum state; hence, light is composed of bosons. These concepts rub on only to undistinguishable particles; since dissimilar

particles are interchanged, the wave function will surely be distinct. Accordingly, particles perform like fermions or bosons only if they are completely indistinguishable. The out-and-out individuality of like particles is one of the fascinating features of quantum mechanics.

So, what a wave function really is and what is intended by making a measurement were strongly discussed in the initial developmental time of quantum physics. By 1930, though, a standard understanding of quantum mechanics had been established by Bohr and his contemporaries. This understanding is termed as Copenhagen Interpretation.

The important elements of the Copenhagen Interpretation are the probabilistic explanation of substance and actions, and an understanding of the wavelike and particle-like landscapes of things through Bohr's principle of complementarity. An essential question in the discussions on quantum mechanics was whether the wave function comprises all conceivable information about a system, or if there might be underlying factors that secreted variables to regulate the consequence of a specific measurement.

The Copenhagen interpretation

The Copenhagen interpretation was primarily exhibited by physicist Niels Bohr in 1920. It articulates that a quantum particle is not in one state or another, but in all of its likely states at once. It's only when we detect its state that a quantum particle is fundamentally involuntary to select one probability, and that's the state that we detect.

This state of prevailing in all possible states at once is named an object's coherent superposition property. The whole of all likely states in which an object can exist, for instance, in a wave or particle form of photons that are transportable in both directions at once, makes up the object's wave function. When we detect an object, the superposition breakdowns and the object involuntary go into one of the states of its wave function.

Bohr's Copenhagen interpretation of quantum mechanics was theoretically confirmed by what turned out to be a well-known understood trial connecting a cat and a box. It's called Schrödinger's cat, and it was initially presented by the physicist Erwin Schrödinger in 1935.

In his theoretic trial, Schrödinger put his cat in a box, along with a bit of radioactive material and a Geiger counter. The Geiger counter is a device for

noticing radiation. The Geiger counter was intended so that when it detected the decay of the radioactive material, it caused a hammer which was composed to break a flask comprising hydrocyanic acid, which, when free, would execute the cat.

To remove any inevitability concerning the cat's destiny, the trial was to take place inside an hour, long enough so that some of the radioactive material could maybe decay, but short enough so that it was also possible none would. Throughout its stay in the box, the cat came to exist in an unknowable state. Since it could not be observed, it could not be said whether the cat was active or departed. It existed instead in the state of both life and death.

Quantum electrodynamics

In the 1940's, three physicists, together with the prodigious Richard Feynman, derived an influential theory named quantum electrodynamics, or QED. It was a simplification of quantum mechanics and provided a novel method of unfolding the way matter interrelates with light. It also puts light on how all matter is apprehended and organized through the electromagnetic force.

QED is an example of what is called a quantum field theory, an impression that drives back to the late 1920's with the effort of Paul Dirac, who inscribed a ground-breaking paper that united quantum mechanics with Maxwell's theory of electromagnetism.

All over the 1930's and 1940's, quantum field theory was overwhelmed by a bothersome mathematical problem like calculations continuously ran to immeasurable answers when they should have been limited. The elementary evidence is that space is not ever really empty, but is, in fact, a fizz of particles and anti-particles unceasingly appearing and vanishing. Nevertheless, this means that even to some degree as candid as the electrical force among two electrons has to be inscribed as an infinite series of ever-more multifaceted procedures captivating place in the space among them.

The subject of the infinitudes was resolved by QED, which is nowadays viewed as the furthestmost precise theory in all of physics.

In the mid-1960's, John S. Bell presented that if secreted variables existed, experimentally experiential probabilities would have to drop underneath certain parameters, called "Bell's inequalities." Trials were carried out by a number of groups, which found that the disparities were desecrated. Their

shared data came down conclusively, contrary to the option of hidden variables. For maximum experts, this resolved any uncertainty about the rationality of quantum mechanics. One of the strange possessions of quantum systems rises from what is acknowledged as entanglement.

For a short time, a quantum system, such as an atom, can be in any one of the several motionless states but also a superposition or sum of such states. If one measures some property such as the energy of an atom in a superposition state, in general, the result is sometimes one value, sometimes another. So far, nothing is strange. It is also conceivable, to understand a two-atom system in an entangled state in which both atoms are joined with each other. If the atoms are unglued, information about one is common, or entangled, in the state of the other. The behaviour is unexplainable apart from in the dialect of quantum mechanics. Entangled states have now been active in quantum communication systems, and entanglement inspires all applications for quantum computation.

Quantum personality: Richard P. Feynman

Richard P. Feynman's technical legacy comprises revolutionary aids in quantum field theory and electrodynamics. The credit also goes to him for producing the widely used Feynman diagrams. Technically speaking, he was the person who took the first steps for quantum computing. In a lecture known as Simulating Physics with Computers, Feynman communicated about why physicists require computers, and what they need of these devices.

Richard Feynman was born in Far Rockway, the USA in 1918. He got his B.Sc. from MIT in 1939 and a Ph.D. from Princeton University in 1942. He turned out to be a Professor of Theoretical Physics at Cornell University in 1945. He contributed to quite a few fields in physics, in specific, quantum mechanics and quantum electrodynamics. He received the Nobel Prize in Physics in 1965, with Schwinger and Tomonaga for his important work in quantum electrodynamics. Quantum electrodynamics had profound significance for the physics of elementary particles. He inscribed quite a few textbooks together with the famous Feynman Lectures on Physics. Feynman expired in Los Angeles, the USA in 1988.

According to Feynman, while natural laws are reversible, the law for computers is not. Founded on this, he said that we require a new model of computer, which is dissimilar from Turing Machine and follows the

philosophies of quantum mechanics, that is, quantum computer. Reversible multiplication is the computation which fulfil the criteria of the rules of thermodynamics. Therefore, its computational procedure is reversible. There are two kinds of reversibility, that is, physical reversibility and logical reversibility. The physically reversible procedure is the method that does not upsurge entropy. The logically reversible method is the process that can hypothesis the inputs from the outputs. In computer science, reversible computation has been already examined. Landauer discoursed the association among physically and logically reversible processes, viewing that though all logically reversible processes may be proficient as a thermodynamically reversible process, some logically irreversible procedures are not. Specifically, it means that if the procedure is physically reversible, then it essentially is also logically reversible.

The fundamentals concepts of quantum computing

In this section we will discuss some building blocks of quantum computers.

Superposition

Visualize if you could toss a coin that could drop not only in a head or tail position but in both positions at the same time. You can name it as a “quantum toss”. Because this is exactly like the quantum behaviour, where one object stays in “up or down” at the same moment. However, the moment you detect this is a quantum coin, it is compulsory to take either heads or tails not ever knowing what position it was in before. This is one reason that one needs to be cautious when measuring qubits since they change as soon as detected.

The qubit maybe 0 or 1 or have any proportion among them. A qubit (a state of atom or electron or ion) can be assumed as an arrow that can point all the way in three-dimensional space; when it points up, the qubit is said to be in the 1 state, down is the 0 state, and any extra direction is a mixture of both. Superposition is an essential feature of quantum computing. In conservative (or traditional or contemporary or classical = these are used throughout interchangeably in this this book) computing, a state of n bits a whole number is defined using n digits in a sequence of 0 or 1. In quantum computing, it's more complex, needing 2^{n-1} complex numbers to define a state of “ n ” qubits. This means that an exponential number of classical bits would be required to store the state of a quantum computer, even approximately.

Two qubits in superposition can signify four states, three represent 8, and 4 represents 16 entangled (associated with each other, so that change in one qubit can incestuously reflect in other) states. Likewise, qubits epitomize $2^{16} = 65,536$, and 32 would be 4,294,967,296 options. With just a few qubits, a huge volume of information processing will be achieved. Assuming that a byte (8 bits) is the elementary unit used to stock information in any system, then the number of values that can be deposited concurrently in a quantum computer would be 2^n where n is the number of qubits. Compare this against the processing capacity of a classic computer and you realize why qubits are powerful.

A coinage has two states and makes a decent bit. But a deprived qubit does not, since it cannot keep on in superposition of head and tail for very long like a conventional one. A solitary nuclear spin can be a very decent qubit since the superposition of being associated with or contrary to an exterior magnetic field can be preceding for an extended time. But then again it can be problematic to shape a quantum computer - nuclear spin since their connection is so minor that it is tough to measure the orientation of a single nucleus. A truthful execution must strike an equilibrium between these constrictions.

In traditional computing, only exactly well-defined outcomes are obtainable, characteristically restricted by an algorithm's design. Quantum responses or amplitudes are probabilistic. It means, because of superposition and entanglement, the likely answers are well-thought-out in a certain computation. Problems are run numerous times, giving an example of likely responses and cumulative sureness in the best response provided. Statistics are used to rank from 0 to 100 percent, the probability of a response to get the correct one. This self-assurance threshold is sensible to deliver the best speed and correctness.

These factors permit quantum computers to resolve certain classes of complex problems much more competently than traditional computers. Though conventional computers would take additional time for each variable added, quantum computers can connect the belongings mentioned preceding to solve complex problems in a method that cumulate the problem size. These reasons for a lesser increase in the time essential to resolve the problem.

Entanglement

It is nothing more than a state of association among two or more quantum objects. They need not even be a similar object. Entanglement is a natural

existence in quantum mechanics. Due to entanglement, anything happening to one of the objects will instantaneously affect its partner. This is referred to as a non-local connection, which is not possible in classical mechanics involving everyday objects because communication faster than the speed of light is forbidden. But non-locality and entanglement are ordinary in the quantum world.

Precisely, this is just the postponement of the impression that particles can occasionally act like spread-out waves. If two particles come into close interaction with each other, they can turn out to be correlated and behave as a single unit. Even more unbelievably, if one of them is put into a quantum superposition of existence in two states on one occasion, then the second particle will also be forced into a superposition. So, by measuring one, we abolish the superposition of its distant companion, instantaneously and irrespective of the distance between them. However, quantum entanglement is not very useful to clarify non-scientific concepts such as telepathy, because it is embarrassing to the subatomic kingdom.

In entanglement, essentially two or more dissimilar qubits can be linked, despite being physically away from each other. Einstein labelled entanglement as ‘spooky action at a distance’. From the QC point of view, as soon as qubits are entangled, the state of each qubit is not autonomous from the rest. Quantum computers feat this phenomenon to deliver a powerful speed-up in calculations. Building a quantum computer with the level of exactness is mandatory to create, manipulate, and measure qubits and it is tremendously challenging, as shown in [Figure 2.8](#):

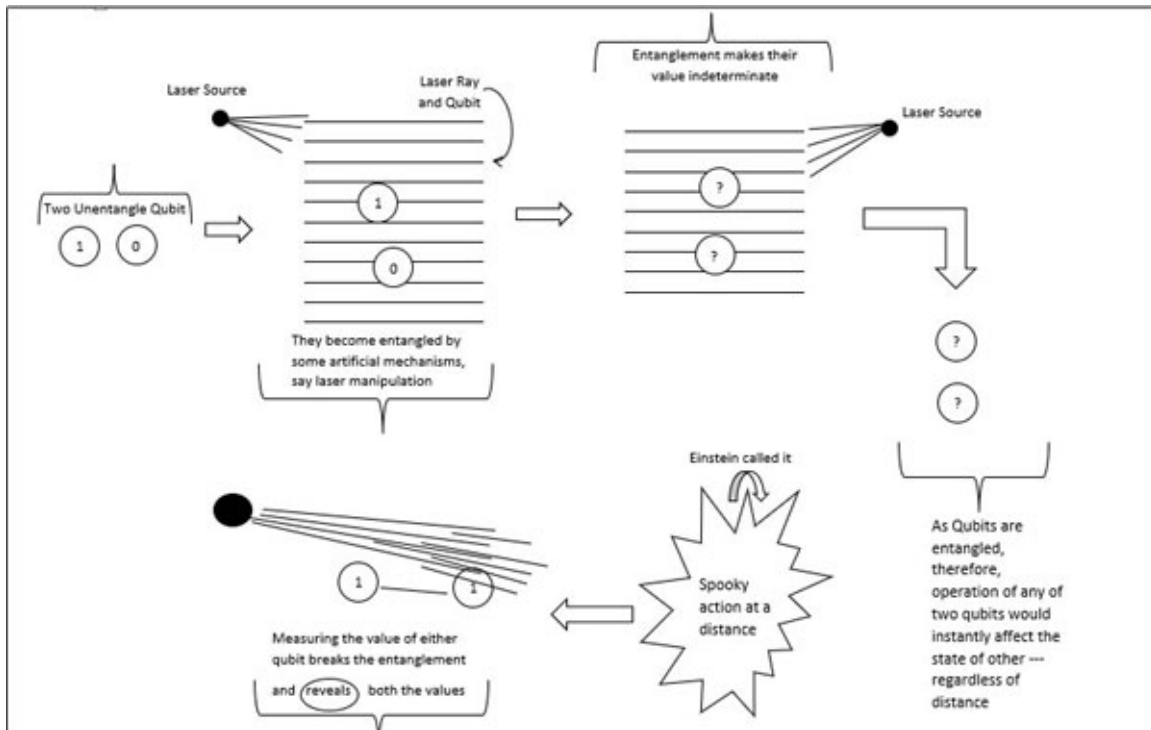


Figure 2.8: How to entangle particles --- photon

Qubits are very sensitive to their environment and any connections can result in decoherence, or loss of information. If a set of qubits are entangled, then individually it will respond to a change in the other instantly, no matter how far apart they are. They may even exist on the opposite sides of the galaxy. Therefore, this property of entangled particles allows instant modification.

This is useful; if we measure the properties in 1 qubit, then we can infer the properties of its companion without having to look. Also, entanglement can be unrushed without looking through a procedure named quantum tomography.

Quantum tomography pursues to determine the state(s) of an entangled set. It computes the likelihood of measuring every possible state of the system. Multiqubit entanglement represents a step forward in realizing large-scale quantum computing. This is an area of active research. Currently, physicists in China have experimentally demonstrated quantum entanglement with 10 qubits on a superconducting circuit. [Figure 2.8](#) explains the entanglement process.

Coherence

In quantum mechanics, particles are defined by the wave function. An essential property of quantum mechanics is termed coherence or the definite phase

relation among states. This coherence is essential for the functioning of quantum computers. However, when a quantum system is in interaction with its surroundings, coherence drops with time due to quantum decoherence. Formally, decoherence is the time that takes for the superposition of states to vanish and is due to the probabilistic nature of the wave function.

Decoherence provides info about the communication of a quantum object and its environment. The higher the decoherence (the time it stays in superposition), the higher the eminence of the qubit will be. Some qubit designs like superconductor loops have very low durability and need to be kept back at tremendously low temperatures (-271 °C) to pledge this effect. Others like trapped ions and diamond (will cover in the next chapter) vacancies have very high durability and can be set aside at room temperatures. Technology companies at work in quantum projects face a discouraging challenge trying to grapple with qubit longevity.

Quantum coherence highlights that all matters have wave-like belongings. If an object's wave-like aspects is fragmented in two, then the two waves may comprehensibly interfere with each other in such a mode as to form a sole state that is a superposition of the two states. This idea of superposition is excellently signified by Schrödinger's cat, which is together dead and alive at a similar time when in its coherent state inside a locked box. Coherence is the central theme and core concept of quantum computing, according to which a qubit is in a superposition of the "0" and "1" states, consequential in a speed-up over numerous classical algorithms. When such a state involves decoherence, though, all of its quantum properties have characteristically vanished and the benefit disappears.

How long can a quantum superposition state continue? That span of time is named the coherence time, and hinge on where a qubit lives. Quantum states are influential for computing and discovering physics, but gentle when fighting the environment. The key is to have a quantum state exist longer than it takes to perform an operation or experiment. Physicists design multifaceted ampules that separate quantum states from the environments, while still permitting for state operation.

Decoherence

All big quantum physics trials for QC are not very fault-tolerant. Because the qubits are incapable of holding their complex coherent state ([Figure 2.9](#)) for a

longer period and the wave-function break downs into one of the eigenstates (a state in contrast to a static value at which the system is detected), the procedure is termed decoherence. The end state bears a resemblance to a time-independent classical state. Decoherence occurs when the qubits are obstructed and hence intermingle with the environment. [Figure 2.9](#) showcases the process of decoherence:

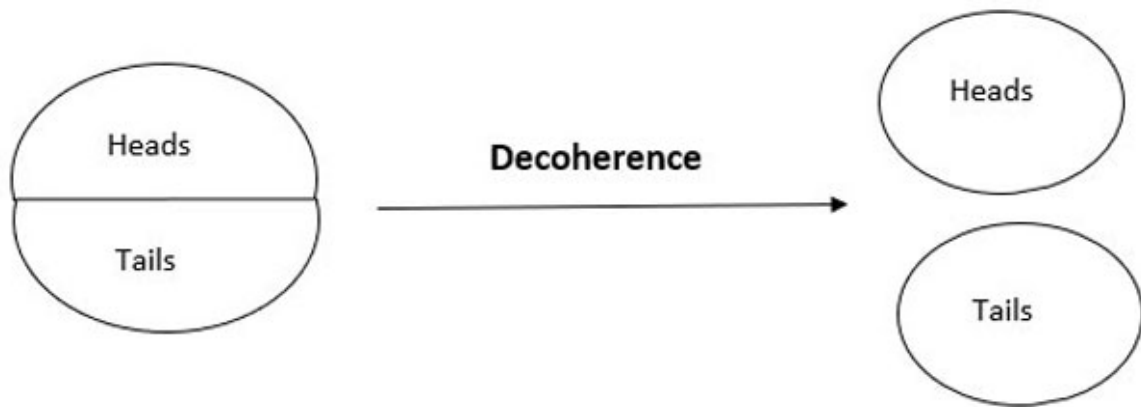


Figure 2.9: Decoherence

[Figure 2.10](#) depicts the spine of electron:

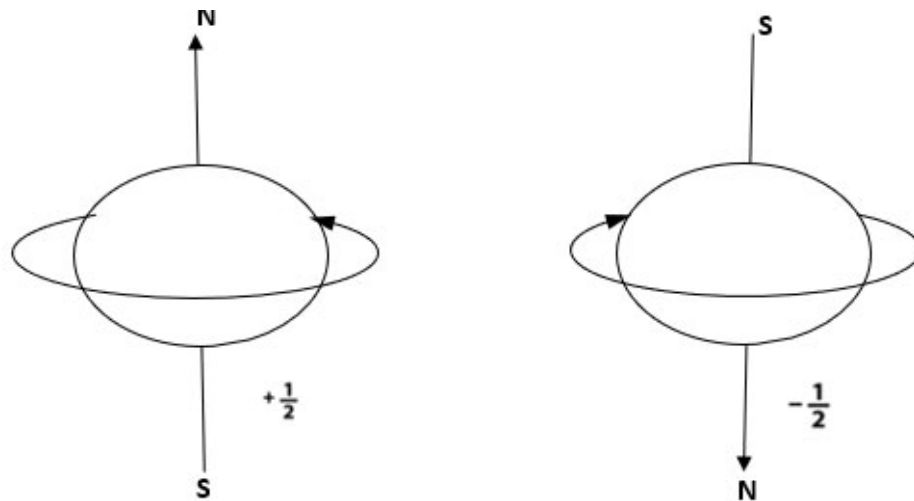


Figure 2.10: Spine of electron

The only way to observe or measure the state of a qubit is when a quantum mechanical system is to characteristically vanish. Also, when a vector or qubit is detected, an unexpected intermittent fuse is formed for one of the eigenstates, distorting the state vector. Now that we know that the state vector

is ruined when it is related to the external environment, we would like to know how this error occurs.

In order to perform useful calculations, we can basically say something about which ground state the quantum mechanical system will end up in. Without this characteristic, the system is not fault tolerant due to decoherence. Recent studies provide methodological methods to improve the results when the error is overcome by internal control and measurement, but not when the error is overcome by decoherence, some fault-tolerant systems.

Decoherence caused by vibrations, temperature changes, electromagnetic waves, and other connections to the external environment ultimately negates the significant quantum properties of computers. Given the prevalence of decoherence and other bugs, modern quantum computers are unlikely to provide accurate answers to programs with uncertain implementation times. Decoherence can be observed as the loss of information from a system into the environment; subsequently, each system is tied the knot with the energetic state of its surroundings.

Decoherence signifies a challenge for the real-world realization of quantum computers since such machines are likely to depend deeply on the uninterrupted evolution of quantum coherences. That is, they need that coherent state to be well-kept-up and that decoherence is managed, in order to really do quantum computation.

The &

defines a quantum computer as a locked quantum system and undertakes that there is a system that implements the circuit but is hidden from the rest. In a practical sense, there is no recognized way of really isolating a quantum system from its environment, once, actual quantum systems are exposed to the environment. Then the exposed quantum systems would pair to their environment and abolish the quantum information in the system through decoherence.

Fault tolerance

The purpose of quantum error correction is to scramble the information stored in a single physical qubit into many qubits. This contributes to the idea of physical and logical qubits. Logical qubits are a group of many physical qubits that contain specific information about the physical qubits. The current ratio of

physical qubits to logical qubits is 10^4 : 1. Computational power increases only when the number of qubits increases exponentially.

Interference

Interference in physics is a phenomena when two waves merge by adding their displacements at each and every point in space and time to produce a new wave that has a higher, lower, or identical amplitude. The interaction of waves that are coherent or correlated with one another, either because they originate from the same source or because their frequencies are similar or almost identical, leads to both constructive and destructive interference. All sorts of waves, including light, radio, acoustic, surface water waves, gravity waves, and matter waves, can exhibit interference effects.

Quantum states can undergo interference due to a phenomenon Known as a phase (Two waves are said to be out of phase if their peaks cross the troughs of the other. Two waves will interact destructively and create zero amplitude if they are exactly out of phase with one another.). Quantum interference can be understood similarly to wave interference; when two waves are in phase, their amplitudes add, and when they are out of phase, their amplitudes cancel.

Tunneling

Quantum physics is probabilistic, rather than deterministic. This means we can never pinpoint where an electron (or any other subatomic particle) really is. We can merely guess with a fairly high probability that the electron is at a certain location. This range of probabilities can be described by a mathematical wave function that allows you to determine all the possible positions a particle can occupy, and the likelihood of it occupying any one of them, as depicted in [*Figure 2.11*](#):

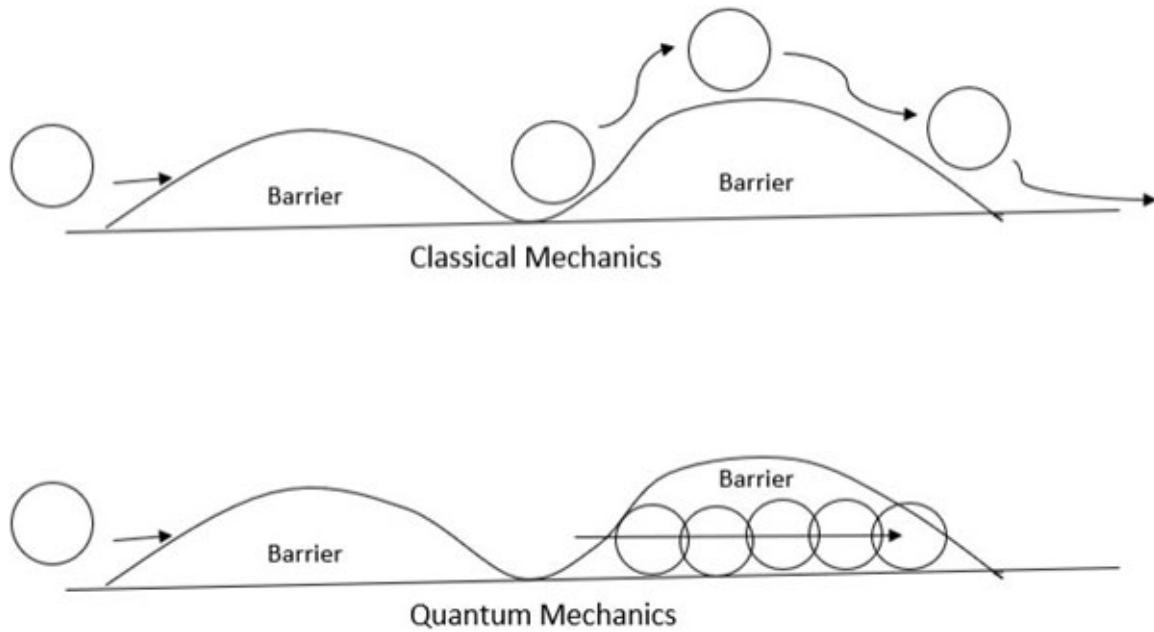


Figure 2.11: Tunnelling

Particle-wave functions can ‘leak’ across barriers if the barrier is thin enough ([Figure 2.11](#)). Instead of there being an abrupt drop-off where the particle-wave hits the barrier, there is a very quick exponential drop-off happening. This is an evanescent wave (Waves with a rapidly decaying amplitude and no transport of energy.). The evanescent wave decays incredibly quickly and lasts only a few wavelengths before disappearing. This ‘evanescent wave’ is the reason why quantum tunnelling is happening. So, if some probability waves make it past the barrier, and our probability wave represents electron, then there is a small, but non-zero chance that electron has surpassed the barrier. The specifics of the who, what, where and whys of this phenomenon is explained through the Schrödinger wave equation, which goes like the following:

$$E\Psi(X) = -\frac{\hbar}{2m} \frac{d^2\Psi(X)}{dX^2} + V\Psi(X)$$

Here, $\hbar$ is the reduced Planck constant, m is the electron mass, Ψ is the wave function, V is the potential energy, E is the energy eigenvalue, as depicted in [Figure 2.12](#):

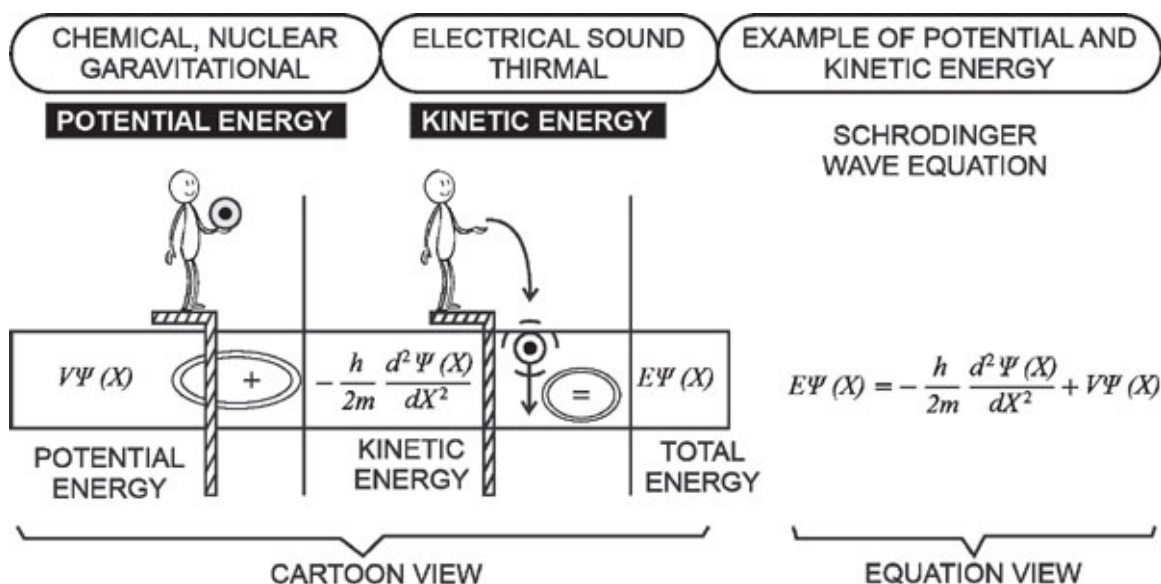


Figure 2.12: Schrödinger wave equation (potential energy + Kinetic Energy = Total energy), classical and quantum view (simplified for the understanding purpose, in quantum world representation of photon, electron or atom is not as easy as shown in the figure)

The Schrödinger equation, or the Schrödinger wave equation, is a partial differential equation based on the idea of energy conservation, which is Kinetic Energy + Potential Energy = Total Energy, to get information about the behaviour of an electron bound to a nucleus (see [Figure 2.12](#) for visual understanding of The Schrödinger equation). It does this by letting an electron's wave function, Ψ , to be calculated. Resolving the Schrödinger equation provides Ψ . With these, we get the quantum numbers and the shapes and orientations of orbitals that characterize electrons in an atom or molecule. The Schrödinger equation provides precise solutions solitary for nuclei with one electron.

Quantum effect is an important phenomenon in quantum physics; in tiny transistors and diodes, this becomes very evident and produces the tunnel effect. The tunnel effect explains with a sure likelihood that quantum particles can overwhelm a fence, even if they don't severely have sufficient energy for that rendering to the laws of classical physics. The particle just tunnels through the energy barrier. Moved to our macro world, that would mean that if we ablaze a thousand rubber missiles at a lead divider, some would seem on the other side, and what's more, we could compute exactly how many missiles that would be. Quantum tunnelling is a strange feature that has very actual and significant consequences in today's technological world.

This is owing to the circumstance that, if the remoteness among the conductive areas of circuits reduces to 10 nm and fewer, problems arise. The electrons will tunnel irrepressibly and reason interference. To stop this, engineers have to come up with all kinds of deceptions. For instance, they syndicate dissimilar constituents so that the electrons are stuck, that is, fewer to be expected to tunnel. In the interim, physicists are even capable to compute the tunnel effect so they can create tunnel-effect transistors (TFETs) whose operative is founded explicitly on the tunnel effect, since even the tunnel current can be controlled.

Randomness in behavior

The quantifiable properties of a quantum system and its progressive growth cannot be resolute. With its capability to be both here and there at a similar time, its noticeable possessions can only be stated probabilistically.

Dependence of a quantum state on measurement: In the micro world, measurements have a straight effect on the restrained object. Even stranger is the circumstance that only observation allocates a certain state to a quantum particle. In crux, this means that quantum particles have no autonomous and impartial properties. Any properties they have are attained by an exterior observer.

Uncertainty principle and how it is useful in QC

The Heisenberg uncertainty principal conditions that the more exactly the location of a quantum particle (There are two categories of quantum particles: fermions, which have a spin multiple of one-half, and bosons, which have a spin multiple of one. Fermions can have spin quantum numbers of $s = +1/2$, $s = 1/2$, or an odd multiple of $s = 1/2$. Fermions include electrons, protons, and neutrons.) is measured, the less exactly its momentum can be known, and vice versa. Uncertainty is also put on to other dependent pairs of quantum properties (known as conjugate variables) but not on all properties. There may be pairs of properties that you can flawlessly measure at the same time, but just not all pairs. Some pairs are weakly related in a way that stops impeccably precise measurements of both properties at the same time.

QC has quantum states which are observables and measurements; hence, the uncertainty principle spread on to their procedure. The uncertainty principle

also serves as a tool that can aid you to design and comprehend portions of quantum algorithms.

Young's double-slit experiment

The uncertainty principle is tremendously significant since it unites the gap between Schrödinger and Bohr by placing down the groundwork of the contemporary quantum theory. This says, the electron is a particle, as Bohr hypothesized, but we do not know precisely where it is, like the uncertainty principle states. However, the likelihood of discovery is given by the wave function (Schrödinger/Born). Therefore, there is a dichotomy like an electron, together as a particle and wave.

Now, if a holistic look is given to this theory, an unshakable understanding of quantum mechanics arises that will further be known as the Copenhagen interpretation. In the commencement, the light was believed to behave as a wave, but Einstein proved that it also shows particle behaviour through photons. The similarity was true about electrons (this fact was established by something called a double-slit experiment) which was experimentally established by Thomas Young.

Young's double-slit experiment ([Figure 2.13](#)) signifies the reflection of an interference pattern consistent with a wave nature for an object that passes through the device (the double slit). The diffraction and interference properties seem at first sight to be due to the ray of electrons, interfering with each other, though the interference pattern still comes out even if only one electron traverses the device at a time.

Here, the pattern is constructed progressively from the statistically associated effects on numerous electrons received independently at the detection system. It is observed that the electron in some sense must pass through both slits at once and then interfere with itself as it travels towards the detector even after performing the test multiple times in dissimilar ways with electrons and other particles.

The inevitable deduction is that each electron is essentially delocalized in both time and space over the device. Therefore, the conclusion is, that the wave-particle duality of quantum mechanics orders that all quantum objects, huge like an atom or otherwise, can act as either waves or particles, as shown in [Figure 2.13](#):

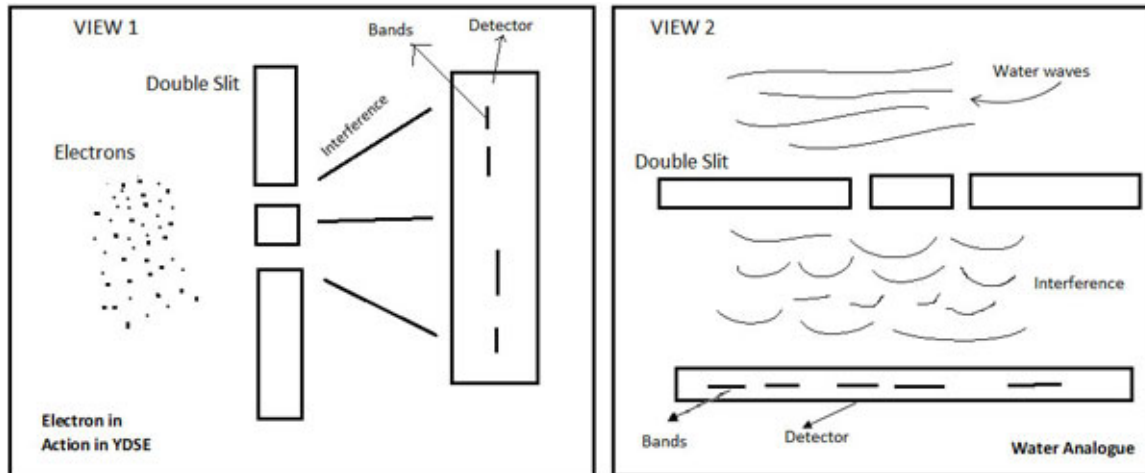


Figure 2.13: Young's double-slit experiment

Moreover, both circumstances were providing an essential idea to get a comprehensive picture, an idea that Bohr called complementarity. So, the question is how is the substance to be understood in the face of these two facts. Bohr thought that the atom, as it is, happened outside our perception. This was more than Einstein could receive as he held on to the idea of space-time at the underpinning of all physical reality and required to spread this concept to the atomic level. Bohr, on the other hand, thought space and time were worthless and reality was incomprehensible, and all that we had were marvels.

Looks messy? Yes, it is. Welcome to the world of quantum physics. Thus, it is good to have this background before we start exploring quantum computers. Now, after having this background, let's move to the arena of quantum computers.

The interrelation between entanglement and coherence

Any nonzero quantity of coherence in a system can be transformed into an equivalent volume of entanglement among that system and another initially incoherent one. This detection of the change among coherence and entanglement has numerous significant consequences. For example, it means that quantum coherence can be restrained through entanglement. Thus, all of the inclusive knowledge that investigators have got regarding entanglement can now be directly useful to coherence, which in over-all is not as well-researched. For instance, the new facts have now allowed for the physicists to settle a significant open question regarding the geometric quantity of coherence.

Before quantum physics, there is no law... right ...!!!

Wrong. Earlier, Isaac Newton (1642–1727) had sketched, in influential mathematical features, what turns out to be recognized as classical or Newtonian mechanics. It comprises Laws of Motion and Universal Law of Gravitation, which are used to compute the forces that are applicable on all objects we see from one place to another. These laws and theories describe the way those objects move. Everything from falling rivers to the paths of the Mangal Yan satellites comes under its arena. Newton's body of work, composed with the explanations and careful experiments of others before him, such as Galileo, had set the scene for the hulks of nineteenth-century physics, and a few others like Michael Faraday and James Clerk Maxwell, to complete the depiction of classical physics.

The definition of quantum computers ... revisited

A quantum computer is a physical device that critically exploits delicate quantum properties, and whose normal progression over time can be understood as the accomplishment of an anticipated and governable calculation.

Strangely, computer science forecasts that quantum computers will be intelligent enough to do certain computational tasks in remarkably fewer steps than any classical computer. Furthermore, quantum properties permit extraordinary errands to be achieved, such as teleporting quantum information, cracking unbreakable codes, producing truly random numbers, and communicating with letters that depict the occurrence of snooping. These abilities are of noteworthy applied importance to banks and government agencies. It is worth noting here that these are not distinct dreams. For instance, a quantum arrangement for transfer and getting ultra-secure information has now been realized.

Why quantum computers ... now

The computational power of conservative computers has been progressively rising over the last thirty years, approximately doubling up every eighteen months. This theory is given by Moor and is hence known as Moore's law. The theory is practically realized by reducing the size of transistors inside a computer processor. However, there is a limit; you can reduce the size of

transistors. Now, with the changing time, that limit is almost reached. If it is further decreased, then the atomic properties come into the picture. And, all other mechanical properties of the material from which transistors are made of are distributed and it becomes unstable. There are worries that the Moore's Law would at some point drop its rationality. The reason is that the cumulative reduction of components is central to the following two glitches:

- Electrons touching through even lesser and further many circuits cause the chips to heat up more and more. Electronic structures have this day and age turn out to be fewer than 10 nm in size. This resembles about 40 atoms. In transistors of this extent, the rules of quantum physics comes in a play, and therefore, the behaviour of the electrons become desperately undependable.
- The circumstance that Moore's Law will mislay its cogency does not mean that we have touched the end of the post concerning additional cumulative competence of information dispensation. This problem leads to looking for some alternative solutions. Quantum computers provide those solutions because it deals with atoms, molecules, and the fundamental particles. Also, additionally, certain glitches are so multifaceted that the finest computers existing would still need a far too long time to compute the correct answer, as shown in [*Figure 2.14*](#):

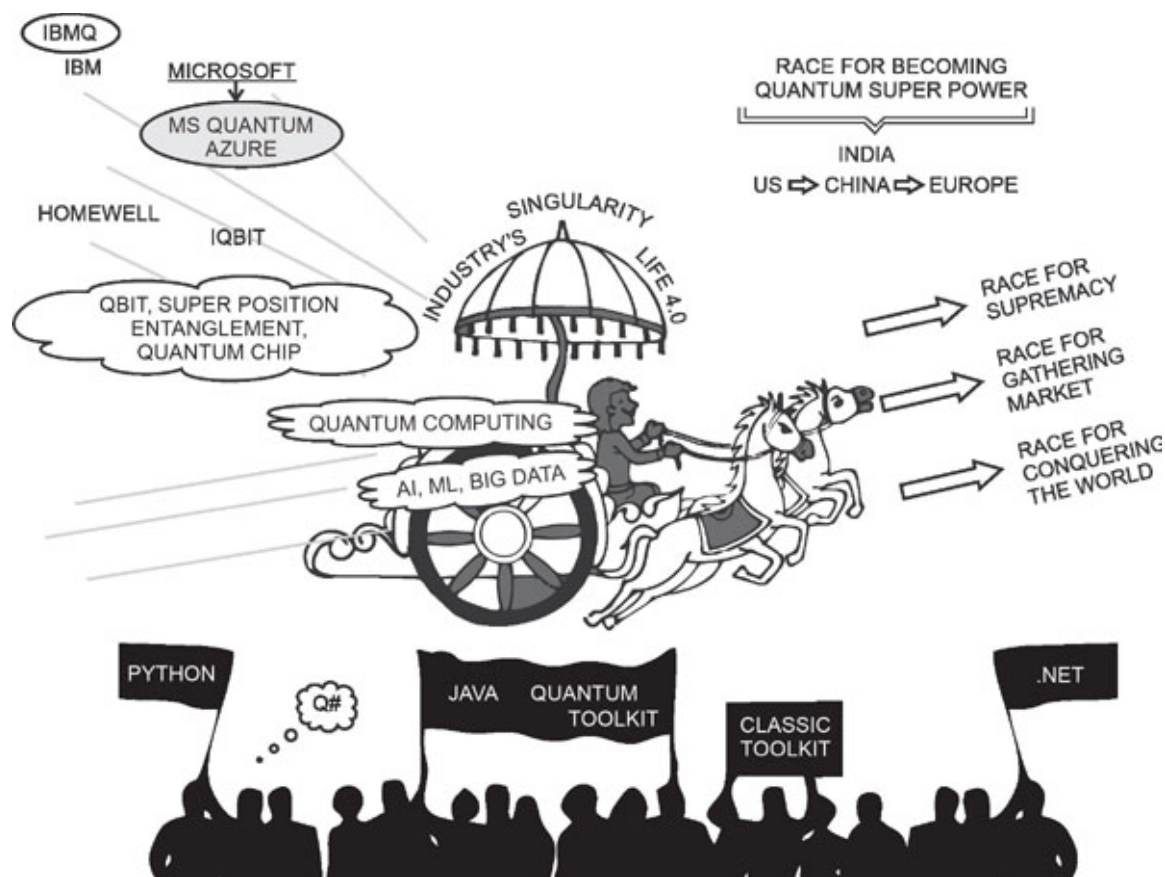


Figure 2.14: TechToon-- Need, relevance, power, supremacy, and quantum computers

Instances of such problems would be to precisely forecast the climate or the keenest conceivable investment on the stock market to make the most of return. The additional application might be to compute the loosest way for a courier company to brand manifold distributions in one tour. Let's take a couple of additional problems that force us to think why quantum computers are needed now. For instance,

Study a courier problem that involves 50 items, such as a courier correspondence with 50 towns or a shift work schedule for 50 persons. Placing the picks in the best order includes working with 30,414,093,201,713,378,043,612,608,166,064,768,844,377,641,568,960,512,000,000,000,000 likely amalgamations.

Historically, humans finished these kinds of choices founded on instinct or previous experience. Conniving the maximum cost-effective (optimal) order would take years on an influential supercomputer, and even then, the response would only be a rough calculation. But finding the best solution on a quantum computer could be completed in minutes. Imagin its impact on a business that

is looking for quick resolutions of their calculation related problems. They will be able to find the best answers in the optimal ways hence, able to save massive sums of money because they would not require to spend on the infrastructure (as quantum computers function on a cloud model) and expertise (because less people required). This new method of delivery could serve clients with intensely additional tailored, exact services, and possibly help the corporation turn out to be an industry leader. Now study an optimization delinquent in financial services.

- Imagine, for each client, portfolio managers could use quantum computing to enhance a sole combination of exponentially better-tailored investment choices in suggestively less time. The time saved would give consultants better capacity and elasticity so that they could provide financial planning that is specific to each client's investing criteria and risk profile. As the degree of such a problem increases, so does the number of parameters which are essential to model such problems. Subsequently, traditional computers struggle to finish the calculation due to the huge number of computational resources required.

Similar types of computation-intensive problems are emerging in the field of chemistry, psychology, physics, business, and other areas. Therefore, there is a need to encounter them. Quantum computers are well suited for these special and particular types of problems. Hence, there is a dire need for it.

Also, QC will possibly deliver us with completely new competencies and consequently change our lives in very unforeseen ways. For instance, we are now harvesting 2.5 exabytes of data per day. One Exabyte is equal to about one billion gigabytes. Also, we're living in an era of data creation machines.

The generated data has to be polished into business understanding to have the correct business value. As the complexity and the absolute extent of our data sets inflatable year after year, hence, there is a need to establish, and excerpt true value from the noise. Quantum computing is the answer.

To understand it in a quicker way, a cartoonist view of the uses, benefits, and implications of quantum computer-related concepts as depicted in [Figure 2.14](#). Also, as we know quantum physics impacts quantum computers at its core. Therefore, [Figure 2.15](#) covers some important aspects of quantum physics which impact the areas like construction, functioning and maintenance of

quantum computers. [Figure 2.15](#) is represented in a form of a mind map for a quick view.

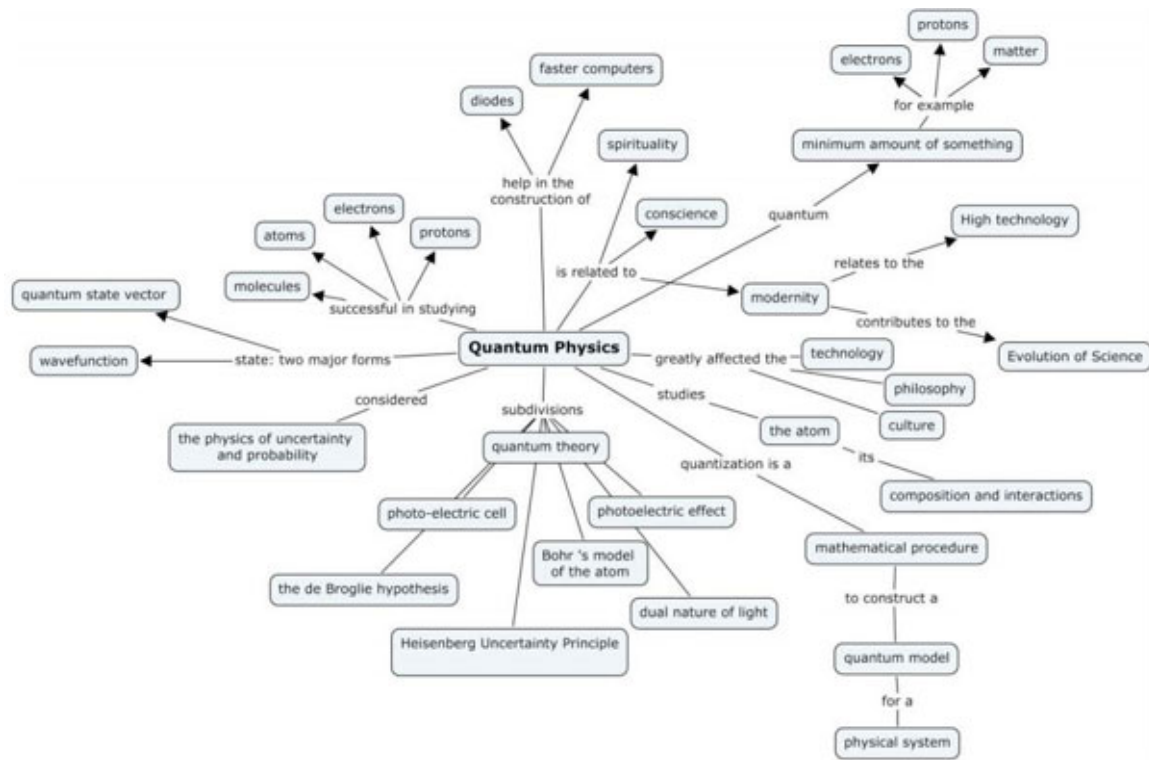


Figure 2.15: The important areas of quantum physics with impacts quantum computers (source: google image, XMin)

Tell me how QC will work with today's technologies

With the incessant growth of technology and the wish to be linked to people and objects alike for an additional effectual, quicker, and calmer way of doing stuff, the **Internet of Things (IoT)** or the network of smart devices that gathers and shares data including humans, produced are becoming popular. Business Insider (an agency which keeps track of the latest happening in the market and does forecasting) Intelligence predicted that “by 2023, customers, businesses, and administrations will put in 40 billion IoT devices internationally.” This enormous number of devices also means an immense volume of data produced every single day. The challenge will then be on how all this data will be analysed and what facts can be consequent. Safekeeping will also be an apprehension given that the devices are linked via the internet and cyberattacks have become more and more urbane as technology progresses.

Handling a big volume of data, carrying out calculations much quicker than traditional computing platforms, and resolving problems concerning cybersecurity are some of the things an influential quantum computer can do. Appreciations to the quantum-mechanical marvels of superposition and entanglement, quantum computers can perform calculations millions of times quicker than the old-style ones.

Sanctuary of the IoT is a principal question. Presently, cryptographic algorithms are being used to make safe communication. But then, since they rely on public-key systems, their encryption could be wrecked by erudite hackers utilizing quantum computers in the not so aloof future.

On the other side of the coin, quantum computing has the aptitude to generate an almost un-hackable network of devices and data. The necessity to steadily encrypt and defend IoT linked devices and power them with exponential speed and analytical capabilities are overbearing for both government and the private sector. This can be done through QC. Finally, to expose cosmos, crack business problems and perform complicated chemical modelling, we need AI with super-fast processing speed (we already discussed details of this in this chapter and would elaborate it further in the upcoming chapters). Quantum computer could potentially provide that.

One of the major theories for quantum computers: The church-Turing Principle (CTP)

Feynman believed for simulating quantum mechanics, there is a requirement for a probabilistic computer or a new kind of a computer. He intends about the quantum computer. It is to be well-known that Feynman used the term “quantum simulator” and projected it as a model of quantum computers. In his original paper, he presented a drawing of quantum computers. Feynman’s view was that if the apparatus founded on the fundamentals of quantum mechanics is interpreted, then it will be able to simulate random quantum systems probabilistically. His insight is looking upon as the source of present investigation on quantum computers.

He deliberated some phenomena to address the practicality of quantum simulators. He insisted that quantum systems cannot be designated utilizing conventional computers. He disclosed that we cannot signify the outcomes of

quantum machines with a conventional universal device owing to the so-called hidden variable problem.

The hidden variable problem addresses that a physical system cannot entirely define nature. If we force to put the quantum equations in a form as close as likely to classical equations, we cannot simulate the quantum equations in the normal way because there are too many variables. Is this a deadlock then? Probably not... here comes something called the **Church-Turing Principle (CTP)** that plays its part.

Discussing the CTP and other specific QC problems makes sense now, as we have a basic foundation of quantum theories. And we know why quantum mechanics are required to understand quantum computers. From here on, we will explore the exciting world of QC. We start our journey with the thoughts of the pioneers of the field, David Deutsch and Richard Feynman.

The discussion continues: Church-Turing Principle (CTP), quantum Turing machine and universal quantum computer

In 1985, David Deutsch was given a physical form of the Church-Turing Thesis. As per the thesis, every finitely attainable physical system can be flawlessly simulated by a universal model computing machine functioning by finite means. Further, Deutsch generalized the Thesis. The generalized version of the thesis says, all computation, which can be appreciated physically, can be performed using a quantum Turing machine. Deutsch has played a revolutionary role in carrying quantum computing into the mainstream of computing.

CTP takes the Turing machine to the next level precisely to the arena of the Quantum world and perform the generalization of universal Turing Machine. Hence, it would able to “compensate” the quantum computers as well. The idea of a universal computer is computationally comparable to a quantum Turing Machine.

In technical terms, quantum Turing Machine could be defined as a computing machine M which is capable of flawlessly simulating a physical system S , under a specified labelling of their inputs and outputs, if there exists a program $\pi(S)$ for M that renders M computationally equivalent to S under the labelling. Here, the labelling represents the physical depiction of data. Quantum

mechanically, the labelling is a set of methodical pairs consisting of the Hamilton operator and its eigenvalues.

In quantum mechanics, a Hamiltonian is an operator equivalent to the totality of the kinetic energies plus the potential energies for all the particles in the scheme. This accumulation is the entire energy of the system in maximum of the cases under analysis. It is usually signified by H , but also $\hat{H}$ or to highlight its purpose as an operator. Its spectrum is the set of likely results when one measures the total energy of a system. Because of its handy relation to the time-evolution of a system, it is of importance in furthermost constructions of quantum theory. To understand it better, let's dig into a universal quantum computer.

Universal quantum computer

In his landmark paper in 1985 titled 'Quantum Theory, the Church-Turing Principle and the Universal Quantum Computer', David Deutsch defined how a computer runs using the bizarre rules of quantum mechanics and why such a computer would vary necessarily from normal computers. In fact, Deutsch presented that it is conceivable to hypothesise reversible quantum gates for any characteristically computable function and proposed that quantum superposition might permit quantum evolution to do numerous classical computations in parallel.

Deutsch's paper described the first true **Quantum Turing Machine (QTM)**. In QTM, read, write, and shift processes were all accomplished by quantum mechanical exchanges and whose tape could be in extremely non-classical states. For instance, it could concurrently encrypt numerous inputs to a problem on the tape and perform a calculation on all the inputs at the same time. In 1997, Samuel Bernstein and Umesh Vazirani exhibited that it is likely to conceive of a universal quantum Turing machine. In such creation, we must assume an adequate number of bits that resemble the tape of a Turing machine. The QTM is adequately modest and at the same time universal to show many theorems about quantum computation.

The universal quantum computer can flawlessly simulate the UTM. In accumulation, it can simulate several physical systems, real and theoretical, which are outside the scope of the UTM. In quantum computers, we can

operate its global state at all times even when we aim to manipulate a precise part of it, even as minor as a qubit.

After the detailed discussion on CTP, Deutsch offered a general quantum model of computation named the universal quantum computer and represented it by Q. It can flawlessly simulate each finite, attainable physical system. It can simulate model closed zero temperature systems, together with all other occurrences of quantum computers and quantum simulators, with randomly high but not faultless correctness. Consequently, upon CTP, Q is a model equal to the quantum mechanical form of the quantum Turing Machine. There is a sense in which Deutsch proposed a quantum Turing Machine, although he did not directly formalize a quantum Turing Machine.

Quantum Computer Q has, similar to a Turing Machine, two components, that is, a finite processor and an infinite memory, of which only a finite portion from each is used. The computation proceeds in steps of secure duration T , and during each step, only the processor and a finite part of the memory interact, the rest of the memory remains static.

Broadly, a universal quantum computer is the hypothetical holy grail of quantum computers. Universal quantum computers are not an exact kind of quantum computer but usually define a quantum computer that is not limited to a collection of limited use cases. Universal quantum is more of a consequence than a specific type of quantum computer or architecture as long as it can proceed with any quantum algorithm. Some critiques reason that it's more of a marketing term than anything else; however, the goal is to produce a quantum computer that can achieve any type of problem you toss at it, be it conventional, quantum, or simulation. A universal quantum computer cannot only complete the broadest diversity of situation problems but can also simulate all the additional types of more limited quantum computers.

The soul of quantum computers: The impact of algorithms

The quantum gates will perform the mathematical operations defined by the quantum algorithm. In between, we have the exact application layers for quantum languages, compilers, and also error amending codes. As classical computers are separate from quantum computers, so are the algorithms. A quantum algorithm is dissimilar from a traditional algorithm in the way that it is not completely deterministic. A quantum algorithm is probabilistic, which means that the precise solution hails from a certain identified probability. A

QC that is a million times quicker than a classical computer. And it is probabilistic in nature. For this probabilistic nature, quantum alternatives of traditional algorithms will be essential to be advanced to be implemented in a QC.

In order to work algorithms in a QC with precise quantum gates, as a prerequisite a quantum form of the classical algorithm would be needed. As of today, not all the traditional algorithms are labelled in the quantum world and only a limited number of them have a quantum counterpart. For instance, clustering and classification algorithms.

Quantum algorithms may make use of superposition and entanglement, and function on the entirety of qubit values all at once. The drawback is that the related superposition collapses on determining (that is, read) the qubits, in its place of a large set of values. The outcome is a single value occupied from the likely values in that superposition with a likelihood. The fundamental query in quantum computing is how to produce expressive outcomes from computations, provided all the superpositions vanish on a measurement, along with all the computed values. As it turns out, there are clever algorithms that can grow unwelcome superpositions to abandon each other out so that only the wanted outcomes continue. They could upsurge the likelihood amplitudes of exact states of interest so that the measured outcome has a high probability of being the required result. Shor's algorithm for factorization and Grover's search algorithm are samples of such quantum algorithms. The detection of such algorithms altered the real scenery of quantum computing.

[Quantum computing supremacy](#)

Quantum supremacy defines the aptitude of a quantum computer to outclass its classical complements. Some corporations, such as IBM and Google, say that they are close to building super high-performance quantum computers, as they last to shove more qubits composed and shape more accurate devices by putting proper error correction mechanism in place. However, not everyone believes that quantum computers are worth the effort. Few mathematicians trust some problems are essentially unbearable to be overwhelmed, putting quantum computing incessantly out of reach. However, mostly the experts are agreeing and converge on the opinion that working quantum computers are definitely a possibility which has the potential to change the complete landmark of computation itself. Below is the point of view of one of the

pioneer scientists in the field Seth Lloyd on the impact of quantum computers on the universe.

Point of view: Seth Lloyd on Universe as QC

Seth Lloyd is one of the pioneer thinkers in the arena of quantum information technology. He's been with the ground from the very beginning to the current day popularity of QC. He is a proponent of the view that the world is a quantum computer. Hence, the "world" computer functions employing the philosophies of quantum mechanics. Seth Lloyd in his book "Programming the Universe: A Quantum Computer Scientist Takes on the Cosmos" which was published in 2006 argues that the cosmos can be viewed as a massive quantum computer. According to him, all experiential spectacles are reliable with the model in which the cosmos is hazy from a quantum computer, for example, a quantum cellular automaton. The cellular automaton is a concept which defines a lively system that is fragmented separately into black and white lattices, and its "cells" wrinkle information from the neighboring cells based on whether or not to alter the color.

However, to start understanding if the cosmos is a massive quantum computer, we have to revisit some of our understanding. Let's start with relearning information. What is information? According to Lloyd, everything in the universe is made of chunks of information named bits. Even the atoms themselves are also bits of information. Information is present all around, just like quantum mechanics. This world and cosmos are built with atoms and the fundamental particles which store and process information all the time. Hence, technically the world is the quantum computer.

As we discussed, QC works on the principal's superposition. A small number of particles in the superposition of both 1 and 0 interpretations manipulates to give us a huge amount of information. For example, a hundred particles in superposition would mean representing every number from 1 to 2100. A traditional computer can be intended to read one mixture of three bits at a time, while a quantum computer will recite all likely amalgamations.

This means that quantum computers can process information in parallel. A system with any number N qubits will process 2^N calculations at once,

giving us a new and extremely fast means of computing, such as factoring great numbers or appraising very complex algorithms used for data analysis in finance, science, or cryptography.

Ed Fredkin originally proposed that the cosmos could be a computer in the 1960's, and Konrad Zuse derived the same idea individually. In their opinion, the cosmos could be a kind of computer like a cellular automaton. This is like the way a line or affecting collection of ants might share information among each other about their environments, signaling to each other whether or not to track a food path.

[Quantum computing use cases](#)

A use case is essentially taking the steps compulsory to complete a process, but doing so in a hypothetical way, you don't essentially need to have the prerequisite acquaintance to appreciate the application of the process, but you do need to know the order of the process and how it might mark the ultimate application. In computer science, we produce use cases as an understood experiment, to comprehend what the physical application of a process in any given circumstances might, or might not attain deprived of physically testing theory. [Figure 2.16](#) depicts some use cases:

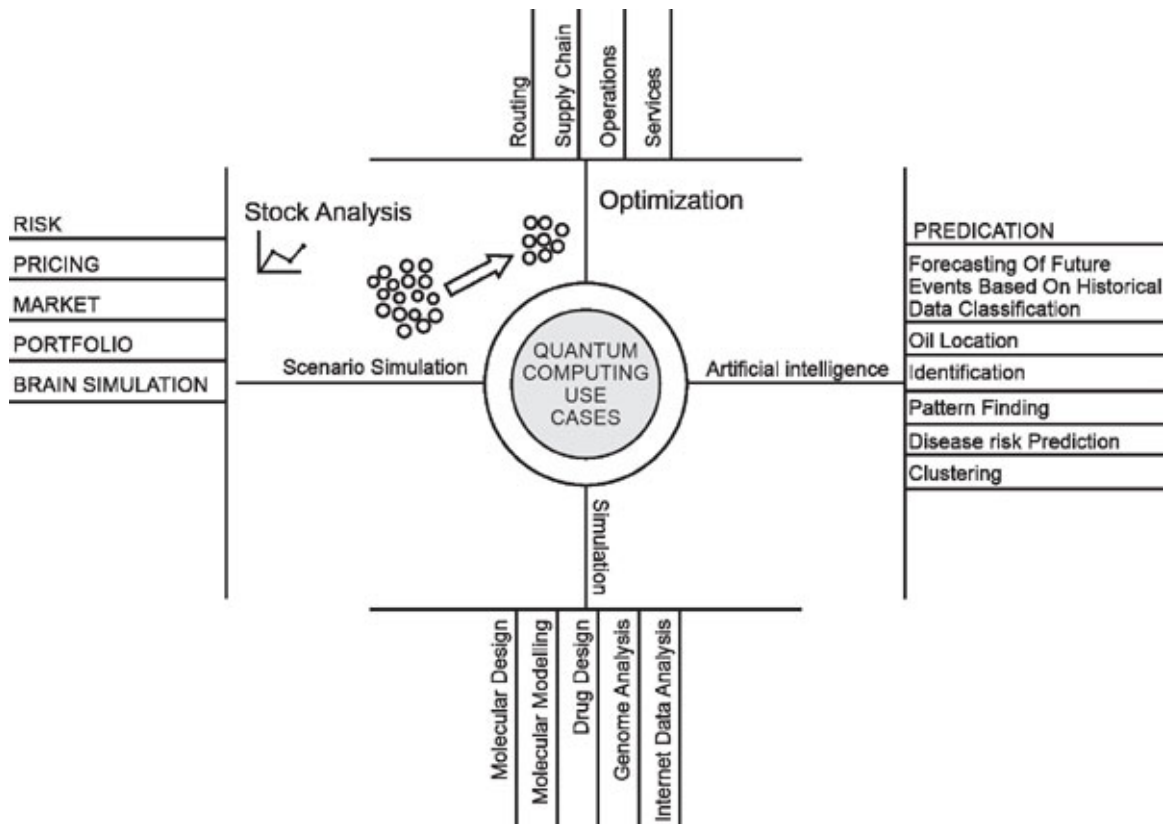


Figure 2.16: Quantum computer use cases

Sample opportunity areas for quantum computing: explanations for few important ones

There are multiple domains of industry, life, business personality, society, and government where quantum computing find its use cases, and applications. This something is what we discussed throughout this chapter and will discuss in the book. However, a few important ones are mentioned as follows and presented in [Figure 2.17](#):

- **Drug discovery:** Simulated annealing is an algorithm, presently used for the forecast of the effects of possible healing methods while enhancing for non-adverse effects. Quantum computing can change some of these methods and may even be able to display developments at gage in the next few years, such as proceeding drug design to the point of providing custom-made treatment drugs for specific patients.
- **Portfolio optimization:** Quantum computing displays potential in serving to regulate striking portfolios, given thousands of resources with

intersecting dependences. Moreover, quantum computing methods could be used to more efficiently classify key fraud indicators.

- **Supply chain optimization:** Glitches related to Supply chain optimization come in numerous dissimilar forms, such as procurement, manufacture, and delivery. As quantum computing improves, it will change from being able to resolve one-time situations like plan-o-grams or lorry loads to large system-wide situations like stock level, local deliveries, and finally global supply chains.
- **Utility delivery optimization:** With quantum computing, optimization of important systems could be a ubiquitous probabilistic endorsement for cost savings. Furthermore, the best product lifecycle and auxiliary could be resolute at a system-wide scale to improve and comprehend the insinuations. Thereafter, it can be fragmented down on a part-by-part basis, as depicted in [Figure 2.17](#):

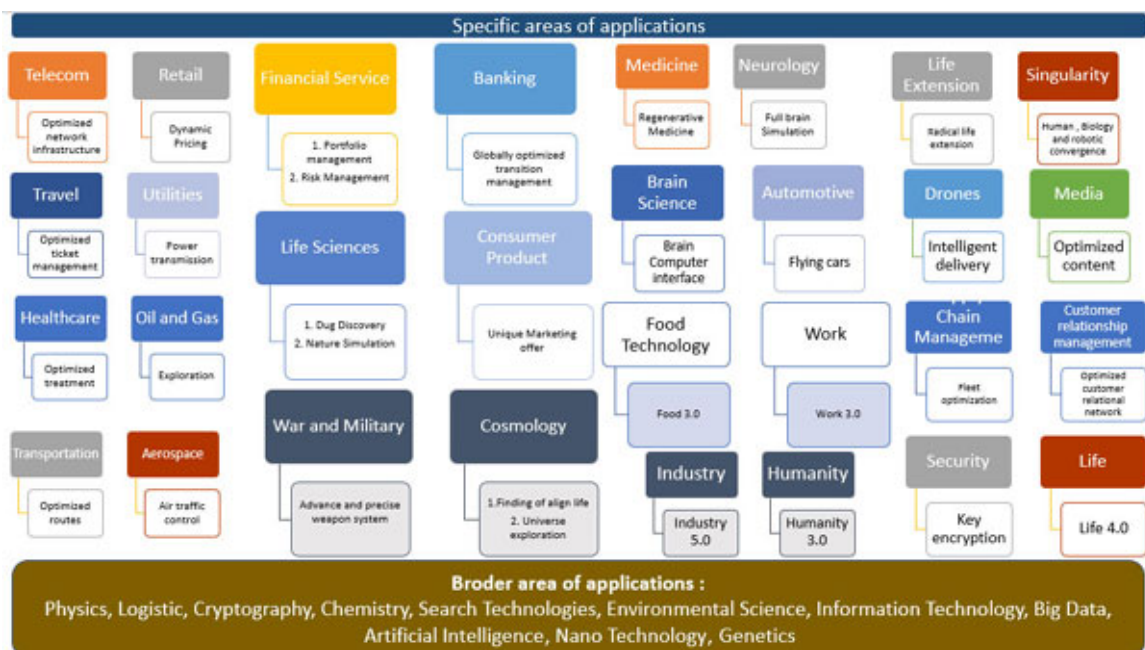


Figure 2.17: Applications of quantum computers (industry and specific)

The spread of QC is wide. Therefore, the companies that are involved in quantum computing developments target numerous enterprises in dissimilar industry sectors.

The social and commercial effects of quantum technology.

The growth of quantum computers and associated areas can help determine and handle the social challenges. This will bring and produce positive changes for all sectors of business, economics, and commerce. Developing a quantum network or coming up with the universal quantum systems would be a couple of milestones in the journey. However, a prodigious deal of advancement in the technological and social landscape is essential before we can understand and create a full-fledged quantum internet or a big universal quantum computer. We are moving in the right direction; there is some progress that has already happened, which is explained as follows:

Secure communication is the backbone of today's information society; hence Quantum security is going to play a major role in this requirement of society, business, and organizations. Quantum security systems are Essentially based on **Quantum Key Distribution (QKD)**. Basic QKD systems are now commercially existing. For instance, in its last elections, the Switzerland government used QKD technology to save the internet connections among vote telling stations and the result collation sites.

Governments, corporations, and other establishments frequently use their IT systems to resolve big search or optimization problems. To discover effectual distributions of resources to the tasks they want to solve, to search through big data files, to design energy-efficient chips or airplanes, quantum computers can bid important speed-ups. Likewise, for search problems, discovering the least or determining a specified function over some finite domain, finding the shortest way among two points on a map, approximately resolving huge systems of linear equations, QC is very effective.

At the moment machine learning is based on classical computers. Typically, in machine teaching, data is provided to the systems for analysis and pattern matching and it provides some model or hypothesis that “fits” this data well. Finding a well-fitting or even optimally-fitting model for the data is an optimization problem that can be solved better or faster on a quantum computer using quantum algorithms.

As we discussed throughout this chapter about the availability of quantum systems, it is not a dream anymore. Several first-generation quantum computers, archetypally partaking ten to twenty qubits, are now obtainable in the cloud to the research community for trying simple algorithms, some on a saleable basis.

Story 1: quantum organization and leadership

Management in the 21st century booked a new positioning. It is progressively originated on the capability to manage with continuous modification. The philosophy of management is changed and it is now prearranged around networks and does not follow pyramids. It is reconstructed on fluctuating corporations and associations and does not reside in the state of self-sufficiency.

It is built on technological benefits and not bricks and mortar. Moreover, it is also imperative to note that new corporation are networks of complicatedly interlaced webs that are based on a virtual amalgamation of thoughts, ideas, technologies, process, and people rather than individuality, and mass customization and then mass production. The new management philosophies and hence the corporations are augmented by the technologies. The new breed of technological revolution is making them smarter and responsive. Industry 4.0 is an example of this.

If a corporation is smart, then its leader has to be extra intelligent to fortify the corporation. He has to think from the quantum way, not from the classical way. He has to be a quantum leader. A quantum leader is the one who handles and acclimatizes the fast transformation in the dynamic circumstances and situation. He is not bothered about uncertainty. He trusts holism with the rapidity of access to data, information, and facts. Also, he can fulfil the purposes of the corporation in an integrative way. To captivate the potentials of an intelligent organization into consideration, the quantum leader should have a robust collaboration with the groups, he should be conscious of the ambiguity and probabilities, and most importantly he should be able to recognize the influence of his leadership on the bigger whole.

By betting on the potential and the assets of a quantum leader, the intelligent corporation should be fast to act and, be adaptive to the nonstop altering circumstances. The quantum element makes a leader elastic in the process, and enable them to be visionary and instinctive. In a nutshell, he has to have a probabilistic mindset, not deterministic. He has to think in a fuzzy way, and not bind him or herself in the rule. He has to first break all the rules. This is the fundamental difference between quantum and classical physics as well.

In an old-style way of management and leadership, administrative glitches are seen from a mechanical viewpoint. We incline to see groups of people carrying

out specific responsibilities as if they were portions of a bigger machine, the corporation. We analyse problems in a reductionist style, examining specific functions and seeing each division of a corporation as if it were a separate entity, even when working on interdepartmental issues. We identify separate glitches, turn our decision-making consideration to the fragments of the organization that displays a strong warning sign, and resolve problems from a righteously applied standpoint.

However, the furthestmost important essential capability for any organization may be elasticity in the expression of alteration. The complexity theory is flowing through officialdoms and moving the properties of apparently unimportant acts between them and transversely the globe with dizzying rapidity. To continue, corporates must be elastic. This understanding left us from strategic, exact solutions and into the direction of employing our stress on business governance which compromises mission, vision, and supervisory philosophies. This is where business expansion becomes glosses, and help corporates to grow evocative ascendancy and actionable goals that are not so unbending as to avert variation. It leads one to release the wheels in a jiffy, trusting the horse to know the method to return home.

Quantum organization or corporation is fundamentally based on innovative and open mindsets. It is all about change in view and perspective and about how we look at things differently. In its place of visualization problems and benefits in a form of charts, structures and teams or people as just mere collections of people. It sees organizations as dynamic modifiable entities where the processes, methods and rules could be changed and evolved continuously.

It is about seeing and visualizing things holistically. Traditionally, well-thought-out procedures are the gears of effective organizations, but in this novel interpretation, the organization is what ensues when procedures carry people systematized. That means, in its place of observing organizations as limited, obviously demarcated things, we have to contemplate each organization as an organic, breathing, conscious entity. That can be a lot to digest for the old-style manager and decision-makers, but it brings tremendous value propositions with a standpoint that permits to beneficially encircle and bring a change and innovation along with hunger for future readiness.

Quantum organizations and leaders believe in probabilities. They know whatever is running today may not work tomorrow and even if it functions, the agility would be less. They are off having a strong mindset of adopting,

changing, and excelling by making themselves future-ready. Quantum computers would not be around for the next few years. But, then why are organizations investing in them? Why are leaders interested in the progress story of it?

Because they are betting on probabilities and trying to make themselves future-ready. They know once it happens and if they are not ready for it, they will be left behind in the race of progress and competition with their counterparts. That “situation” is potential and would be a death sentence for them. History is full of these types of instances. The current list of fortune 500 companies and the list around 5 decades back itself will tell the story.

Here and now is the correct time for business leaders to prepare for quantum. The circumstances are in a place of experimentation and enhance this essential new technology. At this juncture, there are a few early steps that business leaders can take, which are explained as follows:

- Think with a quantum mindset where the rules are not fixed and Boolean in nature but fuzzy which is bound to change if needed, particularly if your organization is tangled in finance, chemistry, optimization, or machine learning. Quantum computing has completed sufficient development to permit some of your time.
- Allocate a quantum expert to explore new opportunities in business and technology. This individual can go to sessions and workshops and measure rival tactics to quantum computing. The assigned person can look at numerous ecosystems that are established to form these methods.
- Assess which precise areas of your business may profit from quantum computing, and which quantum computing method may suit you best.
- Obtain the correct skills. Quantum computing is different from conservative computing, right down to substituting the bit, the base unit of information. Persons who are exposed to novel problem-solving methods should be at the top of your slope. .
- Have a trial with an actual quantum computer today. Free quantum processors are obtainable over the cloud. Explore the lessons and emulators and program your particular algorithms. Start seeing which difficulties you need to resolve in the quantum era. As vital as conservative computing is, it has restrictions.

Quantum computing signifies a fundamental new archetype that may help resolve difficulties we cannot at this time resolve. Organizations that search to be in the front of this transformational change will grab modest leadership.

Quantum computers: just a piece of a holistic system

Just like information technology, quantum information sciences are a very wide field and quantum computers are just one but important part of the bigger integrated system. Quantum information sciences are ever so often separated into four areas: quantum computing, quantum simulation, quantum communication, and quantum sensing. The last two are very close to commercialization and products have already started to appear. For instance, it includes several devices and software components, such as:

- Quantum random number generators.
- Quantum cryptography software's and hardware.

Several companies focus on these kinds of quantum devices in its place of trying to contest in the enormously exclusive field of quantum computers. Quantum information sciences would hopefully open up a variety of applications that will meaningfully do the transformation to our world. The following are the areas of the promised applications:

- Faster optimized searches.
- Better artificial intelligence.
- Better cryptography.
- Secure networking.
- Military uses.
- Improved weather forecasting.
- Improved understanding of the quantum world, astrophysics, and cosmos.
- Perfect privacy systems.
- Better financial modelling (stock trading, derivatives trading, and so on).
- Better financial fraud detection.
- Traffic management for autonomous vehicles.
- Better, longer-lasting, lighter batteries.

- Quantum money.

Facts form research, innovation, and industry

IBM's new computer will boast 53 quantum bits, or qubits, the elements that are the secret to quantum machines' power. Google has a 73-qubit device, but it hasn't let outsiders run programs on it. IBM's machine, on the other hand, will be accessible via the cloud. Few additional insights are provided below

- Quantum computing was started in the early 1980's when physicist Paul Benioff proposed the first quantum mechanical model of the Turing machine.
- Google and NASA reported that their new 1097-qubit D-Wave quantum computer had solved an optimization problem in a few seconds. That's 100 million times faster than a regular computer chip.
- The superposition of qubits is what gives Quantum computers their inherent parallelism. According to physicists, this parallelism allows a quantum computer to work on a million computations at once, while your desktop PC works on one.
- The quantum version of the classical binary bit is physically realized with a two-state device.
- DARPA brought online the world's first operational quantum network in 2003, breaking new ground in the fields of quantum computing as well as secure communications.
- The world's first dedicated quantum computing focused commercial business is done by (as per claim, there are other claims as well) 1Qubit. This startup was established in Vancouver, British Columbia, in 2012
- IBM launched Q, which offers 5 qubit quantum computing services via the cloud in 2016. Last year it upgraded to 20 qubits of quantum processing power.

Some questions and answers

If parallel computing can be achieved by classical computers, then why QC?

The vibrant difficulty of a quantum computer is that it is not deterministic. The precise result can attain an advanced probability of measurement than the others, but their probabilities do not have to drop to zero. The measured result can then be incorrect, which is why the process must be recurrent till you are sure you have the accurate result.

Parallel computing is also conceivable with classical computers, but solitary with an augmented disbursement of hardware as well as a high energy and space need. With quantum computers, the storage volume doubles with each extra qubit. To blow an RSA code, though, a lot of entangled qubits are essential.

Can you give me insights on the historical development and journey of Qubit?

[Figure 2.18](#) showcases the historical journey of Qubit:

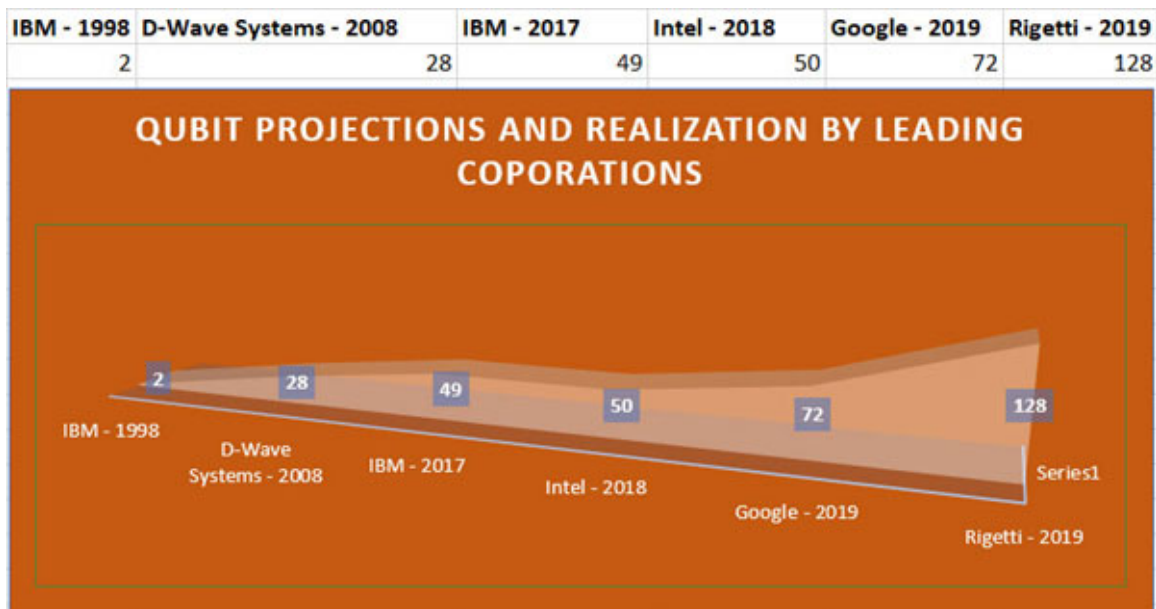


Figure 2.18: Historical development of Qubit

What are the major advantages and disadvantages of Universal Quantum Computers?

Universal quantum computer manufacturers

can use any technology they want to bring quantum processing to the largest and widest number of processing scenarios. The disadvantage is that these computers require the largest number of qubits and are among the hardest types to make. Small quantum computers have now been constructed by many academic and engineering teams. The two platforms are presently proficient in

supporting the biggest systems that make use of ‘ion traps’ or superconducting qubits.

The chief tech companies, such as IBM and Google, are concentrating on superconducting qubits and have come as far as making an initial test version of a quantum computer available via the cloud, complete with a programming language. Intel is following two lines of growth: superconducting qubits and quantum dots in silicon. The latter technology proposals striking scalability potential, since it would be comparatively easy to shape on current chip manufacturing technology.

The semiconductor industry is mainly based on silicon platforms. Microsoft is backing a diverse technology: the topological quantum computer. Although in theory, this technology can harvest a very stable platform, no one has yet thrived in creating a topological qubit. Also, they have a strategic alliance with Honeywell who is working on ion trap technology. In parallel, Microsoft is developing quantum algorithms appropriate for use on several qubit platforms.

Human brain and quantum computers: the QuBrain project

Tech giants like Google and Amazon are sprinting to grow their quantum computing technologies. Recently, teams of Google and NASA experts displayed that a D-Wave chief computer was 100 million times faster than a conservative computer in a test and positively simulated a hydrogen molecule with it.

This brings the point if we can simulate chemical molecules and their behaviour, then why not apply quantum computers to understand the brain. Because ultimately brains also deal with atoms and the fundamental particles. Irrespective of whether human consciousness trusts on some form of quantum phenomena (several speculate that it does but we do not know for a fact). There is no hesitation such a measure will push the limits of the human mind outside its natural competences. Few dedicated types of research are happening in the area. One of them is QuBrain.

The aim of the Quantum Brain Project (QuBrain) is to prove that the brain is a quantum computer, and thus non-deterministic. The motto of the project is very clear, according to them “We are interested in how the brain may use quantum computing for consciousness and cognition. In particular, QuBrain

scientists believe that the brain must use topological computing to overcome the disturbances caused by cerebral dynamics”. Hence, one of the goals of the QuBrain project is to reverse-engineer the brain’s possible use of quantum states in cognition, especially, to examine the nuclear spin of phosphate atoms in the brain.

Scientists selected phosphate for the reverse-engineering approach because phosphate is the only good candidate in the body for qubits. Likewise, Phosphate is very vigorous in biology in general and very significant to the human nervous system, particularly in our mitochondria organelle. In every cell, phosphate allows the making of ATP (adenosine triphosphate, the energy carrier of cells), and phosphate also provisions and issues calcium phosphate, which has a mainly significant role in the function of synapses. Synapses are one of the significant elements of the human memory system.

The project aims to determine whether the brain functions like a quantum computer, with neural qubits, postponed in states of superposition and entangled with phosphate atoms in the brain. The QuBrain team is attempting to confirm its theories of quantum cognition using approaches together with in vitro (test tube), in vivo (in animals, here rats) testing, and quantum measurement via nuclear magnetic resonance (NMR). They hope to prove unambiguously how quantum cognition takes place in the brain by directly rivalling it in the lab.

Based on the present progress and future projection, computers have much further to go to compete with human strengths. Thirty years of computer vision reveals that 1 MIPS can extract simple features from real-time imagery. 10 MIPS can shadow cruise missiles and early self-driving vans show 100 MIPS can shadow abstemiously random features. 1,000 MIPS will be suitable for coarse-grained three-dimensional spatial awareness. 10,000 MIPS can find three-dimensional objects in clutter. Programs require memory as well as dispensation quickness to do their work. The ratio of memory to speed has continued to be constant throughout computing history.

The early electronic computers had a few thousand bytes of memory and could perform a few thousand calculations per second. Early supercomputers did a billion calculations per second and had a billion bytes of memory. The modern, utmost supercomputers can do a trillion calculations per second and can have a trillion bytes of memory. Therefore, we can say that for achieving human capabilities for a machine, the processing speed and the storage capabilities

both are essential. How to achieve the dream of human equivalence through the machine when traditional computers are reaching their limits? Yep... you got it right; probably through quantum computers and other technologies like DNA computing.

In a nutshell, once quantum computers would be realized, the dream of a human-like brain is possible. This could be used to argue humans or create a “real” humanoid or further to achieve the singularity.

Mind Map for the chapter

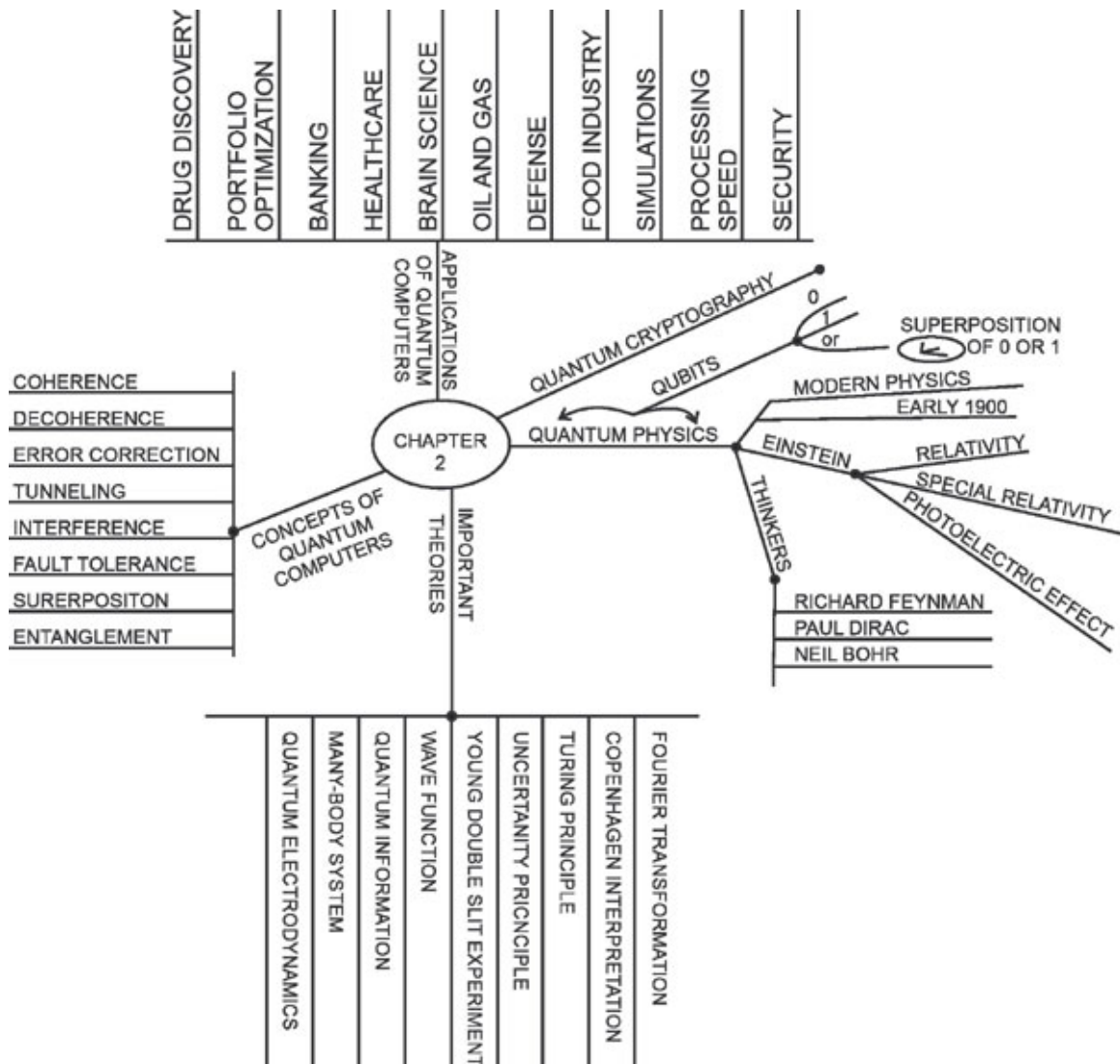


Figure 2.19: Mind Map for [chapter 2](#)

Conclusion

In this chapter, we covered the fundamentals of quantum physics including its history and uses in our daily life. After the brief background of quantum physics, the chapter related those concepts with quantum computers. For instance, the fundamental building blocks of quantum mechanics are the following concepts, however, at the same time, they serve as essentials of QC as well:

- Quantization: Energy, momentum, angular momentum, and other physical quantities of a bound system are restricted to discrete values (quantized).
- Wave-particle duality: Objects are both waves and particles.
- Heisenberg principle: The more precise the position of some particle is determined, the less precisely its momentum can be known, and vice versa. Thus, there is a fundamental limit to the measurement precision of physical quantities of a particle.
- Superposition: Two quantum states can be added together, and the result is another valid quantum state.
- Entanglement: When the quantum state of any particle belonging to a system cannot be described independently of the state of the other particles, even when separated by a vast distance, the particles are entangled.
- Fragility: By measuring a quantum system, we destroy any previous information. From this, it follows the no-cloning theorem that states, it is impossible to create an identical copy of an arbitrary unknown quantum state.
- On top of all these, the chapter has given a snapshot of how and why a quantum organization is needed and how the universe and human brain can be thought of as a QC. It also provided the life history of a few quantum scientists.

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Important terms for the chapter: refer to this when required

IT terminology

- **Artificial Intelligence (AI):** denotes the imitation of human intelligence in machines that are coded to reason similar to humans and impersonate their activities. The term may also be pragmatic to any machine that displays qualities related to a human mind. For example, learning and problem-solving.
- **Machine learning:** It is a sub-branch of AI that offers systems the capability to automatically learn and advance from experience without being explicitly programmed. Machine learning emphasizes the growth of computer programs that can admit data and use it to learn for themselves.
- **Neurons:** This is a specific cell that is used for conveying nerve impulses. It can be thought of as a nerve cell which exists in the brain and does networking.
- **Neural network:** It is a sequence of algorithms that endeavours to identify underlying associations in a set of data by a process that mimic

the human brain functions. In this sense, neural networks denote a system of neurons, either organic or artificial.

- **Big data:** It is an area of study that would focus on the ways to analyze, methodically excerpt the information from a big data store (for example data lake). In other words, it deals with data sets that are too large or complex to be handled by old-style data-processing application software like a **Relational Database (RDBMS)**.

Qubits associated terminologies

Qubits can present three main special properties, which are as follows:

- **Superposition:** It is the capability or phenomenon to be in a state of 0 and 1 or on and off at the same time. It sounds crazy but as per quantum physics, it is possible.
- **Entanglement (Einstein called it spooky action at a distance):** As per the theory of Entanglement, two particles relate to each other, no matter what the distance between them is, immediately. So, if something is on Earth and another associated particle is on the moon, they can change their behaviour immediately if a change happens in one. Even faster than the speed of light!!!
- **Tunnelling:** The finite probability of a particle moving through barriers; classically this is not possible. However, quantum physics proves it to be true. For example, if there is sound in one room, then it's nearby room also gets the same sound even though the two rooms are separated by walls.
- **Qubit:** Similar to bit but bit appears either in one or zero states, whereas a qubit can appear either in 1 or 0 or superposition of them. Meaning a qubit can be 1 or 0 at the same time. Sounds crazy, but as per quantum physics, it is true. We will explore this in the forthcoming sections of the chapter and the book.

Quantum Physics terminology:

- **Quantized:** When something is quantized, it means that it derives in distinct lumps, named “quanta”. For example, photon appears in packets.
- **An electromagnetic field:** It is a traditional (non-quantum) field shaped by moving electric charges. It is the field defined by classical

electrodynamics and is the conventional complement to the quantized electromagnetic field tensor in quantum electrodynamics.

- **Momentum:** In physics, it is the inclination of a moving object to remain moving. For a thing moving in a line, the momentum is the mass of the object multiplied by its velocity (linear momentum). Therefore, a gradually moving, very huge body and a fast-moving, light body can have a similar momentum.
- **Wavelength:** It is the remoteness from one top to another, or from one rack to another, of a wave (which may be an electromagnetic wave, a sound wave, or any other wave). The crest is the uppermost point of the wave while the trough is the lowermost. In other words, the remoteness among points (to be specific, high points) is named as wavelength. The alteration in wavelength is one of the ways to convey dissimilar types of electromagnetic energy separately, as shown in [Figure 2.20](#):

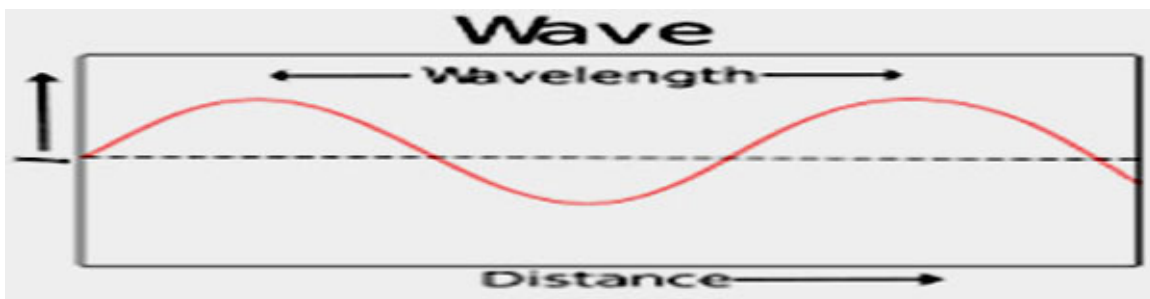


Figure 2.20: Alteration in wavelength

- The **state** of a particle, at any specific time, is engaged to include a sure position in space and a certain motion. Also, the state of a system of particles is engaged to include a fixed position in space and a sure motion for each of the particles.
- The **positron** or **antielectron** is the antiparticle or the antimatter complement of the electron. The positron has an electrical charge of $+1 e$, a spin of $1/2$, and has the identical mass as an electron. When a positron strikes with an electron, annihilation happens.
- A **Hamiltonian** is an operator equivalent to the totality of the kinetic energies plus the potential energies for all the particles in the scheme. This accumulation is the entire energy of the system in maximum of the cases under analysis. It is usually signified by H , but **also** $\hat{H}$ or to

highlight its purpose as an operator. Its spectrum is the set of likely results when one measures the total energy of a system.

- **Laser**, a device that arouses atoms or molecules to produce light at specific wavelengths and intensifies that light, characteristically creating a very thin beam of radiation. The emission usually shelters an extremely restricted range of visible, infrared, or ultraviolet wavelengths. The term laser was invented as an abbreviation for light amplification by stimulated emission of radiation.
- **Ultraviolet** is electromagnetic radiation with a wavelength from 10 nm to 400 nm, smaller than that of visible light but higher than X-rays. UV radiation is existing in sunlight and creates about 10% of the over-all electromagnetic radiation output from the Sun.
- The **electromagnetic spectrum** or **band** is the range of frequencies of electromagnetic radiation and their corresponding wavelengths and photon energies.
- A **magnetic field** is an applied area around the magnetic force. While the electric fields are produced around the particles which attain electric charge, throughout this procedure, positive charges are strained, while negative charges are kept away.
- **Photon Polarization**, before going to details of polarization, be aware that light works as an electromagnetic wave. Hence, it contains both electrical as well as magnetic components (electromagnetic = Electric + magnetic component). For light sources (most) the orders of those components are typically haphazard or random. However, it is probable to choose just a single direction and the outcome is polarised light. This can be realized in different ways, such as reflection, selective transmission, and polarising filtration. But whatever way we can achieve this phenomenon, the electric and magnetic components at all times appear at right angles to each other.
- If the light is polarised, then it trails that we necessarily are capable to polarise the separate building blocks of that light, which is known as photons. In point of fact, any discrete photon will have its electric field component revolving, either clockwise or anti-clockwise around the direction of travel. This is the reason why we use polarising filters to slowly change the way of polarization of a light source. The progressive

change of the phase of the distinct circularly polarised photon basically gives a novel linear polarisation direction.

- **Reflection** is when light rebounds from an object. If the surface is flat and glossy, similar to glass, liquid, or elegant metal, the light will reflect at the identical angle as it knocks out the surface. This is named specular reflection.
- **Selective transmission** occurs when certain color wavelengths are permitted to pass through the material. We can take colored glass as an example of this.
- A **circular polarizer** is intended to eliminate or regulate reflections from surfaces. For instance, liquid, cut-glass, shade, leaves, sky, structures, and even from streets.
- **The universe** is constructed of atoms, which comprise protons (which are positively charged particles), neutrons (which are particles with no charge, hence treated as neutral), and electrons (negatively charged particles). Molecules are shaped when two or more atoms join each other. If molecules encompass atoms of dissimilar elements, that matter which form by this process is known as a compound.

CHAPTER 3

Mathematics of Quantum Computers: The Fundamentals

We must be clear that when it comes to atoms, language can be used only as in poetry.

— Niels Bohr

Quantum mechanics tells us that every feeling is a collapse of some wave functions due to the interference of a matrix of attention functions.

— Amit Ray, *Quantum Computing Algorithms for Artificial Intelligence*

Mathematics is a holistic science which supports all disciplines. It is also thought of as a tool for explaining the ultimate reality because reality can be expressed in the forms of repressions and equations. Therefore, whether it is physics or arts, or any other area of investigation, we need mathematics for sure to get a grasp and understanding of it.

Even though humans search for everlasting inevitabilities, numerous mathematical ideas hint at their roots and help them to find and decode the integrities of life and even more secrets of nature. For instance, astrology and architecture which are fundamentally the branches of mathematics encouraged Egyptians to resolve the complexities of nature and Babylonians to construct complex structures. At the roots of these, essentials of algebra, number theory, and geometry played their role. Furthermore, the teaching of mechanics all through the scientific revolution of the 17th century is primarily based on calculus.

In this chapter, we will explore and recollect the basics of mathematics and give more focus on their interrelation with quantum mechanics. Finally, we will try to depict their role in quantum computation and computers.

Objectives

After studying this chapter, you will be able to understand the concept of the number system, matrix manipulation, Boolean algebra, and Tensors.

- Additionally, you would learn more concepts like quantum operators.
- You will also be able to discuss the need for these and why these are required.
- Also, you would be able to explain why and how mathematical systems are the foundational block of quantum computers.

At the same time, you become familiar with some of its use cases.

Structure

Broadly, this chapter could be divided into two parts. The first part will set the context of the need and desires of the chapter. Whereas the second part will cover the fundamentals of mathematical concepts. It also covers the relevance of mathematics in the world of quantum physics in general and specific to quantum computers.

In this chapter, we will discuss the following topics:

- Objectives
- Structure
- Introduction
- Development of mathematics and physics
- Quantum computers and mathematics
- Number system
 - Some important number systems
 - Structures in the number system
 - Polynomials
 - Number systems in context of computer science
 - Details of specific number systems
- Details of the binary number system
- Laws of Boolean algebra

- Symbols
- Truth tables for the laws of Boolean
- Basic truth tables
- Summary of Boolean algebra
- Implications
- Truth values for implications
- Description of the laws of Boolean algebra
- Boolean algebra functions
- Use of Boolean algebra using its laws
- Operands and operators in quantum mechanics
- Importance of basic probability algebra and theory
- Essential concepts of probability
- Essential concepts of algebra
- Review of Pythagoras theorem
- Essential concepts of vectors
- Vector addition and multiplication
- Multiplying a Bra by a Ket
- Concepts of matrices, different operations on matrices
 - Order of a matrix
 - Transpose of a matrix
 - Various other common operations on matrices and their properties
 - Matrices addition
 - Matrices subtraction
 - Matrices multiplication
 - The determinant of a matrix
- Tensor and tensor product
- Old wine in a new bottle!!!
- Essentials of trigonometry concepts

- An important compilation of results related to trigonometry
- Quantum mechanics, quantum computers, and vectors
- Observables and Hermitian operators
- Jargon buster
- Mind Map for the chapter
- Summary

Introduction

There is a long-standing and close association between mathematics and physics. For instance, to understand the work of Galileo to Feynman and Hawkins, everywhere mathematics plays a key role. Richard Feynman once said that “those who do not know mathematics (put themselves in) problematic situation because, for them to get across an actual sensation to the attractiveness, to (appreciate) the deepest beauty of nature (become difficult). Therefore, if you want to study, admire, and realize nature, you need to study mathematics because it is expressed in it.

He also mentioned *If all mathematics vanished today, physics would be set back precisely one week*, to which a mathematician had the ingenious reply: *This was the week that God shaped the world.*

Development of mathematics and physics

The development of mathematics is bidirectional; what it means is that it takes inputs from the different sciences and the disciplines available but also gives it back to them. For instance, in the 20th century, ideas from quantum theory turned out to transmit marvellous mathematical power as well, even though we have diminutive daily experience dealing with elementary particles.

In the strange world of quantum theory where things can appear to be in two locations at the same time, and the topics of the rules of probability not only signify a more essential explanation of nature, mathematics comes to the rescue to explain the complications of quantum theory in more concise, logical and rule orientated way. As it delivers an ironic context for modern theories.

Further, mathematics has the capability to attach dissimilar worlds. The most unnoticed symbol in any equation is the modest equal sign. Ideas flow through it, the double lines specify that ideas can flow in both directions. Albert Einstein was an out-and-out master of discovery equations that demonstrate this property. For instance, $E = mc^2$, without a doubt, is the most well-known equation in history.

In all its inconspicuous stylishness, it attaches the physical notions of mass and energy that were seen as completely separate before the arrival of relativity. Through Einstein's equation, we learn that mass can be distorted into energy, and vice versa. The equation of Einstein's general theory of relativity, though less attractive and well-known, associates the worlds of geometry and matter in an equally astonishing and lovely way. A concise method to review that theory is that mass tells space how to curve, and space tells mass how to change.

Quantum computers and mathematics

To understand quantum computers, mathematics plays a very crucial role. Whether it is a number system or a complex logic, mathematics is spread across everywhere. Hence, in the upcoming sections of this chapter, we will start covering the fundamentals of the number system and move further to the advanced concepts of mathematics, including tensor algebra. We will also understand Boolean algebra and metrics because they are the essential elements of proceeding further to comprehend quantum physics in general and quantum computers in specific.

Quantum mechanics is founded majorly in linear algebra; this itself highlights how important mathematics is in the field of quantum physics. Quantum computers are based on quantum physics. Hence, obviously, it is important for it.

The general theory uses infinite-dimensional vector spaces. Luckily for us, to define spin or polarization, we require only finite dimensions, which are considerably easier. As is with the case of mathematics, it can appear complex when you first get acquainted with it, but it turns out to be almost second nature after training. The real computations only include the

addition and multiplication of numbers, laterally with an infrequent square root and trigonometric function.

In a nutshell, the language of quantum mechanics is mathematics and for representing it, we use Paul Dirac's notation. Dirac was one of the initiators of quantum mechanics, and his notation is used widely all over quantum mechanics and quantum computing. It is not extensively used outside these disciplines, which is astonishing given how elegant and useful it is.

OK, but to depict Dirac notation and mathematics associated with quantum mechanics, some fundamental mathematical concepts are required. As mentioned, earlier, the foundation of the number system, groups, rings, tensors, and a few additional theories need to be understood to get a clear picture of quantum computation. Hence, in the following few sections, we will cover them.

Number system

To comprehend quantum algorithms, an understanding of the number system is required. For instance, to depict Shor's algorithm, which is one of the most important algorithms, a control over complex number concepts is necessary because it would be tough to describe the algorithm without using complex numbers. This algorithm also desires to explain constant fractions when required which is again an outcome of number theory. There are other examples as well, for instance, while determining the results around the quickness of an algorithm or by signalling out whether a number is prime or not number theory plays an important role.

There would be a substantial jump in the level of mathematical complexity and knowledge required if we want to define Shor's algorithm (a quantum algorithm) holistically. In its place, we will define the elementary ideas that explain the algorithm, representing how these fit together.

Now we are going to deal with an ephemeral explanation of the numbers. Let's start with real numbers, these are the standard decimal numbers with which we are all acquainted. Quantum computation uses complex numbers to explain the concepts associated with algorithms (but not limited to).

Today's numbers, also named Hindu-Arabic numbers, are an amalgamation of just 10 symbols or digits: 1, 2, 3, 4, 5, 6, 7, 8, 9, and 0. These digits were

familiarized in Europe by Leonardo Pisano, an Italian mathematician. The Hindu numeral system is an unadulterated place-value system, which is why you require a zero. Only the Hindus, inside the context of Indo-European civilizations, have reliably used a zero; the Arabs, though, frolicked a vigorous share in the dissemination of this numeral system.

There are different types of numbers that exist; let's start by understanding them one by one. But before we do so, let's define a number line. In mathematics, a number line can be defined as a straight line with numbers positioned at identical intermissions or sections along its length. A number line can be stretched infinitely in any direction and is typically epitomized horizontally. Please refer to [Figure 3.1](#) for a better understanding:

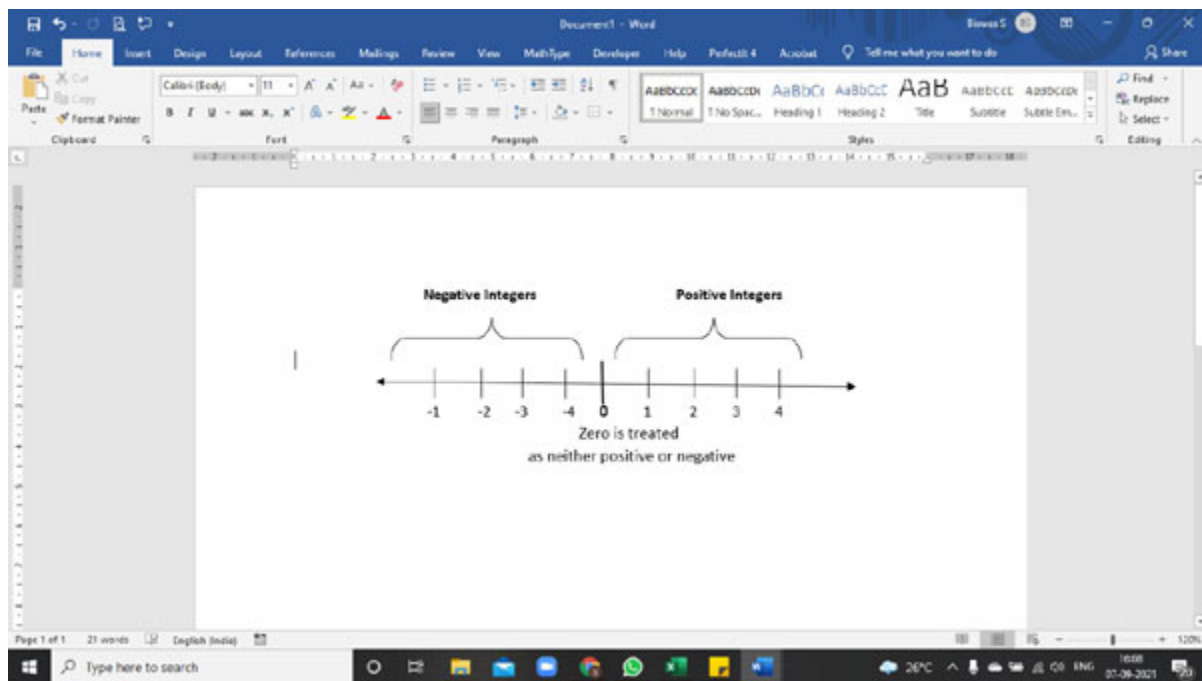


Figure 3.1: Number system

The following are some types of numbers:

Natural numbers: The set of natural numbers is a collection of growing values $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, \dots, 25, 26, 27, \dots, 67, 68, 69, \dots\}$, where we get from one number to the next by adding 1. In this, however, 0 is not included. The braces “{” and “}” indicate we are talking about the entire set of these numbers. The set of natural numbers is infinite. It starts at 1 and

their end is not demarcated. Mostly, Natural numbers are represented by “N”, that is, $N = \{1,2,3,4 \dots 12,13,14 \dots 51,52,53 \dots \infty\}$.

Integers (numbers having no decimals): The numbers which do not contain decimal places in them are usually demarcated or named as integers. All whole numbers together with the -ve number and +ve number come under this category. For instance, $Z = \{\infty \dots -15, -16, -17, -18 \dots -7, -6, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 \dots 21, 22, 23, \infty\}$. The integer may be positive or negative or zero. The group of integers is usually represented by Z. All the operations which happen on integers stay in integers. For instance, operations like addition, subtraction, and multiplication can be done on two (any) integers; the outcome that we get is an integer. Further, when any integer is multiplied, then each of the multiplied integers is known as a factor of the subsequent product which leads to the final result.

Whole numbers: The set of whole numbers is fundamentally the narrator’s numbers and “0”. Whereas when it comes to representation, the whole numbers can be represented by $W = \{0,1,2,3,4,5,6,7, 8,9,10,11,12,13,14,15 \dots 61,62,63, 64 \dots \infty\}$.

Rational numbers: A rational number is demarcated as a number of the prearrangement in the form of x/y where x and y are signifying integers with the condition of $Y \neq 0$, that is, any number which can be articulated as in the arrangement of p/q where “p” and “q” are the integers and $q \neq 0$. The group of rational numbers includes the grouping or (set) of integers and fractions. Also, it is important to note here that rational numbers, that do not possess the quality of integral, will contain characteristics of decimal values. These values can be of the following two kinds:

- **Finite decimal factors:** For instance, $1/6 = 0.16$, $16/5 = 3.2$.
- **Non-terminating decimal fractions:** The non-terminating decimal fractions can be categorized broadly into the following two types:
 - **Non-terminating periodic fractions:** A non-terminating periodic decimal must not be altered to a fraction because it is an irrational (or non-fractional) number. A few of the furthestmost straightforward reiterating decimals are $0.33333 \dots = 1/3$ and $0.66666 \dots = 2/3$.

- **Non-terminating non-periodic fractions:** A non-terminating, non-repeating decimal is a decimal number that persists endlessly. Decimals of this type must not be implied as fractions, and as an outcome are irrational numbers. Pi is the best example of a non-terminating, non-repeating decimal. Terminating and non-terminating periodic fraction numbers have their place in rational numbers.

Irrational numbers: An Irrational number is a non-terminating and non-periodic fraction. Formally, the irrational number can be defined as the number that cannot be represented in the form of a ratio x/y which is commonly termed a fraction. In decimal form, it never ends or repeats. Examples for of rational numbers are $\sqrt{2} = 1.414213\dots$, $\pi = 3.14159265\dots$, $\sqrt{7}$, $\sqrt{11}$, and so on.

Imaginary Number: A number that does not happen to present on the number line (a representation method, where on a straight-line, numbers are mentioned and typically it starts to form $-\infty$ and goes to $+\infty$) is called an imaginary number. There are a lot of interesting and important facts that are associated with an imaginary line. For instance, the square root of negative numbers is always an imaginary number; in the mathematical form it is written as $i^2 = -1$. And, it is signified by i or j :

That is, $\sqrt{-1} = i$

$$i^2 = -1$$

Therefore, we can also derive from the preceding relation that, there is no real number i that exists that pleases the criteria of imaginary numbers. The quantity i is named as the unit imaginary number.

1. **Real numbers:** All numbers that can be denoted on the number line are named as real numbers. The real numbers are the group of numbers comprising all of the rational as well as irrational numbers. Therefore, broadly, the real numbers are all the numbers which can be plottable on the number line. Real Numbers are denoted by capital R that is, R .
2. **Complex numbers:** All numbers in the number system can be visualized as a complex number. By definition, a complex number is

treated as a mixture of a real and an imaginary number. Let's see an example of this.

The examples of complex numbers are as follows:

$2 + i$	$40 + 3i$	$0.9 - 2.3i$	$-7 + \pi i$	$\sqrt{3} + i/4$
---------	-----------	--------------	--------------	------------------

Therefore, a complex number contains a real part as well as an imaginary part. However, any portion can be 0. Hence, all real numbers and imaginary numbers are a candidate for complex numbers, as depicted in [Table 3.1](#):

Complex number	Real part	Imaginary part	
$4 + 5i$	4	5	
6	6	0	exactly real
$-7i$	0	-7	exactly imaginary

Table 3.1: Complex numbers

Complex numbers are more multifaceted or complicated. Complex numbers deliver a sophisticated way of linking trigonometric and exponential functions. Further, a proper understanding of complex numbers is required to decode the secrets of the mathematical model, because it is involved in explaining a lot of fundamental as well as complex aspects of quantum computers. For instance, to derive the spin of a fundamental particle in three dimensions, one must use complex numbers for reaching meaningful decisions. However, at the core, the computations use qubits.

In a nutshell, its main ingredients are described, as shown in [Figure 3.2](#), which depicts the details of complex numbers:

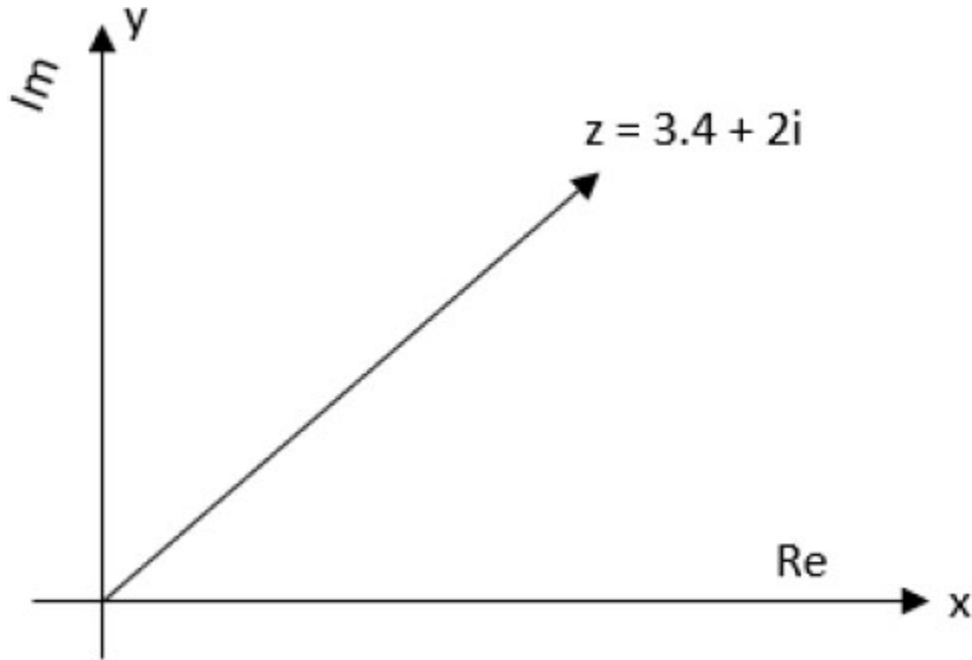


Figure 3.2: Complex number in a nutshell

A complex number can be expressed in the form $z = x + iy$, where $i = \sqrt{-1}$

- $x = \text{Re}(z)$ is the real part of the complex number z .

$y = \text{Im}(z)$ is the imaginary part of the complex number z . Whereas, i is the imaginary unit.

- Arithmetic operators on real numbers extend to complex numbers, noting that $i = \sqrt{-1}$ and $i^2 = -1$.
- A complex number can be represented as a point in the complex plane, associated with the vector from the origin to that point.
- The vector and the positive direction are specified by the angle between the vector and the positive direction of the x -axis called the phase angle.
- The complex conjugate of a complex number is expressed as $z = x + iy$, replaced by z^* and denoted as $z^* = x - iy$.
- The modulus or the magnitude of a complex number is denoted by $|z|$. It is fundamentally derived from the origin and is calculated as $|z| = \sqrt{2z^*}$.

Add: To add two complex numbers, we add each part separately: $(x + yi) + (w + zi) = (x + w) + (y + z)i$

Example: Add the complex numbers $4 + 3i$ and $1 + 7i$.

Add the real numbers, and add the imaginary numbers:

$$\begin{aligned}(4 + 3i) + (1 + 7i) \\&= 4 + 1 + (3 + 7)i \\&= 5 + 10i\end{aligned}$$

Multiply: To multiply complex numbers, each part of the first complex number gets multiplied by each part of the second complex number.

Example: $(4 + 2i)(1 + 7i)$

$$\begin{aligned}(4 + 2i)(1 + 7i) &= 4 \times 1 + 4 \times 7i + 2i \times 1 + 2i \times 7i \\&= 4 + 28i + 2i + 14i^2 \\&= 4 + 28i + 2i - 14 \text{ (because } i^2 = -1) \\&= -10 + 30i\end{aligned}$$

Division: The conjugate is used to help complex division. The best way is to multiply both numerator (top) and denominator (bottom) by the conjugate of the denominator (bottom).

Example: Division:

$$2 + 3i / 4 - 5i$$

Step 1: Multiply top and bottom by the conjugate of $4 - 5i$:

$$2 + 3i / 4 - 5i \times 4 + 5i / 4 + 5i = 8 + 10i + 12i + 15i^2 / 16 + 20i - 20i - 25i^2$$

Now remember that $i^2 = -1$, so:

$$= 8 + 10i + 12i - 15 / 16 + 20i - 20i + 25$$

Step 2: Add Like Terms (and notice how on the bottom $20i - 20i$ cancels out!):

$$= -7 + 22i / 41$$

Step 3: We should put the answer back into $a + bi$ form:

$$= -7 / 41 + 22 / 41 i$$

OK, now lets summarise the concepts of number system and different type of numbers. [Figure 3.3](#) do this for you.

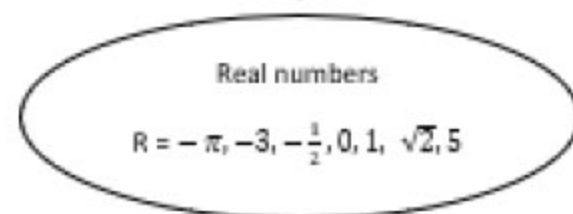
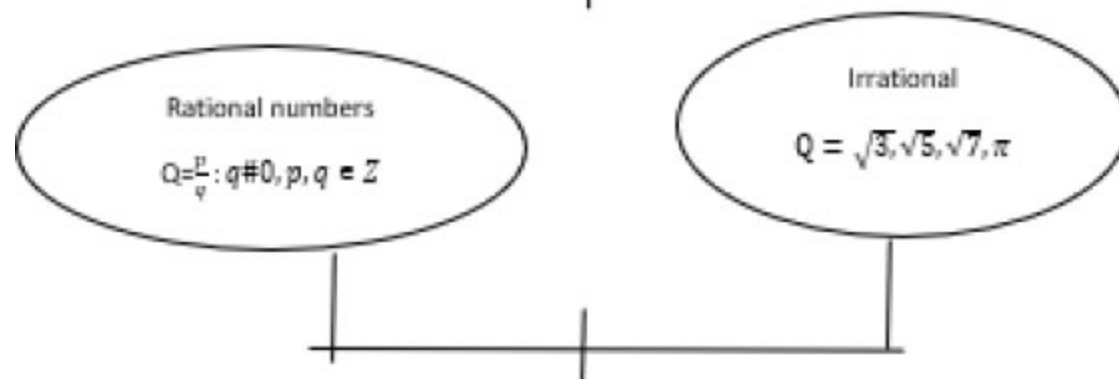
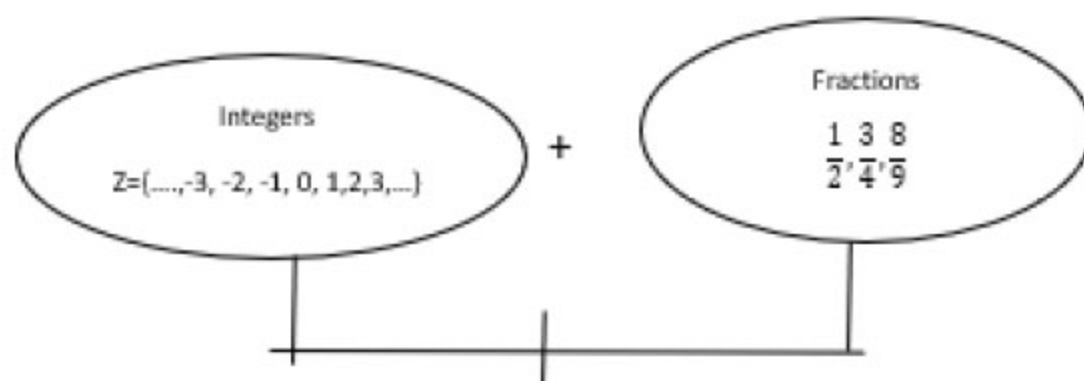
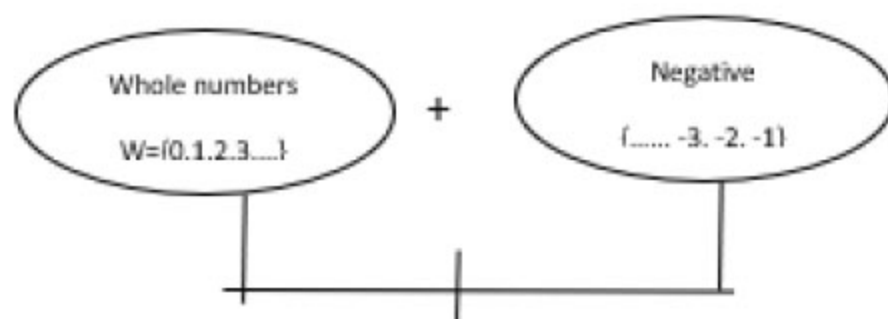
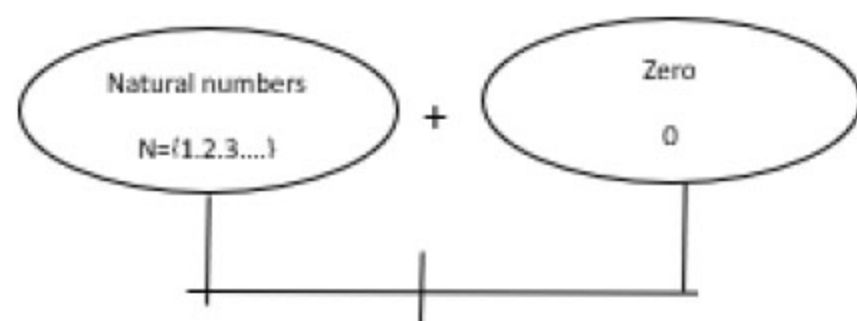


Figure 3.3: The different types of numbers

At this point, it is good to understand Euler's formula to decode a Complex number system.

Euler's formula

It fundamentally established the relation among trigonometric and complex exponential functions. This formula is valid for any real number x which exists in the term of:

$$e^{ix} = \cos x + i \sin x$$

Further, e^{ix} values are shown in [Table 3.2](#):

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	π
e^{ix}	0	$\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}}$	i	-1

Table 3.2: Euler's formula presentation

The general form of Euler's formula is given as follows:

$$e^{iAx} = \cos(x)I + i \sin(x)A$$

Here A and I are Pauli Matrices; further in the preceding equations, the complex exponent ix is replaced by A .

Geometrically, Euler's formula draws a circle in a complex plane, while the generated formula draws a circle in a complex plane that has an additional dimension.

Some additional facts and types of number system

Below are some important number systems are described.

Even and odd numbers: A number that is a multiple of the number 2 is labelled as the even number which plays a significant role in quantum physics. In the coded form, the even number can be signified by $2n$ where n can be any number, or better, we can say that n is a natural number. A

number that is not manifold of the number 2 is named an odd number. The odd number can be characterized by $2n+1$ where n is a natural number.

The operation of even and odd numbers is as follows:

- If odd is multiplied by odd, it produces odd.
- If dd is added by odd, it produces even.
- If odd is subtracted from odd, it produces even.
- If odd is divided by odd, it produces odd.
- If even is added by even, it produces even.
- If even is subtracted from even, it produces even.
- If even is multiplied by even, it produces even.
- If even is divided by even, it produces even or odd.
- If odd is multiplied by even, it produces even.
- If odd is added by even, it produces odd.
- If even is divided by odd, it produces even.
- If odd is divided by even, it is not divisible.

Negative integers: If the integer is smaller than zero, then it is named a negative integer, that is, the set of integers $\{-1, -2, -3, -4, -5, \dots, \infty\}$ is termed negative integers. The product of any two negative integers is at all times a positive integer.

Non-negative integers: A set of numbers zero and natural numbers is termed non-negative integers, that is, $\{0, 1, 2, 3, 4, \dots, \infty\}$.

Positive integers: If the integer is larger than zero, then it is called a positive integer. The set of natural numbers is called non-negative integers. The product of any two positive integers is at all times a positive integer. The product of any two positive and negative integers is always a negative integer that is, the set of numbers $\{0, 1, 2, 3, 4, \dots, \infty\}$.

Non-positive integers: If the integer is less than zero and comprises zero, then it is named non-positive integer. The set of numbers $\{0, -1, -2, -3, -4, -5, \dots, \infty\}$ is called non-positive integers.

Prime numbers: A natural number larger than unity is a prime number and it does not have any additional divisors, excluding unity and itself is termed a prime number. For example, 2,3,5,7,11,13,17,19.

- 1 is not a prime number.
- 2 is the only even and lowest prime number.
- Prime number exactly having two different factors.
- Number of prime numbers from 1 to 50 = 15.
- Several prime numbers from 1 to 100 = 25.
- Number of prime numbers from 1 to 200 = 46.
- Number of prime numbers from 1 to 1000 = 168.

Co-prime numbers: Co-prime numbers mean it is having at least two numbers and its Highest Common Factor(HCF of two or more numbers is the greatest factor that divides the numbers. For example, 2 is the HCF of 4 and 6) is 1. That is, when two or more numbers have no common prime factors apart from the number "1", they are called co-prime or relatively prime to each other.

- Two successive odd numbers are always co-prime numbers (for example, 7 and 9, 11 and 13, 27 and 29)
- Two prime numbers are always prime numbers (for example, 3 and 7, 11 and 17)
- One prime number and another composite number (such composite number is not a multiple of the prime number) are always co-prime numbers. This rule exclusion is only for 17 and 51. (for example, 3 and 16, 4 and 9)
- Three or more numbers being co-prime with each other means that all possible pairs of the numbers would be co-prime with each other. (Thus, if 51, 53, and 55 are co-prime with each other, then the pairs 51 and 53, 53 and 55, 51 and 55 are also co-prime numbers)
- Three consecutive odd numbers are always co-prime numbers (for example, 7,9 and 11 and 2, 5 and 7)

Perfect number: Any number is named as the perfect number if the sum of all its factors except itself is equivalent to the number itself. For example, take numbers 6. Factors of 6 are 1,2,3,6. Some of the factors excluding itself = $1+2+3 = 6$. So, the number 6 is named a perfect number.

Structures in the number system

The operations and the properties of the real numbers and their subsets are important, because these are very common in multiple subparts of mathematics, especially when proper abstraction is required. The structure allows us to learn and prove things and then apply them to new mathematical collections as we encounter them. In this section, we will discuss three structures, groups, rings, and fields, as follows:

- **Groups:** Consider a collection of objects which we call G . For example, G might be $\mathbb{Z}$, $\mathbb{Q}$, or $\mathbb{R}$ as mentioned earlier. We also have some pairwise operations between elements of G , which we denote by $\circ$. It's a placeholder for an action that operates on two objects. These $\circ$ operations could be addition $+$ or multiplication $*$ for numbers but might be something entirely different. Use your intuition with numbers, but understand that the general case is, well, more general. We call the collection together with its operation $(G, \circ)$. We write the use of $\circ$ the same as we would normally with addition or multiplication, between the elements. We write $0 \circ 1$ for 0 and 1 in G . This is called infix notation. Negations like -7 use the prefix notation. The factorial operation $n! = 1 * 2 * \dots (n-1) * n$ uses postfix notation.
- **Rings:** When we have more than one operation, we get a more sophisticated structure called a ring if certain requirements are met. For convenience, we call these two operations $+$ and $*$ but remember they could behave in very different ways from the addition and multiplication we have for numbers.
- **Fields:** A field F is a commutative ring where every non-zero element has a multiplicative inverse. A field is closed under division by non-zero elements.

Polynomials

A polynomial is expressed as an expression which is generally a combination of variables, constants and exponents. These are together using mathematical operations such as addition, subtraction, multiplication and division. Few important facts about polynomials are mentioned below:

Roots of a polynomial: A “root” (or “zero”) is where the polynomial is equal to zero. If we simply put it, the definition goes like this – root is the x-value where the y-value equals zero.

General Polynomial: If we have a general polynomial like the following:

$$f(x) = ax^n + bx^{n-1} + cx^{n-2} + \dots + z$$

Then:

Adding the roots gives $-b/a$.

Multiplying the roots gives:

z/a (for even degree polynomials like quadratics).

$-z/a$ (for odd degree polynomials like cubic), which can sometimes help us solve equations.

How does this magic work? Let's find out. And to get the concepts better it is important to understand factors and quadratics.

Factors: We can take a polynomial, such as:

$$f(x) = ax^n + bx^{n-1} + cx^{n-2} + \dots + z$$

And then factor it like this:

$$f(x) = a(x-p)(x-q)(x-r)$$

Then p, q, r, and so on are the roots (where the polynomial equals zero)

Quadratic: Let's try this with a Quadratic (where the variable's biggest exponent is 2):

$$ax^2 + bx + c$$

When the roots are p and q, the same quadratic becomes:

$$a(x-p)(x-q)$$

Is there a relationship between a, b, c and p, q?

Let's expand $a(x - p)(x - q)$:

$$\begin{aligned} & a(x - p)(x - q) \\ &= a(x^2 - px - qx + pq) \\ &= ax^2 - a(p + q)x + apq \end{aligned}$$

Now let us compare this in [Table 3.3](#):

Quadratic:	ax^2	$+bx$	$+c$
Expanded factors:	ax^2	$-a(p+q)x$	$+apq$

Table 3.3: Comparison

We can now see that $-a(p+q)x = bx$, so:

$$\begin{aligned} -a(p+q) &= b \\ p+q &= -b/a \end{aligned}$$

And $apq = c$, so:

$$pq = c/a$$

And we get this result:

- Adding the roots gives $-b/a$
- Multiplying the roots gives c/a

Example: What is an equation whose roots are $5 + \sqrt{2}$ and $5 - \sqrt{2}$

- The sum of the roots is $(5 + \sqrt{2}) + (5 - \sqrt{2}) = 10$
- The product of the roots is $(5 + \sqrt{2})(5 - \sqrt{2}) = 25 - 2 = 23$
- And we want an equation like: $ax^2 + bx + c = 0$
- When $a=1$ we can work out that:
- Total of the roots $= -b/a = -b$
- Product of the roots $= c/a = c$

Which gives us the following result:

- $x^2 - (\text{sum of the roots})x + (\text{product of the roots}) = 0$

The sum of the roots is 10, and the product of the roots is 23, so we get:

$$x^2 - 10x + 23 = 0$$

Cubic: Now let us look at a Cubic (one degree higher than Quadratic):

$$ax^3 + bx^2 + cx + d$$

As with the Quadratic, let us expand the factors:

$$\begin{aligned} & a(x-p)(x-q)(x-r) \\ &= ax^3 - a(p+q+r)x^2 + a(pq+pr+qr)x - a(pqr) \end{aligned}$$

And we get the following:

Cubic:	ax^3	$+bx^2$	$+cx$	$+d$
Expanded factors:	ax^3	$-a(p+q+r)x^2$	$+a(pq+pr+qr)x$	$-apqr$

We can now see that $-a(p+q+r)x^2 = bx^2$, so:

$$-a(p+q+r) = b$$

$$p+q+r = -b/a, \text{ so:}$$

$$pqr = -d/a$$

This is interesting; we get the same sort of thing, as follows:

- Adding the roots gives $-b/a$ (precisely the same as the Quadratic).
- Multiplying the roots gives $-d/a$ (similar to $+c/a$ for the Quadratic).
We also get $pq+pr+qr = c/a$.

In general:

- Adding the roots gives $-b/a$.
- Multiplying the roots gives (where “z” is the constant at the end).
- z/a (for even degree polynomials like quadratics).
- $-z/a$ (for odd degree polynomials like cubics).

Number of systems in context of computer science

The discovery of zero and the place-value system were discoveries exclusive to the Indian civilization, as was the Brahmi representation of the

first 9 whole numbers. The number set used in the Middle East today is a companion of the modern number set, with a shared antecedent in the ancient Hindu numerals.

The Eastern Arabic numerals (also called Arabic–Indic numerals) are exact numbers currently used to signify the Hindu–Arabic number system in combination with the Arabic alphabet in the republics of the Arab east, and its variant (Persian numerals) in other countries. [Figure 3.4](#) displays different types of number systems:

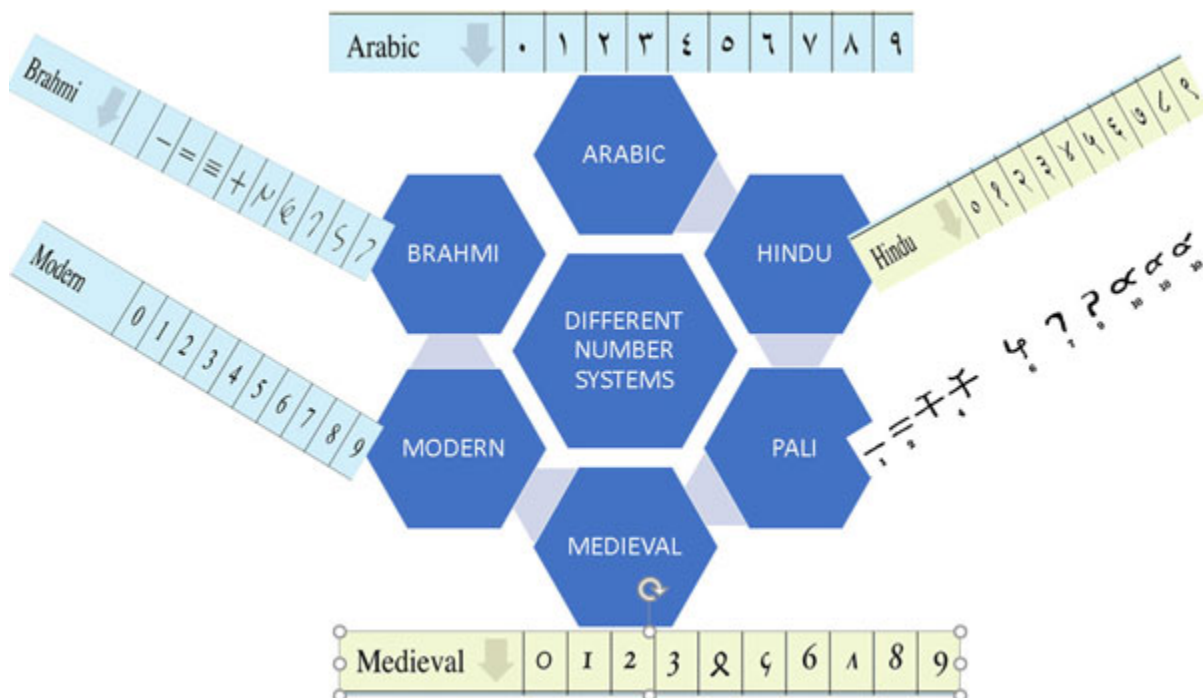


Figure 3.4: Different types of number systems

Number systems are the method to characterize numbers in the computer system architecture; every value that you are saving or receiving into/from computer memory has a definite number system. Computer architecture ropes subsequent number systems, such as the following:

- Binary number system
- Octal number system
- Decimal number system
- Hexadecimal (hex) number system

Details of computer's relevant number systems

Computers typically employ the binary number system. The octal, decimal, and hexadecimal systems are additionally employed on occasion. As a result, a brief description of them is included here for completeness.

- **Binary number system:** A Binary number system has only two digits represented by 0 and 1. All the number (value) in binary number system is represented by 0 and 1. The base of the binary number system is 2 since it has only two digits.
- **Octal number system:** Octal number system has eight (8) digits which starts from 0, the last digit in this system is 7. All the number (value) characterizes with 0,1,2,3,4,5,6 and 7 in this system. The base of the octal number system is 8 since it has only 8 digits.
- **Decimal number system:** The decimal number system has ten (10) digits which starts from 0, the last digit in this system is 9. All number (value) signifies with 0,1,2,3,4,5,6, 7,8 and 9 in this number system. The base of the decimal number system is 10, as it has only 10 digits.
- **Hexadecimal number system:** A Hexadecimal number system has sixteen (16) alphanumeric values from 0 to 9 and A to F. Every number (value) represents with 0,1,2,3,4,5,6, 7,8,9, A, B, C, D, E and F in this system. The base of the hexadecimal number system is 16 since it has 16 alphanumeric values. Here A is 10, B is 11, C is 12, D is 13, E is 14 and F is 15.

Details of the binary number system

The binary number systems are very important in terms of computer processing. So, let's have a detailed look at this.

The number system used by computers is based on 0's and 1's. This system is called binary or base-2 system. This is one of the most popular number systems. A binary number typically contains multiple bits which is represented as 010, 11011 and 010100101. Each bit position in a binary number which is fundamentally associated with a power of two weight, may be represented as 2 positions. The position count starts from the

rightmost bit at position 0 to the leftmost bit at position (n-1), where n is the number of bits in the number.

The rightmost bit is called the least significant bit (LSB) and has weight $2^0 = 1$ and the leftmost bit is called the most significant bit (MSB) and has weight 2^{n-1} . To understand this, let's take an example to convert number (-28) to binary form 111000.

To convert binary to decimal number effectively, all the digits that are 1 in the binary sequence are multiplied by their weights and the results summed up. For example:

$$2^4 + 2^3 + 2^2 = 16 + 8 + 4 = 28$$

A whole decimal number is represented by a unique combination of bits. In general, an n-bit string allows 2^n states and can be represented as 2^n in the range of 0 to $(2^n - 1)$.

We are habitual of doing things in a decimal way. And the computer does it in a binary way. Hence it is important to understand the conversion process.

Well, this is the methodological way in which we would convert decimal number to binary and there are a few ways available where we would convert decimal to binary and binary to decimal. Let's have a look at the following steps for the conversion of decimal to binary:

1. Divide the given number by 2 and note the remainder 0 or 1.
2. Divide the Quotient (result) by 2.
3. If the quotient is not zero, then repeat step 2, otherwise finish.

In the end, the sequence of the remainder is the binary representation of the given decimal number.

Laws of Boolean algebra

Boolean Algebra is the mathematics we use to analyse digital gates and circuits. We can use these laws of Boolean to together decrease and make things easier. A complex Boolean countenance to decrease the number of required logic gates in a circuit. Boolean Algebra is hence a system of mathematics founded on the logic that has its set of rules or laws which are used to describe and lessen Boolean expressions.

Formally, we can say that Boolean Algebra uses a set of Laws and Rules to describe the procedure of a digital logic circuit. It uses the logic symbols 0 and 1 being used to mean a digital input or output, we can also use them as numbers for a lastingly open or closed circuit or contact respectively.

A set of rules or laws of Boolean Algebra expressions have been invented to help reduce the number of logic gates needed to perform a particular logical operation which could be resulting in a list of functions or theorems known commonly as the laws of Boolean algebra.

The variables used in Boolean Algebra only have one of the two likely values, a logic 0 and a logic 1, but the expression can have an immeasurable number of variables, all categorized separately to signify inputs to the expression, for instance, variables A, B, C, and so on, giving us a logical expression of $A + B = C$, but each variable can only be a 0 or a 1. Examples of these individual laws of Boolean, rules, and theorems for Boolean Algebra are provided in the upcoming sections.

This type of table is often called a truth table. A truth table is a list of all the dissimilar amalgamations of truth values that a sentence, or set of sentences, can have. To shorthand our notation further, we will introduce some symbols that are commonly used for and, or, and not.

Symbols

In Boolean algebra “symbols” plays an important role. A few facts about symbols in the context of Boolean algebra are given below:

- The symbol $\wedge$ is used for and: A and B are notated $A \wedge B$.
- The symbol $\vee$ is used for or: A or B is notated $A \vee B$.
- The symbol $\sim$ is used for: not A is notated $\sim A$.

Truth tables for the laws of Boolean

You must remember the first two symbols by relating them to the shapes for the union and intersection. $A \wedge B$ would be the elements that exist in both sets, in $A \cap B$. Likewise, $A \vee B$ would be the elements that exist in either set, in $A \cup B$. In the previous example, the truth table was just summarizing

what we already know about how the or statement works. The truth tables for the basic and, or, and not statements are shown in the following section.

Basic truth tables

The following are some important truth tables, which are frequently used in binary algebra. Fundamentally, it is used for all logic implementation of computer science on the ground level. For union, interaction and negation, refer to [Table 3.3a](#), [Table 3.3b](#), and [Table 3.3c](#):

$A \vee B$

A	B	$A \vee B$
True	True	True
True	False	True
False	True	True
False	False	False

Table 3.3a: Union truth table

$A \wedge B$

A	B	$A \wedge B$
True	True	True
True	False	False
False	True	False
False	False	False

Table 3.3b: Interaction truth table

$\sim A$

A	$\sim A$
True	False
False	True

Table 3.3c: Negation truth table

Practically, truth tables become handy when analysing more complex Boolean statements. For instance, if we want to create a truth table for the statement $A \wedge \sim (B \vee C)$, then it helps to work from the inside out when creating truth tables, and create tables for intermediate operations. We start by listing all the possible truth-value combinations for A, B, and C ([Table 3.4](#)).

Notice how the first column contains 4 “Trues” followed by 4 “False”, the second column contains 2 “Trues”, 2 “False”, then repeats, and the last column alternates. This pattern ensures that all combinations are considered. Along with those initial values, we’ll list the truth values for the innermost expression, $B \vee C$, as shown in [Table 3.4a](#):

A	B	C	$B \vee C$
True	True	True	True
True	True	False	True
True	False	True	True
True	False	False	False
False	True	True	True
False	True	False	True
False	False	True	True
False	False	False	False

Table 3.4a: Truth table for the statement $(B \vee C)$

Next, we can find the negation of $B \vee C$, working off the $B \vee C$ column we just created, as shown in [Table 3.4b](#):

A	B	C	$B \vee C$	$\sim(B \vee C)$
True	True	True	True	False
True	True	F	True	False
True	F	True	True	False
True	False	False	False	True
False	True	True	True	False
False	True	False	True	False
False	False	True	True	False
False	False	False	False	True

Table 3.4b: $\sim(B \vee C)$

Finally, we find the values of C , as shown in [Table 3.4c](#):

A	B	C	$B \vee C$	$\sim(B \vee C)$	$A \wedge \sim(B \vee C)$
True	True	True	True	False	False
True	True	False	True	False	False
True	False	True	True	False	False
True	False	False	False	True	True
False	True	True	True	False	False
False	True	False	True	False	False
False	False	True	True	False	False
False	False	False	False	True	False

Table 3.4c: [Table 3.4b](#): $\sim(B \vee C)$

From the preceding truth table, we can conclude that the complex expression which we solved is only true in the sole case: if A is true, B is false, and C is false.

Till now, we discussed multiple conditions. However, most of them are the type where we would take an action that is founded on the value of the conditions. Moving forward, now we discuss a more generic variant of a conditional, sometimes called an implication.

Summary of Boolean algebra

Fundamentally, Boolean algebra is the way to implement a binary number system. With Boolean algebra, we achieve all the manipulation related task and do the calculations. So if the number system is the body, then Boolean algebra is the blood that is flowing in that. [Figure 3.5](#) showcased the summary of Boolean algebra:

STEP 1:	INPUT				
	A	B	AND	OR	XOR
	0	0	0	0	0
	0	1	0	1	1
STEP 2:	1	0	0	1	1

Figure 3.5: Summary of Boolean algebra

The 2-bit operations can be mentioned as follows:

- AND operations are denoted by $\bullet$ or $\wedge$ or $\wedge$ gives only two bits are 1, otherwise gives 0.
- OR operators are denoted by $+$, $\vee$ or $\cup$ gives 0. If two bits are 0 (input) or will give 1.
- XOR or exclusive OR operations denoted by provides output as 0 when inputs are same and gives 1 as an outcome when inputs are different.

Implications

Implications are fundamentally logical conditional sentences which typically state that a statement p (you could use any other notions as well), called the antecedent, implies a consequence q . Implications are commonly written as $p \rightarrow q$ which means p implies q . Implications are essentially similar to the conditional statements which we have gone through in the previous few sections.

Also, $p \rightarrow q$ is typically written as if p then q , or p therefore q . At this point, it is important to know the difference between implications and conditionals. The most important difference is that conditional statements, which we discussed earlier, fundamentally suggested an action. For instance, if a given condition is true, then we take some action as a result. Whereas, Implications are essentially a logical statement that suggests or

gives an idea about the consequence which must logically follow, especially when the antecedent is true.

In old-style logic, an implication is well thought-out as valid (true) as long as there are no cases in which the antecedent is true and the consequence is false. It is important to keep in mind that symbolic logic cannot capture all the intricacies of the English language, as shown in [Table 3.5](#):

p	q	$p \rightarrow q$
True	True	True
True	False	False
False	True	True
False	False	True

Table 3.5: Truth table for implication

The basic laws of Boolean Algebra that relate to the commutative law allowing a change in position for addition and multiplication, the associative law allowing the removal of brackets for addition and multiplication, as well as the distributive law allowing the factoring of expression, are the same as in ordinary algebra.

Each of the preceding Boolean laws is given with just a single or two variables, but the number of variables defined by a single law is not limited to this as there can be an infinite number of variables as inputs to the expression.

These Boolean laws detailed earlier can be used to prove any given Boolean expression as well as for simplifying complicated digital circuits. A brief description of the various Laws of Boolean is given, following with A representing a variable input.

Description of the laws of Boolean algebra

Now we will look at the fundamental laws of Boolean algebra. The laws are at the laws of mathematics which are extended in Boolean algebra as well, as follows:

- **Annulment law:** When a term is AND'ed with a Zero or 0 equals Zero or 0 or OR'ed with a one or 1 will equal 1(one):
 - $A \cdot 0 = 0$, A is a variable AND'ed with 0 (Zero) is always equal to 0 (Zero).
 - $A + 1 = 1$, A is a variable OR'ed with 1(one) is always equal to 1 (one).
- **Identity law:** When a term OR'ed with a 0 (Zero) or AND'ed with a 1 (one) will always equal (=) to that term:
 - $A + 0 = A$ A variable OR'ed with 0 (Zero) is always equal (=) to the variable.
 - $A \cdot 1 = A$ A variable AND'ed with 1(one) is always equal (=) to the variable.
- **Idempotent law:** An input that is AND'ed or OR'ed with itself is equal to that input:
 - $A + A = A$, A is a variable OR'ed with itself is always equal to the variable.
 - $A \cdot A = A$, A is a variable AND'ed with itself is always equal to the variable.
- **Complement law:** A term AND'ed with its complement equals 0 and a term OR'ed with its complement equals 1:
 - $A \cdot \bar{A} = 0$, A is a variable AND'ed with its complement is always equal to 0.
 - $A + \bar{A} = 1$, A is a variable OR'ed with its complement is always equal to 1.
- **Commutative law:** The order of application of two separate terms is not important:
 - $A \cdot B = B \cdot A$: The order in which two variables are AND'ed makes no difference.
 - $A + B = B + A$: The order in which two variables are OR'ed makes no difference.

- **Double negation law:** A term that is inverted twice is equal to the original term:
 - $A = A$ --- A double complement of a variable is always equal to the variable.
- **de Morgan's theorem:** There are two de Morgan's rules or theorems:
 - Two separate terms together is the same as the two terms inverted (compliment) and AND'ed, for example, $A+B = A. B$.
 - Two separate terms together is the same as the two terms inverted (complement) and OR'ed, for example, $A.B = A+ B$.

Other algebraic laws of Boolean not detailed earlier include the following:

- **Distributive law:** This law permits the multiplying or factoring out of an expression:
 - $A (B + C) = A.B + A.C$ (OR distributive law).
 - $A + (B.C) = (A + B). (A + C)$ (AND distributive law).
- **Absorptive law:** This law enables a reduction in a complicated expression to a simpler one by absorbing like terms:
 - $A+ (A.B) = A$ (OR absorption law).
 - $A (A + B) = A$ (AND absorption law).
- **Associative law:** This law allows the removal of brackets from an expression and regrouping of the variables:
 - $A+ (B + C) = (A + B) + C = A + B + C$ (OR associate law).
 - $A (B.C) = (A.B) C = A. B. C$ (AND associate law).

Boolean algebra functions

Using the preceding information, simple 2-input AND, OR, and NOT gates can be represented by 16 possible functions, as shown in [Table 3.6](#):

Function	Description	Expression
1.	NULL	0
2.	IDENTITY	1
3.	Input A	A
4.	Input B	B
5.	NOT A	$\bar{A}$
6.	NOT B	$\bar{B}$
7.	A AND B (AND)	$A \cdot B$
8.	A AND NOT B	$A \cdot \bar{B}$
9.	NOT A AND B	$\bar{A} \cdot B$
10.	NOT AND (NAND)	$\overline{A \cdot B}$
11.	A OR B (OR)	$A + B$
12.	A OR NOT B	$A + \bar{B}$
13.	NOT A OR B	$\bar{A} + B$
14.	NOT OR (NOR)	$\overline{A + B}$
15.	Exclusive-OR	$A \cdot \bar{B} + \bar{A} \cdot B$
16.	Exclusive-NOR	$A \cdot B + \bar{A} \cdot \bar{B}$

Table 3.6: Boolean algebra function

Use of Boolean algebra using its laws

For visualizing the Boolean algebra in action, it is better to use their important laws. We have already gone through the laws in the previous section. Now we simplify the expression: $(X + Y) (X + Z)$ to understand it in a better way:

$$A = (X + Y) \cdot (X + Z)$$

$$X \cdot X + X \cdot Z + X \cdot Y + Y \cdot Z - \text{Distributive law}$$

$X + X.Z + X.Y + Y.Z$ – Idempotent AND law ($X.X = X$)

$X(1 + Z) + X.Y + Y.Z$ – Distributive law

$X.1 + X.Y + Y.Z$ – Identity OR law ($1 + Z = 1$)

$X(1 + Y) + Y.Z$ – Distributive law

$X.1 + Y.Z$ – Identity OR law ($1 + Y = 1$)

$A = X + (Y.Z)$ – Identity AND law ($X.1 = X$)

Then the expression: $(X + Y)(X + Z)$ can be simplified to $X + (Y.Z)$ as in the Distributive law.

Operands and operators in quantum mechanics

In physics, the idea of space is closely connected to the spreading of matter meaning, two material objects cannot inhabit a similar space at the same time. The view of time originates from change.

Without obvious change, there is no requirement for the notion of time. In math, there are two elementary units: operands (nouns) and operators (verbs). For an operand (noun) to alternate its value, it requires to be functioned on (verbed). Operands have a space-like property in that they can change to a new value.

Operators have time-like stuff since they bring about change. Operators may also display space-like properties when defining the gradient of a supply of operands. In natural languages too, we discover proper nouns also used as verbs and vice versa.

In quantum mechanics, operators serve as substitutes for certain operands, henceforth the commutability of operators plays such a significant role in measurement. Here are a few such operators:

- Time $t \rightarrow t$,
- Position: $r \rightarrow r$,
- Momentum: $P \rightarrow -i\hbar \nabla$
- Energy: $E \rightarrow i\hbar (\partial/\partial s)$

If we briefly see how Schrodinger postulated wave equation, then we can realize that it is instructive. Traditionally, in classical or conventional

physics, at any given time, t , the total energy E of a particle is the total of its kinetic energy and potential energy. The kinetic energy can be defined as the per unit mass given by $p^2 / 2m$ and potential energy per unit mass as some function of position, r , and time t , $V(r, t)$. Hence, the total energy, therefore, can be formulated as:

$$E = P^2 / 2m + V(r, t)$$

If this equation can be written in the operator form, it becomes the form of Schrodinger equation:

$$-\frac{\hbar^2}{2M} \nabla^2 \varphi + v(r, t) \varphi = i\hbar \frac{\delta \varphi}{\delta t}$$

$$\text{Or } \hat{H} \varphi = E \varphi$$

Importance of basic probability algebra and theory

In order to understand quantum physics, the role of mathematics is very important. Especially when it comes to quantum mathematics. probability and linear algebra take the driver's seat. Therefore, in this section, we will explore the fundamentals of probability and algebra.

Essential concepts of probability

Theory of probability are widely used in the field of computer science and quantum physics. In quantum computation, it is used on the large scale. Therefore, we will cover some fundamental and important concepts of probability, as follows:

- The probability of an outcome is generally represented by P . It is defined as the ratio of the number of favourable outcomes to the number of total possible outcomes of an event:

$$P = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

- For instance, rolling a dice has six possible outcomes: 1, 2, 3, 4, 5 and 6. The probability of getting number 5 as the desired outcome is

represented as $1/6$. If there are two possible outcomes, for example, 1 and 2, the probability in that case is represented as $P = 2/6$.

- The probability always exists as a number in the range of 0 to 1, where 0 represents an impossible event and 1 is a certain event. In a nutshell, the fundamental rule of probability says that all the probabilities must add up to 1.
- The experimental probability is the ratio of the number of times the outcome occurs to the total number of experimental trials. Therefore:

$$= \frac{\text{Number of times that outcome occurs}}{\text{Total number of trials}}$$

The XP is always a number between 0 and 1. In case of XP, of all outcomes, some experiments must add up to 1.

Essential concepts of algebra

As probability has a significant role in quantum computer and computation, the same is the case with the algebra / linear algebra. So, there are no surprises that we will cover algebra next.

The word algebra originates from Arabic al-jabr, which means the reunion of broken parts. An algebraic expression is an expression built up from constants, variables, and algebraic operators, as shown in [Figure 3.6](#):

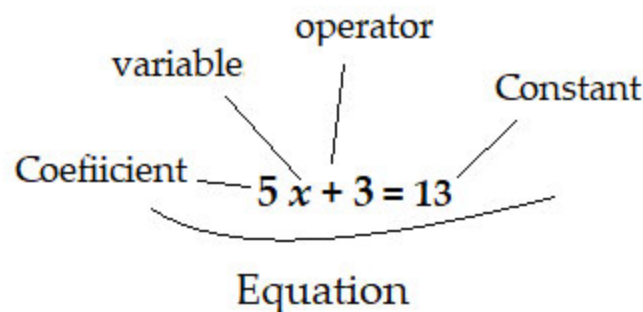


Figure 3.6: Equation

Let us explain [Figure 3.6](#) as follows:

- Constant is a number, which is a fixed value that does not change. For instance, 3 is a constant.
- Variable is a symbol that represents an arbitrary or unknown number. In the example, $5x + 3 = 13$, x is a variable. Variables are usually denoted by letters, that is, x, y, z or a, b, c . If more than three variables are required in a calculation, then single letter with an index or subscript are generally used to distinguish. One of the example is $x_1, x_2, x_3, \dots$
- Coefficient is a number used to multiply a variable. In the given example, 5 is a coefficient of x .
- Operator is a symbol, such as $+, -, /$ and $\times$.
- Term is a constant, a variable or numbers and variables grouped together, for example, in $5x + 3 = 13$, $5x, 3$ and 13 are terms.
- Equation is an algebraic statement that has an equal sign, for example, $5x + 3 = 13$ is an equation. Solution of an algebraic equation is a number or set of numbers that substitute the variables and give a value to the equation, that is, solution of the given example is $x = 2$.
- Identity is an equation that holds a value for its variables. Equation $5x + 3 = 13$ is the identity for $x = 2$.
- Linear combination is the sum of terms. Each term is the product of a constant and a variable:

$$ax + by, ax + by + cz$$

Where a, b, c are constants and x, y, z are variables.

- Linear equations is an algebraic equation that can be expressed in the following form:

$$a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_nx_n = 6$$

Where $a_1, a_2, a_3, \dots, a_n$ are coefficients, 6 is a constant and $x_1, x_2, \dots, x_n$ are variables. Some examples of linear equations are $3x = 10$ and $y_1 + y_2 = 12$.

Five important properties are as follows:

- **Associative properties:** (i) $(ab)c = a(bc)$; (ii) $(a+b)+c = a + (b+c)$

- **Commutative property:** (i) $ab = ba$; (ii) $a+b = b+a$
- **Distributive property:** (i) $(a+b) \times c = a \times c + b \times c$
- **Identity:** (i) $a+0 = a$; (ii) $a \times 1 = a$
- Multiplicative inverse: $a \times = 1$

Review of Pythagoras theorem

One associated term which needs a discussion here is Pythagoras theorem.

Pythagoras theorem states that the sum of the squares on the sides of a right triangle is equal to the square on the hypotenuse. The hypotenuse is the longest side, that is:

$$a^2 + b^2 = c^2$$

This theorem is named by the Greek Philosopher Pythagoras (C. 570 – C. 495 BC). In 3D space, the square of the length of the diagonal x is given as $x^2 = x_1^2 + x_2^2 + x_3^2$ and the vertical edge x^1_2 as follows:

$$x^2 = x_1^2 + x_2^2 + x_3^2 = \sum_{k=1}^3 x_k^2$$

Σ stands for summation. In general, it is given as follows:

$$x^2 = \sum_{k=1}^n x_k^2$$

Essential concepts of vectors

A vector is fundamentally a list of numbers. Whereas the dimension of the vector is the number of items present in the list. If the lists are written vertically, then according to Dirac's notation, we call them column vectors or kets.

However, If the lists are written in a horizontal fashion, they are termed as the row vectors or bras. The numbers which are involved in the making up of a vector are often called entries. Ok, so now to understand this concept

better, we would consider a three-dimensional ket and a four-dimensional bra and perform some operation on them.

So, Ket and Bra could be mentioned as follows:

$$\text{Ket} = \begin{bmatrix} -7 \\ .9 \\ 8 \end{bmatrix} \quad \text{Bra} = [4 \ 21 - \pi \ 30]$$

As mentioned earlier the names bra and ket were proposed by one of the most influential quantum physicists of the classical era, Paul Dirac. Let's consider, a ket where name v is denoted by $|v\rangle$ whereas a bra with name w is denoted by $\langle w|$. So, we might rewrite the complete representation in the following format:

$$|V\rangle = \begin{bmatrix} -7 \\ .9 \\ 8 \end{bmatrix} \quad \text{and} \quad \langle W| = [4 \ 21 - \pi \ 30]$$

Vector addition and multiplication

Given two vectors that have the same type, either both bras or both kets, and having the same dimension, we can easily add them to get a brand-new vector of a similar type and dimension.

The first entry of this vector just comes from adding the first entries of the two vectors, the second entry. Whereas, if we can perform manipulation of a vector by a number (in linear algebra, numbers are often called scalars), it is referenced as Scalar multiplication. We do this by multiplying each of the entries by the given number.

Vector addition is fundamentally a process of totalling two vectors into a vector sum. If two vectors opening at the identical point are watched as sides of a parallelogram, then the sum of these vectors is given by the vector that is the diagonal of that parallelogram. This is the parallelogram rule.

Vectors representing qubit states at all times start at the centre of a unit circle. The sum of two vectors can also be attained by employing them head to tail, and the condition is that their direction must not be changed and sketch the subsequent vector from the free end to the free head.

In terms of quantum computation, vector addition, manipulation or multiplication, and subtractions are performed on a matrix in a very regular way and it is represented by bar/ket notation.

For example, multiplying the ket $|a\rangle = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$ by the number c gives

$c|a\rangle = \begin{bmatrix} ca_1 \\ ca_2 \\ \vdots \\ ca_n \end{bmatrix}$. The same logic goes for addition as well...if we have to add

$|a\rangle$ and $|b\rangle$ where $|a\rangle = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$ and $|b\rangle = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$ then

$$|a + b\rangle = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{bmatrix}.$$

The general form of the vector and scalar addition and manipulation is represented dramatically in [Figure 3.7](#):

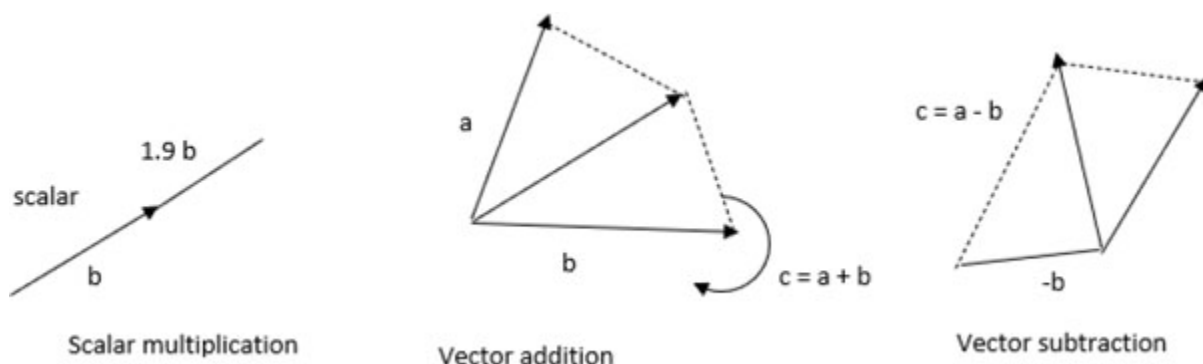


Figure 3.7: vector and scalar addition and manipulation

[Multiplying a Bra by a Ket](#)

If we have a bra and a ket of the same dimension, then to get their product, we would multiply them. By multiplying, the bra on the left and the ket on the right gives us a number. This is very similar to what we performed previously. Let's see this action where we suppose that both a and b are n-dimensional:

$$\langle a| = [a_1 \ a_2 \ \cdots \ a_n] \quad \text{and} \quad |b\rangle = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

We use concatenation to denote the product. This just signifies that we write down the terms side by side with no symbol between them. So, the product is written $\langle a|b\rangle$. By squeezing the symbols even closer, the vertical lines coincide and we get $\langle a|b\rangle$, which is the notation we will use. The definition of the bra-ket product is as follows:

$$\langle a|b\rangle = [a_1 \ a_2 \ \cdots \ a_n] \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n$$

The vertical lines of the bras and kets are pushed together, which helps us to remember that the bra has the vertical line on the right side and the ket has it on the left. The result consists of terms sandwiched between angle brackets. The names bra and ket come from bracket, which is almost the concatenation of the two names. Though this is a rather weak play on words, it does help us to remember that, for this product, the bra is to the left of the ket.

In linear algebra, this product is often called the inner product or the dot product, but the bra-ket notation is the one used in quantum mechanics, and it is the one that we will use throughout the book. Now that we have defined the bra-ket product, let's see what we can do with it. We start by revisiting lengths.

[Concepts of matrices, different operations on matrices](#)

A matrix represents an assembly of numbers organized in the order of rows and columns. It is necessary to encompass the items of a matrix in parentheses or brackets. A matrix with 9 elements is shown as follows:

$$[M] = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

This Matrix [M] has 3 rows and 3 columns. Each element or item of the matrix [M] can be referred to or represented by its row and column number. For example, M₂₃=6 locates the second row's third element or item.

Order of a matrix

The order of a matrix is also defined in terms of its number of rows and columns (this is common for all operations of the matrix). The order of a matrix = No. of rows × No. of columns. In this scenario, the Matrix [M] is a matrix of order 3 × 3.

Transpose of a matrix

The transpose of a matrix is represented by $[M]^T$. fundamentally, the transpose of an m x n matrix [M] is n x m matrix which is obtained by doing a simple operation that is, interchanging the rows and columns of [M]. For example, if $A = [a_{ij}]_{m \times n}$, then $A^T = [b_{ij}]_{n \times m}$ where $b_{ij} = a_{ji}$

Various common operations on matrices and their properties

Now let's discuss various common operations on a matrix that are doable on them, like addition and multiplication.

Matrices addition

The addition of two matrices $A_{m \times n}$ and $B_{m \times n}$ provides a matrix $C_{m \times n}$. The elements or items of C are the sums of corresponding elements or items in the matrix A and B which can be represented as follows:

$$\begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix} + \begin{bmatrix} 10 & 20 \\ 30 & 40 \end{bmatrix} = \begin{bmatrix} 14 & 24 \\ 36 & 47 \end{bmatrix}$$

Key points:

- The addition of matrices is commutative. This means if we add $A+B$ and $B+A$, the result is the same. So, $A+B = B+A$.
- The addition of matrices is by nature associative which fundamentally means $A+(B+C) = (A+B) + C$.
- Whatever the situation would be, the order of matrices A , B and $A+B$ is always the same.
- One important thing is to note that if the order of A and B are different, $A+B$ can't be computed.
- The complexity of addition operation is $O(m*n)$ where $m*n$ is the order of matrices, this property becomes very important in the case of algorithms.

Matrices subtraction

The subtraction of two matrices A_{m*n} and B_{m*n} provides a matrix C_{m*n} . The elements or items of C are the difference of corresponding elements or items in A and B which can be represented as follows:

$$\begin{bmatrix} 10 & 20 \\ 30 & 40 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 9 & 18 \\ 27 & 36 \end{bmatrix}$$

The important points for the subtraction of matrices can be compiled as follows:

Key points:

- Subtraction of matrices is non-commutative which means $A-B \neq B-A$.
- Subtraction of matrices is non-associative which means $A-(B-C) \neq (A-B)-C$.
- The order of matrices A , B , and $A-B$ is always the same.
- If the order of A and B are different, $A-B$ can't be computed.

- The complexity of subtraction operation is $O(m*n)$ where $m*n$ is an order of matrices.

Matrices multiplication

The product of two matrices A_{m*n} and B_{n*p} provides a matrix C_{m*p} . It essentially means columns in A must be equivalent to several rows in B to calculate $C=A*B$. To calculate element c_{11} , multiply elements of first row of A with first column of B and add them ($5*1+6*4$) which can be shown as follows:

$$\begin{bmatrix} 5 & 6 \\ 8 & 9 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} (5 \times 1) + (6 \times 4) & (5 \times 2) + (6 \times 5) \\ (8 \times 1) + (9 \times 4) & (8 \times 2) + (9 \times 5) \end{bmatrix} = \begin{bmatrix} 29 & 40 \\ 44 & 61 \end{bmatrix}$$

Key points:

- The first property is multiplication or product of matrices, which is non-commutative, and means $A*B \neq B*A$ or A (multiply by) B is not equal to ($\neq$) B (multiply by) A.
- The second property is multiplication or product of matrices, which is associative and means $A*(B*C) = (A*B) *C$ or (A (multiply by) B) multiplied by C is equal to ($=$) (B (multiply by) A) multiplied by C.
- For calculating $A*B$ (A multiplied by B), the number of columns in matrix A must be equal to the number of rows in matrix B.
- The existence or presence of $A*B$ (A multiplied by B) does not imply the existence or presence of $B*A$ (B multiplied by A).
- The complexity of multiplication operation among $A*B$ is represented by $O(m*n*p)$ that is, m (multiply by) m multiply by n where $m*n$ and $n*p$ is an order of A and B respectively.
- The order of matrix C computed as $A*B$ is $m*p$ where $m*n$ and $n*p$ are an order of A and B respectively.

The determinant of a Matrix

The determinant of a Matrix is a special number that is defined only for square matrices (typically a matrix that has the same number of rows and columns). Practically the determinant is used at many places in calculus and

other matrices-related algebra, especially matrix algebra. The determinant is representing the matrix in terms of a real number which can be further used for solving a system of a linear equation and finding the inverse of a matrix.

How to calculate?

The value of the determinant of a matrix can be calculated by the following procedure where for each element of the first row or first column, you get a cofactor of those elements and then further multiply the element with the determinant of the corresponding cofactor, and finally add them with alternate signs. As a base case, the value of the determinant of a 1*1 matrix is the single value itself.

Cofactor of an element is a matrix that we can get by removing the row and column of that element from that matrix.

The determinant of 2 x 2 Matrix is as follows:

$$|A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

How it happens



Tensor and tensor product

In mathematics, a tensor refers to objects that have multiple indices. Roughly speaking, this can be thought of as a multidimensional array. The tensor product is a specific method to construct a novel vector space from old vector spaces. This construction would frequently come across as frightening and enigmatic. Here I tried to make it simple. I will not discuss axioms, universal properties, or commuting diagrams. In its place, I'll take a straightforward, real look. Here we go on it.

Old wine in a new bottle!!!

Given two vectors v and w , we can build a novel vector, called the tensor product which can be represented by $v \otimes w$. Now let's try to understand the

tensor product in detail.

To understand Tensor, let's assume V and W are finite-dimensional vector spaces. That means we can visualize V as $\mathbb{R}^n$ and W as $\mathbb{R}^m$ for some positive integers n and m . Hence, a vector v in $\mathbb{R}^n$ is just a list of n numbers, while a vector w in $\mathbb{R}^m$ is a list of m numbers. For practical understanding, $n=3$ and $m=2$ can be represented as follows:

$$\vec{V} = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} \text{ (a vector in } \mathbb{R}^3 \text{)}$$

$$\vec{W} = \begin{bmatrix} 7 \\ 8 \end{bmatrix} \text{ (a vector in } \mathbb{R}^2 \text{)}$$

Assume we want to derive a new, third vector out of v and w . This can be done in two ways: we can pile them on top of each other, or we can first reproduce the numbers together by multiplying and then pile them on top of each other. Refer to [Figure 3.8](#) for a better understanding:

$$\begin{array}{l} \vec{V} = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} \text{ (a vector in } \mathbb{R}^3\text{),} \\ \vec{W} = \begin{bmatrix} 7 \\ 8 \end{bmatrix} \text{ (a vector in } \mathbb{R}^2\text{)} \end{array} \quad \left. \vphantom{\begin{array}{l} \vec{V} \\ \vec{W} \end{array}} \right\} \text{Input}$$

Method 1: $\mathbb{R}^3 + \mathbb{R}^2$ of $\vec{V}$ and $\vec{W}$

$$\vec{V} \oplus \vec{W} = \begin{bmatrix} 3 \\ 4 \\ 5 \\ 7 \\ 8 \end{bmatrix} \quad \text{Output in form of Direct sum}$$

Method 2: $\mathbb{R}^3 \times \mathbb{R}^2$ of $\vec{V}$ and $\vec{W}$

$$\vec{V} \otimes \vec{W} = \begin{bmatrix} 3 \times 7 \\ 3 \times 8 \\ 4 \times 7 \\ 4 \times 8 \\ 5 \times 7 \\ 5 \times 8 \end{bmatrix} \quad \text{Output in form of Tensor product}$$

Figure 3.8: Consolidation of vectors.

The first-choice stretches a new list of $n+m$ numbers, whereas the second selection provides a new list of nm numbers. The first springs a method to build a new space where the dimensions add, the second gives the means to shape a new-fangled space where the dimensions multiply. The first is a vector (v,w) in the straight sum $V \oplus W$ which is similar to their direct product $V \times W$, the second is a vector $v \otimes w$ in the tensor product $V \otimes W$.

Broadly constructing the tensor product $v \otimes w$ of two vectors is a lot like discovering the Cartesian product of two sets $X \times Y$. That's precisely what we're doing if we think of X as the set whose elements are the admissions of v and likewise for Y .

[Essential of trigonometry concepts](#)

Quantum physics deals with angles at a time and involves a lot of trigonometry in it. Hence, it is good to have a fundamental understanding of this. Therefore, in this section, we will discuss the concepts of trigonometry.

Trigonometry is a branch of mathematics that studies angles and their relationships. An angle is a fundamental measure of rotation. A unit of measurement for angles is called a radian and it corresponds to an arc of length 1 in a unit circle. Values for angles are expressed in units of π . A full circle has 2π radians, half a circle has π radian, further a quarter circle has $\frac{\pi}{2}$ radian. [Table 3.7](#) shows frequently used angles and their corresponding values:

θ	0	30°	45°	60°	90°	180°
Sin θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π
	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0
Cos θ	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1
(x, y)	(1, 0)	$\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$	$\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$	$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$	(0,1)	(-1,0)

Table 3.7: Frequently used angles and their corresponding values

An important compilation of results related to Trigonometry and calculus

A compilation of important results for quick reference; these results are very useful while one is dealing with quantum computers, as shown in [Figure 3.9](#):

ESSENTIALS IN TRIGONOMETRY

Basic Trigonometric Results

- ▶ $\sin^2\theta + \cos^2\theta = 1$
- ▶ $1 + \tan^2\theta = \sec^2\theta$
- ▶ $1 + \cot^2\theta = \operatorname{cosec}^2\theta$
- ▶ $\cos 2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta = \cos^2\theta - \sin^2\theta$
- ▶ $\sin 2\theta = 2\sin\theta\cos\theta$
- ▶ $\cos(A + B) = \cos A\cos B - \sin A\sin B$
- ▶ $\cos(A - B) = \cos A\cos B + \sin A\sin B$
- ▶ $\sin(A + B) = \sin A\cos B + \cos A\sin B$
- ▶ $\sin(A - B) = \sin A\cos B - \cos A\sin B$
- ▶ $\sin C + \sin D = 2\sin \frac{C+D}{2} \cos \frac{C-D}{2}$
- ▶ $\sin C - \sin D = 2\cos \frac{C+D}{2} \sin \frac{C-D}{2}$
- ▶ $\cos C + \cos D = 2\cos \frac{C+D}{2} \cos \frac{C-D}{2}$
- ▶ $\cos C - \cos D = -2\sin \frac{C+D}{2} \sin \frac{C-D}{2}$
- ▶ $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$
- ▶ $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$
- ▶ $\tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta}$

ESSENTIALS IN CALCULUS

Differentiation

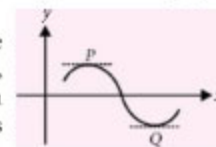
- ▶ $\frac{d}{dx} x^n = nx^{n-1}$
- ▶ $\frac{d}{dx} \log_e x = \frac{1}{x}$
- ▶ $\frac{d}{dx} \sin x = \cos x$
- ▶ $\frac{d}{dx} \cos x = -\sin x$
- ▶ $\frac{d}{dx} \tan x = \sec^2 x$
- ▶ $\frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$
- ▶ $\frac{d}{dx} \sec x = \sec x \tan x$
- ▶ $\frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cot x$
- ▶ $\frac{d}{dx} e^x = e^x$
- ▶ $\frac{d}{dx} a^x = a^x \log_e a$
- ▶ $\frac{d}{dx} e^{ax} = ae^{ax}$
- ▶ $\frac{d}{dx} a^{bx} = ba^{bx} \log_e a$
- ▶ $\frac{d}{dx} \frac{f_1(x)}{f_2(x)} = \frac{f_2(x) \frac{d}{dx} f_1(x) - f_1(x) \frac{d}{dx} f_2(x)}{(f_2(x))^2}$

- ▶ $\frac{d}{dx} (f_1(x)f_2(x)) = f_1(x) \frac{d}{dx} f_2(x) + f_2(x) \frac{d}{dx} f_1(x)$
- ▶ $\frac{d}{dx} (ax + b) = a$
- ▶ $\frac{d}{dx} (ax + b)^n = an(ax + b)^{n-1}$

Maxima and Minima

Let $y = f(x)$ be a function. Then first draw the graph of $f(x)$ as shown.

From the graph we see that at maxima or minima, slope dy/dx of the graph at P and Q respectively is zero.



$\therefore \frac{dy}{dx} = 0$ at maximum or minimum values of y .

Put $\frac{dy}{dx} = 0$ and solve for x . (We may get different values of x .)

At all those values of x for which $\frac{d^2y}{dx^2}$ is negative, we have maximum value of y .

Similarly, at all those values of x for which $\frac{d^2y}{dx^2}$ is positive, we have minimum value of y .

Illustration 1. Find maximum or minimum values of the functions $y = 25x^2 + 5 - 10x$

Sol.: For maximum and minimum value, we can put

$$\frac{dy}{dx} = 0$$

$$\text{or } \frac{dy}{dx} = 50x - 10 - 0 = 0 \quad \therefore x = \frac{1}{5}$$

Further $\frac{d^2y}{dx^2} = 50 > 0$ at $x = \frac{1}{5}$. Therefore, y has minimum value at $x = \frac{1}{5}$. Substituting $x = \frac{1}{5}$ in given equation, we get

$$y_{\min} = 25\left(\frac{1}{5}\right)^2 + 5 - 10\left(\frac{1}{5}\right) = 4$$

Integration

- ▶ $\int x^n dx = \frac{x^{n+1}}{n+1} + c \quad (n \neq -1)$
- ▶ $\int \frac{dx}{x} = \log_e |x| + c$
- ▶ $\int \sin x dx = -\cos x + c$
- ▶ $\int \cos x dx = \sin x + c$
- ▶ $\int \sec x \tan x dx = \sec x + c$

Figure 3.9: Compilation of important results for quick (source: from the notebook of author)

Quantum mechanics, quantum computers, and vectors

In quantum mechanics, vectors in a vector space define states of a quantum system. The state vector conveys both classical and quantum information.

The classical information is available to classical measurement, whereas the quantum information is not. The state-space of a compound quantum system is the tensor product of separate Hilbert spaces $H = H_1 \otimes H_2$. Hilbert spaces simplify the idea of Euclidean space. It outspreads the approaches of vector algebra and calculus from the two-dimensional Euclidean plane and three-dimensional space to spaces with all limited or unlimited numeral of proportions or dimensions. If the state of a compound system cannot be inscribed as $|\Psi\rangle = |\Psi_1\rangle \otimes |\Psi_2\rangle$ of its components (it can be represented in the form of H also), it is an entangled state.

We can understand a vector in terms of quantum machines as the mathematical tool that physicists use to describe the state of a quantum system. If you have a tiny quantum particle, possibly you'd like to identify what its responsibility is. Or what it's skilled for doing. Or the likelihood that it'll be doing to some degree. In crux, you're likely to enquire: What's its status? What's its state? The response to this query is given by a hypothesis of quantum mechanics—which is specified by a unit vector in a vector space; more accurately a Hilbert space, say H^n . That unit vector scrambles information about that particle, as shown in [Figure 3.10](#):

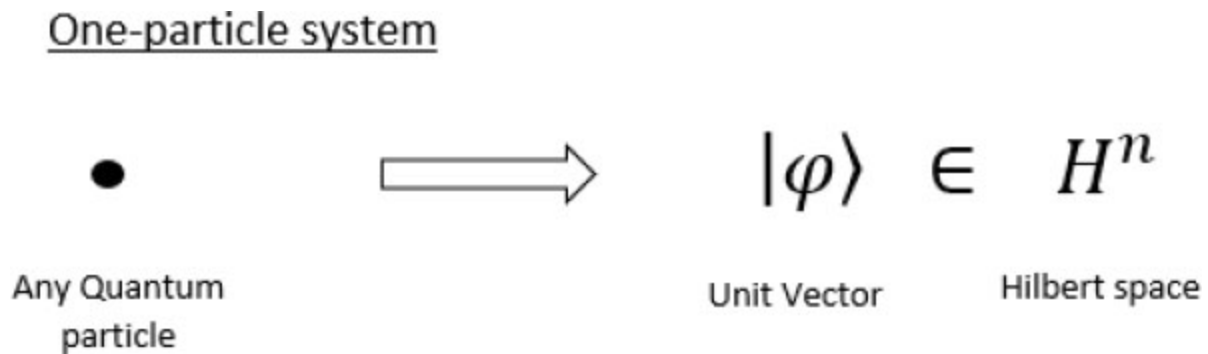


Figure 3.10: One particle system

The dimension n is, roughly, the number of dissimilar things you could detect after the production of a measurement on the particle. But what if we have two tiny quantum particles? The state of that two-particle system can be defined by something named a density matrix on the tensor product of their own spaces $H^n \otimes H^n$. A density matrix is an oversimplification of a unit vector—it books for communications among the two particles. Refer to two particle system, as shown in [Figure 3.11](#):

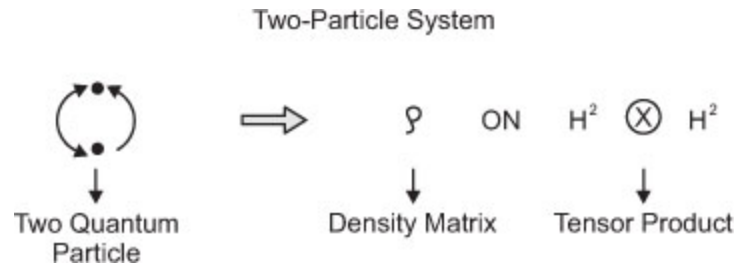


Figure 3.11: Two particle system

As the density matrix is the generalization of unit vector, it holds for N particles as well, that is, the state of an N-particle system, as shown in [Figure 3.12](#), can be defined by a density matrix on an N-fold tensor product:

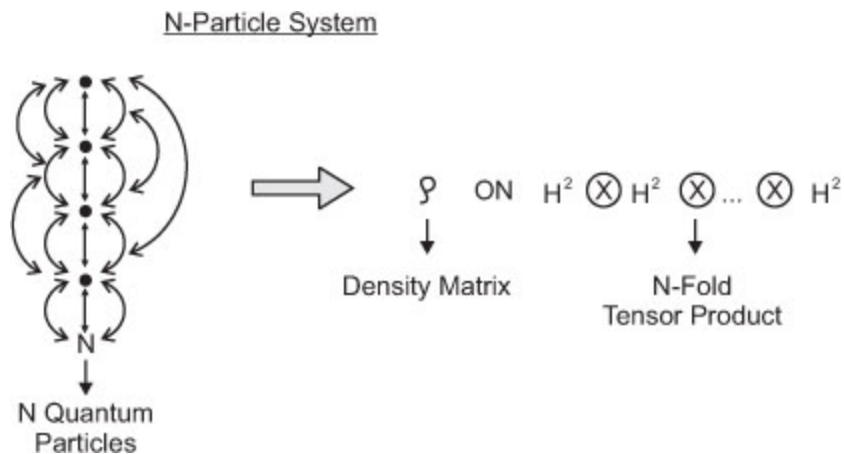


Figure 3.12: N particle system

Lastly, when working with vectors in an n-dimensional vector space, it is very valuable to have a set of n communally orthogonal unit vectors and use them as the foundation (orthonormal basis) to create any vector in that space. Such a vector will essentially be a linear grouping of the base vectors.

Therefore, the dimension of the vector space can be demarcated as the supreme number n of mutually orthogonal vectors that can be found in that space. An orthonormal basis is not exclusive, but n is. Any two bases can be connected for adaptation purposes. For instance, one may select one basis for formulating a quantum system but measure it on another basis. Stipulating an orthonormal basis is theoretically analogous to selecting a Cartesian coordinate system in physical three-dimensional space.

Observables and Hermitian operators

Recent scientific theories pursue validation in observations. Observations need that some outside activity interrelates with the system being measured. Measurements must trail a severe protocol to lessen commotion to the measured object. We hope that the measuring system can accept some minimum disturbance.

A classical measuring device continuously destabilizes a quantum system. Henceforth, we cannot be supposed to find any deterministic fundamental acquaintances using measurements. Though we may accept causality to spread over to uninterrupted systems, for example, we may express the evolution of the system (or subsystem) by differential equations that narrate previous conditions with future ones, but their capability to attach with observations will only be unintended; there will be an inevitable indeterminacy in the calculation of observational outcomes.

Jargon Buster

In math, a differential equation is an equation that relates one or additional functions and their derivatives. In applications, the functions usually signify physical extents, the derivatives characterize their rates of variation, and the differential equation describes an association between the two.

Physical measurements harvest real numbers, so we are concerned with real dynamical variables. A complex dynamical variable imposes two measurements, that is, one each for the real and imaginary portions. This works well only when measurements present insignificant turbulences in the measuring system.

In quantum systems, when Heisenberg's uncertainty theory is put on, one cannot style concurrent measurements if non-commutating observable-operators are tangled. Additionally, the order in which the variables are measured, matters. Observables are also related to a vector space, but they are not state-vectors. They are the effects we measure, and they are signified by linear operators.

In quantum mechanics, measurements are closely connected to the eigenvalues and eigenvectors of observable-operators. Therefore, the

eigenstates imply that of an orthonormal basis. Each measurable quantity, by hypothesis, has a related observable-operator characterized by a Hermitian operator and not a variable.

A Hermitian operator M , has the vital self-adjoint property, that is, $M = M^\dagger$. Also, the complex conjugate of a transposed matrix is named Hermitian conjugate, represented by a dagger superscript.

The set of eigenvalues of M embodies the set of measurement results that M can yield. Each eigenvalue has a conforming eigenstate (or eigenvector), to which the system will break down after the measurement. Some important points to call out here are as follows:

- A Hermitian matrix can be unitarily diagonalized producing an orthonormal foundation of eigenvectors that extends the state space of the quantum system.
- The eigenvalues of a Hermitian matrix are real. The likely results of a measurement are exactly the eigenvalues of the specified observable-operator.
- The matrix in place of the product of two linear operators is the product of the matrices on behalf of the two factors.

If we depict the preceding points, then some implications would be made out of this, which would tell us that any state of a quantum system can be articulated as a linear mixture of the eigenvectors of any Hermitian operator? Equivalently, which signifies that any quantum state can be specified as a linear superposition of the eigenstates of an observable-operator.

Fundamentally, classical mechanics plays with real variables. Further, quantum mechanics deals with operators whose eigenvalues are real. However, when the attention is on operators whose essential property is to bring modification, it is surprising to know that quantum mechanics needs a separate postulate for measurement.

A complex condition rises when there are two or more eigenstates of an observable with the same eigenvalue. In this situation, any state shaped by the linear superposition of the eigenstates will also be an eigenstate of the observable-operator fit into the same eigenvalue. The inference here is that

if two or more states for which a measurement of the observable is firm belong to the same eigenvalue, then any state moulded by their superposition of them will also have its place in the same eigenvalue.

This indicates that it is always possible or likely to discover two orthogonal vectors to form linear combinations. Since two eigenstates of an observable-operator attached to dissimilar eigenvalues are orthogonal, it follows that two states for which a measurement is sure to give twofold different results are orthogonal.

While we cannot formulate an opinion on whether an observable contains a value for a precise state, we can take a stake of the likelihood of it having any stated value for the state. In other words, the possibility of it being gained when one brands a measurement. We can also define an average value for the state.

It is allowable for a quantum state to be concurrently an eigenstate of two observables. The chances for such a presence are most favourable if the two observables go back and forth and somewhat exceptional when they do not. When the observables go back and forth, there exists so many concurrent eigenstates that they form a complete set.

The case when two observables commute is distinct in the gauss that the annotations are non-interfering or intimate in such a way that one can give a meaning to the two observations being completed concurrently and converse the likelihood of their being obtained. Certainly, one may view the two observations as a single observation of a more complex type, the consequence of which is two real numbers rather than one.

A natural simplification of this is that any two or more commuting observable-operators may be preserved as a single observable-operator whose deed produces two or more numbers. The conditions for which this measurement is sure to lead to one specific consequence are the concurrent eigenstates.

After the measurement, the system remains in its collapsed state unless altered. Till altered, any repetition of the measurement will only reproduce the first measurement. This is the deep connection between the set of eigenvalues of an observable-operator and the possible results of measurement associated with that observable-operator. One may, of course,

alter the post-measurement state of the system by measuring a different observable or measuring on a different basis or by applying a quantum (unitary) operation.

Mind Map for the chapter

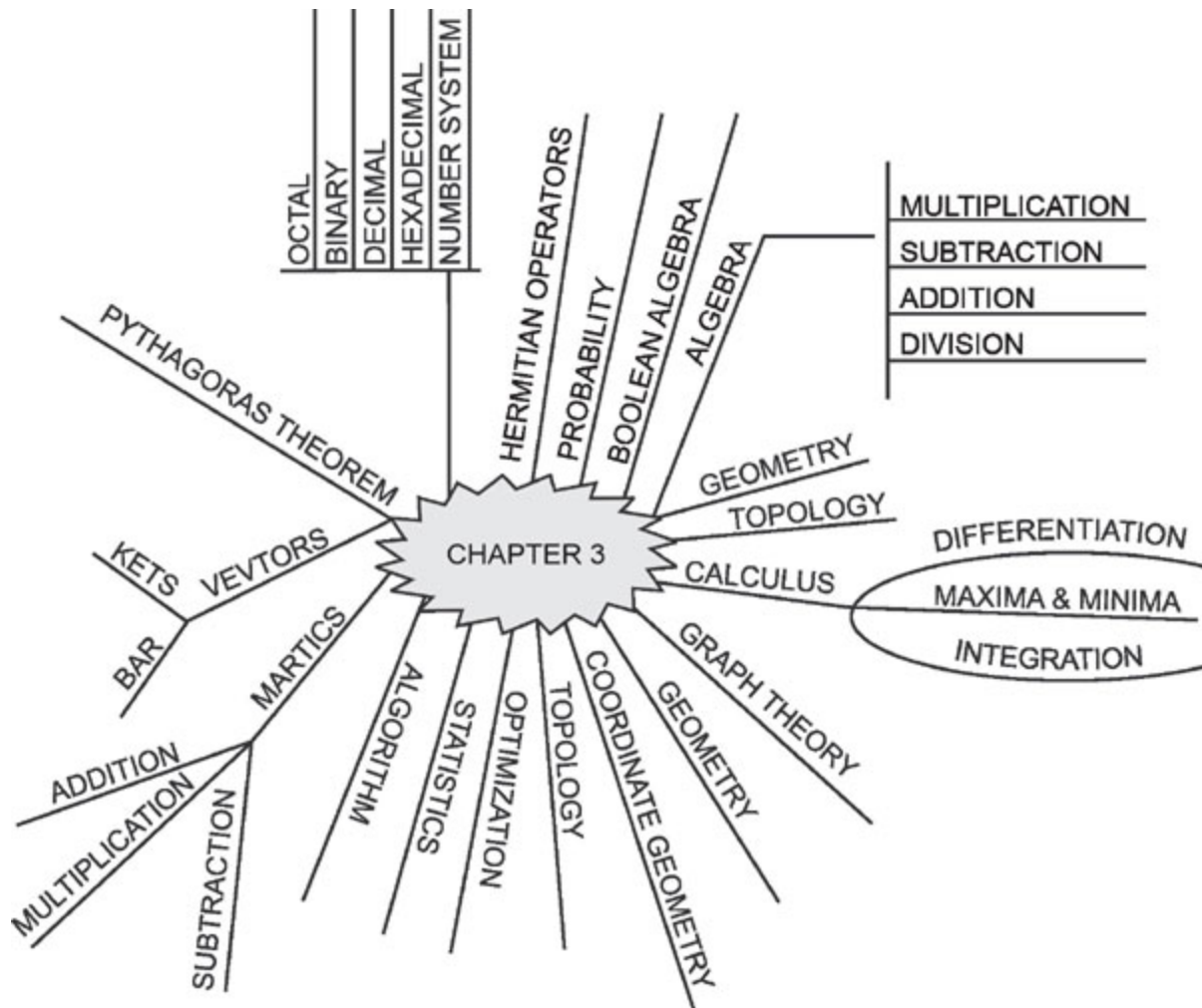


Figure 3.13: Mind map for chapter 3

Conclusion

In this chapter, we went through the fundamentals of mathematics which are relevant to quantum computers. We started our journey with a history of the number system and moved on to advanced concepts like matrix manipulation to tensor algebra. In the next chapter, we will cover the working of quantum computers in detail.

The mathematical model of the Quantum computer is founded on a branch of algebra called linear algebra.

A single qubit state is fundamentally a unit vector in a two-dimensional 2^1 Hilbert space, whereas a 2^2 Hilbert space. In general term, it can be expressed as n-qubit state in a 2^n -dimensional Hilbert space.

Quantum state is mathematically expressed in the term of bar-ket notation which is proposed by Dirac. Fundamentally, the quantum state of a complex vector a is may be written as Ket $|a\rangle$ also.

A linear operator U (quantum gate) is acting as Ket that transforms vector a into another vector b as follows:

$$U |a\rangle = |b\rangle$$

- The bra-ket or the inner product of two state vectors $\langle b|a \rangle$ is the probability amplitude of state $|a\rangle$ transitioning into state $|b\rangle$.
- Further, the probability of observing state $|b\rangle$ is given by the Born rule as $|\langle b|a \rangle|^2$.
- The ket-bra or the outer product $|k\rangle\langle k|$ is a projection operator, that is, an operator $|k\rangle\langle k|$ acting on Ket $|a\rangle$ as $k_1\langle k_1|a_1$ projects $|a\rangle$ in the direction of the state $|k\rangle_1$ exacting the contribution $|k\rangle$ makes as a component of $|a\rangle$.
- $|b\rangle\langle a|$ --- The outer product of two vectors is a matrix acting as an operator.

CHAPTER 4

From Bits to Qubits to Qubytes

“To me, quantum computation is a new and deeper, and better way to understand the laws of physics, and, hence understanding physical reality. The most important application of quantum computing in the future is likely to be a computer simulation of quantum systems because that’s an application where we know for sure that quantum systems, in general, cannot be efficiently simulated on a classical computer.”

— David Deutsch

Quantum mechanics is amazing, and computing abilities accessible by it are just the tip of the iceberg. It has an extensive variety of applicability in biology and computer hardware. And it can even clarify how the anticipation or views of humans can touch the future as enlightened. One of the examples of it is the Global Consciousness Project, which experimentally showed that when human consciousness turns out to be coherent, the behaviour of haphazard systems may change. There is indeed a peculiar relation between the human mind and quantum physics. Furthermore, it can be cast off for Quantum teleportation which was attested by Chinese specialists who had communicated the quantum state of a photon which resides in a laboratory on Earth to an additional photon on a satellite placed somewhere in space in lower Earth orbit. This is just an example that shows that quantum mechanics possesses the potential to alter everything that we know and do characteristically. In this chapter, we will understand the basics of quantum computers including registers, gates, circuits, thermodynamics in the context of QC and more. We will also look at the some of the use cases.

Structure

In this chapter, we will discuss the following topics:

- Introduction
- A miracle called quantum computers
- Information theory, thermodynamics, and computation
- The start of the journey: from then to now
- Before we move further, some additional thoughts on a quantum computer
- Quantum parallelism
- Spin and Qubit
- The physics of atom in nutshell--facts and figures
- Spin states and charges
- Further details on state
- What is this up-and-down state
- How to get up and down state experientially
- Significance of States
- Summary of essential notions, maths, and theory related to Dirac Theory
- Qubit and entanglement
- Spin
- Quantum number and energy state
- One additional point about Qubit and electron states
- Electron spin theory
- Observer effect
- Observer effect, no-cloning theorem, and quantum computing
- From classical bit to Qubit
- Qubit detailed
- Quick facts
- How Qubit is created
- Caution!!! More Qubit is not equal to faster computation
- Systematic expiation of qubits, quantum gates, and circuits

- How to measure Qubit
- Quantum gates
- From quantum gate to quantum circuit
- Quantum register
- Quantum register: working
- Inputs and outputs of quantum circuits: representations, logic, and theory
- Controlled gates
- Measurement operator
- A summary of the key symbols
- Quantum teleportation and its circuit diagram
- Let's go into bit details: mechanics, mathematics, and measurements of Qubit
- Single-Qubit gates
- Few additional key single-qubit gates
- Multi-Qubit gates
- Universal sets: moving from traditional gates to Quantum gates
- Multiple types of Qubits
- Components of a Quantum computer
- How to make robust quantum computers: welcome quantum error correction
- A complete quantum system from hardware to software
- Solving problems on quantum computers
- Hybrid quantum – classical computing
- Real-life working of Quantum Program: Microsoft way
- Case study: Microsoft's journey to quantum computers
- Other companies' technology stacks
- Story 1 : Honeywell
- Questions and answers

- Mind maps
- Conclusion
- Important terms and definitions

Objective

This chapter contains the technical details of quantum computers through proper uses of mathematics, including their building blocks like qubit, logic gates, registers, and circuits. It gives readers a perspective about the complete ecosystems of QC – from hardware to cloud infrastructure to the software stack. After going through this chapter, the readers will become comfortable with most of the quantum technical computing “internals” and understand their work.

At the same time, s/he will become familiar with the thought process and roadmap of a few leading corporations(s) about the technology and be aware of the use cases of it.

Introduction

Old-style binary computers aren't going anywhere anytime soon. We comprehend how conventional computers work at a fundamental level. And we also discussed that it still uses the elements or components that only function because of quantum mechanics. Traditional computers do not have to concern themselves about decoherence, the observer effect, or the no-cloning theorem, and they are not nearly as sensitive to external effects as quantum devices. Therefore, they are quite impressive in the real world. They do not need ultra-super customized environments to run. They provide us a lot of benefits. And we can achieve it by fitting CPU cores into a single chip. The data and consequences they deliver are steady for long periods in memory and on the processors without us having to worry about how the external world will impact them. There are some types of raw, rote calculations that binary computers do very well. Quantum supremacy may sidestep traditional computers one day soon, but it's tough to exhaust the stable binary computers. For this reason, we are likely to have traditional computers in our lives for a long time to come.

A miracle called quantum computers

We practice the benefits of traditional computing every day, though, there are challenges that today's systems will not ever be talented to resolve the highly computationally-intensive problem. The main reason is this – we don't have sufficient computational power on the ground to antagonize them.

To have a chance at solving some of these problems, we fundamentally need a new-fangled kind of computing infrastructure. Universal quantum computers are powered by the quantum mechanical phenomena of superposition and entanglement, and at the same time, it powers back the field of quantum as well to produce states that gauge exponentially with the number of qubits or quantum bits.

The simple idea of engineering quantum computers for commonplace use is flattering and perceptible with novel technological developments in quantum theories. Quantum computing emphasizes the philosophies of quantum theory, which integrates with modern physics that clarifies the behaviour of matter and energy at an atomic and subatomic level.

Quantum computing use quantum phenomena, on which quantum physics is based, such as superposition, tunnelling, uncertainty principle, and entanglement. It uses them for multiple purposes, including data procedures. Hence, Quantum computing in this way fundamentally challenges tremendously tough tasks that normal computers cannot do on their own in an unconventional way, which we are used to with traditional computers.

In conventional computing, a bit is a term to signify information by computers. Quantum computing uses quantum bits or qubits for a unit of memory. Qubits are contained and now realized in a two-state efficient and effective quantum-mechanical system. A quantum-mechanical system is one that can be in any two different quantum states. Superposition is a principle that positions while we do not know the state of a thing at a specified time; it is conceivable that it is in all states concurrently if we do not look at it to crisscross its state. This means that energy and mass turn out to be connected to interrelate with each other irrespective of remoteness.

Entanglement and superposition are tremendously significant in computing and communications that can help us in many ways. These two marvels can be used to process an extremely big number of calculations, whereas traditional computers cannot. The influence of quantum computing is amazing and not many are able to grasp the full competencies it must bid.

We are at least one decade behind to realize the full impact of quantum computing. However, there is a considerable quicker upheaval gathering force and hope of optimism is at play. Hence, it is generating the euphoria of achievements. Therefore, people who sit in the driver's seat of business, promise big changes to the commercial ecosystem in the next ten to twenty years.

However, there is a reason behind it...Researches are underway and are cartelizing the thoughts and science landscape at multiple major so-called technologies, business, consulting, service companies, universities, government agencies, and start-ups for achieving breakthroughs.

IBM, D-Wave, Alphabet (parent company of Google), Righetti Computing, Alibaba (historically China's e-commerce giant, now a technology innovation company), Microsoft, Intel, and Honeywell are the few corporations that are in the race of constructing an ecosystem around quantum computers which includes but is not limited to the services, business models, and technologies. Also, they are essentially leading a series of technological innovations in building effective and practical quantum computing systems and infrastructures.

Few of these extraordinary seamless efforts are complemented by government-funded Research & Development agencies as well because they have a keen interest in becoming a superpower in the field. And for that, research is the key. It makes all but certain that in the near to medium term, we are ready to witness the development of medium-sized, however, still error-prone, quantum computers that can be integrated into the DNA of the business.

This endeavour has the power and capability to produce the super lightning experimental discoveries at a quicker rate. We started seeing them because already quite a few companies are moving aggressively in the value chain by securing intellectual property (IP) rights. Their effort comes in the form

of patents of state-of-the-art business models that are based on quantum computers. Why are they doing this? Because of the belief that these models would potentially position them to be first to market (means more business > profit) by providing a unique value proposition that fundamentally links with their shares of solving technical and business quantum computing puzzle.

Now the time has come when every thriving company that wants to bring uniqueness, needs to understand how future discoveries/innovations/solutions will affect business at its core. Therefore, leaders started to stake out their positions in this emerging technology, which will come up in the next few years.

Information theory, thermodynamics, and computation

Modern information theory is founded on two significant proofs called Shannon's definition of information and Lindauer's opinion & theory. According to their observation, information is physical since it is essentially continuously encoded in a physical system, deprived of which, it is impossible to store, communicate, process, or obtain information. Additionally, the information held by a physical system pays to define the state of the system. It is important to know here that in a mathematical (and henceforth a computational) model of the laws of nature, semantic aspects are immaterial to the interpretation problem.

Shannon's well-known twin papers, A Mathematical Theory of Communication, issued in year 1948, put the basis for information theory. His influential contribution was to describe the idea of information precisely in mathematical form and then to contemplate the broadcast of information as a statistical wonder in a way that provided a technique to regulate the capacity of a communication channel in terms of classical bits.

As per our understanding, information can be articulated and encoded in different but equal ways without degrading its vital nature. The binary form permits suitable automatic manipulation of information in all possible ways. That is, a computer needs to manipulate fairly modest things like integers to perform astonishingly powerful information processing, from document

preparation to integral calculus, to interpreting among human languages, to even imitating human intelligence.

Translation permits us to select appropriate hardware and software technologies for information processing. A legendary physicist known as Shannon demarcated the vital idea of information entropy $H(X)$ analogous to thermodynamic entropy, as follows:

$$H(X) = - \sum_{i=0}^{N-1} p_i \log_2 p_i,$$

This gives a way to guess the average minimum number of bits desirable to encode a string of symbols, founded on the frequency of the symbols or the probability p_i of its appearance. However, Shannon entropy does not measure the degree of correctness, but it measures the degree of the decadence of a system.

Overall, a complex set of instructions can be condensed to n binary picks. We then have a prepared measure of the information content of the object by merely including the number of binary selections or the state of n binary bits. The state of a binary bit, by preceding arrangement, can be randomly mapped to the binary substitutions, such as 0 for spin-down of an electron and 1 for spin-up.

We can, therefore, use the bit as the unit of information and measure the information content of an information resonant object as the scope of the set of orders needed to rebuild the state of the object.

In 1961, Rolf Landauer, introduced a very important concept through his seminal paper on reversible computing, which provided an extraordinary vision to the world. According to him, since information is physical, a fact lately confirmed experimentally, the laws of physics necessarily place bounds on information processing whether classical or quantum. He insightfully turned to thermodynamics and showed that only logically irreversible operations that cause information loss would be able to expend energy. He displayed that there is an important asymmetry in the way nature processes information. Further, he demonstrated the counter-intuitive outcome that all, but one, a process essential in computation could, in

theory, be made reversibly without dissipating heat. For instance, copying classical information can be completed reversibly and deprived of losing any energy. However, when information is removed, there is a minimum energy cost of $kT \ln 2$ per traditional bit, where k is the Boltzmann constant, and T is the absolute temperature in Kelvin of the computer's environment. Also, please mind it, after a system experiences an irreversible action, its previous history, unless archived, irreversibly vanishes.

Landauer visualized computations as engines of changing free energy into left-over heat and mathematical work where deletion of information was unavoidably central to heat production. Landauer's theory and the second law of thermodynamics can certainly be understood as the logical importance of the fundamental reversible laws of physics. The law has multiple uses across thermodynamics and quantum mechanics, for instance, this is usable in the typical Hamiltonian formulation of mechanics and in the unitary time evolution. Landauer's theory attaches physics with information theory. The joining directed individuals to the reason for energy-efficient algorithms at a time when reducing algorithmic intricacy was the fad.

What is extraordinary about Landauer's principle is that it highlights computation as a non-concrete mathematical procedure. According to this principle registering one set of input, bits contribute to an extra set of output bits. It is not at all understood that there occurs an important joining among such a charting and microscopic motion related to heat. It, certainly, seems extraordinary that the rational reversibility of computation (i.e., inputs can be contingent on the outputs) also suggests physical reversibility. The key to physical reversibility is to erase with a copy. That information is adaptable to energy which has been experimentally revealed.

Before Landauer's published paper, it was usually expected that deterministic computation is not essentially logically reversible and characteristic programming is improbable to be so. Lecerf (1963) and Charles Bennett (1973) exhibited that deterministic computation would be simulated through a rationally reversible Turing machine. This means it is likely to reprogram any deterministic computation as an order of logically reversible phases, on the condition that the computation is permitted to save

a copy of its input. Bennett additionally recommended computing with nucleic acids or DNA to realize physical reversibility.

By equivalence with thermodynamics, one may settle that information that has not been compressed to disclose a valuable interpretable pattern is random information, which on average does not provide valuable knowledge for a human mind. In this regard, molecular biologists encounter a severe issue. Till they discover a theory to produce a given DNA sequence, the only way to interconnect comprehensive genetic information to another is to direct the whole DNA sequence. Whereas, the equations of physics have repeatedly advanced in the direction of compression to an astonishing degree. Physics is coded and quantifiable information.

The start of the journey: from then to now

Though the reduction of computer chips has enforced computer scientists to contemplate quantum mechanical laws, the principles and operations of today's computers are majorly based on classical physics except for a few parts. Consequently, the first computers from the 1940's contained tubes and capacitors, and the transistor, at the beginning a classical component, is still a centrepiece in any computer today. The transistor is short for a transfer resistor, which just means that electrical resistance is controlled by an electrical current.

The components have altered, but the elementary procedure of today's computers is still founded on the architecture articulated by John von Neumann in 1945. The fundamental of von Neumann's reference model for a computer is the memory card, which comprises both program instructions and the data to be dealt with. The data is treated successively, that is, step by step, in single binary computation steps, accomplished by a control unit. While transistors and electron tubes have been substituted by minor and quicker field-effect transistors on semiconductor chips, the architecture of today's computers has continued to be similar since it was first expressed.

Alan Turing defined the elementary data blocks and their dispensation theoretically in 1936. The greatest basic information units therein are the binary digital units or bits. Binary means two-valued since a bit can undertake either state 1 or state 0, like a light switch that is either on or off.

Electronic data processing in today's computers involves changing received information in the form of numerous successively organized bits into an output equally in the form of several repeatedly organized bits. A block of eight bits is called a byte. For single bits, there are only two dispensation likelihoods: 0 (or 1) leaves a 0 (or 1), or 0 (or 1) changes to a 1 (or 0). The elementary electronic components of digital computers, known as logic gates, are hence continually the same modest basic electronic circuits, comprehended by bodily components such as transistors.

Through transistors, information is conceded in the form of electric signals. Numerous such gates are then linked, thus allowing more complex actions such as the totalling of two numbers. It does nothing else but sequentially processes information programmed by zeros and ones, transmuting them into an output also encoded by zeros and ones. This straightforwardness in information processing has a charge. A vast number of zeros and ones are essential to be treated to grasp the amount of data indispensable in today's complex computer applications. The calculating power of a computer upsurges linearly with the number of obtainable computational blocks. With double as many circuits, a chip can process information twice as fast.

Today's computer chips function in the gigahertz assortment, that is, a billion operations per second. These need billions of transistors. To be able to pack or fabricate this many transistors on chips, the circuits must be microscopic. Only then can the complete size and energy requirements of such fast-switching systems be kept under control.

Indispensable for the reduction of the basic computing units on integrated circuits in microchips was the change from the electron tube to the semiconductor-based bipolar or field-effect transistors, which were conceived in 1947. These microscopic transistors are constructed on incapacitated semiconductor layers. This is where quantum physics comes into drama. In the direction to comprehend and control what is happening in these semiconductors, we need a quantum mechanical model for the gesticulation of the electrons inside them.

This is the named band model of electronic energy states in metallic conductors. An understanding of quantum physics was not essential for the digital rebellion of the 20th century, but a precondition for the exciting

reduction of integrated circuits was the capability to put on it. The straightforward logic gates that contain the processes are AND, OR, NOT, NAND, NOR, EXOR, and EXNOR. From electronics to quantum, a long journey happened in quick succession. The construction of classical transistors, chips, and integrated circuits (mostly based on the theories of electronics) are an integral part of the classical computer revolution. Similarly, particle spin, states, charges, quantum gates, and qubits are fuelling quantum computers.

Before we move further, some additional thoughts on a quantum computer

Adiabatic Quantum Computer (AQB) are analogue systems supposed to solely be able to run one algorithm at a time, viz. an optimization calculation. Fundamentally, optimization has wonderful applications. The main issue in contradiction of AQC is its absence of set-up to address errors and that there is no improper theory for a quantum benefit on an AQC system. However, without addressing errors, it would be unbearable to scale AQCs to a point where they would contest with classical computers. If investigators can design an AQC that is error-resilient fully connected, it would enable AQCs to run a wider range of algorithms, possibly making it modest with the universal model.

The digital quantum computer is a system named universal QC. These systems function just as a traditional computer does. This is the furthestmost progressive least in the terms of multiplication because, in theory, it would act as a stage that could run any number of algorithms, particularly algorithms like Shor's and Grover's that first prompted general interest in QCs (will discuss these in details in next few chapters).

Also, Universal Quantum Computers (UQCs) have the substructure to error-correct, and though its application still aspects problems, it leaves the greatest feasible platform to understand adequate qubit coherence.

It is significant to comprehend that with all thoughts of quantum computation, separations are not ever binary. Quantum technology is still in its developmental beginning, and different problems are touched best by different kinds of quantum computation.

Research feast that crosses different universities and corporates continue to imagine novel potentials to solve questions. For instance, Microsoft has proclaimed another technique of attaining UQCs with what they call a topological quantum computer, and it will only be through a joint effort that we will be able to bridge the gap between hardware and software.

There is a dream to construct a quantum computer that is totally superior to all others, and hardware will only progress by admitting that different examples each have their support. That they are all parts of a mystery that will eventually converge upon what we envisage to be the quantum computer.

Quantum parallelism

One of the most striking benefits of quantum computers is quantum parallelism. Apart from hardware the algorithms also play a major role in achieving this.

It is thought-provoking to get an instinctive sense of how quantum algorithms can attain Quantum parallelism so effectively. As an analogy, imagine that you have 49 muddy trousers and one spotless pair of trousers, all varied in a mound. The goal of the algorithm is to search this lot of trousers till it can figure out the spotless trousers.

A traditional computer would approach this problem by sightless insights into the mound, by examining if it is spotless, discarding the trousers if it is not, and then moving on. Even though today's computers can work through data sets this small at astonishing speeds, it is still basically a redundant process since there is no intellectual design in the blind choice of trousers.

In the worst-case scenario, this technique would need 50 iterations before arriving at the spotless trousers. Quantum parallelism proposes a technique that would speed up the procedure. Upholding this analogy, one could shape a superposition of all 50 trousers which is representationally lining them up on the ground in front and do the calculations in one go. Because a quantum computer with n qubits would be intelligent enough to procedure 2^n such calculations in parallel. While this appears like a one-shot method, there are some problems with this methodology. Like, the function would do repetition to the principle of superposition for the answers, something along

the lines of a string of answers “no, no, no, yes, no, no ...” for all 50 trousers. The goal would then be to lure the solitary “yes” out of the list of responses and regulate with which trousers it matches. This is all likely and only needs slightly more calculations. In terms of quantum algorithms, this is an amazing achievement that was established by Grover and Shor’s algorithm to attain quantum parallelism.

Spin and Qubit

To understand the fundamentals of quantum computers, it is required to have a better knowledge of spin, the nature of fundamental particles, their states, and some theories associated with them. However, if you want to skip this section you can, after having a look at the following two paragraphs. But the recommendations are you go through them. Broadly, we will cover the following concepts in the next three sections, as shown in [Table 4.1:](#)

1.	Spin	Spin is the whole angular momentum, or essential angular momentum, of a form or body.
2.	States	A quantum state in quantum physics describes the state of an isolated quantum system. Also, a quantum state delivers a likelihood distribution for the value of each observable, that is, for the result of each likely measurement on the system.
3.	Charges	Quantum theory explains that charge is a numerical consequence, but only in combination with a space in which the element occurs.
4.	Fundamental particles	In particle physics which is a branch of physics, an elementary particle, or better, the fundamental particle is known as the subatomic particle with no substructure. Therefore, it is not unruffled by other particles. The total number of fundamental particles is 12.
5.	Quantum numbers	A quantum number is a number which happens in the theoretic countenance for the value of some quantized property of a subatomic particle. The example of the above-mentioned particles is an atom, or molecule which can solely be able to have convinced integral or half-integral values.
6.	Atomic theory	In the world of chemistry and physics, atomic theory is

		known as a scientific theory that explains the nature of matter. It also explains that matter is fundamentally a collection of discrete units called atoms.
7.	Schrodinger wave equation	The Schrödinger equation is termed as a landmark equation. Basically, in terms of mathematics, it is a simple linear partial differential equation that defines the behaviour wave or state function of a quantum-mechanical system. It can be better understood in connection with potential and kinetic energy.
8.	Wave function	A wave function in quantum physics is a scientific explanation of the quantum state of an isolated quantum system.

Table 4.1: Some important terminology

Qubits are signified by the spin (up, down) of electrons or the polarization of photons. Spin and polarization have properties that are quite different from their classical corresponding item which is traditional computation. In QC, to measure spin, we must select a direction and then measure it (spin) in that direction. As we know, the spin is quantized. When we measure, it provides two likely answers. Those answers are not a continuous range of answers.

We can assign classical bits to these outcomes. For instance, if we acquire an N, we can understand it to be the binary digit 0, and if we get an S, we can imagine it to be the binary digit 1. This is precisely how we get answers from the quantum computation. The latter phase of the computation is to take a measurement. The outcome will be one of two things, which will be understood as either 0 or 1. While the real computation will include qubits, the concluding response will be in terms of classical bits.

To understand it better, let's assume that we will generate random strings of binary digits. The trial that produced random strings of Ns and Ss can be redrafted as a string of 0's and 1's. Accordingly, measuring spins of electrons first in the vertical and then in the horizontal direction gives a random string of 0's and 1's. Astonishingly, this is something that cannot be done with a traditional computer.

Classical computers are deterministic. They can compute strings that pass many tests for randomness, but these are pseudorandom, not random. They are calculated by some deterministic function, and if you know the function

and the initial seed input, you can calculate the same string. There are no classical computer algorithms that generate random strings. Thus, already we can see that quantum computations have some advantages over classical ones.

Ok, let's pause for a minute here. We covered some fundamentals of quantum computers and would further explore a lot of ground related to it in the next few sections and chapters. So, it is good to have a look at some core principles of physics specifically in the context of atomic, nuclear, and quantum physics. [*Figure 4.1*](#) will provide a brief look of the same:

The physics of atom in a nutshell – facts and figures

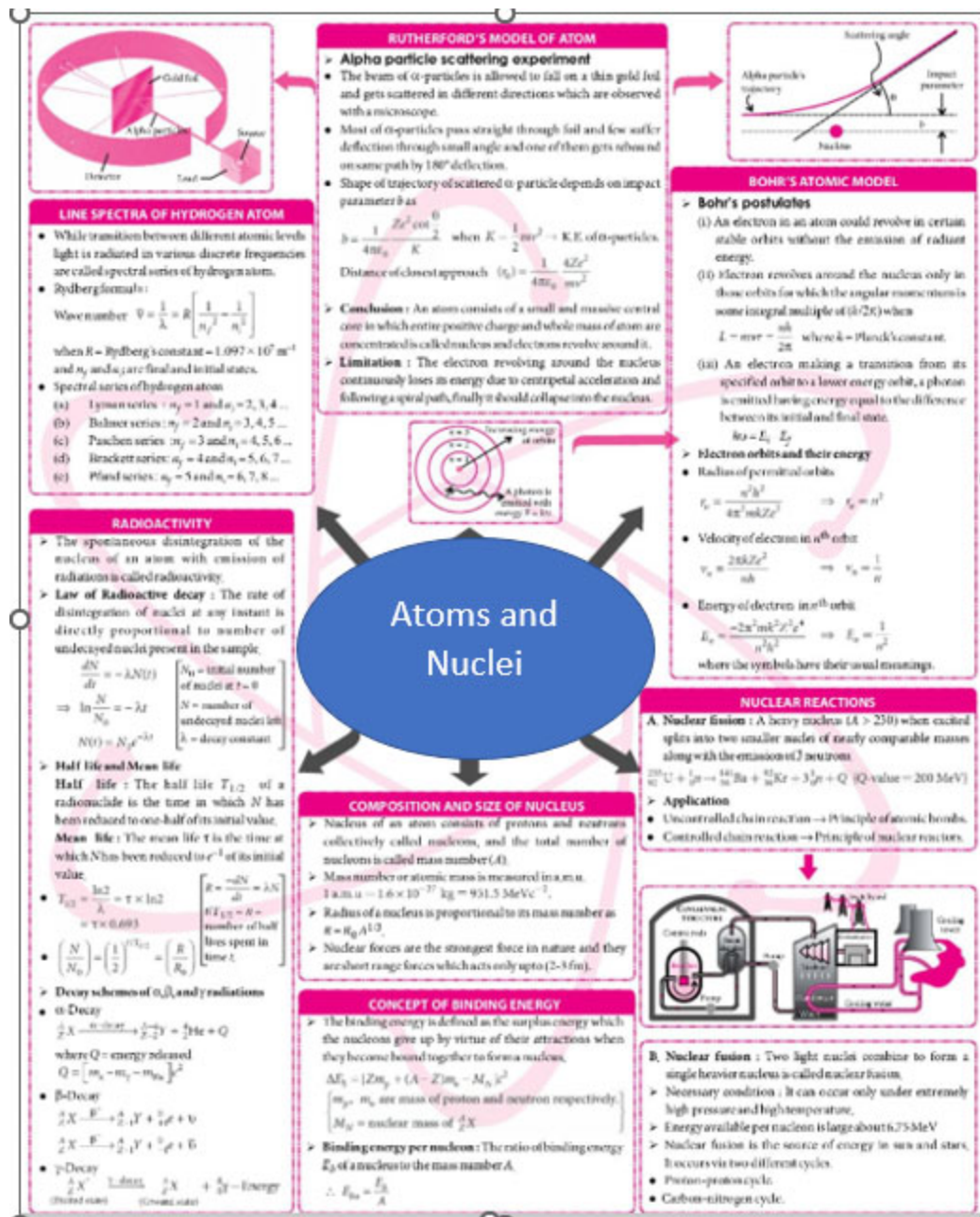


Figure 4.1: Fundamental principles of atomic, nuclear, and quantum physics

Spin states and charges

Twelve essential quantum particles contribute to all substances in the cosmos. Essential particles, can't be fragmented into minor particles. The fundamental quantum particles are the electron, muon, tau, electron neutrino, muon neutrino, tau neutrino (all part of the lepton family), and up,

down, top, bottom, charm, strange (the last six are part of the quark family). Let's start with a quark.

Roughly, quarks may be thought of as a type of elementary or fundamental particle(s) which is an important component of matter. Quarks combine to form particles called hadrons, which are established as the most stable. With the different combinations and arrangements of quarks, protons and neutrons are constructed. Protons and neutrons are in turn the most important components of atomic nuclei.

One can say that particles are made of Quarks and Gluons, which bind them together. But most physicists believe quarks to be the smallest building blocks of matter. Quarks are curbed inside bigger particles, thus they cannot be detached and studied in separation.

Also, the force between two quarks becomes higher as they move far apart, whereas the force between a nucleus and an electron, or two nucleons in a nucleus, grows weaker as their parting surges. Can we go further deep to understand quarks? Probably yes; however, till now, no investigative methods have been discovered to effectively resolve these atomic and nuclear physics glitches. Researches are still on to analyse quarks and gluon's behaviour. Some physicist says quarks may be made of strings if string theory is true. String theory essentially tells us that all particles are minute vibrating strings and each type of vibration links to a dissimilar particle. Other views on quarks are that they may be fundamental; in that case, we can't say what they are made of. Quarks may or may not be the most fundamental, but we can define some of the properties of whatsoever they are constructed.

Quarks come in different types. The different types of quarks in common teams are called flavours and colours. The flavours are maybe up, down, charm, strange, top, and bottom, based on the "state" (refer to [Figure 4.2](#)). The colour spreads are red, yellow/green, or blue. So, if you want to explain to your friend, what type of quark you have precisely – you can tell him that you have a red down quark (in practice, however, it is not possible). The same is true if you want to give him information about antiquarks, for example, you kept an anti-yellow anti-top quark. Now take the example of Meson which contains two quarks of colour and an identical anti-colour, but

they have dissimilar flavours. Baryons, for example, have three quarks which are always of three different colours, but a variety of flavours. But there seems to be no tangible agreement yet, at least on the superficial ground; therefore, to understand the origin of our world in concrete way the composition of atoms and their interplay of mass is required. Refer to [Figure 4.2](#) for SM:

COMMENTS	LEPTONS		Quarks		Remarks
Mass Particles Ordinary particles belong to this group	Electron Electric Chare = -1	Electron Neutrino No electric charge and no mass (possibly)	Up Electric charge +2/3 , Proton 1 , Neutrons 2	Down Electric charge - 1/3 , Proton 1 , Neutrons 2	Each particle has been complemented by an anti-matter. Just like, mirror image.
Existed just after Big Bang , rediscovered in CERN	Muon Havier then electron. It lives for millionth of a second , charge +/- 1	Muon Neutrino No electric charge, created along with Muon	Charm Electric charge +2/3 , Havier then up	Strange Electric charge - 1/3	
	Tau Very Unstable , charge +/- 1	Tau Neutrino No electric charge	Top Electric charge +2/3	Bottom Electric charge - 1/3	
Force Particles	Gluons	Photons	Bosons	Gravitons	
These particles are responsible for four fundamental forces of nature	Transporter of Strong force – The explosive release of nuclear energy is the result of strong force	Building block of light and carrier of electromagnetic force. The disciple of chemistry , magnetism are the results of this.	Transporter of weak force. Some form of radio-activity are the result of weak force.	Transporter of gravity. Experience of Wight is the byproduct of gravitons	

Figure 4.2: The standard model

The fundamental quantum particles are made up of all other subatomic particles. For instance, each proton is constructed by two up quarks and one down quark. A neutron is created by two down quarks and one up quark. Whereas the electron, as an elementary particle, is not made up of anything. It's an electron, with no additional subatomic particles to add up or break it down into. But electrons, protons, and neutrons make up atoms, and atoms construct elements and molecules, and so on.

Each elementary quantum particle has a mass, charge, and spin. A charge is the quantity of current as compared to an electron. For instance, an up quark has two-thirds of an electron charge and a down quark has a negative one-third of an electron charge. Because a proton has two up quarks and one down quark, this means the proton has a 3/3 ($2/3 + 2/3 - 1/3$) charge of an

electron, or precisely equal to one electron. In atoms, the number of protons in the nucleus equals the number of orbiting electrons for this reason.

Elemental particles have spin, which relates inversely to the numeral of revolutions a particle essentially style to return to its original orientation. All elemental particles have a spin of one-half, which means they essentially rotate twice to return to their initial orientation. Quantum charges and spins are important to quantum computers because the responses that quantum computers give us are often a consequence of charges and spins. Dissimilar quantum properties and states are used to deliver answers in different types of quantum computers. Let's understand it a bit deeper because it is important. We will cover the spin, charges, and state of the particles in the following section.

Jargon box

- **Euclidean geometry is an accurate system credited to the Alexandrian Greek mathematician Euclid. He discussed this geometry in his textbook on geometry in the book called the Elements. Euclid's technique contains a small set of instinctively attractive axioms, and presuming several other propositions (theorems) from these. Though many of Euclid's outcomes had been specified by previous mathematicians, Euclid was the first to confirm how these propositions could fit into a completely reasonable and common-sense system.**
- **Geometry (geo-earth, -metron measurement) is a division of mathematics concerned with queries of form, magnitude, the comparative situation of figures, and the properties of space.**

Further details on state

In quantum physics or specifically in quantum machines, a quantum state would be thought of in terms of the state of a distant quantum system. A quantum state delivers a likelihood spread or distribution for the value of each observable, that is, for the consequence of all probable measurements on the system. Knowledge of the quantum state composed of the rules for the system's evolution in time that uses all that can be prophesied about the

system's behaviour, that is, a mixture of quantum states is over a quantum state.

Quantum states that cannot be inscribed as a mixture of other states are named pure quantum states; all additional states are called mixed quantum states. Scientifically, a pure quantum state can be signified by a ray in a Hilbert space over one of the very important types of sets of numbers known as complex numbers (for details on complex numbers, please refer to the previous chapter).

The ray maybe thought of as the set of nonzero vectors that is divergent by just a complex scalar factor. Further, any of them can be selected or chosen as a state vector to signify the ray and thus the state. A unit vector is typically chosen, but its phase factor can be selected freely anyway. Yet, such factors are significant when state vectors are added to form a superposition.

Hilbert space is a simplification of the ordinary Euclidean space and it covers all possible pure quantum states of the specified system. The mathematical idea of a Hilbert space, called after David Hilbert, simplifies the concept of Euclidean space. It spreads the approaches of vector algebra and calculus from the two-dimensional Euclidean plane and three-dimensional space to spaces with any limited or unlimited number of dimensions. A Hilbert space is a non-concrete vector space possessing the structure of an inner product that permits length and angle to be measured. Also, Hilbert spaces are complete, meaning there are enough limits in the space to permit the techniques of calculus to be used.

What is this up and down state

The functions representative elements of the Hilbert space are called wave functions. Further, when dealing with the energy spectrum (in physics, the energy spectrum of a particle is the number of particles or intensity of a particle beam can be described as a function of particle energy) of the electron in a hydrogen atom, the applicable state vectors are recognized by the principal quantum number n . The principal quantum number which is one of the four quantum numbers assigned to all electrons in an atom is used to describe that electron's state. Additionally, the other quantum

numbers are the angular momentum quantum number which is generally denoted by l , the magnetic quantum number symbolized by m , and the spin z -component.

To understand it in more detail, let's take a more complicated case in the context of bra $\rangle$ and ket $\langle$ notation which is written in the singlet state, as follows:

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle \right),$$

The preceding mathematical expression mentioned in the form of an equation fundamentally involves and expresses the idea of the superposition of joint spin states for two particles with spin $1/2$. Here, the singlet state satisfies the quantum property which says that if you would like to measure the particles' spins along the same direction, then finally what you get is the spin of the first particle in the upward direction and the second one in the down position. In other words, the singlet state satisfies the property that if the particles' spins are measured along the same direction, then either the spin of the first particle is observed in an upward direction and the spin of the second particle is downward, or the first one is observed down and the second one is observed up.

Therefore, what we derive from these statements is that both possibilities occur with equal probability. Remarkably, this phenomenon holds even if the measurements are made or conducted at the same instance of time and the quantum particles are arbitrarily put far apart, for instance, one on Earth and the other one on the moon (entanglement principle of quantum physics).

[How to get up and down state experientially](#)

Now, let's discuss one very interesting state known as a mixed quantum state which corresponds to a probabilistic mixture of so-called pure states which can be thought of as the quantum state where one is able to squeeze information about a given quantum system. The different distributions of pure states can generate equivalent physically indistinguishable mixed states, which can be further described as a quantum state, which is a

statistical group of pure states that stand up from pure states. Mixed states can further be understood in detail by using the so-called density matrices; however, this is not required at this juncture of our journey.

Now, we can take a pure state as a recast of a density matrix; this approach enables us to represent a pure state as a subset of the more general mixed states. For example, if the spin of an electron is measured (in any direction) with a Stern–Gerlach experiment, which is one of the experimental arrangements to identify “spin direction by using deflection and magnetic field combination”, it would help us in finding the two possible results: up or down.

A pure state has been represented by something called a two-dimensional complex vector which is denoted as (α, β) , its length is one; mathematically it can be expressed as $|\alpha|^2 + |\beta|^2 = 1$, where $|\alpha|$ and $|\beta|$ are the absolute values of α and β .

A mixed state, in this case, has taken the mathematical structure of a 2×2 matrix which is fundamentally a Hermitian, positive-definite, and can be traced.

Significance of states

Before a particular measurement is performed on a quantum system, the quantum theory usually gives only a probability distribution for the outcome if it is applied correctly. The form that this distribution takes place in is completely determined by the quantum state and the observable fundamentally describing the measurement.

It is important to note here that these probability distributions arise for both mixed states and pure states. However, it is impossible in quantum physics (unlike classical mechanics) to prepare a state where all properties of the system are fixed, deterministic and certain due to the obvious reason of probabilistic aspects like quantum theories and the overall subject itself. This is something that can be easily exemplified especially by the Hilgenberg uncertainty principle. This is probably one of the important differentiators which reflects a core difference between age-old classical or traditional physics and novel quantum physics. According to the quantum theory, if understood holistically together with the border field of quantum

physics, in every observable, some states have an exact and determined value for that observable to exist.

The paradigm of Quantum computing isn't only limited to two states of the fundamental particles. Rather, they encrypt information in the form of quantum bits more specifically known as qubits which reside in the state of superposition. Qubits essentially signify atoms, ions, photons, or electrons and their controlling devices. Qubits and in turn controlling devices are occupied in the composed manner and act as computer memory and a processor.

Summary of essential notions, math, and The Stern–Gerlach experiment in context of Dirac theory

In this summary, some common notations and their uses are provided. Also, some additional facts are given related to Hilbert space and the Stern–Gerlach experiment, as follows:

- The dot ($\bullet$) inside is the symbol of placeholder for tables.
- The bra and ket with the same label are Hermitian conjugates.
- Writing bras, kets and linear operators next to each other signifies Matrix Multiplications. However, the numbers based on their place in an expression are used as operands or operators.
- $\langle u|$ or $|v\rangle$ symbolizes Row and Column vectors, respectively.
- The Bra-ket $\langle u||v\rangle = \langle u|v\rangle = \langle v|u\rangle = z$ represents the product of two vectors that tilt towards each other. For any vector, V : $\langle v|v\rangle \geq 0$; $\langle v|v\rangle = 0$ iff $v = 0$.
- Their direction corresponds to the states of a dynamical system at a particular time. $|u\rangle$ and $|v\rangle$ must belong to the same vector space.
- Linearity in the second argument represents $\langle u|c_1 v_1 + c_2 v_2\rangle = c_1 \langle u|v_1\rangle + c_2 \langle u_2|v\rangle$. In the equation, c_1 and c_2 are complex variables.
- Tensor product of $|\emptyset\rangle$ and $|\phi\rangle$ is represented as $|\emptyset\rangle \times |\phi\rangle = |\emptyset|\phi\rangle = |\emptyset\phi\rangle$
- Hilbert space is about arranging matrices. For instance, it could be used for generating bigger matrices by mending together minor ones

which may be thought of as the tensor product. The Hilbert space is the same thing as an inner product space. Also, it must be noted that what seems like a variable in traditional mechanics may look like an operator in quantum mechanics. For example, in classical physics, momentum is a variable but it is treated as an operator in quantum physics.

- The Stern–Gerlach experiment established that the three-dimensional orientation of angular momentum is quantized. Thus, an atomic-scale system was exposed to have essentially quantum properties. In the original experiment, silver atoms were directed through a spatially varying magnetic field, which when bounced earlier, struck a detector screen which is one of the components of the experimentation. This screen may be a glass slide or some similar arrangement(s). Further, particles with non-zero magnetic moments are bounced, due to something called the magnetic field gradient, from a straight path, not the zig-jag path. A magnetic field gradient is fundamentally a discrepancy in the Magnetic Field (MF) which is always relative to the position. A one-dimensional Magnetic field gradient (MFG) is dissimilar to one direction. Whereas a two-dimensional gradient is the difference between two.

In addition, the screen shows individual buildup points instead of a permanent post-quantized spin distribution. Historically, according to prominent physicists, this experiment was decisive for the realism of angular momentum quantization in all atomic scale systems. The process, which was then initiated by Otto Stern in 1921, was first actively managed by Walther Gerlach in early 1922.

Qubit and entanglement

The word entanglement was made known by Schrödinger in the initial years of quantum mechanics. To get a clue of entanglement in a clearer way, assume a two-qubit system with the qubits characterized 1 and 2 with individual state vectors.

$$|\Psi_1\rangle = \alpha_1|0\rangle + \beta_1|1\rangle \text{ and } |\Psi_2\rangle = \alpha_2|0\rangle + \beta_2|1\rangle.$$

So, any state of the form:

$|\Psi_1\rangle \otimes |\Psi_2\rangle \equiv |\Psi_1\Psi_2\rangle$ in this composite two-qubit system can be written as follows:

$$|\Psi_1\Psi_2\rangle = \alpha_1\alpha_2|00\rangle + \alpha_1\beta_2|01\rangle + \alpha_2\beta_1|10\rangle + \beta_1\beta_2|11\rangle$$

Here, the coefficient of the four base vectors which are $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ is understood as a probability amplitude. For instance, $|\alpha_1\beta_2|^2$ provides the probability of discovery of qubit 1 in state $|0\rangle$ and qubit 2 in state $|1\rangle$ if the two-qubit system is measured. In this two-qubit system, the so-called Bell state is represented by:

$$|\Psi\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Bell state has the extraordinary property that there are no single-qubit states $|a\rangle$ and $|b\rangle$ such that $|\Psi\rangle = |a\rangle \otimes |b\rangle$. A state of a compound system that cannot be depicted as a product of states (factorized) of its constituent systems is named an entangled state. Entanglement is an arrangement of quantum superposition.

Meanwhile, an entangled state cannot be factorized, hence it is not possible to lay down a pure state of its constituent components. Only a group state would be specified. This group state is pure in the Hilbert space. Alternatively, we have information on the correlation among measurement consequences on qubits 1 and 2 but we cannot, in theory, classify a pure state with each of the qubits 1 and 2 separately. Entanglement is a vital variance between classical and quantum physics.

Spin

A spin of a single electron in a magnetic field can either be in the spin-down (low energy) or the spin-up (high energy) state. Relating to a traditional bit, the spin downstate will be the analog to zero and spin up to a one. For graphical representation of spin, refer to [Figure 4.3](#):

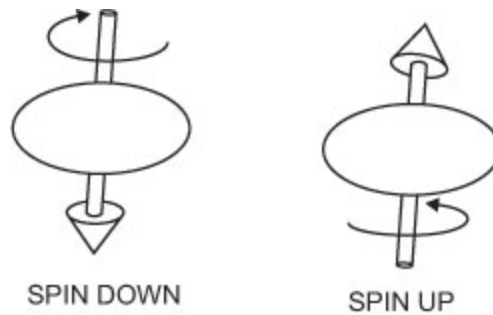


Figure 4.3: Graphical representation for spin refers.

An original electron's spin relays in a state where it spins up and down at the same time. The electron on observation shows differences in its state either an up spin or a down spin with a likelihood for each result. The particle's state is labelled mathematically as wave functions. The wave function expresses likelihoods for dissimilar likely consequences of an observation. The collapse denotes element purpose in one of those states with possibility. Such probabilistic behavior is constructed into quantum mechanics with its results of observation (measurement) capable together with chances. Nature seems to be astonishingly probabilistic and non-deterministic at its genuine level.

The spin of a particle has an actual tangible consequence on what happens when it's physically rotated around another identical particle. When you revolve two particles in a way that they alter location, you find that their quantum wave function is affected.

Without going into too much detail, for particles called fermions, this leads to the "Pauli Exclusion principle" which is responsible for the matter not being allowed to be in the same state (which includes place) at the same time. For all other particles, which are known as bosons, it does not affect at all.

Before going into the details of spin, we need to understand the quantum numbers and energy states in an atom. Let's have a look at them.

Quantum number and energy state

The physical features of the fundamental particles are quantization. We can consider here energy, charge, and angular momentum. Whereas, the values of quantized entities are articulated in terms of quantum numbers. The rules

governing them are of the utmost importance in determining what is its nature and what it does. Further, if we minutely observe the quantization of fundamental particles, then we come to know that the energy states of bound systems are quantized.

For instance, the hydrogen atom, for which the permissible energies are articulated as $E_n \propto 1/n^2$, where $n = 1, 2, 3, \dots$, we describe n to be the principal quantum number that tags the elementary states of a system. The lowest-energy state has $n = 1$, the first excited state has $n=2$, and so on. Thus, the allowable values for the principal quantum number are $n = 1, 2, 3, \dots$. This is more than a numbering arrangement, as the energy of the system, such as the hydrogen atom, can be stated as some function of n , as can other features (such as the orbital radii of the hydrogen atom).

Here we must be very clear about electron spin and the electron spin quantum number. They are frequently treated as one and the same thing as both are quantum numbers. However, they are different, but meticulously connected concepts. For instance, different from the electron's mass and charge, spin has an orientation. The spin's orientation is exposed by the spin quantum number.

Okay, now have a look at the behaviour quantization in the hydrogen atom. The magnitude of angular momentum it quantized was discovered by Bohr, concerning the hydrogen atom. It is now known to be true in a universal way. With the growth of quantum mechanics, it was established that the magnitude of angular momentum L can only have the values, as follows:

$$L = \sqrt{l(l+1)} \frac{h}{2\pi} \quad (l = 0, 1, 2, \dots, n-1)$$

Here, l is demarcated to be the angular momentum quantum number. The law for l in atoms is specified in the brucites. For specified n , the value of l can be any integer from zero up to $n - 1$. For instance, if $n = 4$, at that point, l can be 0, 1, 2, or 3. Refer to [Figure 4.4](#):

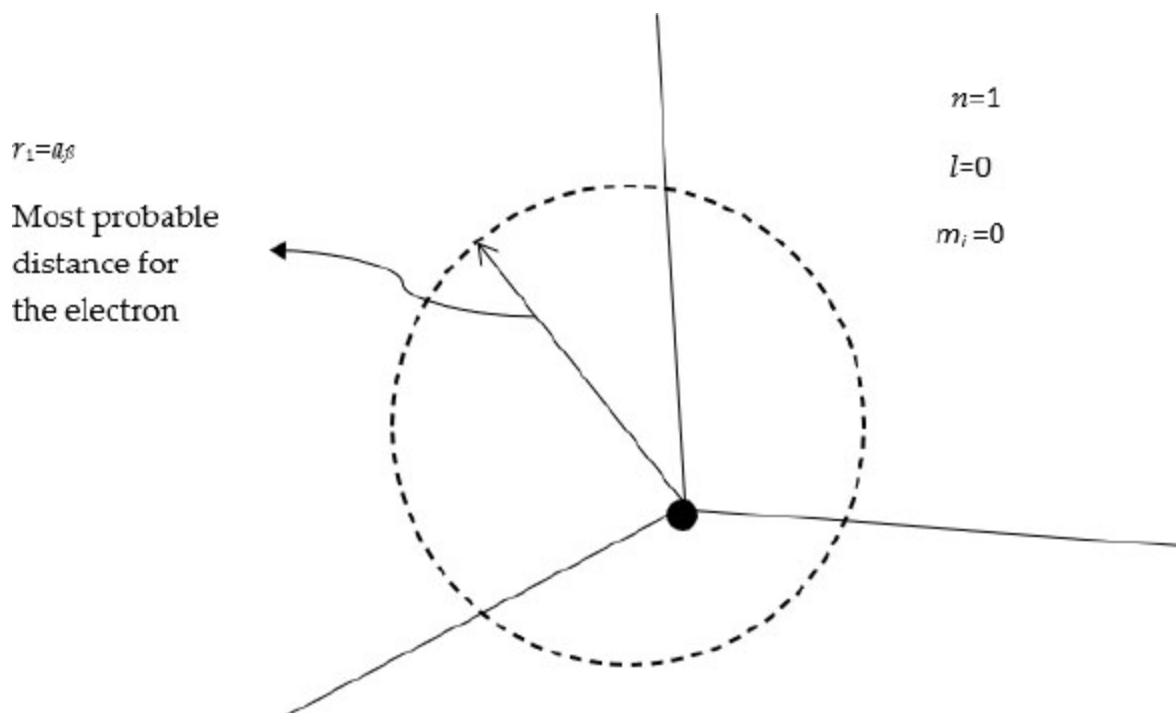


Figure 4.4: The ground state of a hydrogen atom has a likelihood cloud explaining the location of its electron. The likelihood of finding the electron is relative to the darkness of the cloud. The electron can be nearer or farther than the Bohr radius, but it is improbable to be a great distance from the nucleus.

For $n = 1$, l can only be zero. This signifies that the ground-state angular momentum for hydrogen is zero, not $h/2\pi$ as Bohr anticipated. The image of circular orbits is not valid since there would be angular momentum for any circular orbit. A more effective image is the cloud of likelihood given away for the ground state of hydrogen in [Figure 4.4](#). The electron truly devotes time in and around the nucleus. The reason the electron does not continue in the nucleus is connected to Heisenberg's uncertainty principle—the electron's energy would have to be greater to be limited to the minor space of the nucleus. Now the first excited state of hydrogen has $n=2$ so that l can be either 0 or 1, conferring to the rule in:

$$L = \sqrt{l(l+1)} \frac{h}{2\pi}$$

Likewise, for $n=3$, l can be 0, 1, or 2. It is often most suitable to state the value of l , a simple integer, rather than scheming the value of L from:

$$L = \sqrt{l(l+1)} \frac{h}{2\pi}$$

For instance, for $l = 2$:

$$L = \sqrt{2(2+1)} \frac{h}{2\pi} = \sqrt{6} \frac{h}{2\pi} = 0.390h = 2.58 \times 10^{-34} \text{ J.s}$$

An electron has a typical angular momentum which is a magnetic moment that is recognized as spin. In atomic physics, the distinguishing angular momentum of a specific particle is parametrized by spin quantum number (QN). The spin quantum number is the fourth number. The other three are a principal quantum number, azimuthal QN, and magnetic QN. The spin quantum number defines the exclusive quantum state of an electron. This is chosen as 's'. The Spin plays a distinguished role in quantum computing. Also, it plays an important role in defining the features of elementary units like electrons. The direction of spin of the particle controls relatively a few things like a spin quantum number, angular momentum, the degree of freedom, and so on.

Formally, the electron spin is one of the three characteristic properties of the electrons. The others are the mass and charge of the electron. The electron spin is demarcated as the spinning of the electron about its axis. It is articulated as follows:

$$\|S\| = \sqrt{s(s+1)}\hbar$$

Where:

- s is equal to a quantized spin vector.
- The spin vector is articulated as $\|s\|$.
- The spin quantum number (s) is associated with the spin angular momentum and $\hbar$ is Planck's constant.

The spin quantum number can be articulated as $S=n/2$:

- Any non-negative integer can be n .
- The allowable values of the spins are 0, 1/2, 1, 3/2, 2, and so on.

- The inherent angular momentum of Electron is signified by quantum number $\frac{1}{2}$.
- The total angular momentum s is articulated by:

Where:

$$S = \sqrt{n(n+1)}\hbar$$

$$S = \sqrt{\frac{1}{2}\left(\frac{1}{2} + 1\right)}\hbar = \sqrt{\frac{3}{4}}\hbar$$

The reduced Planck's constant is $\hbar$, $\hbar = h/2\pi$.

One additional point about Qubit and electron states

As mentioned, earlier, in the atom model, the electron can exist in either the 'ground' or 'excited' states, which we'll call $|0\rangle$ and $|1\rangle$, individually. By putting light on the atom, with suitable energy and for a proper length of time, it is conceivable to change the electron from the $|0\rangle$ state to the $|1\rangle$ state and vice versa. But more fascinatingly, by reducing the time we shine the light, an electron originally in the state $|0\rangle$ can be moved 'halfway' between $|0\rangle$ and $|1\rangle$, into the $|+\rangle$ state.

Electron spin theory

The electron spin theory defines the electron as a quantum particle in its place of the meek sphere as in the classical theory. The quantum philosophy says that the electron spin direction and its effect depend on the induced properties like magnetic properties of the atom. The interrelation among the magnetic field spin direction is as shown in [Figure 4.5](#):

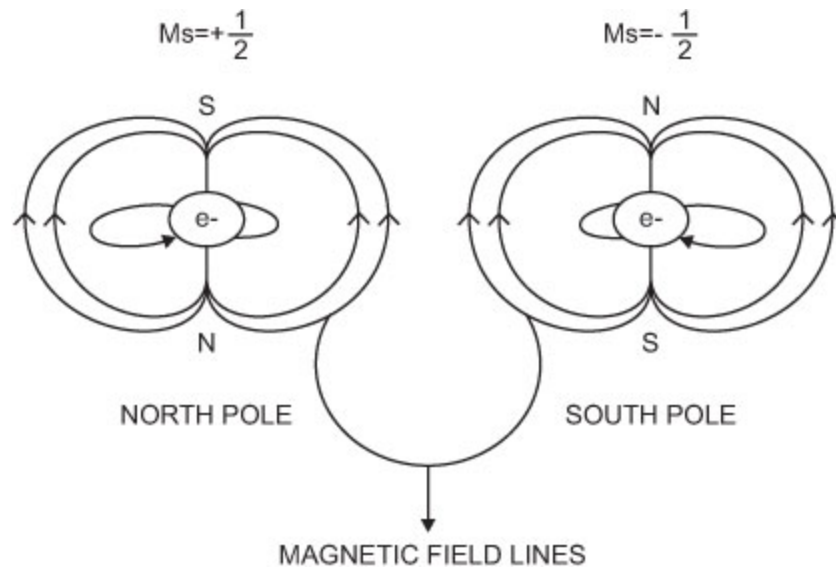


Figure 4.5: *Interrelation among magnetic field spin direction*

The spin up and spin down the path are equal to the spinning in the +z or – z-direction. These spins are associated with the particles that contain spin s equivalent to $1/2$, that is, for electrons.

In quantum theory, the electron is made up of the tiny magnetic bar and its spin points to the north pole of the minute bar. If two neighbouring electrons have a like spin direction, the magnetic field is formed by them which strengthens each other and therefore a robust magnetic field is expanded. If the adjacent electrons have an opposite spin direction, the magnetic field is formed by calling off each other and then no magnetic field exists.

If the nearby electrons have an opposite spin direction, the magnetic field formed by them removes each other and no magnetic field is present.

Observer effect

In the quantum world, just witnessing a quantum system changes it, though quantum physicists don't know why. However, it is established after decades of experimentation, hence this property is actual and correct. Experts aren't speculating it is true; only why or how it is true is their area of investigation. For instance, in each double-slit experiment, when experts place a photon detector to measure which of the two slits a photon goes through, the photon always acts only as a particle and the consequential

wave bands do not occur. Whereas if they turn off the detector, the wavebands come back.

It looks like, nature understands the measurement happening, cares about it, and changes its behavior accordingly. This may not be what occurs or why, but it's how we define what is happening based on our experimental explanations and its consequences. Since we don't have a lot of other ways to communicate what we are seeing, we doesn't know yet what is happening.

Therefore, it led to several different competing explanations. One interpretation says that it's impossible to observe a system deprived of somehow interfering with it. For example, to just observe something often needs light (that is, a photon) or some hypocritically introduced gear to the imprisonment of the consequence, and trappings influence the likeliness to provide quantum results. With the photon example, the photon necessity has hit the object being measured and rebounded to the detector (or our eyeball) for us to notice it, and that hit must cause some sort of interaction.

Another popular interpretation (the Copenhagen interpretation) says that when a quantum wave function of the several likelihoods is measured and observed, the wave function breaks down which is commonly known as wave function collapse into a final state. This means the observation making the resultant collapse is the interference. The act of measuring the quantum situation decreases all the states or answers into a single concluding answer or state.

Measuring it breakdowns all the simultaneous, possible responses into one final, enduring answer which may or may not probabilistically be the correct response and may not have been a similar response if measured or observed otherwise in any way. The Copenhagen interpretation has the major support in the quantum world for elucidation-- why detecting something changes it. Though its characteristic strangeness is why Schrödinger created his cat-in-the-box paradox thought experiment, Schrödinger wanted to point out exactly how counterintuitive the Copenhagen interpretation was to what we previously believed.

Observer effect, no-cloning theorem, and quantum computing

The observer effect has an enormous influence on quantum computing. Because quantum computers are supposed to give us perfect responses for otherwise tremendously hard-to-solve complications, they must be manufactured and function in a way that reduces or uses the observer effect so that we can get accurate replies when we want them.

The no-cloning theorem declares quantum states cannot be directly copied. Because measuring a quantum state changes it from a quantum state to its classical, perpetual state. Likewise, according to the observer effect, just observing or measuring a quantum state changes it. Hence it is not possible to copy the quantum state. This is not to say that “copying” can’t be done, but it is necessary be done meanderingly. The no-cloning theorem has various consequences for quantum computing. On the other side, it means you can’t back up a quantum state in the middle of a quantum computation like you can with a classical computer. Its brand’s copying and error correction are more problematic on quantum computers and networking devices. On the optimistic side, it is a prodigious property for quantum cryptography and stops several inquisitive scenarios that are distantly relaxed in the classical world.

From classical bit to Qubit

In quantum computers, a quantum byte is composed of eight qubits. However, a qubit that can be both a 0 or a 1, or a 0 and a 1, depends on the state of the particle involved in the processing. This is owing to the wave function of a quantum particle and its characteristic set of likelihoods. A common metaphor is to liken a qubit to a coin that is being used to choose which side kicks off a football game.

As the referee describes the coin for heads and tails sides of the coin, they request the individual calling the coin to call the outcome as heads if they see the heads side of the coin or to call the outcome as tails if they see the tails side of the coin. But after the referee handsprings the coin into the air, in place of it flatly mooring on a heads or tails side, it lands precisely on its thin edge and stays there spinning for a few seconds. There is a potential,

earlier the coin is called (that is, measured), for the calling person to see all three sides before it rests to one smooth side.

The coin's decisive final answer when it finally falls might be heads or tails, but at the instant of its edge-spinning, it could be read as both at the same time (that is, superposition). This is not a perfect allegory, but it does link the clue of superposition against the final, measured response.

The classical computer is based on the concept of high and low voltage for processing, then passes them through a sequence of gates called logic gates which can manipulate the data to get the result. Logic gates like AND, OR, NOT, XOR, and so on can be organized in different ways to process the bits and produce the output.

It can perform very basic operations like addition or subtraction or even help solve complicated encryption problems. Logical gates are practically achieved by using transistors, which nowadays hinge on the possessions of silicon semiconductors to perform an operation instead of using mechanical switches (for details, refer to the previous chapter).

Due to the different nature of qubits, calculations on a quantum computer vary essentially from those on a traditional computer. Different from a conservative logic gate, for instance, a quantum gate (or quantum logical gate) is not a technical construction unit that translates distinct bits into one another in a well-defined way, but somewhat defines an elementary physical manipulation of one or more (entangled) qubits. Scientifically, a specified quantum gate can be labelled by a conforming unitary matrix that function on the states of the qubit ensemble (the quantum register). How precisely such a procedure and the movement of information will look in each case is contingent on the physical nature of the qubits.

As we know, a single qubit can be in two states at once (that is, 0 and 1), and a 2-qubit system can represent four states at once (that is, 00 and 01 and 10 and 11). Whereas, the 3-qubit system can represent eight states at once (that is, 000 and 001 and 010 and 100 and 010 and 011 and 110 and 111), and so on.

[Qubit detailed](#)

A qubit has some resemblances to the usual bit such as it can be measured to either zero or one. States of a qubit are denoted using ket 0 and ket 1 and its representation is written as $|0\rangle$ and $|1\rangle$ and the basic measured state is similar to classical 0 and 1. A qubit can be an electron with spin, a photon with polarisation, contamination spins, trapped ions, neutral atom, semiconducting circuits, and so on (details are in a later chapter). The superposition of fundamental levels of the qubit can be signified by using the following Bloch sphere. Bloch sphere gives a valuable means of envisaging the state of a single qubit and frequently aids as an outstanding testbed for thoughts about quantum computation and quantum information, as shown in [Figure 4.6](#):

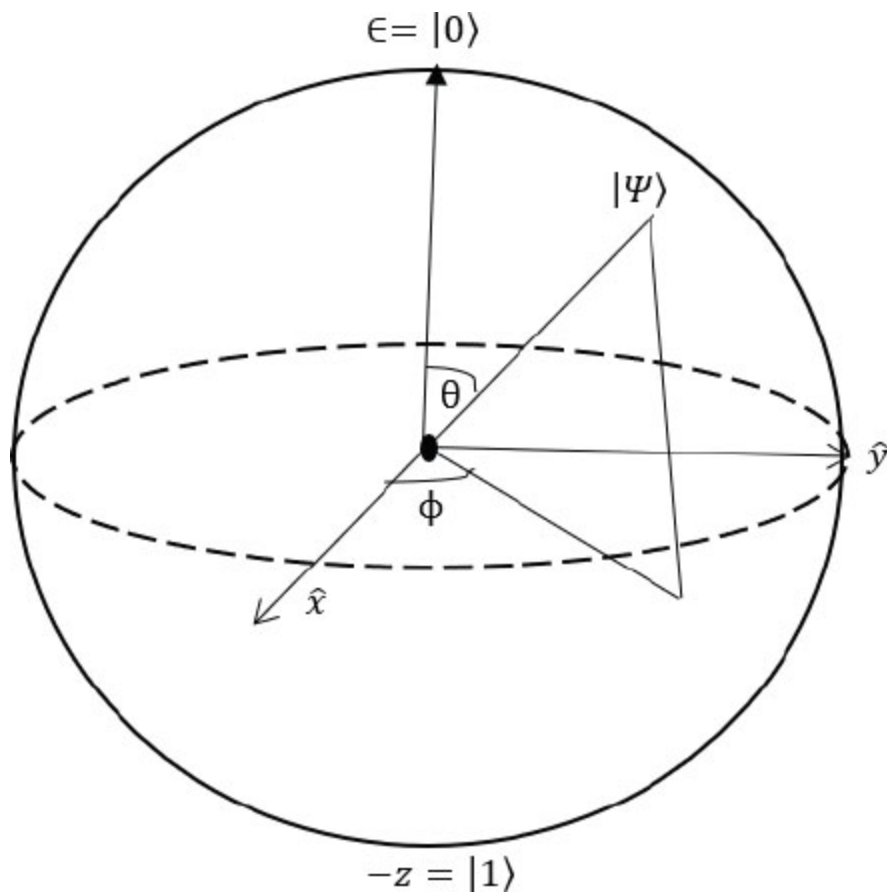


Figure 4.6: The superposition of fundamental levels of the qubit can be represented by using the Bloch sphere

It may appear a bit complex but the key point that we need to retain in mind is that the solitary qubit at any time can be in a superposition of $|0\rangle$ and $|1\rangle$ and it can be expressed as $a|0\rangle + b|1\rangle$ where a and b are amplitude

(proportional to probabilities) of qubit being measured to 0 and 1 correspondingly and $a^2 + b^2 = 1$.

The 0 and 1 classical states are stated using ket and bra as $|0\rangle$ and $|1\rangle$. Whereas the complete state of the qubit can be inscribed as

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. Here α and β are complex coefficients that satisfy the normalization condition. $|\alpha|^2 + |\beta|^2 = 1$.

The qubit has a hundred percent chance that it will be found in either of these two states at a particular time, and 0% chance that it will be measured to be in any other state at that time.

Now, any unitary transformation we do on $|\psi\rangle$ can be visualized as just a moving point around the surface of the sphere. For instance, if the state were $|\psi\rangle = |0\rangle$, the point would take a seat on the z-axis at the position marked $|0\rangle$. However, this visualization can alone be used for single-qubit states, as there is no known (simple) generalization that applies to multi-qubit systems. Superposition is the wonder where a quantum system occurs as a probabilistic distribution of states. A single qubit can exist in a

superposed state and is mathematically represented as $\frac{|0\rangle + |1\rangle}{\sqrt{2}}$.

Entanglement needs two or more qubits in this one particle. Whereas one can mark the state of others. The entanglement happens irrespective of distance or physical parting from one another. And this could be quicker than the speed at which light communicates.

One common example of an entangled state is known as Bell State, which is explained as $|\phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$.

Any two-dimensional column vector of real or complex numbers with standard 1 signifies a likely quantum state held by a qubit. Therefore, $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ characterizes a qubit state, if α and β are complex numbers satisfying $|\alpha|^2 + |\beta|^2 = 1$.

$|\beta|^2 = 1$. Some samples of effective quantum state vectors representing qubits comprise:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \text{ and } \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

Quick Facts

- As we know a qubit can exist in the superposition of two states, and when it is measured it will do the repetition in one of the two states founded on the likelihoods of each state. Therefore, in simple words, we can understand that measuring a qubit itself has a consequence on the system. Thus, the measurement of a qubit can be visualized as the analogous to a gate in the traditional computation or standard electronic systems that marks the state of the qubit.
- On similar lines, two, as we discussed earlier as well that the qubit states are 00, 01, 10, 11 but the qubits can be in a superposition of all four states at the same time. Therefore, this should be represented as $a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$.
- Hence, it is natural that to represent two qubits, we need 4 probabilities/amplitude (a, b, c, d); if we have three qubits, we will need 8. So, if we have n qubits, we will need 2^n numbers to represent the overall state of that quantum system. So, with the insignificant upsurge in the numeral of qubits, we will be capable to produce systems that can signify huge states and this is different from classical computers.

A significant point to comprehend about qubits is that any similar number of bits in a binary system has an analogous number of likely states but is only one state at any one time. A binary coin cannot ever land on the edge, so to speak. Thus, a 3-bit system can come back only one state/answer at one time. A 3-qubit system can have all 8 states all at the same time tangled in solving a problem. A 3-qubit system is an 8-fold state augmentation over a 3-bit system.

A 4-qubit system is a 16-fold state augmentation over a 4-bit system, and so on. Now imagine you have thousands of qubits, each encompassing a concurrent two-state system, and because of superposition, they are all in the possible states of thousands of qubits at the same time.

What one needs to understand at this point is that the volume of information is compulsory to define the superposition when it grows exponentially with several qubits. However, it may happen that we are not able to access all this information because of the ultimate boundaries of quantum measurement. For instance, the spin of an electron in superposition can be in all directions but when we measure, it is in one direction, either up or down. Also, we will not be able to predict the outcome, because it is probabilistic and depend on the probability associated with the state (half times up and half times down).

That means, to use the bursting potential of a quantum computer, we need to grow quantum algorithms that discover the being of the enormous amount of information stored in the superposition of qubits. And, at the end of the computation, dispense the system in one of the base states which we can detect with inevitability.

Here, one comparison helps us to decode the bit and qubit-based processing and speed requirements. To understand the quantum of speed comparison between bits and qubits, we can take a wheat grain/chessboard example.

Imagine that we want to put a wheat grain in every square of a chessboard in such a way that one grain would be placed on the first square, two on the second, four on the third, and so on. This is nothing but doubling the number of grains on each following square. Now assume a hypothetical question-- how many grains of wheat would be required on the chessboard at the end (when all the squares are filled with grains as per the previously mentioned logic).

The problem may be solved using simple addition. With 64 squares on a chessboard, if the number of grains doubles on consecutive squares, then the number of grains on all 64 squares is: $1 + 2 + 4 + 8 + \dots$ and so forth for the 64 squares. The whole number of grains equals 18,446,744,073,709,551,615, a much huge quantity than most expect.

The same logic can be thought of when quantum computation happens. However, this peculiarity gives the edge to quantum computation because if a traditional bit-based computation would be done over, it will take a long time to calculate because of its sequential nature of computation.

But as for the preceding problem, the conventional system would take a long time whereas a quantum computer, with a few hundreds of qubits, can produce the correct response in under sixty seconds. That's the power of quantum computers. because it will not do the computation sequentially but parallelly.

Qubits are conceivable since the base objects being used by a quantum computer are the quantum states of quantum particles. A traditional binary computer may use electrons (or even photons), but they are shaped, manipulated, and measured as binary objects or states.

They are either 0 or 1, that is, either on or off. A quantum computer, utilizing quantum particles, measure those particles' quantum states. Thus, when measuring an electron's state, it sees all likely states of all thinkable quantum properties. When measuring a particle, it sees all probable states, like charge, spin, polarity, and so forth. The base state of a true quantum computer is that it can epitomize all likely particle states at once, not just a binary state.

In a quantum computer, the logical comparative unit is known as a quantum logic gate (or quantum gate). Quantum gates are characteristically capable of more options and complex problems than classical gates. Even though superposition is most likely states, at the same time, this doesn't mean quantum computers can demonstrate all conceivable answers for all possible problems promptly.

All computers, as well as quantum computers, still must compute and run programs to resolve complex problems. They don't just get each answer instantly. Some essential computations and algorithms must be tailed to get the final answer. But it is precise to say that because of the qubit superposition, quantum computers are probable to resolve numerous types of problems far quicker than traditional computers, and in some cases, the solutions will be provided so quickly as compared to traditional binary computers that it appears instantaneous.

Another important aspect is how much information is signified by a qubit? Typically, there is an immeasurable number of points on the unit sphere, so that in theory, one could store a complete text of civilization in the infinite binary expansion of θ (Bloch sphere), though, this deduction turns out to be ambiguous, because of the behaviour of a qubit when observed.

The measurement of a qubit will give only either 0 or 1. Additionally, the measurement vicissitudes the state of a qubit, collapsing it from its superposition of $|0\rangle$ and $|1\rangle$ to the precise state consistent with the measurement result. For instance, if the measurement of $|+\rangle$ gives 0, then the post-measurement state of the qubit will be $|0\rangle$. Why does this type of collapse happen? Nobody knows. This behaviour is just one of the important postulates of quantum mechanics. What is applicable for our purposes is that from a single measurement, one gets only a single bit of information about the state of the qubit, thus determining the seeming paradox. It turns out that only if infinitely many identically ready qubits were measured, would one be talented to regulate α and β for a qubit in the state given in Equation $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. But an even more thought-provoking query-- how much information is signified by a qubit if we do not measure it?

This is a hoax question, since how can one count information if it cannot be measured? Yet, this is rather conceptually significant here, because when Nature evolves a closed quantum system of qubits, not executing any 'measurements', she seemingly does retain track of all the unremitting variables describing the state, like α and β . Therefore, in the state of a qubit, Nature hides a countless deal of 'hidden information'. And even more fascinatingly, the potential quantity of this additional 'information' raises exponentially with the number of qubits.

So how is the Qubit created?

It is essential to produce qubits at a tiny scale in an extremely controlled style so that superposition and entanglement can happen. It can be applied in algorithms until the calculated qubits can be readout. The precondition for near-complete isolation of qubits and quantum circuitry (called quantum gates) point toward extreme manufacturing and immense challenges.

At this juncture, taking a few examples of evolving technologies will be useful. Trapped atomic ions (such as Ca^+) steadied, utilizing Coulomb Force (An electrostatic force or the Coulombic force is defined as the force of attraction or repulsion between two like and unlike charges, respectively.) among them can function as qubits. RF electric field (Radiofrequency electromagnetic fields are a type of non-ionizing electromagnetic radiation found on the electromagnetic spectrum covering the range of frequencies below 300 GHz) limits these ions. Lasers, or electrodes driving microwaves, are used to operate the qubits.

Few additional methods attempt to use a version of traditional silicon engineering processes to produce qubits. For instance, qubits could be also generated by manipulating electrons in a superconducting electronic circuit, with the necessary criteria of keeping the system close to absolute zero (degree Kelvin). This is desired to attain superconductivity. And therefore, the qubit produced by this method is called a superconducting qubit.

Companies like IBM and Google are focusing on superconducting qubits-based QC. The primary reason for adopting superconducting qubits is because they are easier to style, tune, and interrelate. The fact that superconducting qubits can be interconnected without much difficulty, it permits for opportune schemes for quantum error correction, which is vital for superior quantum computers. We will discuss the types of quantum computers in the forthcoming chapters in detail.

Also, spin qubits-based quantum computers are a good option to realize quantum. There may be multiple ways to achieve robust qubits and hence QC. But for any type of QC, the quantum circuit is necessary.

Caution!!! More Qubit is not equal to faster computation

It is vital to spot that just having extra qubits does not directly associate with quicker and improved quantum computers. However, having more qubits is good. Quantum computers' aptitude to resolve something is resolute by additional variables than just the pure number of qubits. Some quantum computers, no matter how many qubits they have, cannot solve specific types of problems.

The reason behind this is QC are special-purpose computers, not general-purpose (at least for now). They come in different variants and are based on different technologies. Each major type of quantum computer is maximized to solve particular types of problems. And quantum computers intended to solve a wide variety of glitches won't as rapidly solve particular types of problems that certain kinds of quantum computers emphasize on to the individuality of other types.

A higher number of qubits has the dormant to style a specific quantum computer quicker, but it doesn't promise it, just as advanced horsepower doesn't always mean that a car will compete in a race. A race car's victory is contingent on the whole shebang that goes into creating the car moving forward quicker than the other cars, together with the engine, fuel inoculation, exhausts, transmission, and torque conversion.

Systematic explanation of qubits, quantum gates, and circuits

A quantum computer, is to be expected to have two qubits with reverse values, but the value of distinct qubits is anonymous till we measure. This is thinkable because of quantum entanglement that is quantum phenomenon. To understand it in a better way, assume we have two electron qubits which have zero states, then we put the first electron/qubit into superposition by spreading an electromagnetic wave on to them. The spreading of an electromagnetic wave would be put on with a certain frequency which is proportionate to the energy alteration among zero and one state.

When we put effort to control the second electron/qubit by putting on an electromagnetic wave of frequency which was mandatory when the primary electron/qubit is in zero states, the first electron/qubit in superposition will have a significance in the spin of the second qubit, and the second one will change to a superposition (not a stable state). Therefore, we can infer that the state of these two qubits will be entangled. Further, it can be proved practically as well. It is significant to consider that if we measure the first state and get up-spin, then the other will natively provide down-spin. Remember happens in vice versa (natural property of entanglement) way.

This entanglement will be persisted at any extended distance. Till we measure them, the two qubits can only be well thought-out as a single system with likely values. This is a unique property that could be realized in the working behaviour of quantum computers and it is an impossible process to achieve in a classical computer, to have two bits with no value but the opposite values.

The upsurge in the number of qubits will exponentially surge the number of likely entangled states. One key point to notice is that realizing quantum entanglement is not easy and it is very fragile (due to the occurrence of quantum decoherence) as well and any system that uses this should have a very minimum outside interference.

How to measure Qubit

A measurement resembles the casual knowledge of looking at a qubit, which directly breaks down the quantum state into one of the two traditional states. When a qubit given by the quantum state vector is measured, we get the result 0 with likelihood $|\alpha|^2$ and the result 11 with probability $|\beta|^2$.

Note that these likelihoods sum up to 11 because of the normalization state $|\alpha|^2 + |\beta|^2 = 1$. Negating a vector is equivalent to $\alpha \rightarrow -\alpha$ and $\beta \rightarrow -\beta$. Since the probability of measuring 00 and 11 rests on the magnitude squared of the terms, introducing such signs does not vary of the likelihoods of any kind. Such phases are usually named global phases and more generally can be of the arrangement of rather than just ± 1 .

This can be represented in 3D utilizing the Bloch sphere depiction. The Bloch sphere gives a method of recitation of a single-qubit quantum state (which is a two-dimensional complex vector) as a three-dimensional real-valued vector. This is vital since it permits us to envisage single-qubit states and thus develop reasoning that can be vital in understanding multi-qubit states. But in multi-qubit states, the Bloch sphere depiction breaks down. The Bloch sphere can be envisioned in [Figure 4.7](#):

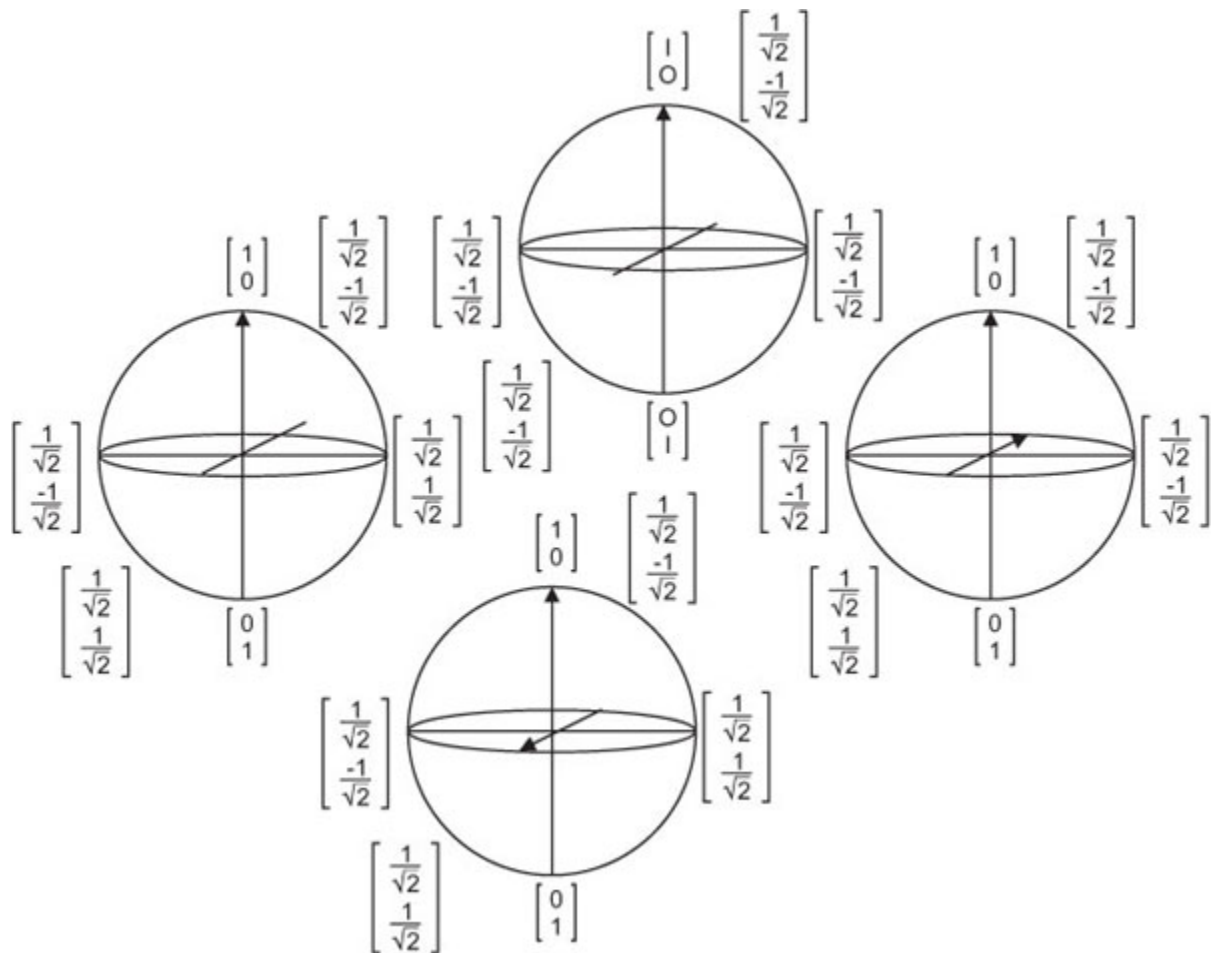


Figure 4.7: The Bloch sphere representation of a single-qubit quantum state

The darts in this drawing show how the quantum state vector is directing and each alteration of the dart can be assumed as a rotation about one of the cardinal axes. While being rational about quantum computation as a sequence of rotations is a powerful perception, it is challenging to usage this intuition to design and label algorithms.

Quantum gates

David Deutsch bids a similarity between Computation and Physics. Let's see what he explained – we all know that one of the important fundamentals of any type of computation is a system which does the calculation. Further, for explaining, he considered a system called the computer that does computation on a provided input.

Now, by putting on a set of instructions or rules to the Input, it gives the output. Likewise, in physics, a physical system is a portion of nature that experiences a motion from a primary state. Then, followed by the application of the laws of motion on the Initial state, it takes it (system) to the final state. As experimentation and measurement are forms of motion, they require a system to trial upon and a measuring tool. Thereafter, the system may be in an initial state, or we formulate it in the initial state. Therefore, the measuring instrument will measure the outcome, which is the final state. [Table 4.2](#) portrays this similarity:

Computation	Physics
Computer	Physical system
Computation	Motion
Input	Initial state
Rules	Law of motion
Output	Final state

Table 4.2: *Measuring instrument—parallel among computation and physics*

For numerous reasons like interactions with external systems, it is problematic to uphold a system of qubits in an organized state for long before they ‘decohere’ because of unwelcome associations developing among the system and its unidentified and uncontrolled environments. When a quantum system decoheres, the quantum bits trail a statistical delivery when measured of 0 or 1 rather than a quantum delivery. Once decohered, no quantum operations can be used to re-cohere the state. The time taken for our system to decohere must thus be much lengthier than the time needed for gate operations.

A universal set comprises all the gates desirable to do any quantum computation. At a minimum, we must be able to move single qubits to all locations on the Bloch Sphere by using single-qubit gates, as well as present entanglement in the system. However, for this, a multi-qubit gate would be required. For instance, the Hadamard, phase, CNOT, and gates form a universal set (we would lean on it in a while), from which any quantum computation on any number of qubits can be generated.

From quantum gate to quantum circuit

Let's first understand the circuit and then we move further in understanding the role of the quantum circuit. In the mechanical world, a circuit is a fully enclosed circle through which electricity travels. A current source, conductors, and a load make up a straightforward circuit. In a broad sense, the word "circuit" can refer to any permanent path through which electricity, data, or a signal can pass. Electrons leave the power source, move along conductors, pass through a load to do work, and then return to the power source in an electronic circuit. Because of the electrons' round direction of travel, this system is known as a circuit. Ohm's Law explains the connection between electrical flow and load (is the mathematical relationship between electric current, resistance and voltage). The electrons in a circuit move from the negative side of the power supply to the positive side. Whereas a quantum circuit is a type of computational procedure that combines simultaneous real-time classical computation with coherent quantum operations on quantum data like qubits. It consists of a series of ordered quantum gates, measurements, and resets that can all conditionally use information from real-time classical computation. Any unitary transformation of the quantum data can be efficiently approximated arbitrarily well as a series of gates in a set of quantum gates, which is considered to be universal. A series of conventional computing and quantum circuits can be used to describe any quantum programme. The elementary elements of quantum circuits are quantum logic gates. As we know, fundamentally the quantum logic gates are thought to be like logic gates of traditional computers. However, unlike traditional logic gates, quantum logic gates could be reversible.

Why is this so? Because quantum mechanics requires an arrangement in quantum system which enable them to not ever drop information against time and it is not possible every time to reconstruct the past.

Traditionally, only NOT classic gates will make use of reversibility. Fundamentally, reversibility is not possible for the gates like AND gate because if we see both 1 AND 0, 0 AND 1, they always provide the output as 0. Whereas while we would take NOT under consideration, if we provide 0, we get 1, and when 1 is supplied, the output would be 0. Likewise, this is

reversible; if the output is zero, then the input is one, if the output is one, the input is zero. This type of gate is recognized by the term Pauli-X gate in the quantum world. It maps $|0\rangle$ to $|1\rangle$ and $|1\rangle$ to $|0\rangle$. Hadamard gate also acts on a single qubit and creates a superposition.

The swap gate swaps two qubits. That is $|10\rangle$ to $|01\rangle$ and so on. Controlled gates act on 2 or additional qubits, wherever one or more qubits turn as a control for some act. For instance, the controlled-NOT gate (or CNOT) turns on 2 qubits, and does the NOT operation on the second qubit only when the first qubit is $|1\rangle$, or else leaves it unchanged.

It is not sufficient to only generate qubits, it is also required to work on Computational Logic to achieve results.

For doing the quantum computations, quantum circuit is required and on the similar lines quantum register become very important. So, it is high time to look on these concepts.

Details of quantum circuits

Quantum computation employing a quantum logic circuit archetypally ends with a measurement, which crops the output into a classical state. Also, quantum gates and circuits as mentioned are reversible. A gate that abolishes information (like the AND gate) is equivalent to a quantum measurement, which reasons a wave function to collapse.

There is no annihilation or loss of information in the quantum world. In a quantum computing system of logic gates, one would not, overall, continue with a measurement till the data has conceded through various gates of the circuit and we are equipped to detect the result. Thus, quantum gates need to be reversible.

This further suggests that all such gates would have an equivalent quantity of outputs against inputs, which raises the query of how one can contrivance as an AND gate.

The answer is to have an extra complex logic gate with an equivalent number of inputs and outputs, and one of the outputs delivers the compulsory AND operation. The unwelcome outputs can be using clever tricks. Quantum computing circuits, which are reversible, trail from a belief

of quantum mechanics known as unitary evolution. Though, all conventional computing logic can also be applied using quantum logic gates and circuits.

Non-classic logic circuits can provision superposition and entanglement. Hadamard gates can be used to produce a superposition of values of one or more qubits. CNOT and Hadamard's gates are combined to form entangled qubits or quantum registers. These methods are widely used in Shor's algorithm for prime factorization.

Given that the outcome of every quantum measurement is probabilistic, it may be necessary to ascertain the likelihood of the measured event(s) in order to conclude the computation by iterating the calculation several times and making a conclusion. However, there are also quantum algorithms whose output is entirely deterministic. In these circumstances, other superpositions call one another out, resulting in an outcome with a probability of 1.

Before going into further details of gates and circuits, it is necessary to understand the quantum register. Let's quickly go through it.

Quantum register

A quantum computer at least for the moment has both classical and quantum parts. The quantum program is completely classical, kept in a classical controller. A quantum algorithm works in cooperation with classical and quantum data. We can visualize the complete system to treat the quantum portion as a coprocessor to a traditional system. Like the GPU (graphics processing unit) in a desktop or laptop computer, a quantum computer is used to perform precise tasks in the setting of a bigger program.

The quantum register is a system including numerous qubits. It is the quantum analog of the classical processor register. Quantum computers perform calculations by operating qubits inside a quantum register.

There's a theoretical variance among the quantum and classical register. A classical register refers to an array of flip flops. Whereas a n size quantum register is collection of qubits. Moreover, while an n size classical register is able to store a single value of the possibilities spanned by classical pure bits, a quantum register is able to store all. An n size classical register refers

to an array of n flip flop(Flip-flop is a 1 bit memory cell which can be used for storing the digital data. To increase the storage capacity in terms of number of bits, we have to use a group of flip-flop. Such a group of flip-flop is known as a Register. The n-bit register will consist of n number of flip-flop and it is capable of storing an n-bit word.). An n size quantum register is just a group of n qubit. Furthermore, though an n size traditional register can hold a single value of the 2^n likelihoods reached over by n traditional pure bits, a quantum register can stock all 2^n likelihoods spanned by quantum pure qubits in a similar time. For instance, think of a 2-bit-wide register. A traditional register can stock only one of the likely values signified by 2 bits 00,01,10,11 accordingly. Whereas, if we deliberate 2 pure qubits in superpositions, it can be represented as:

$$|a_0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \text{ and } |a_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Using the quantum register description would be as below:

$$|a\rangle = |a_0\rangle \otimes |a_1\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$$

Further, a pictorial representation of the quantum register is shown in [Figure 4.8](#):

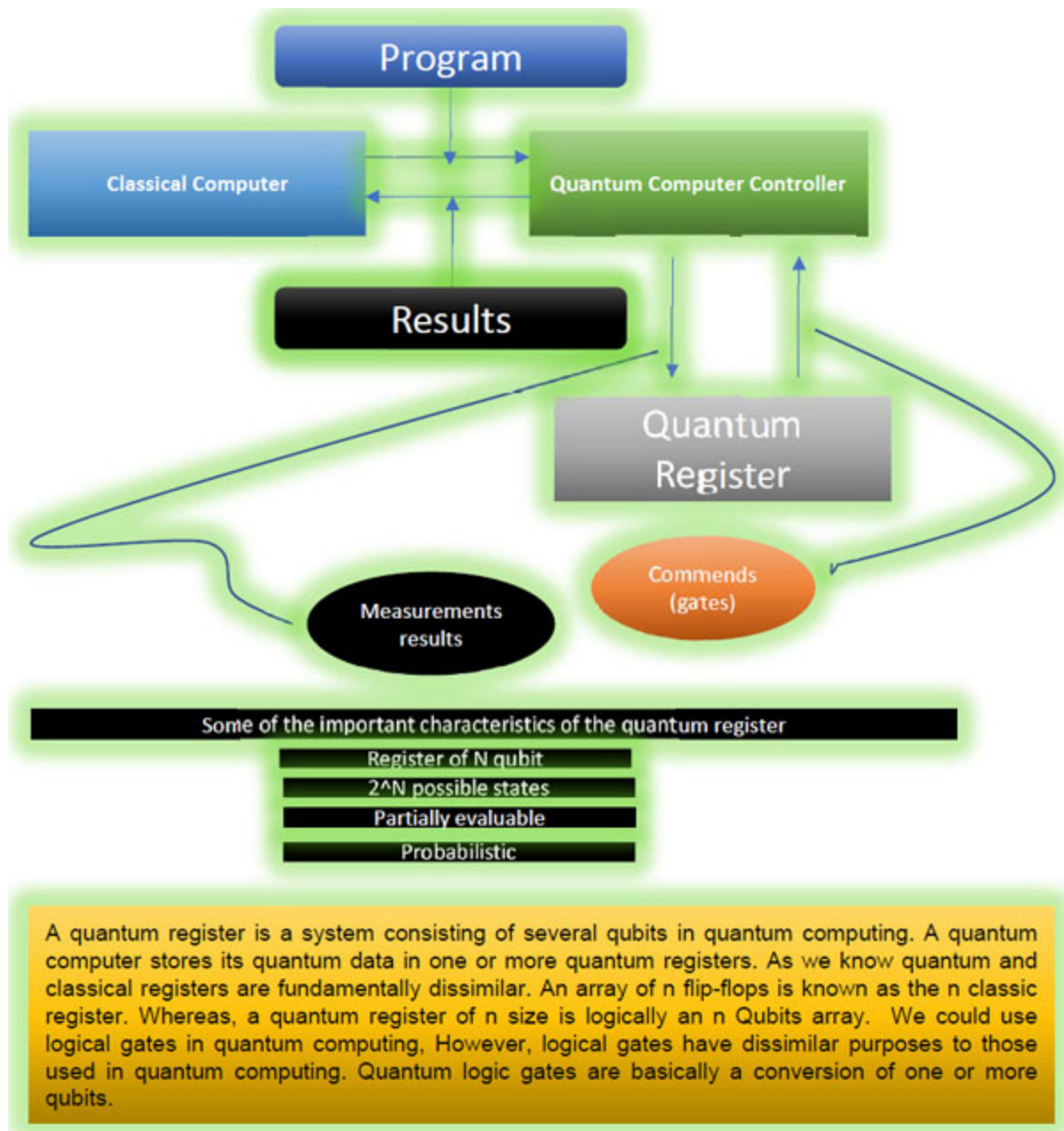


Figure 4.8: The basic model for the place quantum computer register

Quantum register: working

The quantum register affects our quantum data for the algorithm. For most kinds of quantum hardware, the quantum actions are performed on the quantum data where the register sits. All exciting quantum algorithms need more than one variable. The quantum register can be made to hold quite a

lot of data values by subdividing it into quite a few registers or sub-registers.

Classical computers usually have numerous kinds of memory, such as registers for holding small amounts of data that are being vigorously used, and main memory (RAM) for greater amounts of data being kept around for later use. Most future quantum computer architectures do not mark a difference between the register(s) and a more inactive memory comparable to a classical computer's RAM.

The traditional controller grips the program and instructs the devices controlling each qubit on what to do at each step. Reliant on the kind of hardware, the specific devices may regulate a voltage, produce a spurt of microwave radiation, or aim a you will find the brife description of it in the end of the chapter for quick reference.

These very low-level particulars will be unseen by the system so that programmers can grow applications using the CNOT, Hadamard, and other gates. The sequence of gates is approximately static; sometimes in programs, we must put on a gate based on the result of some measurement. This occurs when we are performing error correction, and because some gates efficiently perform different operations, depending on the outcome of a measurement.

[Inputs and outputs of quantum circuits: representations, logic, and theory](#)

It might at first seem sensible that quantum circuits could have more, or less, outputs than inputs overall. This is incredible, though, all quantum operations, save measurement, are unitary and henceforth reversible. If they did not have the same number of outputs as inputs, they would not be reversible and hence not unitary, which is a flaw. For this reason, any box haggard in a circuit diagram must have the same number of wires incoming from it. Multi-qubit circuit diagrams follow similar pacts to single-qubit ones. As an instructive example, we can describe a two-qubit unitary process B to be $(HS \otimes X)$ and express the circuit homogeneously as represented in [Figure 4.9a](#).

We can also view B as consuming an act on a single two-qubit register rather than two one-qubit registers dependent on the background in which the circuit is used. Maybe the greatest valuable property of abstract circuit diagrams is that they permit complex quantum algorithms to be labeled at a high level deprived of having to compile them down to essential gates. This means that you can get a perception about the data flow for a big quantum algorithm without demanding to comprehend all the particulars of how each of the subroutines within the algorithm work.

Controlled gates

The other concept that is constructed into multi-qubit quantum circuit diagrams is about control. The action of a quantum singly controlled gate denoted $\Lambda(G)$, where a single qubit's worth controls the request of G , can be understood by observing the following example of a product state input

$$\Lambda(G)(\alpha |0\rangle + \beta |1\rangle) |\psi\rangle = \alpha |0\rangle |\psi\rangle + \beta |1\rangle G |\psi\rangle$$

That is to say, the controlled gate put on G to the register covering ψ , if and only if the control qubit takes the value 1. Overall, we label such controlled processes in circuit diagrams, as shown in [Figure 4.9b](#).

Here, the black circle means the quantum bit on which the gate is controlled and a perpendicular wire means the unitary that is functional when the control qubit takes the value 1. For the distinct cases where $G=X$ and $G=Z$, we make known to the subsequent notation to define the controlled version of the gates that can be diagrammatically scratched in [Figure 4.9c](#).

Measurement operator

The outstanding procedure to visualize in circuit diagrams is measurement. Measurement taking a qubit registers measures it and outputs the outcome as classical information. A measurement operation is denoted by a meter symbol and always takes as input a qubit register denoted by a solid line and outputs classical information denoted by a double line. See [Figure 4.9d](#). Similarly, the subcircuit could be visualized as [Figure 4.9e](#).

It signifies a classically controlled gate, where G is functionally accustomed to the classical control bit being value 11. Refer to [Figure 4.9\(a, b, c, d, e\)](#):

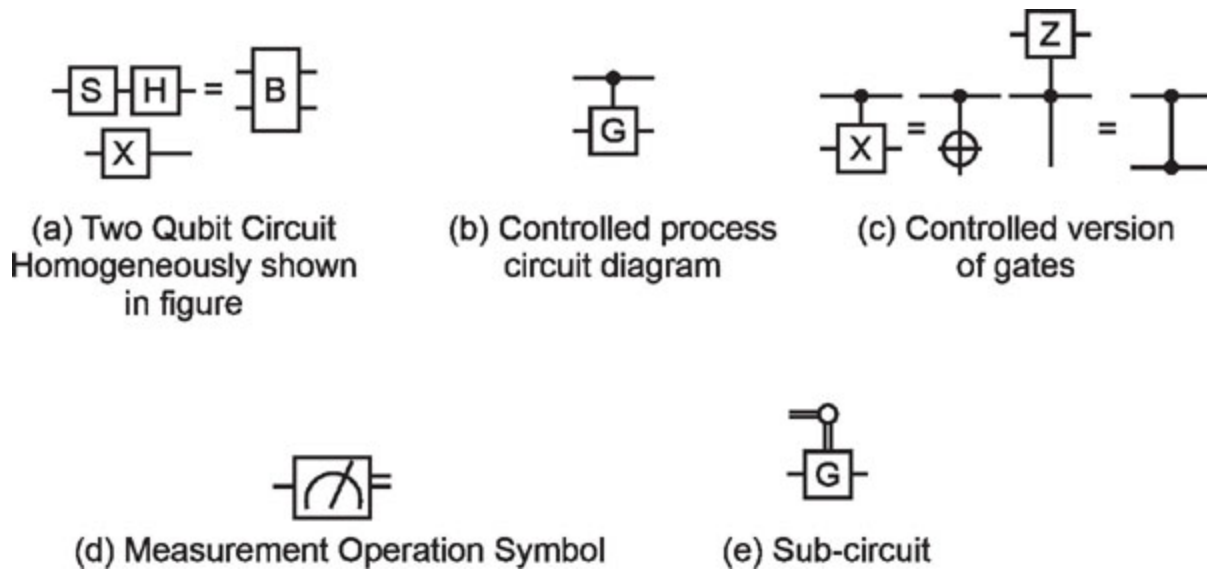


Figure 4.9: Representation of circuits, gates, measurement, and sub-circuits, including control one

A summary of the key symbols

To understand the quantum logic gates, these symbols are important to understand. We must understand some notations and symbols presented earlier because they play critical roles in the overall representation of quantum gates and circuits. The following are the few important symbolic depiction tools (s) related to quantum logic gates or circuit diagrams, as shown in [Figure 4.10](#):

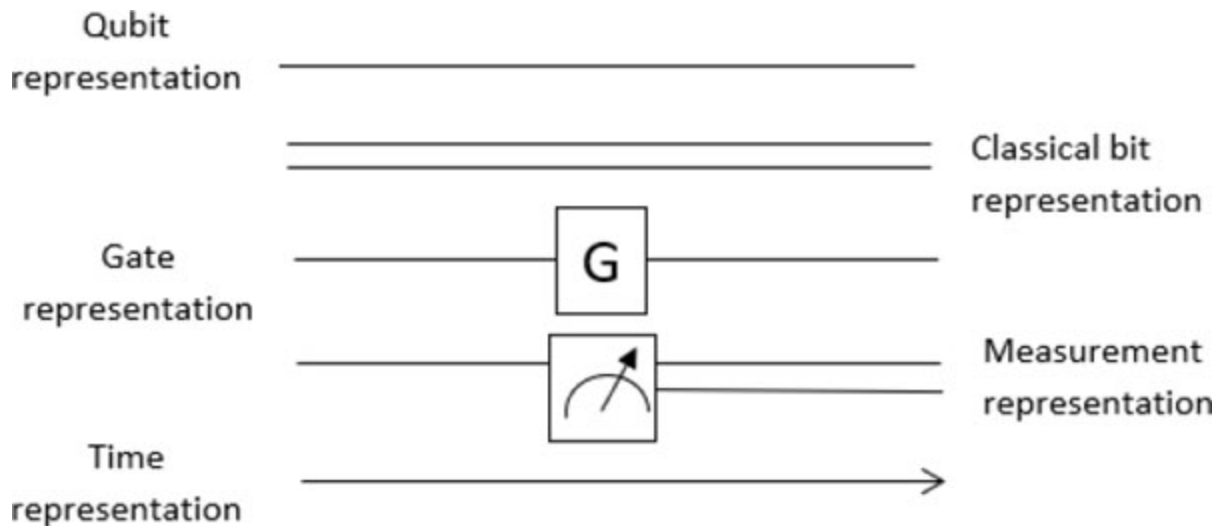


Figure 4.10: Depiction tool(s) related to quantum logic gates or circuit diagrams

These are important for completing the cycle of understanding, as follows:

- In the circuit diagram, the movements happen from left to right.
- Qubit is represented by a single, horizontal line.
- Generally, gates are signified by boxes, the letters/symbols inside give us information about what kind of gate it is. The exceptions to this rule generally arise because there's already a classical circuit element for the gate in question (e.g., the NOT gate).
- Some gates can have multiple diagram elements (for example, NOT).
- The performance of measuring qubit reasons any superpositions to fail and its quantum properties to disappear, inveterate classical information.

Quantum teleportation and its circuit diagram

Quantum teleportation is maybe the finest quantum algorithm for exemplifying these components. Quantum teleportation is a technique for moving data inside a quantum computer or even among distant quantum computers in a quantum network through the use of entanglement and measurement. Fascinatingly, it is proficient at moving a quantum state, say the value in each qubit, from one qubit to one more, deprived of even knowing what the qubit's value is. This is essential for the protocol to work rendering to the laws of quantum mechanics. The quantum teleportation circuit is specified underneath. We also deliver a glossed form of the circuit to exemplify how to read the quantum circuit. [Figure 4.11](#) represents the teleportation circuit diagram (taken from the Microsoft website):

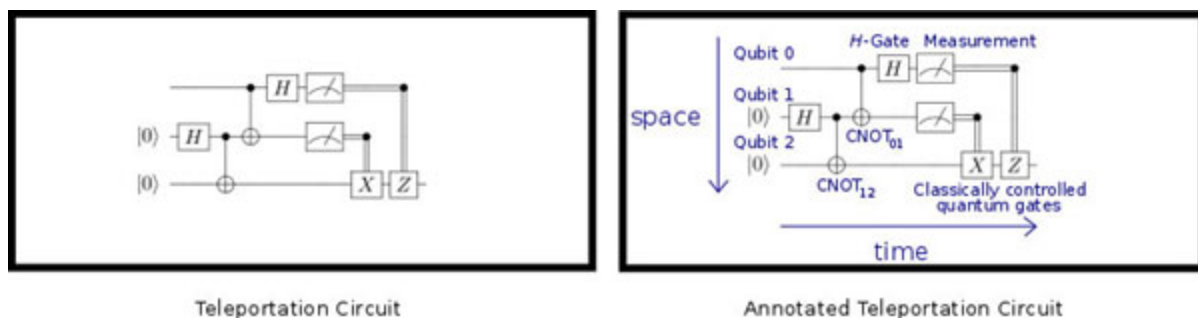


Figure 4.11: Represents teleportation circuit diagram (taken from Microsoft website)

Let's go into a bit detail: mechanics, mathematics, and measurements of Qubit

In quantum computing, we deal with the state of the particles and their rotations. Rotations are measured in radians. As per high school mathematics, radians are just a way of measuring angles in terms of π rather than in degrees, For Instance, there are 2π radians in a full circle. Angles are measured counter-clockwise. [Figure 4.12](#) shows the key adaptations.

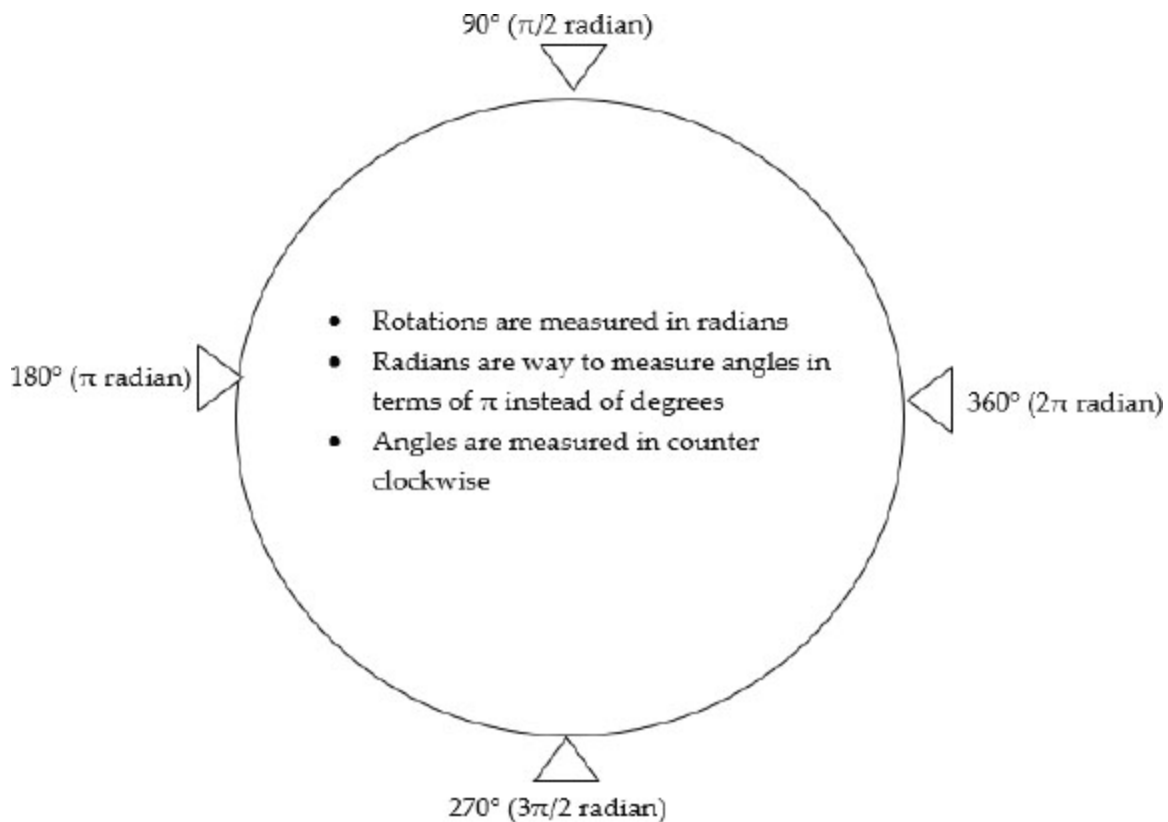


Figure 4.12: Key applications along with radians and angles representation

Single-Qubit gates

We now look at some qubit transformations (gates) which are central to quantum computing. The reader should commit these transformations to memory and become familiar with their use. The 1-qubit transformations (gates), namely the Pauli gates, and the Hadamard gate, and the 2-qubit CNOT gate are particularly important. They are all unitary matrices, as represented in [Table 4.3](#):

--	--	--

$\sigma_0 \equiv I \equiv \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	$ 0\rangle\langle 0 + 1\rangle\langle 1 $	$ 0\rangle \rightarrow 0\rangle$ $ 1\rangle \rightarrow 1\rangle$
---	---	--

Table 4.3: Unitary matrices

In matrix form, by convention, the basis states are lexicographically ordered, for example, 3-qubit gates are ordered according to $|000\rangle$, $|001\rangle$, $|010\rangle$, $|011\rangle$, $|100\rangle$, $|101\rangle$, $|110\rangle$, and $|111\rangle$. A few additional qubit systems are represented in [Table 4.4](#) along with Bar-Ket notation and comments:

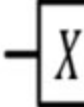
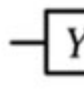
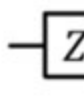
Name(s)	Matrix	Circuit Symbol(s)	Bar-Ket Form	Comments
Pauli X, X, NOT, bit flip, σ_x	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	 	$ 0\rangle\langle 0 + 1\rangle\langle 1 $	$ 0\rangle \rightarrow 1\rangle$, $ 1\rangle \rightarrow 0\rangle$.
Pauli Y, Y, σ_y	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$		$ 0\rangle\langle 1 - 1\rangle\langle 0 $	$ 0\rangle \rightarrow - 1\rangle$, $ 1\rangle \rightarrow 0\rangle$.
Pauli Z, Z, phase flip, σ_z	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$		$ 0\rangle\langle 0 - 1\rangle\langle 1 $	$ 0\rangle \rightarrow 0\rangle$, $ 1\rangle \rightarrow - 1\rangle$.

Table 4.4: Pauli, X, Y, Z representation

Mind it $|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

Single-qubit gates are the simplest. You can visualize the process of any single-qubit gate as the revolution of qubit's state to a different point on the Bloch Sphere. The three most elementary single-qubit gates are the Pauli X, Y, and Z gates.

The following are a few observations:

- The X gate is directly like the conventional NOT gate.

- The X gate transmutes $|0\rangle$ to $|1\rangle$ and $|1\rangle$ to $|0\rangle$. In the relation of the Bloch Sphere, this is equal to rotating $|\psi\rangle$ around the x-axis by π radians (180°).
- The Y gate signifies a revolution of/around the y-axis by π radians.
- The Y gate transmutes $|0\rangle$ to $|1\rangle$ and $|1\rangle$ to $|0\rangle$.
- The Z gate is essentially a special case of the phase shift gate R_ϕ where $\phi = \pi = 180^\circ$.
- The Z gate embodies a rotation of/around the z-axis by π radians. It has no consequence on $|0\rangle$ but transforms $|1\rangle$ to $-|1\rangle$.

On similar lines, let's discuss CNOT gates in some details, without much math to understand the concept in a better way, as follows:

- In the case of the CNOT gate, it is observed that each output is distinct or different and there is no ambiguity that exists, and further its states could be restored easily.
- Quantum computers use multiple technologies to become functional; one of the promising approaches is an electron spin-based quantum computer. Also, the CNOT gate can be implemented easily by using the same technology. Since in this approach, the control bit and target work fantastically well because the bit is close-together and control bits spin have some effect on the target bit. Therefore, it is easy for them to decide when the target can be flip with a certain electromagnetic wave.
- CNOT gate can be utilized in multiple ways. For example, it can also be used to produce the “so-called” entangled state of Control and Target. Hence, if we apply the Hadamard gate to control first and then apply the CNOT gate, both control and target will come in a state of superposition and become entangled to each other.
- As per the preceding-described technical superiority, we can safely say that the CNOT gate is used in quantum computing to generate entangled states which lie in the heart of quantum computers of specifically qubit technology.

- CNOT together with arbitrary qubit rotation can successfully implement any quantum logic functions in the realm of quantum computing.

Measurement of a qubit can also bring changes in the state of the quantum system and possibly to their functions. For a quantum computer to work efficiently and effectively, we must be able to change the properties of arbitrary qubits. Further, we must be able to run quantum logic gates on one or more qubits by making the interaction or communication between them.

In general:

It is vital to note that smearing the similar Pauli gate to the identical qubit, twice in a row abandons out the process since you have now done a full revolution of 2π (360°) over the shallow to the Bloch Sphere, thus arriving back at the starting point.

The preceding mathematics and operations tell us

$$XX = YY = ZZ = \mathbb{I} \text{ Because}$$

$$XX = X^2 \text{ and so on, this is equivalent to } X^2 = Y^2 = Z^2 = \mathbb{I} \text{ Where } \mathbb{I} \text{ signifies the}$$

identity matrix: $\mathbb{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$

Just a reminder, the identity matrix is the matrix equal to 1. Because you can always reproduce any matrix (M) by the identity ($\mathbb{I}$) to get the same matrix back again: $M\mathbb{I} = \mathbb{I}M = M$.

Operationally, the identity matrix can be thought of as an operation that does nothing to a quantum state. This can be seen on the Bloch Sphere, as shown in [Figure 4.13](#):

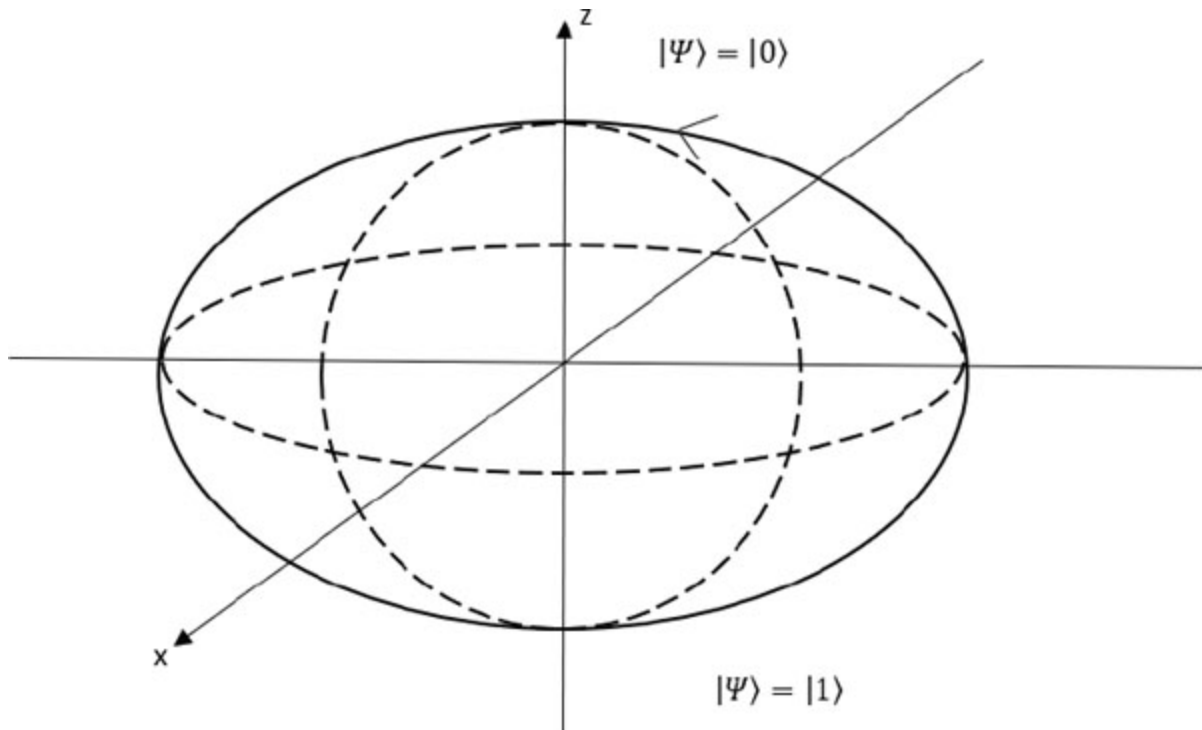


Figure 4.13: Operationally the identity matrix

Because of this relationship, we say that the Pauli matrices are the square roots of the identity matrix.

Few additional key single-qubit gates

The Hadamard gate, the phase gate, and the $\pi/8$ gate are commonly used gates in the world of quantum computers. Few of them are mentioned in [Table 4.5](#). The Hadamard gate is often used in quantum computations as the primary footstep in generating input data at the jolt of computations. It is treated as a special instance of the overall Fourier transform is a mathematical model which helps to transform the signals between two different domains, such as transforming signal from frequency domain to time domain or vice versa. It converts a qubit in state $|0\rangle$ into the superposed state $H|0\rangle$ and similarly state $|1\rangle$ into $H|1\rangle$. It means, H rotates the state of a qubit in a manner so that the outcome may also be observed as just a change in basis:

$$(|0\rangle, |1\rangle) \rightarrow (|+\rangle, |-\rangle)$$

And the overall outcome is held if the qubit is restrained in the foundation. We will get $|0\rangle$ or $|1\rangle$ with equivalent likelihood, but if it is measured using the $(|+\rangle, |-\rangle)$ base. We will measure $|+\rangle$ if the qubit's earlier state was $|0\rangle$ and $|-\rangle$ if the qubit's former state was $|1\rangle$. For details, refer to [Table 4.5](#).

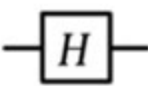
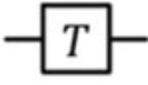
Name(s)	Gate (Matrix Form)	Circuit Symbol(s)	Operation	Gate (bar – Ket form)
Hadamard, H	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$		$ 0\rangle \rightarrow (0\rangle + 1\rangle)/2$ $ 1\rangle \rightarrow (0\rangle - 1\rangle)/2$	$[(0\rangle + 1\rangle)\langle 0 + (0\rangle - 1\rangle)\langle 1]/2$
Phase, $\frac{\pi}{4}, S$	$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$		$ 0\rangle \rightarrow 0\rangle$ $ 1\rangle \rightarrow i 1\rangle$	$ 0\rangle\langle 0 + i 1\rangle\langle 1 $
$\frac{\pi}{8}, T$	$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$		$ 0\rangle \rightarrow 0\rangle$ $ 1\rangle \rightarrow e^{i\pi/4} 1\rangle$	$ 0\rangle\langle 0 + e^{i\pi/4} 1\rangle\langle 1 $

Table 4.5: Hadamard and phase shift

The Hadamard gate is mainly significant as it can be used to make a superposition of the $|0\rangle$ and $|1\rangle$ states. In the Bloch Sphere depiction, it is laid-back to the reason of this as a revolution of $|\psi\rangle$ round the x-axis by π radians (180°) followed by a (clockwise) rotation around the y-axis by $\pi/2$ radians (90°). Hadamard on the Bloch Sphere 13 is depicted in [Figure 4.14](#):

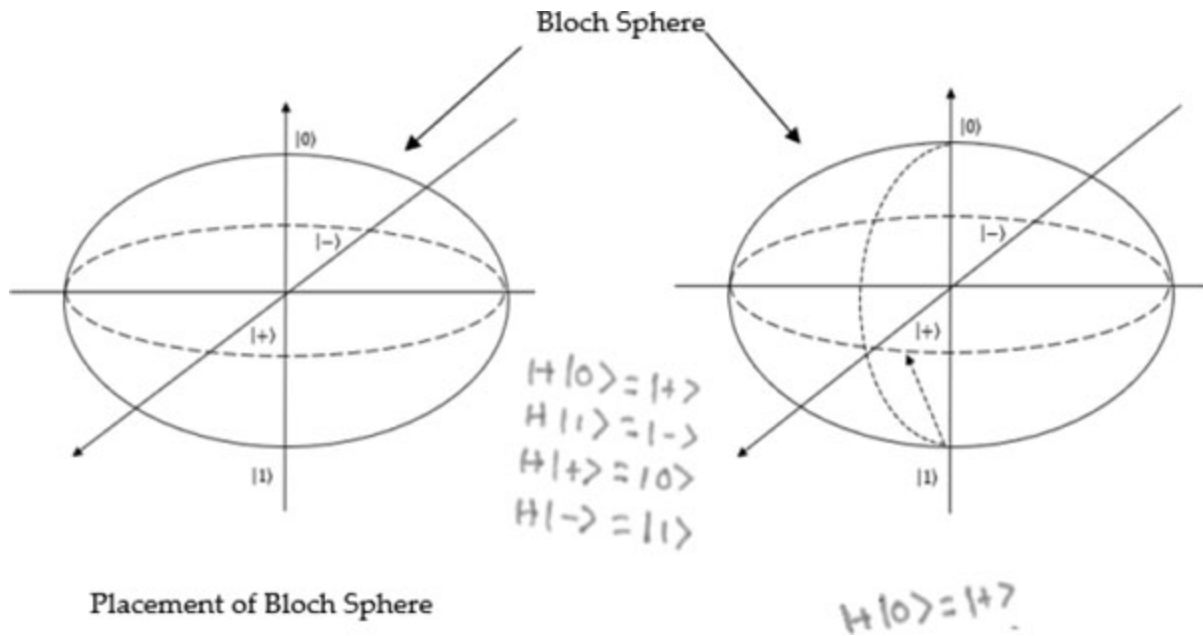


Figure 4.14: Hadamard on the Bloch Sphere 13

The phase shift gate is a general gate that has several valuable applications, most remarkably the $\pi/4$, $\pi/8$ and Pauli-Z gates, which are made by setting ϕ to $\pi/2$, $\pi/4$, and π , correspondingly.

Normally, the application of Hadamard transform on n qubit (represented by $H^{\otimes n}$) is initially as follows:

$$H^{\otimes n}|00\cdots 0\rangle = |00\cdots 0\rangle + |00\cdots 1\rangle + \cdots + |11\cdots 1\rangle = \frac{1}{\sqrt{2^n}} \sum_x |x\rangle,$$

Where the sum is overall conceivable if c is in mutually orthogonal states of n qubits or values of x . Therefore, the Hadamard transform produces an equal superposition of all 2^n possible computational basis states which can be mapped to the number from 0 to $2^n - 1$. That is, only n qubits are needed to carry simultaneously an exponentially large number, Hadamard transform, integers are in superposed states. Remember Hadamard gate is as follows:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Multi-Qubit gates

The 2-qubit controlled-not gate, CNOT, is a generalization of the irreversible classical XOR gate. Here, the state of the bit chosen as the control bit decides what happens to the other called the target bit. Let $|a\rangle \oplus |b\rangle \equiv |a, b\rangle$ represents a 2-qubit system on which it acts. Then, if the first bit is the control bit:

$$\text{CNOT } |a, b\rangle = |a, a \otimes b\rangle$$

Where $\oplus$ signifies the exclusive-or (XOR) operation (that is, modulo 2 addition)⁹³. It means that the output is “true” if and only if exactly one of the operands has a value “true”. This gate cannot be decomposed into a tensor product of two 1-qubit transformations, that is, one cannot find two unitary operators O_1 and O_2 such that $\text{CNOT} = O_1 \otimes O_2$. The list of state changes, as shown in [Table 4.6](#), is the analog of the truth table for a classical binary logic gate. Note that $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ form an orthonormal basis for the state space of a 2-qubit system. By convention, they are associated with the standard 4-tuple basis as shown in the table.

Multi-qubit gates work on binary qubits or further. For a better understanding, the SWAP gate is taken as one of the simple examples, as shown in [Table 4.6](#):

Name of the gate	Matrix representation	Circuit symbol
Swap gate	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	

Table 4.6: SWAP gate

The SWAP gate swaps the dual input qubits from one place to another. For instance, $\text{SWAP}|0\rangle|1\rangle = |1\rangle|0\rangle$ and $\text{SWAP}|0\rangle|0\rangle = |0\rangle|0\rangle$.

Another class of multi-qubit gate is the controlled gates. Controlled gates need at least one control and one target qubit. The gate will only function on the target qubit if the control qubit is in a definite state. If the gate needs the control qubit to be in the $|1\rangle$ state, this is signified by a filled-in circle on

shown as follows:

$$T |x, y, 0\rangle = |x, y, x \wedge y\rangle$$

$$T |1, 1, x\rangle = |1, 1, \neg x\rangle$$

The classical Toffoli gate is, therefore, a classical universal gate. The quantum version, which simply permutes computational basis states in the same way as the classical Toffoli gate, nevertheless cannot carry out the FANOUT operation because of the no-cloning theorem. The quantum Toffoli gate is not universal in quantum computing, but it can implement all possible classical computations on a quantum computer. The Toffoli gate allows the irreversible classical gates to become reversible. This means that quantum computers can mimic the Universal Turing Machine despite being restricted to reversible, unitary gates. For details on CCNOT, refer to [Figure 4.16](#):

NAME	MATRIX	CIRCUIT SYMBOL	OPERATIONS	OPERATION REPARATION IN TERMS OF BRA AND KET
C NOT TOFFOW	$\begin{bmatrix} 10000000 \\ 01000000 \\ 00100000 \\ 00010000 \\ 00001000 \\ 00000100 \\ 00000001 \\ 00000010 \end{bmatrix}$		$\begin{aligned} 1000 &\rightarrow 1000 \\ 1001 &\rightarrow 1001 \\ 1010 &\rightarrow 1010 \\ 1011 &\rightarrow 1011 \\ 1100 &\rightarrow 1100 \\ 1101 &\rightarrow 1101 \\ 1110 &\rightarrow 1111 \end{aligned}$	$10 \rightarrow \langle 01 \otimes I \otimes I + \bar{1} \otimes II \rangle \langle 11 \otimes \text{CNOT}$

Figure 4.16: CCNOT representation

Universal sets: moving from traditional gates to Quantum gates

Classical computers calculate utilizing rules of Boolean algebra and the expressions derived out of them. Any Boolean expression, however, may be constructed or derived out of a non-unique set or group of universal logic gates. The combination of so-called AND, OR, and NOT gates typically form a universal set. Alternatively, AND and NOT (also known as NAND),

or OR and NOT or AND and XOR form three other possible usable effective, and efficient universal sets.

In addition to logic gates, one requires extra ancilla or working bits (which is the absence or presence of electric current), to allow extra working space throughout computations. Further, Muller's theorem tells us that the selection of a universal set has little effect on the complexity associated with computing.

As the operations of AND, OR, and XOR gates are many-to-one, they are logically irreversible (cannot be reversed). Only the NOT gate is rescindable or reversible since it is a one-to-one operation. Henceforth, the information contained on the right-hand side of the expression is less than on the left-hand side:

$$(a, b) \rightarrow \text{NOT } (a \text{ AND } b)$$

In the quantum world, however, any unitary operation which is performed on n qubits can be realized precisely by winding composed operations which is self-possessed of single-qubit and controlled-NOT gates. For instance, for the Hadamard phase, controlled-NOT gate delivers a set or group of universal gates, which is a fault-tolerant construction. This output set is the standard set of universal gates.

There is a second set of universal gates that is considered which comprises the Hadamard, phase, controlled-NOT, and Toffoli gates. These gates can be built or constructed in a fault-tolerant way. Additionally, in this, there are many more picks that can be taken for the universal quantum gate than in classical reversible computing due to the superior power of quantum computing and there are a lot of theories behind them. For instance, DiVincenzo presented two-qubit universal quantum gates model which is certainly possible and Barranco protracted this knowledge to display that nearly any two-qubit is universal. Further, Lloyd and Deutsch, et al, have revealed that nearly any two-qubit or n -qubit ($n \geq 2$) quantum gate is also universal.

Also, Rolf Landauer, Charles Bennett, Tommaso Toffoli, Edward Fredkin, and others presented that reversible computing on a Turing machine was certainly likely and computationally inexpensive, and it was clear that quantum computers can simulate Turing machines by safeguarding that

they remain on a computational basis state at the end of each step, given that they start in one. To ensure unitarity, it is necessary and sufficient that the mapping from one state to another is bijective. The universal quantum computer can perfectly simulate any Turing machine and can simulate with uninformed exactness any quantum computer or simulator. In theory, any quantum particle, e.g., a photon or an electron, can function as a gate. The knowledge that quantum computers can be constructed to do any computation reversibly both logically and thermodynamically by a physical device dissipating arbitrarily little energy is no longer an imaginary concept.

We essentially are capable of implementing a universal set of gates for the physical system we are using as our quantum computer. This means that any computation likely on our system must crumble back into a finite system of known gates. For instance, the Hadamard, phase, CNOT, and $\pi/8$ gates composed form one such universal set.

It means that any alteration allowable by quantum physics can be applied on a quantum computer. This not only means we can run any quantum program, we can also use universality to run any physics or chemistry problems. Universality hence permits us to use a quantum computer to impersonate molecules, superconductors, and all way of bizarre and pleasing quantum systems.

Multiple types of Qubit

Whatever type of quantum computer we think qubits are, sit in the core of all the ideas. Qubits can be realized with numerous technologies, for instance, quantum dots are structures that can restrain and operate a single electron to be acted as a qubit. But broadly, there are two diverse methods to quantum computing in practice, as follows:

- Universal gate-based
- Quantum annealing

But the qubit is something which is the soul of both types of quantum computes. The distinguishing aspect among the two is the procedure of how they come to an outcome. In the gate model style, quantum information (kept in qubits) follows consecutive orders, doing step by step to attain the

anticipated outcome. These systems are seen as the general-purpose quantum computer.

With quantum annealing, one must decipher a problem into an energy landscape, which is determined by the parameters of the qubits and the communications among them, and the system discovers the lowermost optimal point. Visualize, after a rainfall a landscape with valleys, crags, and basins, where water will run to the lowermost points of the basins. This is similar to the method quantum annealing works in – discovering those lowermost points, with the advantage that the quantum state of the annealer can “tunnel” crossways fences in the landscape, letting the system to pursue out the global least. These types of systems are intended to resolve exact optimization and probabilistic sampling problems.

It can be helpful to think of a quantum computer like a GPU which is a specialized processor used for appropriate situations. Likewise, as old-style computers run computations, the assignments, or glitches best suited for quantum could be treated by the quantum computer. From development to deployment, multiple companies like IBM and Microsoft are allowing the quantum revolution with a whole compressive approach towards quantum systems. This approach uses topological qubits, a kind of qubit that permits the system to scale. The outcome is an integrated, scalable solution that syndicates both quantum and classical computing.

An additional way of operating the spin of an electron is done via nitrogen-vacancy centres (or NV centre in diamond have few of the utmost hopeful applicants to instrument room-temperature quantum computing) in a diamond.

Transmon qubits are one type of superconducting qubits that fundamentally uses Josephson junctions (a quantum mechanical tool which is built by sandwiching two superconducting electrodes separated by a block) to harvest a single magnetic flux for use as a qubit.

Ion-based qubits use ion traps to stock qubits in separate ionized atoms on hold in a vacuum and controlled using LASER and electric and magnetic fields. Qubits can also be accomplished by governing the polarisation of photons, or the number of photons.

Remember, different to classical computing, qubits can be in 1, 0, or a superposition of 1 or 0 quantum states up until we measure them. Hence, by extrapolation, we can take a broad view that n qubits will be in a superposition of up to n unrelated quantum states concurrently.

It is this simultaneousness property that chaperons us to possible quantum benefit. If we can use this enormous parallelism, we can calculate many operations at the same time, thus inferring a time advantage. Besides qubits, to operate a quantum computer, we need the jam-packed quantum hardware and software layer stack.

Currently, there are a few qubit implementations that look quite promising. The most prominent examples are superconducting qubits, ion traps, and spin qubits. All the platforms mentioned earlier fulfil the so-called DiVincenzo conditions. These conditions, demarcated in 2000 by David DiVincenzo, need to be satisfied for any physical implementation of a quantum computer, as follows:

- A scalable physical system with well-considered qubits.
- The capability to set the states of the qubits to a simple state, such as $|000\rangle$.
- Extended relevant coherence times, considerably longer than the gate process time.
- A universal set of quantum gates.
- A qubit-specific measurement competence.

Components of a Quantum computer

Like qubit is an integral part of the computes, irrespective of what kind of architecture a quantum computer usage, nowadays they all have fairly alike involved components such as the following:

- Naturally measured, protected, clean computer room.
- Refrigeration system.
- Gas storage and distribution systems.
- Good cabling.
- Excellent and effective piping arrangements.

- Classical computers to monitor, control, and manipulate the quantum computer.
- Sophisticated circuitry.
- Electromagnetic shielding.
- High capacity storage devices.
- Electricity backup and fault-tolerant classical machine infrastructure.
- **Quantum Processor Unit (QPU)** with good physical packaging.
- Quantum data plane (comprises QPU and all other quantum components).
- Control and measurement area.
- Hardware networks, far-off connections, and interfaces.
- Conventional computer components to stock outcomes and subsequent data.
- Networking.
- Operating system (start-up code, control, monitoring, compiler, and so on).
- Software interfaces.
- Algorithms.
- Application software.
- Exposer to cloud infrastructure.
- Quantum computing specific Software libraries.
- Knowledgeable technicians.
- Research scientists, programmers, and technology managers.

Nowadays, in a practical quantum computer, all the components are involved to give the best return on investment. Most computer system components are bulky and costly, at least at the moment. However, over time each quantum component and the complete system of components are expected to get smaller and smaller and less resource exhaustive.

How to make robust quantum computers: welcome quantum error correction

As per the present informal agreement among the experts' 72 qubits and beyond, quantum supremacy could be achieved. However, the biggest challenge to achieve quantum supremacy is early qubit decoherence. As we know, decoherence is a quantum particle's states going from its easy-to-see superposition to its settled, measured, single, classical state before all the ensuing entangling brands receive useful information. Once decoherence has occurred, premature or not, it cannot simply be inverted. With faultless quantum computers, decoherence would occur only and precisely at the point where a "response" is desirable and measured. It would always be on purpose.

For an ideal quantum computer, a qubit would stay in its cohered state if it is desirable for the quantum calculation and then, and only then, decohere when measured. How long the qubit will stably maintain the state of the qubit is called its coherence time. It is ever so often measured in milliseconds, but some quantum computer types can last seconds to many minutes. The first instruction of business for most quantum computer makers is to increase coherence time. Increased coherence time means fewer errors and more time to compute and return answers.

The current fact is that quantum computers are filled with premature quantum decoherence and just absolute errors. Both can happen because of qubit building, high temperature, contamination, sound, vibrations, faulty gates, faulty measurements, defective initial state groundwork, background nuclear spin, and countless other events.

Any contact with the external world is a hazard. Dipping errors and noise is the number-one quantum challenge and has produced a field that has its name, quantum error correction. Error correction is making an effort using a group of different arrangements, including both quantum and classical methods. No one has achieved it yet, but every quantum computing vendor is trying. Error rates are typically reported as a ratio of quantum operating time to decoherence time.

The quantum error threshold theorem grasps that any quantum system that modifies errors faster than it makes them is practical. As the number of

qubits upsurges, so does the natural error rate. For quantum computers to be very beneficial, the error rates need to be below 1 percent, precisely below 0.002 percent. As a comparison, a traditional CPU can do trillions of calculations short of an error.

In the quantum world, we are just eager to contract the error rate down to one error per thousand calculations. Attaining that, along with some good error correction, will permit some serious quantum work. Till now, we are searching for this with no tangible result.

However, the efforts are on. Quantum computing experts are trying to decrease quantum errors by recognizing and fixing the most important rate-limiting components or issues. Common answers comprise refining coherence times, stringently isolating quantum components from the outside world, utilizing supercooling.

There are several different types of collaboration and integrations required to make the quantum initiative a success for corporate, government or holistic research. But it does not end at this point; there are several additional features of the quantum computer that are also important to make QC efficient and usable. For instance, Calibration Errors actions how well the electronic controls for software design and measuring qubits are standardized to guarantee precise procedure and error resolution.

Whereas, circuit Optimization takes care of how well an optimizing compiler recovers the plan and acts of a circuit across its penetration and qubit register breadth. Coherence, as discussed, is significant for how the qubit stays in an operational state. The coupling map tells us how qubits are linked and how the actions of qubits are associated with other qubits and in what patterns. Qubit crosstalk is providing insight about how the state of one qubit marks nearby qubits. And, gate fidelity is the error rates that gate processes to move qubits from their present state to new states.

As these are the important aspects of quantum computing, let's discuss a few of them in a bit more detail.

It's not just a competition for accumulating new-fangled qubits, but a race for totalling error-tolerant qubits and then appraising the development of technology. This needs benchmarking both metrics. The present goal is a 0.5% coherence rate. At those levels, computational power is predictable to

upsurge exponentially. The development of performance cuts suggestively at advanced error rates, and outside 1% error rates adding more qubits does not add any computational power at all. Hence, a lot of effort is going in this direction.

A lot of companies are coming with innovative solutions to cater this. For instance, IBM has proposed a metric called Quantum Volume that scales with both error tolerance and qubit count. [Figure 4.17](#) explains this phenomenon in a simplified manner. It reduces errors as qubits are added. In other words, the reduction in errors is as important as the addition of computing power. Refer to [Figure 4.17](#) for a better understanding:

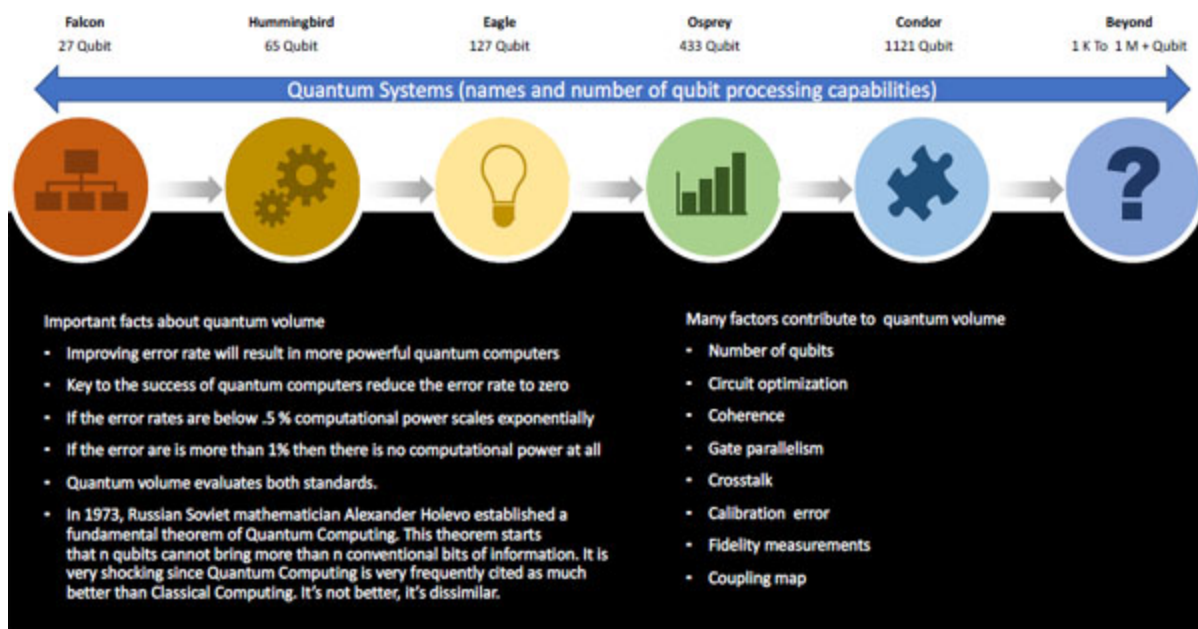


Figure 4.17: Quantum volume – error rate and qubit quantity must be considered for reaching out to any conclusion (Source: IBM website and google image)

A few other additional points related to this are described as follows:

- Refining coherence periods:** This is one of the techniques of error rectification. It helps advance the quality and control of each qubit's cohered state to be longer than the desirable calculation time. It can achieve this by improving whatsoever is the error-limiting component of the quantum computer, such as quantum gate sound or connectivity speed. As we know, the longer the qubit can stay cohered, the fewer errors it is likely to have.

- **Surrounding separation:** Computer scientists have known the benefits of operating computers in controlled surroundings that isolate them from the immoderations of the outdoor world. This is not a new phenomenon and is applicable to classical computers as well, at least in its initial days. That's the reason all computer rooms are temperature-controlled (heat is the enemy of all computer components), air-filtered, humidity-controlled, cleanser environments.

You don't find a lot of computers running very long left in the outdoor world unprotected from normal weather and events. But most of today's classical computers have mellowed physically to a point where they can live operating in normal weather environments, except they are bare to truthfully extreme conditions. For instance, the most general types of computing devices work every day in the real world. Most laptops, tablet devices, and personal computers function fairly fine outdoors in a very controlled computer room.

Quantum computers are not there yet. They are still very delicate machines and are essentially protected from not only weather extremes but also very usual conditions. Most of them function most proficiently by running (at least the quantum components) in specific types of weather conditions, such as very cold temperatures. They essentially are protected against radio waves, normal circumstantial radiation, electromagnetic interfering, brash noises, and vibrations. But most quantum experts do imagine a day when, as is the situation with classical computers, quantum computers will be constructed in a way so that they are ample resilient and are in less requirement of different environmental isolation.

- **Supercooling:** Most quantum computers essentially supercool their qubits (and other nearby components) to near zero degrees Kelvin, which is close to -460°F to minimize early decoherence issues. Heater temperatures are exposed to permit more errors and to produce more unwelcome stray quantum particles with nearly all the quantum computer technologies, even with the few kinds that allegedly don't require ultra-cool temperatures. They may not "want" super cold temperatures, but even they appear to do better with fewer errors in lesser temperatures. Therefore, qubit chips and closely related

apparatus are supercooled using outside cryogenic or dilution refrigerators.

Till now, absolute zero (0K) is not achieved. Also, at absolute zero all particles' energy and momentum would be at a standstill to the bare minimum physical conceivable (something called zero-point energy). At absolute zero, most moving and even solid-state gears would flop to function as usual. With that said, the quantum computer of the upcoming is likely to be able to endure advanced temperatures, perhaps equal to the necessities of today's classical computers, because necessitating supercooling is expensive and restricts where and how they can be used.

But for now, lower temperatures typically advance coherence times and help in the reduction of errors. Lower temperatures also produce a quantum property called superconductivity, which is zero or near-zero electrical confrontation in things cooled below critical temperature thresholds. Superconductivity upsurges electron flows and permits stronger electromagnetic connections. Many quantum computers use superconductivity to create their qubits, and it is used in many other applications, such as superfast maglev trains, medicinal gear, and super-strong magnetics.

- **Repetitive calculations:** One way to contest errors is to run a similar calculation at least three to four times and store the outcomes in the classical world. After running the same calculation many times, the computer will look at all the kept results and, if there is a difference, take the outcome that seems more than the others. Though this error correction technique also decelerates the computer down in direct relation to how many duplicative processes are used, and there is no guarantee that the furthestmost signified answer is the right answer.
- **Using Quantum Entanglement for error correction:** In the traditional computer world, if a lot of faults are to be expected, a bit's value can be derivative and stored to one or more "backup" bits at the identical time. If there is a difference among bits, then the most popular value is taken. But in the quantum world, because of the no-cloning theorem and the observer principle, qubits cannot be copied directly while in their quantum states. In its place, entanglement can

be used to produce indirect copies, although entanglement promises are subtle and easy to lose track of as decoherence happens.

- **Check Qubits:** Additional error correction technique uses extra qubits as check qubits, which is like how to check bits that are used in the classical binary world. The check qubits are applied in some sort of logical inspection method, which notices errors and helps to correct them. For instance, an extra check qubit is used to guarantee that the subsequent qubyte adds up in a specific way. In this case, the computer can add a 0 or a 1 to the check qubit location of the qubyte to make sure the sum of all qubits ends up as an even value.

If the quantum computer notices an adverse sum inveterate from a qubyte sum, an error can be acknowledged and the quantum operation can be recurrent. You might be familiar with a like general error correction traditional technology called Redundant Array of Independent Disks (RAID) in the binary computer world. The big problem with this sort of simple even-or-odd checking is that there is no way to guarantee that errors don't occur in a way that isn't noticed only by even or odd values.

Quantum computer manufacturers are using quantum error-checking approaches to make quantum computers more dependable. Error-checking qubits added to a system only to deliver that fault tolerance are recognized as ancillary (or ancilla) qubits. Quantum computer vendors are even now besieged to make the best use of the number of qubits, so partaking to "left-over" some to deliver error checking isn't best. Right now, the present state of several quantum computers and devices is that it desires many, many ancillary qubits to ensure one stable qubit. The number of ancillary qubits desirable to make one stable qubit has been measured from a few up to millions. That's a strange range and scaling problem. Still, occasionally the best performance that can be attained includes adding more qubits and using them as ancilla.

- **Increase performance of other components:** An applied way to conquer decoherence errors is to upsurge gate, connectivity, and state groundwork. By decreasing the time it takes to calculate a quantum outcome and read it, the wanted computation and subsequent value

read can be done before an error makes the qubit hurriedly decohere. For instance, let's suppose that a quantum computer has a load of decoherence errors that jolt trendy around 100ms. If computations can be finished in below 100ms, then that quantum computer can circumvent the nastiest effects of decoherence and get more precise consequences.

The large takeaway from this section of the book is to understand that the problem-solving power of a particular quantum computer is not just a function of the number of its qubits. Nowadays, several of the qubits are probable to be involved in error correction. The complete performance of QC hangs on a diversity of some other additional factors, such as gate preparation, connectivity between gates, error correction, and read performance. Even when several qubits and error correction rates are identical between quantum computers, there may be different types of quantum computers made to solve different types of problems. Therefore, any direct judgments grounded on qubits alone is foolish. It is very difficult to say that a 90-qubit computer must be better than a 60-qubit computer, particularly for all types of problems.

Now after having this discussion, it is a good time to take a look at the important parameters of quantum computers. [Figure 4.18](#) shows the performance factor for quantum computers.

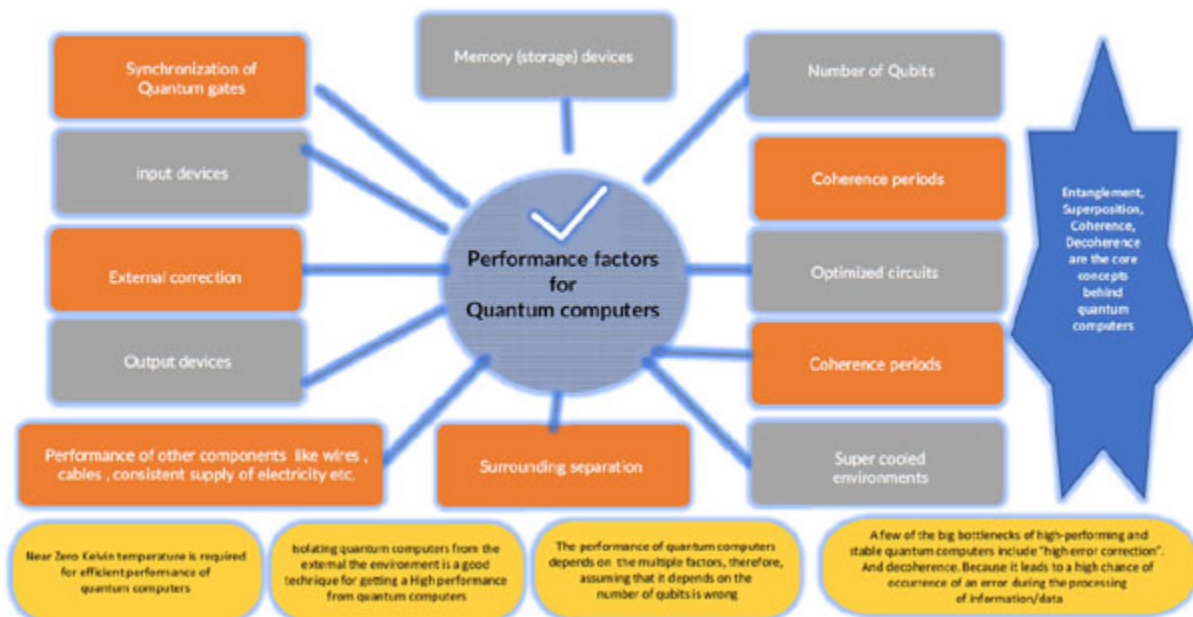


Figure 4.18: Important parameters of quantum computers

A complete quantum system from hardware to software

The enthusiasm for quantum computing stems from its potential to qualitatively increase regarding what computational problems we are accomplished in resolving. This is permitting us to discover the solutions to significant but hard and complex problems that are stubborn with contemporary classical computing. Though these are initial days, the business is now in a serious and hurrying retro in which the approaching years will determine its marketable success or disappointment.

Over the next few years, engineering will see the most important progressions in quantum hardware and software infrastructures. The promise is that this technology may one day have a ground-breaking consequence which would change our perceptions in the ways we comprehend and communicate with complex systems, also, the range of the applications fluctuating from portfolio-management to traffic optimization to custom-made healthcare and drug design.

The procedure of construction of a quantum computer comprises helping the raw materials become desirable to make topological quantum devices, thus, making the cold electronics and cooling systems, and developing the complete infrastructure desirable to carry the solution to life. In accumulation, the system must include everything which the user needs to program the quantum computer, together with a control system, software, development tools, and cloud-based services.

A cloud-based service is required because QC is a special purpose computer and mostly in an evolving state. It needs a specific condition to operate. Hence, any vendor who is offering its service must go through cloud infrastructure. Because in that mode of offering, the giant quantum system would be kept in the desired environment, and the software and application development toolkit would be exposed through the cloud.

Now with this background and a combination of all the bits and pieces which we discussed here, it would lead to an understanding of a full quantum stack.

Several vendors deliver a network of resources, together with tutorials, coding tools, simulation tools, and admittance to their cloud resources. Several quantum computing vendors will also talk about having the full quantum stack. A quantum stack is the quantum-based assemblage of hardware, qubits enabling devices, quantum software development kit, APIs, and applications. Numerous vendors, together with IBM, Google, Microsoft, D-Wave, IonQ, and many others, offer all or parts of the stack.

Most of the companies who are now pioneering in QC are investing and participating in essential quantum physics investigations for more than a decade or two intending to empower a quantum-computing revolution. Advances in quantum and computational theory across the years have led to innovations in experimental physics and the realism of a path forward for scalable quantum computing. For instance, take the case of Microsoft. They give focus on topological quantum computing through Majorana fermions, which promise to harvest fast, steady quantum bits. To attain the goal, they started by producing the raw materials essential to make topological quantum devices. Further, it nurtured focused nanowires that can be used to make qubits, followed by the construction of atomically thin conducting coatings with striking topological properties using integrated-circuit engineering methods to realize a completely scalable quantum computer.

Taking the materials and turning them into working devices is not a trivial task. Microsoft runs quite a few laboratories around the world that are attentive to the fabrication of quantum devices. The set-up they built permits for the well-organized mounting, wiring, and cooling of these devices, which in turn supports rapid prototyping, characterization, and exploration. These devices function at temperatures nearly 200 times cooler than the uttermost spreads of outer space. Transferring and getting signals to nanoscale devices is also a frightening job. A scalable quantum computer needs several reckless and dependable signals. Hence, Microsoft designed traditional integrated-circuits that run at cryogenic temperatures where usual chips will not function. They changed the wiring system to handle signals that are in the super-conducting management. The whole set-up is integrated to permit scalability, including the cooling system itself. So, in conclusion, we can say that innovation in the quantum system is not enough; a complete ecosystem needs to be evolved and be mature. A

generic ecosystem for a corporate, however, ideally contains the following elements, which needs to be developed to get a perfect environment for QC. Few important components are mentioned as follows:

- Collaboration with universities
- Application developers
- Software, process, tools, programming languages, run times, compilers
- Collaboration with startup's
- Interaction with research lab for innovation
- Technology infrastructure providers
- Interaction with the organizations facing similar challenges
- Organization vision and strategy
- Access to the talent pool

[Figure 4.19](#) depicts the important components of an effective quantum company that creates a dynamic, effective, and efficient quantum ecosystem:

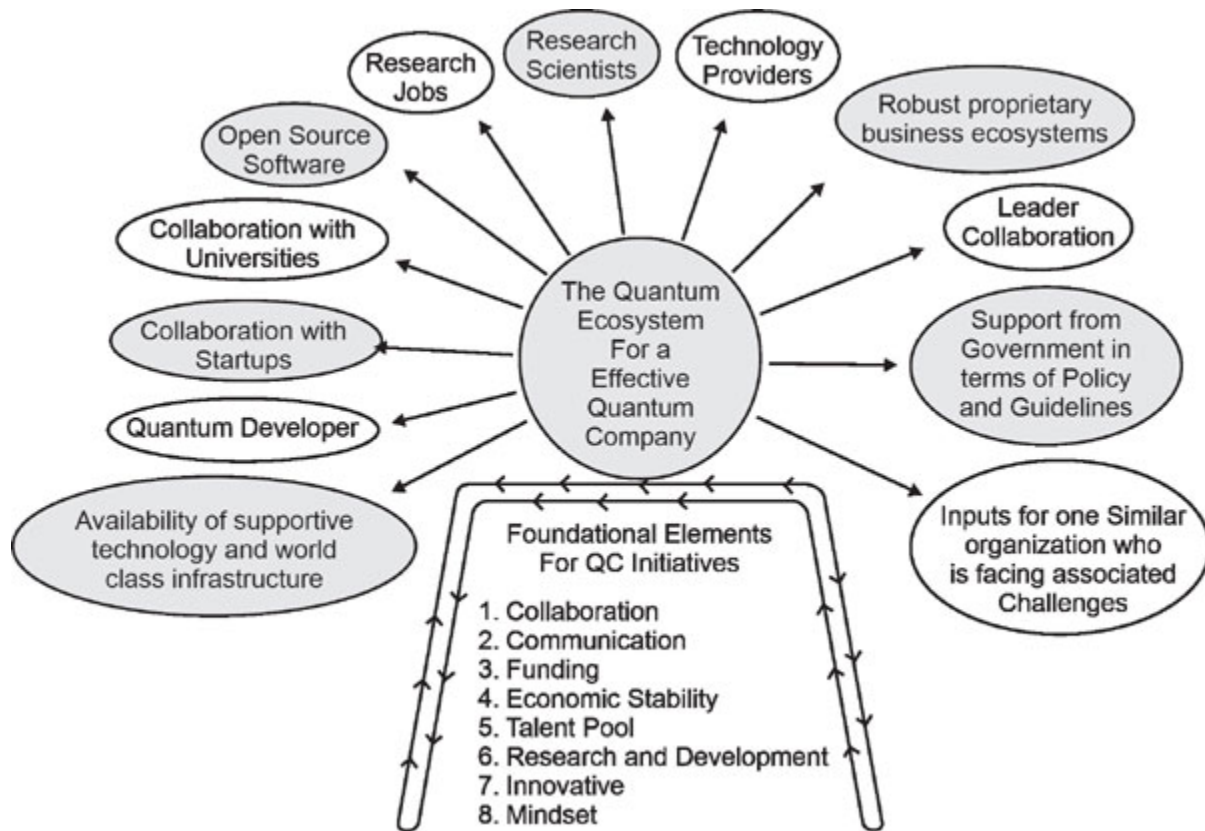


Figure 4.19: Important components of an effective quantum company

Solving problems on quantum computers

It takes more than hardware and qubits to make a quantum computer intelligent enough to resolve problematic problems. Each quantum device comes with one or more operating systems, algorithms, interfaces, and application programs. The quantum computer essentially has the precondition firmware or control software that permits the quantum computer to produce, prepare, measure, control, error check, and withdraw qubits. Each quantum device must implement one or more quantum algorithms, which knob the base layer manipulations and math based on qubit operation. Not all quantum devices provision all algorithms, though the universal gate quantum computers are reportedly meant for this. Most quantum computers have compilers, programming languages, and script languages to allow the coder to write their particular quantum computer programs. Many quantum vendors provide their preoperatory software to their clients.

Some quantum device vendors have shaped or inspired their customers to work with open-source quantum software. Others offer their exclusive quantum software for free as a way to persuade developers to learn and develop on their quantum computers. It's a model that has worked quite well in the classical computing world.

At the moment, a quantum computer is a hybrid machine containing a quantum device (hardware) and a classical computer, which directs the instructions to the hardware and obtains and processes the outcomes (software). Quantum languages link with this classical software. Some of them are suitable programming languages as are the classical ones. Others are in fact libraries coded utilizing some well-known language like python, C++, Matlab, and C# that aid the user to code quantum algorithms.

To resolve problems on a quantum computer, the prerequisite is a runtime that implements a quantum algorithm while upholding the state of the machine, functioning the control system in a parallel real-time environment, and cooperating from the device to the external world. The runtime layer is the firmware and operating system of the quantum computer. To help quantum developers shape applications and algorithms, few companies and the open-source world are providing the Quantum Development Kit which is essentially a set of enterprise-grade gears to transcribe, debug, and optimize quantum code. For instance, from the open-source world, Python quantum development software tool kit is available. These kits comprise all the frameworks and gears one requires to get started. [Figure 4.20](#) shows end to end software and hardware stack.

Vendors like IBM and Microsoft are now providing quantum services in a cloud environment which gives the user a fast path from simulation to optimization to deployment on quantum hardware. Refer to [Figure 4.20](#) for a better understanding:

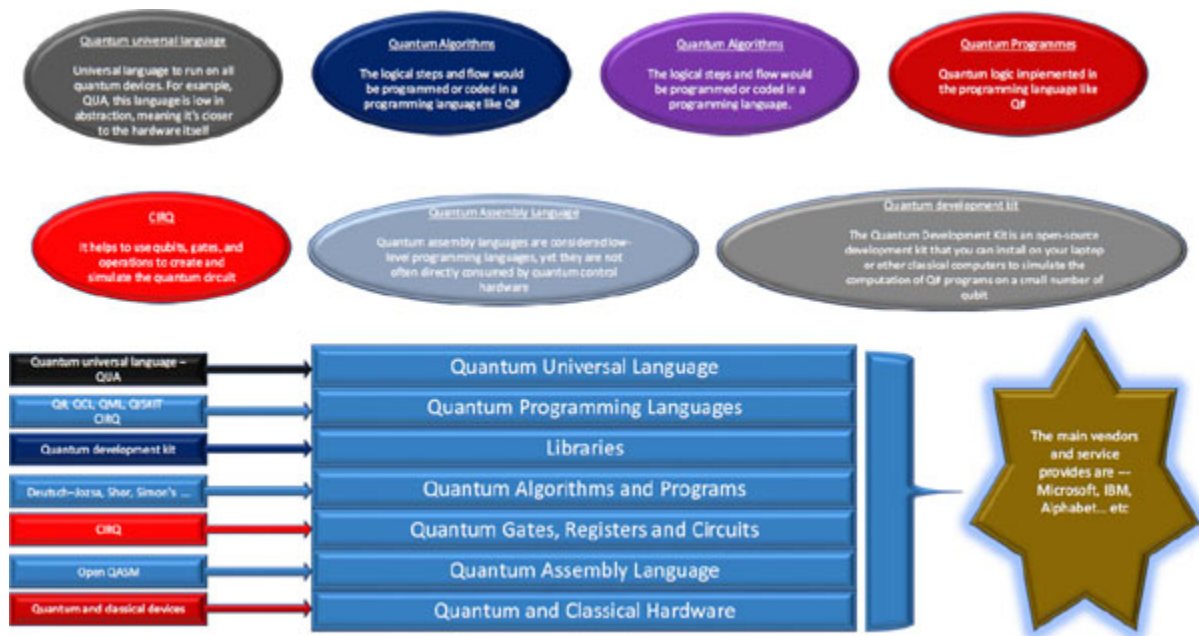


Figure 4.20: End to end software and hardware stack

At present, mostly quantum and classical work are composed; therefore, vendors are providing an environment for quantum processing and deployment which can handle the need of both the worlds. Hence, with data stored in a cloud environment, developers will be able to access quantum processing together with classical processing, creating a streamlined experience.

Hybrid quantum – classical computing

Utilizing the power of QC to solve increasing complex problems does not mean abandoning current applications built for classical computing. With the D-Wave Hybrid™ platform, optimizing current applications in a quantum computing environment using hybrid workflow of D-Wave is a good use case. It provides the result in the best of both worlds, allowing the user to obtain the results from both classical computing and quantum computing and better solutions to the complex problems, as shown in [Figure 4.21](#):

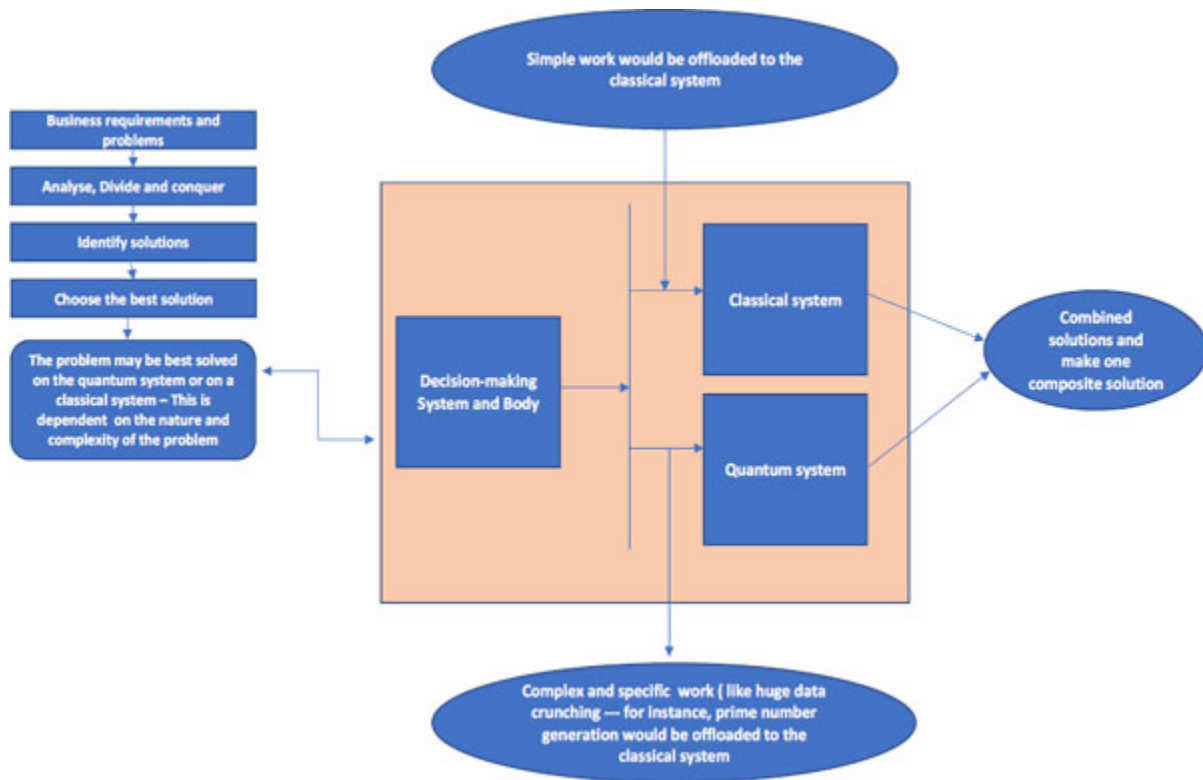


Figure 4.21: Hybrid approach quantum and classical way (integrative approach)

Real-life working of Quantum Program: Microsoft way

Fundamentally, writing a quantum program is not different from the way to write the traditional one. However, tools and technologies are different. Here we would understand that how to write one in azure quantum. Mentioning a familiar scenario will help the user to understand the complete flow of the process. For example, let's assume that we are using Microsoft Azure or precisely quantum Azure to perform quantum processing for a given problem by using a quantum algorithm. The following are the steps for the same:

- Build the solution in visual studio by using the tools like the Microsoft Quantum Development Kit.
- Do coding using Q#, a language that is specifically created for quantum program development.
- Use readymade quantum libraries in your program wherever required. Microsoft provides extensive quantum libraries along with Q#.

- When the coding is completed, fire a quantum simulation to check for bugs and confirm that your solution is equipped or ready for deployment.
- After validation, you would be prepared to run your solution on the cloud-based framework (Microsoft Azure Quantum) which is an abstraction layer on a quantum computer. It hides internal details.
- Hence, a developed quantum code or solution would be positioned from within Microsoft Azure, utilizing the quantum computer as a co-processor.
- As you will use both quantum and classical processing, Azure will rationalize workflows as real-time or batch applications, later linking results directly into your commercial procedures.

A typical abstracted view of Microsoft Technology Stack is depicted in [Figure 4.22](#):

Application Areas	Machine learning optimization, cryptography, Health care, Finance
Solutions	Quantum Solutions
Software tools and services	Q#, Python, Javascript, Simulations
Cloud infrastructure	Microsoft infrastructure
Environment control Like Room temperature	Room Temp Controller
Quantum Device	Microsoft and collaborative vendors

Figure 4.22: Microsoft quantum technology stack

[Case Study: Microsoft's journey to quantum computers](#)

Microsoft is investing heavily in quantum computing. Its Azure cloud computing service offers select customers to access three sample quantum computers. It ranges from engineering corporation Honeywell and start-ups like IonQ and CI. Microsoft has a three-pronged goal line with Azure Quantum. It can be used for learning, developers can write programs with Q# and the QDK along with testing of their programs with the help of simulators, and organizations can use them to resolve complex business problems utilizing solutions and algorithms running in Azure.

The new platform is hardware-independent. Therefore, it can be used with any kind of quantum computing hardware. Honeywell, IonQ and a few others have joined with Microsoft to leverage the aids of Azure Quantum. This will benefit these companies to bid quantum computing as a service running quantum applications on the cloud. Microsoft did a few upgrades, which, according to them, brings a breakthrough in quantum engineering. They believe these breakthroughs will open the potentials of scaling beyond the physical limits of current quantum systems. For example, the new system now permits users to control up to 50,000 qubits using three wires. Microsoft has also advanced a new-fangled cryogenic CMOS design and a 1-cm / 2 chip for computing at near absolute zero temperatures.

Apart from others, Microsoft is also betting on topological qubits, which is one of the key elements in the Microsoft plan to carry an influential, scalable quantum computing solution to the world. Topological qubits produce a more stable footing than conventional qubits. This augmented stability permits the quantum computer to do computations that can uncover solutions to some of the world's hardest problems. Topology is a division of mathematics recitation structures that experience physical variations such as being bent, warped, compressed, or strained, yet uphold the properties of the original arrangement. When applied to quantum computing, topological properties produce a level of fortification that aids a qubit to hold information despite what's happening in the environment. The topological qubit attains this additional fortification in two different ways.

First is by excruciating the electron. This way quantum information is stored in both splits, acting similar to data redundancy. If one half of the electron runs into interference, there is still sufficient information kept in the other half to permit the computation to last. This is called Electron

fractionalization. Whereas in another method, Topological qubits are caused to have two ground states, known as ground state degeneracy. The ground-state degeneracy makes qubit much more unaffected to environmental sound. Usually, attaining this fortification isn't practicable since there's no way to distinguish between the two ground states. Though, topological systems can use interweaving or measurement to differentiate the alteration, permitting qubits to attain this extra protection.

Whereas qubits can be advanced in diverse ways. One way is to use it is the topological qubit. For the topological qubit, Microsoft brings together experts from condensed matter physics, mathematics, computer science, electronics, nano-Technology, and materials science to invent an exclusive method that produces specific crystals with the properties desirable to make the topological qubit a reality. Also, the experts from many fields of knowledge have come together to understand the topological qubit, together with instrumentation and measurement technology, information technology, computer science, natural computing quantum informatics and algorithms, quantum error correction, and software application development.

Linking these diverse areas of knowledge has led to advance methods that intersect multiple technical features of the subject stacks in order to realize a topological qubit. Just have a quick look at how different fields contributes to the development of topological qubit, as follows:

- **Theory and simulation:** This field of study is useful in bringing vision into reality because with this creation of a rapid design (in almost no time), simulation of the complex systems, and enhanced prototyping process is achieved.
- **Fabrication:** This field becomes handy in bringing unique fabrication approaches onboard. Additionally, it is very useful in finding new-fangled techniques to map different properties.
- **Materials growth:** This field of study is of tremendous use when it comes to developing innovative methods to construct materials using special growth techniques and properties to produce the exact and customized chattels required at the nanoscale.
- **Measurement and quantum control:** This area of investigation is very important when it comes to tuning devices to achieve specific

needs with accuracy.

At Microsoft, the development of the topological qubit continues, bringing us closer to scalable quantum computing and finding solutions to some of the world's most challenging problems by use of the novel as well as traditional areas of study, research, and innovations.

Other companies' technology stacks

Almost similar steps you need to follow while you are developing software on IBM cloud infrastructure. The tools, framework, and programming language would be different.

IBM's quantum computing ecosystem is another example of the type of ecosystem that present-day quantum vendors are offering. The IBM Q Network is a comprehensive ecosystem of Fortune 500 corporations, leading academic organizations, start-ups, and national investigation labs, empowered by IBM's quantum computers, experts, engineers, and consultants. Members already work together on initial commercial applications. In addition to the IBM Q Network, the IBM Q Experience and Qiskit permit free-tier access to IBM's quantum computers and simulators, and educational resources. IBM Q Industry Consulting allows customers to make quantum roadmaps and recognize industry-specific applications. IBM can also aid customers to define their cryptography approach for the quantum era. It offers, in a nutshell, the entire quantum stack from top to bottom, that is, from hardware to software.

QC is an evolving field and continuous research and developments are happening in the area. Hence, any organization that is considering QC for their business needs to adhere to the following guidelines:

- Innovate with quantum technologies
- Cooperate with a community of pioneers
- Build next-generation apps today
- Lead with responsible and accountable innovation

Question and answers

Can we ever get quantum computers down to a very small size (like a desktop computer or laptop) and workable in hot, noisy, heavily externally influenced environments like classical computers do today.

Yes, it can be done... For one, an analogy will help in understanding this. Our brains work on quantum mechanics, and it's as warm, drizzly, and ill-treated by the exterior environment as it comes. Nature figured out a method to do it. One day, humans may be capable to figure it out as well for QC. Humans seem particularly able to shrink gears once we figure out how to shape them. Two of the components, the application software, and what is called the stack, justify extra attention.

Story 1: Why Honeywell?

Honeywell is a proficient corporation in most of the component fragments required to shape and function a quantum computer. They advanced their know-how while resolving real problems for numerous businesses across Honeywell. The technology development by Honeywell is well-versed by real-world applications and restraints. If we broadly visualize the problems of quantum computers, most of them are the control problems. Honeywell gets the advantage here because they are the world's leading controls corporation. Henceforth, Quantum computing is a natural postponement of uniting Honeywell's proficiency across multiple technical disciplines. Few important points related to Honeywell quantum journey is mentioned below.

Honeywell has the skill and a long history of fabricating effective products and services using the same technology essential in shaping a quantum computer. These comprise state-of-the-art hardware and software control systems, progressive electronics, photonics, lasers, modulators, fibre optics, ultra-high vacuum surroundings, and cryogenics.

Honeywell has put their experience and expertise to work. It fabricates one of the most critical components, the ion trap, in its microfabrication production facility in Minneapolis. It also makes its control system and user interfaces.

Honeywell Quantum Solutions is one of the world's greatest progressive quantum computers, founded on trapped ion technology. Honeywell's best

teams of scientists and engineers with influences and skill in multifaceted hardware and software control systems, advanced optics and lasers, magnetics, cryogenics, atomic physics, and ultra-high vacuum environments, who are experts in design and building systems with best-in-class performance, are involved in the development of quantum systems. They are working in terms of both fidelity and qubit count. Honeywell trusts that quantum information systems will transform the way several industries work. As a foremost technology company, they mean to help shape the adoption and integration of those systems into the markets for customers.

While the big technology companies like IBM, Google, and Intel, use superconducting qubits, the trapped ion is Honeywell's technology of choice. This choice provides them several technical advantages. Trapped-ion quantum computers use many, distinct, charged atoms (ions) to affect quantum information. Honeywell systems use electromagnetic fields to hold (trap) each ion so it can be manipulated and determined using microwave signals and LASERS.

Due to atomic structure, trapped-ion qubits can be consistently manufactured and controlled more effortlessly and rapidly compared to other qubit technologies that use atoms; however, to achieve this, they developed a proprietary technique. As it is based on the natural behaviour of the atoms and ions, they use the term Nature's Qubit, for their systems. Also, they claim the qubits which they generate through this process are all identical and defection-free.

Further, selecting a trapped ion for the QC is because they gained prior experience with the technology when Honeywell previously worked on a program with **Intelligence Advanced Research Projects Activity (IARPA)** in 2014.

Honeywell Quantum Solutions has demonstrated record-breaking high fidelity quantum operations on their trapped-ion qubits. It is a major step towards producing the world's most powerful quantum computer.

The quality of qubits, and their quantum operations, is designated by a metric named fidelity. It is a significant feature of any quantum computer since it describes how big of a problem may be cracked while upholding its

quantum advantage. They attained separate qubit operation fidelities of 99.997 percent, presently the best-reported performance of any addressable qubit technology. For using ion-trapped successfully, Honeywell formed three parallel operating zones, which permits sole quantum gates to be achieved on dissimilar qubits in each zone concurrently. The procedure of applying parallel, independent zones delivers faster quantum algorithm execution times as well as better options for qubit connectivity. Parallel operating zones are an important distinguishing feature of the trapped-ion system and will allow exceptional capability when joined with high-fidelity quantum gates.

Honeywell also achieved the previously mentioned success by assimilating several classical sub-systems together with exact optical control. To date, Honeywell has formed qubits in each operating zone and has already started optimizing the so-called quantum logic gates in each region(s). Also, several techniques are used to develop and determine parallel operations to become successful in the laboratories. These can be further scaled to support larger qubit counts and computational steps. All these achievements provide Honeywell with the tremendous capability not only to come up with a first-generation of effective quantum systems but also quickly move towards constructing future systems.

Currently, Honeywell has progressed its testing with qubit devices based on the innovation which uses trapped-ion. They called it first generation micro-fabricated trapped-ions which will be ready for commercial purposes. This includes multiple novelties along with a technology roadmap provided by **Honeywell Quantum Solutions (HQS)**.

Traps include different arrangements, or architectures, of qubits. The dissimilar architectures under evaluation include one-dimensional (1D) chains of ions as well as an ascendable design of two-dimensional (2D) arrays. Also, this eradicates qubit count limitations forced by a linear qubit arrangement. The success of these devices provide an enabling component for Honeywell's first quantum computer.

The fabrication of traps also demonstrates a unique feature of our end-to-end quantum computing technologies. This includes all components from the traps that form computational qubits to the augmented control

electronics and user interfaces which are already in place for managing critical infrastructure around the globe.

Honeywell challengers

Currently, the start-up IonQ is also one of the leading trapped ion quantum computers developing company. Apart from this, there are commercial quantum annealing systems from D-Wave Systems with 2000 qubits that exist somewhere on the horizon. Moreover, there are superconducting quantum computers with 16-72 qubits from Google, IBM, Intel, and Rigetti Systems that are also waiting for their chance to turn the market around.

Applications

Why is all this happening? One of the reasons out of several others is that global macro trends show a desire for more computational power. This is happening while classical computing is showing signs of stalling, where power density and quantum effects are limiting the performance of ever-shrinking transistors. Now and in the coming times, the world is going to build up a gigantic set of data. The **Internet of Things (IoT)** and the **Industrial Internet of Things (IIoT)** are providing a rich data set. This needs to solve important problems through machine learning. Along with that, Quantum computing provides the promise of exponential performance improvement and data crunching capabilities to meet these needs.

It's expected that many industries will benefit from quantum computing. I believe that chemical, materials, and pharmaceuticals will be the first to see benefits, as they can extract meaningful insights from quantum computers with even small numbers of qubits. If a pharmaceutical company can reduce development time, for example, they can potentially save billions of dollars. As the qubit count increases and algorithms advance, we expect artificial intelligence and machine learning applications to enable a variety of other industries.

However, it's early days. Classical computers have been around for six decades or more, and we couldn't have predicted they would wind up with the capabilities they have today. We expect quantum computers to be able to solve more complex problems as time goes on. The quantum computers

of the next few years will provide a testbed to develop sophisticated quantum algorithms to handle the real world and more precisely the natural problems in real-time. With this optimistic note, I conclude [chapter 4](#) and move towards [chapter 5](#), but before that let's have a look at the mind map of [chapter 4](#) to get a snapshot of concepts discussed in a visual manner. [Chapter 4](#) mind map is given in the [figure 4.23](#)

Mind Map of the chapter

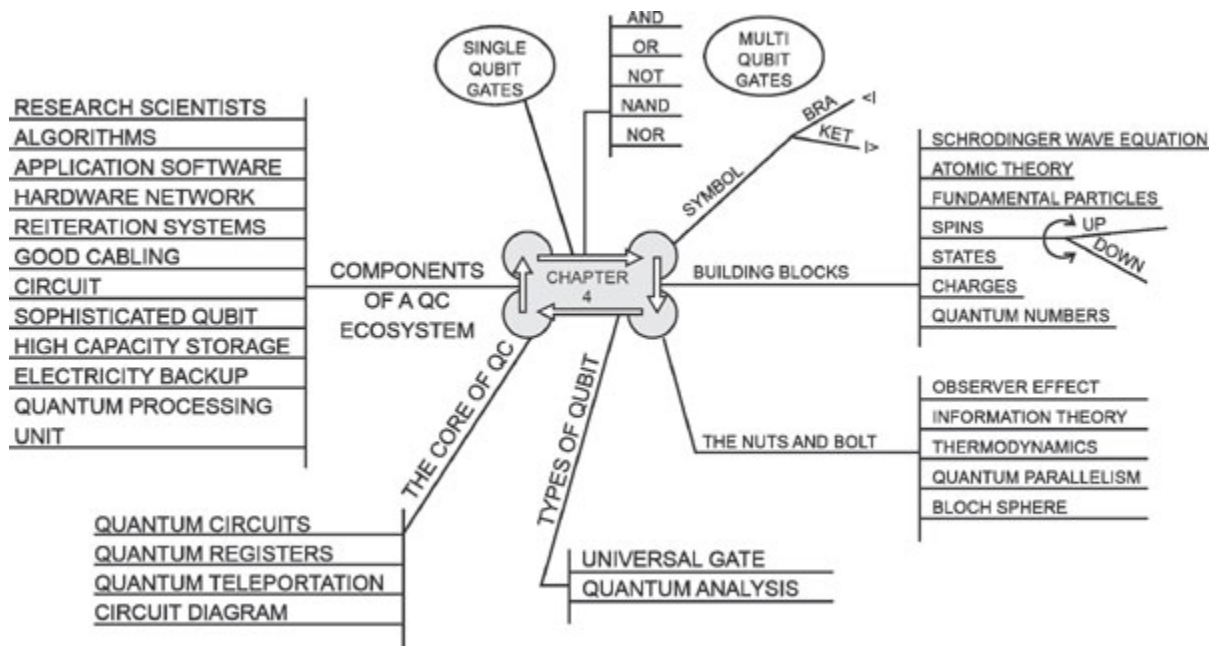


Figure 4.23: Mind map for [chapter 4](#)

Conclusion

This chapter started with providing the details on qubits, qubytes, registers, and ends with covering some research and the prospects of QC given by corporations like Microsoft and Honeywell. In between, it covered a lot of ground related to quantum circuits, gates to superposition, and entanglement in detail. In the next chapter, we will cover QC in association with AI.

Broadly, this chapter was divided into three parts. The first part discussed the technical aspects of quantum computers. We discussed a few of the concepts broadly in the first chapter but those were not covered in a detailed

fashion. Also, the mathematical elements were missing from them. This was intentional due to two reasons. First, I wanted to give the reader an overview so that s/he would be aware of the bigger picture. The second reason was that I assumed that the reader did not have a foundation of basic mathematics which is relevant in the context of quantum physics and computing.

The second part of the chapter covered some important frameworks and discussed how they are used for realizing quantum programs for practical purposes. For instance, how different libraries are organized and placed on the overall technology stack of a quantum computing system and framework.

Finally, the third part of the chapter gave the reader a view about how business is using the ecosystem of QC. For example, how they would be able to benefit by using “collaboration” effectively and efficiently.

Important areas of quantum computing ecosystem were discussed in length in this chapter. Concepts like quantum volume in reference to error correction was also discussed in details. Further, hybrid approach of classical and quantum computers was discussed, which is greatly popularized by the company like D-Wave.

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Important terms and concepts used in the chapter

Some important terms used in the chapter are mentioned below for the quick reference

- Spin, or occasionally nuclear spin or intrinsic spin, is the quantum version of angular momentum. Dissimilar to regular angular momentum, spin has nothing to do with real spinning.
- The Boltzmann constant establishes the relationship between temperature and energy. It is one of the crucial theories or in the field of thermodynamics. Further, Thermodynamics is the area of the study which investigates heat and its relationship to different types of energy. Boltzmann constant was named after Austrian physicist Ludwig Boltzmann who lived between (1844–1906). He is considered as one of the forerunners of statistical mechanics.
- The entropy of a thing is a measure of the quantity of energy that is unobtainable for effort or work. Entropy is also considered as a measure of the numeral of likely preparations of the atoms in each system. In other words, entropy can be considered as a measure of indecision or haphazardness.
- An irreversible process destroys the act of a thermodynamic system, which may be intended to do work or harvest refrigeration or cooling, and result in entropy production. The entropy cohort throughout a reversible process is 0.

- The information entropy, frequently termed as entropy, is an elementary amount in information theory related to any haphazard variable, which can be understood as the usual level of information, astonishment, or indecision characteristic in the variable's thinkable consequences.
- Absolute is the lowermost temperature at which the system is in a state of bottommost likely energy. The grade or strength of heat existing in a material or thing is termed as temperature.
- Joule is the standard unit of energy in computer electronics and overall technical or scientific applications. One Joule is demarcated as the quantity of energy used when a force of one Newton has pragmatically made a movement of one meter.
- A CPU core is fundamentally a CPU's processor. Initially, all processors had a single core that could emphasize a solitary job at a time. But now, CPUs like dual and 18 cores exist, respectively, which can put effort or work on a different job or jobs. Hence, a multi-core processor is nothing but a common computer processor integrated circuit with dual or further distinct dispensation or processing units, which is called cores. Each core can read and perform program commands as if the computer had numerous processors.
- Chip is synonymously used for a microchip, the extremely multifaceted yet minute modules that stock computer memory or deliver logic circuitry for microprocessors. The most common chips are the Pentium, which is a product of Intel corporation.
- An integrated circuit or massive integrated circuit is a group of microelectronic circuits on one minor plane, a part of semiconductor material that is usually made of silicon.
- Intellectual property rights comprise patents, copyright, industrial design rights, trademarks, plant diversity rights, trade clothing, topographical signs, and a few authorities' trade confidences as well.
- Copyright is geared the way storybooks and creative works, such as books and videotapes. A trademark defends substances that help describe a business brand, such as its symbol. Whereas a patent is a method of intellectual property that stretches its proprietor the lawful

right to eliminate others from creating, consuming, vending, and introducing an invention for a restricted period of years, in exchange for publication and allowing public revelation of the invention.

- DNA or deoxyribonucleic acid is an extended molecule that comprises our sole genetic code – the genetic code that regulates all the physiognomies of a living thing. Like a procedure book, it grips the commands for creating all the proteins in our bodies. Finally, the genome is made of a biochemical named deoxyribonucleic acid, or DNA. It contains nucleotides. Each nucleotide encompasses a sugar and a phosphate molecule, which make up the ‘backbone’ of DNA, and, one of the four biological bases. The bases are adenine (A), guanine (G), cytosine (C), and thymine (T).
- Euclidean space is an important space of classical geometry. Originally, it was thought of as the three-dimensional space of Euclidean geometry, but as time passed and understanding of contemporary arithmetic enhanced, it led to the notion of Euclidean spaces of any nonnegative integer dimension, which coexists with the three-dimensional space and the Euclidean plane (having two-dimension). It is used to differentiate from other spaces that were later revealed in physics and modern mathematics.
- By concept, a density matrix is a mathematical arrangement of rows and columns (matrix) that expresses the statistical state of a system in quantum physics. The likelihood for any outcome of well-defined measurements upon a system can be calculated or derived out from the previously described density matrix for that system. The extreme points in the group of density matrices are known as the pure states, which can also be inscribed as state vectors or wave functions in quantum mechanical systems.
- Earth rotates around its axis, and a ballet dancer with his arms rounded revolves around his axis. These are some of the utmost common examples of rotation around an immovable or static axis. The speed of the ballet dancer changes when his stretched arms are crumpled inwardly. Why does this happen? Well, the reason is the altering angular momentum in the circular motion of the ballet dancer. Linear momentum is the product of mass and velocity, $p=mv$. Angular

momentum is about the impetus of a revolving body. For instance, if a body revolves at r distance from the center of rotation, we compute angular momentum by multiplying the linear momentum $p=mv$ by a static radial distance r . Henceforth, angular momentum $L= r \times mv$, that is, Angular momentum is a vector quantity. Similarly, the subsequent vector is a cross product. In the case of the cross product (against dot product), the resultant acts along the direction perpendicular to the location comprising the radius r and object of the motion of the linear momentum (mv).

- The Hadamard gate is a single-qubit operation that maps the basis state $|0\rangle$ to $\frac{|0\rangle+|1\rangle}{\sqrt{2}}$ and $|1\rangle$ to $\frac{|0\rangle-|1\rangle}{\sqrt{2}}$, thus creating an equal superposition of the two basis states $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$.

CHAPTER 5

Artificial Intelligence and Associated Technologies: A Review

According to Kaplan and Hoenlein: “*Artificial intelligence (AI)—defined as a system’s ability to correctly interpret external data, to learn from such data, and to use those learnings to achieve specific goals and tasks through flexible adaptation.*”

Artificial Intelligence is a division of science, that deals with serving machines to find answers to complex problems in a human-like way. This usually includes features from mimicking human intelligence. The vital goals of AI investigation comprise intellectual, knowledge, planning, learning, natural language processing, insight generation, and the capability to move and operate objects.

All-purpose intelligence is among the field’s long-term objectives. Numerous tools are used in AI, together with forms of search put, mathematical optimization and logical approaches based on probability and economics. The AI umbrella contains multiple disciplines including computer science, engineering, information technology, electronics, physics, mathematics, logic, psychology, sociology, linguistics, philosophy, neuroscience, and artificial psychology. Hence, without any doubt, we can conceivably say that currently Artificial Intelligence or AI is the most thrilling technology, and also potentially, it will become more exciting in the near future. This chapter will focus on artificial intelligence, deep learning, and other associated technologies.

Structure

In this chapter, we will cover the following topics:

- Introduction
- Intelligence and AI

- The AI hypes
- Can machines think?
- Let's define AI
- Why AI now
- The relevance of AI
- Types of AI
- Sometimes hard sometimes soft: AI computing paradigm
- What is Computational Intelligence (CI) and how it fits in the overall picture of hard soft, strong, conventional computing?
- AI research and its impact on the associated subsystems
- Machine learning revisited
- Machine learning types
- Deep learning
- Let's build AI projects
- AI, Business of intelligence and the projects
- The essential skills for AI projects
- Does it mean managing AI projects are difficult?
- The relation between AI, business, and perfect decision
- How to create an AI solution?
- Well understood, now, tell me how AI is related to Cloud, IoT, and big data:
 - Cloud computing
 - Internet of Things (IoT)
 - What is big data?
- Data, data and data: I want to know it in detail but technically
 - Schema-on-write ecosystems
 - Schema-on-read ecosystems
 - Data lake
 - Data vault

- Hubs
 - Links
 - Satellites
- Data analytics
- Data analytics
- Think holistically
- Building IoT enabled AI
- A pictorial real-life case study for IoT
- Building Big data-based AI solution
- The technology for Big data and AI
- The advantages of HDInsight
- Where is a quantum by the way: quantum computing and artificial intelligence?
- OK now I want to understand the interrelation between AI, ML deep learning, and quantum machine learning
- Use cases for AI
- Story 1: future of humanity-- integration of brain science, AI and quantum computers
- Story 2: AI, culture, arts, and quantum computers
- Story 3: the security of an AI is an emerging topic
- Quantum use case 1: Quleap -- First integrated AI platform for small molecule drug design
- Quantum use case 2: detecting market instabilities
- Quantum use case 3: optimizing radiotherapy treatments, fighting with cancer and the quantum computing algorithm
- Mind Map for the chapter

Objective

After studying this chapter, you will be able to understand, explain and relate with the concepts of artificial intelligence, big data, and Internet of Things

(IoT), cognitive and quantum computing as this chapter describes these concepts in a lucid fashion.

This chapter will also discuss AI and its interrelation with quantum computers and present a synopsis of the future pathway. It will explain how QC along with cloud computing, big data, AI, and IoT will shape the next few decades. This chapter will present the case strongly on why AI is here to stay and, in the future, why it will be integrated with quantum computing for good. This chapter also presents some practical use cases and case studies.

In a nutshell, this chapter will:

- Give users of the book a brief understanding of the AI-related concepts.
- And bring it in from the perspective of IoT, big data, and cloud computing.

Introduction

AI application areas are as varied as finance, retail, marketing, utilities, accounting, manufacturing, journalism, and law. AI systems are working together with specialists to allow us to do our best. But it is also a fact that before reaching this point, AI follows a long past of technological growth. Technology has not ever really substituted people. In its place, technology enhances people's capabilities. Technology like AI provides us superpower capabilities. For example, the cell phone allows talking to people thousands of miles away at a fast speed. AI helped us to become specialists on our first day at work. Likewise, instead of learning wisdom from multiple places, documents, and manuals, and doing trial and error to decode the facts, we can now make complete decisions with nominal training.

Intelligence and AI

The way AI is progressing, the day is not too far when an AI has the free will, intelligence, means, and motivation to excel but never be dangerous for us. AI is a technology used by people. Similar to any technology, there is a possibility of misuse. But if cast off sensibly, AI will bring filthy assistance to society, individuals, governments, and businesses. As AI grows, the complete industrial, personal, social, and even spiritual landscape will change. AI will not only be empowered to take improved and improved

decisions, but it will also allow us to do our work more proficiently. Therefore, in the future, those who accomplish to set free the powers of this technology, have a true potential to flattering the leaders of the forthcoming novel world.

After the discovery of computers, their competence to do numerous everyday jobs effectively is rising exponentially in recent times. Humans have advanced the power of computer systems, especially in the context of their varied working domains and cumulative rapidity. This brings drastic changes in humanity as a whole.

All this becomes possible due to the subdivision of Computer Science called **Artificial Intelligence (AI)** which chases making computers or machinery as intelligent as human beings. Definitely artificial Intelligence is a method of making a intelligent system, a computer-controlled robotic system, or software to reason intelligently, in a similar way to intelligent humans.

AI is proficient by studying how the human mind contemplates, and how humans study, decide, and work while trying to resolve a problem, and then utilizing the results of this study as a foundation of evolving intelligent software and systems.

Artificial intelligence is fundamentally a science and technology founded on corrections such as Computer Science, Biology, Psychology, Linguistics, Mathematics, and Engineering. A main push of AI is in the growth of computer functions related to human intelligence, such as mental, learning, and problem-solving. Mind it, AI is a multi-disciplinary field that takes inputs from different areas of knowledge. Also, AI has applications in different areas as well. Please refer to [Figure 5.1](#) for details:



Figure 5.1: AI is a multi-disciplinary field that takes inputs from different knowledge areas and finds its applications in different industries and corporates

The AI hypes

Fresh developments in AI have given computers the skill to program themselves. Whereas conventionally, computers were mechanical and had to be fed a rule for each likely possibility by a human programmer; these new AI procedures allow computers to learn their own rules from data. Historically, people learned by practising through internships and experience. For example, young untrained people working together with more knowledgeable people or enabling the transmission of competencies and facts through peers. But after the advent of the printing press, this way of learning is altered. Specialists were now able to write down their knowledge, and limitless pupils could learn from it. However, learning from a book is hard, too. AI systems removed this hurdle. AI systems can perceive specialists, excerpt patterns, and interpret skilled people's behaviour. Thereafter they skilfully coach beginners to perform at proficient dexterity almost naturally. AI is impacting all walks of life. For example, in medicine, AI empowers inexperienced doctors to identify cancer with the same precision as most experts. The achievements of AI are endless. For

instance, Cresta, a Palo Alto-based start-up, has built an AI system that turns beginner sales agents into celebrity performers. Alphabet's (Google parent company) self-driving car is a reality now.

Can machines think?

Alan Turing in 1950 asked this fundamental question in his pivotal computing research paper. Even though he was not aware that he is kick-starting a new field that potentially could change everything including the way we live, work, interact, and die... This was the starting theme of the field of AI/ML. To his arguments on the question, "Can machines think?", Turing shaped one of the initial descriptions of artificial intelligence. The Turing test, as it is acknowledged, determines that a computer can think if throughout a series of double-blind communications and chats, a human interrogator cannot distinguish between the machine and the human being.

The development of AI/ML through the twentieth century has not been persistent. As growing figures of computer scientists started developing algorithms and programs that would empower thinking computers, the data requirements for broad-based applications became a preventive condition. Huge investments in AI/ ML, particularly by the Defence Advanced Research Projects Agency and additional federal agencies in the United States, produced an improved understanding of the possible applications and uses, but software development far outstripped hardware. As an outcome, the quicker developments in AI/ML, and investments in fundamental research started to dry up. At the same time, automation through robotics developed as a method to upsurge efficiency, particularly in manufacturing. Though these robotic machines were power-driven by complex software for a task, their capability to learn, and therefore be observed as intelligent, was inadequate. The arrival of quicker computing and processing rapidity in the late 20th century and the amplified use of cloud-based computing in the 2010's has given rise to a rebirth for AI/ML, as these systems are now capable to do more complex calculations on bigger data sets, at advanced speeds, and with superior accurateness, than had ever been visualized. However, the question of what AI/ML is and how it can be functional leaves an open question. Hence, there is no sole or extensively acknowledged meaning of AI/ML that has emerged. But if we go through some reputed

books and opinions of experts of the field. We are able to derive some important characteristics of AI/ML systems. Fundamentally these systems can be imagined as:

- Think like humans by simulating biological brain-like behaviour.
- Act like humans by replicating human functions.
- Think rationally by modelling common sense.
- Act rationally in complex situations.

These characteristics of AI/ML lead towards the subject of levelheadedness and human-like behaviour into the essence of the field. The consequence is that intelligent machines are supposed to have the competencies to think and perform in ways that permit them to work with, or even in place of, the humans that shaped them.

Let's define AI, ML

One definition given by Wikimedia Foundation says that AI investigations can be thought of as the study of 'intelligent agents' so that any device that observes its environment can take actions that make the most of its chance of effectively attaining its goals.

Along with AI, machine learning is another field that has grabbed a lot of attention in recent times. And at a few places, they are incorrectly used simultaneously. Hence, there is a need to distinguish between them and also understand the right context of these technologies.

There are also numerous viewpoints on the association between AI and machine learning (method of finding patterns in huge data sets). The following are a few views; however, it is important to note that none of them is accepted by all the schools of thought related to AI:

- AI is a broader ground of study than ML. It is a superset of concepts which ML derives from it. For instance, while ML includes working with algorithms, just as AI does, AI is as well deeply used with robotics, behavioural science, and so on.
- AI pacts with more up-to-date algorithms and methods than traditional ML. In these definitions, novel know-hows (such as deep learning),

when pragmatic to visual recognition or natural language processing tasks, are a well-thought-out share of AI.

- The connotation between AI and ML undergoes alteration over time.
- Machine learning enables AI to Learn without any kind of supervision needs, and a capability to recognize patterns in streams of inputs. Learning with satisfactory supervision includes classification and numerical regressions. Classification regulates the class an object fits to and regression contracts with gaining a set of numerical input or output examples, thereby determining whether functions will enable the generation of appropriate outputs from particular inputs.

In a nutshell, we can say that **Machine Learning (ML)** is an amalgamation of problematic formulation, optimization, and result assessment. The solitary share of ML that AI can do on its own is optimization. Human beings have been found to express the problem and understand and put on the consequences in the background of their business, as well as describe suitable evaluation metrics.

Well, so now we have a border understanding of AI. But how come we say this system is an AI system? There must be some criteria that exist for that. [Figure 5.2](#) showcases the components of AI:

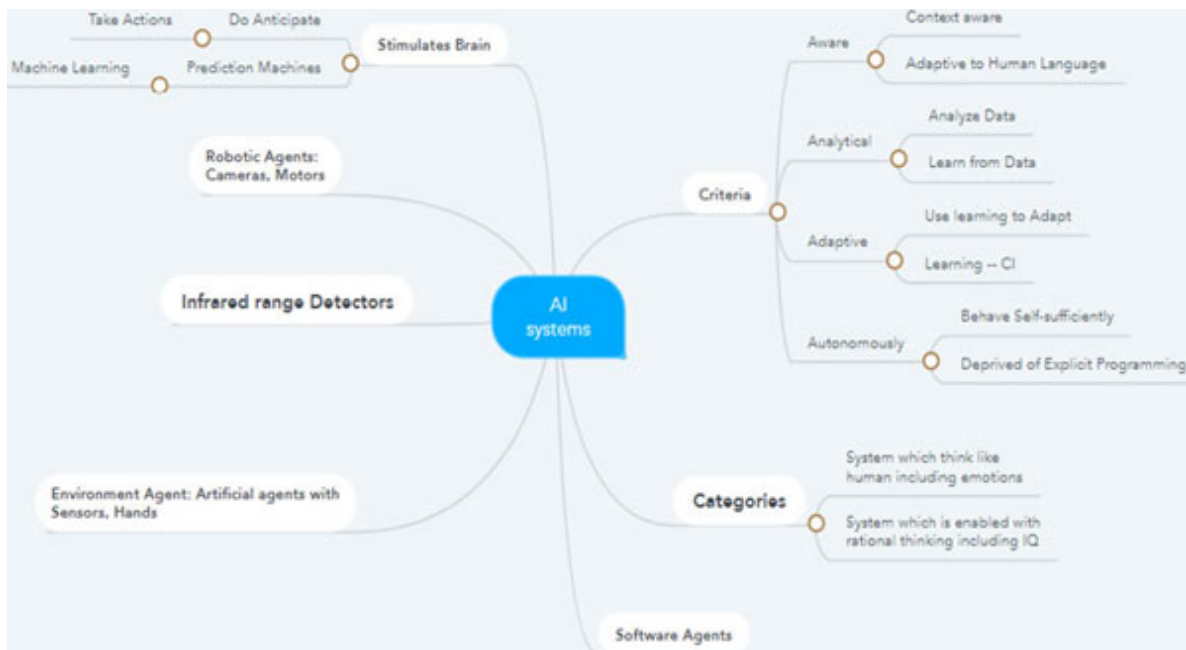


Figure 5.2: Components of AI

It is important to discuss at this point as to why the whole world is crazy about AI now. Also, it is equally important to enquire how AI is relevant when we are looking for the new paradigm of computation called quantum computing. We will discuss all this in the latter part of the chapter, but before that, let's go ahead and figure out why there is so much buzz around AI at the moment.

Why AI now?

Data generated by smartphones, multiple types of sensors, industrial drones, cube satellites, smart meters, intelligent watch, fitness tracker, smart bulbs, homes, cities, tracks, wearables, surveillance cameras, and millions and millions of other things are uploaded on public, private, and hybrid cloud networks and on the Internet in a lightning speed. Enterprises, businesses, and individuals find it difficult to manually manage this BOOM in data. Practically, they can't manage this huge data in the traditional technical ways. Therefore, they need an automated system that manages and analyses data in real-time on a need basis. In the era of the **Internet of Things (IoT)** and the connected world, continuous, effective, efficient, and innovative self-learning automation is the key to solving the challenge thrown by the business world. Enterprises need a smarter and intelligent system that would efficiently interact with their environment, users, and systems. Moreover, they need systems to become more intelligent so that the systems are able to infer based on the history of their interactions with users and environments. Enterprises need efficient and tuned technologies and tools that can help intelligent devices and systems to learn from their experience (data). Enterprises need a mechanism so that instead of asking analysts to generate insights from data and make decisions that drove changes in the system, the system itself learns from data automatically and responded appropriately and makes changes wherever required.

Also, enterprises do not want the system to heal manually. They want the system to be capable enough to heal themselves based on the symptoms. AI-based systems potentially fulfil all the criteria and requirements of a system to achieve self-healing capabilities. It also provides a clear case that enterprises need AI to make IoE / IIoT (internet of everything or Industrial

internet of everything)*, big data systems, and cognitive systems more and more usable.

Enterprises have already started taking benefit of this all-new technical ecosystem and launching devices and systems. For example, A French designer duo has unveiled a new concept for a future machine learning-based smartphone. The technology used in the phone is holograms a specific kind of laser-created image or photograph where the things appear solid and realistic-looking rather than flat, voice control, and face tracking (artificial intelligence and deep learning) with glowing gelatinous features. The smartphone is called Alo. Alo is designed in a way that is appropriate for the palm. Also, a combination of luminous skin emits vibrations that are produced by heat depending on the activities of the individual. Its skin repairs automatically as soon as it is damaged.

Murqan, Dad and Mom

Do not worry...his nappy is smart. It would learn his habits automatically and inform you accordingly



AI is the backbone of all these innovations. And, we are not limiting ourselves here and going beyond all this. Riding on the data revolution, AI becomes more and more relevant with every passing day and embedding itself in our daily lives. Let's give a detailed look at its importance.

[The relevance of AI](#)

Technology companies are investing heavily in AI and associated technologies. They are enhancing existing products to sustain in the market, and at the same time, few companies are involved in creating altogether new product categories. The by-product of this comes in the form of offerings related to intelligent home robots and vehicles. The new generation AI solutions is powered by big data technologies, virtual reality, IoT, cloud, and cognitive computing (which is a subsystem of AI). And becoming part of our life by bringing products and services that make our life smarter and easier. It works based on the intelligent and efficient use of data by which we are surrounded nowadays (Facebook feeds, Amazon reviews, travel history and much more). For instance, Alexa now becomes a necessary household item and Uber is our first choice for booking cab services. They gear up our life by efficient use of data or rather big data and algorithms. Hence, data is given a new meaning by the levels of intelligent processing. Therefore, offering like the intelligent bed, truth brush smart automobiles are not only solving day-to-day needs but also enabling and enhancing human capabilities. The business benefits of AI are much broader than the traditional way of cost savings that are broadly associated with the term “automation.” It is the automation of automation that travelled a long journey from tribal automation to the automation of automation.

AI ecosystem improves almost all-important parameters of productivity of enterprise if applied correctly at the right place with right data sets and algorithms. The following are some of the areas where it helps in the border way. If you keenly observe, it affects all levels of Maslow’s hierarchy of needs (he defined five classes of human requirements starting with physiological followed by safety, love, self-esteem, and self-actualization and arranged in the order of pyramid). It enabled top management (CXOs) for taking lightning-speed actions and decisions to the level of mere automation of reparative tasks. For example, its uses in anomaly detection are getting better, more efficient, and effective outcomes. These are the useful insights which would help top management and decision-makers (CXOs). For example, it uses sales, demand, and supply forecasting. There are a few other benefits of an AI mindset, as follows:

- To achieve efficiency and quality across levels, starting from the lowest level of organizational process tuning to advance smart self-healing

automated process. This type of automation is helpful for the mid-manager to provide effective and efficient delivery on time with the quality in the centre. It would apply to agriculture and space exploration.

- To achieve lower costs machine learning would be looked for the solution. As machine learning effectively does automation, the organization would take its help to do the same and reduce labour/staff. This reduction of skilled workers in one highly automatable area of the enterprise would further be redeployed in another area where their need is more. This strategy is an effective way to get benefit from the intellectual workforce.
- It provides capabilities to perform tasks on a bigger scale (that is, performing large-scale tasks that are difficult to perform manually). Knowledge workers are benefited from that as they get real-time on-time assistance. Also, monotonous work is done by smart machines. They would channel their energy into more knowledge-based work. Machine learning and associated technologies are guiding a new wave of innovation.
- AI helps less-skilled workers in multiple ways. There are times when low-skill workers are engaged in the mass production of life-threatening products. Hence there are chances that their safety may be threatened due to multiple adverse reasons, like high heat, working in night or nuclear reactors. Automated robots become handy there.

Technology sits in the heart of the AI ecosystem and is an amalgam of a lot of other technologies, domains, areas, and fields. It cannot be seen in isolation. We must see machine learning as one of the technologies in the wide spectrum of existing technology suites (Big data analytics, IoT, cognitive computing). But the sky is not limited here; it goes beyond all these technologies.

AI exists in multiple flavours and types. It enhances our intelligence on the fundamental level from narrow to general to superintelligence. Hence, discussing the types of AI in context of intelligence make sense here.

Types of AI

Broadly, on the present, past, and future conditions, AI can be categorized into the following three types:

- **Artificial Narrow Intelligence (ANI):** Artificial Narrow Intelligence is the ability of a computer or program to display intelligent behaviour in a unique useful area only. Imagine it to be like a genius child. Like a kid who exceeds normal human intelligence level for a specific show of talent but is typically not so clever in the other mental areas (like physics, arts, political science, and so on). The transformation between a human prodigy and a computer ANI is that the latter doesn't have even elementary intelligence in arenas which do not tumble in its domain. We have long since attained excellent achievements in ANI and have kind of tried its application areas. We are now moving gradually but progressively to the second ability of AI, which is Artificial General Intelligence.
- **Artificial General Intelligence (AGI):** This is an intimidatingly tough part to attain but several researchers believe we are on the verge of producing an AGI. AGI or Artificial General Intelligence is when an appliance has an Intelligence level equivalent to a human. The reason this is termed general intelligence is that such a program will be conscious of all human behaviours and ideally work the way humans do tasks and therefore, will be intelligent in more than one areas.

AGI-capable equipment will work at a much quicker DPR (data processing rate) than a normal human being. Also, it can unceasingly advance its source code to brand itself improved and better, and this is where it is thought to be creepier, for when an AGI develops extra intelligent capabilities and Celtic across the edge of the utmost intelligent human, it goes into the kingdom of something known as **Artificial Super Intelligence (ASI)**.

Such a system is thought of as dangerous as it has all the capabilities of humans and it contains all the characteristics including emotions (thanks to science fiction films). However, in reality, AGI doesn't hazardous at least till a computer operates at an AGI (which is far ahead from now) level and will have all the intelligence of a human mind, viz - the power of reasoning, ability to do focused work, perform intellectual activities, analysis, problem-solving and more importantly

self-awareness. Even after that, it will definitely be bound by rules, regulations and probably morality.

- **Artificial Super Intelligence (ASI):** This is demarcated as that level of intelligence that exceeds human intelligence level in every field, starting from originality, problem-solving skills, and societal skills. It can go from an intelligence level that is just to some extent higher than that of the utmost intelligent human to billions of times more intelligent. That would brand the ASI brain equated to our brain as associated with the brain of ants. However, what will occur when ASI is attained? We are not sure about that, but whenever it happens, quantum computers would play a significant role in the journey. We have to be mindful here.
- ASI is not a miracle; it will slowly happen. And, it depends on the development and collaboration of the other fields. Just take an analogy here to understand how humans see, interpret, analyse, and take action on multiple events through or their understanding of objects' states through sensors like eyes, mouths, and so on. The same is applicable to an AI system; however, all these, have artificial elements that are derived from several disciplines and sciences. For example, for logical reasoning, they take help from brain science and emulating locomotion from life sciences and so on. [Figure 5.3](#) will showcase the interaction of multiple disciplines with each other:

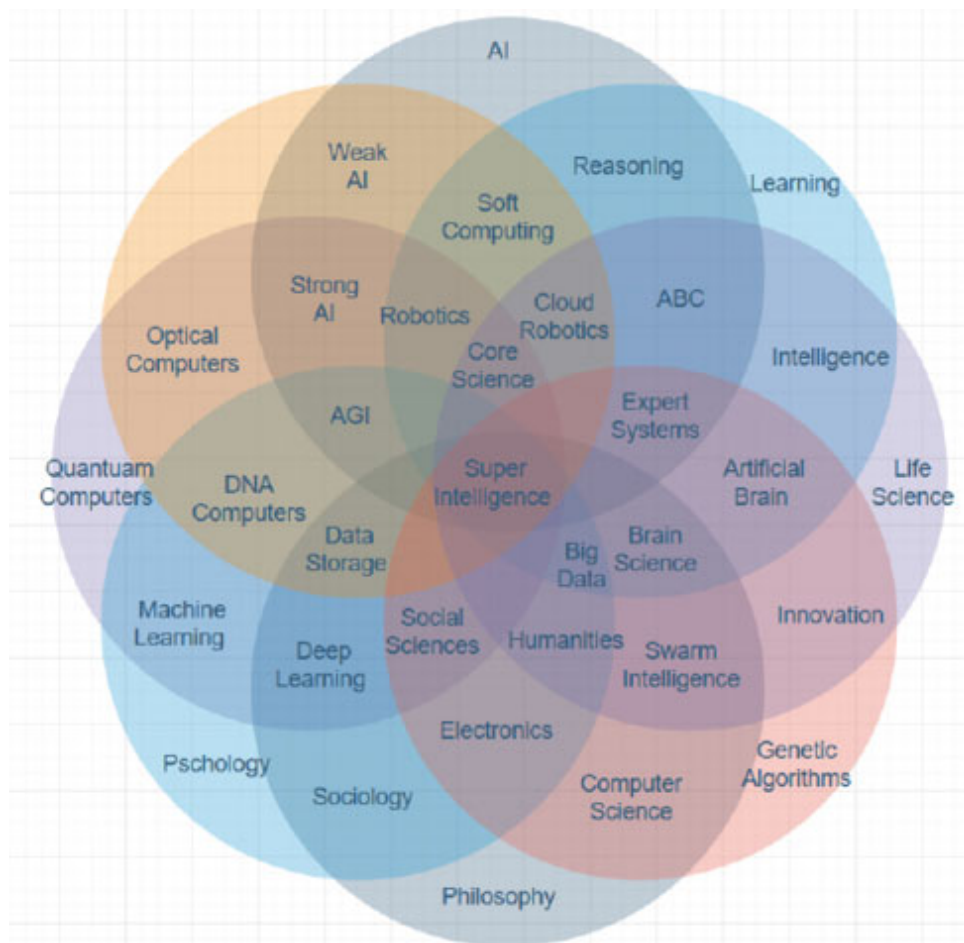


Figure 5.3: Interaction of multiple disciplines to achieve artificial superintelligence. The figure also shows that AI is fundamentally a multi-disciplinary field.

- Multiple paradigms of AI exist today and they are confusing at times. Let's clarify them.

Sometimes hard, sometimes soft: AI computing paradigm

Hard computing or traditional computing, need an exactly specified analytical model and generally a lot of computation period. Soft computing varies from conservative (hard) computing in a way that, it is accepting fuzziness, uncertainty, fractional facts, and guess. Therefore, the ideal comparative model for soft computing is the human mind. Soft computing methods enable adaptation to many situations. The following are some of the facts related to hard and soft computing:

- In hard computing, fuzziness and indecision are unwanted possessions; in soft computing, the tolerance for fuzziness and indecision is used to attain controllability, lesser cost, and high Machine Intelligence.
- Hard computing necessitates programs to be written; soft computing can develop its programs.
- Hard computing customs two-valued logic; soft computing can use multivalued or fuzzy logic.
- Hard computing is deterministic, whereas soft computing includes randomness.
- Hard computing needs precise input data, whereas soft computing is fine with vague and noisy data.
- Hard computing is severely sequential, and soft computing permits parallel computations.
- Hard computing harvests exact answers, whereas soft computing can produce approximate answers.

In literature, sometimes the term conservative AI is used which essentially represents Hard computing AI. The hard-computing AI investigation emphasizes efforts to copycat human intelligence through symbol operation and symbolically structured knowledge bases. This method bounds the circumstances to which unadventurous AI can be applied. Conservative AI mostly includes approaches now classified as machine learning, characterized by formalism and statistical analysis. This too is named symbolic AI, logical AI, neat AI, and Good Old-Fashioned Artificial Intelligence. The chief approaches that are tangled with this form of AI are as follows:

- **Expert systems:** The procedure whereby cognitive abilities are applied to conclude. Such a system is capable to manipulate big volumes of identified information to deliver a set of conclusions.
- **Case-based reasoning:** Some case or real-life problem is provided to the system and it supposes to solve what humans does.
- **Bayesian networks:** Behaviour-based AI is a flexible technique of construction of AI systems by hand. Soft AI, which should really be called "Custom AI," as it resolves a particular issue. For instance, a

Voice Recognition AI (Alexa, Siri, etc.) is referred to as a Soft AI since it only recognizes voice and lacks the capacity to detect images or answer via writing, among other things.

What is Computational Intelligence (CI) and different variant of intelligent computing

In recent times, the appearance of **Computational Intelligence (CI)** also became popular. This typically denotes the capability of a computer to learn a precise job from data or investigational observation. It is usually considered an alternative expression of soft computing. Further, computational intelligence is a set of nature-inspired computational procedures and methods to discourse complex real-world complications to which mathematical or old-style modelling can be unusable for a simple reason, as the procedural complexity for mathematical reasoning or its existence of uncertainties throughout the procedure. CI's prevalent methods consist of biologically encouraged algorithms such as swarm intelligence and artificial immune systems (which is a computing system that is influenced by biological neural networks that create animal brains). In general terms, systems that “learn” to do everyday jobs by seeing samples, usually without being programmed and guided with task-specific rules, qualify for CI. These can also be considered as a part of evolutionary computation (which is a group of algorithms for optimization stirred by biological evolution), image processing, data mining, and natural language processing.

Computational Intelligence can be seen as a subsection of Artificial Intelligence. Further, to draw a connection with other techniques, we can remember that hard computing methods work following binary logic grounded on only binary standards (the 0 or 1) on which contemporary computers are built. Whereas soft computing techniques are founded on fuzzy logic, which works by combining data to limited truths or Crisp or fuzzy systems, and this logic is the main private feature of CI.

Fundamentally, crisp logic is a portion of artificial intelligence ideas and contains decisions of either including an element or component in a group or not, whereas uncertain or fuzzy systems allow elements to be part of a group. Succeeding with this logic, each component or element can be

specified as a grade of association (from 0 to 1) and not entirely as one of these 2 values. Due to the similarity of the conceptual framework between AI and CI, people are confused with these concepts. But there's a strong difference between CI and AI. Artificial and Computational Intelligence pursue an analogous long-term area of attainment of general intelligence and can be represented as follows:

- Hard computing AI = conventional AI
- Soft computing AI = computational intelligence (CI)

As soft computing techniques are based on fuzzy logic, it can be useful in the simulation of brain functions. They can go much closer to understanding the way the human brain works by combining data to incomplete truths; this logic is one of the key exclusive features of CI. In summary:

- **Strong AI:** This is the field of artificial intelligence study which theorizes or hypothesizes that both human-level (AGI) and non-human level (ASI) are possible. Its ultimate vision is a machine that is similar to humans, especially in cognitive aspects.
- **Weak AI:** This is the field of artificial intelligence study which theorizes or hypothesizes that AI cannot achieve human-level intelligence. It sees AI as a mechanism to achieve specific expertise. For example, a champion chess-playing machine.
- **Superintelligence:** A machine that is more powerful than humans, especially in cognitive aspects.

AI is a continuously evolving field because of continuous research and development happening in the field. Therefore, it is natural that it is impacting the holistic ecosystem of AI. This is an ongoing field of study because it is asking fundamental questions. It is worth taking a look at them here.

Further, AI has different branches. Please refer [Figure 5.4a](#) for a detailed view:



Figure 5.4a: Different branches of AI

We will cover a few important ones in the upcoming sections.

AI research and its impact on the associated subsystems

The unification of thought is lacking in AI research. Due to the fluidity of the field, there is no standard unifying theory or model that guides AI research to date. Investigators disagree on many matters. Some age-old questions still sustained and a few more come. For example:

- Can artificial intelligence produce natural intelligence by learning and understanding the brain, and conducting a comprehensive study of neurology?
- Can intellectual behaviour be demarcated utilizing simple principles and theories? Does AI manifestly dictate determining a big number of completely unrelated problems?
- Can intelligence be simulated utilizing high-level symbols, identical arguments, and thoughts? Does AI able to do sub-symbolic processing?

AI is reinforced by computational psychology, computational philosophy, and computer science to deliver wisdom on these questions. For instance,

computational psychology is used to generate computer programs that reproduce human behaviour, or computational philosophy is used to the development of an adaptive, free-flowing computer brain. Likewise, putting on computer science would help the goal of the creation of computers that can-do activities that solitary individuals could attain. Thus, broadly, we can say that the humoresque behaviour, brain, and actions make up artificial intelligence. But the fact is fundamentally the AI research is associated with areas like a human way of knowledge gathering, doing reasoning, solving problems, perception, planning, learning, and the ability to move the object and take intelligence including satiation-based decisions.

Apart from ML, AI has multiple subsystems. A few are mentioned as follows:

One of the integral subsystems is **Knowledge engineering**. Knowledge engineering is an indispensable chunk of research and development efforts helping to take AI to the next level. Machines can ever so often act and respond like humans only if they have ample information relating to the world. Artificial intelligence must have admittance to objects, groups, properties, and relations among all of them to contrivance knowledge engineering. Originating common sense, reasoning, and problem-solving power in machines is hard and tedious to achieve. But through knowledge engineering and other new breeds of technologies like big data, it is obtainable.

Machine insight pacts with the capability to use sensory inputs to deduce the unrelated faces of the world, while computer vision is a nutshell provides the power to analyze visual inputs with few sub-problems such as facial, object, and speech recognition. **Robotics** is also a chief field connected to AI. Robots require intelligence to understand tasks such as object manipulation and steering, along with sub-problems of localization, motion planning, and mapping.

Most of the data humans produce is unstructured. Typical computer programs require structured data to process to give meaningful output. Whereas cognitive computers or AI systems need unstructured or semi-structured or structured data or a combination of any of these. Because they are supposed to behave like humans and humans make their decisions based mostly on unstructured data.

Therefore, AI systems also expected to make a reason form unstructured data by learning, rationalising, and portrayal of acquaintances the way humans do. 80% of accessible data is unstructured in the world at the moment. Old-style computers or programs can only make intelligence of an inconsequential portion of the unstructured data. Luckily, companies like IBM have developed cognitive computing technologies that can help humans make logic of the unstructured data.

Cognitive Computing is a computer program that can contemplate similar to a human. Or rather, a variant of artificial intelligence that can reason, learn, and make connections. Cognitive AI can make intelligence out of enormous unstructured data and reformulate it into modest analytics that can be requested by humans. A product like IBM Watson can help humans to make sence of unstructured data. IBM Watson is a cognitive computing system. It is a product that is useful in businesses like healthcare but other domains are also not untouched by it. Its next mission is to make sense of the IoT. Google has its own cognitive AI DeepMind. A cognitive computer like Watson can obtain a lot of data and make intelligence out of it quickly. However, to process every possible combination of data available today instantly, Watson will be likely to necessitate moving beyond the boundaries of classical computing. Cognitive computing uses classic computers because we don't have fully error free professional quantum computers yet but in order to make cognative computing more effective it is needed to integreat classical computers with quantum computing. [*Figure 5.4b*](#) showcases AI subsystems:



Figure 5.4b: AI subsystems and research

In addition to these subsystems, areas like the Internet of things, knowledge graph blockchain, and quantum computing are closely related to it. We can discuss them in the latter part of the chapter. [Figure 5.5](#) describes the relationships among them:

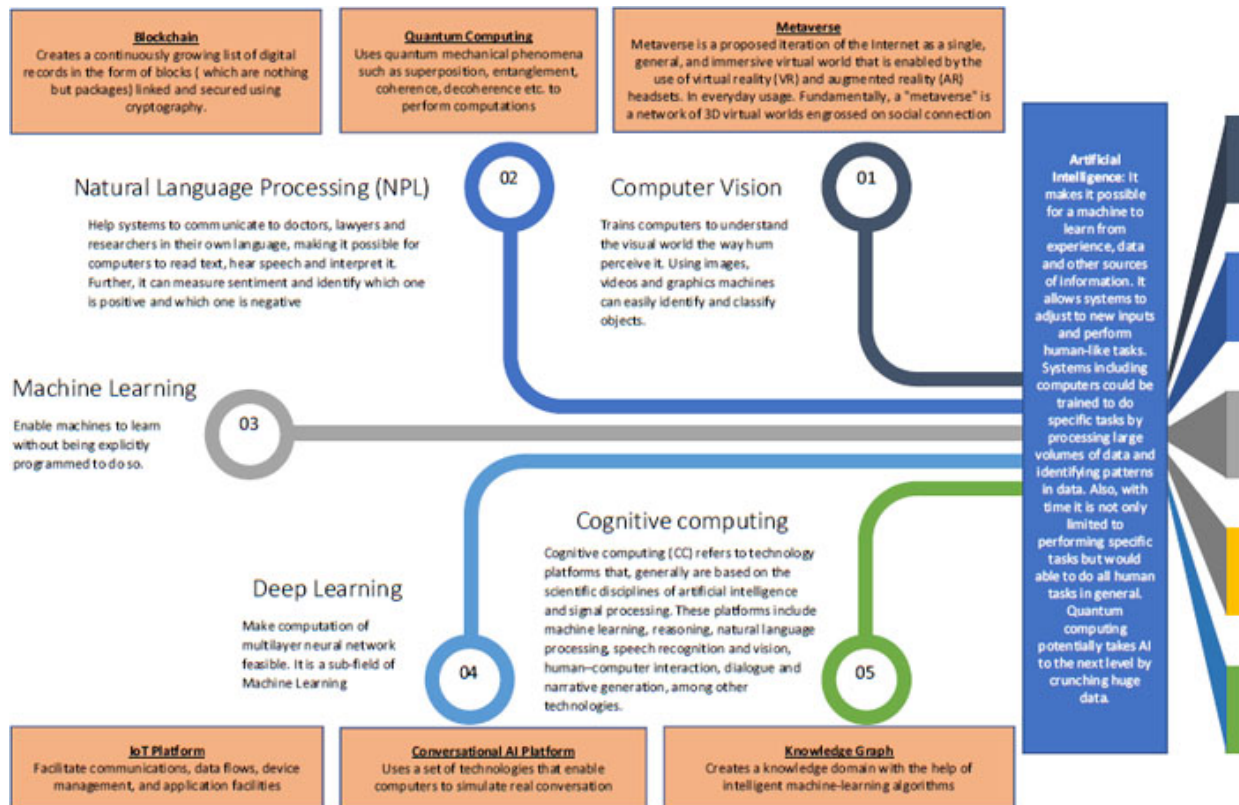


Figure 5.5: How IoT, knowledge graph, NLP, and a few other technologies work with AI

How a typical organization creates an AI-based solution, there is a process for that. We will discuss that in a bit of detail. But before that, it is important to understand Machine learning.

[Machine learning revisited](#)

The commercial ecosystem is lively with actions and thoughts about machine learning and its application to many business environments. Machine learning is the ability of systems to learn deprived of explicit coding. It originates from the idea of pattern identification and computational learning philosophy. At this juncture, it is good to differentiate between ML and data mining. Data mining is all about processing data to separate unique patterns and create relations among data entities within the data lake (or any data repository). For data mining to be effective, there are a few data-mining theories that one must recognize for data patterns. In a nutshell, the field of data mining is broader than the ML but on the ground level, both are generally used for doing the same set of

tasks. [Figure 5.6](#) explains the process of how a typical machine learning process works (architectural view).

However, it hardly matters if you are using data mining or ML; with the gathering of suitable skills, one can enhance their individual data competences. Please refer to [Figure 5.6](#) for machine learning architecture:

Machine Learning Architecture

Machine learning in statistics can be defined as an application of "Intelligent" data analytical tools which deals with the problem of extracting features from data so as to solve many different predictive tasks.

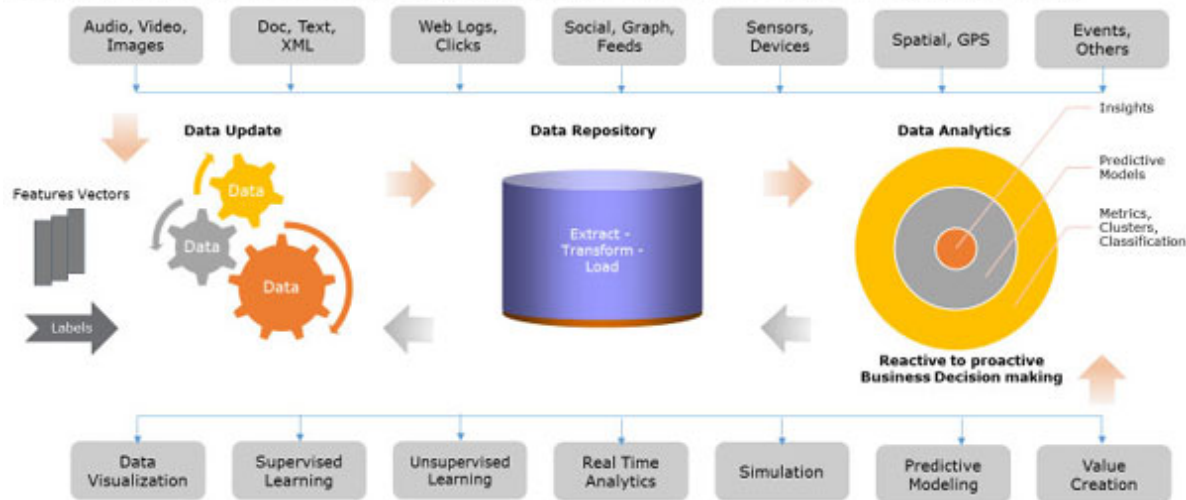


Figure 5.6: Machine learning architecture

Machine learning uses statistics as its language. Because fundamentally it is the study of the assortment, examination, clarification, performance, and organization of data. Statistics contracts with all features of data, counting the preparation of data gathering, in relationships to the design of reviews and trials. Data science, ML, AI, and statistics are thoroughly connected. ML has a few important types like supervised and unsupervised learning. Let's have a look at them.

Machine learning types

It is broadly of three types, known as supervised, unsupervised, and reinforced learning, as explained as follows:

- **Supervised learning** is the category of learning that takes place when the training cases are characterized by an accurate outcome, which gives a response about how learning is rolling. This is similar to having a supervisor who can say to the representatives, whether or not

something was right. In unsupervised learning, the goal is firmer because there are no pre-determined classifications. In simple words, supervised ML is a type of ML where, when training an ML model, an ML algorithm is provided with the input data features and the answers, also known as labels. A dataset with labels is called a labelled dataset. You can think of supervised ML as learning by example.

- **Unsupervised learning** has shaped numerous achievements, such as world-champion calibre backgammon programs and even machines accomplishing the task of driving cars. It can be a powerful procedure when there is a relaxed way to allocate values to activities. In simple words, unsupervised learning doesn't assume that data is labelled; instead, its goal is finding similarities in the data, like grouping similar activities. It's often used for self-organizing dimensionality reduction and clustering, such as K-means algorithm. For instance, if you train an unsupervised model with sufficient data containing images of sportsmen performing actions in different sports, such a model should be able to predict what group or sport a given image belongs to. This method is great if you don't have a labelled dataset, but sometimes you have some labels in an unlabeled set, this scenario is often called a semisupervised learning.
- **Reinforcement learning** is a reward-based learning system. Supervised learning or clustering is typically quicker than Reinforcement learning techniques. Clustering and classification are generally associated with unsupervised and supervised learning. Clustering can be beneficial when there is sufficient data to form groups, particularly when additional data about affiliates of a cluster can be used to harvest further outcomes due to dependences on the data. Whereas, classification is influential when they are recognized to be right or when the grouping is just random things that we would like the computer to be able to identify for us. Classification is generally essential when the decisions are taken by the algorithm. Else, it wouldn't be easy for whoever needs that input to figure out what it means. However, both the methods can be valuable and applicable to specific conditions depending on the type of problematic situation it is

pertaining to resolve and most importantly how much time it takes to resolve.

Along with machine learning, there is something called deep learning which is very popular nowadays. Let's have a look at that.

Deep learning

Deep learning is a branch of machine learning, which relays to the education of deep neural networks in the human brain. Deep learning efforts to compete with the roles of inner layers of the brain and its actual applications are identified in image recognition, linguistic conversion, email security, and so on. Deep learning produces knowledge from many layers of information dispensation. The deep learning technology is modelled after the human brain, and each time novel data is transferred in, its competencies get better. Deep learning is also using machine learning algorithms that try to model high-level constructs in data. Generally, there are two sets of neurons – ones that accept an input signal and others that send an output signal. Process wise when the input layer obtains an input, it passes on an altered form of the input to the next layer. In a deep network, there are several layers among the input, permitting the algorithm to use multiple processing layers which are in turn self-possessed of manifold linear and non-linear renovations.

In deep learning, an observation (for example, an image) can be characterized in numerous ways such as a vector of concentration values per pixel, or additionally in an abstract way as a set of ends, areas of a specific shape, and so on. One of the potential areas of deep learning is replacing handcrafted structures with effective algorithms for unsupervised or semi-supervised feature learning and hierarchical feature removal. Numerous deep learning architectures such as deep neural networks, convolutional deep neural networks, deep belief networks, and recurrent neural networks have been implemented in arenas like computer vision, automatic speech recognition, natural language processing, audio recognition, and bioinformatics where they have been made known to harvest advanced consequences on many activities.

[Figure 5.7](#) shows sequential evolution and association among artificial intelligence, machine learning, and deep learning along with the use cases in

different industries:

Artificial intelligence, Cognitive computing, Machine and deep Learning based use cases and applications			
Retail	Consumer packaged goods	Travel	SCM and Logistic
• Customer preference mapping • AI-powered email marketing, campaign and automation • Automation of planning and buying • Use of social media through Automation for targeting a specific audience • Image recognition system • Product recommendation system • Intuitive chat system • Intelligent bot functionality • Behavioral analytics • Recommendation engines. • Inventory optimization • Predictive pricing • Smart merchandising • Visual conversation among shoppers and retailers catalogue for buying • Robotic carts • Minimization of stack-out and stack-in self situations • Customer churn prediction • Customer inflow prediction (in-store) • Behaviour analysis for digital payment and security • Better relationships with vendors and suppliers • Higher inventory accuracy	• Product affinity(Analyzing shopping behavior to identify what products are generally bought together) • Customer segmentation • Customer experience enhancement • Analysis of the influence of weather on the customer's buying decision • Up-sell and cross-sell campaigns to an existing customer • New design launches to market • Customer voice and feedback • Inventory management • Robotic cart • Online order fulfilment • Sales forecasting • Fundraising campaigns • Personalized product recommendation • Revamp the business model with data-driven insights • Lean supply chain by using predictive analytics • Harnessing new customer touchpoints for sales and marketing actions	• Smart rooms, motels and hotels • Smart mirrors and floors • Personalized recommendations • Disruption management handling such as adverse weather predictive maintenance • Whether-based holiday destination planning • Chatbot for customer support • Robotic/automated customer support services • Banquet forecasting and staffing • Travel Deals • Confirmations • Itinerary Updates • Reminders. • Cart Abandonment Notifications • Personalized marketing and customer segmentation • Customer sentiment analysis • Recommendation engines • Route optimization	• Fleet and capacity planning and management • Inventory management • ERP integration • Drone delivery • Self-driving cars and trucks • Self-driving aeroplanes • Robotic process automation • Demand Forecasting. Let AI remove the guesswork in forecasting and avoid supply chain surprises • Supply Forecasting • Text Analytics • Price Planning • Inventory Planning • Inventory Price Balancing • Stock Analytics • Warehouse Management

Figure 5.7: Artificial intelligence, machine learning, and deep learning use cases in different industries

After having this understanding, let's move to the data lake and a few other important concepts which are important and related to data analytics, AI, and ML. Apart from this, it is important to understand "what is project and why corporates would care about them". It is important because ultimately all the mentioned technologies and concepts are professionally executed under projects not only to get monetary benefits, but achieving excellence in government agencies and taking strategic quantum jumps.

Let's build AI projects

But why we are discussing all these things? What is the motive to discuss these concepts? The answer is simple, complex, and philosophical; in simple terms, we are discussing them because we want to understand the concepts to figure out the future of humanity (philosophical), we want to explore the impact of these technologies on our daily business and life (complex), and we want to make products, applications, and services to help the corporate world and earn profit out of them (simple). Further, for making products or services, we have to run projects and seal them. This chapter will help you because it will explain in the upcoming sections technologies and ways to build products and services. Understanding technologies like ML, AL, big data/analytics / QC and a few more of the pack is necessary as they would come in upcoming sections of the chapter. Also, their interrelation with quantum technologies is need to be called out for corporates, individuals and teams to create futuristic products and services.

If you observe we are using a team called "projects" repeatedly here, what is it... as per definition, a project is a temporary endeavour which can be thought of as any undertaking, carried out individually or collaboratively and possibly involving research or design, that is carefully planned to achieve a particular aim. Corporations are executing multiple projects for their clients or for themselves for generating revenue and sustainability.

Hence proper understanding of a project and what it means is essential here. Because now we are going to discuss its impact, importance and significance in upcoming sections in the context of quantum computing, AI, the business of intelligence, projects, team skills and corporate success. These are needed for business acceleration and excellence.

AI, business of intelligence and the projects

AI is typically about the discovery of correlative patterns. Profits can be made through business if they are used mindfully and correctly. However, broadly, profit isn't made when AI delivers a response to a research question. If you don't know how you'd take a business action based on a response provided by the AI project, you and your projects might be in trouble. Because in changing time in corporations or in general life AI already took centre stage. From delivering an effective project to booking your cab through an app (Uber)... AI is everywhere.

This dependency leads to the point where finding answers to the specific business problem in a given context is very much required. However, there are a few pitfalls associated with AI if we talk about context mainly because of dependencies on data and lack of sophisticated systems; if those are avoided, then there are high chances of success of any AI initiatives. Further, instances of those downsides are overconcentrating on infrastructure, going into the particulars of algorithms, and data collection.

One can use cloud infrastructure and practice standardized workflow in every AI project to get rid of these problems. That workflow contains triaging the likely business actions, defining research questions, business metrics to track, and estimating project trouble and expenditure. For running an AI project statistic, algorithms are very important. Therefore, let's have a look at them.

As we know, an algorithm is an independent step-by-step set of procedures to attain an exact consequence. Further, algorithms perform calculations, data dispensation, or automated reasoning tasks with repeatable consequences. Also, as we discussed, algorithms are the mainstay of the data science procedure. One should collect a sequence of approaches and events that will comfort the difficulty and processing of specific data.

The essential skills for AI projects

To execute an AI project, only the proper understanding of data, business problems, and technology is not enough. It is also equally important how the involved stakeholder would understand the overall scenarios. This will make

it easy to get the most important aspect of any project which is securing funding.

However, to convince stakeholders storytelling ability is required in the data scientists to project managers to technology managers. Data storytelling is the procedure of interpreting data analyses into amateur's footings, to affect a business choice or act. One can have the hand-picked data science concepts, but deprived of the business story to translate answers into business-relevant activities, one will not do well.

Running an AI project needs multiple skills, different roles, and knowledge.

[Figure 5.8](#) depicts interrelations:

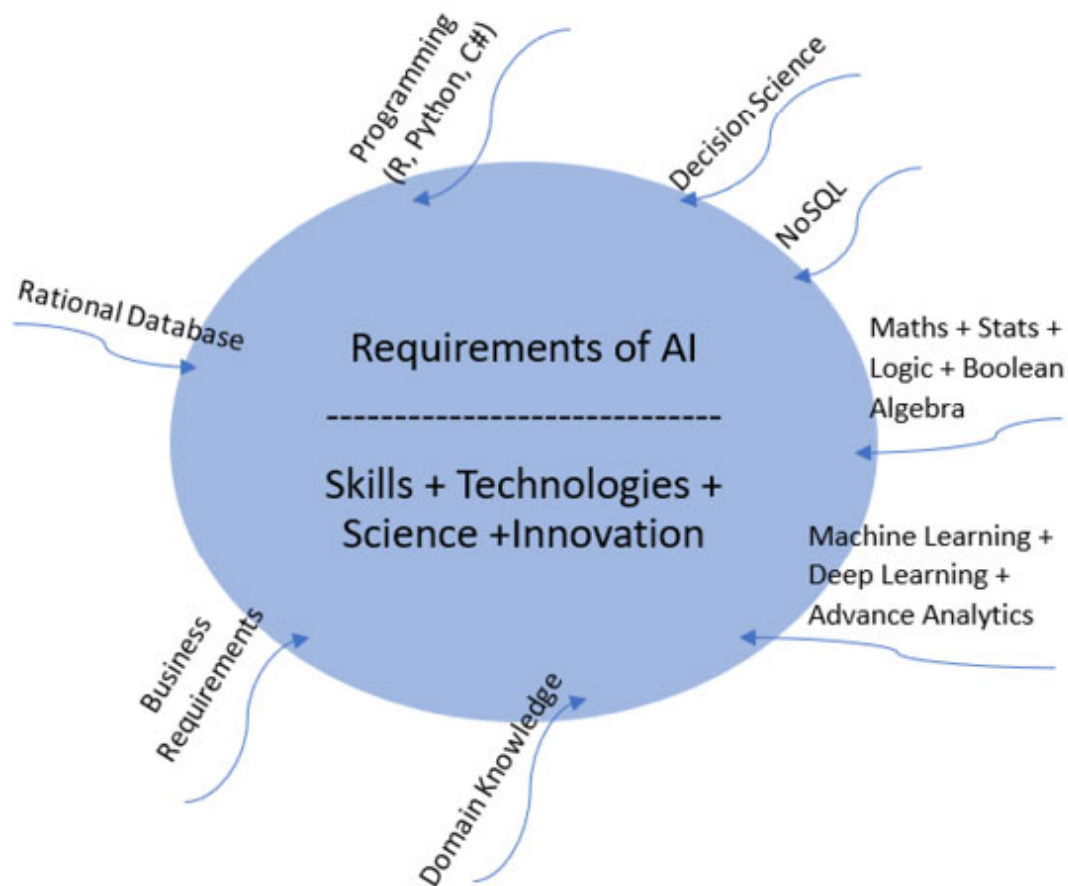


Figure 5.8: How multiple skills, technologies, and sciences are required for AI

Does it mean managing AI projects are difficult?

Yes and no... if you onboard the right skilled people, appropriate processes, and best-fitted technologies, understand the requirements correctly, respect

timelines, onboard good management and take care of your stakeholders and incorporate the feedback in an intelligent way.... managing, executing, running and maintaining an artificial intelligent project is easy, else this endeavour would be a nightmare. further, handling AI projects or systems doesn't need know-how in the particulars of AI algorithms. In its place, your prerequisite is to distinguish how to clarify the reimbursements of an AI project in business standings. For instance, what business problem is being cracked? What business benefit is provided by AI? How is that advantage measured?

AI can assist businesses by the execution of the analysis that notifies some real business action. AI prospects rise when you can smear an Intelligence/Examine/Respond loop with the Analyse part based on AI competence and the Respond part on real business actions you can take. No solitary individual is skilled in all themes of AI, data science, and data engineering. Therefore, for successful implementation and outcome, an expert has to identify and close any pertinent breaks in the knowledge and competences of a group who is interested in using AI for the benefits, in terms of IT – it may be a team.

Relation between AI, business, and perfect decision

Every AI project uses some form of the ML pipeline. That ML pipeline starts by collecting data and finishes by presenting results. The ML pipeline is one of the primary determinants of how the system as a whole will be perceived by the user, as shown as follows:

- The ML pipeline starts to ossify the moment it is constructed. Choosing the wrong ML pipeline, or worse, letting it directly emerge from **Proof-of-Concept (POC)** is a costly mistake.
- The success of the AI project is based on the entire system as opposed to the individual ML algorithms used.
- The user sees the output of the whole system, not the output of the individual AI algorithm. Think about the ML pipeline as a critical architectural artifact and about AI algorithms as important (but evolving) pieces of that pipeline.

- To get perfect decisions and insights, there is a stepwise flow of information involved, which converts data to insights. [Figure 5.9](#) depicts the same:



The primary ingredient for building machine learning solutions in the current time is data. The source of data is the IoT devices, social media, and so on. The data gathered from sensors through edge devices is stored in the cloud or on-premise or a mix of both infrastructures, which must be designed to scale and handle huge data of all types. You need machine learning specific algorithms and models (output) which are capable of learning from the data to make implications or sense about the new data. You need the mathematical and machine learning libraries, frameworks, and programming languages like R, Python, or C# for realizing your requirements through code. Speed and best utilization of underlying hardware are also one of the important criteria. Hence, you need development systems (hardware and

software) equipped with processors with multiple integrated cores, faster memory subsystems, and architectures that can parallelize the processing and compatible software for the same. At last, scalable clusters of these systems is required (if the complex solution needs to be built) that would use to train a complex machine learning model. There are a lot of technologies involved in building machine learning solutions and we would discuss them in detail in this chapter. However, it is a good idea to have a running look at the overall flow of AI from a technical perspective, especially in the context of available technologies in the market from a different vendor. If you don't know a few of the technology, please refer to the definition section in the book. If you are interested in the detailed outlook, then refer to Hotonworks and IBM web site. For H₂O (The H₂O AI Cloud solves complex business problems and accelerates the discovery of new ideas with results you can understand and trust) and Google, refer to their respective sites. [Figure 5.10](#) showcases the interplay of technologies that helps make a data-driven forecast:

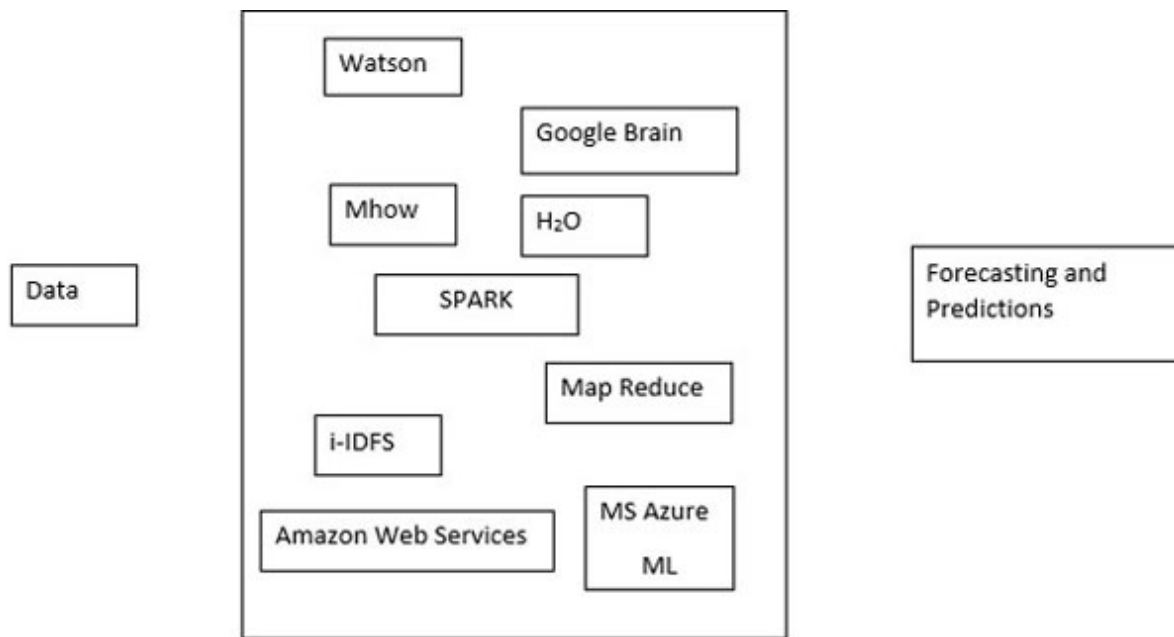


Figure 5.10: Data-driven forecast--- important technologies

How to create an AI solution

One way to classify AI solutions for business and scientific requirements is by the level of complexity of the request: simple, complex, or very complex.

But before going into the details of these, let's understand a few important concepts. We all know, modern AI relies heavily on data. Therefore, it is natural that the AI solutions and architecture are based on the data. It implies that for Data, sources are one of the important elements of any AI solution, and those data must be digested by the technical framework. In technical terms, this process is known as data ingestion. Multiple AI systems use them (please refer to [Figure 5.11](#)) which is supported by multiple systems. Finally, the product is used by different users.

Few guidelines for the categorization are given by the Schloer Consulting Group, as follows:

- **Simple:** Solutions and platforms for slim business requirements, such as eCommerce, network integration, or resource management.
- **Complex:** Includes the organization and examination of the precise roles of a system. For example, it could comprise optimization of work systems, forecasts of actions or situations founded on historical data, security monitoring, and management.
- **Very complex:** Working over the complete information assemblage, analysis, and management processes, the system wants to know where to look for data, how to gather, and how to analyse, and then suggest resolutions for nearby and mid-term prospects.

Refer to [Figure 5.11](#) for AI design and architecture:

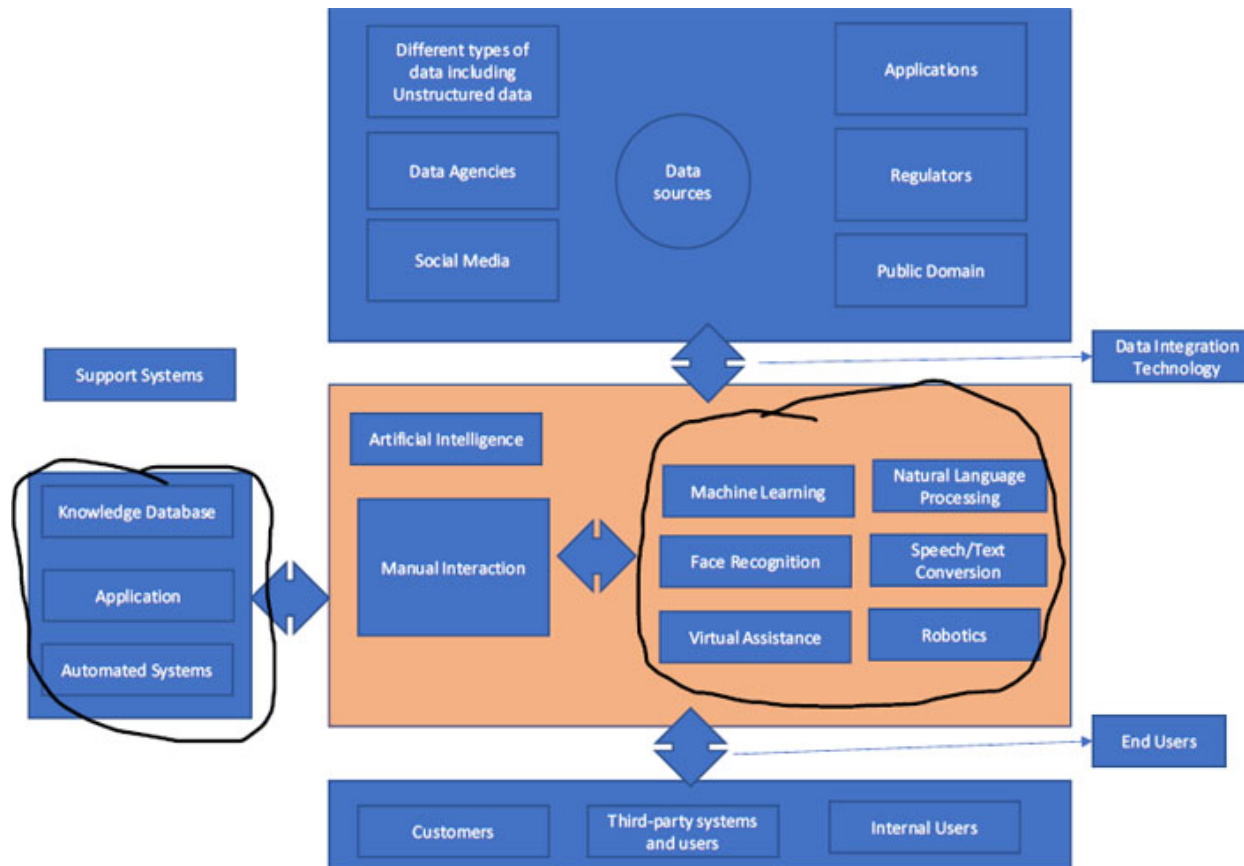


Figure 5.11: AI design and architecture

Typically, an AI solution takes data from different sources, store them in some underlying systems or on cloud infrastructure (we will cover that in the next chapter), run analytics on top of that by using multiple algorithms and generate insights for user. Users may be categorized under business or general. An AI solution works iteratively, meaning it is continuously taking inputs from the environment and improving itself. A child analogy can be taken for better understanding – when a child is born, his brain is a clean slate; then he starts observing or gathering information that is spread around him through sensory organs like eyes and skin. Afterward, analyze gathered data in his brain and make decisions, first small, followed by complex ones.

Different options are available to build an AI/ML solution, including:

- Different programming language like C#, R, Python.
- Different data storage option available like data lake for raw data.
- Options available to build a solution on Azure cloud for data SQL server or cosmos.

- Are multiple frameworks available to build AI solutions?

Well... understood, now, tell me how AI is related to Cloud, IoT, and Big data?

We covered a bit of it in the preceding sections. But it is important to further explore these concepts because these concepts are the foundational element of modern AI systems.

Cloud computing, the internet of things, and AI are very closely connected to each other. Hence, to get a holistic picture, their understanding is required. Covering them in detail is not possible here due to space constraints and the scope of the book. However, we will go through the most fundamental ideas. Let's start with cloud computing.

Cloud computing

Cloud computing could be defined as the practice of utilizing a web or network of remote servers hosted on the internet to store, manage, and process data, instead of a local server or a personal computer. Cloud computing is fundamentally a collection of dissimilar services provided by diverse companies like Microsoft and Amazon in their own customized way. This customization may happen in terms of infrastructures, services, products and software.

It primarily hinges on resource sharing using internet empowered devices that permit the function of application software. The Cloud can aid in a wide range of jobs over the Internet, such as storage from virtual servers (is a software-based representation of a physical server), virtual applications (Applications that have been optimised to function in a virtual environment are known as virtual applications. A virtual application, in other words, operates on a computer without needing to be installed), authorization of desktop applications, and so on. By applying resource sharing, cloud computing is intelligent enough to attain reliability and economies of scale. There are four types of cloud computing models are available. They are public, private, community and hybrid as shown in [*Figure 5.12*](#):



Figure 5.12: Different cloud options

Cloud computing deployment models are chosen and used based on multiples factors and parameters including location, uses and requirements. To know which deployment model would best suit your organization's necessities, typically one has to analyse the requirements in detail. Now, let's have a look on the deployment models briefly:

- **Public Cloud:** It is a kind of hosting in which cloud services are brought over a network for public usage. Clients do not have any control over the site of the infrastructure. The price is shared by all users, and are either free or in the form of a license policy like pay per user. Public clouds are great for organizations or companies or individuals that need handling the host application and the many applications users use.
- **Private Cloud:** It is a cloud infrastructure that is exclusively used by one organization or company. It gives organizations better control over safety and data which is safeguarded by a firewall and managed internally. It can be hosted internally or externally. Private clouds are great for organizations or companies that have from head to toe security requirements, high management demands, and uptime requirements.
- **Hybrid Cloud:** This uses both private and public clouds, but can keep on as distinct units. Resources are managed and can be provided either internally or by external providers. A hybrid cloud is useful for scalability, flexibility, and security. An example of this is an organization that can use the public cloud to intermingle with customers while keeping their data protected through a private cloud.
- **Community Cloud:** This is an infrastructure that is jointly shared between organizations that fit a specific community. The community members usually share-alike privacy, performance, and security

apprehensions. An example of this is a community cloud at banks, government in a country, or trading firms.

Refer to [Figure 5.13](#), for the key highlights for the different cloud service models:

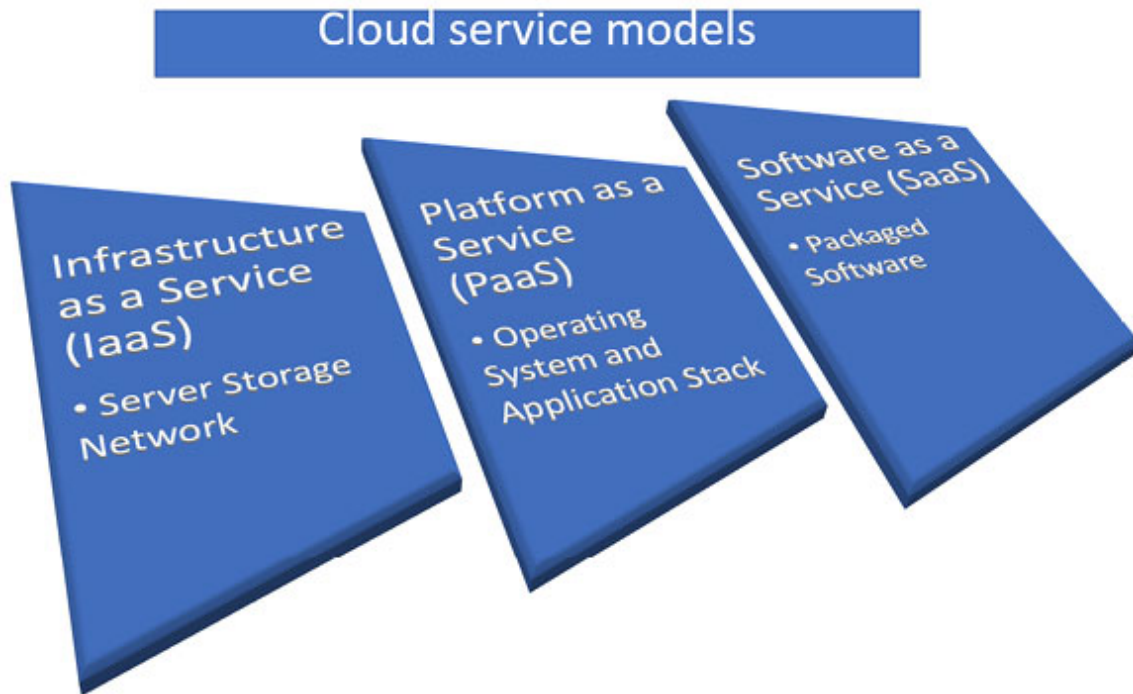


Figure 5.13: The key highlights of the different cloud service models.

Focus on providing multiple types of offering to their clients, as follows:

- **Software as a Service:** It is a kind of cloud solution that offers an “application” to customers or organizations through a web browser. The data for the applications runs on a server on the network, not through an application on the user’s computer. Software is typically traded via subscription. Examples of SaaS are Salesforce, Google Docs, Office 365, Basecamp, and so on.
- **Infrastructure as a Service:** It offers the hardware and typically virtualized OS (logical separation of the hardware and software resources) to their clients. Software is charged only for the computing power that is used, typically CPU hours used in a month. Examples of IaaS are Amazon EC2, Rackspace, Google Compute Engine, and so on.

- **Platform as a Service:** It offers networked computers running in a hosted environment, and also enhances support for the development environment. PaaS aids usually support a precise program language or development environment. Installing your applications in this environment, you can take the benefits of dynamic scalability, automated database backups without the need to explicitly code for it. PaaS is billed as an extra cost on top of the IaaS charges.

The cloud impacts the multiple dimensions of individual, technology, society, and economy and help to achieve excellence by:

- Building cloud businesses and ecosystems: This helps in being quicker and cost-effective. As a business does not need to invest heavily in infrastructure and software before the start of business. Also, they have the flexibility to pay per use.
- Creating: one can innovate fast as he has to focus only on creativity and need not to waste time on other things
- Changes in society and politics
- Enable Fog computing which is a decentralised computing environment. For computing is a decentralised computing environment. In fog computing data, compute, storage, and applications are distributed between the data source and the cloud. Fog computing brings the benefits and power of the cloud closer to where data is created and used. This is similar to edge computing.
- Bring new paradigms and enable disruptive things to do quickly. Because you do not need to purchase software, infrastructure and networking devices. For these you can just pay and use

Cloud computing is a wide field of study. Hence, it is not possible to cover everything here. However, [Figure 5.14](#) gives oversight on cloud computing:

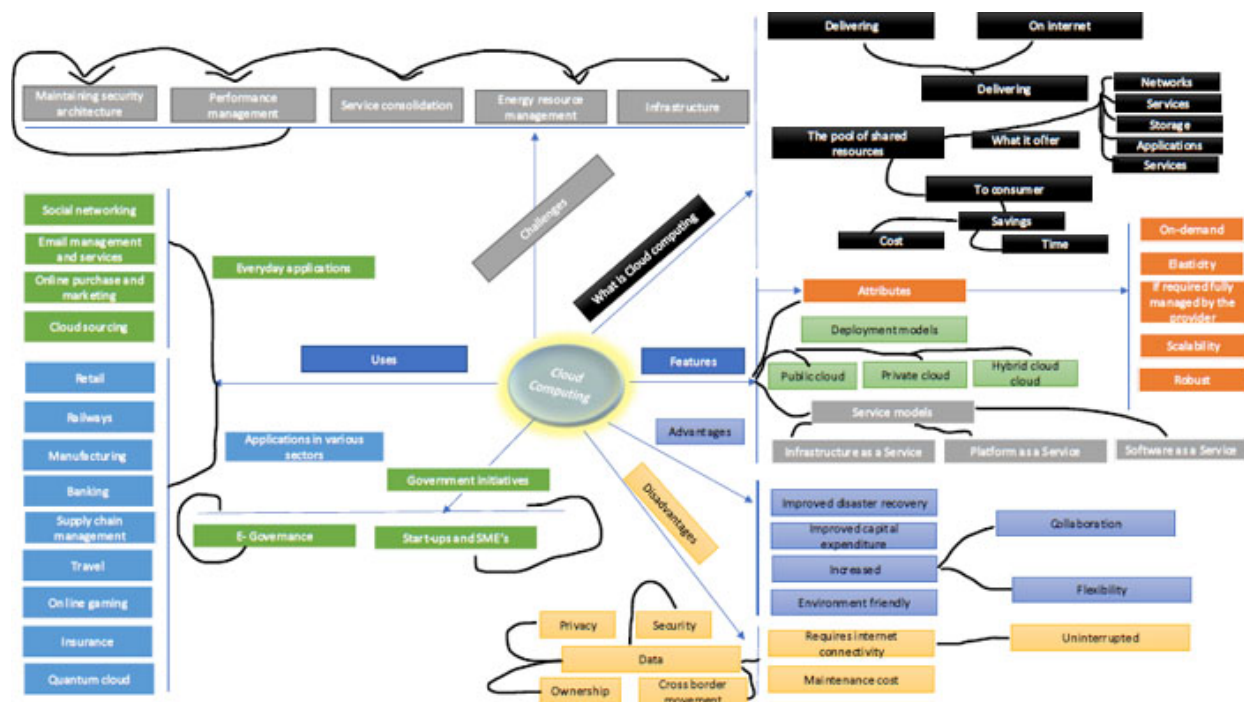


Figure 5.14: Cloud computing in a nutshell

Internet of Things (IoT)

The Internet has extensively altered from being a decently informational system to one that has socio-technological transformational capabilities. It established that it has enormous power for bringing changes in social, creative, and political dimensions of life, work, and governments. Nevertheless, the position of its non-technological sides is gratifying even further, apparent in the growth of society along with the new generation of technologies like the Internet of Things, AI, and data analytics. These technologies are adding the fuel of intelligence to the internet which is shaping humanity in an entirely new way. Intelligent objects can accrue a huge volume of data, as shown in [Figure 5.15](#), just to aid us in the finest conceivable way:

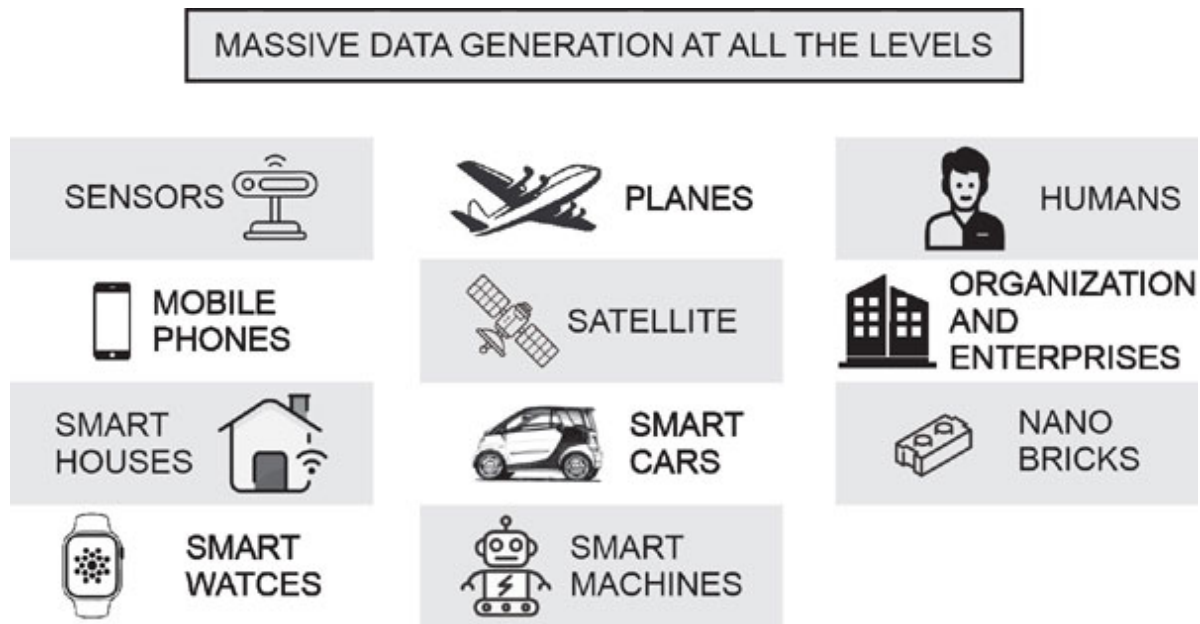


Figure 5.15: Data generation sources

However, it is bringing some issues as well; along with the data protection problems, there is also the questions of who would own the common people data as it is captured, interpreted, and potentially significant commercial or social value. There is also a serious aspect of dependency on the bigger mass of people on technologies. To understand it better, take an analogy of electricity in corporate and also in society; usually we have now become very reliant on the overall availability of electricity. For instance, even rare blackouts have put society in general to the serious consequences of not delivering commitments on time or lead life in a proper way. But if everyday stuff only functioned correctly with an Internet connection in the upcoming time, this would lead to an even larger dependence on the technology. If the technology infrastructure fails for whatever reasons like, design faults, material defects, sabotage, overloading, natural disasters, or crises, it could have a catastrophic consequence on the economy and society. Remotely controlled things could also source us to turn out to be dependent and drop our sovereignty on a personal level. And even with no account of ill intent, our intelligent objects might not act as we would wish, but rather as they “trust” is best for us, foreshadowing an understated type of technological authoritarianism. The quick response that smart or intelligent things provide about themselves or that cooperative tools such as smartphones and augmented reality glasses can give us around our atmosphere is also a mixed

consecration. Although it can inspire us to do good, beneficial things, it can also seduce us into creating needless instinct procurements.

IoT devices and AI systems must play well together. AI and data are fundamentally reliant on each other. The architecture of the AI solution is intended in close collaboration with the device's arrangement. IoT is about structuring physical devices that are linked to the internet. Those devices can also have an AI constituent governing its behavior. AI is characteristically trained using data composed across the whole inhabitants that use the device. IoT and AI naturally suit in a composed manner since AI can deliver additional functionality to the device.

On a greater scale, we can find fantastic use cases of IoT, Big data (will discuss shortly), and AI in the areas of smart cities where IoT sensors* gather data and then use AI to accomplish assets more efficiently. Producing IoT devices needs a comprehensive variety of know-how. For instance, in the IT world reliant on the kind of device, a development team might need know-how in software, electrical, and mechanical engineering.

As IoT devices are matured, most of the data AI will use originates from sensors on the device itself. If the device's design is finalized before considering AI, you've already locked yourself into which data you'll have access to; before you've even considered the dataset, you'll need to produce actionable analytical results! This significantly reduces your options once you do start thinking about AI because you'll then commonly fail to collect the data the AI needs.

Gathering data in the IoT environment is practically at all times more luxurious. Therefore, to collect data, effectively and more importantly cheaply, there is a need to embrace a sensor in the device. At first, using such sensors upsurge the cost of your device, but in the longer run, it will pay off. Also, when put on AI to work in the IoT domain, one has to comprehend that the physical device can be problematic or unbearable to change after it's positioned in the field. Hence, embedding all the hardware sensors needed in advanced versions of the AI side of the system needs to be considered well in advance.

What is big data?

Big data is not a slogan any longer. Enterprises are accepting, structuring, and applying big-data solutions. By description, big data labels any big body of digital information. It can be historical or in real-time, and varieties from streams of tweets to customer acquisition history, and from server logs to sensor data from industrial gear. It all cascades under big data. As far as the meaning goes, there are numerous different definitions and opinions present and still continue to evolve. It can be defined as the big data is high-volume, high-velocity, and/or high-variety information possessions that request cost-effective, ground-breaking forms of information dispensation that permit better insight, decision making, and process automation. Another way it can be defined is as methods and knowhows that make the management of data at an extreme scale in the economical form. Also, as data is one of the important elements of AI, here big data plays a very significant role. Any data set which can be called big data must qualify for three fundamental characteristics. Based on standard definition, the following are the three Vs of big data:

- **Volume:** The quantity of data that cannot be stored even by using techniques like scale-up/vertical scaling. There may be multiple reasons for that, including physical and software boundaries or limits. However, when a scale-out or a horizontal scaling approach would be used, it generally works.
- **Variety:** Once a novel set of data is impending in, it carries a different assembly and arrangement than what is already stockpiled, or it is unstructured or semi-structured. This kind of data is well-thought-out as a data variety problem.
- **Velocity:** The frequency at which data come to or vicissitudes matters in the processing of data. When the window of dealing out with data is moderately small, then it is called a data velocity problem.

Typically, if one is tackling with more than one V, there is a need for a big data solution, or else old-style data management and dispensation tools can do the work very well. As far as different Vs are concerned, few kinds of literature talk about 6, whereas others have mentioned 9 or 12 or 14 or even 17, but the 3 most important ones where all experts agree are velocity, verity,

and volume. These are technically the most significant and accepted. The major Vs are depicted in [*Figure 5.16*](#):

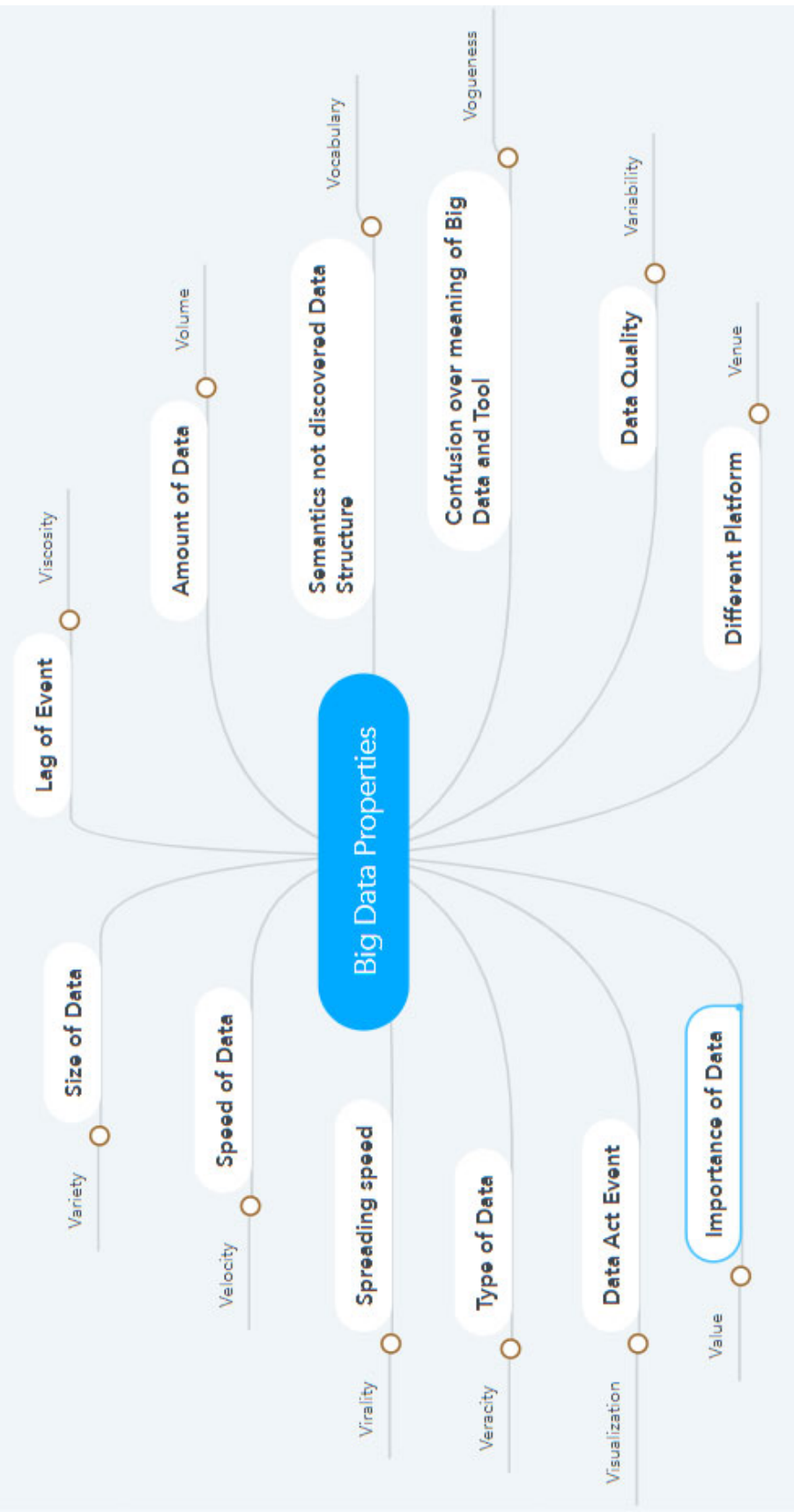


Figure 5.16: Major Vs of big data

Data, data and data: I want to know it in detail, but technically

Modern IoT, Big data, and AI solutions revolve around data. So, in this section, we will cover data-related concepts to understand the jargon and make our understanding more robust. Fundamentally, two basic methodologies are supported by data processing tools. The following is a brief outline of each methodology and its advantages and drawbacks.

Schema-on-write ecosystems

A traditional **Relational Database Management System (RDBMS)** requires a schema before you can load the data. To retrieve data from structured data schemas, you may have been running standard SQL queries for several years.

Schema-on-read ecosystems

This alternative data storage methodology does not require a schema before you can load the data. Fundamentally, you store the data with minimum structure. The essential schema is applied during the query phase.

Data lake

A data lake is an example of schema - on -read methodology. A data lake is a storing repository for a huge quantity of raw data. It stores data in the innate arrangement, in expectation of upcoming requirements. One will obtain insights from the data lake by running advance analytics over it. This can be achieved by using ML and AI algorithms.

While schema-on-write systems store data in predefined databases, tables, and records structures, a data lake uses a less limited schema-on-read-based construction to stock data. Each data component or entity in the data lake is allocated a characteristic identifier and marked with a set of wide-ranging metadata tags. A data lake is classically deployed using distributed data object storage, to allow the schema-on-read structure. This means that business analytics and data mining tools access the data deprived of a

multifaceted schema. Utilizing a schema-on-read practice one can get the capability to load data as-is and start to get value from it promptly.

Data vault

Data vault modelling is designed by Dan Linstedt. It is a database modelling technique that is deliberately organized to regulate long-term past storing of data from manifold operational systems. The data vaulting procedures alter the schema-on-read data lake into a schema-on-write data vault. The data vault is intended to handle the schema-on-read query request and then be performed against the data lake. The elementary structures are mandatory to express a data vault. The structure is constructed from three basic data structures: hubs, links, and satellites. Let's inspect the exact data structures, to illuminate why they are compulsory.

- **Hubs:** Hubs comprise a list of sole business keys with a low tendency to modification. They comprise a replacement key for each hub item and metadata classification of the source of the business key.
- **Links:** Relations or transactions among business keys are modelled using link tables. These tables are fundamentally many-to-many join tables, with precise additional metadata. The link is a particular relationship among hubs to safeguard the business.
- **Satellites:** Hubs and links form the structure of the model but store no sequential physiognomies or expressive features of the data. These faces are stored in appropriated tables recognized as satellites. Fundamentally, satellites are the constructions that store complete levels of the information on business physiognomies and are usually the main volume of the whole data vault data structure.

Data analytics

Data analytics is the discipline of fact-finding examination of raw data, with the goal of sketch deductions from the data lake* . Data analytics is determined by specialized algorithms to statistically describe relations between data that produce insights.

Data science

Peter Naur was the first person who used data science for the first time in the decade of 1960. He started using the term data science in place of computer science. His argument was that to work with data, you need more than what is called computer science. Now Data science is visualized as an interdisciplinary science and rightly so because it includes practices and methods with actionable facts and understandings from data in varied schemas (such as structured, semi-structured, or unstructured). It merges the technical fields of data examination with stimulating investigation fields such as data engineering, information science, computer science, statistics, artificial intelligence, machine learning, deep learning, data mining, and predictive and prescriptive analytics. We will learn more about descriptive, predictive and prescriptive analytics during our detailed discussion of the data lake and AI technologies.

Data Science is not a solitary area of study, but a dexterity at the connection of several fields. So to understand how the different bits will fit together, the Data Science Process could be visualized into six portions, as follows:

- **Construct the use case:** This must be based on who are you serving? What do they want?
- **Gather raw data:** What are the volume, type velocity of data obtainable? Out of those, which portions are valuable?
- **Process the data to get insights:** What do the variables mean? What scrubbing is essential?
- **See the sights in the data:** What are the patterns present in the obtained data? Are they important, valuable, and useful?
- **Do an in-depth examination:** How can the historical information be useful in the future context? If it is, then to what grade?
- **Communicate outcomes:** Why are the statistics important? What must be finished otherwise?

Data science is the practice of using data to try and understand and solve real-world problems. This concept isn't exactly new; people have been analysing sales figures and trends since the invention of the zero. In the past decade, however, we have gained access to exponentially more data than what existed before. The advent of automated systems, social media and

explosion in video footage have contributed in the generation of huge data. With computer code, a data scientist can transform or aggregate data, run statistical analyses, or train machine learning models. The output of a specific code may be a report or dashboard for human consumption, or it could be a machine learning model that will be deployed to run continuously.

Also, a successful data science project needs well-skilled people. [Figure 5.17](#) will provide some important factors related to good skill generation:

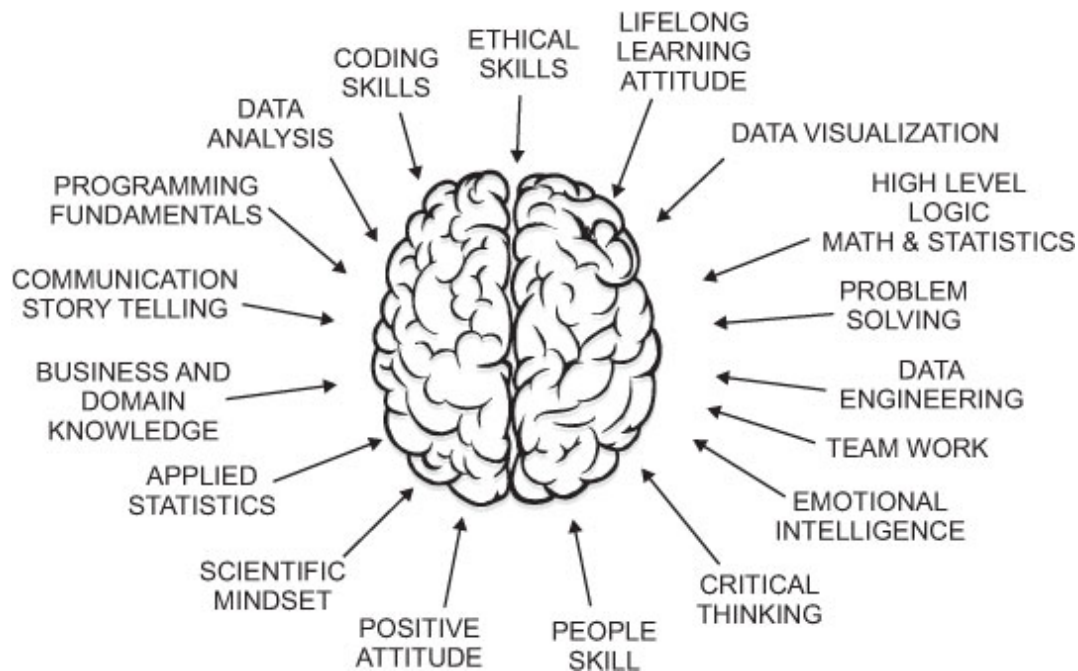


Figure 5.17: Some important factors related to good skill generation

Good skills are one of the major criteria for building an AI solution. However, producing a workable AI system is one thing and constructing a robust, optimized, and effective solution is a different ballgame altogether. But for constructing a highly efficient solution of an environment, multiple factors need to be taken under consideration and most of them are related to data and therefore to AI. A few important aspects consisting of an integrated, modular environment to manage application data and optimize data-driven over their lifetime are mentioned next. [Figure 5.18](#) shows an integrand data environment component, with pinch of insights on **data platforms present and future considerations**. Because selecting the correct data platform is very important for the success of AI projects / solutions.

For choosing or constructing a good data platform six essential considerations need to be understood in the better manner. They are the six essential considerations: on-premises infrastructure vs. cloud infrastructure, scalability, flexibility, serviceability, safety and automation. Driving all of them is the indispensable contemplation that you want to be able to work with any data in the organization; irrespective of source. It is desirable to handle data sets of bigger velocity, variety and volume, while permitting a variety of users from data engineers to data scientists to technology managers — to carry real-time data to each question, decision and act. A data platform necessarily permits users to examine, monitor and analyse data — and take efficient action based on the insights revealed. And they must have the below-mentioned capabilities.

Essential characteristics:

- **Data security:** Data security management is a way to maintain data integrity and to make sure that the data is not accessible by unauthorized parties or susceptible to corruption of data.
- **Data quality:** Set of characteristics of data-completeness, validity, accuracy, consistency, availability, and timeliness fulfill requirements.
- **Master data management:** Processes, policies, standards, and tools that consistently define and manage the critical data to provide a single point of reference.
- **Metadata:** Managing data about other data generally referred to as content data (catalog, dictionary, taxonomy).
- **Organization of data:** People management and costs control.
- **Data life cycle management:** Managing information throughout its lifecycle, from requirements through retirement. Including archiving and lineage.

Provide Governance capabilities which includes:

- Maintaining a full audit history across all data in a single place.
- Tracking, classifying, and locating data to comply with governance and compliance rules.

- Visualizing the upstream and downstream lineage of data to verify the reliability.
- Defining and automating complex data lifecycle activities with integrated metadata policies.
- Verifying access privileges.
- Searching metadata and visualizing lineage.
- Encrypting or decrypting data.

Administration:

- Monitoring and scheduling system.
- Hardware failure, disk or server crash, rack failure.
- Data deletion, data corruption.
- Site failure, disaster (fire, water, network, power).
- Backup and restore management. [Figure 5.18](#) gives an integral view of data platform.

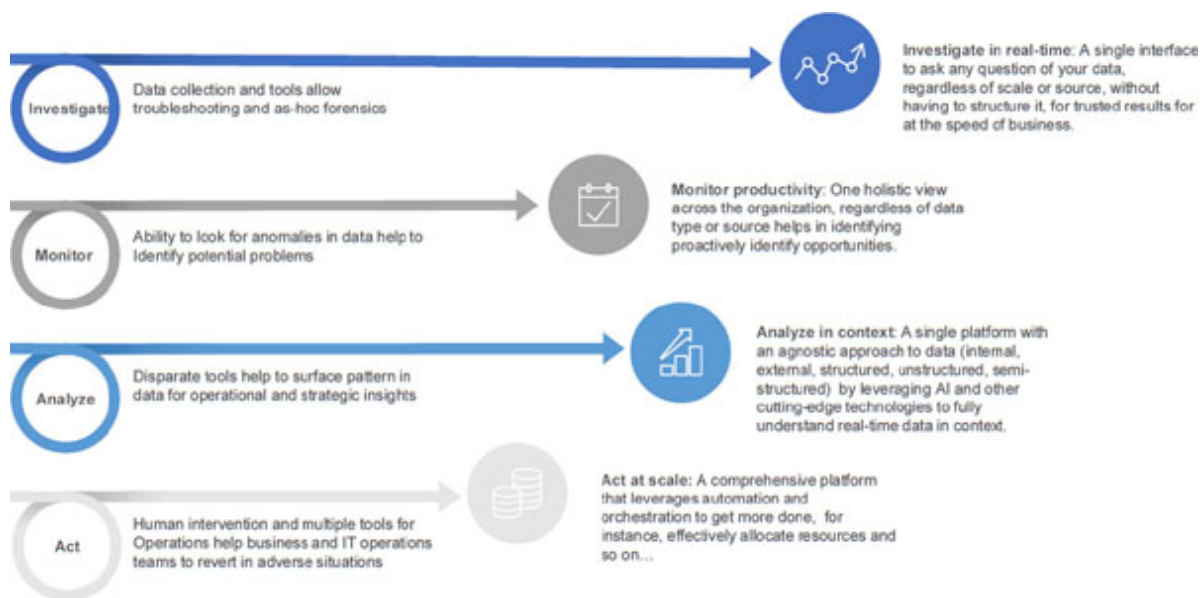


Figure 5.18: An integrated view and components for the robust data environment

Nowadays, the AI applications which we develop are incorporated the elements of machine learning in it. ML depends heavily on data. Also, for building good AI applications, a proper understanding of machine learning is needed. We touch-based ML in the earlier section of the chapter and one

complete chapter would be provided on it in this book. But for the time being, as we learned about the foundation of data technologies and ML. In AI thinking holistically is very important. Because as mentioned earlier, is not an isolated term but is thought to be an "ecosystem" of concepts, technologies and terms. Let's discuss a few additional accessories of AI.

Think holistically

To attain a confident level of accuracy and speed, AI programs need access to huge volumes of training data and processing power. Now, this is possible due to big data technologies, including big data analytics along with the internet of things. Big data is a comprehensive and growing term for a hefty number of datasets. The data could be structured, semi-structured, or unstructured. An example of structured data is an RDBM table where it is organized in row and column format. Whereas unstructured data is the content available on Facebook, black box, or sensor data. Finally, semi-structured is the data that is available in Excel or Word documents.

Big data analytics along with ML is the procedure of analysing big data to recognize unseen patterns, prevalent trends, exclusive associations, and other critical and advantageous information. For instance, an e-commerce corporation will put on big data analytics to inspect customer or customer behaviour and mentality, and procurement patterns. Whereas big data is all about the huge volume of data and their management, including their impacts.

If we think from a more rational perspective, traditional projects mostly require the uses of the left-brain, whereas data-based projects use both parts of the brain. Just for the reader's information, the human brain is divided into two parts, out of which the left is responsible for logic and mathematics, whereas the right is dedicated to visualization and imagination. If both parts would be utilized appropriately in a composite fashion, it would do miracles.

[Figure 5.19](#) highlights this fact:

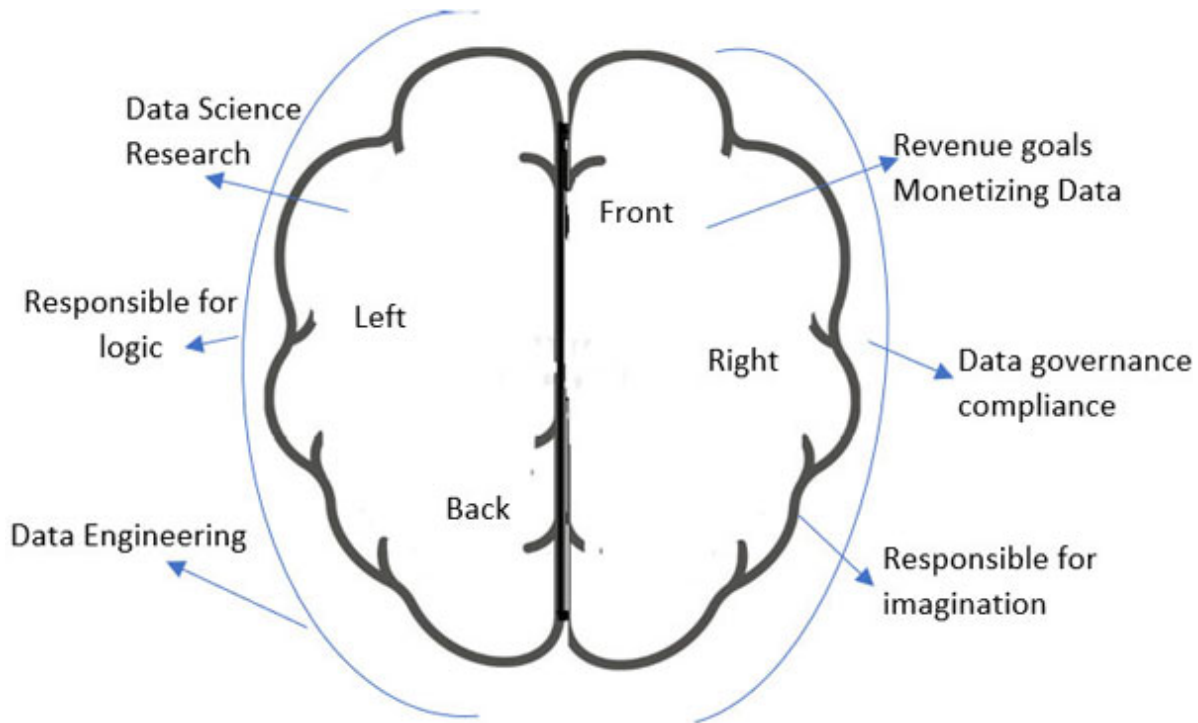


Figure 5.19: Holistic brain functions at work

When it comes to thinking holistically, we have to also consider where the data is coming as input and going as output. As we know there is a boom in the area of machine-generated data. And thanks to IoT for this.

When it comes to thinking holistically, we can not leave IoT out of discussion. As we know there is a boom in the area of machine-generated data. And thanks to IoT for this. The Internet of things is about data, devices, and connectivity. Analytics technologies are examining large volumes of typically human-generated data. IoT wholes and pads enormous quantities of low latency, low duration, high volume machine-produced data impending from an extensive variety of sensors to support real-time use cases. Stacking sensor data (the output of a device that detects and responds to some type of input from the physical environment) into a data lake (technology for storing all types of data) and accomplishment data science (a subfield of mathematics which deals with data analysis, storage, processing, and presentation) to create predictive analytic models qualifies as doing IoT analytics. [Figure 5.20](#) showcases the conversion of data from inception to visualization:



Figure 5.20: Showcase the conversion of data from inception to visualization

As adoption of the IoT is increasing, more and more data presented are produced, not to mention all the digital blogs, social media stories, user interactions on the online sharing including videos, websites, user click stream (the information collected about a user while they browse through a website or use a web browser) and server log (a simple text document that contains all activities of a specific server in a given period of time) are fulfilling this. They will only complement more and more data production. So, what are the issues and opportunities as both are basically data? The opportunities are the benefits we get from the amount of data that gets analysed with the use of the existing technologies. The issue is large amounts of data are not easy to analyse with traditional tools and technology. Hadoop, a technology ecosystem provided by Apache fundamentally changed all of this and enabled individuals and corporations to analyse massive amounts of data using commodity hardware which was not in fashion before its advent... at least for the business world. Until the cloud became mainstream, it was not economical, efficient, and effective for small and medium-sized businesses to buy all the hardware compulsory by an abstemiously sized Hadoop cluster.

As discussed big data fundamentally changed the way we manage and use data. It also changed the landscape of technology infrastructure, the way CPU or computing power is utilized, the way data is structured, scaling of the resources, storage of data, and so on. [Figure 5.21](#) showcased this in a nutshell. For instance, before the advent of big data (it is a relative term), people only cared about structured data but after this semi-structured and unstructured data came into fashion, the world was exposed to the potential of huge volume data. The terms like NoSQL have become more and more popular instead of SQL, because of its capabilities to query unstructured data. Refer to [Figure 5.21](#) for a better understanding:

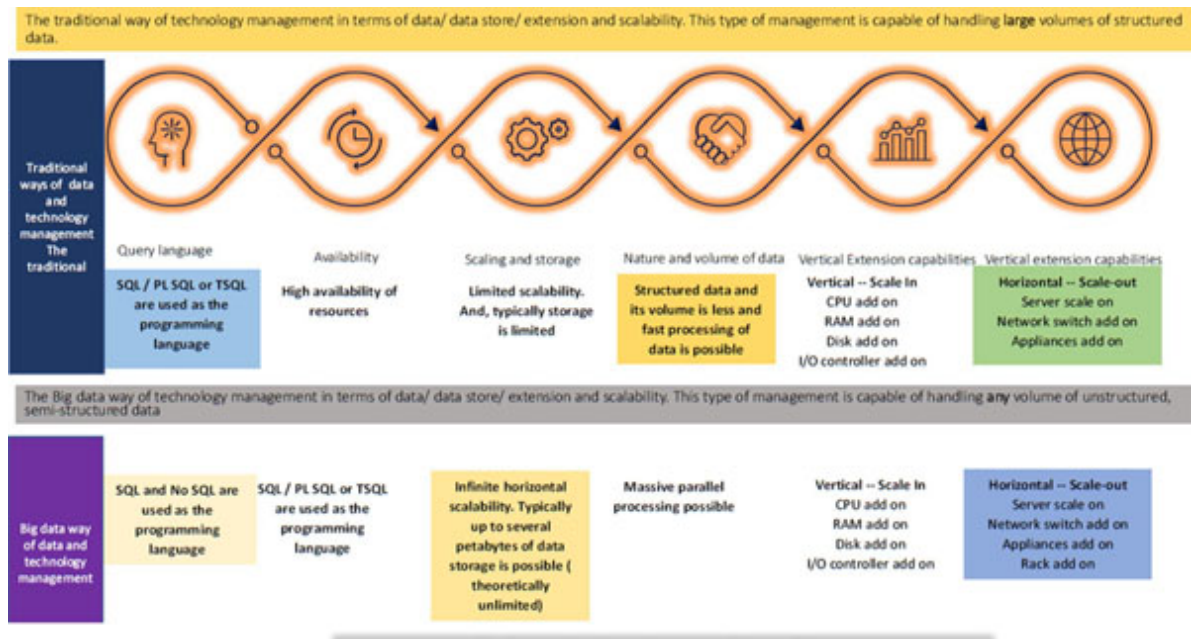


Figure 5.21: Traditional and modern ingredients of data

IoT along with big data and cloud computing allowed and enabled everybody, including individuals and corporations to take benefit from on-demand scaling. Now, if one needs to analyse terabytes of data, s/he needs to just to spin up a cluster and slash it down when finished processing, and pay only for the time that the hardware is used. This model of doing business has condensed the complete cost of data processing and storage which everyone can afford. These technologies also fasten the adoption and incorporation of AI. And, it actually made AI more democratising i.e., now it is not confined to the big corporate's project plans or in research laboratories but become mainstream and available to almost all. Let's see how it work on technical level.

Building IoT enabled AI

To build a complete IoT system or solution may be a costly and timely business. It requires demanding skills and expertise, time, and money. If not planed properly and if resources are not utilized properly, it may go through the long IT project cycles and end up getting a low return on investment. Effective IoT platforms would simplify the development of IoT solutions. The criteria like scalability, robustness, and performance are essential. So, that whenever needed, the application or product easily accommodates

continual growth and change. These “must require” requirements enable enterprises to develop systems and solution which would help industries to easily connect to devices and sensors. So, that they would take benefit from seamless information flow between systems and operational assets and deliver solid business value to their customer through IoT/IoE (Internet of Everything)/IIoE (Industrial Internet of Things). As today’s customers or clients are well educated and connected to their needs through the internet of things, they are empowered through data, as shown in [Figure 5.22](#):

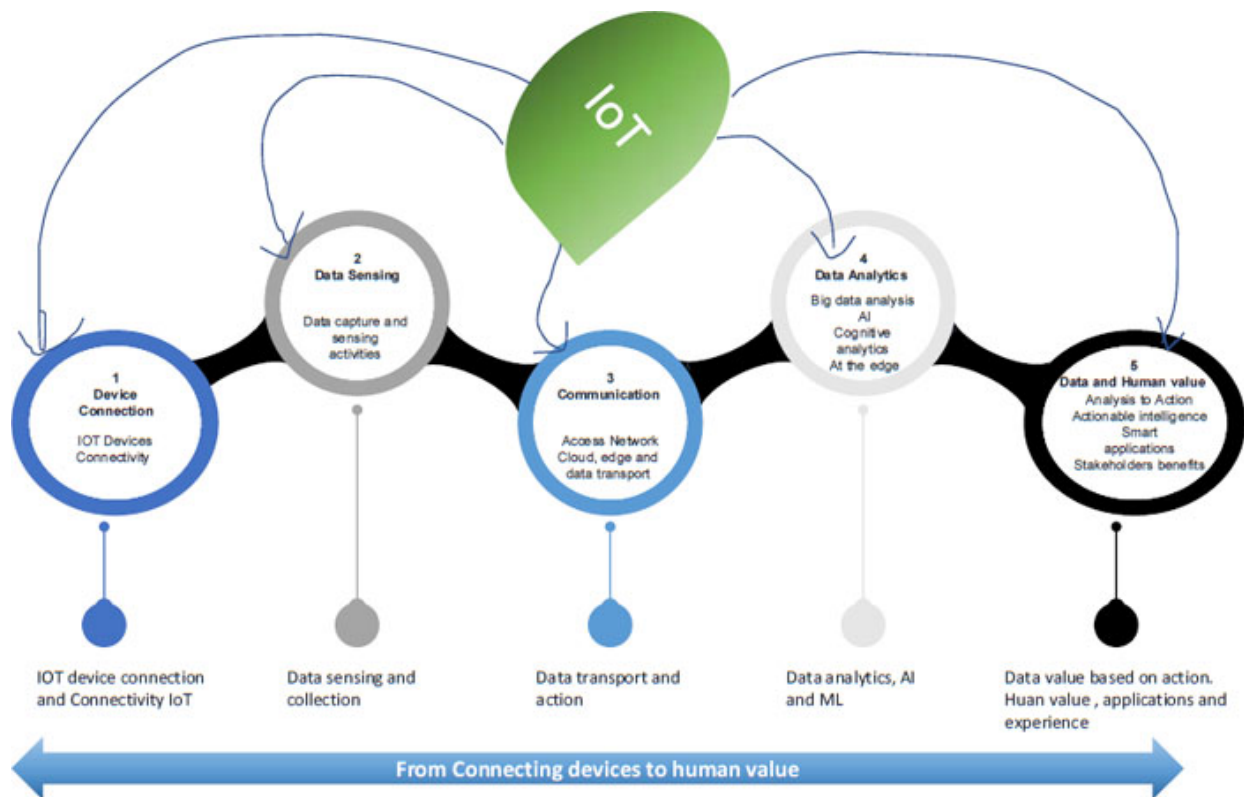


Figure 5.22: A look at the Internet of things based on connected devices and systems

As we discussed earlier, IoT along with other technologies like AI and big data are changing our lives in a very constructive way. It can be a boon if used correctly. For instance, for elderly or handicapped people, this technology potentially helps them to lead their life in a far better way than now. [Figure 5.23](#) highlights it in the form of a case study:

[A pictorial real-life case study for IoT](#)



Figure 5.23: IOT case study. snapshot about the future patient care and the role of sensors. Source: <http://www.chcf.org/~media/MEDIA%20LIBRARY%20Files/PDF/PDF%20M/PDF%20MakingSenseSensors.pdf>

Building Big data-based AI solution

In the previous section, we saw how IoT, big data and AI solutions would be built on the cloud. In the next couple of sections, our focus would be on big data and AI-based solution creation. We will see the product solutions, technology stacks and their execution along with cloud computing. You may find some repetition here but it is worth discussing these fundamentals here for completeness. Let's start with a hypothetical scenario. Think of a coffee

stand, where replacing a sluggish coffee maker with a further skilled person who makes coffee in a reduced amount of time, but for higher remunerations, improves the overall efficiency. Yet, it can be better-quality up to an only convinced point, since the worker has to take their time to make the coffee no matter how long the line is and he cannot attend the subsequent client in the line till the present one is attended. Also, there is no control over customer behaviour. The customers can modify their orders, and reimbursement takes each customer a different amount of time. So, scaling up can take far, but in the end, it will spasm to blockage.

So, if the resource is totally busy, add an individual to the job, but not at a higher wage. You should double the act, thus linearly scaling the throughput by allocating the work across different resources. The same approach is taken in large-scale data storage and processing scenarios.

One can add additional commodity hardware to the network to advance performance. But the addition of hardware to a network is a bit more complex than adding more employees to a coffee stand. These new units of hardware should be taken into account. The software has to provision separating processing loads across many machines. If one permits a solitary system to process all the data, even if it is stored on numerous machines, you will hit the processing power based on the load distribution.

This means that there has to be a method to assign not only the data to new-fangled hardware on the network but also commands on how to process that data and get outcomes back. Usually, there is a master node that instructs all the other nodes to do the processing, and then it totals the outcomes from each of them.

The scale-out method is very common in real life. You can find its instances in overfull coffee stands and grocery store lines – everybody uses this method. So, in a way, big data glitches, and their answers are not so new.

For building big data solution, we have to take different approaches that we used to take for developing the traditional applications. [Figure 5.25](#) shows how big data solutions would be built. Also, it explains the uses and pacing of different types of analytics like predictive, precative, and descriptive (described next; also refer to [Figure 5.24](#)) which could be performed on big data through artificial intelligence technologies. Here, it is worth noticing that technology suits AI, ML, Big data, and IoT do have overlap and it is

“not” necessarily true that they are different from each other. For instance, data storage solutions, tools, and frameworks for building either big data or AI or ML or IoT specific solutions are more or less the same. Before moving further, let’s take a look at different types of analytics first, as follows:

- **Descriptive:** Analytics is about analysing historical data and getting simplistic insight based on previous activities and records. Further, it can be used to understand previous behaviour and outcomes.
- **Diagnostic:** Analytics analyse the patterns of failures (product, service, and strategies) and find root cause based on the historical data. Once the root cause is established, understood, and diagnosed, the cause is attempted to be cured through the use of various tools.
- **Predictive:** Analytics uses data mining, modelling, and machine learning technologies for analysing data gathered in real-time/near real-time and historical data to predict the outcome. It does that based on running algorithms on the data iteratively. Ultimately, it anticipates future outcomes based on identified patterns.
- **Prescriptive:** Analytics helps in identifying the next step based on identified patterns and outcomes; also, it automatically enables them to get better results.
- **Cognitive:** Analytics works like the brain and helps in understanding user behaviour in real-time. Also, it depicts the thinking patterns of the users so that more aligned products, services, and applications are made for the targeted users.

[Figure 5.24](#) summarises the different types of analytics:

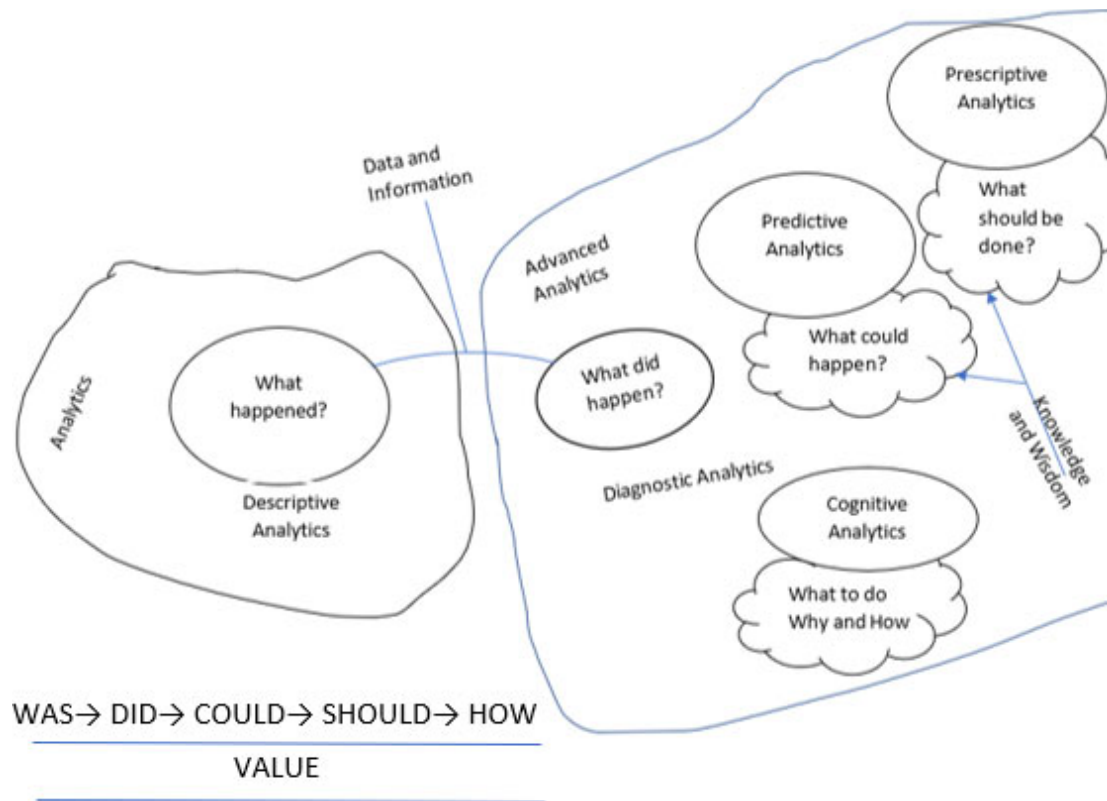


Figure 5.24: The different types of analytics

In the race for offering the best big data framework, software, customized innovative algorithms companies, and businesses are coming up with multiple strategies and providing unique and sophisticated solutions. However, the basic principle and practices remain the same for all the successful projects and offerings. The narration of core philosophy in modern context tells us that big data uses learning algorithms to build models that are mainly based on data. [Figure 5.25](#) depicts how big data solutions would be built:

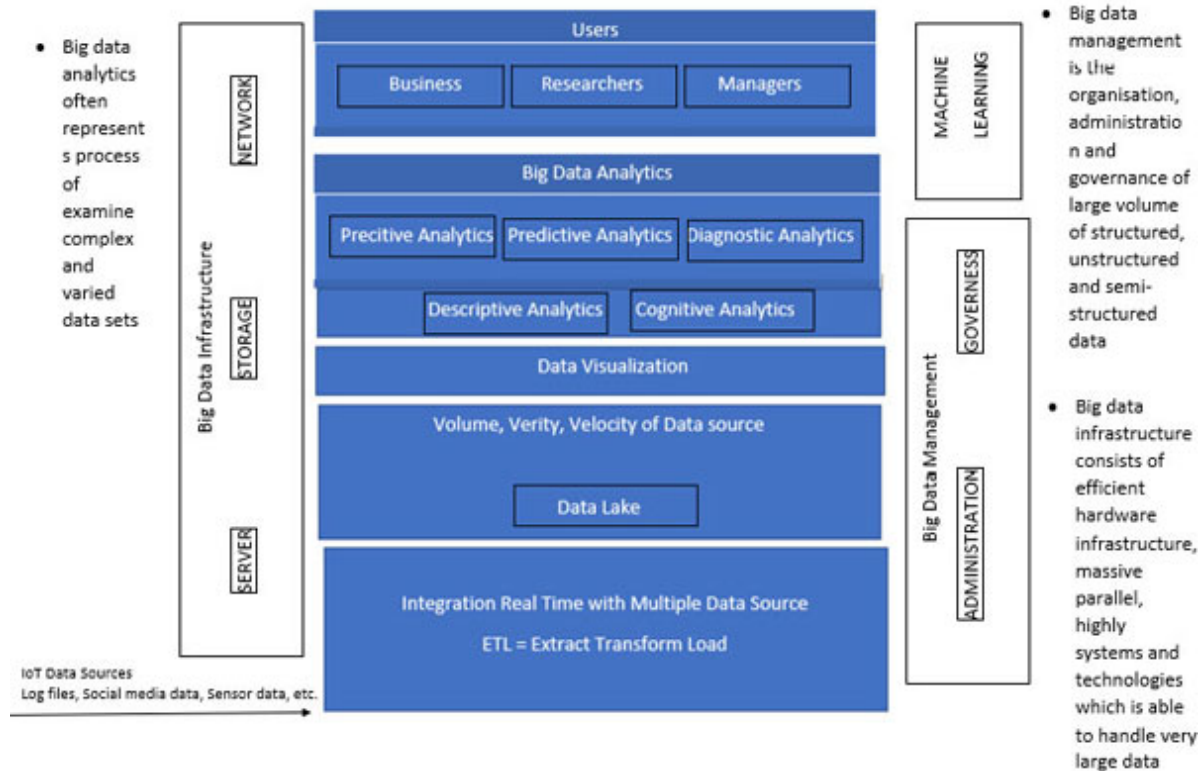


Figure 5.25: Shows how big data solutions would be built

The technology for Big data and AI

Hadoop is usually favored by the Hadoop ecosystem of components, technologies, frameworks, and libraries, which are used to realize AI, Big data and IoT related technological landscape. It comprises Apache HBase, HDFS, Pig, Hive, Spark, Apache Storm, and others, although there is a separate suite of technologies that exist for ML and AI. But as I mentioned earlier, it is possible to build big data solutions without AI and ML. However, in current times, typically big data solution revolves around AI. The following are some important technologies for big data. But we have to keep in mind that they may be thought of as a technology for ML and AI as well. Even now, Hadoop and big data are synonyms to each other; therefore, there are a few of the utmost suitable components under the Hadoop umbrella, which are mentioned as following:

- **Hadoop Distributed File System (HDFS):** A **Distributed File System (DFS)** is any file system that allows access to files from multiple hosts sharing via a computer network. A distributed file

system may include facilities for transparent replication, capabilities to deal with large data set, write once, and read many times, and fault tolerance. It is used for storing large volumes of data in inexpensive commodity hardware. HDFS is used as the main data storage system by the Hadoop suite of applications. HDFS creates multiple replicas of data blocks and distributes them on compute nodes throughout a cluster of enabling reliable, extremely rapid computations.

- **Hadoop MapReduce:** Hadoop Map/Reduce is an excellent software framework for processing large data sets in a distributed computing environment. MapReduce is the heart and soul of the Hadoop framework. It is this programming model that permits and facilitates massive scalability across hundreds or thousands of servers in a Hadoop cluster. It works on compute clusters of commodity hardware. The framework is used by organizations, enterprises, and companies for scheduling tasks, monitoring, and re-executing any failed tasks. As it is one of the sub-projects of HDFS, its working style is naturally inherited from the basic philosophy of HDFS.
- **HBase:** It is a non-relational database that runs on top of the Hadoop Distributed File System. Apache HBase is an open-source **NoSQL** (**Not only SQL**) database. NoSQL database technology is designed to support the changing requirements of cloud applications that contain and deal with a very large volume of unstructured data.
- **Ambari:** Apache Ambari is a rummage-sale for provisioning, supervision, and monitoring Hadoop clusters. It makes things easier for the management of a cluster by giving an easy web UI. Likewise, it delivers a healthy API to permit developers to an improved way of integration to other technologies and applications. Note that web UI is only available on Linux clusters. For Windows clusters, REST APIs can be viewed as an option.
- **Hive:** Maximum developers and BI folks now recognize that SQL and Apache Hive were created to allow those with SQL knowledge to submit MapReduce jobs utilizing an SQL-like language named HiveQL. Hive is an abstraction layer over MapReduce. HiveQL requests are internally interpreted into MapReduce jobs. Hive is theoretically closer to relational databases; henceforth, it is appropriate

for structured data. Hive also ropes user-defined functions on top of HiveQL for special-purpose processing.

- **HCatalog:** Apache HCatalog is an abstraction layer that offers a relational view of data in the Hadoop cluster. We can use Pig, or Hive, or any additional higher-level processing tools on top of HCatalog. It provisions the interpretation or writing of any file for which serializer-deserializer can be written.
- **Oozie:** Oozie is one of the important elements of the Apache data product ecosystem. It is a Java web application that takes care of the workflow organization. It achieves it by doing coordination for Hadoop jobs. In Oozie, a workflow is well-defined as intensive acyclic graphs (DAGs) of activities. It provisions dissimilar kinds of Hadoop jobs, such as MapReduce, Streaming, Pig, Hive, Sqoop, and more. Not only these, but also system-specific jobs, such as shell scripts and Java programs.
- **Pig:** Apache Pig is a high-level platform for analyzing big data sets. It needs multifaceted MapReduce transformations that use a scripting language named Pig Latin. Pig translates Pig Latin scripts to a sequence of MapReduce works to run in the Hadoop environment. It automatically enhances the implementation of complex tasks, permitting the user to emphasize business logic and semantics rather than competence. Also, you can produce your **User-Defined Functions (UDFs)** to outspread Pig Latin to do special-purpose processing.
- **Spark:** Apache Spark is a quick, in memory, parallel-processing framework that increases the performance and efficiency of digital applications (ML, AI, Big data). Spark empowers high performance querying on big data. It takes advantage of parallel data processing framework that persists data in-memory and disk if needed. It runs on top of the existing Hadoop cluster and can access the Hadoop data store (HDFS), and also process structured data from Hive. It streamlines data from HDFS, Flume, Kafka, Twitter, and other sources. It is receiving a lot of consideration from the digital community since it can deliver enormous performance improvements over MapReduce jobs. Also, it is a big data technology through which you can perform flowing analytics. It works with SQL and traditional applications as well.

Spouts are thought as the sources of data streams. Its role is usually to read data from external sources (for example, Twitter, Facebook API) or physical/virtual disk and emit or pass them in the topology.

- **Storm:** Apache Storm permits to process big amounts of real-time data that is impending at a high velocity. It can process up to a billion records per second. It is also obtainable as a managed service. An application would be defined and identified in Storm through a Topology. The topology would explain its logic as a DAG of operators and data streams.
- **Bolts** process input streams and (eventually) produce output streams. They represent the application logic.
- **Apache Solr:** Apache Solr is an open-source enterprise search platform that is based on Java search servers. Solr was developed by using Standards Based Open Interfaces XML. Solr is widely used with enterprise content management systems and to provide the power of full-text and near real-time indexing to some of the big internet sites. For example, Alfresco and Drupal use Apache Solr to incorporate search capabilities and provide benefits to the end-user.
- **Sqoop:** Apache Sqoop is an instrument or tool for the transmission of bulk data to and from Hadoop and relational databases proficiently. It is used to import data from RDBMS. An example of database systems is Oracle, MySQL, SQL Server, and MySQL systems. It then does process over them and ultimately completes the transformation on the so-called data using Hive or MapReduce. Thereafter, it exports the data back to the relational databases.
- **Tez:** Apache Tez is an application framework constructed on top of YARN to permit high-performance batch and cooperative data processing. Tez is a Hindi word that means quick, and as the term suggests, it makes Hadoop jobs are exceedingly quicker while satisfying MapReduce's scalability for petabytes of data. It is an inheritor of the MapReduce framework. It is more elastic and delivers better performance for data-intensive processes, such as Hive and Pig.
- **ZooKeeper:** Apache ZooKeeper organizes distributed processes on servers through a common ranked namespace of data registers (node),

comparable to a file system. ZooKeeper is an imitation over several hosts (called an ensemble) and the servers are conscious of each other. Distributed applications use ZooKeeper to stock and arbitrate appraises to significant configuration information.

Before concluding this section, an end-to-end technology Stack of Hadoop products are presented in pictorial form here, which would give the reader an overview of the flow of data from "raw form to insight". Hortonworks Data Platform (HDP) is an open-source platform used for the distributed archiving and processing of massive, multi-source data collections. In addition to assisting in generating new revenue streams, enhancing customer experiences, and reducing expenses, HDP helps in modernizing IT infrastructure and maintains the security of data in both the scenarios i.e, on-premises and in the Cloud. Hadoop products are open-sourced therefore, multiple vendors provide their customized versions around them and offer as a framework to cater to market needs and demand. Cloud. Hadoop products are open-sourced therefore, multiple vendors provide their customized versions around them and offer as a framework to cater to market needs and demand. For instance, Apart from HDP, Cloudera also provides their customizations of Hadoop products in the form of the Cloudera Hadoop framework that offers a number of new features and improvements, such as a resource management module, deployment using a wizard, and dashboard management, which simplifies the expansion and capacity management strategy. [Figure 5.26](#) gives a perspective on how multiple technologies that were described earlier would work in the practical world:

The latest version of HDP delivers new capabilities for the enterprise to enable agile application deployment, new machine learning/deep learning workloads, real-time data warehousing, & security and governance. It is a key component of the modern data architecture.

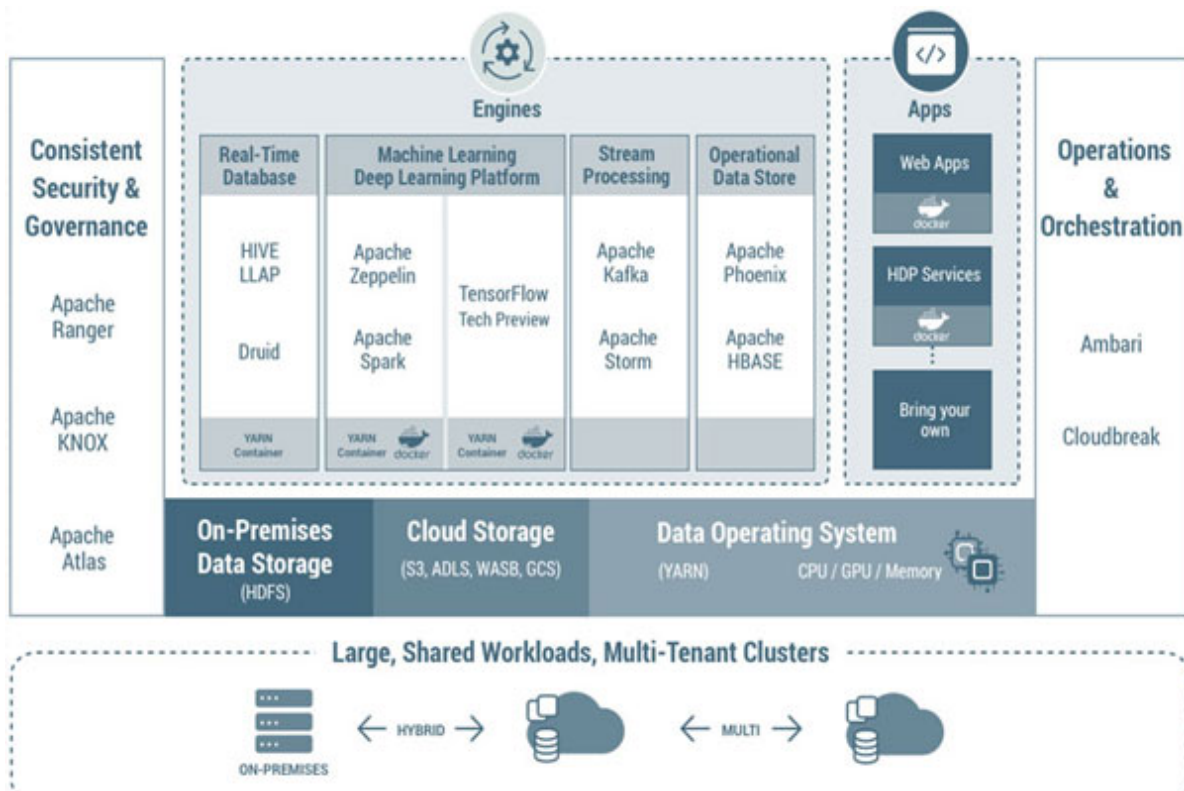


Figure 5.26: Hadoop technology offering for data-based solution

An excellent decision tree graph shown in [Figure 5.27](#); the graph world depicts the some standers challenges, facts and the briefs of big data ecosystem.

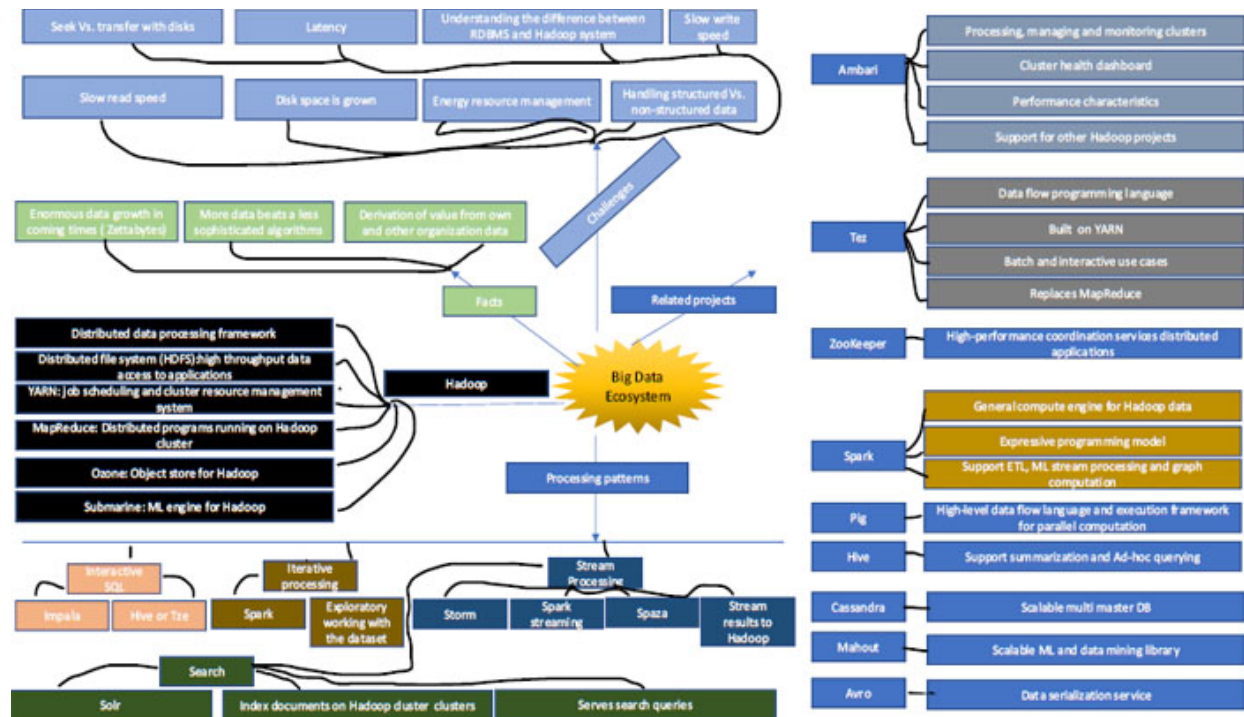


Figure 5.27: Decision tree graph for picking right big data solution

As we discussed, apart from Apache, multiple other companies provide end to end solutions for the big data, AI and ML, which include IBM, Google, and many more. Discussing offers and solutions from all is not possible here due to the obvious reasons, like space. However, it is good to give a quick look at a couple of big offerings. It will help while deciding on “go / no-go”.

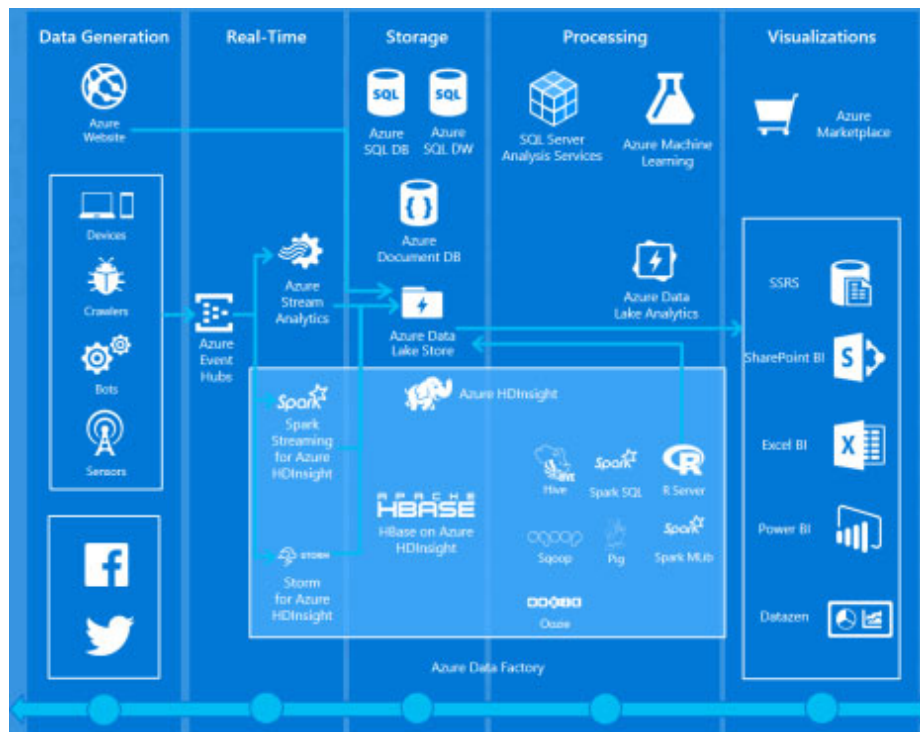


Figure 5.28a: Microsoft way of doing the data analytics (legacy way) (Source: Google image)

In order to get a holistic picture on big data offerings, let's have a look what Microsoft offers. [Figure 5.28a](#) represents Microsoft's way of doing the data analytics (legacy way) whereas [5.28b](#) showcase Microsoft's way of doing the data analytics (End to End), in modern perspective through Azure Synapse in the context of big data. Having said that, Spark is also part of later for analytics purposes. For completeness, we would also discuss Databricks in brief. But first, start with Azure Synapse.

By combining Big Data Analytics, Data Lake, Data Warehousing, and Data Integration into a single unified platform, Azure Synapse offers an End-to-End Analytics Solution. It can conduct intelligent distributed queries across nodes in the backend in a fault-tolerant way, allowing it to query relational and non-relational data at a petabyte scale. Fundamentally, Synapse SQL, Spark, Synapse Pipeline, and Studio make up the Synapse architecture. Synapse SQL aids in the execution of SQL queries, whereas Apache Spark handles batch/stream processing for Big Data. Synapse Studio offers a secure collaborative cloud-based analytics platform with AI, ML, IoT, and BI in one location, while Synapse Pipeline offers ETL (Extract-Transform-Loading) and Data Integration capabilities.

Azure Synapse uses Azure Data Lake Storage Gen2 (Data Lake Storage Gen2 combines features from Azure Data Lake Storage Gen1, such as file system semantics, directory, and file level security and scale with low-cost, tiered storage, high availability/disaster recovery capabilities from Azure Blob storage) as a data warehouse and to provide a unified data model with sections for administration, monitoring, and metadata management. In terms of security, you may use features like single sign-on and Azure Active Directory integration to safeguard, manage, and administer your data and analysis solutions. Since it incorporates additional stages of the process and gives users the option to also build reports and visualisations, Azure Synapse essentially completes the entire data integration and ETL process. It is also considerably more than a typical data warehouse. It also provides support for a variety of computer languages, including SQL, Python, C#, Java, Scala, and R. Because of this, it is ideally suited for various analysis workloads. The Synapse Analytics Studio includes everything, which makes it simple to combine AI, ML, IoT, intelligent apps, or business intelligence within the same unified platform.

Data bricks and Azure Synapse are frequently confused. Let's have a look at the brief difference. While Databricks is a cloud-based data engineering tool for processing, converting, and examining massive volumes of data to construct Machine Learning models naturally, Azure Synapse offers End to End analysis service, making the goals of both services extremely obvious. The Databricks platform now supports AWS, Microsoft Azure, and Google Cloud as its three main cloud partners. With just one click on Azure Portal, users may access Azure Databricks, a first-party service from Microsoft that was jointly created.

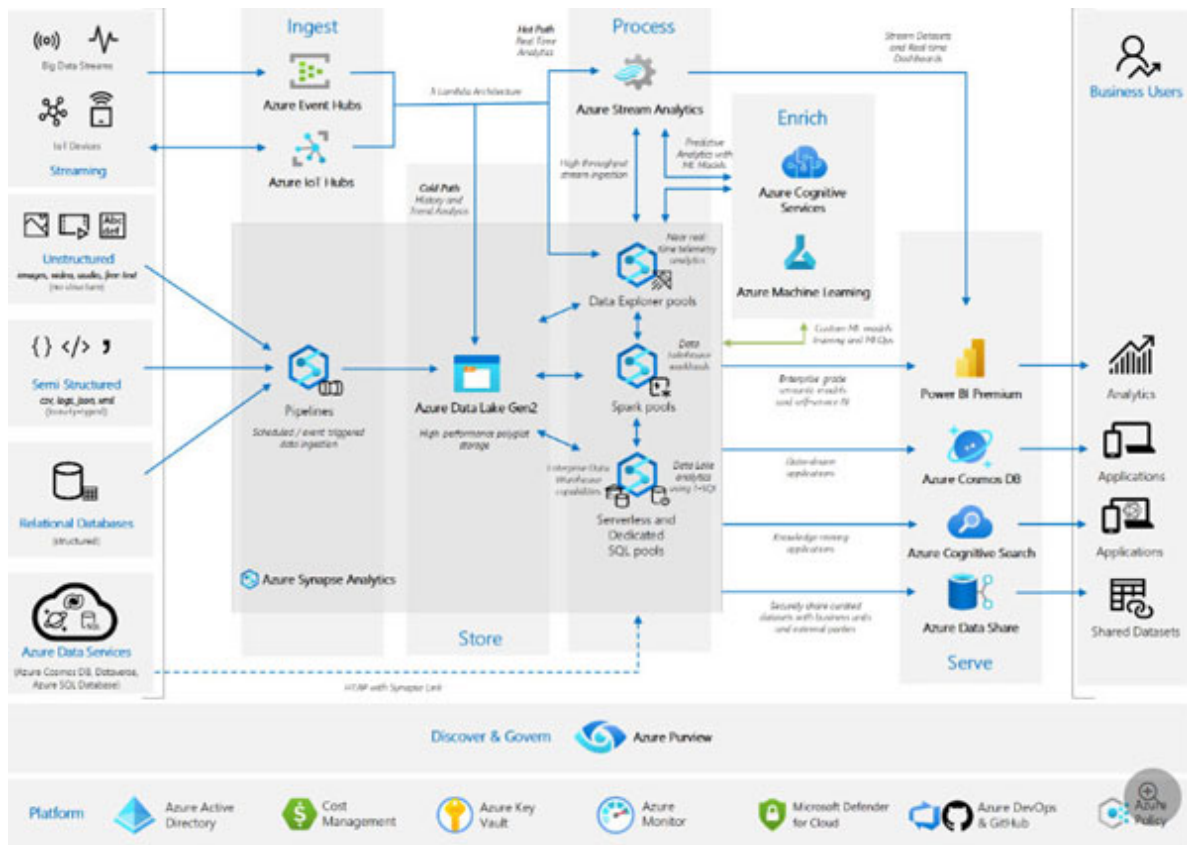


Figure 5.28b: Microsoft way of doing in the data analytics End to End, in modern perspective (Azure Synapse, Source: Microsoft website)

As we call out a few offerings provided by different vendors. Now quickly look at more details related to them. HDP and the Microsoft data platform on Azure are significant solutions for end-to-end data analytics. As we know Microsoft provides an open-source analytics solution HDInsight, it is a customizable, enterprise-grade offering. Azure (Microsoft cloud solution) does data processing at a global scale, crunch enormous volumes of data with ease and enables users to enjoy all the advantages of the robust open-source project ecosystem. It helps big data workloads and processing can be moved quickly to the cloud.

HDInsight is a Hadoop distribution provided by one of the leading vendors, Hortonworks. It is a common misapprehension that HDInsight is enclosed around Hadoop or is the part of core Hadoop by Microsoft. HDInsight is Apache Hadoop solution which runs on Azure (Microsoft's cloud platform). It is unvarying, open-source Hadoop, and must not be mistaken as a distinct Microsoft form of Hadoop. Henceforth, one can run any Hadoop

application over HDInsight, even without doing minor changes. HDInsight is the obtainable framework on the cloud platform as a go-to solution for big data, AI, and ML solution development. It comprises the employment of Apache products like Spark, HBase, Storm, Pig, Hive, Sqoop, Oozie, Ambari, and so forth. The HDInsight can be effortlessly combined with business intelligence tools, such as Power BI, Excel, SQL Server Analysis Services, and SQL Server Reporting Services (please refer [figure 5.28a](#) for the details).

HDInsight delivers a preconfigured Hadoop cluster with additional components. There is a number of choices, together with Apache Hadoop, Spark, Storm, HBase, Spark, R Server, Interactive Hive, and Kafka, which are available. Not all cluster kinds are obtainable on Windows. Only Hadoop, HBase, and Storm are obtainable on both operating systems, which will alter in the future. One can also modify their cluster if required. For doing this, HDInsight offers script actions. Utilizing script actions, one can install components, such as Hue, R, Giraph, Solr, and so forth. These scripts are bash scripts, and can run throughout cluster formation on a running cluster, or when totaling more nodes to a cluster using dynamic scaling.

The advantages of HDInsight

Hadoop in HDInsight offers several benefits. A few of them are listed here.

- Hassle-free provisioning: Rapidly shapes cluster. Takes data from Azure Blob storage and sobbing the cluster when it is not desirable. Uses the correct cluster size and hardware volume to decrease the time for analytics and cost as per requirements. Provide facility to choose among a Windows or a Linux cluster, a sole litness that only HDInsight delivers. It turns current Hadoop workloads without adapting a single line of code. Also, it takes care of another pain area in the structure cluster. That is, mixing different components, such as Hive, Pig, Spark, HBase, and so forth. HDInsight offers seamless integration deprived of perturbing about which version works with a specific Hadoop version.
- A persistent data storage choice that is dependable and inexpensive. With old-style Hadoop, the data stored in HDFS is demolished when

you tear down your cluster, but with HDInsight and Azure Blob storage, since your data is not destined to HDInsight, the same data can be fed into several Hadoop clusters or different applications.

- Automate cluster errands with easy and elastic PowerShell scripts or from an Azure command-line tool. Cluster scaling allows you to vigorously add and eliminate nodes without reproducing your cluster. You can scale the cluster using the Azure web portal or using PowerShell/Azure command-line script. It can be used with the Azure virtual network to provide isolation of cloud resources or hybrid scenarios where you link cloud resources with your local data center.

Here, we saw multiple options from different vendors, broadly concentrated on Hortonworks, Apache, and Microsoft based technology stack. However, ultimately it is the choice of the user and the demand for the project on which a choice would be made.

Quantum computing and artificial intelligence?

Though we have crafted programs that are superbly intelligent in some areas, even surpassing human intelligence and cognitive powers, we have not yet been able to make a computer that is even as intelligent as the small kid. Though ASI seems to be the penultimate goal for AI enthusiasts, most AI experts believe achieving AGI is the real challenge. Why is achieving general intelligence so incredibly difficult? To understand this, we need to understand the basic difference between which a computer operates and how our brain works. A computer works with distinguishable pieces of data. It performs operations faster than we can even process the information. But the human mind can draw inferences and make decisions that may not necessarily be logical (you decide to jump out of a flying plane with a parachute bag on your back at an altitude of 3000 feet or higher during (sky diving). The computer will reeve its brains...sorry, circuits out, figuring out why you chose to do it. A program can calculate the nth root of a complex number to mind-boggling precision, but show it a movie and ask it whether it liked it, and it becomes mute. These are only peripheral examples of the difference between a computer brain and a human brain. From here, it gets even more complex.

While exploiting the power of the computer systems, the curiosity of humans lead them to wonder – Can a machine think and behave as humans do? Thus, the development of AI started with the intention of creating similar intelligence in machines that we find and regard high in humans. Let us take the example of the ANI incorporated in your smart camera which takes a snap whenever someone smiles at the lens (or it detects a smiling face) -- pretty impressive. Now, say a hideous person is smiling at you. Something tells you his intentions are not right and he may con or harm you. In effect, the smiling man is not smiling. It's easy for you to tell the difference, but a computer, at this level, won't be able to do so. Consider another scenario - you have 7 bucks and you want to buy something worth 8 bucks. The computer will decide immediately that you cannot buy for 7 what is to be sold at 8. But you, endowed with human intelligence, will try to bargain with the seller and try to convince him to sell the product at a discounted price of 7 bucks. This is something that will never occur to a machine level intelligence (MI).

Old-style computers think by analyzing big groups of data. Apart from other benefits, a quantum linear system algorithm could aid to process data and resolve problems linked to numerous industries together with merchandise pricing, social networks, and chemical structures. Utilizing the influence of a big data matrix, a linear system algorithm can process bulky sets of data to infer into the upcoming events or brand forecasts. The matrix syndicates a large set of historical data, the features that may affect that data, as well as how thoroughly connected this information is, associated with each other. By doing this on a big scale, a linear system algorithm could attract moderately precise deductions about the future.

The increased traction of quantum computing has paralleled AI/ML becoming a deployed technology, leading to increased interest in whether AI/ML can be enhanced with advances in this type of computing. It was only after computing power and speed increased that AI/ML started attracting sustained investments, with a ramp-up in such investments taking place over the past decade. As increased processing capabilities develop, including through new technologies like quantum computing, it stands to reason that, so should the effectiveness of AI/ML systems.

In theory, quantum computing should allow for a fundamentally different way to address computation. Rather than looking at quantum computing as being faster computing, it needs to be looked at as a different way of computation. Given the technical architecture and types of computations that these machines perform, taking advantage of the benefits of quantum principles in computing requires new coding and algorithms specifically designed for quantum computing.

The ability to create stable and usable machines that can utilize quantum effects is still more theoretical than actual. Quantum effects are known to take place at atomic scales, but scaling that up to the larger scales required for computing in a stable and useful way is less well established. A lot of research in both the public and private sectors has focused on developing not only such capabilities but also the needed algorithms to use with this type of technology.

In terms of quantum computing, only recently has research started to address this issue. Experts are optimistic that quantum computing will improve machine learning by improving reinforcement learning—one of the key areas of machine learning. Huge data crunching capabilities is the key outcomes of quantum computing due to this feature there is an increase in AI/ ML capabilities is encountered. The ability for quantum computing to compute much more data in parallel than classical computing allows as a major advantage of this technology. The relationship between quantum computing and AI/ML is an area of research with major commercial and security implications. There is great potential for collaborative international work to develop new AI/ML algorithm approaches to take advantage of this powerful technology.

OK, now I want to understand the interrelation between AI, ML deep learning, and quantum machine learning

Big data has a large share in the success of any AI project or solution, and most AI activities occur in the context of big datasets. For instance, Big data is certainly necessary if you are going to build facial recognition solution because, for this, there is a need for storing and analyzing hundreds of pictures, at high resolution, taken by millions of people/machines/ apps. How you will analyze them without the use of big data. Again, not only big

data, you certainly need deep / machine and in future quantum (machine learning). [Figure 5.29](#) showcases the interrelation among them.

But big data is just one type of data your AI algorithms can use, so you shouldn't think solely of big vs. small data. Think instead of all the data necessary to make decisions. Sometimes you don't need big datasets.

Some datasets may be small, but they still hold information about important outcomes and may be expensive to collect. To regain data from structured data schemas, one may have been running typical SQL requests for several years. But in the changing time, it is not enough to have only structured data, as data is available in multiple sources, including social media and logfiles. To make an AI solution to be effective, we need all kinds of data and after performing transformation, we would run the desired algorithm on top of that. Further refer to [Figure 5.29](#) that depicts the interrelation between ML, AI, deep, and quantum machine learning:

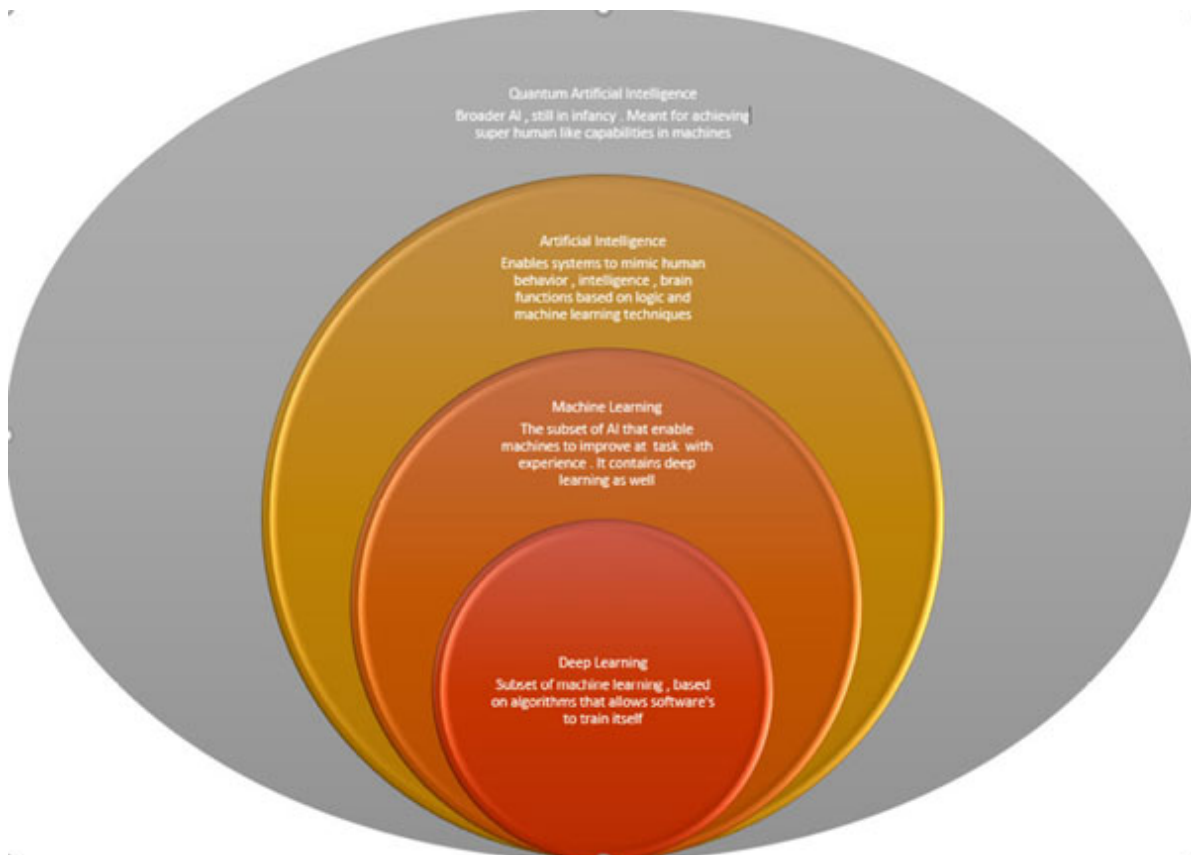


Figure 5.29: Interrelation between ML, AI, deep, and quantum machine learning

Use cases for AI

Now, we understood most of the concepts related to AI, ML, Big Data, IoT and quantum computing at least on the broader level. So, now is a good time to look at some typical use cases. In the practical world, technology has no use until we utilize it either for profit generation, business growth, or making someone's life easier. For AI, enormous use cases exist in almost all walks of life, business, and research. However, it is not possible to cover all of them in this chapter. But few of them are covered in [Figure 5.30](#):

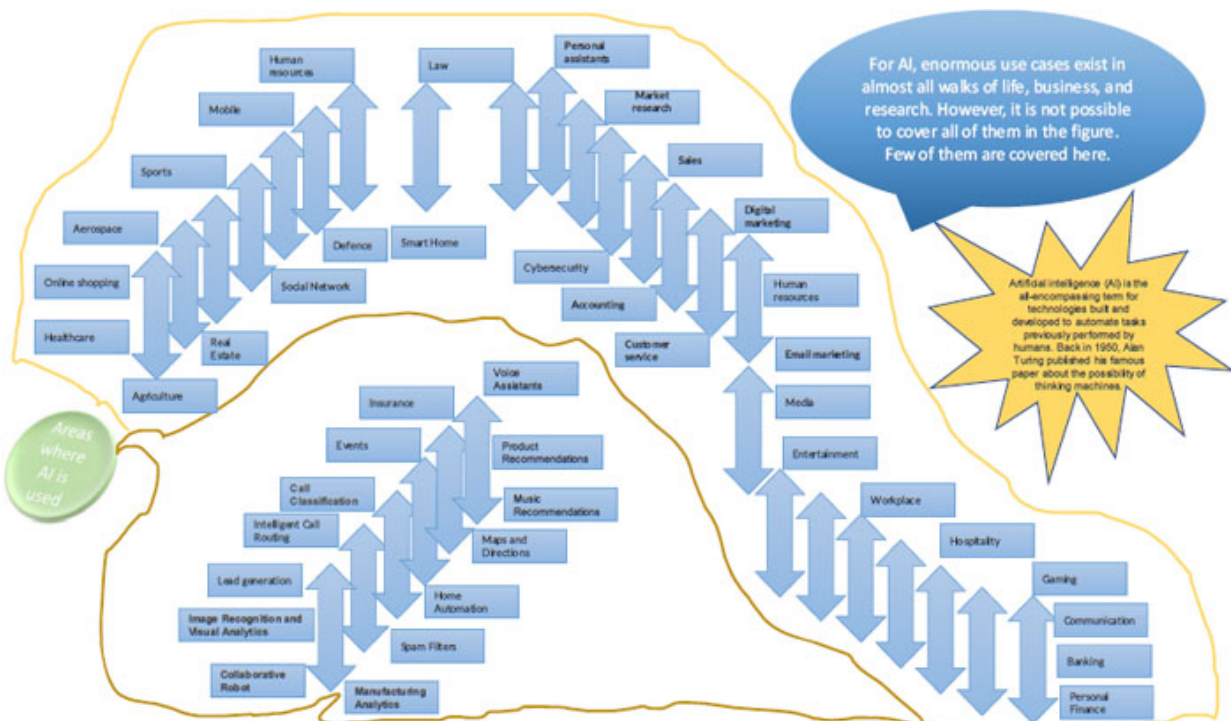


Figure 5.30: A fresh look at AI use cases

Story 1: Future of humanity-- integration of Brain science, AI and quantum computers

The theoretical foundation of Quantum neural networks has been present since the 1990's, and there is extensive research since then. Quantum Neural network is one of them. The optimism is that landscapes of quantum computing such as concurrency or the effects of interference and entanglement can be used as assets for attaining effectiveness in artificial neural research. Quantum physics and its merger of the brain or more precisely neurophysiology through quantum computers will potentially take

possibilities of excellence way beyond the boundaries of science. For instance, there are pioneer works happening in the theoretical field of quantum biology which has been gaining momentum by discoveries like, efficient energy transport in photosynthesis due to quantum effects. The chip industry along with biotech is also riding on the boat, companies are developing a circuit-based quantum computer that simulates associative memory utilizing execution of an algorithm which writes memory states into superpositions, and then it uses a Grover-like quantum search to retrieve the memory state closest to a given input with the ultimate goal being to simulate geographies of the human brain. Also, ideas have been proposed about modeling black holes as **Quantum Neural Network (QNN)** because theories say that black holes and brains may store memories in similar ways. Additionally, there is a lot of research happening to create a network of AI-based quantum computers to bind humanity in the future. These technical directions are also leading a type of social transformation.

[Story 2: AI, culture, arts, and quantum computers](#)

Computational art is a creative area that designates an innovative impression of artificial intelligence, notwithstanding the shared trust that a machine is incapable to produce art. However, current developments and instances in computational art present a novel form of art. Attainment to a comprehensive diversity of artistic dimensions, artificial intelligence programs are producing poems, tune, visual art, architecture, and design. Artificial intelligence can be used as an artistic wonder and analyses the artifacts that are shaped by computer algorithms. It delivers a theoretical framework and a logical debate on the creative aptitudes of artificial intelligence.

The analysis of computational artworks and the computer code that produce artworks show that the purpose of artificial intelligence is far outside being just a tool to create art; it is somewhat an actor that has an artistic and creative action.

Considering AI art as a human/non-human hybrid, it is likely to define it as an autonomous art category. So, it is conceivable to identify AI art as an art form that hangs the art maker's position in self-expression and includes artistic productions without the initial meaning to express a sensation, an

aesthetical point of view, or any individual declaration. In that case, the artwork itself seems like an initial object that stretches the liberty of the viewer to brand a clarification of their own emotional or aesthetic love with the artwork. Consequently, the viewer can shape a separate joining with the artwork. Regardless of the nonappearance of an initiative emotional countenance of the creator, the movement of effect will happen through a bottom-up viewpoint, and the feeling of artwork will be formed through the emotional add-on of the human spectator.

Above and beyond the query of can artificial intelligence be measured as an artist, the ‘artiness’ of AI artworks was an additional chief autopsy that can be thought of. In this sense, the assessment of artiness is an additional feature that mentions the autonomous nature of AI art. The AI artworks are eventually assessed by human spectators as the viewers. Thus, artificial intelligence should be able to harvest the artworks that petition to human discrimination, human emotions, and human aesthetics.

Story 3: The security of an AI is an emerging topic

The security of AI systems is quickly improving the field of study. Now we’ve learned a lot of astonishing things about the security of AI algorithms. For instance, deep learning systems can be deceived into misclassifying imageries by adding minor vicissitudes to the image that is unnoticeable to the human eye but which can reason an AI to misclassify the image completely. This technique is known as **Generative Adversarial Networks (GAN)**.

Security problems aren’t limited to deep learning algorithms. For instance, model stealing could be done with numerous additional types of often used AI and ML algorithms. It’s not clear what future developments await in the area of AI security, and it’s a decent stake that new developments will arise by the time this book is in print. What’s also vibrant is that the security of AI systems is a new-fangled interests that hold numerous orders in the future. Indeed, some of the aforementioned security problems of AI came as an amazement to everybody, including AI investigators and specialists.

In its place of enquiring, if a specific AI algorithm is safe (for fairly an insufficient of the modern ones, the precise answer, is unknown), one has to

comprehend that this is a quickly developing field and that the responses received today can alter tomorrow.

Quantum use case 1: Quleap -- first integrated AI platform for small molecule drug design

Qulab, a pharma-chemical corporation recently developed Quleap. They claimed that it is the first integrated AI founded platform for small molecule drug design. Qulab's mission is to reshape R&D in the pharmacological and biochemical industries by integrating forward-thinking computing technology and automation into the discovery phase. And this is a thrilling time for the pharmacological and chemical industries because of multifaced research happening around the world in the area. Quantum computers are one of the areas out of multiple others. The quantum computers R&D is evolving quickly and corporations essentially are preparing their roadmaps to quantum-readiness. Qulab and Rigetti's partnership is a noteworthy step in the direction of structuring a scalable network around quantum computing. [Figure 5.31](#) showcases Quleap framework:

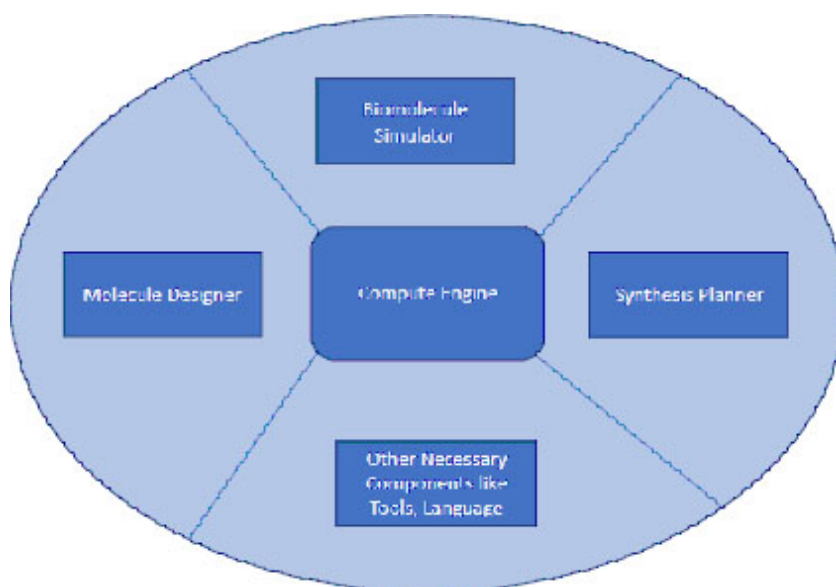


Figure 5.31: Quleap framework (source: Qulab, official website)

With the quick expansion of sequencing technologies, data is produced at a quicker and sooner pace, pouring the biomedical investigation into the age of big data and custom-made genomics. The goal of the Qulab lab is to decode,

understand, and manage the digital life at a single cell steadfastness. The laboratory is assimilating the knowhow in Bioinformatics, Immunology, Genomics, big data, internet of things, and Artificial Intelligence (BIGAI) to develop new untried and computational approaches to forecast, and control the large-scale solitary cell gene controlling programs, and to deliver insights into exact diagnosis and therapeutics for human cancer and autoimmune diseases.

Presently concentrating on small molecule drugs, Qulab is yoking computational chemistry, artificial intelligence, and quantum computing to mechanize molecule design.

The QULAB current investigation areas comprise the following:

- Exactness drug & personalized tumor regulome.
- Bioinformatics, computational natural science, and growth of cloud-based software for big data in life science.
- Transcript factor, lncRNA and epigenetic alteration in gene directive.

Quantum use case 2: Hyrax – the quantum discovery framework

Rahko is a London-based **Quantum Machine Learning (QML)** company which provides a platform for chemical simulations. The corporation was created in 2018 by Leonard Wossnig, Edward Grant, Miriam Cha, and Ian Horobin. Rahko originates from UCL's Computer Science program, reinforced by Conception X, an undertaking manufacturer for deep tech start-ups. Rahko's partners comprise IBM, Microsoft, the Nvidia Innovation program, Amazon Web Services Activate program, and the **Quantum Business Network (QBN)**.

Rahko's team focuses on 2nd generation QML algorithms; they are parametrizing quantum circuits that can be trained similar to deep learning models and be extremely vigorous to noise (we know noise is a roadblock for the development of QC).

Rahko is using QML for chemical simulation which is the instant investigation area for Rahko's QML technology. Chemical simulations take time for intense and affluent research. In areas like chemical engineering,

drug discovery, and material science, out of the lab and onto a computer is a time-consuming process. Chemical simulation utilizing QML permits the development of batteries, superconductors, therapeutic drugs, solar technologies, and advanced materials, among others at a faster rate.

If we compare classical ML-based methods to traditional processing, they do offer a speedup. Rahko uses a hybrid approach where part of the work, which is suitable for development utilizing classical ML approaches to deliver faster classical solutions that work, is done on the traditional cluster of systems using old-style processing. Whereas quantum computers are used to generate training data to improve classical ML algorithms.

For the second type of calculation (high-cost, high-precision), quantum computers will carry far superior correctness at abridged cost. Most outstandingly, quantum computers will be able to harvest precise simulations where traditional methods fail. This will be a game-changing enhancement when working with powerfully correlated materials, which play an enormous role in batteries and room temperature superconductivity.

[Hyrax](#)

Hyrax is Rahko's platform that uses an exclusive pipeline for the integration of quantum and classical hardware with QML algorithms. With Hyrax, Rahko works with clients on business valuable problems in a chemical simulation that was earlier almost difficult to solve by using classical computers. Hyrax permits clients to speedily test and archetype algorithms, and professionally access the best obtainable quantum computers.

Several algorithms have projected that QML methods are 10,000,000 times quicker than old-style variational quantum algorithms. This means that QML methods will allow applied improvements months, even years before other variational quantum approaches flourish.

Variational quantum algorithms – These algorithms permit users to alter control strictures of the quantum computer till outcomes bout a target property, such as the energy of a molecule. Variational quantum algorithms have now been used to effectively simulate small chemical systems on quantum computers over the last couple of years.

Rahko is working hard to bring these improvements for the last couple of years. Hyrax, a QML platform is a by-product of that, which allows us to quickly shape, test, and deploy QML algorithms. The platform has a high-level python interface and is hardware-independent. It can be integrated with numerous backends together with IBM Qiskit, Google Cirq, Rigetti Forest, and TensorFlow.

Hyrax Quickens

- **Catalysts:** Quantum simulation to determine improved catalysts.
- **Drug discovery:** Revealing the quantum potential for simulating possessions of drugs and proteins.
- **Batteries:** Prevailing novel tools for determining and modeling progressive materials.

The Hyrax framework

- **Hyrax application toolbox:** Hyrax bids pre-built models for area definite applications in quantum machine learning, and quantum-powered chemical simulation.
- **Hyrax quantum computing API:** Quantum and standard subroutines are associated with their control logic in a computational graph for effectual hybrid computation.
- **Hardware backends:** Extremely well-organized CPU and GPU founded simulations, with an easy switch among traditional simulation and deployment on major quantum computers.

Quantum use case 3: Detecting market instabilities

The finance sector has been one of the very early adopters of digital computers. However, there are so many intractable problems that are waiting to be solved by mathematicians, programmers, and economists. The existing solutions, if any, shall take digital computers years to solve. Scientists from different fields such as finance, computer science, quantum computing, and quantum mechanics have come together to address these problems. Hence, the term quantum finance was coined for this new interdisciplinary field.

Classic computers have empowered economists with technological tools. These tools range from basic spreadsheets application to complex enterprise resource planning solutions. The finance sector has benefitted substantially from such advanced computational power. These tools assisted in addressing problems related to financial modeling, complex analysis, time value of money, cash flow analysis, loan repayments, interest rates, assets valuation, future growth forecasting, business process workflow modeling and so on. Here we have only named a few applications of classic computers and the list shall go on indefinitely.

In a fast-paced economy, this still was not good enough to meet the unbounded ambitions of decision-makers. Some problems are still intractable such as the optimal trading trajectory, multi-period profile optimization, risk management, regression analysis, scenario analysis, path-dependent option pricing and so on. Here comes up the notion of using a quantum computer to solve intractable financial problems. Finding scalable solutions to these problems shall empower decision-makers with insights and analytics. These problems were previously expected to be solved in years, but with the advent of quantum algorithms, they shall be solved in only a matter of days if not less.

Quantum Computing will enable massively parallel processing, which is needed for handling matrices and tensors. Predicting the stock markets is the job of Machine Learning algorithms, not Quantum Computers, which is the underlying hardware that as and when practically implemented, will make Machine Learning algorithms exceedingly fast, due to the massively parallel processing capabilities of Quantum Computers. Therefore, QC would be beneficial here.

Creating any computer algorithm to track and develop outcomes-based upon all of the factors that go into determining prices on the stock market, is theoretically possible with enough computing power, and the ability to capture all of the data needed is more than a technical challenge. This challenge can be handled by QC with ease. Further, we have to be mindful about the fact that no one can predict the Stock Market with 100% accuracy; however, 80–85% prediction is possible due to the simple reason that no one knows anything can happen in Stock Market just like a Cricket match, but crunching data in real-time will help.

Quantum use case 4: Optimizing radiotherapy treatments, fighting with cancer, and the quantum computing algorithm

Microsoft's quantum dazzling algorithms are mainly beneficial for optimization problems. This includes going through an enormous number of likelihoods to discover the best or effectual solution. Generally, these are so difficult and need so much computing power that current technologies fight to resolve them.

Typical instances might comprise safeguarding traffic flows easily across a whole metropolitan area, apportioning gate and tarmac space at a demanding international airport, or shaping how to best sequence complex manufacturing processes across numerous different bits of gear.

To detect cancer and other diseases rapidly and dependably, Microsoft's quantum team is also joining with the Dubai Electricity and Water Authority, which is consuming quantum-inspired algorithms to figure out how to preferably balance resources from different energy sources across its whole electric grid. Also, nowadays, if somebody has breast cancer, then she or he may have to undertake numerous circles of chemotherapy and devote months in limbo before medical scans can demonstrate if that specific concoction of toxic drugs is decreasing the tumor. Case Western Reserve University investigators are at work to alter that. They have initiated a new method called Magnetic Resonance Fingerprinting, which uses more delicate scanning methods that they think could sense whether actions are working after just one dose of chemo. The new-fangled technique has unbelievable potential, but scheming the scans to rapidly and precisely identify the disease is a massively challenging computational problem necessitating ground-breaking methods. In this day and age, the Case Western Reserve investigators have started a solution to that problem and understood affected developments utilizing algorithms advanced by Microsoft's quantum computing team.

Willis Towers Watson, a worldwide advice-giving, brokerage, and solutions enterprise, is also discovering how Microsoft's quantum enthused algorithms might progress the complex scientific models the company uses to compute risk and inform speculation plans.

Microsoft investigators advanced the algorithms as part of a greater exertion to fashion the business's furthest steady and ascendable quantum computer utilizing quantum information particles named topological qubits. As we know, when it's built, the investigators say the quantum computing platform might license specialists to do calculations in minutes that would take current computers billions of years.

The quantum founded algorithms simulate how those systems work but can be run on current traditional style computers. As the advancement of a general-purpose quantum computer endures to improvement, corporations today can join the Microsoft Quantum Network.

[Mind Map for the chapter](#)

The following is the mind map for the chapter, which graphically summarizes the concepts explained in the chapter:



[Conclusion](#)

This chapter explained in brief the technologies available for IoT, Big data analytics, and machine learning in the context of AI. It has also given the perceptive of technology in the current business scenarios. It explains how QC along with cloud computing, big data, AI, and IoT will shape the next few decades.

In the next chapter, we will cover quantum machine learning in detail.

CHAPTER 6

Quantum Algorithms for Everyone ...!!!

“We must be clear that when it comes to atoms, language can be used only as in poetry.”

— Niels Bohr

“Quantum mechanics makes absolutely no sense.”

— Roger Penrose

As quantum computers turn out to be accessible to a wide-ranging sample of the public, the requirement has stood up to train a unit of quantum programmers. Also, the fact is that most of them have been engaged in developing classical computer programs for years. Though currently existing quantum computers have fewer than 100 qubits, quantum computing hardware is extensively predictable to produce more in terms of qubit total, excellence, and connectivity. Multiple quantum algorithms exist which can be implemented through frameworks and libraries provided by multiple vendors including Microsoft and IBM.

Quantum Algorithms can be visualized as the means of unlocking future additional value in the quantum IT world. There is no limit to the number of possible quantum algorithms, but due to the immense complexity of multi-valued logic behind quantum algorithms, the number of present-day mathematicians that can develop them successfully is very limited.

The potential markets will open up in the coming time as and when QC becomes robust and the requirement for speed increases. When the quantum machine becomes efficient and practical from the hardware perspective then as a by-product of this the quantum algorithms as well develop. For instance, the best choice for a Datacentre Company may no longer just be based upon price but could be aligned to the ones that can most effectively perform your application’s tasks. Choosing between conventional suppliers and a supplier that offers ten times the speed of operation for mission-

critical tasks is a no-brainer and these are excellent fields for quantum algorithms to flourish.

As time passes, quantum algorithms find their applicability in all fields. For instance, the finance sector opens up a whole range of compliance, improvements, savings, and new markets. Besides compliance (for example, security and cryptocurrency markets), there is a lot of potential in pattern recognition, real-time risk analysis, and financial forecasting. Also, for banks with large client databases, overheads can be reduced through the use of improved searching quantum algorithms.

Structure

This chapter is organized in a very intuitive way. It starts by explaining the basic ideas behind computation and goes further to explain what algorithm is all about. Then, it deep dives into the quantum algorithms and discusses around 15 algorithms. However, the chapter focuses more on the two main algorithms called Shor and Grover. Why? Because these are the most fundamental and widely used algorithms.

In this chapter, the following topics will be covered:

- What is computation
- What is quantum computation
- Moore's law
- The Deutsch algorithm
- Shor's algorithm
- Grover's algorithm
- Quantum computation and algorithms – a collection of vital facts
- Story 1: Semiconductor spin qubits for construction of reliable quantum computers
- Story 2: Quantum compatible areas for algorithms
- Story 3: Quantum computing and corporate traction
- Provocative thought 1: selecting the right quantum computer for your business

- Provocative thought 2: selecting the right quantum computer for your business
- Provocative thought 3: industry transformation and quantum computers
- Latest research
- Quantum computation and algorithms– Important questions and answers
- Mind Map

Objective

The stride of growth in quantum computing reflects the quick developments made in machine intelligence and artificial smartness. So, it is normal to inquire whether quantum technologies possibly will provide enhancement to learning algorithms or not. This turf of investigation is named as quantum-enhanced machine learning.

The objective of this chapter is to show what benefits current and future quantum technologies can provide to computation, focusing on algorithms that are challenging with classical digital computers. The chapter will discuss quantum computing models and implementations. It will also throw light on the limits of the current and future quantum algorithms. And showcase the kind of responsibilities where they outstrip or predict to outdo traditional computational intelligence. It will explain the important algorithms and different types of quantum computers.

Introduction

Quantum computers would endeavour the bizarre laws of quantum physics to process information in conducts that are dreadful on a standard computer. They would solve precise problems, such as factoring integers, in an exceedingly quicker way than what we expect. Now, we can distinguish how to resolve a few specific problems with QC more efficiently whereas others with classical computers. The instances where the QCs are coming out from the lab suggest that if we are mindful (with the help of analysis), then the problems where quantum computers are involved would surpass

the conventional “time-consuming way” to solve problems due to its extraordinary processing power. Bizarre modifications to the identified rules of physics would permit the building of computers that could resolve large classes of tough problems efficiently and effectively. But those modifications appear unlikely on a classical computer. In the real world, possibly the castle in the air of proficiently resolving these problems should be engaged as an elementary physical principle, hence everyone started looking out for more.

Adventures in quantum algorithms and problem-solving

Quantum computing adventures will showcase its effect in all walks of human endeavour, but especially in quantum-mechanical effects, along with superposition, entanglement, and quantum tunnelling to achieve more efficient execution computation. Compared to traditional, electronic computing, quantum computing bids the possibility to sensationally decrease at both implementation time and energy consumption. These potential recompenses are the by-product of steady developments in nano-manufacturing, and the slow-down of old hardware scaling laws, for example, Moore’s Law. Therefore, it led to considerable saleable and national-security attention and outlay in quantum computing technology.

The problem undertaken by the quantum computer is not one with any direct real-world application. However, this is a turning point for quantum computing and is extensively seen as a significant step in the direction of building quantum computers that will bid applied speedups when resolving real-world problems. Even though the scientific basis of quantum computing, the software design model, and most quantum algorithms have been available for ages (since the 1990's), they have been of interest only to a small dedicated community.

Now the time has come to make quantum algorithms and their implementations accessible to a broad swath of researchers and developers across computer science, software engineering, and other fields.

The quantum programming model is fundamentally different from traditional computer programming. It is dominated by physics and algebraic notations that at times are not required for normal computer experts and

other more statistically trained researchers. For instance, one of the most common quantum algorithms called Shor's algorithm is an excellent pick not only because it delivers exponential speedup over the dissolute traditional algorithms, but also because it is very effective in public-key cryptography, together with the commonly used RSA algorithm. Further, this is way ahead in doing factors of integer into the prime numbers than its counterpart classical algorithms. Shor proved that the comprehension of a full-scale quantum computer would have possibly been able to provide a truthfully noteworthy upsurge in computing speed, and at the same time, direct the likely insinuations of such an intensification in computational power to the arena of cybersecurity. For these reasons, Shor's finding flashed a prodigious transaction of attention in the design of quantum algorithms and computers that bears today.

As we saw in the earlier chapter(s), quantum algorithms are frequently represented diagrammatically as circuits in popular literature. In the circuit depiction, qubits are characterized by horizontal lines. Gates are then strained on the qubits they act on. This is done in order from left to right. The early state of the qubit is signified at the commencement of each of the qubit lines. When we inscribe down a mathematical countenance for the circuit, the gates are written down from right to left in the order of their application.

A case of quantum algorithms

Technically, an algorithm is a step-by-step technique to achieve a calculation, or an order of instructions to resolve a problem, where the steps can be individually completed on a computer. Consequently, an algorithm is a quantum algorithm when it can be done on a quantum computer. In theory, it is conceivable to run all standard algorithms on a quantum computer. Though, the term quantum algorithm is functional to algorithms of which at least one of the stages is definitely 'quantum', using superposition or entanglement.

Problems that are deeply impenetrable by classical algorithms cannot be resolved by quantum algorithms. The extra value of quantum algorithms is that they can crack some problems expressively quicker than classical

algorithms. The best-known cases are Shor's and Grover's algorithms. Shor's algorithm is a fundamental quantum algorithm which is used for integer factorization or prime number calculation. So, when an integer N is provided, it will calculate or find its prime factors. Also, it is notable here that it can resolve this problem exponentially sooner than the best-known classical algorithm can potentially do. Similarly, the other well-known quantum algorithm called Grover's algorithm can perform a search on an unstructured database or rather unordered list of records quicker than the best classical algorithm meant for this purpose.

The fundamental question when we considered QC is how a quantum computer could deliver quickness for some problems, such as contravention codes, but is not that efficient for the other set of problems. Isn't a quicker computer just a faster computer? The response is no, and to clarify this, imagine how computer science and scientist work and think.

For computer scientists, the vital thing about a problem is how rapidly the time required to solve a problem grows as its extent upsurges. The time is calculated in the number of basic steps essential by the algorithm to get a solution. For instance, utilizing the grade school method, we can multiply dual n -digit figures in an amount of time that produces the number of digits squared, n^2 (a quantity of time said to be a polynomial in n). But for factoring a numeral into primes, even the greatest forward-thinking approaches recognized to date, would take an amount of time that grows exponentially with the figure of digits (in specific, like 2 to the cube root of n power). Thus, factoring appears inherently tougher than multiplying. Also, when we get up to thousands of digits, this change matters much more. The type of problems that computers can resolve in a judicious amount of time, even for big values of n , are those for which we have an algorithm that uses numerous steps that increase as n is extended to a fixed power, such as n , n^2 or n^3 .

Computer scientists call such an algorithm effectual, and problems that can be resolved by a well-organized algorithm are said to be in the difficulty or complexity of class P , which stands for "polynomial time." A simple example of a problem in P is: If a road map is provided, then is all town reachable from every other town? P also comprises a few problems whose effectual answers are not so understandable, such as: providing a whole

number, is the given number prime (like 13) or composite (like 12)? Another example would be “provided with a list of which men and women are eager to wed one another, is it likely to pair everybody off with a keen partner? Quantum computers use different computer algorithms for solving these types of problems.

The algorithms like Grover and Simon are important landmarks for efficiency and performance improvement. Also, these two algorithms are decent models for the present understanding of quantum computation. Apart from these, several other quantum algorithms use comparable techniques to attain things which were never thought of before, like, they are used for cracking complex linear or differential systems of equations, or rapidly computing unthinkable numbers. For instance, Grover’s algorithm hunts for a quantified entry in an unordered database, using an imperative method in quantum algorithm design to recognize amplitude amplification to attain a polynomial speedup over the best classical algorithms.

These are best for any quantum algorithm for the execution of such type of search. Also, searching for an exclusive element in an unordered group can be generalized to put on to extensive diversity of problems in computer science. One of the examples of such a case is the problem of Boolean satisfiability (SAT).

Likewise, defining whether there occurs a set of truth values that gratify a given set of divisions is equal to looking for an entity, or the collection of truth values, that contents a sure condition, the group of clauses, in an unorganized or unordered search space or sample. The set, therefore, of all possible truth values of Boolean variables (who stay either in 1 or 0 state) comes under the umbrella of these algorithms. The speedup is still surely important with large input for all the quantum algorithms.

An algorithm called Daniel Simon’s algorithm is used for defining the exclusive-or (XOR) mask over which an assumed black-box function is invariant. Simon’s algorithm was the first quantum algorithm initiate to have exponential speedup over any corresponding popular algorithm of today, better termed as classical or traditional algorithms, and the runtime of his algorithm is optimal. Though the problem may at first appear very non-concrete, Simon’s problem resembles a significant problem with some

applications in computer science known as the concealed Abelian subgroup problem, which comprises factoring integers into primes and scheming separate logarithms.

What is an algorithm by the way?

An algorithm is a portrayal of a process that dismisses an outcome. It has the following five basic physiognomies:

- **Input:** An algorithm should have an input.
- **Process:** Computation should be complete on the specified input to get the desired output.
- **Finiteness:** An algorithm should come to an end after a limited number of steps.
- **Effectiveness:** Each step of an algorithm should be exact and precise in such a way that it should be executable in a certain period on the target machine.
- **Output:** An algorithm should give an anticipated output.

Types of algorithms

There are multiple types of algorithms; a few important ones are explained as follows:

- **Recursion or iteration:** A recursive algorithm is unique because it prays (refers to) itself repeatedly. This is characteristically a technique that is specifically vigilant to functional programming languages. Iterative algorithms generally use repetitive hypotheses like loops and occasionally bet on data structures like stacks to resolve the specified problems. Also, the fact is few problems are certainly suitable for one implementation or the other. For instance, something called towers of Hanoi is a decent applicant for recursive implementation. Every recursive version has a corresponding iterative form and vice versa.
- **Logical:** An algorithm may be experiential as measured logical inference. This idea may be articulated as Algorithm = logic + control; in other words, we can say that an algorithm is nothing but the

combination of logic and flow of controls. The logic component especially delivers the axioms that may be used in the computation and the control component recast off how inference is pragmatic to the axioms.

This is the foundation for the logic programming paradigm. In unadulterated software development by the use of logic-based programming languages, the control component is immovable and algorithms are stated by providing only the logic component. The petition of this method is the sophisticated semantics; an alteration in the axioms has a well-defined change in the algorithm.

- **Serial, parallel, and distributed:** Algorithms are typically deliberated with the supposition that computers perform one instruction of an algorithm at a time. Those computers are occasionally termed serial computers. An algorithm intended to run over the serial computers is called a serial algorithm. The serial algorithms are fundamentally opposed to something called parallel algorithms or **distributed** algorithms.
- **Parallel:** algorithms are the algorithms that take benefit of a special type of computer architecture where processors (typically more than one) can work on a given problem at a similar time (in a concurrent fashion), whereas dispersed algorithms use manifold machines associated with a network. Parallel or distributed algorithms split the problem under consideration into further proportioned or asymmetrical subproblems (by doing multiple logical divisions) and accumulate the results back together.

The resource feeding in such special types of algorithms is not only limited to something called processor cycles on each processor but also the communication above amid the processors. Sorting algorithms can be parallelized competently, but their communication directly above is exclusive. Iterative algorithms are usually parallelizable. Some problems have no analogous algorithms and are named integrally **serial** problems.

- **Deterministic and non-deterministic:** Deterministic algorithms resolve the problem with the meticulous decision at each step of the

algorithm while non-deterministic algorithms resolve problems via predicting, although typical guesses are more precise through the use of heuristics.

- **Exact and approximate:** While many algorithms provide an exact solution, approximation algorithms pursue an estimate that is close to the factual solution. Approximation may use either a deterministic or a haphazard plan. Such algorithms have applied value for numerous security problems.
- **Quantum algorithm:** Quantum algorithm runs on a truthful model of quantum computation. The term is typically used for those algorithms which appear characteristically quantum. The quantum algorithms most frequently use vital features of standard quantum computation. Those features include quantum superposition or quantum entanglement. However, it is not limited to only these features.

Examples of an algorithm:

- Algorithm to find the greatest common divisor of a number:
 1. Step1: Start
 2. Step2: Input two numbers greater than zero in the variable A and B
 3. Step3: IF B=0 THEN GOTO 9
 4. Step4: IF A > B THEN GOTO 7
 5. Step5: LET B=B-A
 6. Step6: GOTO 3
 7. Step7: LET A=A-B
 8. Step8: GOTO 3
 9. Step9: PRINT A
 10. Step10: Stop
- The algorithm that reads a list of 50 integer values and prints the total of the list:
 1. Step1: Start

2. Step2: Total = 0
3. Step3: Input a number in VAL
4. Step4: $C = C + 1$
5. Step5: Total = Total + VAL
6. Step6: Do a comparison between C and 50
7. Step7: If C is fewer than 50, rerun steps 3 to 6
8. Step8: PRINT Total
9. Step10: Stop

What are quantum algorithms

It is a sequence of commands as any normal algorithm, but this time it's a circuit constructed which comprises quantum gates. Ok, then what is a quantum gate? As we already discussed, but just for completeness, it's like a traditional gate, like OR, AND, NOT, that acts on qubits in place of bits. Well then what is a qubit? As we know, it's a quantum bit that is technical of being 1 or 0; it can be a bit of both. For example, it can be in the superposition state $20\%|0\rangle + 80\%|1\rangle$.

We use the representation $|0\rangle$ and $|1\rangle$ to present these as quantum states. Further, the state of a qubit is $\alpha|0\rangle + \beta|1\rangle$, where α and β are complex numbers named amplitudes. With 2 qubits, one can have the four amalgamations $|00\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$ at a similar time. The eight amalgamations with 3 qubits, and so on, continuously exist in parallel.

Practically, a qubit can be an atom, a photon, an ion, or any quantum system that contains two states. Hence, the input of a quantum algorithm is constructed of qubits. In a nutshell, if you have n qubits, you have concurrently 2^n formations. So, what it means is that with 300 qubits, you have 2^{300} states. technically, these are big volumes of atoms that exist in the universe. As we know, to achieve these, quantum computers use the superposition principle. A small number of elements in superposition states can transport a huge amount of data. For instance, a simple 1,000 elements can be in a superposition that signifies every number from 1 to 21,000, and a quantum computer would operate all those numbers in parallel, for instance, by striking the particles with laser throbs. When the elements'

states are measured at the end of the computation, however, all but one arbitrary version of the parallel states disappear. Ingenious manipulation of the elements could nonetheless resolve specific problems very rapidly, such as factoring in a large number.

We have now had an idea of all the fundamental elements required for the understanding of practical quantum algorithms. So now let's define the Quantum algorithm. A quantum algorithm consists of three basic building blocks or steps:

- Its capability is the encoding of the data, which could be classical or quantum, into the state of a set of input qubits.
- An order of quantum gates functional to this group of input qubits.
- Measurements of one or more of the qubits at the end to get a classically interpretable outcome.

How Quantum Algorithms are different from the classical algorithm

So, based on the discussion, we can say that quantum computers solve a problem more quickly than a classical computer. But what does it mean? Let's figure out; as per typical computational complexity theory, we will contemplate asymptotic mounting of complexity measures. These complexity measures are thought of as runtime or space utilization along with problem size, instead of individual problems of a fixed size.

In both forms of algorithmic implementation, either the classical or quantum settings matter, as it helps us to measure runtime by using the number of basic processes influenced by an algorithm. However, when we use quantum computation, this can be controlled by utilizing something called the quantum circuit model (which we already covered in the previous chapters).

As we know, a quantum circuit is nothing but an order of elementary quantum operations called quantum gates, each functional to a minor number of qubits (quantum bits). To get a feel of the performance of a computer, the best way is to compare the performance of algorithms. The most fundamental way of getting the result of this is to use computer

science style notation $O(f(n))$. The $O(f(n))$ is popularized as ‘asymptotically upper-bounded by $f(n)$ ’.

Difficulties with the implementation of a quantum algorithms

The decrease of the wave packet is an important notion of quantum mechanics and thus of quantum computing. This leads to the fact that, after the measurement of a quantum system, a physical system’s state would be reduced to the one that has been measured.

This notion of wave packet reduction involves many difficulties in the implementation plane of quantum algorithms, and more particularly the parallelization of intermediate computations, which is the main gain of the quantum. Concretely, to be able to implement a classical algorithm on a quantum computer, it is necessary to find an implementation that takes advantage of the power of the parallelization only in the intermediate calculations and giving only one result. Also, the quantum computer gives a simple answer even if it does multiple parallel calculations.

Computational complexity

In computer science and quantum computing, it is significant to assess how effectual an algorithm is. One of the measures of it is to find out how many steps an algorithm under consideration would take to run a specific algorithm. We may use big-O notation to characterize the upper bound of the vilest case of running an algorithm. The O in big-O notation originates from the order of the algorithm.

We practice big- (Omega) notation to designate the lower bound of the worst-case scenario. So, though Deutsch’s problem receives the worst $O(n)$ steps to resolve on a traditional computer, Deutsch-Jozsa’s algorithm resolves the problem in one stage on a quantum computer. A big-O notation will be helping through this book in revealing the differences between the classical and quantum algorithms.

In 1993, Bernstein and Vazirani (BV) made public a paper that explained an algorithm that exposed perfect quantum-classical departure even when little

oversights were permitted. Why is this important? Because while Deutsch-Jozsa revealed a deterministic quantum benefit, even after that, if small errors were allowed in the computation, both classical and quantum versions could be run at the worst steps, showing no separation. By contrast, the Bernstein-Vazirani (BV) algorithm established separation even when tiny errors were permissible, thus displaying non-deterministic quantum benefit. The problem modelled in BV can be resolved in $O(n)$ time on a classical computer and in $O(1)$ using the BV circuit on a quantum computer.

BV crafted a foster involvement in their 1993 paper. They designated a quantum version of the Fourier transform. This **Quantum Fourier Transform (QFT)** would assist as a critical module for Peter Shor when he advances his algorithm to factor large numbers.

The publication of BV was quickly followed by Daniel Simon, then a postdoc candidate at the University of Montreal. In 1994, Simon sketched a problem that a quantum computer would undoubtedly resolve exponentially quicker than a classical one. To be further precise, Simon's algorithm has an upper bound of $O(n)$ on a quantum computer. Subsequently, the lower bound on the traditional computer is of a greater order than the upper bound on the quantum computer; there is a vibrant display of quantum benefit that would be observed. [Table 6.1](#) provides a classic definition of the Class of Computational complexity:

Class	Informal definition
P	This class of problems can be resolved by a deterministic traditional computer in polynomial time.
BPP	This class of problems can be resolved by a probabilistic classical computer in polynomial time.
BQP	This class of problems can be resolved by a quantum computer in polynomial time.
NP	This class of problems can be resolved by a deterministic classical computer in polynomial time.
QMA	Answers or solutions can be checked by a quantum computer in polynomial time.

Table 6.1: Class of Computational complexity

Complexity theory and Quantum computers

We use fundamental ideas from computational complexity hypothesis which is detailed to the concept of complexity classes, to group the problems by difficulty level. Refer to [Table 6.1](#) for informal metaphors of around noteworthy complexity classes. If a problem is unproven to be comprehensive for a complexity class, then it means that it is one of the toughest problems within that class. It is limited inside that class.

To start to comprehend the conceivable power of quantum computing the computational power of quantum computers concerning their traditional complements need to realize. Think of problems in P as decision problems that can be resolved in polynomial time (a function of the size of the provided input in terms of polynomials) by a deterministic Turing machine (it is fundamentally a theoretical machine).

The equivalent for space efficiency, the group of decision problems that can be resolved by a Turing machine utilizing polynomial space, is raised to PSPACE. Whereas NP problems are those that need a nondeterministic Turing machine to be resolved competently, and therefore, the exponential time needed on a so-called “deterministic machine” is theoretical.

The class of NP-complete problems is shortened to NPC. The NPC includes the toughest problems which reside within the boundaries of NP. Each problem that falls under NP can be further reduced to a problem in NPC. Think like this; if one NPC problem was initiated to be in P, then all of the further problems that are in NP would “naturally” fall under P. This can be further explained in the form of the long-open problem of whether $P = NP$. Having said that, still, a few theoretic computer researchers and scientists believe that P is not equal to NP. But till now, nobody has been able to effectively show the guesswork either way.

There is one more significant complexity class named BPP which means Bounded-error Probabilistic Polynomial-time. BPP labels decision problems that can be resolved in polynomial time by a probabilistic Turing machine, with a chance that the resolution could be wrong. Probabilistic Turing machines are those with the capability to flip coins to make arbitrary decisions, that is, machines with straight access to few sources of truthfully random input.

The disadvantage of using haphazardness in an algorithm is that it frequently means that there will be a fortuitous that the algorithm will produce an improper outcome. In BPP, the error of the solution is restricted in that the likelihood that the response is right must be at least two-thirds formally, that is, if the response to the decision problem is yes, then at least two-thirds of the likely computational tracks will need to agree, and if the answer is no, then a maximum one-third can accept.

The BPP was introduced in the 1970's, though there are current problems resolvable in BPP that are not in P, the quantity of such problems has been lessening subsequently. While it is not yet been established whether $P \subset BPP$, it is estimated that $P = BPP$. Quantum computation presents a number of novel complexity classes to the polynomial grouping. Conceivably the greatest study of complexity class is Bounded-error Quantum Polynomial-time or BQP.

BQP is the quantum postponement of BPP. BQP is fundamentally the class of decision problems resolvable in polynomial time by a naturally probabilistic quantum Turing machine, with a similar error restraint as demarcated for BPP. However, unlike BPP, it is supposed that $P \subset BQP$, which would mean that quantum computers are accomplished by resolving some problems in polynomial time that cannot be resolved competently by a classical Turing machine. The estimated relationships among all of these complexity classes, as well as a few of the major problems recognized to be in individual classes, are portrayed in [Figure 6.1](#).

[The quantum case of complexity and computation](#)

As we saw earlier, a quantum computer is a computer design which uses the philosophies of quantum physics to upsurge the compute capabilities of system or systems outside what is achievable by a traditional computer. Quantum computers have been constructed on a small scale and work endures to upgrade them to more practical models. A quantum computer looks like this, taking n input qubits, the register V , and producing n output qubits, the register W . The graphical representation of this concept is explained in [Figure 6.1](#):

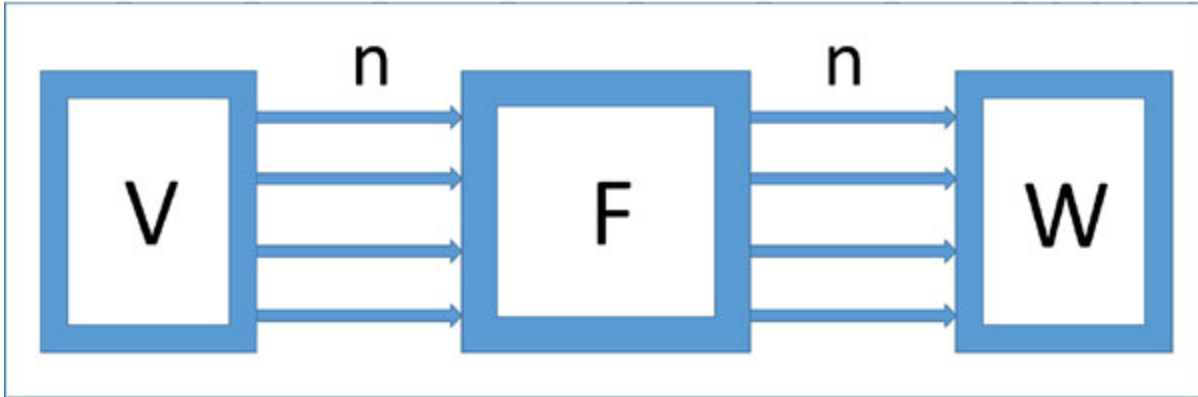


Figure 6.1: Input register prepared as a superposition of states

$$V = \sum_i^{2^n} \frac{1}{\sqrt{2}} (|0_i\rangle + |1_i\rangle)$$

The computer then computes in parallel the function applied to entirely n^2 integers concurrently. From QMP (Quantum Measurement Postulate), once we measure W, it will pick a Boolean for each bit of the yield register according to the subsequent entangled wave function of the output qubits.

Mathematically, if F is designed so that it exploits the likelihood that the output, we measure, is the answer we need, then measuring the output ruins the wave function and enables us to get Boolean values for all the qubits in W. The outcome is one of the possible outputs.

Imagine that F is the (integer) square root $W = \sqrt{V}$. For the best solution, let V be equipped as the superposition of all integers from 0 to 2^n so that when we run the computer, W can be measured. F calculates the square roots of all the integers in parallel, while in QMP, we can only find out about one. For practical problems, F should be decided so that the probability amplitudes of the output state powerfully favour the anticipated output from F.

To understand it in a better way, assume that quantum computers are like enormous multidimensional arrays of gashes that produce interference patterns in the wave functions. If the array is intended properly, the pattern explains the problem. A quantum computer is probabilistic, henceforth we may require to run it several times before we get the response we want.

The quantum computers would solve BQP problems proficiently and might narrate to other important classes of computational problems. The uneven boundary means that BQP does not seem to appropriate gracefully with the other classes. The BQP class comprises all the P problems and also limits other NP problems, such as factoring and the so-called discrete logarithm problem.

Most other NP and all NP-complete problems are thought to be outside BQP, denoting that even a quantum computer would need more than a polynomial number of ladders to solve them. Further, BQP might overhang outside NP, meaning that quantum computers could resolve certain problems quicker than classical computers could even check the answer.

A classical computer can professionally confirm the response of an NP problem but can proficiently resolve only the P problems. To date, though, no substantial instance of such a problem is known. The computer experts do know that BQP cannot spread outside the class known as PSPACE, which also covers all the NP problems. PSPACE problems are those that a conservative computer can solve utilizing only a polynomial quantity of memory but probably needs an exponential sum of steps.

[Let's program a quantum algorithm: Qiskit way](#)

You can program a quantum algorithm in multiple ways through the available frameworks and libraries. Multiple vendors are providing these tools including IBM and Microsoft. For example, Qiskit is an open-source quantum computing library advanced under the sponsorship of IBM.

Qiskit permits users to inscribe and run programs on either IBM's quantum processors or a local simulator, deprived of the use of the graphical interface. This is a significant feature since the graphical interface becomes unreasonable as the number of qubits turns to be large. Users can use Qiskit to access quantum processors with up to 16 qubits. Lesser processors are also available.

Qiskit is a very influential **Software Development Kit (SDK)** that has several elements in it that confront a diversity of problems related to real-world quantum computing. Qiskit is additionally split into four modules

named: Terra, Aer, Aqua, and Ignis. Further, each of these modules indentures with a precise portion of quantum software development.

Qiskit is fundamentally a Python library for quantum circuit implementation. An elementary Qiskit code has two fragments, scheming the circuit and running it. In the circuit design phase, we produce an instance of Quantum Circuit with the compulsory number of qubits and classical bits.

Then gates and capacities are added to this blank circuit. Gates and measurements are applied in Qiskit as methods of the Quantum Circuit class. After the circuit has been designed, we essentially need to select a backend to run the circuit. This can either be a simulator called the qasm_simulator or it can be unique to IBM's quantum processors. To use a quantum processor, you need to have IBM Q (IBM quantum provide an account to the interested users on request) account info into Qiskit.

For a complete overview of Qiskit and its various capabilities, the reader is encouraged to visit the authorized website (www.qiskit.org).

Quantum algorithms in action

Initially, computers had a single kernel. Later, the number of kernels increased. However, intrinsically the logic of dealing out the codes rowed couldn't be prohibited. For example, an f function within the codes couldn't be treated by segmenting. It was treated by a rowed logic. For example, by calculating the function of $f = \sin(\pi / 5) + \log_{10}(3)$, the primary value of $\sin(\pi / 5)$ was calculated, then the value of $\log_{10}(3)$ was calculated, and later the sum of these dual value was calculated and the value of f function was found.

Let's reason how a quantum computer would process this type of function. When the program stops working, $\sin$ value is directed to an operator, and the $\log_{10}$ value is directed to another operator. There is an additional operator that adds these dual values together and another which inscribes the value of f function. Thinking this way, an arrangement of movements is harm to dual actions at the lowermost rank and these dual actions work up by connecting with each other.

Hereby, all the operators wanted for the work, function at a similar time, and run for a deduction. Running for a conclusion means there is no total deduction. In other words, for a quantum computer, there is no certain deduction but accomplishing it is dissolute. For instance, no computer can compute the real, present, or upcoming value of the f function.

Though, a quantum computer devoted to adequate hardware power is going to compute the Taylor expansion at $\sin$ and $\log$ dissolute in an indispensable resolve. Now let's stretch an instance with a total consequence. Let's deliberate a quantum computer which inscribes the lowermost value in a number series like 12, 9, 4, 2, 7, 8. These numbers are discrete to the operators, and for example, suppose 7 is the lowermost, and then 4 is thought to be its bottommost and then 2 is said to be its lowest. For quantum computers, there is no complete value of the lowermost or no meaning in reality. The lowest touching pertinent states are the subject here.

In the start, in the quantum computer, there is no joining between states but physical assembly is done completely. Just like the neurons in the human brain take no connection at a certain time (the moment of birth or before) but communicate with the start of learning, software joining is constructed when the actions that quantum computer will do comes up and results are shaped. Our human brain labels what kind of connections will come up with the start of learning.

[A brief journey in the universe of quantum algorithm and computing](#)

One of the most overlooked aspects concerning quantum computers is the design, use, and implementation of quantum algorithms. People know that quantum computers can do wonderful things, but they can forget that it is not the computers, but the algorithms that they execute which do useful and interesting work. Neglecting algorithms in your quantum business strategy is a risky mistake.

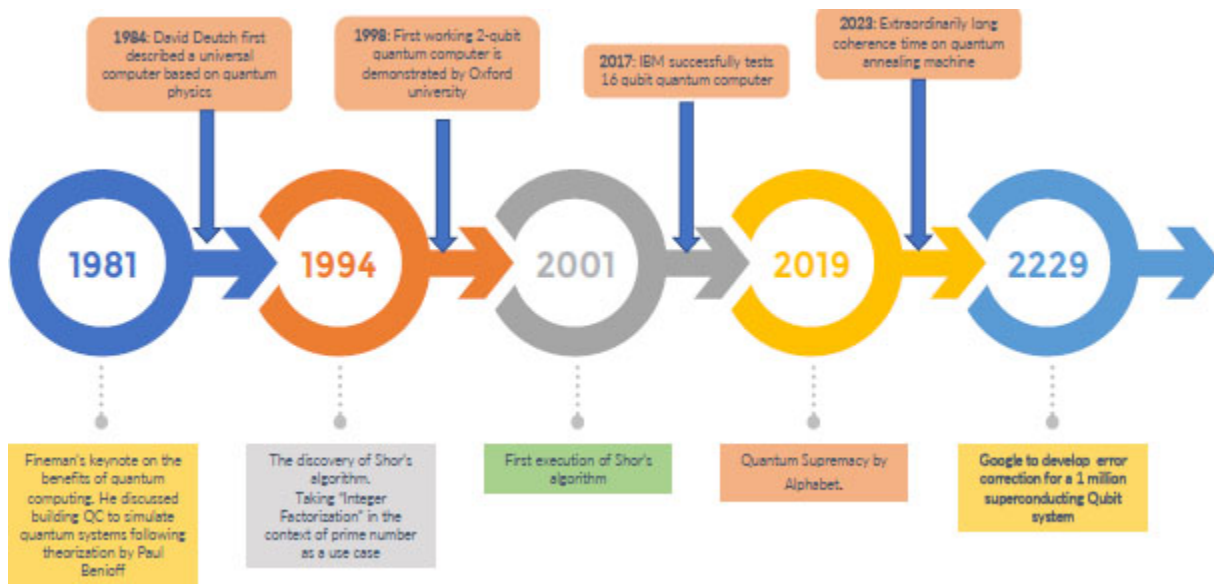


Figure 6.2: Journey of quantum computers

Likewise, anticipating and working in the algorithm market today can give businesses a clear head starts in the exploitation of Quantum Computing potential. The known, the unknown and the possible Current Quantum Computing techniques are particularly good at solving certain types of mathematical problems – these typically involve searching for prime factors and unstructured data searching. But to reach to a level, where quantum computers and algorithm changes current business, innovative and educational scenarios, it has travelled a long way (not that long although). The Journey of quantum computers is captured in [Figure 6.2](#).

Quantum Computing can be applied to more general problems and so it is important to focus on areas where it is likely to demonstrate material benefit compared to classical computing techniques. There are three areas to be recognized; the known algorithms, the unknown algorithms, and the possible algorithms. The most important algorithm sets are known algorithms. Known algorithms are those with proven application and benefit, and they include. Factoring algorithms (breaking factoring cryptography), Discrete-log algorithms (Breaking elliptic curve cryptography), Pell's equation algorithms, Principal ideal, Gauss Sum, and other classically impossible algorithms. Ordered and unordered searching algorithms (Quadratic speed up), Fourier transform algorithms, large logistical tree computations, minimization and learning algorithms, pattern

matching algorithms, statistical difference algorithm. [Figure 6.3](#) explains the classical classification of quantum algorithm:

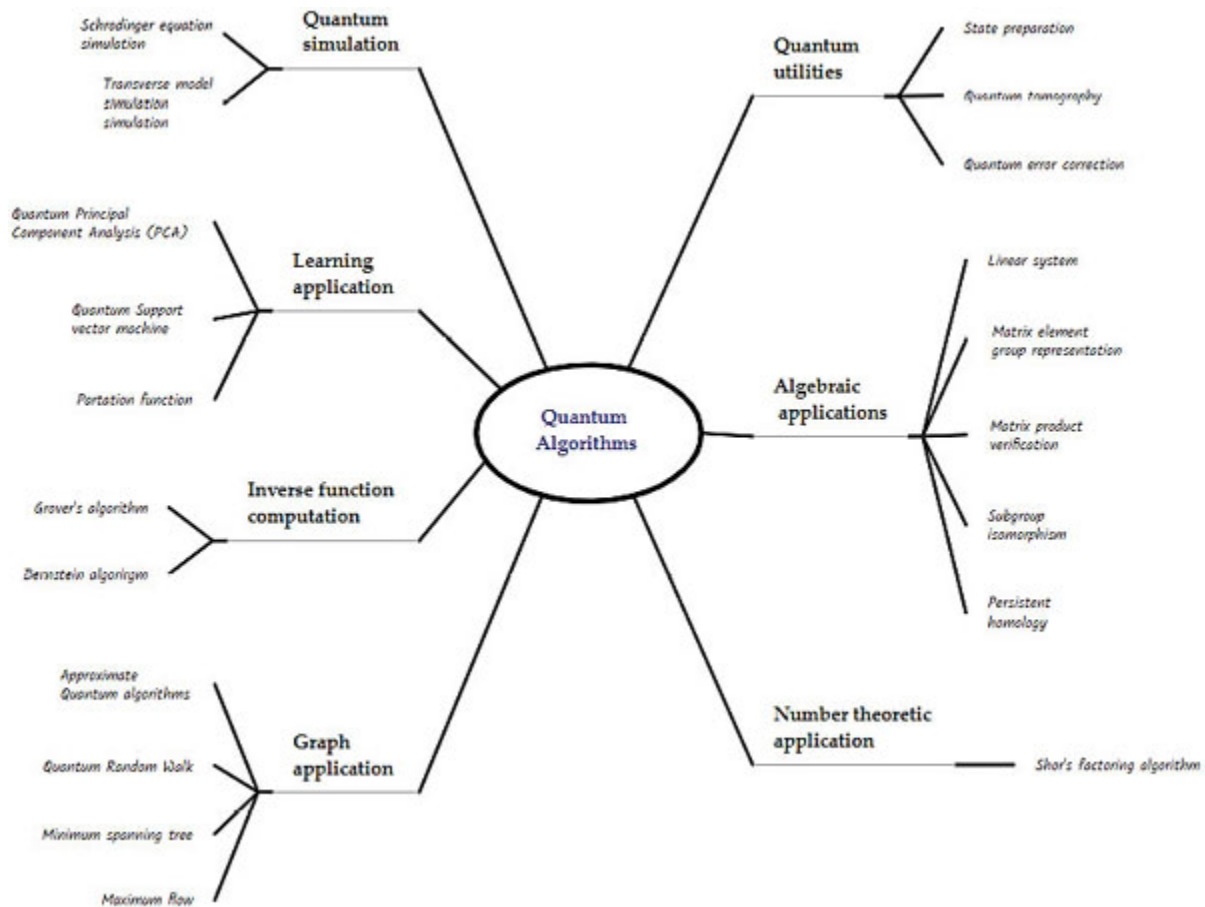


Figure 6.3: Classification of quantum algorithm

We can generally classify quantum algorithms conferring to their area of application and uses. There are a lot of algorithms that exist for quantum computers; however, a few important ones are described in the chapter. You can also refer to the Quantum Algorithm Zoo website for a brief and complete list of quantum algorithms.

In classical computing, algorithms are frequently intended by making use of a single or more algorithmic patterns like dynamic programming or local search. Similarly, most recognized quantum algorithms also use a mixture of algorithmic paradigms that are intended for quantum computing. The common paradigms are the QFT, the **Grover Operator (GO)**, along with something known as **Harrow Hassidim-Lloyd (HHL)** which is used as a method for linear systems.

Additionally, **Variational Quantum Eigenvalue solver (VQE)**, and direct Hamiltonian simulation (SIM) are also used. However, the number of recognized quantum algorithmic paradigms is much lesser compared to the amount of identified classical paradigms.

The most common restraint of unitarity on quantum operations and the castle in the air of non-intrusive measurement which make it problematic to design in context of quantum paradigms to current classical paradigms.

How to input data into a quantum computing algorithm using the quantum gate model

OK, first understand the problem and the requirements. The prerequisite to process 2^n states in the quantum algorithm is the need of the n -Qubit. For instance, the 4 states (00, 01, 10, 11) need two 2 Qubits, and the 8 states need (000, 001, ..., 111) which is the states of three Qubits. Now, if we have a great list with millions of items, (for example, a network of bees or travel salesman problems), using the quantum gate model, how to input these items to the algorithm for processing becomes the first bottleneck to proceed further.

However, this is not as difficult as it sounds. There are dissimilar methods to access classical data with a quantum computer (to know more about travel salesman problem reader could refer [chapter 11](#)).

Let's simplify the problem and get resolution, as follows:

- Accept that the data can be kept in a form of binary vectors, each of identical proportions 2^m .
- Accept that all those binary vectors are representative of the numerous data in 2^n binary vectors.
- Utilizing a standard computer (that is, electronic computer) to process such data, it will need 2^n phases only to read it (that is, $O(2^n)$), assuming that this computer can recite a solitary vector of size 2^m at a time.
- Now on the quantum side, we have a quantum CPU that can-do calculations quicker than traditional CPU. We also have a quantum memory.

- For the quantum CPU to process this enormous quantity of data, this data desires to be overloaded into quantum memory. For this, we need to have a quantum filling mechanism that is communally known as Quantum Loading Scheme or QLS.
- Loading N-dimensional vector into quantum memory from traditional memory with $O(\log N)$ phases, which was advanced used in creating a quantum search algorithm for a group of operations on classical data was achieved in 2012, by Pang. Also, in 2013, BenQiong et al. given a quantum loading scheme which was based on entangled states.

The method was capable of loading grayscale imageries of dimensions $2^p \times 2^p$ in quantum memory of proportions 2^{p+8} qubits in just $O(p)$ steps. Their method of QLS would allow quantum CPUs to be well-matched with electronic memory and electronic data, and produce likely quantum image compression and quantum signal processing that has traditional (electronic computers) like input and output.

We could use QLS to load our classical data into quantum memory of size $O(n+m)$ qubits, with only $O(n)$ steps.

How quantum algorithms work

As we know, in the decades of the 1970's and 1980's, based on the ideas of P. Benioff (1980) and R. Feynman (1982), the quantum theory of information, essentially information processing utilizing quantum systems, has been advanced. Also, the likelihood of contrivance computing machines that can simulate accurately and effectually of a particular quantum system has evolved. This theory bids much meeker solutions for certain applications of modern communications and informatics.

Superposition, entanglement, and interference are the chief physiognomies of the quantum world that make possible quicker cracking of certain problems that need exponential computing time with a classical computer. Utilizing superposition and interference, a function $f(x)$ can be assessed concurrently for all values of x . This asset of quantum computers is named quantum parallelism. The significant property of an entangled couple is that as soon as the state of one particle is known,, the state of the other particle

is known promptly, no matter the detachment between the particles at the time of the measurement.

One more significant feature of quantum information is the measurement process which, in overall, vicissitudes the state of the system being measured. Any quantum computation process ends with system measurement. The process is explained in [Figure 6.4](#):

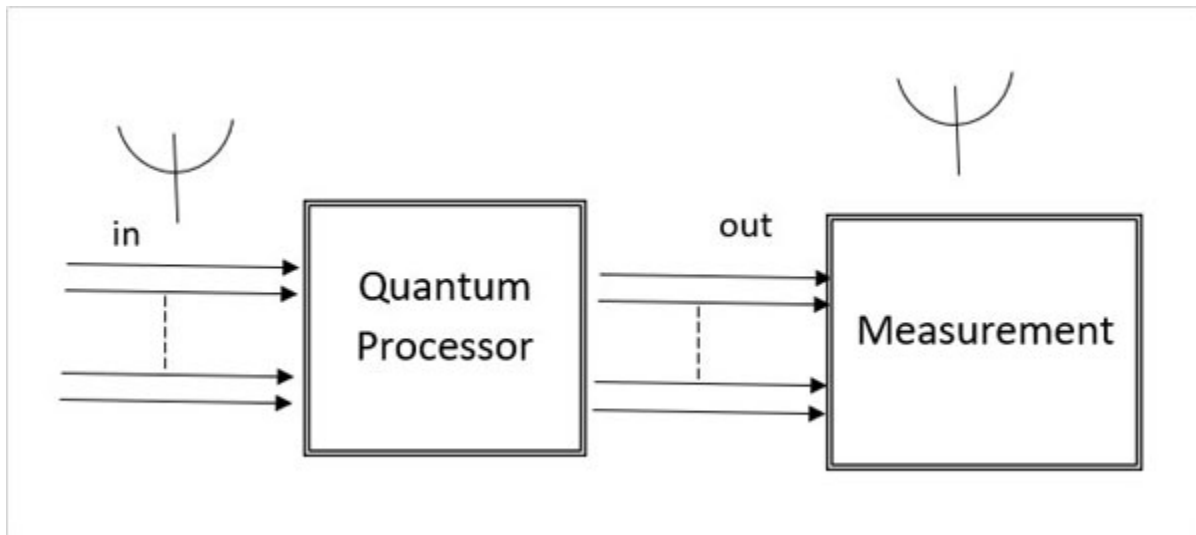


Figure 6.4: Quantum measurement

Before applying quantum algorithms, the accepted step is to simulate them on a classical computer. The two chief motives for this are the discovery of design faults prior to manufacturing and improved understanding of designs.

Since quantum computing is based on a concept that is different from that of governing classical computing, the algorithms that are constructed are fundamentally different which inevitably affects the structure as well as the function, implementation, and execution. The original quantum algorithms were intended to the competence of quantum computation to solve complex and time-intensive computational problems that include oracles. Oracles are nothing but the devices that are used to provide answers to questions with a humble yes or no.

[The big classes of quantum algorithms](#)

There are multiple types of algorithms. Few important ones are mentioned below:

Search algorithms: This type of algorithm essentially founded on Deutsch-Jozsa, Simon, and Grover.

- **Algorithms that try to find an equilibrium point of a complex system:** Algorithms that hunt for a balance point of a complex system comparable in the training of neural networks, the exploration for an optimum pathway in networks, or procedure optimization.
- **Algorithms founded and take advantage of QFT:** Such as Shor's Integer Factorization, that generated a deliberation among persons who are deficient to produce quantum computers accomplished of violation of public RSA-type security keys, and individuals being the ones looking to defend digital communications with algorithms resilient to the reckless factorization of integers.
- **Simulation algorithms of quantum mechanisms:** These aid in specific to simulate connections among atoms in numerous molecular constructions, and include inorganic and organic structure as well.

Some important quantum algorithms

It's important to understand the run-time efficiency of the algorithms we use. When a computer takes too long to solve a problem, it costs more money and deprives the computer of the time to spend on other valuable problems. Plus, if the algorithm is used inside a user-facing app, it may result in frustrated users who give up on the app entirely.

We can categorize the run time of an algorithm according to how the number of steps increases as the input size increases. Does it always take the same amount of time? That's a constant increase, a very fast run time. Does it always require looking at every possible permutation of the input? That's an exponential increase, a very slow run time. Most run times are somewhere between.

- **Constant time:** When an algorithm runs in constant time, it means that it always takes a fixed number of steps, no matter how large the input size increases.

- **Logarithmic time:** When an algorithm runs in logarithmic time, it increases proportionally to the logarithm of the input size.
- **Linear time:** When an algorithm grows in linear time, its number of steps increases in direct proportion to the input size.
- **Quadratic time:** When an algorithm grows in quadratic time, its steps increase in proportion to the input size squared.
- **Exponential time:** When an algorithm grows in super polynomial time, its number of steps increases faster than a polynomial function of the input size.
- **Polynomial versus super polynomial:** Computer scientists often classify run times into the following two classes:
 - Polynomial-time describes any run time that does not increase faster than n^k , which includes constant time (n^0), logarithmic time $\log_2 n$, linear time (n^1), quadratic time (n^2), and other higher degree polynomials (like n^3).
 - Super polynomial-time describes any run time that does increase faster than n^k and includes exponential time (2^n), factorial time ($n!$), and anything else faster.

Now, let's look on few interesting quantum algorithms

The Deutsch Oracle

Assume we have a function $f: \{0, 1\} \rightarrow \{0, 1\}$, which can be either constant or balanced. Further, the function would be treated as constant if $f(0) = f(1)$ and it can be termed as balanced if $f(0) \neq f(1)$. One can make the input qubits of the Deutsch oracle as the superposition and this would be expressed as $(|0\rangle + |1\rangle)/\sqrt{2}$ (using the Hadamard gate on $|0\rangle$) and the superposition $(|0\rangle - |1\rangle)/\sqrt{2}$ (using the Hadamard superposition) $(|0\rangle - |1\rangle)/\sqrt{2}$ (using the Hadamard gate on $|1\rangle$). The final output of the algorithm is the following state:

$$\pm |f(0) \oplus f(1)\rangle (|0\rangle - |1\rangle)/\sqrt{2}$$

Note: An oracle O is a black box process that is used as input to one more algorithm. Frequently, such processes or operations are

demarcated utilizing a classical function $f:\{0,1\}^n \rightarrow \{0,1\}^m$ which receives an n -bit binary input and produces an m -bit binary result.

Shor's algorithm: Theoretical interaction

Shor's algorithm was established in 2001 at IBM; it's very first version was capable of factored 15 into 3 and 5. And for this, a quantum computer with 7 qubits was utilized. It is a quantum algorithm for factoring a numeral N in $O((\log N)^3)$ time and $O(\log N)$ space. It is named after Peter Shor. The quantum factoring algorithm takes asymptotically:

$$O((\log n)^2 (\log \log n)(\log \log \log n))$$

This calculation includes a polynomial in logarithmic n amount of time which is required after the processing time on a classical computer under consideration. The classical computer is used to renovate or convert the output provided by the quantum computer to the factors of n .

Shor's algorithm uses the number theory to change the problem of discovering prime factors into an equivalent problem which reveals an overall structure in the solution space. The algorithm is significant since it suggests that public-key cryptography might be effortlessly fragmented, provided a sufficiently large quantum computer with a well-written algorithm is used for the same.

RSA, for instance, uses a public key N which is fundamentally the product of two big prime numbers. One method is to simplify RSA encryption by factoring N , but with traditional algorithms, factoring becomes progressively inefficient as N grows large. Further, no traditional algorithm is identified that can factor in time $O((\log N)^k)$ for any k . By difference, Shor's algorithm can blow RSA in polynomial time. It has also been protracted to about numerous other public-key cryptosystems, similar to every quantum computer algorithm.

Shor's algorithm is probabilistic; it gives the exact response with high probability. Also, the probability of disappointment or failure of the algorithm can be decreased as we repeat the algorithm multiple times. Fourier transform plays an important role in the working and applications of the Shor algorithm. Therefore, before moving the actual working of the Shor algorithm, let's have a look at it.

The Fourier transform: an introduction

The Fourier transform happens in numerous different forms throughout classical computing, in parts fluctuating from signal processing to data compression to complexity theory. The QFT is the quantum employment of the discrete Fourier transform over the amplitudes of a wave function. It is part of several quantum algorithms, most particularly Shor's factoring algorithm and quantum phase estimation. It can be further explained as follows:

The discrete Fourier transform acts on a vector $(x_0, \dots, x_{N-1})$ and maps it to the vector $(y_0, \dots, y_{N-1})$.

According to the formula:

$$y_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j \omega_N^{jk}$$

Where, $\omega_N^{jk} = e^{2\pi i \frac{jk}{N}}$.

Similarly, the quantum Fourier transform acts on a quantum state to $\sum_{i=0}^{N-1} x_i |i\rangle$ and maps it to the quantum state $\sum_{i=0}^{N-1} y_i |i\rangle$.

According to the formula:

$$y_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j \omega_N^{jk}$$

With ω_N^{jk} defined as earlier. Note that only the amplitudes of the state were affected by this transformation.

This can also be expressed as the map:

$$|x\rangle \mapsto \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} \omega_N^{xy} |y\rangle$$

Or the unitary matrix:

$$U_{QFT} = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} \omega_N^{xy} |y\rangle \langle x|$$

The QFT transforms between two bases, the computational (Z) basis, and the Fourier basis. The H-gate is the single-qubit QFT, and it transforms between the Z-basis states $|0\rangle|0\rangle$ and $|1\rangle|1\rangle$ to the X-basis states $|+\rangle|+\rangle$ and $|-\rangle|-\rangle$. In the same way, all multi-qubit states in the computational basis have corresponding states in the Fourier basis. The QFT is simply the function that transforms between these bases.

$$|\text{State in Computational Basis}\rangle \xrightarrow{\text{QFT}} |\text{State in Fourier Basis}\rangle$$

$$QFT|x\rangle = |\tilde{x}\rangle$$

(We often note states in the Fourier basis using the tilde ($\sim$)).

[Shor's algorithm prime numbers: Generic interpretation of working](#)

It is a lot easier to compute the product of two prime numbers or prime factors than to take a composite number and break it down into its prime factors. For example, let N be a composite number having factors N1 and N2. If N1 = 879651 and N2 = 685741, then N = 603212756391. If you look closely, you will deduce that it is relatively easy to compute the products of those large numbers than to take the result and find the factors that make it up. The competence of Shor's algorithm is owing to the effectiveness of something called quantum Fourier transform and modular exponentiation achieved through squaring.

Hypothetically (and theoretically proven), if a quantum computer is supplied to a sufficient number of qubits and it can be consumed by that computer, then Shor's algorithm potentially would be able to break public-key cryptography schemes which are most prevalent nowadays for a security mechanism. The widely used RSA-129 and so on scheme is also using the same cryptography technique.

[Working of Shor's algorithm: Step wise explanation](#)

Suppose we want to find the prime factors p and q of N . We proceed as follows:

1. Pick a random number a less than N . Repeat until you've found some a which is not a factor of N .
2. Find r , the period of $a \bmod N$. (We'll come back to this in just a moment.)
3. Check if r is even, and that $a^{r/2} + 1 \not\equiv 0 \bmod N$.
4. Let $p = \gcd(a^{r/2} - 1, N)$, and let $q = \gcd(a^{r/2} + 1, N)$.

Step 2 is both the tricky part (that is, where we need to take advantage of quantum computation) and also the part that enables us to find and exploit a global property of the solution space. We construct a sequence $a \bmod N, a^2 \bmod N, a^3 \bmod N, \dots$. Let's try it with $a = 2$ and $N = 15$

$2 \bmod 15, 4 \bmod 15, 8 \bmod 15, 16 \bmod 15, 32 \bmod 15, 64 \bmod 15, 128 \bmod 15, 256 \bmod 15,$

$= 2, 4, 8, 1, 2, 4, 8, 1, \dots$

The resulting pattern of digits (2,4,8,1) repeats with a period of 4. So, $r = 4$ in this case, and $q = \gcd(4+1, 15) = 5$. Further, 3 and 5 are the prime factors of 15. Steps 1, 3, and 4 we can do quite efficiently on a classical computer, but in step 2 we can't.

If a big, perfect quantum computer would face most of the similar limits as our present-day classical computers encountered, then definitely physicists and mathematicians would not work on the extremely tough task of constructing even elementary quantum computers. Quantum computers are different and revolutionary, thus, attracting so much attention due to multiple reasons.

Adiabatic algorithm

According to the adiabatic theorem, under some specified conditions, a quantum system remnant in its lowest energy state, branded as the ground state. In an adiabatic transformation in which the system is distorted slowly and effortlessly from a preliminary Hamiltonian to a final Hamiltonian, the

utmost significant condition in this theorem is the energy gap among the ground state and the subsequent excited state.

Being inversely proportional to the evolution time T , this gap controls the latter. If this gap happens during the entire evolution, the theorem commands that in the adiabatic limit (when $T \rightarrow \infty$), the system will continue in its ground state (Farhi, 2001). In practice, of course, T is at all times finite, but the longer it is, the less likely it is that the system will diverge from its ground state during the process of the time evolution.

Topological-Quantum-Field-Theory (TQFT) Algorithms

An exotic model called Topological Quantum Field Theory model for quantum computing is attracting a lot of consideration, particularly from Microsoft.

In contrast to the candid and normal circuit model, this model exists in the most abstract influences of theoretical physics. TQFT defines topological states of matter. The model was shown to be competently simulated on a standard quantum computer. Its excellence lies in its high acceptance to errors which is resultant from any likely understanding of a large-scale quantum computer. Topology is particularly accommodating since several global topological properties are, by explanation, invariant under deformation, and given that utmost errors are local, data arranged in topological properties is vigorous against them.

Discrete-log

If we are provided three n -bit numbers as a , b , and N , with the condition that $b = as \pmod N$ for some s , also, a task is given to identify the values of s , then as per Shor, this can be attained on a quantum computer in a $\text{poly}(n)$ time. The quickest identified classical algorithm needs time for super polynomial in n . By comparable methods, quantum computers can resolve the discrete logarithm problem on elliptic curves*. The super polynomial quantum speedup has also been protracted to the disconnected logarithm problem on semigroups.

Verifying matrix products

If we provided three $n \times n$ matrices, A , B , and C , the matrix product authentication problem is to resolve whether $AB=C$. Then classically, the finest recognized algorithm attains this in time complexity of $O(n^2)$, whereas the best recognized traditional algorithm for matrix multiplication runs in time complexity of $O(n^{2.373})$. Ambainis et al. exposed a quantum algorithm for this problematic situation with runtime $O(n^{7/4})$. Then, Buhrman and Špalek enhanced upon this, gaining a quantum algorithm for this problem with runtime $O(n^{5/3})$. Historically, this algorithm is based on outcomes that are known as quantum walks (the quantum mechanical counterpart of classical random walks, is an advanced tool for building quantum algorithms that has been recently shown to constitute a universal model of quantum computation.).

Subset-sum

Provided a list of integers $x^1, \dots, x^n$, and a target integer s , the subset-sum problem is to find out whether the summation of any subset of the provided integers adds up to s . This problem falls under the category of NP-complete, and thus is improbable to be resolvable by traditional or quantum algorithms with polynomial worst-case complexity. In the tough cases, the given integers are of order 2^n and abundant investigation on subset-sum emphasizes on average case examples in this regime.

A quantum algorithm is certain that it resolves such cases in time $2^{0.241n}$, up to polynomial factors. This quantum algorithm works by putting on a variant of Ambainis's quantum walk algorithm for element clearness to quickening up an urbane traditional algorithm for this problem due to Howgrave-Graham and Joux. The dissolute recognized classical algorithm for such cases of subset-sum runs in time $2^{0.291n}$ up to polynomial factors.

Grover's algorithm

This quantum algorithm was invented by Lov Grover in 1996. The analogous problem in old-style computation cannot be determined efferently by classical computers. After Grover issued his algorithm,

Bennett, Bernstein, Brassard, and Vazirani showed that any quantum solution to a problem needs to assess the function times.

Unlike other quantum algorithms, which may bring exponential quickness over their classical counterparts, Grover's algorithm offers only a quadratic speedup. Grover's algorithm could brute-force a 128-bit symmetric cryptographic key in crudely 264 iterations, or a 256-bit key in jaggedly 2128 iterations. As an outcome, it is sometimes advised that symmetric key lengths be crumpled to defend in contradiction of future quantum attacks.

It is a quantum algorithm for searching an unsorted database with N entries with $O(\sqrt{N})$ time and using $O(\log N)$ storage space. In models of classical computation, searching for an unsorted datastore would not be done in less than linear time.

Grover's algorithm demonstrates that in the quantum model, searching can be completed quicker than this. Its time complexity $O(\sqrt{N})$ is asymptotically the reckless likely for searching unsorted datastore in the quantum model. It delivers a quadratic quickness, dissimilar to further quantum algorithms that offer exponential speedup over their classical complements. Though, quadratic is substantial when N is big.

Similar to several other quantum algorithms, Grover's algorithm is also probabilistic; it gives the means it needs to bring an outcome in high probability. [Figure 6.5](#) explains about Quantum speed-up for Grover's algorithm:

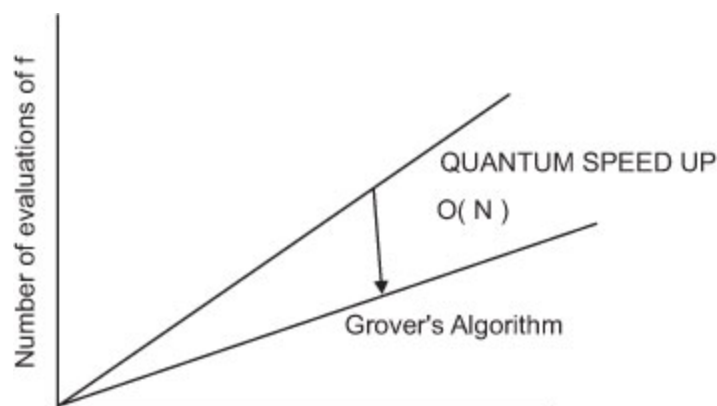


Figure 6.5: Quantum speed-up for Grover's algorithm

Repetition decreases the probability of disappointment. An instance of a deterministic quantum algorithm is the Deutsch-Jozsa algorithm. It at all

times provides a precise outcome to the problem. The probability of getting the correct response using the Deutsch-Jozsa algorithm is 1 (100%). [Figure 6.6](#) talks about Quantum Gate Operation in the presence of Grover's Algorithm:

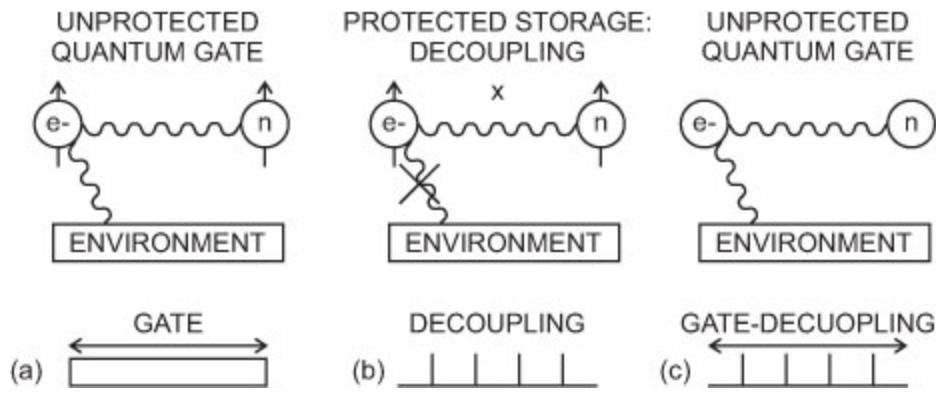


Figure 6.6: Quantum gate operation in the presence of Grover's algorithm

Uses of Grover's algorithm

Though the drive of Grover's algorithm is typically defined as “searching a database”, it may be more precise to label it as “inverting a function”. Crudely speaking, if we have a function, $y=f(x)$ that can be assessed on a quantum computer, Grover's algorithm permits us to compute x when specified y . Inverting a function is connected to the searching of a database since we could come up with a function that produces a specific value of y if x equals an anticipated admission in a database and a new value of y for other values of x . Grover's algorithm can also be Predicting for guesstimating the mean and median of a set of numbers, and for resolving the crash problem. Further, it can be used to crack NP-complete problems by executing comprehensive searches over the set of likely resolutions. This would result in a substantial quickness over traditional solutions, even though it does not deliver the “holy grail” of a polynomial-time answer.

Setup steps of Grover's algorithm

Imagine an unsorted database with N entries. The algorithm needs an N -dimensional state space H , which can be completed by $\log_2 N$ qubits. Let us number the database entries as $0, 1, \dots (N-1)$. Select an observable, Ω , acting on H , with N different eigenvalues whose values are all identified.

Each of the eigenstates of Ω encrypts one of the entries in the database, in a way that we will define. Represent the eigenstates (using bra-ket notation) as:

$$\{|0\rangle, |1\rangle, \dots, |N-1\rangle\}$$

And the equivalent eigenvalues by:

$$\lambda_0, \lambda_1, \dots, \lambda_{N-1}$$

We are provided with a unitary operator, U_ω , which acts as a subroutine that compares database entries deliberating to a few search standards. The algorithm does not lay down how this subroutine functions, but it must be a quantum subroutine that works with super-positions of states. Additionally, it essentially acts on one of the eigenstates, $|\omega\rangle$, which resembles the database entry similar to the search criterion. To be exact, we need U_ω to have the following effects:

$$\begin{aligned} U_\omega |\omega\rangle &= -|\omega\rangle \\ U_\omega |x\rangle &= |x\rangle, \text{ for all } x \neq \omega \end{aligned}$$

Here, we are trying to recognize this eigenstate $|\omega\rangle$.

Steps of Grover's algorithm

The steps of Grover's algorithm are as follows:

1. Prepare the system to the state.
2. Do the subsequent "Grover iteration" $r(N)$ times. The function $r(N)$ is designated as follows:
 - a. Put on the operator U_ω
 - b. Put on the operator $U_s = 2|s\rangle\langle s| - I$.
3. Do the measurement Ω . The measurement outcome will be λ_ω with a probability impending 1 for $N \gg 1$. From λ_ω , ω may be got.

Explanation of the algorithm

Our initial state is given as:

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_x |x\rangle$$

Imagine the plane crossed by $|s\rangle$ and $|\omega\rangle$. Let $|\omega^\times\rangle$ be a ket in this plane perpendicular to $|\omega\rangle$. Since $|\omega\rangle$ is one of the basis vectors, the overlap is as follows:

$$\langle \omega^\times | s \rangle = \frac{1}{\sqrt{N}}$$

In geometric terms, there is an angle $(\pi/2 - \theta)$ among $|\omega\rangle$ and $|s\rangle$, where θ is specified by:

$$\begin{aligned} \cos\left(\frac{\pi}{2} - \theta\right) &= \frac{1}{\sqrt{N}} \\ \sin \theta &= \frac{1}{\sqrt{N}} \end{aligned}$$

The operator U_ω is a reflection at the hyperplane orthogonal to $|\omega\rangle$; for vectors in the plane spanned by $|s\rangle$ and $|\omega\rangle$, its performances as a reflection at the line finished $|\omega^\times\rangle$. The operator U_s is a likeness at the line through $|s\rangle$. So, the state vector leaves in the plane spanned by $|s\rangle$ and $|\omega\rangle$ later each request of U 's and after each application of U_ω , and it is to traverse that the operator $U_s U_\omega$ of each Grover iteration step rotates the state vector by an angle of 2θ toward $|\omega\rangle$.

We need to stop when the state vector passes close to $|\omega\rangle$; after this, subsequent repetitions rotate the state vector away from $|\omega\rangle$, dipping the probability of gaining the right answer. The number of times to iterate is given by r . To align the state vector exactly with $|\omega\rangle$, we need:

$$\begin{aligned} \frac{\pi}{2} - \theta &= 2\theta r \\ r &= \frac{\left(\frac{\pi}{2} - \theta\right)}{2\theta} \end{aligned}$$

However, r must be an integer, so generally, we can only set r to be the integer close to $(\pi/\theta - 2)/4$. The angle between $|\omega\rangle$ and the final state vector is $O(\theta)$, so the probability of gaining the wrong answer is $O(1 - \cos^2\theta) = O(\sin^2\theta)$.

For $N \gg 1$, $\theta \approx N^{-1/2}$, so:

$$r \rightarrow \frac{\pi\sqrt{N}}{4}$$

Also, the likelihood of gaining the incorrect answer becomes $O(1/N)$, which goes to zero for large N .

Deutsch-Jozsa

We are provided an oracle whose input is n bits and whose output is one bit. We are assured that out of the 2^n likely inputs, either all of, none of, or half of them harvest outcome. The task is to differentiate the stable case (half of all inputs yield output 1) from the constant case. It was revealed by Deutsch that for $n=1$, this can be resolved on a quantum computer utilizing one query, while any deterministic classical algorithm needs two. This was factually the primary well-defined quantum algorithm attaining a quickness over classical computation. Though probabilistically easy to resolve with $O(1)$ queries, the Deutsch-Jozsa problem has exponential worst-case deterministic query complexity classically.

Matrix commutativity

We are given oracle admission to k matrices, each of which is $n \times n$. Given integers $i, j \in \{1, 2, \dots, n\}$ and $x \in \{1, 2, \dots, k\}$ the oracle returns the ij matrix element of the x^{th} matrix. The job is to select whether all of these k matrices travel. This can be attained on a quantum computer using $O(k^{4/5}n^{9/5})$ $O(k^{4/5}n^{9/5})$ queries, while classically this needs $\Omega(kn^2)$ queries.

Group commutativity

Let's imagine that we are provided a list of k generators for a group G and access to a Blackbox applying group multiplication. By querying this Blackbox, we would like to determine whether the group is commutative.

The best-identified classical algorithm identified to date needs $O(k)$ queries. Magniez and Nayak have exposed that the quantum query complexity of this. The task is $\Theta(k)$. For these types of scenarios, group commutativity is best fitted.

Quantum simulation

It is thought that for any physically truthful Hamiltonian H on n degrees of freedom, the conforming time evolution operator e^{-iHt} can be applied using $\text{poly}(n,t)$ gates. Except for $\text{BPP}=\text{BQP}$, this problem is not resolvable in normal conditions on a traditional computer architecture based system in polynomial time. Several methods for quantum simulation have been advanced for general classes of Hamiltonians chemical dynamics, for instance, the Dirac and Klein-Gordon equations, open quantum systems and quantum field theory.

The complexity of classically simulating quantum systems is exponential. This is directed by Richard Feynman to chiefly recommend that quantum computers might outdo traditional computers on a few jobs. Though the problem of detection of ground energies of local Hamiltonians is QMA-complete. Therefore, it perhaps desires exponential time on a given quantum computer, especially in the worst-case scenarios. The quantum algorithms have been progressive to approximate ground and thermal states for few classes of Hamiltonians. Well-organized quantum algorithms have also been gotten for making certain classes of tensor network states.

Machine learning algorithms

Machine learning includes a wide variety of computational problems and can use a variety of algorithmic techniques. This entry recapitulates quantum algorithmic methods for better-quality machine learning results. The quantum algorithms for resolving linear systems are functional to speed up cluster-finding, excellent in principal component analysis, very good in doing binary classification, exceptional in the training of neural networks, and it is useful in many forms of regression analysis including linear and logical. It works provided the data gratify certain conditions. Though,

several quantum machine learning algorithms founded on linear systems have afterward been dequantized.

Finally a summary of important facts about quantum algorithms

A quantum algorithm is a comprehensive set of orders for resolving a specific problem on a quantum computer. Algorithm complexity is resolute by the number of quantum processes obligatory to solve the problem. Quantum algorithms usually have a sublinear complexity, for which the quantity of operations is relative to the square root of the data input size N . In Big O notation, this is expressed as $O(\sqrt{N})$.

The Grover algorithm, developed by Lov Grover in 1996, is a quantum algorithm for searching an unsorted database with N entries. While a classical solution requires $O(\sqrt{N})$ steps to find a required entry, the Grover algorithm finds it in $O(\sqrt{N})$ steps.

The super-dense or dense coding protocol enables transmitting two classical bits of information by sending one qubit of an entangled pair, provided that the other one is in the possession of the receiving party.

Quantum teleportation transports a quantum state by using two qubits that are entangled and two classical bits of information that are used as a key to reassemble the teleported state.

The Deutsch-Jozsa algorithm is one of the first quantum algorithms to demonstrate the advantage of quantum over classical computing. Suppose we have a black box that takes an input set of 0's and 1's and maps it to an output set of 0's and 1s in one of two ways: constant or balance. For the constant way, all outputs are 0's or all outputs are 1s regardless of the input. The balanced way returns 0's for half of all inputs and 1s for the other half. The Deutsch-Jozsa algorithm determines which of the two mappings the black box performs in one query only, a task impossible to achieve with a classical computer.

Quantum computing software platforms

Few famous quantum computing software platforms are Cirq from Google, Rigetti, PyQuil, QISKit, ProjectQ, and Microsoft's Quantum platform. PyQuil is an open-source Python library developed from Rigetti computing that constructs program form for the quantum computer. PyQuil produces a program in the quantum instruction language Quil.

Story 1: Semiconductor spin qubits for construction of reliable quantum computers

Upcoming universal quantum computers resolving glitches of practical significance are predictable to necessitate at least 10^6 qubits, which is a huge scale-up from the current numbers of fewer than 73 qubits functioned together. There are different types of qubits techniques, out of which, the solid-state qubits are painstaking to be feasible applicants for this scale-up, but interfacing to and controlling such a big number of qubits is a multifaceted contest that has not been resolved yet. One option to address this challenge is to use qubit control circuits situated close to the qubits at cryogenic temperatures.

Quantum computing aptitudes exponential speedup and retention (memory) improvement for specific types of computational problem is also on rise. Though, applications such as Shor's algorithm will perhaps need many millions of qubits for problem sizes where a benefit over traditional computers can be anticipated. This big number mostly results from the essentials of the creation of quantum systems in contradiction of errors due to decoherence, that is, contrary to the forfeiture of the information stowed in the qubits produced by the communication with the environment. The furthestmost recognized method to counter these errors trusts on the laid-off encoding of the delicate quantum information to allow error correction, resulting in an overhead of 10^2 – 10^5 in the number of physical qubits. This leads to construction on production methods that allow chips with billions of transistors, and solid-state qubit employments are capable applicants to support scaling to such big numbers. Still, linking and regulating such a bulky number of qubits signify an engineering challenge. Present trials on the most ascendable solid-state qubits and processor protestors are functioned at temperatures of twenty to hundred mK and operated using

room temperature signal generators, whose output is running to the qubits via coaxial cables.

A tremendously integrated control system provides all compulsory functionality in the instant neighbourhood of the qubits, either on the same chip or with some form of chip-to-chip interconnection know-how. The significant advantage of this technique is that microfabricated interconnects could be used between the qubit and the control system.

Story 2: Quantum compatible areas for algorithms

Quantum computing is top matched to resolving glitches using three types of algorithms categorized optimization, sampling, and machine learning.

Optimization: The main area of emphasis in quantum computing are optimization problems. Optimization glitches are characteristically tough yet appreciated for solving real-world problems. And they occur in nearly all fields in which businesses function. Instances of optimization problems comprise finding the furthestmost cost-effective way for delivery of things, determining the furthestmost effectual way to excerpt resources, coming up with innovative pharmaceutical drug discovery approaches, or classifying an improved method to accomplish risk in financial portfolios.

The processing time required by a traditional computer to deliver high quality answers that an optimization problem can upsurge exponentially with the size of the problem, and quantum computing will deliver a much quicker response. The consequences for such alteration will be melodramatic as enterprises with augmented optimization competencies will be talented to get novel cost savings and revenue generation prospects utilizing such methods.

Sampling problems: Additional job that adiabatic quantum computers can do is in the field of sampling. Fundamentally sampling is the method of drawing values from a solution space that permits simplifications about the delivery of the populace. Traditional computers can fight to competently produce random examples of certain kinds of spectacles that specimen can hold well. Though, if complex quantum states can be measured and controlled, sampling from these states can be done abundantly more professionally.

Machine learning: This uses sampling and optimization approaches; the aptitude to advance these methods will lead to improved machine learning competence as a consequence. The sampling technology in quantum computers can deliver more dispersed, dependable input data for the machine learning algorithms. Each repetition of the new-fangled data would aid the ML to learn. As for the optimization competences of quantum computing, numerous machine learning methods boil down to thought-provoking optimization problems. For instance, a corporation could shape a probabilistic depiction of a certain feature of the world (customer behavior, for instance), use the examples from an adiabatic quantum computer to change to information about what the model looks like now, and then use an ML algorithm to regulate how to advance the model over time.

Story 3: Quantum computing and corporate traction

Accenture Labs and IBM have charted numerous likely use cases for quantum computing for several industries including Aviation, healthcare, and BFSI. Their objective is to recognize and validate the glitches where a quantum algorithm will outstrip current computing approaches and expand outcomes. The organization that starts business experiments with quantum technology today will be well equipped for the main industry vicissitudes that could come through the outline of quantum computing.

Quantum hardware and software progressions are happening at a quick rate in the field of quantum computing research. Therefore, a number of thrilling growths is predictable on the marketable side. A few leading corporations are even now using numerous methods to brand quantum hardware which is presented for procurement and shared use. Besides, a few companies like Microsoft and IBM are at work to offer cloud-based quantum computing platforms and software applications to enhance quantum computing power.

Provocative thought 1: Industries, quantum computers and way forward

In reply to the foretold demand for quantum services, software corporations are creating hardware-independent quantum platforms and applications. For instance, Accenture Labs is monitoring the quantum computing network

and working in partnership with leading businesses such as Vancouver, 1QBit. 1QBit is a software company devoted to the construction of development tools and software to resolve the world's most difficult computational tasks. The firm's platforms permit the growth of hardware-agnostic applications that are well-matched with both traditional and quantum processors. Division of 1QBit is developing an intellectual property using mathematical algorithms, solvers, and embedders that can use to interface with quantum computing. Instances of these tools comprise the following:

- **Graph similarity:** Equates two or more graphs to perceive Similar patterns.
- **Maximum quasi clique:** Classifies a group of thickly connected actors in a set. This can be used to find a group of objects that are pseudo alike to each other.
- **Balanced graph partitioning:** Breaks problems into equal-size parts.
- **Linear knapsack:** Governs the amalgamation of substances that holds the total utmost value given a restricted space.

1QBit has applied these and additional algorithms in its quantum computing exploration to resolve stimulating industry problems. Examples in the financial services industry comprise dynamic portfolio optimization, clustering, and portfolio analysis. [*Figure 6.7*](#) is showcasing quantum processing power through the API way:

processing power through an API from 1QBit.

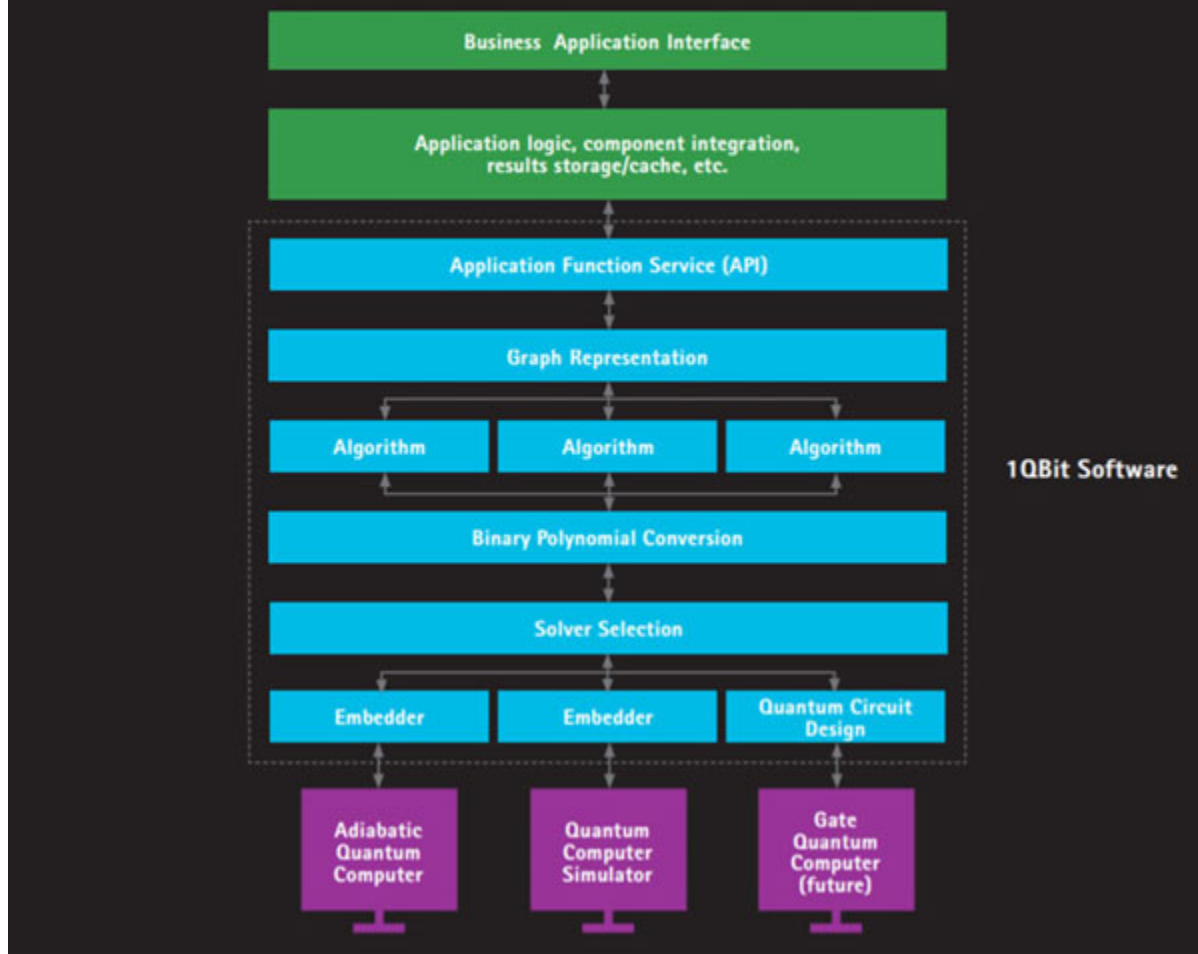


Figure 6.7: 1Qbit technical stack

Provocative thought 2: The case of the right quantum computer for business

Quantum computers are not equal, also they are not meant for solving the same problems. Quantum computers are characteristically classified into 3 groups: quantum annealing, **Noisy Intermediate-Scale Quantum (NISQ)** computing, and fault-tolerant universal quantum computing.

The technical community agrees that quantum annealing will not bid an eloquent speed-up over traditional computing. Also, quantum annealers are not on the growth track that is central to fault-tolerant universal quantum

machines. Therefore, quantum annealers cannot be well-thought-out as true quantum computers.

On the other hand, NISQ computers have the best possibility to bring commercial benefit, and numerous novel algorithms are being improved for them. Furthermore, as NISQ computers scale up, they advance to the final goal of quantum computing that is a fault-tolerant universal quantum computer that would be able to handle significant classes of business and technical problems often exponentially quicker than a traditional machine. There is not at all any hesitation in saying that the quantum revolution is coming. It is domineering for leaders to make sure that their trades are prepared for this cutting edge of technology innovation.

Innovativeness can be enhanced by learning more about the reckless developing market, and classifying where quantum will influence the business. And then making quantum-ready applications to cater those identified needs. Quantum computing's potential for noteworthy speedup over traditional computers, conferring to IBM internal analysis are explained in [Figure 6.8](#):

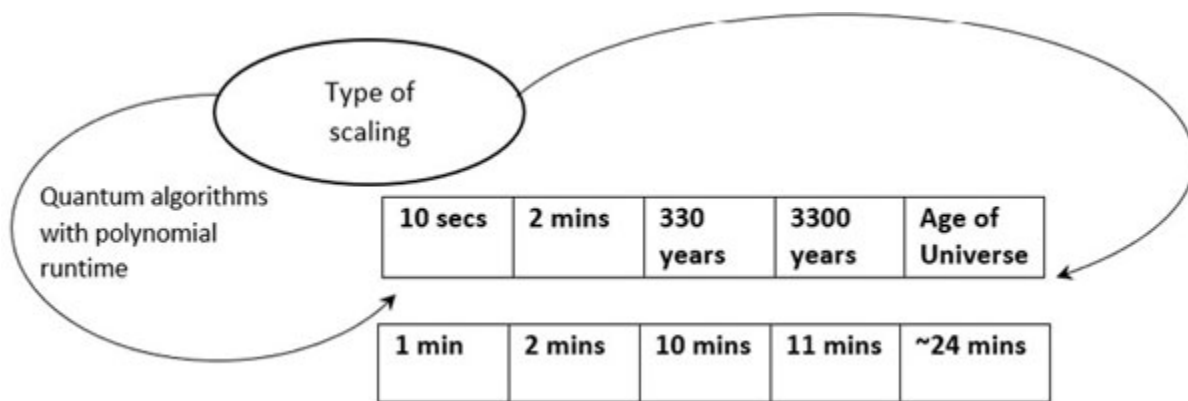


Figure 6.8: Quantum algorithms with polynomial runtime

Moreover, quantum computing's expected aptitude to resolve today's dreadfully complex logistics optimization problems could get-up-and-go to substantial cost savings and carbon footprint decrease. For instance, imagine refining the global ways of the trillion-dollar distribution industry. If quantum computing could advance ampule use and delivery volumes by even a minor portion, this could save carters hundreds of millions of dollars. Therefore, now forward-looking businesses are developing their

expertise in the field of QC and discovering use cases that may benefit their business. The trials are intended to determine feasible quantum problems using a first-rate set of preprogrammed quantum algorithms, to classify if these algorithms are operative substitutes for current classical computing applications and grow a quantum application to establish the functionality.

Provocative thought 3: Industry transformation and quantum computers

Quantum computers are seen as a device that would be able to transform industries since they have the potential to do so. Future quantum computers could aid to attain product advances in fields such as Physics, nanotechnology, traditional chemistry, computational and another variant of biology, healthcare, protein analysis, finance, Insurance, artificial intelligence, and materials science, allowing quick market share improvements and superior effectiveness for the visionary corporations that adopt them. Quantum computing's problem-solving competences could exceedingly redefine modest benefit, transmuting commercial operating models and value chains that transfigure whole industries.

For instance, the problem of optimally scheming individual point-to-point ways of filling numerous supplies on a very big logistic network is complicated and can quickly turn out to be out-of-reach for traditional supercomputers. If one were to discover all the potentials in such an optimization problem, it could take billions of years even with just a few hundred stations in the network. Quantum computing may be talented to travel the space of potentials much earlier. For instance, in specific use cases like airline scheduling optimization, quantum computing may be talented to produce daily ad hoc flight timetables, exactly custom-made for the thousands of travelers flying to hundreds of terminuses on a specific day, plummeting client travel time, air traffic cramming, and airline fuel prices. If an initiative were to grow a quantum answer for logistic network design optimization, it could quickly aim to develop a market leader in each industry where logistics are critical for accomplishment. The transformation plan could be called out as follows:

- Start learning about quantum computing and the tools obtainable to harness it.
- Recognize fields of the business where today's quantum computers can have a chance.
- Test early use cases.
- Produce a timeline for how these use cases will scale with quantum computing progressions.
- Design a quantum computing roadmap for the commercial sector and revisit all over the year.
- Employ a worker(s) to monitor inclinations and report monthly.
- Shape quantum-ready requests on top of a hardware-agnostic interface, to permit for unified swapping among different kinds of quantum computers as they advance.
- Organizations may take the following directions or steps to build quantum competencies.
 - Select Champions or brand ambassadors, who will participate in different conferences and keep track of the latest happening in the field of QC. Further, provide his or her inputs to the leadership of the company.
 - Figure out specific relevant use cases for you.
 - Do the experimentation with the use cases.
 - Repeat the preceding step until you are not successful or fail. If you are successful, move ahead towards quantum readiness. Else, rethink and go back to the reidentification of the use case.
 - Try to be mature and grow with a flexible attitude of continuous learning and improvements.

[Figure 6.9](#) depicts the transformation roadmap of industries through quantum computation and computers

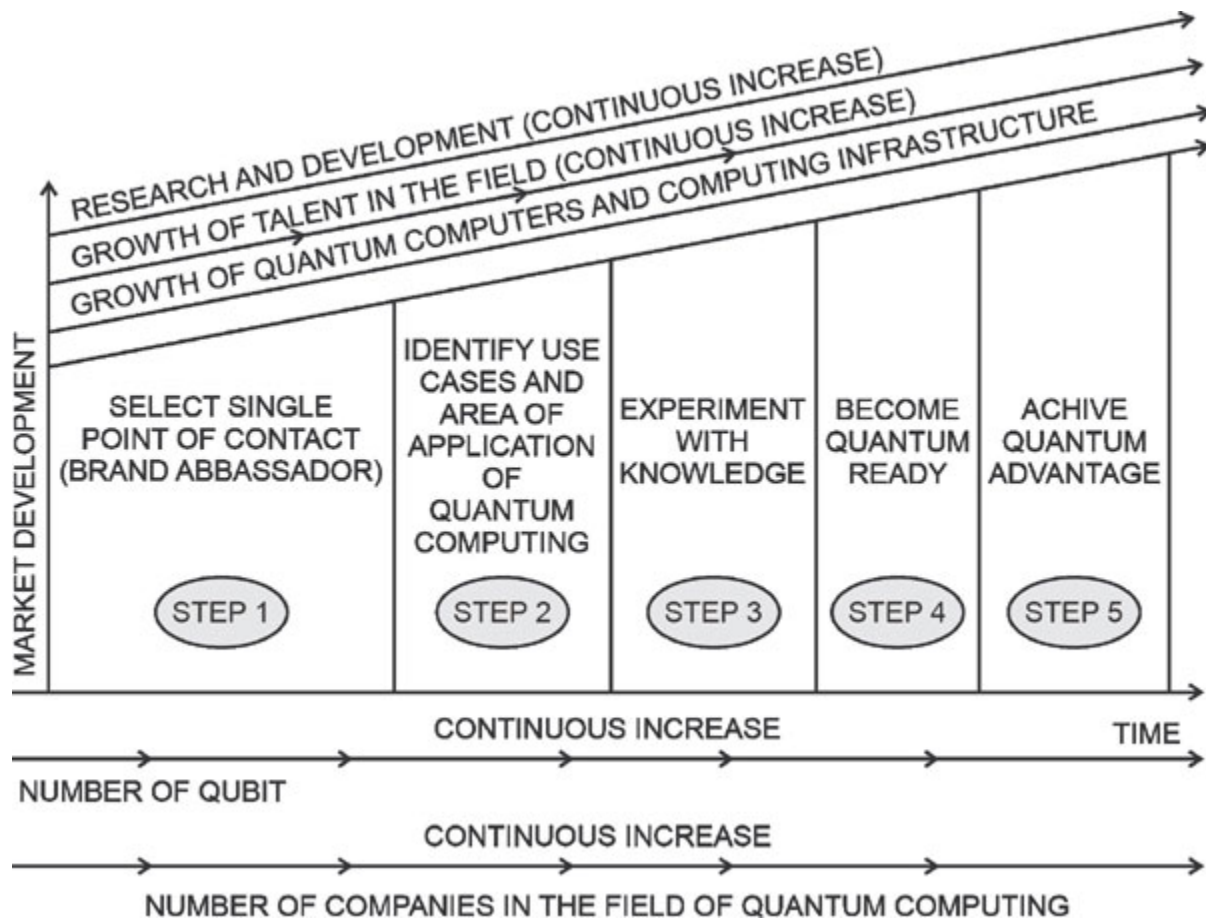


Figure 6.9: transformation roadmap of industries through quantum computation and computers

Mind Map

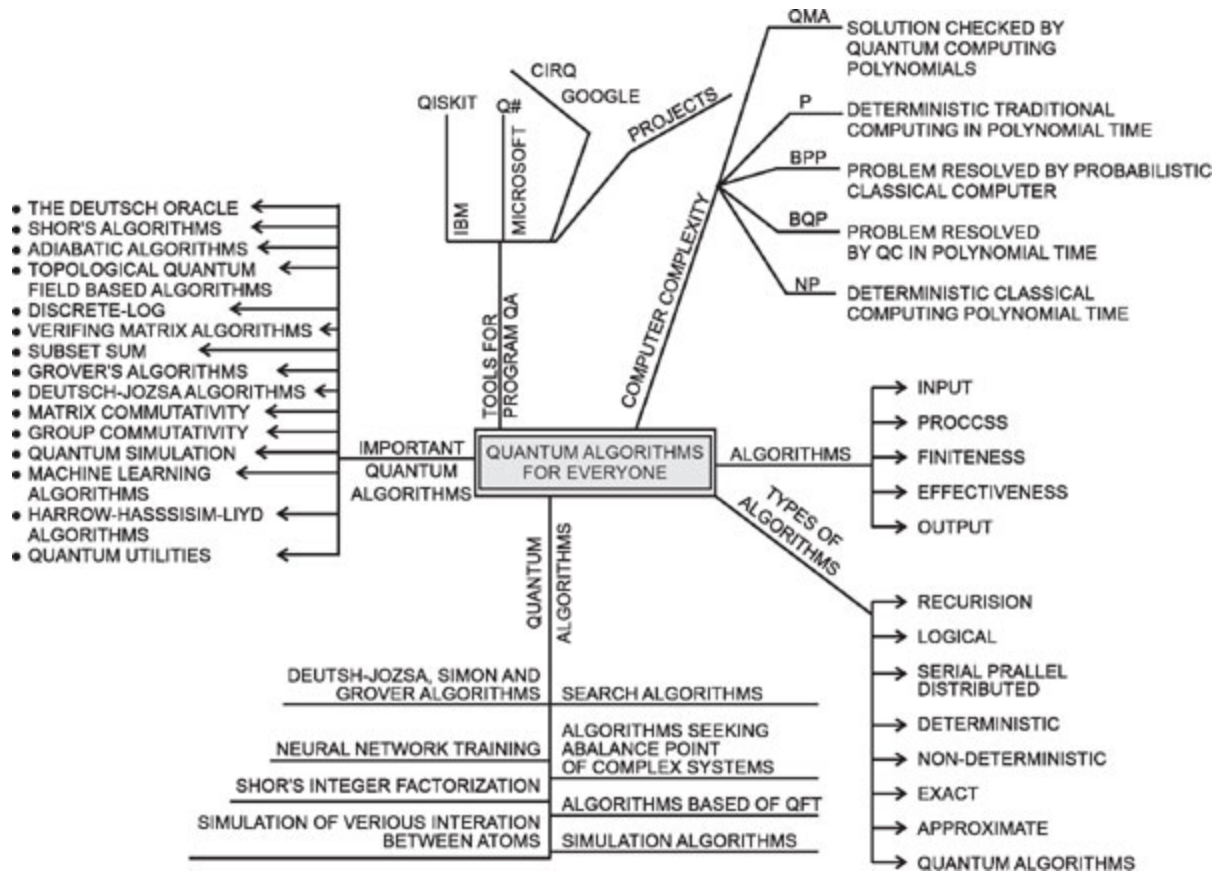


Figure 6.10: Mind map for [chapter 6](#)

Conclusion

In this chapter, we covered quantum algorithms, complexity, and their uses. We also touched base with some of the important categories and class of quantum algorithms. Nowadays, there are multiple software platforms available to do coding, managing, and maintaining the quantum algorithms. In this chapter, we were given a brief look at them as well. Further, in this chapter, some of the most important and fundamental algorithms were covered in detail, like Shor and Grover.

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CHAPTER 7

Quantum Machine Learning

Fuelled by snowballing computer power and algorithmic developments, machine learning methods have turned out to be influential tools for discovering patterns in data. Subsequently, quantum systems harvest counter-intuitive patterns held to be efficiently produced by classical systems, and it is reasonable to postulate that quantum computers may outstrip classical computers on machine learning errands. The stride of growth in quantum computing reflects the quick developments made in machine learning and artificial intelligence. Therefore, it is expected to question whether quantum technologies could bring improvement in the learning algorithms. This idea is sometimes referred to as quantum-enhanced machine and deep learning. The base of quantum machine and deep learning discovers how to invent and instrument tangible quantum software that bids such advantages. Recent work has made clear that the hardware and software challenges are still substantial but they have also unlocked paths in the direction of solutions.

In this chapter, we will learn about quantum machine and deep learning. We will also walk-through traditional software stacks, products and services available for deep and machine learning along with artificial intelligence. In order to have a solid background in these technologies.

Structure

In this chapter, the following topics are covered:

- Big data and quantum computing
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- Building a business case for artificial intelligence
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- Quantum deep learning
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Objective

After studying this chapter, you will be able to have a holistic idea of machine and deep learning and its relationship with quantum computing. The machine learning enthusiasts are heading into a forthcoming time where multipart quantum algorithms can chart out the head circuitry, decipher the genetic conception, and shape a focused infrastructure that syndicates biometrics and IoT devices to permit high-level safety devices, and even crack some extraordinary novel detections about the massive mysterious cosmos. Technically, quantum machine learning could enable charting out of these possibilities. In line with this thought, the chapter will focus on the following themes:

- This chapter will explore the combined potential of machine learning, deep learning, artificial intelligence and QC. It sees these technologies in a holistic framework products and services and explains the technical, functional, and process-based details of in the boundaries of available products and services.
- The chapter would show how Quantum machine learning is an interdisciplinary method that syndicates machine learning and associated technologies with the philosophies of Quantum Physics.
- The chapter also highlights how Quantum chips can be used to chart out computer algorithms to resolve multifaceted problems.

Introduction

The building of digital computers in the late decade of the 40th / early decade of the 50th century permitted the automation of data analysis methods. Over the past half-century, the quick development of computer power has permitted the application of linear algebraic data analysis techniques such as regression and principal component analysis and led to additional multifaceted learning methods such as support vector machines. Over a similar time, frame, the growth and quick development of digital computers laid novel machine learning approaches. Artificial neural networks such as perceptions were applied in the 1950's, as soon as computers had the power to comprehend them. Deep learning constructed on neural networks such as Hopfield networks and Boltzmann machines, and training methods such as back propagation were made known to and realized in the 1960's to 1990's.

Building a business case for AI

Machine learning and associated technologies are serving businesses to gain a viable benefit by transporting in operational competencies, attainment to novel customer segments, and serving to distinguish themselves. Forrester Research foretold a better than 300 percent upsurge in investment in artificial intelligence in 2017 associated with 2016. Two key ways machine learning is used in business are for sensing and predicting. Sensing is seeing massive amounts of data from sensors and learning to distinguish what's there. Applications and analytics platforms use a big amount of data to predict future behaviour. Cheers to automation and digitization, the important upsurge in the amount of data shaped in the mainstream of industries can be curved into a modest benefit and can lead to achieving strategic forecast goals.

Machine-learning algorithms are used to forecast maintenance requirements in manufacturing, inventory requirements in retail, disease risk in healthcare, potential credit risks of suppliers and buyers in economics, traffic patterns in travel, and power practice in energy sectors. Revenue teams are utilizing machine learning for enhancing promotions, rewards, and rebates, forecasting the tendency to purchase crossways all channels, making custom-made recommendations to customers, and forecasting long-term customer faithfulness. In healthcare, doctors and nurses are using computer vision and machine learning for identifying diseases. **Natural-Language Processing (NLP)** is used to fold insights from medical reports and literature. Now with the increasing traction on Quantum machine learning and Quantum enabled AI, the complete game plan is going to change. After having this background in traditional ML and AI, let's move toward the quantum world.

[The deal of quantum machine learning](#)

In 2009, a team led by Hartmut Neven, a computer expert at Google who broke new ground for augmented reality, co-founded the Google Glass project and took that to an altogether new height, and displayed how an initial D-Wave machine (a quantum machine company which produce quantum computers) could do a reputable machine-learning job. His team used fundamentally, a single-layer neural network that organized images into two classes: car or no car in a collection of 20,000 road scenes. The machine had only 52 employed qubits, far too little to take in a whole image. Neven's team united the machine with a traditional computer, which analysed various statistical numbers of the images and intended how sensitive these quantities were to the presence of a car. Some mixture of these quantities could, composed, spot a car reliably.

The experiment was not ground-breaking; however, it provided a hint on what **Quantum Machine Learning (QML)** can code.

QML is the technology of the future and a lot of efforts are going in that direction. For instance, Alphabet Inc. is giving a lot of focus on developing a hybrid quantum-classical machine learning technique(s), so that it could work seamlessly on near-term quantum devices. Alphabet Inc. algorithms are fundamentally understanding universal quantum circuits and applying their learning for the classification and clustering of quantum and traditional data. They are also attentive in multiplicative and discriminative quantum neural networks.

Roadmap for building, executing, and running quantum machine learning

Building a quantum machine learning and associated technologies solution needs a focused technical, procedural, and governance-based method. Therefore, the required action is essential to be taken on time. It wishes appropriate evaluation of the current process, metrics, design thoughts, and measurements. Correspondingly, beforehand, during, and after the employment of quantum machine learning, the solution required to follow some standard guidelines, or else the complete effort would go in vain. These strategies comprise risk management, making sure of funding, compliance arrangement, and lots of additional influences. The picture portrays the roadmap of retrieving, designing, applying and measuring a machine learning solution. Some of the prerequisites of quantum machine learning and associated technologies solution implement strategies that are as follows:

- Designing the Quantum machine learning employment process (from Idea Generation stage to rapid prototyping stage).
- Designing the Quantum machine learning project and organization governance structure, which comprises roles and errands of the people tangled.
- Bringing composed the design enablers.
- A provision in applying the Quantum machine learning solution and sustained program management.
- Allowing the development of ideas with a rapid prototyping factory.

For applying any Quantum machine learning solution, some guideline needs to be followed. [Figure 7.1](#) would highlight some of them. Also, it would present a

different style of implementation of the ML solution. Fundamentally, there are different workstreams that exist where the Quantum Machine Learning (QML) solution needs to implement. But how do we come to know that this area and situation is suitable for the implementation of quantum machine learning, machine learning, or artificial intelligence-based solution? Or is there no need for all this, and only simple traditional technology would solve our problems?

We need to evaluate the need and then design the solution accordingly. Once the solution is designed, its proper implementation is necessary. However, only implementation is not enough, it needs to be measured, and accordingly reinvention and reimplantation is required. Refer to [Figure 7.1a](#) and [Figure 7.1b](#) that illustrate these:

	Access	Design	Implement	Measure
Process Workstream	Access the current service management process	- Align Process - Design IT standards, policies, and templates as per industry standards	- Implement process across identified groups - Implement practical communication channels and training methodologies - Implement the agile process components	Measure the effectiveness of the implemented process and its success based on the KPIs and defined plan
Technology Workstream	- Analyze the sufficiency of present tools - Access tools for monitoring, building, testing and deploying	- Understand tools dependencies - Design / identify integration components for tools and requirements	- Provide functional and technical support for tools and frameworks implementations - Make changes in the process to align with the tools and implement new tools if needed	- Measure the framework's / tools' efficiency and effectiveness. - Gather feedback
Governance and People workstream	Analyze programme, vision, risks and compliance controls (are they in place) Further analyze project plans, templates, and governance structure along with roles and responsibilities	- Design management reviews, team composition with RACI, governance structure and matrices. - Validate findings with informal groups for service management process findings	- Implement governance structure and mechanism - Implement reporting, KPI's and matrix mechanism	- Measure and tune the defined matrix and KPIs - Measure data quality - Suggest implementation plans

Figure 7.1a: People, process and technology: interaction

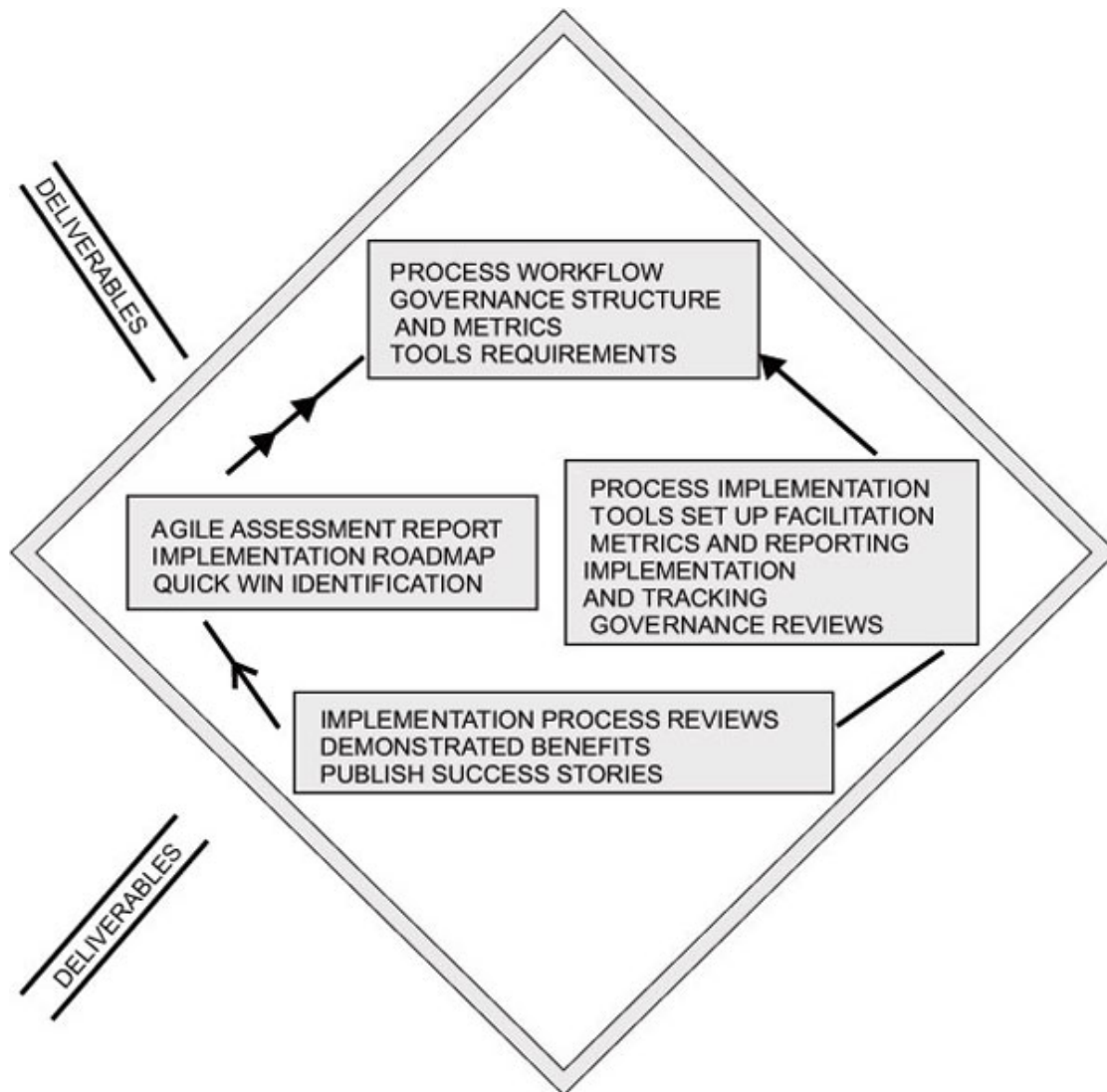


Figure 7.1b: Agile process in the context of deliverables

[Figure 7.1](#) are the guidelines for machine learning-based solution implementation across all the areas. [Figure 7.1a](#) describes the relationship of people, process and technology. Whereas [Figure 7.1b](#) describes the agile process of deliverables.

The current state of AI, ML, and deep learning technology

Well, after having a background of the roadmap of AI and quantum ML, now there is a need to visualize the present state of ML and AI solutions, so that this knowledge would help us to create a proper solution.

The democratization of artificial intelligence

The questions like how do I get started in the Quantum artificial intelligence, deep learning and Quantum Machine learning space? How do I turn out to be a data scientist or quantum machine learning expert? Do you see the way? Experts continually catching these inquiries know that making sagacity of the valuable technologies originates in the dizzying array of novel expansions, and it is certainly a problematic delinquent, for both persons and organizations of all dimensions.

There is a real gap in machine-learning talent sets, and it does help sagacity to request how to get started in this swiftly-changing field. Even though getting started with QAI, Quantum deep learning and quantum machine learning can be firmer than it appears, there are numerous different learning tracks to choose from; organizations are working tough to help ease this learning curve. The term democratization of artificial intelligence denotes a determination to brand machine-learning growth available for the greater set of developers.

Once it derives from democratized AI, we can effortlessly categorize machine-learning and deep-learning aids into three logical groups: cloud-based, on-premises, and hybrid models. Let's first analyse the machine-learning libraries and SDKs before looking into the platform offering. It is required to have knowledge of AI, Deep learning and ML in general and what they are bringing in terms of offering to the table. Because the market is flooded with products and solutions. Then we would move to the specifics of QAI and quantum machine learning.

Machine learning: libraries

At the moment, numerous different open sources of deep-learning, ML, and AI frameworks have been presented. The subsequent is by no means a comprehensive impression of all these toolkits or machine-learning methods. Now let's have a brief look at them.

In the context of deep and machine learning, Theano was the primary extensively accepted deep-learning library, which is upheld by the University of Montreal. Now, TensorFlow appears to be the most-used deep-learning library grounded on the figure of GitHub as well as based on Stack Overflow action. However, other libraries are increasing fervent user bases. PyTorch is a distinguished example and was presented in January 2017 by Facebook. It was a harbour to Python of a general Torch framework inscribed in Lua. PyTorch's acceptance is the consequence of its use of dynamic computation graphs as opposite to static

graphs. In totalling to PyTorch, Facebook also open-sourced Caffe2 and professed it as a New Lightweight, Modular, and Scalable Deep Learning Framework.

Microsoft hurled its cognitive toolkit, or **Computational Network Toolkit (CNTK)**, an open-source deep-learning toolkit shaped by Microsoft Speech investigators, in 2012. It has been obtainable on GitHub since January 2016 and is being used by Bing Cortana, HoloLens, Office, and Skype. Over 80 percent of Microsoft's interior deep-learning assignment is founded on CNTK which likes a first-class inhabitant rank on both Linux and Windows. CNTK has ironic API support, is frequently applied in C++ (training and evaluating), and delivers both low- and high-level Python APIs as well as R and C# APIs for training and evaluation. CNTK also supports UWP, Java, and Spark support, while the Keras and CNTK ropes model firmness and is part of the ONNX ecosystem.

Between all these different developments that look like branded silos, Facebook and Microsoft presented a novel open ecosystem for interchangeable AI frameworks called the **Open Neural Network Exchange (ONNX)** format. ONNX has been presented as a standard for the representative of deep learning models that allows models to be moved between frameworks. This aids developer, who can now train their model in one framework but then help it in production in a different one. ONNX is an open format that signifies deep-learning models and is now supported by CNTK, PyTorch, Caffe 2, and MxNet to allow interoperability among frameworks. While ONNX is a neural network exchange format for models, Keras is efficiently an interface that hush-up multiple frameworks. François Chalet, a deep-learning researcher at Google, shaped and upholds Keras. Keras is a high-level neural networks API, is inscribed in Python, and is capable of running on top of TensorFlow, CNTK, or Theano. Google proclaimed Keras was selected as the authorized high-level API of TensorFlow.

[Machine learning and deep learning platforms](#)

Microsoft, and other vendors, as well as IBM, Apple, Facebook, and Google, bid on machine-learning platforms to make life cooler for developers. Out of multiple offerings in the market, TensorFlow is very popular; one of the reasons is the largest ecosystem that it provides. By characterization, a complete machine-learning platform delivers algorithms, APIs, starter kits, development and training toolkits, training and testing datasets, as well as **Machine Learning Development Lifecycle (MLDLC)** procedures.

The democratization of the platform also makes it relaxed for developers to get up to rapidity. It's a one-stop-shop providing the computing power to design,

unsoiled up data, train models, and organize in staging and construction to containers, applications, services, and so forth. An additional feature of such a democratization platform is the provisioning of AI-optimized hardware. GPU-, TPU-, and FPGA-powered computing are not for the faint-hearted. The hardware, designed and architected to proficiently run deep-learning-based computational jobs, is affluent to build and preserve. So, it is characteristically in the best interest of individual developers and start-ups to exploit the already-built systems.

Microsoft AI Portfolio comprises four main areas: agent, applications, service, and infrastructure. For the agent, Microsoft plays on Cortana because of its intuitive nature. The application pillar of the four key areas comprises almost all desktop applications, office apps, and .net or core applications. Instead of services, Microsoft's AI ecosystem includes Bot Framework, Cognitive Services, Cortana Intelligence Suite, and Cognitive Toolkit (CNTK). The agent in knack being Cortana, the Microsoft bot environment is serendipitously named Bot Framework. The Cognitive Services offered cover vision, speech, language, knowledge, and search APIs, among others. The machine-learning tools comprise Azure ML and Cognitive Toolkit (CNTK), and the augmented reality counterpart is HoloLens. Lastly, the infrastructure services cover Azure Machine Learning, Azure N Series, and FPGAs to deliver the best-in-class training support for complex machine-learning and deep-learning models.

Amazon also has a huge AI offering and uses Alexa as its agent. Amazon Lex is the bot framework, while the cognitive offering of gratitude is absorbed in deep learning-based image analysis. Amazon Polly is the text-to-speech engine, while Apache MxNet is the deep-learning framework driving the cognitive services.

Apple's Siri is perhaps one of the most well-known digital assistants as it is obtainable with almost all Apple products, together with phones, tablets, and computers. With SiriKit's existence as a combined bot framework, Apple's CoreML library helps integrate machine-learning models into the application. Apple's method for Core ML supports Vision for image analysis, natural language processing, and Gameplay Kit, which assimilates well with the iOS application ecosystem.

Facebook bids Facebook Messenger as its agent or digital assistant, ParlAI as a bot framework for dialogue research, FastText and CommAI as their cognitive offering, and PyTorch is the machine-learning and deep-learning library.

IBM gambles deeply on Watson's ecosystem as its cognitive computing platform. With Watson Virtual Agent as the agent and Watson Conversation as the bot framework, Watson also bids vision, speech, language, and IBM data insights for

cognitive API and services. For machine learning, Watson ML services and Apache System ML are used.

Alchemy, the language-understanding framework, has also lately been superannuated in favor of Watson Discovery or Watson Natural Language Understanding.

Google broadcasts the intelligent and omnipresent Google Assistant as its agent, while its bot framework is partly founded on API.ai technology, a corporation it acquired in September 2016. In the cognitive space, Google bids a cloud vision API, video intelligence, a speech API, natural language, knowledge graph, custom search, and ML advanced solutions lab. For machine learning, the offering includes a cloud ML engine and the marquee ML library, TensorFlow, now synonymous with deep learning. Open-sourced in 2015, TensorFlow is a computation that uses data-flow graphs for scalable machine learning.

Evolution of QML

In the past decade, mainly in the previous five years, the amalgamation of influential computers and special-purpose information processors skilled in applying deep networks with billions of weights, composed with their application to very great data sets, has exposed that such deep learning networks are skilled to learn composite and delicate patterns in data. Quantum mechanics is recognized to produce poker chip instinctive patterns in data. Classical machine learning approaches such as deep neural networks often have the feature that they can together know statistical patterns in data, and harvest data that own the same statistical patterns. They know the patterns that they harvest. This observation proposes subsequent hope. If minor quantum information processors can harvest statistical patterns that are computationally tough to be shaped by a traditional computer, then possibly they can also know patterns that are correspondingly tough to distinguish classically. The understanding of this hope is contingent on whether effective quantum algorithms can be found for machine learning. A quantum algorithm is a group of instructions for resolving a problem, for example discovering whether two graphs are isomorphic, that can be achieved on an upcoming QC. Quantum machine learning software makes use of quantum algorithms as a fragment of a greater application. Analysing the stages that quantum algorithms recommend, it turns out to be clear that they have the latent to outdo classical algorithms for precise problems. This potential is recognized as quantum speedup.

Holistic approach towards QML and QC

Quantum machine learning is at the junction of two of the most thrilling present zones of investigation that is, quantum computing and classical machine learning. It discovers the collaboration among quantum computing and machine learning, examining how consequences and methods from one field can be used to resolve the glitches of the other. Among the ever-growing quantity of data, existing machine learning systems are quickly reaching the limits of traditional computational models (in terms of data processing). In this sense, quantum computational power can offer an advantage in such machine learning tasks. The field of quantum machine learning discovers how to invent and contrivance quantum software that could empower machine learning that is quicker than that of the so-called classical computers.

Powered by cumulative computing power and algorithmic developments, machine learning procedures have developed as influential tools for finding patterns in data. Quantum systems harvest uncommon patterns those conventional systems are thought not to produce efficiently, so it is sensible to hypothesize that quantum computers may outstrip classical computers on machine learning and artificial intelligence-related tasks. Here, we review the previous literature on quantum machine learning and provide the current status of it.

Training the machine to learn from the algorithms applied to hold data is the core of machine learning. This field of computer science and statistics employs artificial intelligence and computational statistics. The classical machine learning technique, through its subsections of deep learning (supervised and unsupervised) aids, categorizes images, recognises patterns and speech, knob big data, and numerous other things.

At present, due to the enormous amounts of data with which we work every day, novel approaches are needed to automatically achieve, rearrange, message, organize, and classify the provided huge amount of data. Classical machine learning, which is a flexible procedure, can know patterns efficiently, but some of these problems cannot be efficiently determined by these algorithms. The corporations, whose labour contains in big database management, are conscious of these boundaries and are absorbed in novel methods to achieve this. They have originated in the quantum machine, learning one of these methods.

Though, the interest to implement these methods through quantum computation paves the way to QML. However, quantum machine learning is a goal to tool QML in quantum systems. This can be achieved by utilizing quantum properties such as superposition and entanglement to resolve these glitches efficiently and

effectively. Quantum computers use the numerous superposition states $|0\rangle$ and $|1\rangle$ to agree with any computation process at the same time. This gives an edge over the traditional machine learning technique in terms of the speed of operative and data handling. In the quantum machine learning techniques, we grow quantum algorithms to function the traditional algorithms with the use of a quantum computer. Thus, data can be classified, organized, and analysed using the quantum algorithms of supervised and unsupervised learning approaches. These approaches are again implemented by finished models of a quantum neural network or support vector machine.

In the present time and in the future, for solving certain classes of prediction problems, machine learning fused quantum algorithms is the way forward. However, the randomness of combining quantum computing into the machine learning workflow grants a barrier. At the moment, broadly two tracks, that is, machine learning and quantum computing, are largely independent. The holistic approach is missing. For instance, for machine learning experts, it's extremely easy to make sense of how to program the system. However, fitting that into a machine learning workflow is an extra challenge because subsequently, machine learning programs are getting very complex.

Currently, ML experts require a different person to deal with the quantum computing portion. But this is going to change (already started) because in all cases, the connection between these two grounds goes much more than the standalone way. Also, it's not just AI bids that can take advantage of this merger but the other areas as well. There is an assembly area where quantum computers do machine learning algorithms and normal machine learning plans are applied to survey the quantum computers. This area of investigation is creating at such overflowing speeds that it has shaped an entire novel field named QML.

This interdisciplinary arena is very new, though, new work has shaped quantum algorithms that could go about as the structural chunks of machine learning programs; up till now the hardware and programming problems are as yet noteworthy and the growth of completely functional optimized quantum computers is five to ten years away.

Quantum computing will be proficient in captivating multifaceted AI issues and obtaining manifold solutions for complex issues all the while. This will help artificial intelligence to become more effective in executing complex tasks in human-like ways.

For instance, it can help robots to become optimized decisions in real-time in practical conditions. Now the natural question is how far are we from this kind of

future? Well, seeing a group of the world's topmost organizations and universities, possibly we are not so far away. Quantum computing and machine learning are extremely well matched. Because it's difficult to reproduce "something" with a traditional computer, it is easy to get in the vicinity of the quantum computer.

Quantum machine learning and Quantum information processing

Quantum machine learning is closely related to quantum information processing. But before going into the details of it, let's try to find out why quantum information processing is different from the traditional one.

Certain kinds of quantum superpositions in correlated particles remain inextricably linked: an operation or measurement on one particle determines the state of the other, even when the particles are far apart. The first application in which quantum entanglement was specifically exploited was in cryptographic communication. This was first developed in the UK in the early 1990's. Since then, entanglement has become a core resource in the development of QIP. QIP is providing the conduit for the birth of a new scientific field: the physics of quantum entanglement.

Quantum algorithms power up the speed of information processing; IBM's Blue Gene supercomputer would take millions of years to factor a 256-bit binary number, whereas a quantum computer working at the same clock speed could do it in a few seconds. (This is because if a classical computer factors a number in a certain time, it would need that time squared to factor a digit number twice as big, but a quantum computer could do it in just twice the time.)

As we saw in the previous few sections, there are a lot of technologies that are available which are directly or indirectly intersect QML. [*Figure 7.2*](#) provides an idea of the positing of QML in the bigger ecosystem of technology:

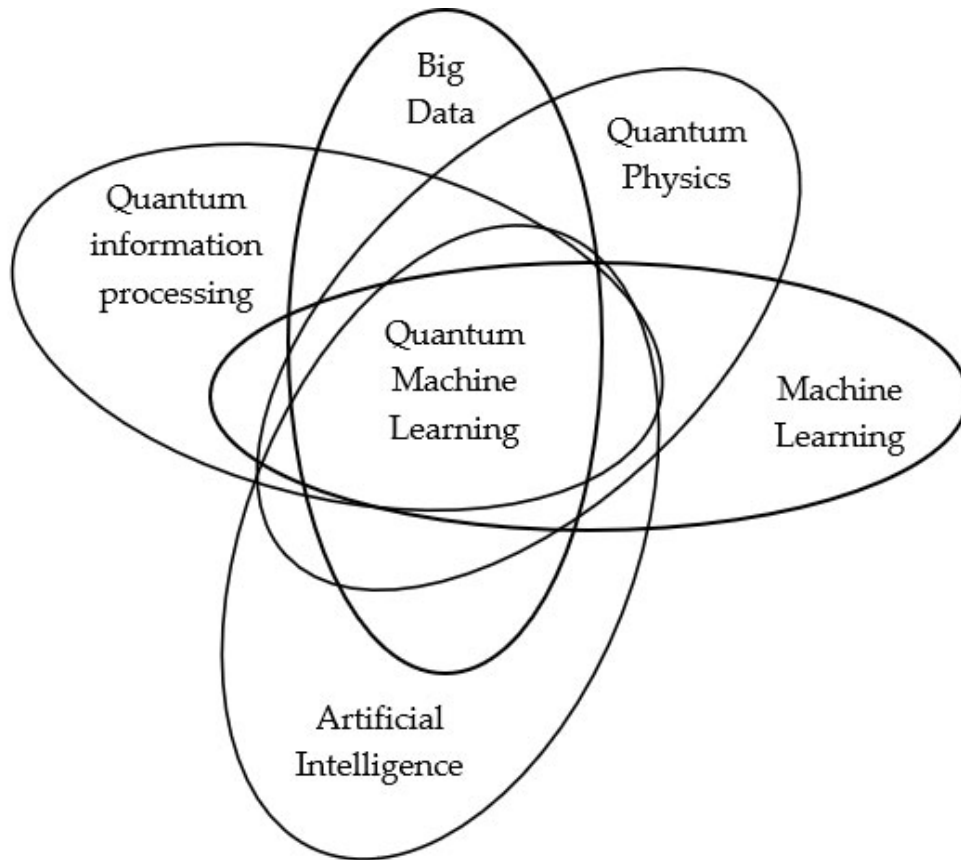


Figure 7.2: Positing of quantum machine learning in the jungle of technologies

Classical machine learning versus quantum machine learning

Machine learning is a way to teach a computer to learn from the set of data and produce patterns. While QML provides a mechanism where quantum computers or any other quantum device which is capable of processing information can learn patterns in data that cannot be learned by traditional machine learning algorithms, as follows:

- If we need to study a machine learning algorithm on a conventional data set utilizing quantum computing, we are essentially equipped enough to be talented to change this dataset into quantum data which is nothing but quantum states. Thereafter, put on the quantum algorithms on it. The main benefit which we get here is the Timeliness. To have a grasp on quantum machine learning, it is good to have seen the comparison between traditional and quantum machine learning. [Figure 7.3](#) projects this fact:

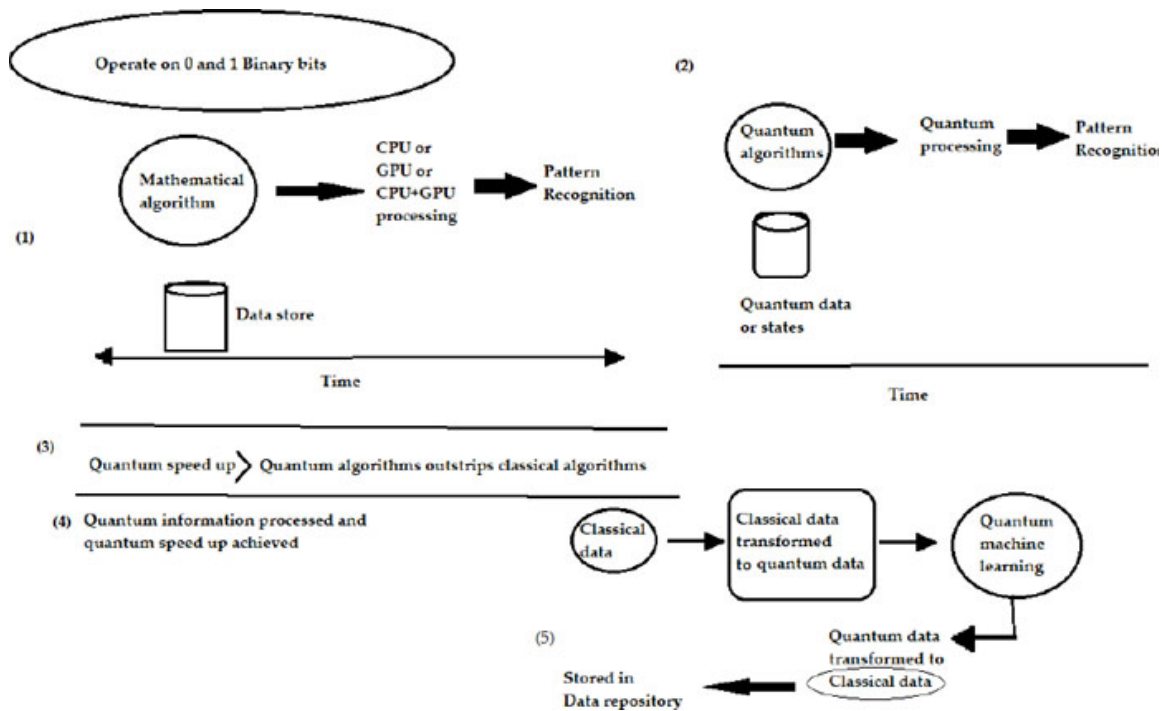


Figure 7.3: Comparison between traditional and quantum machine learning (1 and 2). And graphical representation of quantum speed up and performance improvement (3 and 4). Quantum data processing (5)

- These quantum algorithms will decrease the computing power of the mainstream of self-learning algorithms. However, practically it is not that easy because quantum storage is a challenge. At the moment, there is a requirement of a sophisticated device that successfully takes classical data which is for instance stored in the hard drive or DVD and mapped them with a quantum state. Presently, we are using something called a **quantum Random Access Memory**, or **qRAM**.

Let's explore QML

Quantum machine learning has taken an approach, in which the quantum system does not just simulate the network but it is the networks. Each qubit positions for one neuron. It is not like conventional computing. Because, in its place of starting with input data, performing a sequence of actions, and showing the output, it works by finding interior steadiness. Consequently, each of its qubits is a superconducting electric loop that turns as a minute electromagnet oriented up, down, or up and down that is, a superposition. Qubits are reinforced together by permitting them to interrelate magnetically.

To run the system, the first state is to enforce a horizontal magnetic field, which primes the qubits to an equal superposition of up and down. Further, there are a few unique ways to input data. For instance, in some cases, the fix of a layer of qubits can be done to get the anticipated input values. The incorporation of the input is the strong point of the connections that happen among qubits. The qubits search to bring into line in the same direction, whereas some go in the conflicting direction.

Due to the effect of the horizontal field, qubits flip to their desired orientation. However, in doing so, they activate other qubits to flip. Originally that occurs a lot since several of them are askew. Over time, though, they calm down, and the horizontal field can be turned off to lock them in place. At this juncture, the qubits are in a pattern of up and down that safeguards the output trails from the input. It's not at all comprehensible, what the concluding preparation of qubits will be, and that's the point. The arrangement, just by doing what originates naturally, is solving a problem that a normal computer would battle with. It's different from conservative programming. Nature solves the problem.

The qubit-flipping is determined by quantum tunnelling, a natural propensity that quantum systems have to search for their best arrangement, rather than settle for the second best. One could shape a classical network that functioned on analogous theories, using random wiggling rather than tunnelling to become bits to flip, and in some cases, it would work well. But, fascinatingly, for the kinds of problems that arise in machine learning, the quantum network appears to spread the optimum quicker.

QML is founded on two concepts: quantum data and hybrid quantum-classical models. We will explore these concepts one by one. Let's start with quantum data.

Quantum data

Quantum data is any data source that arises in a natural or artificial quantum system. This can be data produced by a quantum computer, like the samples collected from the Sycamore processor for Google's demo of quantum supremacy. Quantum data displays superposition and entanglement, leading to combined probability distributions that could need an exponential volume of classical computational resources to signify or store. The quantum supremacy trial displayed is likely to sample from a tremendously complex joint probability distribution of 2^{53} Hilbert space.

The quantum data produced by NISQ processors are deafening and characteristically entangled just earlier than the measurement happens.

Experiential machine learning methods can produce models that exploit the removal of valuable classical information from deafening entangled data. The Quantum library provides primitives to grow models that disentangle and simplify correlations in quantum information. It opens up opportunities to advance current quantum algorithms or determine new quantum algorithms. The following are a few examples of quantum data that can be produced or simulated on a quantum device:

- **Natural simulation:** Excerpt information about chemical constructions and changing aspects with possible applications to material science, computational chemistry, and a lot of other fields.
- **Quantum substance simulation:** Prototype and design high-temperature superconductivity which displays many-body quantum properties.
- **Quantum communication networks:** Use ML to distinguish between non-orthogonal quantum states, with application to blueprint and building of designed quantum repeaters, quantum receivers, and cleansing components.
- **Quantum metrology:** Quantum-enhanced excessive meticulous measurements such as quantum sensing and quantum tomography are integrally done on examinations that are small-scale quantum devices and might be proposed or be of better-quality by variational quantum models.

Hybrid quantum-classical models

A quantum model can signify and simplify data with a quantum mechanical source. Since near-term quantum processors are still impartially minor and deafening, quantum models cannot generalize quantum data utilizing quantum processors unaided. NISQ processors must work in collaboration with classical co-processors to turn out to be efficient. As most of the cloud platforms already provision heterogeneous computing across CPUs, GPUs, and TPUs, it is used as the base platform to trial with hybrid quantum-classical algorithms.

A **Quantum Neural Network (QNN)** is used to define a parameterized quantum computational model that is best performed on a quantum computer. This period is often substitutable with a **Parameterized Quantum Circuit (PQC)**.

Noisy Intermediate-Scale Quantum (NISQ)

Quantum Computing (QC) has involved inordinate attention from both the academic world and industry throughout the last decades owing to its strong

potential in hurrying numerous significant applications, for example, integer factorization, database search, molecule simulation. The transition from quantum theory to quantum engineering is leading us in the direction of the **Noisy Intermediate-Scale Quantum (NISQ)** age. In this era, QC devices have lesser than 1000 qubits and are not high enough to backing Quantum Error Correction.

To make use of such NISQ devices that grieve from incomplete qubit lifetime and flawed operations, more care is given to NISQ system design and optimization in recent years, fluctuating across NISQ compiler, quantum control hardware architecture, NISQ device, and so on.

Preferably, all these novelties should be assessed on truthful NISQ hardware. Though NISQ devices need a thrilling execution environment and the greatest of them continue in physics workrooms. Current QC cloud services, for example, IBM Quantum Experience, Rigetti's QPU, only deliver restricted access which cannot please the ever-increasing request for trials for assessing new NISQ system designs. These limitations are obstructive for investigators from getting into this area.

Simulation can be a probable solution to this problem as NISQ system novelties can be proposed and evaluated deprived of retrieving truthful hardware. Subsequently, a whole NISQ system contains two major components, the quantum processor and its traditional control system, a simulator for NISQ schemes needs to meet the succeeding chunks, as follows:

- **Simulating a noisy quantum processor:** Quantum processors in the NISQ age hurt from numerous sound possessions. A simulator desires to be talented to model the noises on genuine NISQ devices. The simulated production faithfulness can help guide future NISQ system design.
- **Simulating a classical control system:** The design of a control system can meaningfully distress the complete NISQ system act, for example, execution time. Such properties also affect the activity of the quantum processor. For instance, lengthier execution time may outdo the qubit lifetime, which limits the scope of a QC program that can be performed.

[Quantum speed-ups in machine learning](#)

If we trust theories, experiments, and hypotheses around quantum computers, then we conclude that QC based systems (algorithms) can search an unsorted database with N entries in time proportional to $\sqrt{N}$, that is, where a classical computer has given Blackbox access to the same database, or generically, datastore would take

the time that is proportional to N (entities). As we understand, the quantum computer exhibits a square root speed up improvements over the classical Von-Neuman architecture-based computer. Similarly, a quantum computer can perform Fourier transforms over N data points, invert sparse $N \times N$ matrices, and find their eigenvalues and eigenvectors in time proportional to a polynomial in $\log_2 N$ where the best-known algorithms for classical computer take time proportional to $N \log_2 N$. The quantum computer exhibits an exponential speed-up over the best classical computer algorithms. [Figure 7.4](#) would give us an idea about the level at which the quantum speedup will be achieved for the different quantum-enhanced machine learning algorithms or subroutines.

The speedup showed in this figure are taken concerning their classical counterparts. Hence, means quadratic speedup and $O(\log(N))$ means exponential relative to their counterparts. The details of quantum speedup are shown in [Figure 7.4](#):

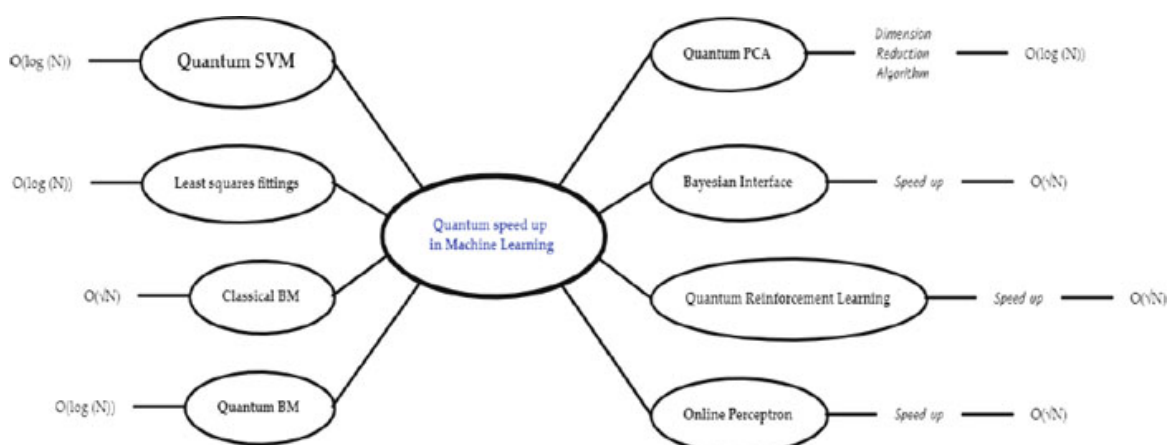


Figure 7.4: The details of quantum speedup

[Author's POV on building machine learning / Quantum machine learning solutions](#)

How to architect and build a quantum machine learning solution? let's go generic here. Due to the nature of ML and QML, the same hygiene, analysis and implementation factors are applied to both viz., ML and QML, especially when it comes to building ML or QML solutions. Further, when we decide on a few important parameters in QML, again follow a similar path of ML. For instance, cloud strategies need to chart out for QML as well because a practical QML solution would be feasible only on the cloud infrastructure, the same is true for ML. Likewise, either the need to do blueprinting, or require to take some

prerequisite steps like defining the scope of the project and taking architectural considerations into the account, specialists generally follow ML way. In this section, we would give a brief look at these aspects.

Organizations are looking to technologies like QML / ML, IoT, and big data to enable themselves to be more relevant in the market and grab customers in the competitive age of technologies, processes, and war of talent. However, the only way to achieve this is to provide effective products and responding to market needs on time. To effectively and efficiently take advantage of these technologies, an organization must have a strong real infrastructure to store data, execute analytical jobs, and defend its data assets from unforeseen modification or compromise. A cloud-based infrastructure is best suited to the organization and provides them a flexible platform for data storage, management, and processing. It would also cater to scalability (horizontal and vertical) and availability needs with ease. However, to build a good machine learning solution clarity of requirements, needs, vision, and practice-specific use cases are the mandatory criteria; for good architectural design, cloud-based infrastructure has the added benefits of easily adding new capabilities. As the cloud providers increase their feature range, as well as gain benefits from leveraging the cavernous know-how in scale and security the cloud provider facilitates, there is another side of the coin as well; in recent times, machine learning implementation becomes more inexpensive through the use of cloud infrastructure. Therefore, chances are high that technology would be misapplied. If your go-to cloud provider always highlights that machine learning would provide you with huge benefits, then this is a very subjective statement and depends on multiple factors. Therefore, taking proper, rational decisions is the key because the value and benefit of a machine learning solution would not be realized if it is applied to systems where it is not required. For instance, if it is applied to a system where prediction capabilities are not required at all or appropriate data is not available.

Generally, a machine learning solution is neither implemented to replace existing systems nor to replace an existing data store. However, for building a machine learning solution, organizations need solid data strategy, infrastructure, architecture, and workflows. These would help ensure high-quality data availability across the organization (LOB, UNITS) which is linked to rapid analysis and does not expose the organization to risk through data compromise. It also helps them to cater to compliance challenges.

Some common steps an organization has to take before and after kicking off ML/ QML projects. These must have a checklist for an effective and efficient

machine learning project implementation. And, same can be extrapolated for quantum machine learning solutions as well:

- Define scope.
- Review the existing system and data governances' structure.
- Revisit security policies and tune them as per current needs.
- Generate a map of existing workflow and relationships between applications, and data stores.
- Create infrastructure, process, and technology maps and highlight strong and weak relations among them.
- Define a business vision for machine learning.
- Find out the suitability of technologies for the solution, define business drivers, call out risks and establish a business case.
- Define migration strategies for the application, infrastructures, and services.
- Secure budget and resources.
- Define operational model.
- Create an implementation roadmap.
- Chart out the implementation plan.
- Implement it.
- Evaluate the implementation.
- Take feedback on the iterative way.
- Modify and finetune the strategies based on the analysis of feedback.
- Follow the usual cycle.

The primary value of big data and ML/ QML is to bring flexibility and insight through the analysis of complex data sets. Many associated technologies will be part of this suite, including predictive analytics tools, IoT and cloud technologies data modelling, data quality, and cognitive computing frameworks. However, the most important step of any analytical workflow is the efficient data process. **The following are the steps which would be generally followed to convert data to insight to its consumption;** completing these steps would be confirming that high-quality data is taken under consideration. It also helps data scientists to analyse the prepared data correctly. Below are the steps:

- **Gather:** Gathering data from heterogeneous sources of data.

- **Discover relationship:** Discover the relationship in the existing data set.
- **Organize:** Organize and then reorganize data set for efficient and effective utilization of it.
- **Analyse:** Identify and analyse the relationship.
- **Generate insight:** Generate insight after the analysis of data.
- **Report:** Report the insight in user understandable format.
- **Consume:** Consume the insight for the business purpose.

Architectural considerations for a machine learning / quantum machine learning solution

The technical architecture for a machine learning solution must be built around the need and requirements; therefore, while designing the ML/ QML solution, the key design considerations must be aligned with the following factors:

- **Know your need:** Identify your need and define the use cases accordingly. This would enable proper prioritization of work, a selection of technology, scalability considerations, seamless system, and data integration.
- **Define operational strategies:** Define the operational strategies beforehand because it is a must to execute the solution effectively.
- **Be optimistic about scalability and performance:** While creating a ML/QML, an optimistic approach is required. You have to believe as time passes, your organization would grow and evolve. Therefore, while you are designing the solution or architecture, your platform would be capable enough to incorporate the growing demand for data and must take care of analysis without much change. In a nutshell, when data expands, it would be able to handle it effortlessly.
- **Strong data access and retrieval strategies are the keys:** Easy to understand interfaces, effective tools for storing data, optimized platforms, and excellent handling capabilities for unstructured data are some of the parameters which must be considered during the design. These parameters facilitate efficient and effective ingestion and processing of data / quantum data.
- **Security controls, logging and auditing:** Security is a key consideration for machine learning solutions. Identity management, auditing, and access controls must be designed to cater to the risk levels of the organization and efficient enough to handle compliance needs. Access control implementation must be consistent between access methods.

- **DevOps and analytics Ops:** Incredible operational value comes from the heterogeneous sources of data for storing and processing for a machine learning solution, especially in the cloud environment. Therefore, it is difficult to manage them manually. Hence, the preferred solution is automating deployment and recovery. This would lower the operational problem on the IT team when making changes.
- **To be in synchronization of advanced capabilities:** Implementing advanced capabilities, like APIs, parallelization, cognitive capabilities are the key to effective machine learning solutions. Therefore, synchronization with the latest happening is a must.

Cloud adoption of the ML / QML solution

When organizations are planning to put their QML / ML solution on the cloud, there are a few specific steps and strategies that they have to adopt. In a cloud-based environment, making sure about the availability, reliability, and scalability of the environment is important. It is more relevant in big data/machine learning scenarios because of the real-time demand for insight or on-demand insight. Therefore, to attain analytics and ML/QML as a service, the preceding parameter is of high priority.

Blueprinting is a very important activity for designing any architecture. Therefore, a brief description is provided next. However, these are outline views and if someone wants to go into detail, they may consider dedicated literature for the same. The details of **Blueprinting** are shown in [Figure 7.5](#):

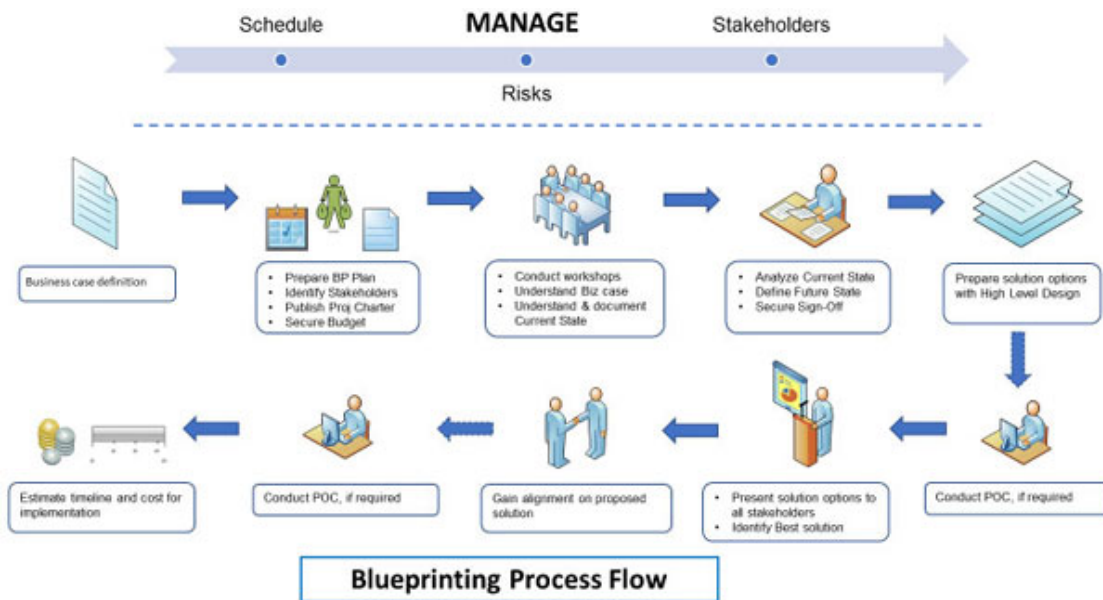


Figure 7.5: The details of blueprinting

A blueprinting exercise in an IT project involves the following:

- Brainstorming of business and technology stakeholders to elaborate, align and document the scope of a stated business problem.
- Translate it further to high-level requirements.
- Come up with various possible technical solutions (including high-level cost & timeline).
- Align and secure sign-off the best solution option.
- After this exercise, the budget is secured for the implementation project and a project team is identified and instilled.
- The team refers to the Blueprinting documents as high-Level requirements to furnish the technical requirements, followed by subsequent SDLC phases.
- It is extremely important to spend the initial time in the brainstorming sessions to understand the business problem in detail from the business stakeholders.
- Business folks tend to state the problem at a very high-level. So, ensure to have a questionnaire to probe and find the necessary details.

The questionnaire should cover aspects like the following:

- What is the base issue(s)?
- How is it impacting their business or the performance of their duties?
- Are there any financial impacts on the issue?
- Is the stated functionality they are considering going to solve the base issue or create new ones in the future?
- Have they tried to solve the issues earlier?
- If yes, what have been the shortcomings of the solution?

For a successful Blueprinting exercise, it's imperative to include all the relevant stakeholders.

The best practices for ML / QML are as follows:

- Identify your need, understand the business requirements.
- Be specific to your requirements and tune it with your need.
- Keep your development approach simple.

- Identify a reliable source of data. Gather desired data and make that noise-free.
- Normalize data before using algorithms.
- Identify appropriate language, framework, and libraries as per your need.
- Define strategies to train models including hardware infrastructure, licensing.
- Define execution strategies like the implementation of parallelism and iterations.
- Test infrastructure, software independently, and then in combination. This would enable holistic testing.
- Design and create an appropriate model. Also, do not forget to ensemble models.
- Start with an independent model it would help in testing.
- Train multiple single-threaded models in parallel.
- Make your code portable and try to be environment independent. A cloud environment is the best way to achieve this.
- Iterate your model to purify and make it accurate based on the change in underlying data. So that its training would be up to date.
- Measure the outlet between the models.
- Business problems change over time, so keeping the model in tune with the requirements is the key to success. Monitor models for decreasing accuracy.
- Keep track of predictions and forecasts.

Languages, Tools, Platforms, and QML

The quantum versions of machine learning can learn from data quicker and with advanced exactness, and therefore be able to do more precise forecasts. Also, Machine learning help physicists connect and control quantum effects and its astonishing properties in laboratories to improved understanding of quantum systems and their behavior. As we understand, quantum physics also helps to augment the act of machine learning algorithms that are hard to use with classical computers. There is some reason behind it.

One of the most important reason is due to the strange nature of how a quantum computer works; it is stimulating to input standard data such as financial transactions or pictures. Out of the multiple ways, the most intuitive way to

handle this issue is to use “quantum data”. It is much easier to deal with data that is already quantum, such as info about the internal working of a quantum computer or data that is shaped by quantum sensors. A Quantum sensor is nothing but a device that harvests data in a quantum state. Another approach is to do what is named a quantum state preparation of the classical data. Further, what experts can do in laboratories is using polarization of photons to encrypt zeros and ones of classical data into physical degrees of freedom compulsory for quantum processing of the data.

The other significant challenge is that quantum computers do not talk in the languages that most program writers are acquainted with like, Python, C#, F#, Java, C++, and JavaScript. Program writing for quantum computers for optimized processing of data is generally complex than in traditional programming language (at least at the moment). The multiple basic quantum programming languages are under development; Microsoft Q# is one of them, which is technologically advanced by Quantum Architectures group. This is enhanced by the Microsoft Research Computation division.

Birth of Q#: The quantum programming language

The multi-paradigm quantum software development language Q# is optimized for training and producing new programs. However, the purpose and nature of the quantum and traditional programming language are different but one thing is always true that Q# (any other quantum programming language) code go beyond the competencies of any traditional code.

This language was constructed by Microsoft to thrive LIQUi|> and is a Domain-Specific Language that can be used inside Microsoft’s Quantum Development Kit. The quantum programs are run on a quantum simulator which holds possessions that an actual quantum computer does at least from the behavioral feature. It uses the .NET Code SDK, permitting for laid-back formation, structure, and running of quantum projects via the command line.

IBM Q, an IBM ingenuity to shape saleable quantum computers, has its graphical language platform named Quantum Experience, whereby anybody can run algorithms on IBM’s quantum processor and be able to work with separate qubits. The computing paradigms are also changing. For example, D-Wave’s quantum annealer is not a universal quantum computer, which implies that it would not execute on any of the stated quantum programming languages; however, the hardware is valuable for learning from data quicker than with traditional

computers. As subsequently scientists have effectively fed quantum machines with data that it can read, and also inscribed and applied algorithms to the development of their data, they still have a problem, as it is tough to read and provide the outcomes precisely.

The outcome of quantum operations is a quantum state, which can be supposed to as a superior probability spreading. Gaining classical information about the quantum state needs “measurement”. To completely describe a quantum system or in other words to get the outcomes, an investigator would have to repeat the measurement and moderately perhaps all computations that led to the state. This is needed to do this tiresome task to comprehend what the quantum system did to the data, and to be capable to have a classical interpretation of the outcomes. These challenges will need functional work as QML transfers from the research test site and into science-based start-ups. Also, to resolve problems like error correction in quantum computers, QML has the potential to bring about the next wave of technological surprise and produce a type of revolution in the field of the quantum computer itself to take that to the next level.

At the moment, both machine learning and quantum information processing are quickly rising and have their reservations. Quantum machine learning is interplanetary where these two components of research turn as accompaniments of each other and each step forward in one is a prospect of development in the other. Some of the difficulties and challenges are slow to accomplish when the goal is to shape intelligent systems that learn from data. In accumulation, machine learning may itself permit scaling up quantum information processing, which can finally lead to building scalable quantum computers. While gathering and storing massive quantities of data are cheap, that it is no longer a concern. The machine learning community is coming up with quicker and well-organized approaches and techniques to learn from this data. What governs how reckless we exploit our data and models are computational power, and quantum machine learning is screening the much potential desire to be our finest stake yet for building both intelligent systems and quantum computers.

Quick question

Question: Why should we hope that quantum mechanics might help in learning patterns in nature?

Answer: Quantum computers could recognize strange and counter-intuitive patterns that cannot be recognized by any classical computer. So, quantum weirdness might allow us to help in understanding the weird features of nature itself.

Quantum deep learning

Quantum Neural Network model is the method of deep supervised learning to train the machine to categorize data, acknowledge patterns or imageries. Deep learning networks learn by regulating the strengths of their connection in a better way to convey input signals. This communication of the network is done through the passing of input signals through multiple layers of artificial neurons. The typical flow starts when data is fed into the network of artificial neurons (or connections), each neuron that fires transmit signals to certain neurons in the next layer, which are likely to fire if multiple signals are received. The process filters out the noise and retains only the most relevant features.

The elementary principle behind the quantum neural network is to design the circuits with qubits and spin gates to function the network corresponding to the neurons and weights as used in a traditional neural network. The network learns from a set of training samples. Each input string originates with a label value. The persistence of the network is to come to be the label value of the data set and minimize the unconventionality of the found label from the true label. The emphasis is to acquire the training parameter that gives the least error. The training parameter is rationalized through every iteration. Error minimization is complete by the backpropagation method, which is based on the gradient descent principle.

The foremost job of a neural network, be it traditional or quantum, is to know patterns. Enthused by the human brain is a network of basic computing units, neurons. Though each can be as modest as an on-off device. A neuron displays the productivity of manifold extra neurons as if captivating a ballot, and changes to on if sufficient of them are on. Characteristically, the neurons are settled in coatings. An early layer receives input (such as image pixels), middle layers produce numerous mixtures of the contribution (instead of structures such as boundaries and geometric shapes), and a concluding layer produces harvest (a high-level explanation of the image content). The process of quantum deep learning is shown in [Figure 7.6](#):

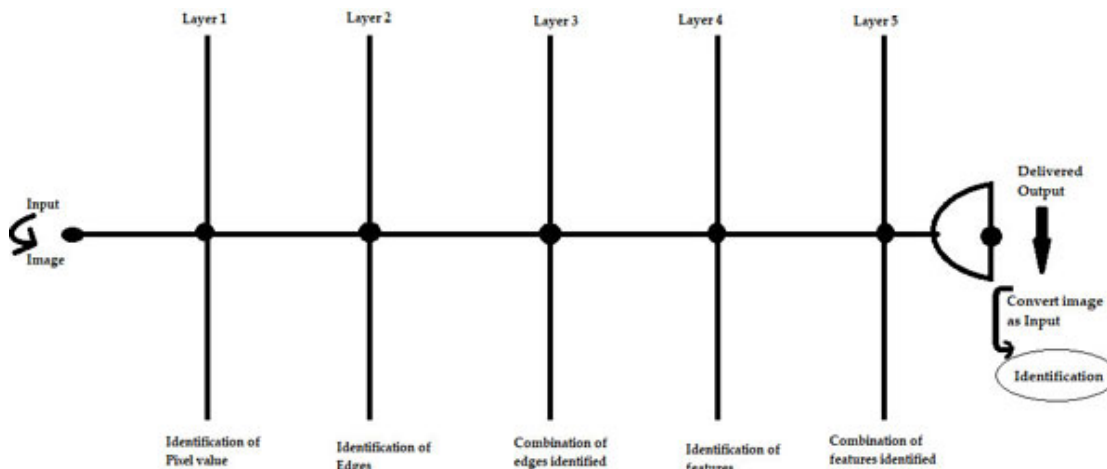


Figure 7.6: Process of quantum deep learning

Significantly, the connection of neurons is not decided from before but familiarizes itself in a procedure of trial and error. The network might be fed images labeled kitten or puppy. For each image, it assigns a label, checks whether it was right, and tweaks the neuronal connections if not. Its guesses are random at first, but gets better. For instance, after analysing perhaps 10,000 examples of pet animals, it knows it's a pet. A serious neural network can have a billion interconnections, all of which need to be tuned.

On a traditional computer, all these connections among node or artificial neurons are signified by a vast matrix of numbers and running the net what fundamentally implies as undertaking matrix algebra. Conservatively, these matrix processes are subcontracted to a dedicated chip such as a graphics processing unit. Nonetheless, nobody does matrices like a quantum computer. Manipulation of huge matrices and big vectors are exponentially faster on a quantum computer. One of the reasons is that quantum mechanics itself is based on the foundation of linear algebra and matrix manipulation. Also, quantum computers are good at benefiting from a quantum system's exponential nature. The massive majority of a quantum system's information storage volume exists not in its separate data units that is, qubits, but the quantum complements of traditional computer bits. However, the joint properties of those qubits also play a major role there. As we know, two qubits have four joint states: both true, both false, true/false, and false/true. Each has a sure allowance, or amplitude, that can signify a neuron. If one does the addition of a third qubit, he/she can characterize eight neurons, a fourth, 16. The volume of the machine produces exponentially. As a result, the neurons are spread out over the complete system. When you act on a state of four qubits, you are dealing with 16 numbers at a stroke, while a traditional computer would have to

go complete those numbers one by one. Understand the impact 60 qubits would have to encode a quantity of data equal to that fashioned by us in a year, and 300 could transmit the traditional information matter of the noticeable cosmos. And that's presumptuous to each amplitude is a solitary traditional bit. Amplitudes are incessant quantities and would be explained in the form of complex numbers. Therefore, for a reasonable investigational exactness, one force stores as many as 15 bits. Remember a quantum computer's aptitude to stock information efficiently doesn't make it quicker.

Limits and discoveries in Quantum Deep Learning

The main obstacles limiting quantum growth in the deep learning area are as follows:

- The primary difficulty of quantum neural networks was the absence of an actual quantum computer to trial with. However, newly numerous corporations have brought quantum computers including IBM, who has made them available to researchers for free over the Internet.
- The subsequent difficulty was the castle in the air of training quantum networks. A novel algorithm now resolves that problem using two chief steps: Simple Quantum Neural Network and Quantum Neural Network training.
- The third problem is that the classic neuron/perceptron has nonlinear functions, tapping it in a struggle with the quantum qubits that function only with unity and linearity. This problem has been resolved with a novel quantum perceptron utilizing a different or more accurately special quantum circuit, better known as the **Repeat-Until-Success (RUS)** circuit.

Benefits of Quantum Machine Learning (QML)

The core of many ML application is the repeated computation of a complex mathematical and statistical expression. For example, matrix manipulations, and for almost any type of quantum computation, math is one of the mandatory criteria. One step further, for doing quantum processing like gate based or even for annealing, math plays an important role. For these types of processing, quantum physics integrates itself with ML which is based on the solid foundation of math. In the present day, the technological landscape goes the new generation of applied technologies becoming mainstream like big data and artificial intelligence, which again takes the majority of inputs from the internet of things

devices. So, this complete ecosystem of science and technology creates a platform where quantum machine learning becomes effective and more efficient every day. In the following sections, we will cover some of the benefits of quantum machine learning. This will give the reader an idea on how QML would potentially change the future.

Quantum computers use quantum properties such as quantum coherence, superposition, and entanglement to process in ways that classical computers can do. Quantum computer exhibits a square root speed up over a classical computer. For instance, QC is capable of searching unsorted database or datastore with N entries in time proportional to the square root of N , algorithmically $O(\sqrt{N})$ whereas a classical computer does the same thing on the same database, that is, it takes proportional to N , in algorithmic terms, it is $O(N)$. Also, the quantum adiabatic computing approach allows deep learning network topologies to much more complex than what is feasible with traditional classical or chip integrated circuit-based computers.

In deep learning, we create models that have model parameters like learning rate, number of hidden layers, hidden nodes in each layer, kernel size, and so on. This is called the hyperparameter (model parameters). The complexity of hyperparameter scales exponentially with the size of the system. However, what level of complexity it may be, with the help of a quantum computer, it can be adjusted automatically. Imagine, in the deep neural network where a lot of parameter presents, the system would automatically become slow and less efficient, but when you apply QC on hyperparameter, its performance and efficiency will automatically increase. As we know, the Tensor network represents the weight of the neural network structures. The number of parameters needed to describe quantum neural network layers scales linearly with system size rather than exponentially as in the conventional tensor network. With the help of quantum computers along with quantum machine learning complex matrix and tensor, manipulations can be done with high speed.

There are multiple platforms available to support quantum machine learning from vendors like Microsoft and IBM. These tasks can be achieved with the help of them.

Few important QML algorithms

Models, simulations, and experiments have been at the vanguard of exploration in both technology and natural sciences for the reason of the big amount of data they harvest. Removal and dispensation of this big amount of data are likely owing to

developments on the way to add complex and powerful computational competencies. These comprise application of linear algebraic methods such as regression, principal component analysis, support vector machines, development of statistical frameworks such as Markov chains, and the appearance of new machine learning and deep learning methods such as artificial neural networks, CNN, and Boltzmann machines. A few quantum machine learning algorithms are shown in [Figure 7.7](#):

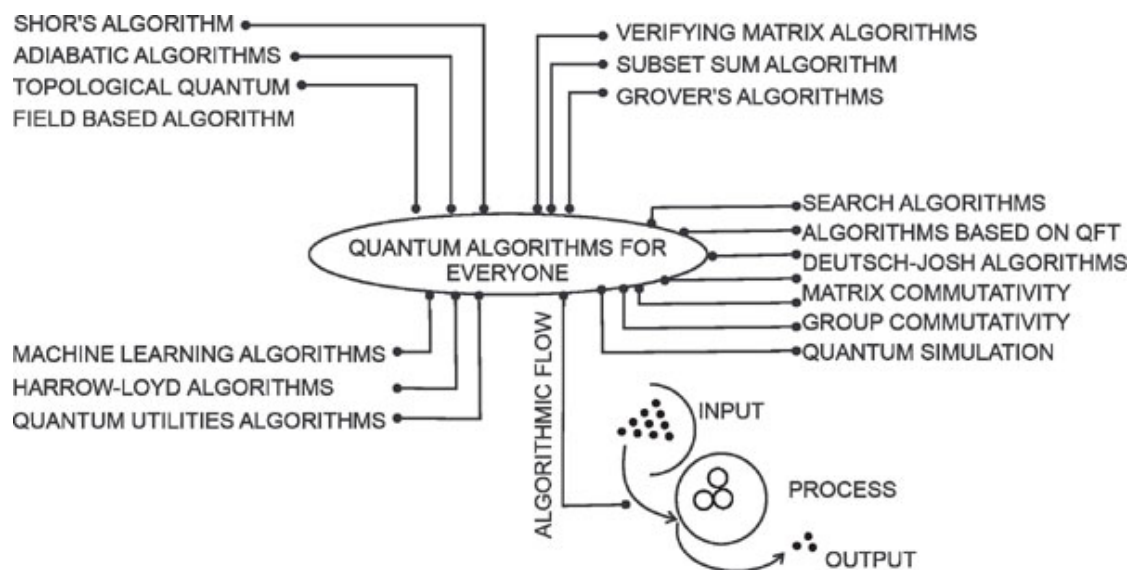


Figure 7.7: Important quantum machine learning algorithms

These methods have been long used in big datasets in both industries and academic research. A huge number of HPCs (with GPUs, TPUs, and so on) have used their billions of clock cycles to train deep learning networks on valuable datasets to classify multifaceted and understated patterns in data. These have caused the growth of computational programs useful in illness classification in scientific diagnosis, novel materials and amalgam discovery, path planning in autonomous robots, and so on. Though, training these networks to do somewhat useful tasks takes a lot of computational resources contingent on the difficulty of the job at hand. This is because classical/deep machine learning approaches either need algebraic routines (SVM, PCA, and so on), or identifying statistical patterns in data (deep neural networks), or frequently both. As we will be learning in the following section, this observation is central to an expectation that quantum computing could be a way onward in development, but before hopping there, let us first fleetingly re-enter quantum computers.

Using the present generation of quantum computers in machine learning, it must be known that their achievement depends on whether a well-organized quantum

algorithm (such as QBoost) can be found for this task. This impression is being discovered under a cooperative framework of QML.

The quantum algorithms like QSVM, QPCA, QBoost, Q-Means, are proved to be effective in the controlled environment. These algorithms were intended to speeding ML, and hence vital to the existence of both fault-tolerant quantum devices and QRAM to reach a quantum benefit. The following are the incentive and potentials of the algorithms. [Figure 7.8](#) shows the applications of a few algorithms like QSVM, QPCA, QBoost, Q-Means:

Classical Problems	Quantum Tools	Applications	Methods
BLAS Matrix inversion, inner products, eigenvalue decomposition, singular value decomposition	QBLAS/HHL Quantum phase estimation post selective amplitude update, Hamiltonian simulations	Support Vector Machine Gaussian processor Linear regression Principal component analysis	QSVM, QPCA, LES Regression Quantum Gaussian process
Search Finding the closest, neighbors, Markov chains	Grover Search Amplitude amplification, quantum walk	K-nearest neighbour, Page ranking Clustering Associative memory Perceptrons Active learning agent	Quantum K-means Clustering
Sampling Sampling from the model distribution	Quantum Sampling Quantum annealing Quantum rejection sampling	Boltzmann machines Bayesian nets	Bayesian Interface Quantum Boltzmann machines
Combinatorial Optimization QUBO problems (variables taking values 1 (TRUE) and 0 (FALSE))	Ground State optimization Adiabatic quantum computing Quantum annealing Quantum simulation	Associative memory Boosting Debugging EM algorithms	Bayesian Interface Perceptrons

Figure 7.8: Applications of a few algorithms like QSVM, QPCA, QBoost, Q-Means

A few important applications of QML

At the moment, mostly hybrid approaches are famous due to obvious reasons. These include training parameterized quantum circuit models for a diversity of supervised and unsupervised machine learning tasks on both classical and quantum data. Overall, a few instances of these tasks could be as follows:

- Identifying patterns to categorize classical data.
- Learning the likelihood distribution of the training data and producing novel data succeeding like a probability distribution.

- Discovery of short-depth hardware-efficient circuits for high-level algorithms.
- The solidity of the quantum states.

These tasks allow using quantum machine learning in a diversity of fields fluctuating from chemical simulations, combinatorial optimizations, finance, quantum many-body simulations, the energy sector, healthcare sector, and so on.

Algorithmic case study - Quantum Support Vector Machine (QSVM)

Fundamentally, Machine learning is a set of methods that are used to analyze and discover patterns in input data to brand forecasts. It has applications in zones such as image recognition, NPL, robotics, spam cleaning, drug finding, medical analysis, financial analysis, bioinformatics, marketing, and vote pattern.

A support vector machine is a supervised machine learning technique that is trained to utilize a dataset and will forecast if a specific observation is in a certain class founded upon what it has been trained on. Original SVM was familiarized by Vapnik in the 1990's. SVM is similar to a linear classifier in that it uses a hyperplane to distinct classes. Support vectors are generally used along with the hyperplane.

These are fundamentally data points that are used to maximize the border and therefore the remoteness among the classes. Support vector machine learning is one of the most used “readymade” supervised learning algorithms. SVM resolves glitches in both classification and regression. Its uses the principle, practices, and theories of something called maximum margin classifier to separate data. SVM would use a line for classification. The principle, practices, and hypothesis of maximum margin classification guarantee that there is a maximum parting among positive results ($y = 1$) and negative results ($y = -1$). The margin, in this case, is the remoteness between the decision border and the support vectors. However, as per the condition, support vectors are data points neighboring to the decision boundary.

One of the key compensations of support vector machines is that unlike other supervised learning algorithms, its forfeiture (loss) function is a global optimization problem. Therefore, it is not prone to something called local optima (comparative finest resolutions inside a neighbor resolution set). Additionally, SVM also shows very robust behavior against over-fitting, which is a modeling fault that happens once a function is too meticulously suitable to a restricted set of

data points. Hence it is appropriate for the construction of generalizations even with a minor dataset. To conclude, utilizing a technique branded as a kernel trick, SVM can disperse data which is not linearly detachable in its input space. This procedure empowers SVM to transmute input data into advanced dimensional space, where an unraveling linear hyperplane can be found.

In the context of the quantum world, the Quantum Support Vector Machine is using a feature map to map data opinions to a quantum circuit.

Story 1: Quantum imaging

Imaging is a significant technological tool in several disciplines, such as biomedical research, biomolecular technology, nanotechnology, and elementary physics investigation. Quantum imaging offers a practical improvement over classical imaging techniques; this opens the door to a renewed effort in constructing practical protocols that can be used in biomedical research, nanotechnology, and basic physics research. The impact in academia will consist of an increased understanding of aspects of quantum mechanical information processing other than imaging and in the long-term practical imaging devices. Quantum Imaging Technologies is the Powerful Sources of Entangled Photons. This makes Quantum Imaging technologies very important. Because due to this, one can take benefit from the quantum mechanical features of the light and of the essential and inherent parallelism of optical signals to create new procedures in the dispensation of information at the quantum level. A few popular quantum imaging protocols are as follows:

- Two-photon microscopy and spectroscopy
- Quantum holography
- Quantum lithography
- Quantum illumination

There are two fundamental aspects to comprehend the exact difference between traditional and quantum imaging. Number one is that it will help recognize new-fangled approaches for better imaging, possibly leading to innovative technology. Number two is that it will disclose an important feature of physics that has previously remained indefinable, specifically what makes quantum optics more influential in imaging than traditional optics, as follows:

- Quantum Imaging Systems Quantum
- Optical Coherence Tomography (QOCT)

- Quantum coincidence (or Ghost) imaging
- Quantum laser radar
- Quantum lithography

The driving forces for quantum imaging are fundamentally divided into four broader areas, as follows:

- Acquisition of knowledge.
- Public understanding and appreciation of nature.
- Physical and biomedical imaging applications.
- Nano-technology via the increasing miniaturization of devices.

Story 2: Quantum sensors a brief look

Quantum sensors use quantum states for measurements. They exploit the aspect that quantum states are fundamentally very sensitive to disorders. But at the same time, it creates an opportunity for possibilities to advance tremendously sensitive measuring devices. Measurement devices that feat quantum properties have been about for a while, such as atomic clocks, laser distance meters, and magnetic resonance imaging used for medical diagnosis. The term Quantum sensing tags the practice of a quantum system, quantum properties, or quantum process to do a measurement of a physical quantity. Historic instances of quantum sensors comprise magnetometers founded on superconducting quantum interference devices and atomic vapors. Nowadays, quantum sensing has developed as a separate entity and is quickly rising as a branch of investigation inside the area of quantum science and engineering. This development is happening along with the utmost common platforms like spin qubits, trapped ions, and flux qubits. The field is anticipated to deliver new prospects. The high probabilities of prospects are visualized particularly concerning high sensitivity and precision as it is applicable in practical physics and other areas of science.

Nowadays, a new set of applications has become popular that projects quantum mechanical systems as sensors for many physical quantities, which includes and ranges from magnetic and electric fields to time and frequency. It is also extended to rotations, to temperature and pressure-oriented applications. Quantum sensors further take advantage of the fundamental feebleness of quantum systems that is their robust sensitivity to exterior turbulences.

This tendency in quantum technology is inquisitively evocative of the history of semiconductors. The concepts of sensors are not new and have found marketable

applications decades before computers. For instance, light meters are based on selenium photocells. Although quantum sensing is established as a separate field of exploration in quantum science and engineering, it is well-known in the physics community and has been caused by decades of expansions in from top to bottom resolution of spectroscopy, particularly in atomic physics and magnetic resonance. Distinguished instances comprise atomic clocks, atomic gas magnetometers, and superconducting quantum interference devices. In terms of quantum computers, quantum sensors will consider mostly qubits which are fundamentally two-level quantum systems. Though an impression over real applications of qubits are mostly observed.

The possible influence of quantum sensors is comprehensive and substantial – from ultra-high-precision microscopy, positioning systems, clocks, gravitational, electrical, and magnetic field sensors, to optical resolution beyond the wavelength limit. Quantum sensing will certainly aid development and contributes to research in numerous fields of science. For instance, the gravitational-wave observatory Ligo is now applying cuddled quantum states in its following generation of sensors. This quantum improvement will permit for gravitational-wave observation at ten times longer reserves than before. Quantum sensors are also foretold to allow full mapping of the subversive, self-directed driving, medicinal progress, brain-machine interfaces, discovery of minor suggestions of explosives and venoms, and better-quality imaging technologies both at brief and extended remoteness.

Story 3: Quantum simulation

The simulating quantum process is recognized to be a hard-computational problem, particularly when dealing with huge systems. Though, this trouble may be overawed utilizing some controllable quantum system to study additional fewer controllable or accessible quantum systems.

Quantum simulation aptitudes have applications in the study of several problems, especially in the field of condensed-matter physics, high-energy physics, atomic physics, quantum chemistry and biology, and space science. Quantum simulation could be applied to utilize quantum computers, but also with meeker, analogue devices that would need less control, and so, would be easier to construct. A number of quantum systems such as neutral atoms (photons), nuclear spins, and ions have been planned and used as quantum simulators. Quantum simulation will deeply influence physics research. It will offer a novel tool for testing physical models or forecasting the behaviour of physical systems in numerous likely

circumstances, in addition to permitting admittance to novel physical regimes that are presently outside our reach. Even in the nonappearance of a theory to be verified or realistically interested conditions under which the behaviour of a physical system is to be foretold; quantum simulators will also bid experts a novel kingdom of exploration. It might very well occur those unexpected detections are complete through curiosity-driven research with upcoming quantum simulators. Quantum simulation will deliver a valued tool that investigators from many fields will wish to complement their toolbox.

Use case 1: LightOn's optical processing unit

LightOn's technology uses light to do some multiplications of interest to ML and AI. Their analog calculation devices accurately produce natural physical processes at unparalleled rapidity, magnitude, and power competence. Their product, a hardware co-processor named the Optical Processing Unit or OPU, recently came to the market. It is intended to achieve improvement in some of the maximum compute-intensive tasks in ML. It is very easy to use and needs to just be plugged onto a normal server or workstation, and retrieved through a humble toolbox that is flawlessly combined within acquainted programming environments. Full-scale OPU examples are previously obtainable to designated users through the LightOn Cloud. It could also be effective in providing better results for the following areas:

- Image and video classification
- Recommender systems
- Anomaly detection
- Natural language processing

The LightOn Appliance can be flawlessly combined with the on-premises datacentre. Further, as per the claim, the LightOn Appliance is the best method for ML and data engineers to appreciate at negligible latency and the power ingesting of LightOn OPU. The LightOn Appliance will be obtainable underneath a leasing package that will comprise a 2U rack-mounted Aurora 4.5 OPU. This OPU is connected to the server through PCIe, training, maintenance, support, as well as Machine Learning referring and software information.

Use case 2: Quantum control

In traditional engineering, working to stabilize an unstable system against disturbance from its environment is frequently the domain of control engineering. Control engineering is fundamental to nearly all of our technological industries. This is fundamental to most of the major industries including aviation, heavy machinery, and automobile.

Now with the changing times, quantum control will become essential in the development of the developing quantum computing industry. One of the examples of the advantage of expertise in control engineering is Honeywell. Honeywell is taking lead in the quantum computing space because they already have the infrastructure ready for control engineering.

Predisposition to sound and error ruins the Achilles heel of quantum computers and steadying hard-ware in contradiction of noise leftovers, the main driver of study in the field. Understanding applied, functional machines needs sustained expansion of techniques to increase the error robustness of quantum computing hardware. Providentially, that's exactly where quantum control originates.

Fundamentally the field of quantum control emphasizes on how to operate quantum devices effectually, and in a way, that's vigorous against decoherence and error. Its responses to the central challenge of quantum technology, while also knowing the challenge of hardware constancy, is not exclusive to quantum technology.

Suitable incorporation of quantum control solutions at the hardware layer can alleviate the harmful belongings of environmental alarms and recover the logical error rate for quantum hardware. Since these methods usually include direct actuation on the devices used in executing computations, they cooperate with the physical layer in a general quantum computing software stack, preceding any form of rational encoding used in fault-tolerant quantum error correction.

This class of protocols is roughly referred to as firmware as they are typically software-defined but entrenched at the physical layer and imperceptible to advanced layers of abstraction. Significantly, these methods allow similarity to other forms of firmware in computer engineering, such as DRAM revive working to alleviate traditional memory hardware against charge leakage. Our approach leverages best practice in the computer engineering community to help quicken the path to the first suitable quantum computers.

Use case 3: Chatbot and virtual assistants

Chatbots frequently bomb to uphold context over the progression of a conversation and cannot comprehend client intent in several circumstances. Whereas Virtual assistants, service lively dialog that is, the flow can switch between being meticulous by the user or the technology. Businesses employ bots to reduce their cost and protect margins. If bots are put to work, businesses can save a lot of labor costs. Chatbots can allow organizations to work more efficiently by minimizing repetitive work for agents. If they are used properly, they can reduce wait times by automated commonly asked queries. The process of the chatbot is shown in [Figure 7.9](#):

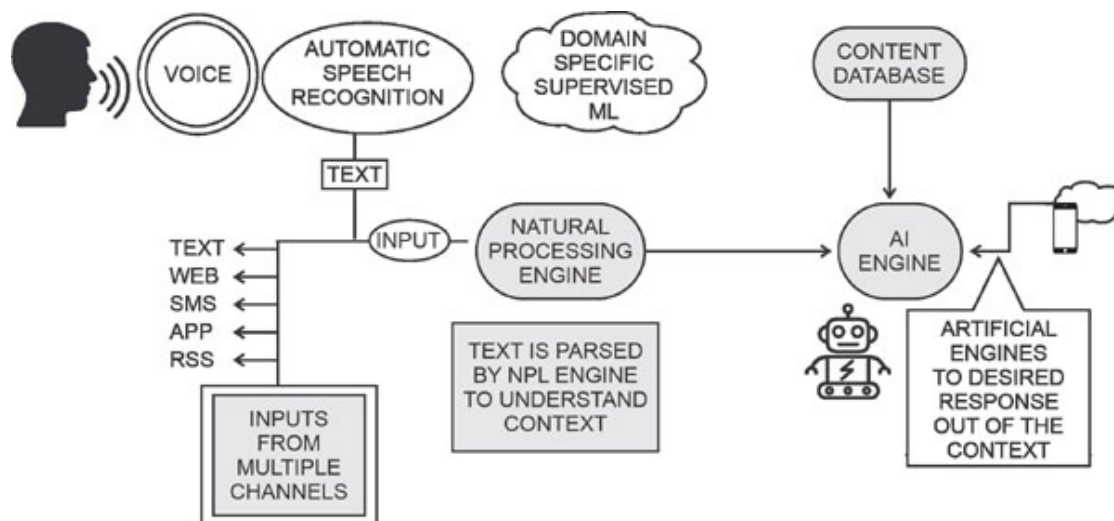


Figure 7.9: The process of chatbot

They can even reduce the average handle time by suggesting responses while the agent is speaking with the customer, leading to a faster resolution. With Alexa, Siri, Cortana, Google Now, and S Voice, tech companies have thrown their hat in the ring and shown what they believe is the user interface of the future. A virtual assistant is working as a virtual person and provides all the assistance to the individuals as a real individual would provide. In future for their data processing and intelligence, need chatbots will depend on quantum systems.

Well now, let's see some of the differences between chatbots and virtual assistance, as shown in [Figure 7.10](#);

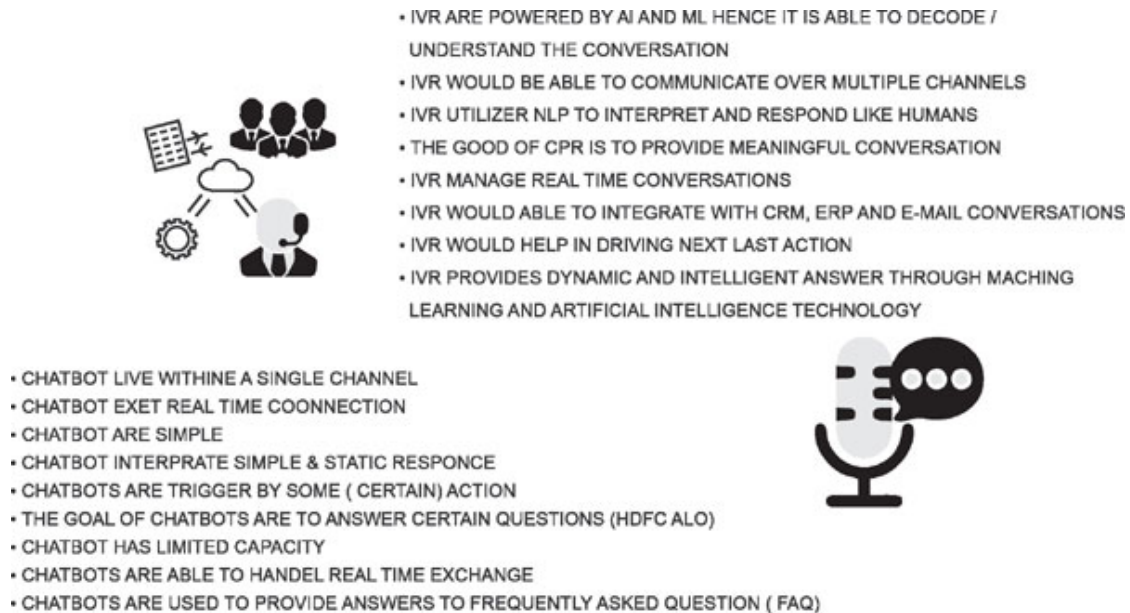


Figure 7.10: Differences between chatbots and virtual assistance

Use case 4: Robotics in the context of AI and ML

Though everybody is still processing the fact that robots with the addition of AI can be intelligent, quantum computing has also loomed over robotics to teach some new shades. As we arrive at a novel era of computing, the standard shift will also influence the work of robotic scientists and the applications of robotics. Quantum computers will enable the formation of even more influential and brainy robots that make use of quantum computing cloud services or co-processors. Its application area is huge. For instance, contemporary autonomous robots essentially need reckless vision competencies to observe and measure their environment. Computer vision and image recognition algorithms are computationally luxurious as they have to calculate outcomes on millions of pixels. So, the hopes lie with the quantum competence to improved understanding of the nature of graphic information, and save, accumulate, and process them possibly with the aid of quantum properties entanglement and parallelism.

Such efforts lead to an area of a study named **Quantum Image Processing (QIMP)**. The elementary impression is that properties of an image, like the colours at certain locations, can be prearranged as qubit-lattices, which were extensively acknowledged and officially protracted by numerous depictions and likely applications, together with videos.

There are multiple branches of robotics along with ML and AI that have emerged; almost all the areas will contain the implication of robotics. Branches,

applications, and areas of study of robotics are shown in [Figure 7.11](#):

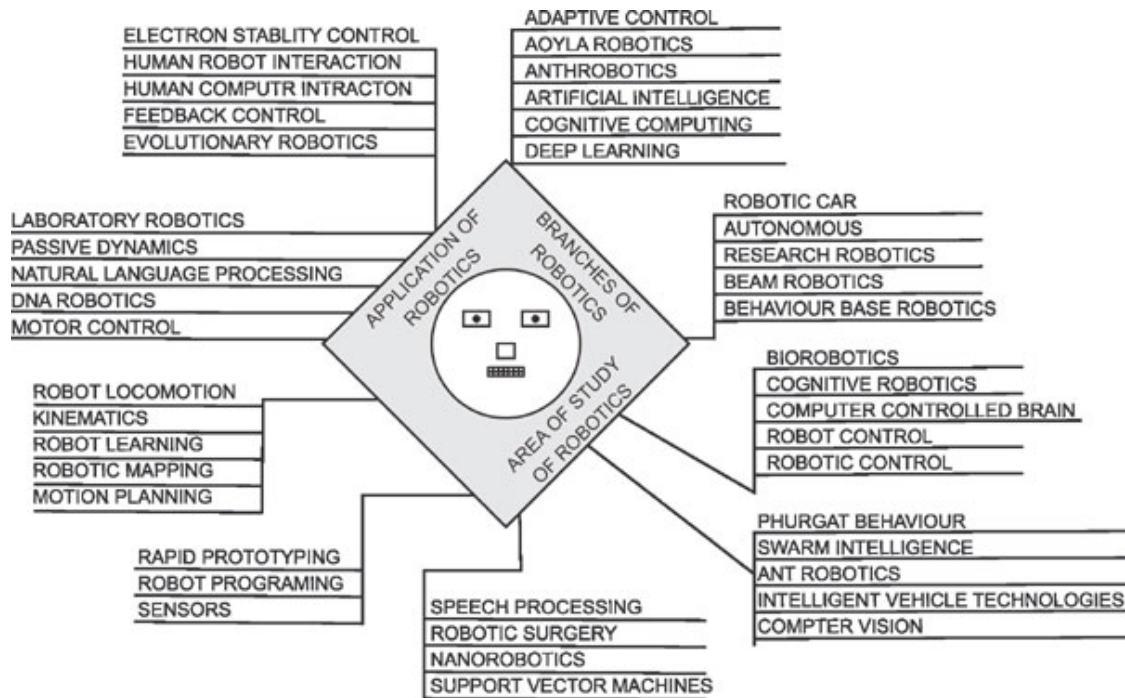


Figure 7.11: Branches, application, and area of study for robotics

Use case 5: 5G -- Present and future applications

5G is the futuristic wireless network. This next phase of wireless growth potentials up to 100X the data and speed likened to 4G, which we are currently using today. Just headfirst into the elementary research on this market is nearly irresistible. The forthcoming influence of 5G is inconceivable. There's just so much going on, moving nearly every façade of our lives.

Beginning from self-driving cars to the online drug, Smart Cities, totally linked drone navies, hyper communications, investment markets, building probabilities, armed applications, homeland security, serious medical connections with hospitals & surgeons, emergency reply services, water supply, food delivery channels. Almost in all the fields, a tidal wave of ground-breaking novel goods and services all determined by real-time big data, machine learning, and AI, everything is going to be affected by this. That's what 5G means. In other words, 5G bursts the current data channel by one hundred times. It is like substituting a causeway with a data superhighway. The following are some of the statistics. [Figure 7.12](#) highlights the present and future applications:

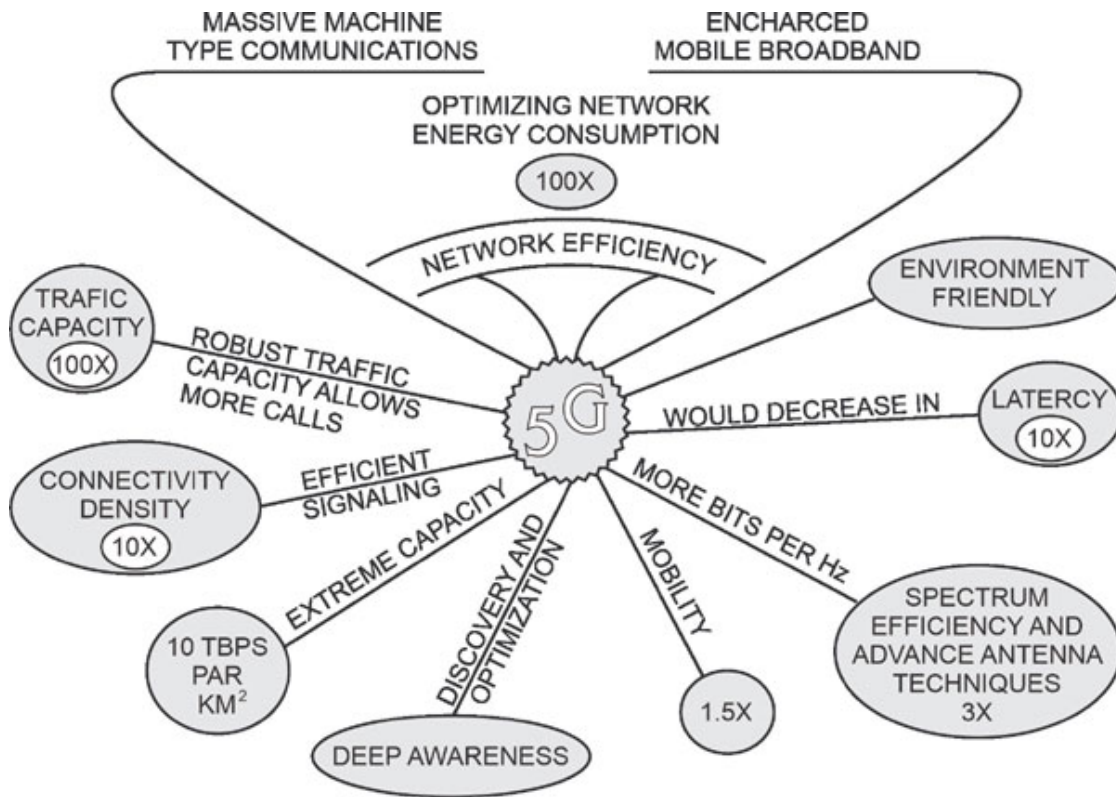
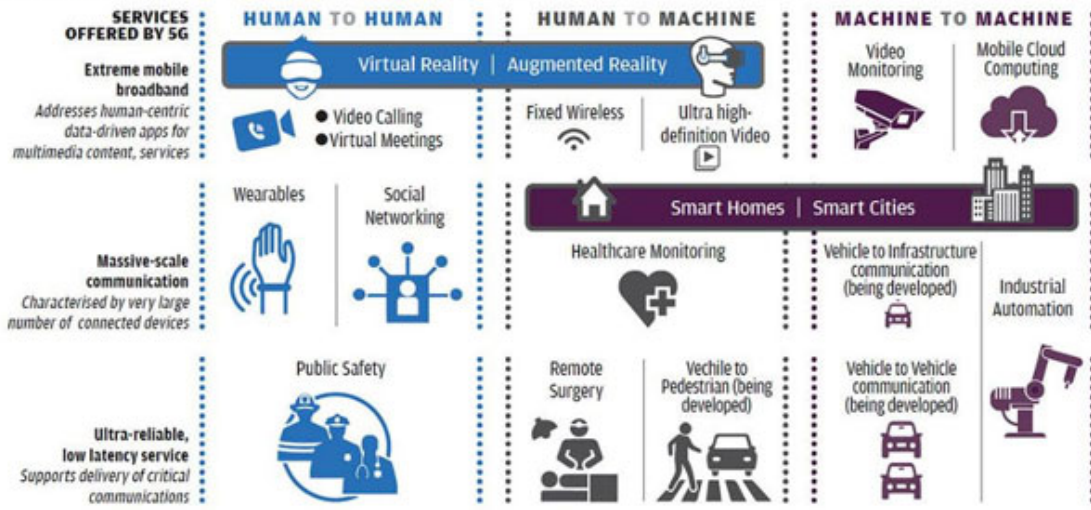


Figure 7.12: Present and future applications of 5G

Fundamentally, 5G will permit a matrix of technologies that will let us attach to everything and anything in real-time. And it will do so in a new way that we cannot yet foreknow. Further, imagine if you associate it with quantum computers, then how would the data, communication, collaboration, and technology dynamics change in the coming time?

This will provide new avenues and create opportunities for innovation and investment. Further, it will crown the prospect for the predictable and sometime unpredictable future. Savvy investors stand to make fortunes, this would provide tremendous growth opportunities. Particularly to those who are early adopters. Areas that 5G would affect and bring change are sketched in [Figure 7.13](#):

Telecom networks are set for a massive overhaul, which will make interaction with machines more immediate and real



Sources: Enabling 5G in India, White Paper by the Telecom Regulatory Authority of India

Figure 7. 13: areas affected by 5 G technology (Source: google image)

In a nutshell, the coming of 5G has long been anticipated by a variety of businesses. The absence of ultrafast data transport has slowed the development of artificial intelligence (AI)-based technologies and the Internet of Things (which includes smart wearables and virtual reality headsets). One illustration is self-driving cars. Automakers have already created autonomous smart vehicles. Only when they can communicate with other vehicles and traffic lights will they be useful. This demands extremely quick data transport with little lag (a latency as low as 1 millisecond). All of these low-latency applications, from industrial robots to remote surgery, are supported by the way 5G is being designed. For details, please refer [Figure 7.13](#)

Mind Map

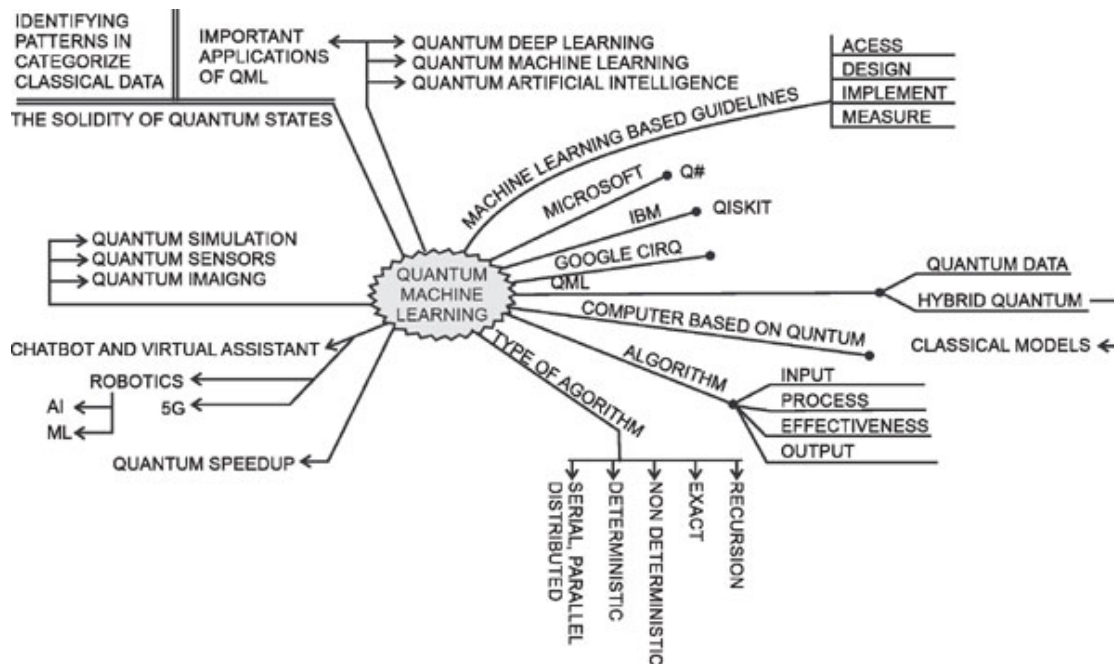


Figure 7.14: Mind map for the [chapter 7](#)

Conclusion

Quantum mechanics is strange. Quantum systems can produce strange patterns that are very tough to produce classically. However, quantum systems can also learn and identify bizarre patterns that can't be recognized classically. And the reason that they can do this is that quantum computing and quantum mechanics is using linear algebra to depict. With a very minor number of quantum bits, which have states that represent vectors in a very high-dimensional vector space, the quantum computers can be used to do linear algebra operations to simplify the patterns with Fast Fourier Transforms, Finding Eigenvectors, and Eigenvalues, and Inverting Matrices. And it does in an exponentially faster way than we can do classically.

In the next chapter, we will discuss about quantum cryptography, which is one of the most important topics of the quantum world.

Important terms and concepts used in the chapter

- **Serverless:** According to the Serverless computing concept, you can develop, build and run applications and services without tacking headache of infrastructure management to run the code for example configuration of web and application server, memory management capacity planning, and so

on. However, in a serverless environment (cloud infrastructure) as well, applications still run on servers, but all server management is done by the vendors or cloud service providers. Amazon is a pioneer of serverless architecture. Amazon web service (AWS) implemented serverless computing through AWS Lambda. AWS Lambda runs a user's code without provisioning or managing servers by them. Lambda provides you the facility to virtually run any type of application or backend service and it takes care of everything required to run and scale (up and down) your code with high availability.

Advantage of serverless

- **No Servers to Manage:** No infrastructure management headache for the user; they have to concentrate on just on “functionality”. It is one of the reasons that the serverless paradigm is also known as function as a service
- **Automatic Scaling:** Based on need application scaling would be done based on application trigger (for example, event)
- **Pay per user pricing:** If you use services for 10 seconds of the vendors. Pay for 10 seconds.
- **Microservices:** Tire-based architecture tells you to divide your code into multiple tires so that you have clear segregation of code. Hence its execution and management are easy. For example, in three tire architecture, you divide your code into three tires, presentation, business logic, and data, and achieve cleanness and separation. However, the fact is different; even after dividing your code into different tires, they are dependent on each other for passing, processing, consolidating, and presenting holistic information.

Microservices provide you a mechanism to cater to this problem, it says that you divide your application based on functionality and develop and deploy them in one self-sufficient module so that they would work independently. These independent services must not be dependent on other services. For instance, if the project has a customer, invoice, CRM, SCM, and user interface modules, you must break each service as an independently deployable module. Each module would contain its maintenance, monitoring, application servers, and database. Therefore, with microservices, there is no concept of the centralized database; as each module is self-sufficient, so, they contain their database.

- DevOps: It is a mindset that thinks in terms of a set of patterns and practices that syndicates software development (Dev) and IT operations (Ops). It aims to condense the SDLC and deliver continuous delivery with high software quality. DevOps is balancing with Agile software development. Some DevOps features came from Agile methodology.
- Operational analytics: It is the procedure of using data analysis and business intelligence to advance effectiveness and rationalize normal operations in real-time. A subsection of business analytics, operational analytics is reinforced by data mining, AI, deep learning, and ML.

CHAPTER 8

Quantum Cryptography: The Future of Security

We must be clear that when it comes to atoms, language can be used only as in poetry.

— Niels Bohr

The first quantum revolution gave us new rules that govern physical reality. The second quantum revolution will take these rules and use them to develop new technologies and offer the next level of opportunities.

— Ian R. McAndrew, PhD

There is no hesitation in saying that quantum computers, cryptography, and the quantum internet will have a prodigious influence on our world. But we don't yet recognize fairly how. But we definitely will understand the effects in the decades to come. This chapter will provide you with a basic understanding of quantum cryptography, quantum internet, and protocols, which include the protocols like BB84. The reader will be able to foresee the potential influence of quantum computing and the quantum internet on business.

Quantum mechanics is strange. Quantum systems can produce strange patterns that are very tough to produce classically. However, quantum systems can also learn and identify bizarre patterns that can't be recognized classically. And the reason that they can do this is that quantum computing and quantum mechanics is using linear algebra to depict. Machine learning is known for finding patterns. In the previous chapter, on Quantum machine learning, this is what we covered. Cryptography is very important in the present world. Because day in and out, hackers are trying to bypass personal and corporate information. In order to find a defence mechanism to counter this, quantum cryptography is born because conventional methods have

failed to handle the threat. This chapter makes sure that the user would get an idea of Quantum cryptography.

The arrival of quantum processors that have the competence to break our present cryptographic primitives is an actual threat on the skyline. Accenture, one of the leading consulting firms, believes that quantum processor capability will be able to compromise existing cryptography within the next eight years. Businesses have a complex task ahead to detect, evaluate, and prioritize the business remediation to defend their data from the cryptanalysis cracks and compromise. Much like the call to action from dealing with legacy systems, the changes needed are deep in the fabric of our infrastructure. Replacing cryptographic methods across our business processes is a complex effort that will require a concerted campaign of technology and business transformation. Companies need to start preparing now for the transformation ahead, ensuring that they are resilient to the threats of quantum processing before this tipping point.

This chapter is meant for a comprehensive and varied spectator including policy-makers, people with a scientific or personal interest, business executives, and students at all levels. A few of the important headlines of the chapter are described in the following section.

Structure

In this chapter, the following topics will be covered:

- Introduction
- Need, shift, and desire of cryptography
- The quantum computing threat to cryptography
- The thesis of quantum cryptography
- Classical cryptography
- Secret key cryptography
- Public key cryptography
- Limitations of classical cryptography techniques
- Problem with cryptography
- Why quantum cryptography is important

- Quantum cryptography
- Quantum cryptography versus traditional cryptography
- Photon polarization and quantum cryptographic
- Cryptography and polarization
- Measurement, cryptography, and polarization
- Quantum cryptography Procedure
- Quantum key distribution
- Lattice-based cryptography
- Hash-based cryptography
- Fully distrustful protocol
- Distrustful protocol
- Bit commitment protocol
- Some important protocols
 - BB84 protocol
 - BB92 protocol
- Cryptography attack
 - Brute Force Attack (BFA)
 - Known Plain Text Attack (KPA)
 - Replay Attack (RPA)
 - Man-in-the-middle Attack (MITM)
 - Denial of Service Attack (DoS)
- Quantum internet
- How to create a quantum network
- Quantum internet consists of three essential quantum hardware elements
- Quantum communications and Internet
- The Quantum Xchange Trusted Node and QKD handling
- Teleportation
- Blockchain

- Blockchain technology: How does it work?
- Blockchain: the perfect partner for AI data
- Meeting the challenge--capturing the opportunity
- Quantum Blockchain
- How likely is a quantum attack on bitcoin?
- Quantum networks and classical digital networks
- Future-ready: what is after quantum?
- Story1: Quantum computing-based recommendation systems
- Story 2: Disease spread, bacterial growth, and QC
- Story 3: How quantum computers will transform travel
- Story 4: Quantum computer and personalized customer experience
- Use case 1: Quantum prediction
- Use case 2: Caixa Bank journey to grow risk classification model through quantum computing
- Use case 3: Quantum computing and the automobile industry
- Latest research: Path to quantum computing at room temperature

Objective

After studying this chapter, you will be able to summarize the current state of quantum cryptography and highlight the actual application implementation of quantum cryptography which is useful in the real-world. As this chapter would give the theory of quantum cryptography, it will explain in what way this technology plays a key role in security. It will also take some well-known examples of quantum cryptography under consideration and provide insights on how the use of quantum communication will happen to securely exchange a key (quantum key distribution) and the (hypothetical) use of quantum computers that would allow the breaking of various prevalent public-key encryption and signature schemes like Rivest–Shamir–Adleman is a public-key cryptosystem that is widely used for secure data transmission. Finally, it will throw some light on the forthcoming time and provide directions in which quantum cryptography is moving.

After completing this chapter, you would understand the field of cryptography, and be able to answer a few very important questions like the following:

- Is my communication private?
- Has my communication been altered?
- Who am I dealing with?
- Who sends and receives messages when it is broadcasted and when?

Hence, this chapter will try to answer them as well. These may be termed as the objective of cryptography. Either it is conventional or quantum cryptography. Therefore, the chapter will broadly try to fulfil these objectives by various means which include tools, methods, theories, models, and so on. So let's begin our quest is to figure them out.

Introduction

The two chief techniques that cryptanalysis attackers use to break these algorithms are reverse engineering of the mathematical operations performed in the algorithm, or brute-force predicting of the secret key/s. The first technique is characteristically the outcome of human error on the software development side; when producing encryption and decryption programs, developers may unintentionally make an error in the execution of the mathematical operations, opening a door for attackers to evade the cryptographic approaches through reverse engineering. The second method of brute-forcing, a properly applied algorithm with an adequately complex key, is often unreasonable for attackers, even when equipped with hardware accelerators such as **Graphical Processing Units (GPUs)**, **Field Programmable Gate Arrays (FPGA)**, and **Application-Specific Integrated Circuits (ASICs)**. Above all, brute-forcing by examination of all possible key values could take eras of computing time. The idea of cryptography can be summarized in the following six footings:

- **Encryption:** Alteration of data into a pattern, named ciphertext, making it incomprehensible.
- **Decryption:** Procedure of changing ciphertext data back into its original form so it can be recited.

- **Algorithm:** Formulation used to transmute the plaintext into ciphertext; also named a cipher.
- **Key:** Complex order of alphanumeric characters shaped by the algorithm that permits data encryption and decryption.
- **Plaintext:** Decrypted or unencrypted data.
- **Ciphertext:** Data that has been encoded.

Need, shift, and desire for cryptography

Brute force occurrences continue to be the mainstay of supercomputers. Quantum computation takes a completely novel approach to cryptographic methods by transforming the number theory problems into problems a quantum computer can resolve with an importantly condensed computational effort. The efficiently whirling problems that would take aeons of computing time classically, are possible with QC in much more practicable compute times like days or weeks. Lately, investigators have shown that quantum computing is accomplished by breaking the strong cryptographic primitives, such as the Diffie-Hellman key exchange.

Classic cryptography trusts the strength of the secret key to deliver cryptographic security. Certain number theory problems like the Prime Factorization problem, the Discrete Log problem, and the Elliptic Curve methods fundamentally reinforce present cryptographic schemes. These were selected because it is computationally easy to syndicate two large numbers together. However, it is computationally problematic to go the conflicting way and control what the original numbers were if only one is given or present.

For example, it is easy to compute $148 \times 179 = 26492$, but it is tougher to find out which two numbers when multiplied together to give 26492. Presently, there are no known mathematical shortcuts to these algorithms, which could in a sense provide every solitary likely amalgamation (or brute force) that must be tested to find the key number that will unlock the algorithm. However, with QC, along with an algorithm in progress, cracking these types of problems are possible. The objective of cryptography is depicted in [Figure 8.1](#):

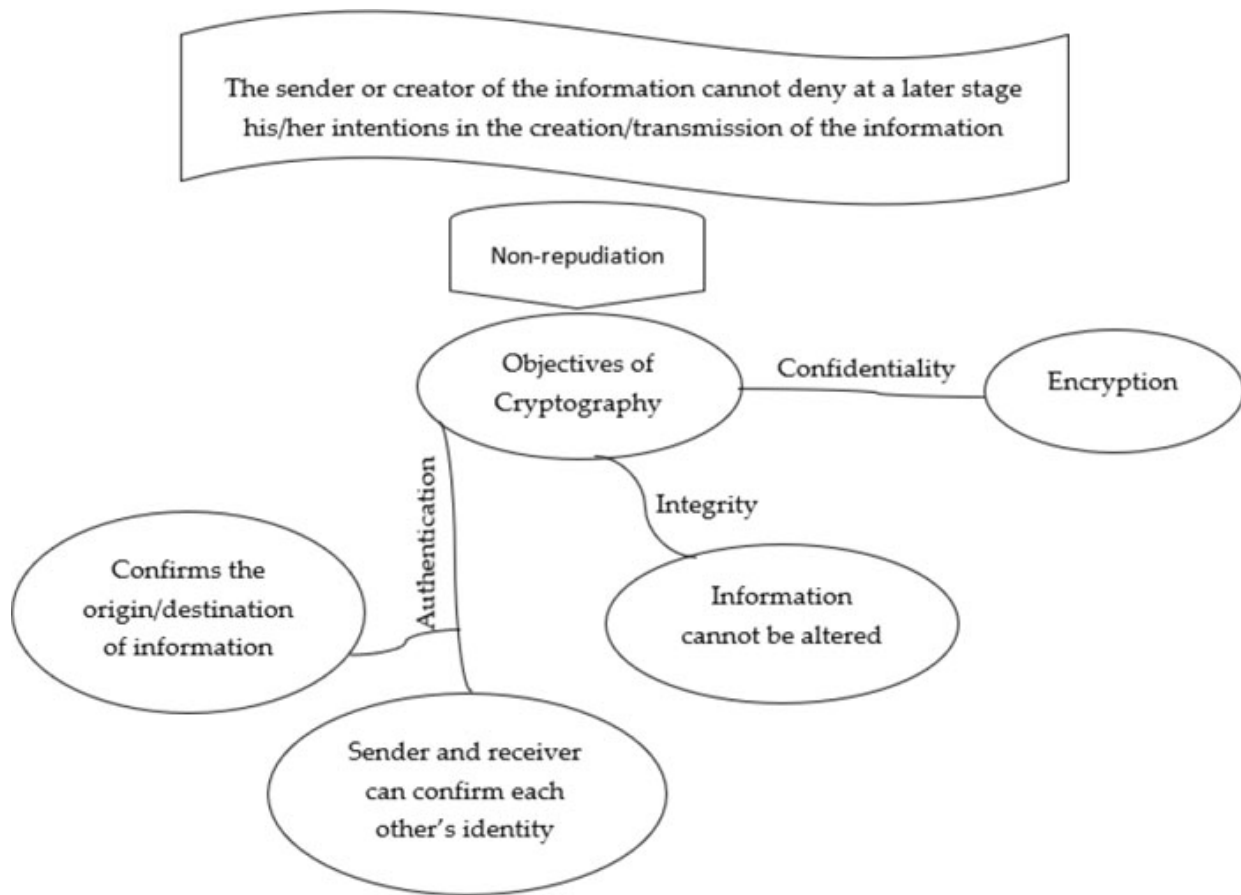


Figure 8.1: Objective of cryptography

Further, in the world of cryptography, the following parameters are very important, especially in the context of quantum cryptography:

- Integrity
- Encryption
- Authentication
- Non-repudiation

Utilizing present classical computation, and even with hardware accelerators, could take hundreds of years to resolve some cryptographic, mathematical, physical or any other type of problems. In contrast, one of the universal classes of problems that quantum computers resolve best is phase approximation, which can be defined as classifying where two different frequencies overlay. Both the Prime Factorization problem and the Discrete Log problem can be transformed into phase estimation problems. In 1994,

Peter Shor displayed that quantum computers could use quantum physics physiognomies to professionally resolve these problems without trusting on brute force, thus interpreting any information encrypted by our public key system susceptible to decryption.

The quantum computing threat to cryptography

As quantum computational technology rises, it will progressively endanger the security and strength of the public key cryptographic paradigm. A quantum computer could be used competently for the brute-force methods for each of the three cryptographic principles (confidentiality, integrity, and authentication), next to breaking much of the security systems used today.

This would endanger any data or messages that are presently endangered by these cryptographic schemes, permitting anyone with an adequately advanced quantum computer to read any information or communications encrypted by the schemes. It could also depict any encrypted historical data composed by third parties, and even allow threat actors to impersonate authenticated users, legitimate organizations, and web applications.

Though quantum cryptanalysis occurrences are still practical, NIST and other entities have been monitoring progressions in quantum computing and working to advance the present cryptography standards by recognizing novel quantum-safe algorithms. They define the implementation approaches and gain broad agreement on the approach so that it will be adopted. Quantum computing research has advanced to the point where a functional quantum computer accomplished of breaking current-grade cryptography will arrive within the next few years. Therefore, to uphold secure communications and encryption, Global 2000 corporations in all industries must start now to revive the technologies behind cryptography. This is the only way to formulate a future full of QC that can destroy classical or traditional cryptography.

Quantum cryptography is developing the know-how in which two parties can secure network communications by smearing the marvels of quantum physics. The security of these spreads is based on the fundamental laws and rules of quantum mechanics, theories, and engineering. Quantum cryptography relies on two deadly significant constructs of quantum mechanics and engineering:

- The Heisenberg Uncertainty principle and practices along with the dynamics of photon polarization. The Heisenberg Uncertainty principle says that it is not likely to measure the quantum state of any system without disturbing that system.

The Heisenberg uncertainty principle, which fundamentally highlights that particular couples of physical properties are connected in such a way that determining one property stops the observer from concurrently being aware of the value of the other. This rule was put forward in 1927 by German physicist Werner Heisenberg. At the macroscopic stages, it is not evident.

We know from quantum physics, that a shifting photon has one of the positionings mentioned: vertical, horizontal, or diagonal in conflicting directions. Quantum cryptographic devices produce photons one at a time, and each photon has a specific orientation. Photon sniffers are capable of recording the orientation of each photon, nonetheless, be in harmony with Heisenberg's uncertainty principle; ensuring so will alter the orientation of various particles which in future will caution both the sender and the recipient that their network is being checked or monitored.

Heisenberg's uncertainty antecedent is of enormous advantage to information security in that, if quantum cryptography is used to direct keys via photons, then faultless encryption is certain. If it is found that the keys have been observed and are consequently in jeopardy, then it is a modest matter to generate a novel set of keys. In an old-style key exchange, it is not likely to know if a key has been inhibited with a similar degree of inevitability as with quantum cryptography. Numerous quantum cryptography advocates and vendors openly state that quantum cryptography delivers complete security.

The principle of photon polarization states that an observer cannot copy unidentified qubits that is, unfamiliar quantum states, due to the no-cloning theorem (it is not possible to produce an indistinguishable duplicate or copy of a random unidentified quantum state).

The details of quantum cryptography with well-known Alice and Bob example are shown in are shown in [Figure 8.2a](#) and [Figure 8.2b](#):

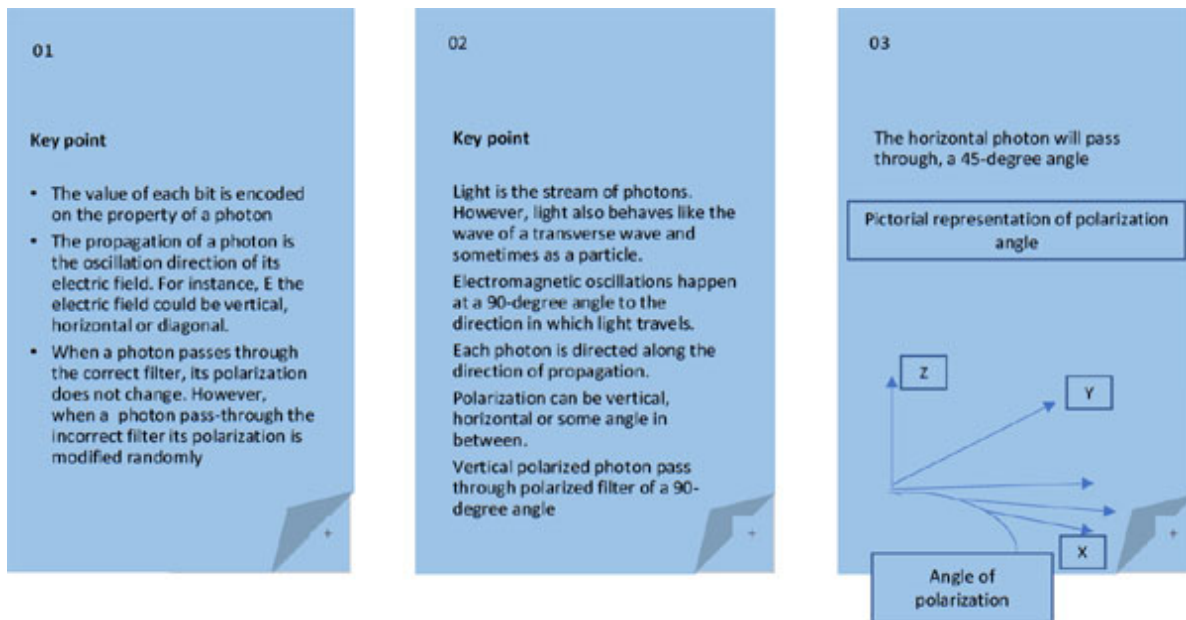


Figure 8.2a: Quantum cryptography process – theoretical foundation

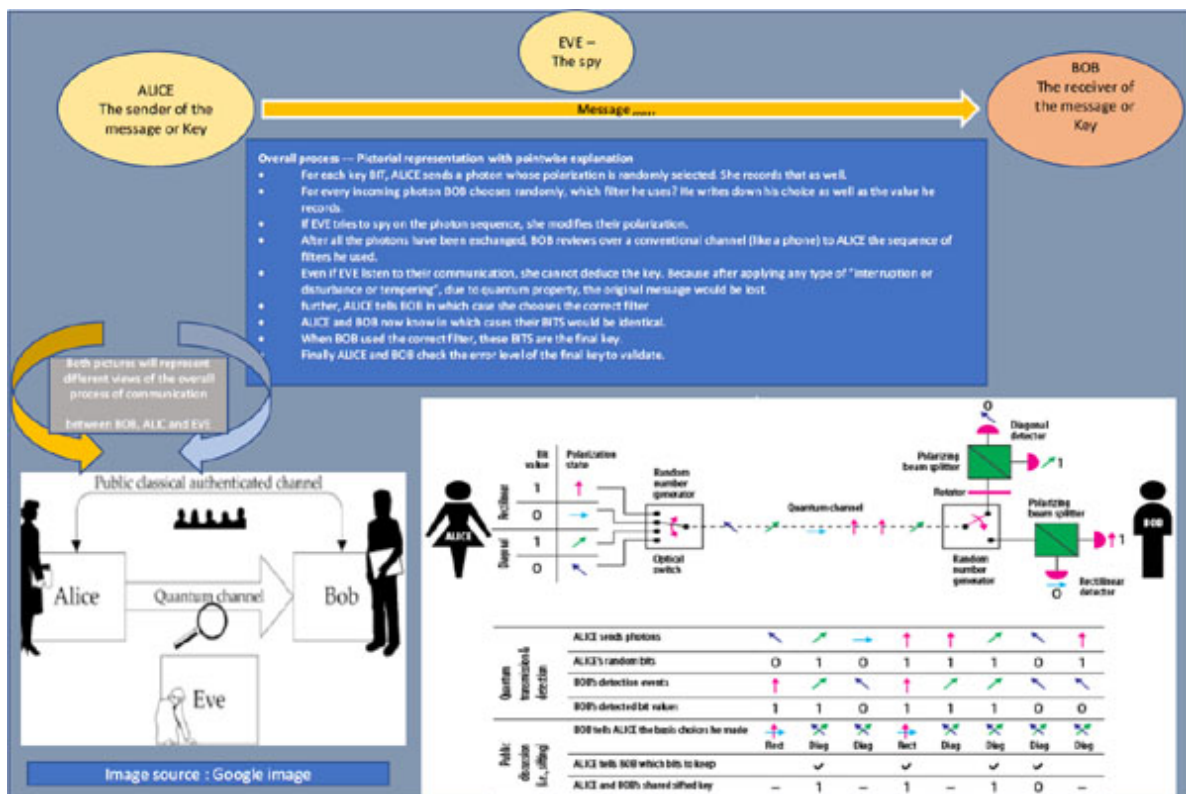


Figure 8.2b: Quantum cryptography process – Explanation of cryptography process

The thesis of quantum cryptography

Quantum physics permits bits to be in a superposition of 0 and 1, the same way that a Schrödinger's cat can exist in a superposition of alive and dead. This occasionally lets quantum computers discover likelihoods further rapidly than traditional computers. For instance, in a general search problem, such as trying to discover the key to a secret code by experimenting with all of them, quantum computers are predictable to have quadratic speed-up.

There is a reason for this, because of the expectations set by different experiments and research happening across the world. For instance, the Advanced Encryption Standard, approved by the United States government, has gone up to 2256 or about a 1 trailed by 77 zeros keys on classical computing for a complicated search. Whereas for similar search problems, a quantum computer could brand that similar search only with 2128 keys, about a 3 followed by 38 zeros; that's a lot faster in comparison with traditional ones. But there is still a dreadful lot of searching to do.

There is a different type of cryptography, called public-key cryptography, which was conceived in the 1970's. As the name suggests, these are systems where two people can settle on a key, or portion of a key, by swapping over only public information. This is extremely advantageous in modern electronic commerce. For instance, if anyone who requires to transmit his or her credit card number securely over the Internet to Flipkart or any other eCommerce site.

They don't want to have to physically go to their main centre to have a secret meeting to discuss how to transfer the number first. Public-key systems trust the reality that certain mathematical procedures appear to be one-way and therefore, they are simple to accomplish but problematic to unfasten or undo. For example, multiplying two huge whole numbers is comparatively effortless but is the reverse also true? As we discussed earlier, captivating the outcome and factoring it into the original numbers typically seems a much tough or harder endeavour. This specific one-way job is used in RSA cryptography, one of the most prevalent public-key systems.

Tactlessly for RSA, not all one-way purposes are shaped equally. The factoring problem cascades into a group identified as hidden subgroup problems. A group is a certain kind of mathematical structure and a hidden subgroup is an additional structure inside it, unidentified to the codebreaker. In the factoring example, the product harvests the group and the unidentified

factors crop the unseen subgroup. On unseen subgroup problems, quantum computers are predicted to get exponential speed-up.

Typically, factoring is a quicker and faster way than searching. Hence it is natural that an ordinary computer could factor several sizes 215360 in the time it has to search 2256 keys. Instead of a contemporary, a quantum computer could factor (as we know from the previous read) that same number in more like the time it takes to search 20,000 keys. That's a huge speed-up. It has the capabilities to destroy RSA. And, the is applied to the other public-key systems currently in common use.

Quantum cryptography is an alternate security solution for a computer network. In its place of using all-purpose encryption and decryption methods, quantum cryptography can confirm that the key is conveyed deprived of interruption from an eavesdropper. In the case that the key is interrupted, both sender and receiver, simply dewdrop the key and re-send the novel key.

BB84 is the protocol that familiarizes the way to transmit the key with the quantum method. The BB84 protocol fundamentally needs multiple polarization states (if we want to be precise: four in numbers). These polarization states are termed as horizontal $|h\rangle$ polarization state, its vertical counterpart called vertical $|v\rangle$, left circle polarized $|lcp\rangle$, and its counterpart called right circle polarized $|rcp\rangle$.

Whereas in the BB92 protocol, the two non-orthogonal quantum state exists, where $|h\rangle$ represents a 0 and $|rcp\rangle$ represent a 1 in BB92. Also, the process of setting up the shared key for both sender and receiver, BB92 is simplified with a more effective transmission.

This quantum cryptography can be applied to the application where the need to communicate in an ultra-safe way exists (like transmission of banking information). And therefore, a need for superior security is the channel required. In the future, this protocol is also ready for the impending high-performance computer, quantum computer, photonic or quantum-hybrid computers.

[Understanding cryptography](#)

Cryptography is the fine art and science of encoding and decoding messages. It is existing and will exist because people have been suspicious of each other and therefore required a form of secure communication. Cryptography determines to convey information such that only the envisioned receiver obtains it. Though the arena of cryptography is old, it is not static. Cryptographic methods have grown over some time, and this was not the small timeframe but in the ranges of centuries. It can be linked back to the desire of the code-makers to stay ahead of the code-breakers. The next main stage in this evolutionary process may be at hand.

Cryptography is where security engineering encounters mathematics. It delivers us with the tools that cause utmost contemporary security protocols. It is perhaps the key empowering technology for shielding distributed systems, yet it is astonishingly tough to do so correctly. Cryptography has frequently been used to defend the incorrect things or used to guard them in the incorrect method.

The elementary vocabulary is that cryptography mentions the science and art of scheming cyphers. Cryptanalysis is the science and art of infringement them, while cryptology, frequently condensed to just crypto, is the study of both. The input to an encryption procedure has usually named the plaintext and the output of the ciphertext. Afterwards, things get somewhat more complex. There are numerous cryptographic primitives or uncomplicated building blocks, such as block ciphers, stream ciphers, and hash functions. Block ciphers may either have one key for both encryption and decryption, in which case they're named shared keys or have distinct keys for encryption and decryption, in which case they're named public keys or asymmetric. A digital signature scheme is a distinct type of asymmetric crypto original. Within the background of any application-to-application communiqué, there are around exact security necessities, together with the following:

- **Authentication:** The procedure for demonstrating one's uniqueness. We can relate it to the prime forms of host-to-host verification that happen on the Internet. In its current form, it is more like name-based or address-based authentication, both of which are disgracefully weak.
- **Privacy/confidentiality:** Safeguarding that no one can read the message excluding the envisioned receiver.

- **Integrity:** Guaranteeing the receiver that the received message has not been changed in at all methods from the original.
- **Non-repudiation:** A mechanism to demonstrate that the dispatcher sent this message. Though the final goal of cryptography, and the mechanisms that make it up, is to fleece information from unsanctioned persons, the utmost algorithms can be fragmented and the information can be exposed if the assailant has sufficient time, wish, and resources. So, a more truthful goal of cryptography is to make gaining the information too work-intensive to be worth it to the attacker. When data is deposited on a computer, it is typically protected by rational and physical access controls. When this similar subtle information is sent over a network, it can no longer take these controls for being settled, and the information is in a much more susceptible state.

Fundamentally encryption is a technique of altering original data, named plaintext, into a form that seems to be random and unreadable, which is termed ciphertext. The plaintext is either in a form that can be unstated by a person (a document) or by a computer (executable code). When it is transformed into ciphertext, neither human nor machine can correctly process it till it is decrypted. This allows the broadcast of intimate information over insecure channels without unauthorized revelation. Please refer to [Figure 8.3](#) to understand this process:

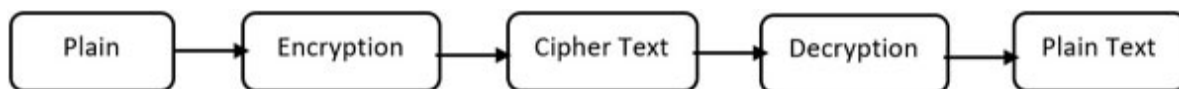


Figure 8.3: The process of encryption transforms plaintext into ciphertext and the process of decryption transforms ciphertext into plaintext

A system that offers encryption and decryption is denoted as a cryptosystem and can be shaped through hardware components or program code in an application. The cryptosystem uses an encryption algorithm.

Most encryption approaches use a secret value. In technical terms, it is named a key which is usually a long string of bits. Thereafter, an algorithm could be applied to encrypt and decrypt the text. Refer to [Figure 8.4](#):

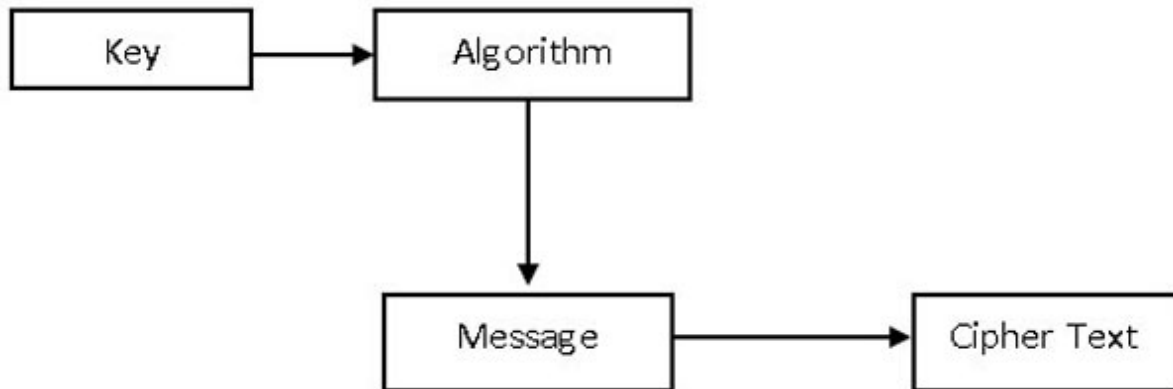


Figure 8.4: The process of encryption transforms key into ciphertext

It can be thought as the key is inserted into the mathematical algorithm and the result is applied to the message, which ends up in ciphertext.

If a listener captures a message as it passes among two people, s/he may be able to see the message, but it seems in its encrypted form and is therefore ineffectual. Even if the attacker recognizes the algorithm that the two individuals are using to encrypt and decrypt their information, deprived of the key, this information is useless to the listener.

Classical cryptography

There are several ways of classifying cryptographic algorithms. They will be characterized by the foundational idea of the number of keys that are put into the work for doing encryption and decryption. Further, they are also demarcated by their specific and important applications and uses.

The two most famous classical algorithms are mentioned as follows:

- **Secret key cryptography:** On the most basic level, in this type of cryptography, a solitary key is used for both encryption and decryption.
- **Public key cryptography:** In this type of cryptography, fundamentally one key is used for encryption, whereas others are for decryption purposes.

Now let's look these in detail. For deeper understanding

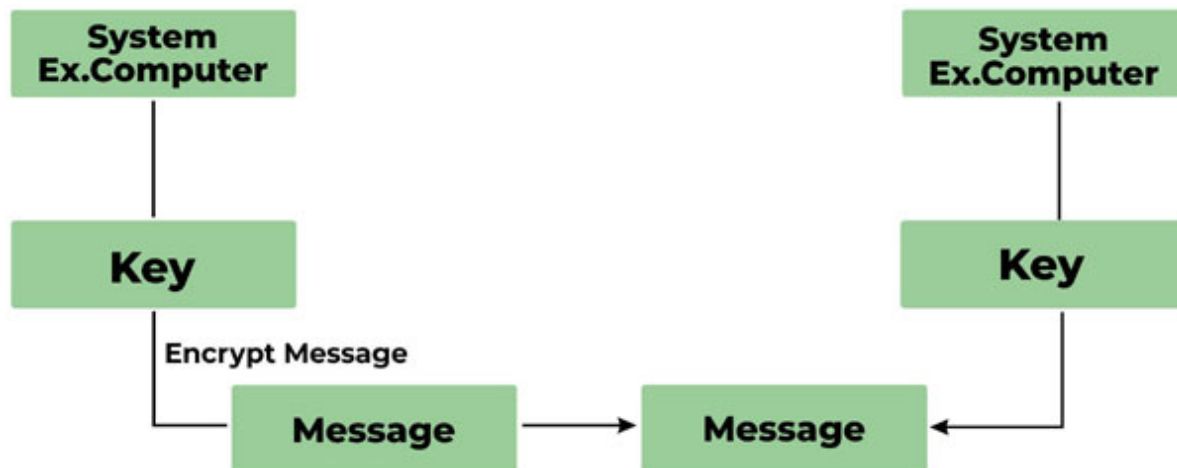
Secret key cryptography

Secret-key encryption needs that two users first develop and firmly share a secret key, which is essentially a long string of randomly-chosen bits. The users then use the secret key along with some particular so-called public algorithms to do the encryption of the given string and further follow the process of decryption of the messages as well. Doing this type of work makes the algorithms very complex because it can be intended to get each bit of output which is reliant on each bit of input. For a simple visualization, imagine that a key of 128 bits is used. Further, assume that brute force is in play, along with some element of parallelism, also working.

The encrypted message should be innocuous: a billion computers doing a billion operations per second would need a trillion years to decrypt it. Secret key cryptography arrangements are usually characterized as being either stream ciphers or block ciphers, as described as follows:

- Stream ciphers function on a solitary bit at a time and implement some form of response mechanism. Hence, the key is continually altering its progress and other important information.
- In a block cipher, the arrangement encrypts one block of data at a time utilizing a similar key on each block. Overall, the similar plaintext block will continuously encrypt to a similar ciphertext when using a similar key in a block cipher while the same plaintext will encrypt to the dissimilar ciphertext in a stream cipher.

Refer to [Figure 8.5](#) for the working of the symmetric algorithms:



Symmetric Key : Uses The same key

Figure 8.5: In symmetric algorithms, the sender and receiver usage a similar key for encryption and decryption purposes.

The security of the symmetric encryption technique is reliant on how well users defend the key. Since both users are using a similar key to encrypt and decrypt messages, symmetric cryptosystems can offer confidentiality, but they cannot deliver authentication or nonrepudiation. There is no method to show who sent a message if two people are using the same key. The following are examples of symmetric key cryptography algorithms; these concepts will be explained in the latter part of the chapter:

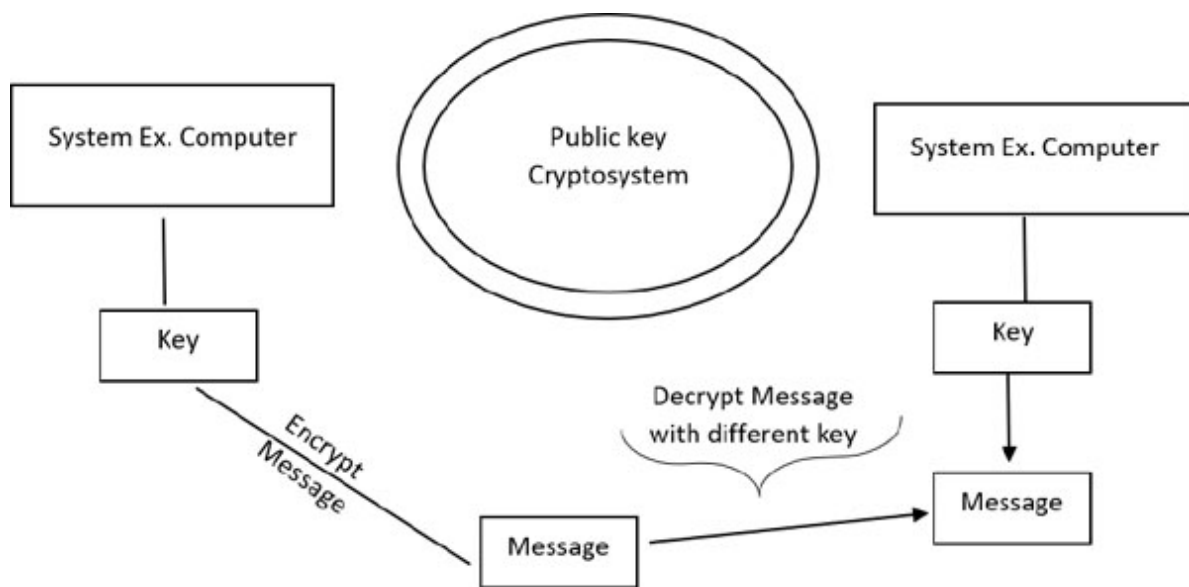
- Data Encryption Standard (DES)
- Blowfish
- IDEA
- RC4, RC5, and RC6.
- Triple-DES (3DES)

Public key cryptography

Public-key encryption is founded on the concept of a safe with two keys: a public key to latch the safe and a private key to open it. Utilizing this technique, anybody can send a message since the public key is used to encrypt messages, but only somebody with the private key can decrypt the

messages. Since the encrypting and decrypting keys are different, it is not essential to securely allocate a key.

The security of public-key encryption is contingent on the expected trouble of certain mathematical operations, such as factoring very large prime numbers. The public and private keys are mathematically connected, but cannot be consequent from each other. This means that if one gets a copy of Madhav's (an entity, in this case, he is a human) public key, it does not mean he can now use mathematical magic and discover Madhav's private key as well. Please refer to [Figure 8.6](#) to understand the working of public-key cryptography:



Asymmetric systems: Uses two different keys for encryption and decryption process

Figure 8.6: Public key cryptosystem

If privacy is the most significant security facility to a sender, he or she would encode the file with the receiver's key which is essentially a public key. This is named a secure message format since it can only be decrypted by the person who has a consistent private key. If authentication is the greatest significant security service to the sender, then she or he would encode the communication or message with their acquired secret private key.

This guarantees the receiver (the individual who gets the information) that the solitary being who could have encoded the message is the individual who

has possession of that private key. If the sender encrypted the message with the receiver's public key, authentication is not on condition that this public key is obtainable to anyone. Encrypting a message with the sender's private key is named an open message format since anyone with a copy of the consistent public key can essentially decrypt the message. Consequently, confidentiality or privacy is not fortified.

For communication or message to be in a protected and signed arrangement, the sender would scramble the message with his "own" private key and then consequently encrypt it over with the receiver's public key. The receiver would then require to decrypt the message with his private key and then decrypt it once more time with the sender's public key. This delivers privacy and authentication for that brought message.

Each key category can be used to encrypt and decrypt, so do not build an impression and reason the public key is only for encryption and the private key is only for decryption. They both can encrypt and decrypt data. Nevertheless, if data is encoded with a sequestered or private key, it cannot be decrypted with a private key. If data is encoded with a private key, it necessarily is decrypted with the corresponding provided public key. If data is encoded with a public key, it essentially is decrypted with the corresponding private key.

Limitations of classical cryptography techniques

Classical cryptography techniques are great; however, they suffer from a few limitations. The following are some of them in the context of secret-key cryptography:

- **Key distribution:** It needs a secure mechanism to distribute keys correctly.
- **Scalability:** To each twosome user requires a sole pair of keys, so the number of keys raises exponentially.
- **Limited security:** It can deliver privacy, but not authenticity or nonrepudiation.

Problem with cryptography

There are two chief problems with secret-key encryption.

The first problem is that by examining the publicly-known encrypting algorithm, it occasionally turns out to be easier to decrypt the message. This problem can be rather counterbalanced by the cumulative length of the key. **The second problem** is to securely allocate the secret key in the first place. This is the well-known “key-distribution problem”. Users must either decide on the secret key when they are organized in the same location or when they are in diverse locations.

The disadvantages of developing the key when they are in a similar location are that it is not at all times real for the users to encounter; a big database would be desired to store the prearranged keys, and such storage is not secure. The disadvantage to presenting a key when the users are in different locations is that all traditional methods of conveying the key are subject to snooping that cannot be detected by the users.

Public key cryptography contribution towards security:

The main weakness of Public-key cryptography is it works much slower than symmetric systems. This further contributes to two problems with founding security on the presumed trouble of mathematical problems, which are explained as follows:

- The first problem is that the trouble of the mathematical problems is presumed, not established. All security will disappear if well-organized factoring algorithms are exposed.
- The second problem is the danger of quantum computers. The theoretical capability of quantum computers to fundamentally procedure large amounts of information in parallel would eliminate the time limit to factoring large numbers. Therefore, public-key encryption, though secure at the moment, faces a thoughtful threat as quantum computing arises closer to realism. Presently, though, this technique is still extensively used, particularly for the encryption of financial information sent over the internet.
- After having an understanding of cryptography in general and quantum cryptography in particular, along with its limitations, drawbacks and advantages, it is time to go a bit deeper into quantum cryptography. And for achieving this motive we will explore the importance of quantum cryptography, and its relationship with physics. Also, we will

touch base on the importance of polarization. Finally, as a practical implementation of it, we would see how one could be able to securely transmit photons from one place to another. This is not very different from the classical quantum transmission mode by the use of the Bob and Alice analogy.

Importance of quantum cryptography

The problems encountered with classical cryptography lead to quantum cryptography. The following are a few additional points:

- Quantum cryptography offers an effective cryptography method that is future proof and dependable. Since it is not endangered by quantum computing, it is innocuous from debauched brute-force bouts of fast computers of the future. Also, intrusion detection makes it steadfast.
- Quantum cryptography can be used to exchange keys among parties that do not believe each other and neither their apparatus.
- As quantum cryptography allows the detection of an eavesdropper, the key exchange can be securely made, without the threat of the key being compromised during the initial key exchange.

Quantum cryptography : in context of physics

Today's utmost public encryption methods are endangered by the possible formation of the QC. But even now, the elementary quantum cryptography has been in use and somewhat industrialized to the closest group of corporations which potentially is the additional secure communication than any current procedures and cannot be compromised by QC.

Quantum cryptography exploits the benefit of the sole and unusual behaviour of macroscopic objects to permit users to firmly develop secret keys as well as to perceive eavesdropping.

The growth of quantum cryptography was interested in the shortcomings of classical cryptographic approaches, which can be confidentially bucketed as the public-key or secret-key methods.

More general Quantum cryptography resolves the glitches of secret-key cryptography by as long as a method for two users who are in different

locations to firmly find a secret key and to detect if eavesdropping has occurred. Also, subsequently, quantum cryptography does not become contingent on problematic mathematical problems for its security, as it is not endangered by the growth of quantum computers. Quantum cryptography achieves these extraordinary achievements by exploiting the properties of microscopic foundational elements of quantum physics and Einstein's dear photons.

Quantum cryptography uses our present knowledge of physics to grow a cryptosystem that is not bright to be overpowered, that is, one that is protected in the inconsistency of existence liaised deprived of the acquaintance of the sender (one who broadcast the info) or the receiver (one who get the info) of the messages. The word quantum mentions the most important behavior of the least particles of matter and energy.

The quantum theory clarifies all that occurrences and nonentity can disrupt it. Fundamentally, quantum cryptography is founded on the custom of separate particles/waves of light (photon) and their inherent quantum properties to grow an indestructible cryptosystem. It is unbelievable to measure the quantum state of any scheme deprived of troubling or disturbing that system. It is hypothetically conceivable that other particles could be used, but photons bid all the essential abilities required. Their actions are moderately well-understood, and they are the statistics or information transporters in optical fibre manacles (cables). The optical fibre manacles are the greatest capable medium for tremendously high-bandwidth communications.

Quantum cryptography versus traditional cryptography

A central difference between conventional cryptography and quantum cryptography is that conventional cryptography chiefly uses tough mathematical methods like integer factorization in RSA as its central mechanism. Whereas Quantum cryptography usages the principles of quantum physics to secure data. Further, conventional cryptography positions on a basis of robust math, while quantum cryptography has a fundamentally different ground in that the security should be founded on known physical laws instead of mathematical problems.

Precisely, quantum cryptography is customary of protocols, systems, and measures that is thinkable to make and allocate secret keys. Quantum

cryptography can be cast-off to produce and allocate secret keys, which can then be used together with old-style cryptography algorithms and protocols to encrypt and transmit data. It is significant to note that quantum cryptography is not used to encrypt data, transference encrypted data, or stock encrypted data.

As mentioned earlier, the requirement for asymmetric key systems gets up from the subject of key distribution. The quandary is that it is necessary to have a secure channel. To set up a secure channel, quantum cryptography resolves the key distribution problem by permitting the exchange of a cryptographic key among two remote parties with whole security as ordered by the laws of physics.

When the key exchange takes place, conservative cryptographic algorithms are used. For that reason, numerous desire the period quantum key distribution as opposed to quantum cryptography. The subsequent is an elementary and too naïve explanation of how quantum cryptography can be used in a commercial setting. The flow is as follows:

- Two parties want to exchange data by electronic means in an extremely secure way.
- They select ordinary cryptography algorithms, protocols, systems, and transportation technologies to interchange the data in an encrypted method.
- The use of a quantum cryptography channel to produce and interchange the secret keys needed by the algorithms.

They use the secret keys produced by quantum cryptography and the traditional algorithms to encrypt the data. They exchange encrypted data using the selected traditional protocols and transmission technologies. Inside quantum cryptography are two separate channels. One channel is used for the broadcast of the quantum key substantial via solitary photon light pulses.

The additional channel transmits all communication circulation, counting the cryptographic protocols, encrypted user traffic, and more. According to the laws of quantum physics, when a photon has been observed, its state changes. This brand of quantum cryptography is perfect for security determinations since when somebody stabs to snoop on a secure channel, it will reason a commotion in the current of the photons that can be effortlessly

recognized to offer extra security. Quantum algorithms are orders of scale improved than present systems.

It is assessed that quantum factorization can factor a number of million areas longer than any used for RSA in a millionth of the time. Further, it can crash a DES (The Data Encryption Standard is a symmetric-key algorithm for the encryption of digital data. Although its short key length of 56 bits makes it too insecure for modern applications, it has been highly influential in the advancement of cryptography.) in lesser than four minutes! The amplified rapidity is due to the superposition of numbers. Quantum computers can do calculations on numerous superpositions concurrently, which produces the result of an enormous parallel computation.

At this point, it is good to understand the pros and cons of quantum cryptography. Please refer to [Figure 8.7](#) for the same:

Quantum key distribution	Pros / Cons	Public / Private Key	Pros / Cons
Requires dedicated hardware and communication lines	Con	Can be implemented in software / portable in nature	Pro
Mathematically proven and based on the laws of quantum physics	Pro	Mathematically undecided	Con
Will be secure even when the quantum computer would be built	Pro	Will be broken, when quantum computer would be build	Con
Still under development and in the prototyping phase. Works on limited distance and uses mainly optical fiber	Con	Work at any distance with any type of network connection	Pro
Can be used with a one-time pad , the only mathematically proven secure cryptographic algorithm	Pro	Cannot be used with a one-time pad	Con
Very expensive	Con	Affordable by anyone	Pro

Figure 8.7: Pros and cons of quantum cryptography

[Photon polarization and quantum cryptography](#)

After having this background now, it is the right time to look at photon polarization. For lightwave, we presumed that the route of the electric field is immovable and the magnitude differs sinusoidally with space and time. The electric field in a light wave propagating in unrestricted space is perpendicular to the direction of propagation. Though there is an infinite number of directions perpendicular to the direction of propagation and the electric field may be vertical in any of these directions. For instance, if the light propagates along the X-axis, the electric field may be the vertical Y-axis, or along the Z-axis or any direction in the Y-Z plane.

If the electric field at a point at all times stays parallel to an immovable direction as time passes, the light is named linearly polarized along that direction. For instance, if the electric field at a point is continuously parallel to the Y-axis, we say that the light is linearly polarized along the Y-axis ([Figure 8.8](#)). The same is also termed plane-polarized light. The plane encompassing the electric field and the direction of propagation is named the plane of polarization. Refer to [Figure 8.8](#) for clarity:

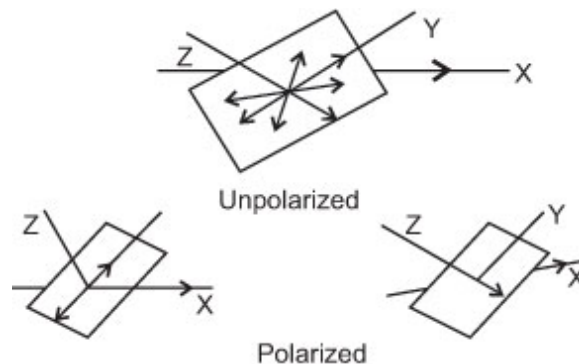


Figure 8.8: Photon polarization

Polarization and polarizer

There are numerous methods to harvest polarized light from unpolarized light. A gadget used to crop polarized light from unpolarized light is called a polarizer. Plane sheets in the shape of circular discs called polaroids are commercially obtainable which transport light with E-vector parallel to a special direction in the sheet. These polaroids have long chains of hydrocarbons which turn out to be conducting at optical frequencies. When light falls perpendicularly on the sheet, the electric field parallel to the

chains is absorbed in setting up electric currents in the chains but the field perpendicular to the chains comes to be communicated.

The direction perpendicular to the chains is called the transmission axis of the polaroid. When light passes through the polaroid, the transmitted light turns out to be linearly polarized, especially with an E-vector parallel to the axis under consideration which is the transmission axis.

If linearly polarized light is incident on a polaroid with the E-vector parallel to the transmission axis, the light is completely transmitted by the polaroid. If the E-vector is vertical (in mathematical terms perpendicular) to the transmission or broadcast axis, the light is at a standstill by the polaroid. If the E-vector exists at an angle θ along with the transmission or broadcast axis, light is partly communicated or transmitted. The intensity (strength or concentration) of the transmitted light is represented in the form of $I = I_0 \cos^2 \theta$; in this equation, I_0 signified the intensity or strength of light, when the incident E-vector is parallel to the transmission or broadcast axis. This is famous as the law of Malus. Refer to [Figure 8.9](#) for more clarity:

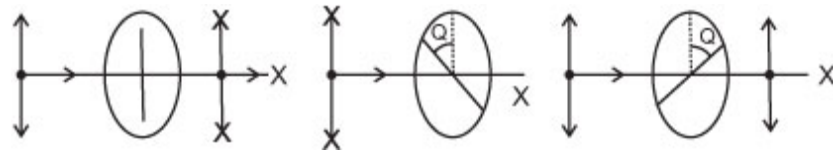


Figure 8.9: Polarization and polarizer

Photon polarization measurements form the basis for the greatest shared quantum cryptographic methods; however, it is significant to first comprehend their properties. The three selected or chosen bases of polarization (which happens on multiple angles) and the likely outcomes of a measurement, according to the bases are mentioned as follows:

- Rectilinear (horizontal or vertical)
- Further the other one is circular (left-circular or right-circular)
- And lastly diagonal (45° or 135°)

Out of three bases, only two bases are superfluous in any specified protocol for what we call as quantum cryptography. Photons can be unhurried to regulate their orientation comparative to one of these headquarters of polarization at a time.

Characteristically, one would imagine the photon to have a certain polarization (this can be achieved in multiple ways, the simplest one is to put a photon filter between the light source and detector at a certain angle say 45°). Further, it can be restrained but is not altered by the measurement. Photons, though, are quantum objects, and in the quantum world, an object can be well-thought-out to have property only after you have measured it, and the kind of measurement influences the property that you find the object to have. This suggests that a photon can only be well-thought-out to have a specific polarization after you measure it and that the foundation you select for the measurement will have an influence on the polarization that you find the photon to have. [Figure 8.10](#) would explain this in an easy-to-understand way; Also, we would take a scenario to understand it in a better way.

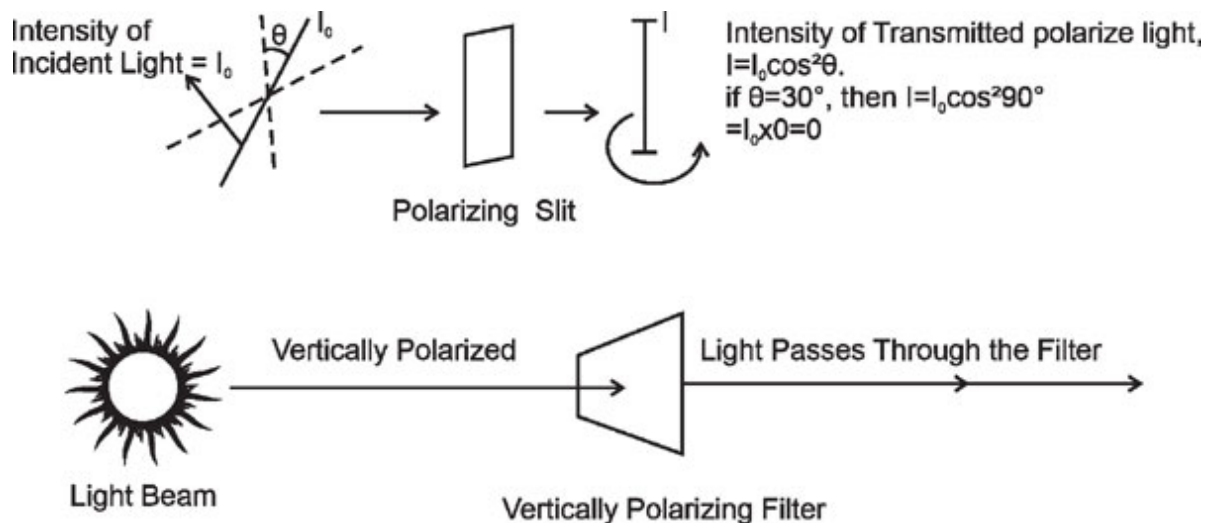


Figure 8.10: Cryptography and polarization

When a photon sends through a device to measure its alignment comparative to a rectilinear coordinate system, you are requesting the question, how is the photon disturbed in comparison to a direct coordinate scheme? To get the answer to this question, you will figure out that in the choice of being either vertically polarized or horizontally polarized, it would stay in one, as there are only two likelihoods possible where the photon would reside. Supposing, you measure this photon as horizontally polarized.

Next, you send this same photon through a device to measure its orientation comparative to a slanting coordinate system. The photon is focused relative

to a diagonal coordinate system and it is 45° polarized or 135° polarized. Fundamentally, only these two likelihoods exist.

Measurement and polarization

The kind of measurement does certainly influence what property you find. This is an astonishing difference to the classical situation, where something flat is concerned, it would be probable to have an element in the crosswise or diagonal direction.

A horizontally-oriented photon may then be measured to have a 45° polarization and this would happen because the state of horizontal polarization is essentially a superposition of the two so-called diagonals (a line that connects from one bend to another with a given condition that it would not join the edge) polarization states. All polarization states are a superposition of other polarization states. To see the measurement of polarization in action, refer to [Figure 8.11](#):

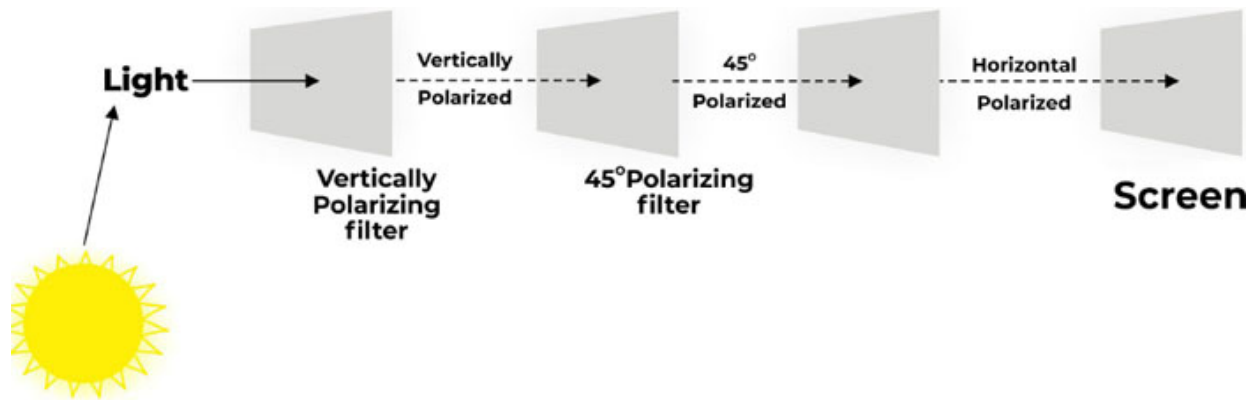


Figure 8.11: Measuring polarization

It is significant and very important to note that once the measurement, especially diagonal measurement, was made, all information about the preceding property of horizontal polarization of the photon disappeared. As a consequence, it is impossible to regulate a photon's rectilinear and diagonal polarizations at the same time. This is comparable to the castle in the air of specifying a particle's location and momentum at an identical period. Further, information about one outcome is less information about the other. Refer to [Figure 8.12](#) for a better understanding:

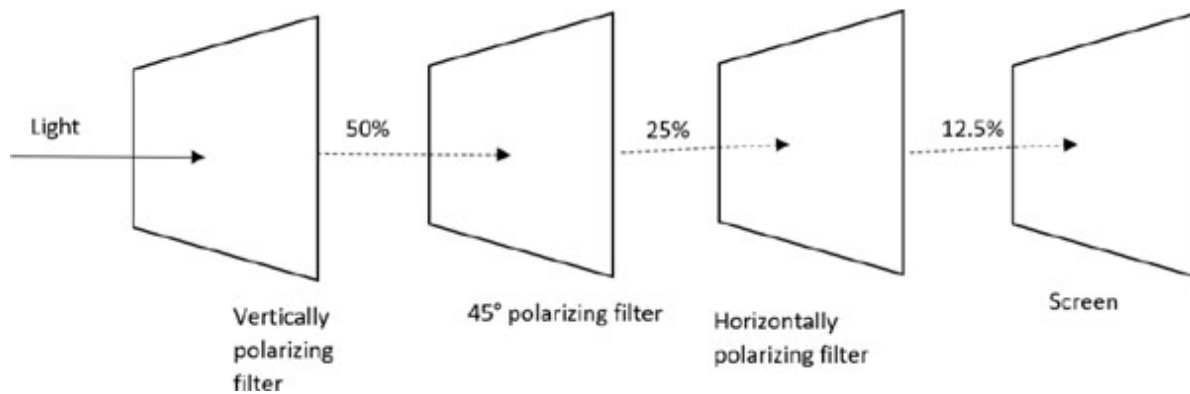


Figure 8.12: *Measuring polarization and its distribution*

The behavior of photons sent, done by a series of the polarizer is exemplified next. Before going into details, let's have a look at legends and then move forward.

Legend

+ = a device that measures rectilinear polarization

V = vertical polarization

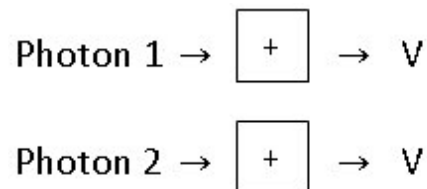
H = horizontal polarization

O = a device that measures circular polarization

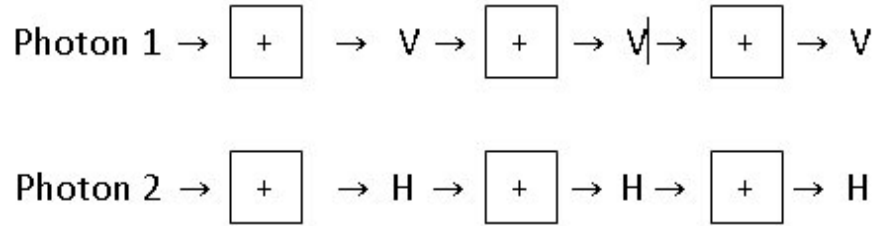
L = left-circular polarization

R = right-circular polarization

A photon is directed through a rectilinear (+) measurement device. The photon has equivalent likelihood of being vertically or horizontally oriented.



A photon that is recurrently sent through the identical measurement device will continuously give a similar answer.



A photon that was unhurried to be vertically polarized is guided to be done by a gadget to measure its circular polarization. The photon (generally available in the form of energy packets) will derive out either left-circular polarized OR right-circular polarized, with equivalent likelihood.

$$V \rightarrow \boxed{\circ} \rightarrow L \quad \text{OR} \quad V \rightarrow \boxed{\circ} \rightarrow R$$

Equivalent outcomes would arise if a circularly polarized photon was sent or directed over a rectilinear measurement apparatus.

To understand in a better way, we can use the standard agreement and the metaphor of Alice, Eve, and Bob. Alice is sending the information, Bob is on the receiver end and receiving the information, and Eve is short of playing the role of the spy. Alice is fundamentally making photons randomly with either rectilinear or circular polarization in the overall process. Further, Alice chronicles the polarization of every photon (energy packets) and thereafter directs them to Bob who stays on the receiver end.

Thereafter, Bob obtains each photon and arbitrarily measures its polarization, giving it to the rectilinear or circular basis. He annals the measurement type (basis used) and the subsequent polarization is measured. It is important to recall that the polarization which is directed or sent by Alice may not be the same polarization information that Bob (who is the receiver) finds, if he does not use the same basis used by Alice. Bob openly expresses to Alice what the measurement forms were, but not the consequences of his measurements. The step-by-step depiction of the quantum cryptography process is shown in [Table 8.1](#).

Alice openly expresses Bob in which measurements were of the precise type. Correct measurement is the correct type if Bob used a similar basis for measurement as Alice did for preparation. Alice and Bob each lob out the

data from measurements that were not of the precise type and change the enduring data to a string of bits using a convention such as:

left-circular = 0, right-circular = 1

horizontal = 0, vertical = 1

Refer to [Table 8.1](#) which depicts this:

Step	Description	1	2	3	4	5	6	7	8	9	10	11	12	Components
1	Filters were used by Alice to prepare photons.	+	+	0	+	0	0	0	+	+	+	0	0	+ represents filter
2	Polarization of photons sent by Alice.	V	H	L	H	R	L	R	V	H	H	R	L	
3a	Measurement types made by Bob.	+	+	+	+	0	0	0	0	+	0	0	-	
3b	Result of Bob's measurements.	V	H	H	H	R	L	R	L	H	L	R	H	
4	Bob publically tells Alice which type of measurement he made on each photon.	+	+	+	+	0	0	0	0	+	0	0	+	Measurement

Step	Description	1	2	3	4	5	6	7	8	9	10	11	12	Components
5	Alice publically tells Bob which measurements were the correct type.	Y	Y	N	Y	Y	Y	Y	N	Y	N	Y	N	
6	Alice and Bob each keep the correct measurement and convert to binary.	1	0		0	1	0	1		0		1		Binary digits

Table 8.1: The step-by-step depiction of quantum cryptography process

The next sample data was made assuming that Alice directs 12 photons and the detector never fails.

The string of bits now possessed by Alice and Bob is 1 0 0 1 0 1 0 1. This string of bits procedures the secret key. In duplication, the number of photons sent and the subsequent length of the string of bits can be taken under consideration. Nonetheless, what if Eve has been snooping on Alice and Bob's broadcasts?

To comprehend why eavesdropping does not find the middle ground in the security of keys advanced using quantum cryptography, it is significant to note that broadcasts between Alice and Bob take place on two different kinds of channels. The photon broadcasts are quantum and happen on what will be named a quantum channel, such as optical fiber. The deliberations among Alice and Bob about the kinds of measurements made, happen on a traditional channel, such as telephone or email. The properties of these two channels are very different.

Eve can interrupt information conveyed on the traditional channel without being noticed. Though this was some problem in preceding contemporary cryptographic approaches, it is not a problem in quantum cryptography. This is because, in quantum cryptography protocols, the only information Bob and Alice discuss on a classical channel is the kind of measurements they made, which tells Eve nothing about the consequences of the measurement, and consequently nothing about the key that was advanced.

Supposing Eve attends on the quantum channel. One way she may do this is by scanning some photons from the spurt sent from Alice to Bob. Now, Eve has photons that are undistinguishable in polarization to those acknowledged by Bob, but she has to arbitrarily select her measurement type subsequently as she doesn't know what measurement kind Bob is going to use. Therefore, about half the time, she will choose a different basis than Bob for measurement.

For instance, supposing Alice directs a burst of circularly polarized photons, a few of which are acknowledged by Eve and Bob. Eve decided to use the rectilinear basis for measurement, and Bob decides to use a circular basis. Alice and Bob keep the resulting data because they both used a similar basis, but since Eve used the incorrect basis, she doesn't know what their outcome was.

Now assume in its place, Eve had decided to measure it conferring to the circular basis, but then again Bob decided to measure it according to the

rectilinear basis. Now Eve would know the polarization that Alice directed, but as Bob did not select an accurate basis, Alice and Bob would toss the outcomes out. Eve will not end up with the string of bits that Alice and Bob generate.

We can do it in other ways as well. For instance, assume, Eve could snoop on the quantum channel to interrupt the photons, measure them, and then direct them on to Bob. When she selects a different basis for measurement than Alice had used for preparation, she will alter the photon's polarization through the act of measurement, instigating Bob to obtain a photon that does not have a similar polarization as that sent by Alice. To understand, consider the following scenario.

Alice directs or sends a right-circularly polarized photon, which is captured by Eve. Eve measures it according to the rectilinear basis, discovers that it is vertically polarized, and sends it on to Bob. Bob measures it according to the circular basis and has an equivalent likelihood of finding it to be right-circularly polarized and left-circularly polarized. This will introduce faults in Bob's concluding string of bits. Alice and Bob can distinguish these faults when they run a key validity check. Therefore, although Eve could get a few accurate bits of the concluding key by interrupting photons, Alice and Bob will know that the security of their key has been compromised.

In reality, sensors flop some of the time, there is sound in the channel, and Eve could be snooping. All of these issues can reason Bob to end up with a different string of bits than Alice. Thus, Alice and Bob need to do a key validity check to settle that they have the same string of bits.

They cannot relate the complete string of bits over a conventional channel since this would compromise the security of their key. One method, however, is to associate a big random subset of their string of bits, assuming that if these match up, then the others that they are not associating also match up and can be used as the key. The bits that they have compared are used since this information was shared over a public channel and could have been interrupted. This method is useful for noticing eavesdropping since any activity by Eve would introduce a large number of errors.

There is an additional technique as well. For instance, in one method, Alice and Bob come to an agreement on certain blocks of bits of their key and

compute and equate the parity of each block. If an odd number of faults exist, the parity calculated by Alice and Bob will be different.

Alice and Bob checked several overlapping blocks and make lesser and lesser block sizes to find the errors. Each time the assessment is performed, Alice and Bob abandon the last bit in the block. This is one of the methods for removing errors that can successfully eliminate faults, and Alice and Bob can be sure that they have the identical string of bits for use as the key.

Significance of quantum key distribution

As we know **Quantum Key Distribution (QKD)** is a novel progressive technique for secure key exchange. Through QKD, parties will be able to allocate shared secret keys directly deprived of the possibility of hidden snooping or tampering. Therefore, founded on the properties of quantum mechanics, any efforts to disturb or observe the communication will leave physical traces, that is, impressions of the tampering effort. Quantum communications are tamper-proof. Throughout the quantum message, information is encoded into photons. Each photon has a quantum state that controls its position and way in physical space. Please refer to [Figure 8.13](#) for the process of quantum key distribution.

The turn of measuring a quantum state will alter the position of a photon and interrupt the system. As a consequence, if an opponent efforts to measure or eavesdrop on a quantum key exchange channel, the alteration will later be sensed. In practice, the QKD key would be castoff and the members informed of the compromise. At the very least, the key would fail in usage as observing or measuring the key vicissitudes of the value. This defence could be willingly applied using **Wegman-Carter Authentication (WCA)** and would render eavesdropping bouts unsuccessful.

Currently, countries such as Austria, China, Japan, Switzerland, and the US are at work to instrument QKD to provision **One-Time Pad (OTP)** generation. OTPs are big keys that are pre-shared among the sender and recipient.

Various other forms of cryptography in which keys are used; an OTP is a single use key to defend a single communication and must be as long as the communication itself. As the rapidity of QKD upsurges, QKD technology

bids a means of steadily transmitting OTPs. Using fiber optics and FPS 3000 technology, the current standard for transportations transport is 100 Gbit/s in freestyle and drifting networks. Present swiftness of the quantum-based communications must be meaningfully augmented before adoption. Teamwork among the University of Cambridge and Toshiba using the BB84 protocol (it is a type of a rule, we will know more about this shortly) established an exchange of secure QKD keys at 1 Mbit/s (over 20 km of optical fiber) and 10 Kbit/s (over 100 km of fiber).

Now let's look at the different types of cryptography.

[Lattice-based cryptography](#)

Currently, researchers are in the process of modifying the existing asymmetric public-key encryption schemes so that the underlying algorithms used to encrypt and decrypt data are resistant to phase estimation transformations. These newer encryption algorithms are similar to algorithms that take advantage of the Prime Factorization and the Discrete Log problems, but because of aspects of these particular problems, they still maintain currently unsolvable complexity from attacks using quantum algorithms.

The Shortest Vector problem is one of the leading replacements for the Prime Factorization problem and Discrete Log problem. A form of cryptography known as lattice-based cryptography uses it to provide currently quantum-safe encryption. Lattice-based cryptography can be used for most current cryptographic services, like encryption, message signing, and hashing, making it a popular candidate for post-quantum cryptography. Research into the Shortest Vector problem has shown that it resists one of the key techniques (periodicity finding) used in Shor's Algorithm. Until proven otherwise, lattice-based cryptography remains a safe alternative. However, lattice-based public-key encryption algorithms do not scale particularly well.

There are multiple lattice-based cryptographic systems currently available for distribution, albeit the efforts are mostly from the open-source community. The most mature of these offerings include NTRUEncrypt and NTRUSign (GNU General public license for the development, commercial

license for production), RLWE-KEX Encryption (available as an OpenSSL modification), and SWIFFT Hashing (GNU General public license).

Hash-based cryptography

Hash-based cryptography is an alternative quantum-proof cryptographic scheme that is primarily focused on digital signatures that verify that the document or message originated from the initial sender of the document. There are currently no hash-based cryptographic schemes for encrypting and decrypting messages using asymmetric public key exchange (PKE), so additional cryptographic methods would be necessary for the remaining cryptographic services.

The digital signatures that hash-based cryptography produce are not number theory problems and are impervious to mathematical optimization techniques that quantum attacks use to break current cryptography. Because signatures generated from hash-based cryptography are only suitable for one-time use, keys must be regenerated for each new message sent and old keys must be tracked to prevent recycling.

Popular hash-based cryptographic schemes typically take advantage of Winternitz's **One-Time Signatures (OTS)**. Winternitz-OTS uses a tuning parameter that must be set. This tuning parameter trades off between signature size and key generation, signing, and verification speed. Hash-based cryptographic implementations need to optimize this parameter to operate effectively.

After having this background of different types of cryptography techniques, let's look at the protocols.

Fully distrustful protocol

A quantum protocol is said to be gear independent if the dependability of its enactment can be certain without creating any assumptions concerning the inside workings of the underlying device. The key impression is that the certification of an adequate quantity of nonlocality safeguards that the underlying systems are quantum and entangled.

A gear-independent protocol, in theory, remnants safe even if the gear were made up by an adversary. So far, gear-independent protocols have been

projected for quantum key distribution, haphazard or random number generation, state estimation, and the self-testing of quantum computers.

Distrustful protocol

The distrustful protocol is a protocol in which two parties do not believe or trust each other. Anybody may attempt to shark, so to safeguard dependability, an additional party must be able to observe duplicitousness. It is not a priori clear, whether the scope of the apparatus's unprejudiced method can be prolonged to conceal cryptographic problems with distrustful parties. For protocols in the distrustful cryptography model, truthful parties can bank only on themselves. In circumstances (for example, the usage of credit cards on the internet, safe identification, digital signatures), we essentially safeguard sanctuary not only in contradiction of an eavesdropper but significantly counter to malevolent parties partaking in the protocol, that is, when Alice and Bob do not believe each other.

Several significant outcomes in quantum cryptography are linked to the important primitives in this background. While, on the one hand, quantum feeble coinage flicking with subjectively minor bias is likely, randomly hiding and binding quantum bit-commitment is impossible. However, less secure but non-trivial bit - commitment is possible with trusted devices.

Bit commitment protocol

It is a distrustful protocol. A bit-commitment protocol consists of two phases:

- **Commit phase:** In the commit phase, Alice interacts with Bob to commit to a bit.
- **Reveal phase:** In the reveal phase, Alice discloses the worth of the bit, probably trailed by some test that each party transmits out to safeguard that the further party has not been cynical.

In the time which may be sandwiched between the two stages, which may be of any period, no actions are taken. The security of a protocol is at all times analyzed here as the supposition that one of the parties is truthful.

Based on the discussions until now, we easily understand the importance of protocols in quantum information theory. So, let's have a brief look at a couple of fundamental other protocols as well.

Some important other protocols

It must be considered that quantum computing and quantum cryptography are separate areas sharing a shared term. Quantum computing is now in the prototype stage, however quantum cryptography is a practical, marketable solution. There are already few protocols available. Protocols are a set of rules. Hence in terms of quantum cryptography, these rules enable secure communications. Let's have a look at the most important ones.

BB84 protocol

BB84 protocol is the first and foremost imperative quantum cryptography protocol and it was proposed by Bennett and Brassard in the year 1984. This protocol is typically used in the communication of quantum information. This protocol uses two polarization foundations of single-photon. This first base is called the rectilinear basis, whereas the other one is known as the diagonal basis. Further, the single-photon may be polarized utilizing four states which are known as the horizontal $|h\rangle$, vertical $|v\rangle$, left circle polarized $|lcp\rangle$, and right circle polarized $|rcp\rangle$ respectively.

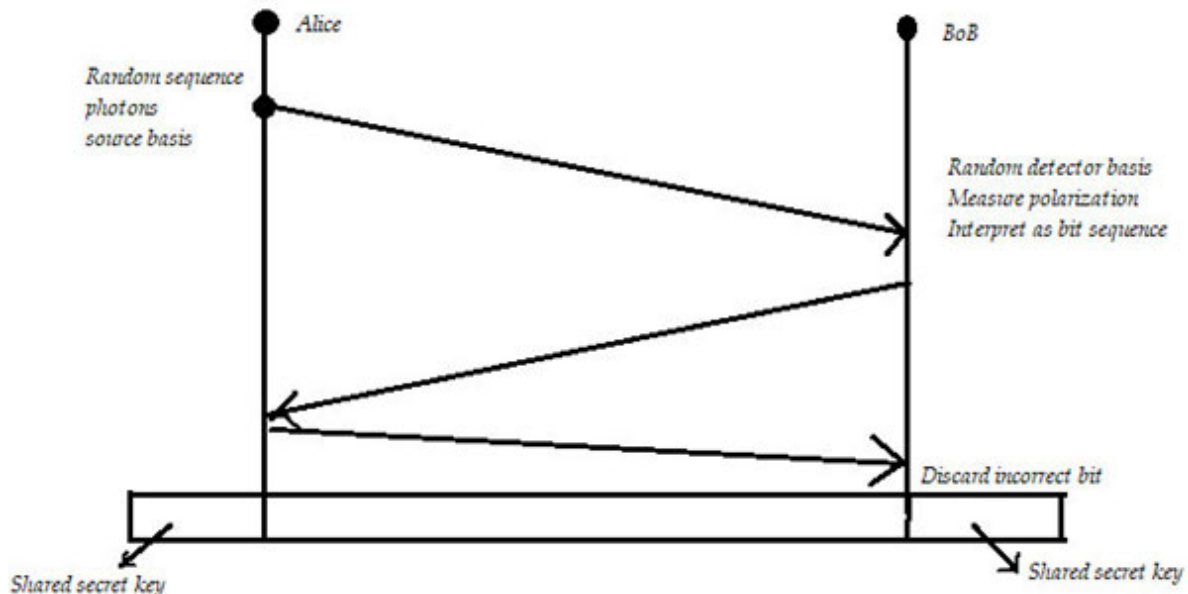
The two states out of four, the Polarization state horizontal $|h\rangle$ and the left circle polarized $|lcp\rangle$ signify a 0. On the other hand, polarization state vertical $|v\rangle$ and right circle polarized $|rcp\rangle$ characterize a 1. Further, to exchange secret keys among dispatcher or sender and receiver or getter, in this case they are Alice and Bob respectively, requires two channels for doing the communication. Also, out of them, the requirement is that the first channel is a quantum channel. whereas the other one is a public channel.

Consecutively, the process goes as follows:

- Alice directs a random order to polarize photons to Bob via the quantum channel.
- After the communication is finished, Bob arbitrarily selects his detector, either rectilinear based or diagonal based and then reports his detector to Alice via the public channel.

- Subsequently, Alice replies to the precise basis to Bob. Lastly, Alice and Bob share the right bits which are used as the key for secure encryption.

The process of the BB84 protocol is shown in [Figure 8.13](#):



BB84 Protocol: Key change between Alice and Bob

Figure 8.13: The process of the BB84 protocol

BB92 protocol

BB92 protocol was put forward by Charles Bennett and Gilles Brassard in 1992, comparable to BB84 protocol but uses only two non-orthogonal quantum state $|h\rangle$ which signifies a 0 and $|rcp\rangle$ that denotes a 1. This broadly 50% of phase reduction in the comparison of the BB84 protocol uses to communicate the key. The technique of secret key transmission comprises of steps like the following:

- Alice directs or sends an arbitrary order of photons, $|h\rangle$ and $|rcp\rangle$.
- Bob arbitrarily selects his detector basis from $|lcp\rangle$ or $|v\rangle$ and understands as a quantum bit order, 0 and 1.
- Alice and Bob share the identical quantum bits, dumping all other bits.

The BB92 protocol is shown in [Figure 8.14](#):

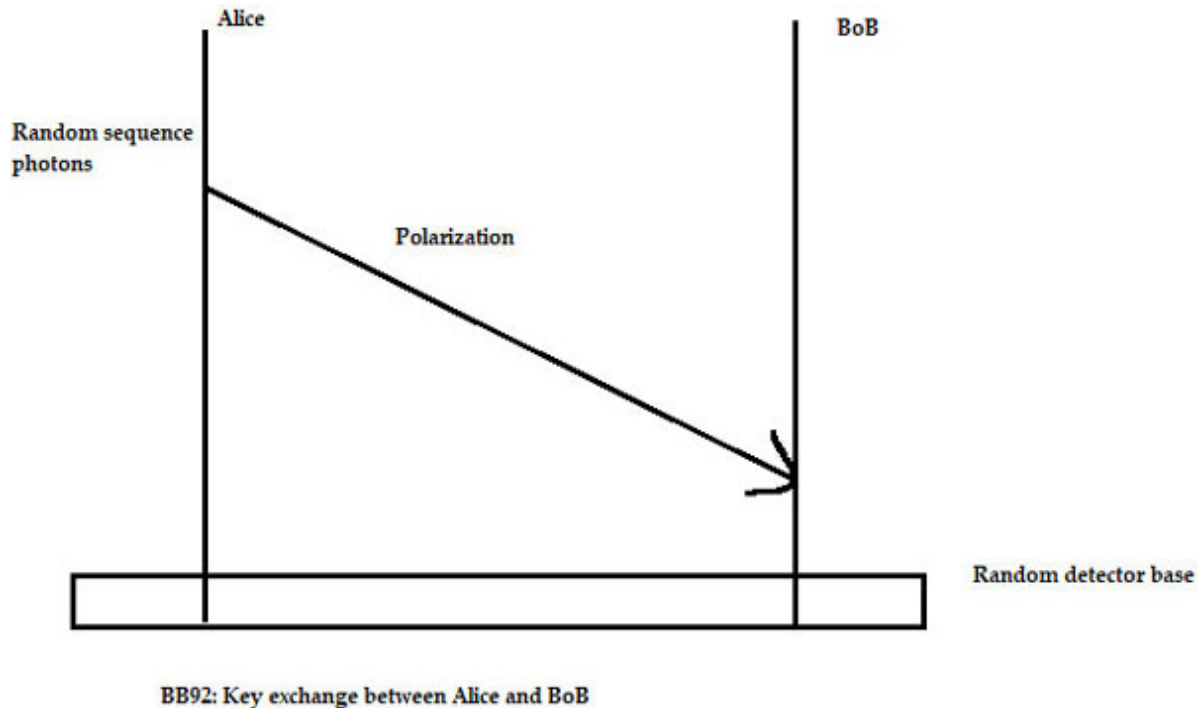


Figure 8.14: The process of the BB84 protocol

Cryptography attack

In general, cryptography, key, or password is the bull's eye of an interloper that efforts to disclose the encrypted data. There are quite a few procedures to attack by using the prospect in communicating data in the network. Out of which a few are mentioned in the following sections.

- **Brute Force Attack (BFA):** This technique overthrows cryptography by tiresome each likely key. It supposes to discover the right key approximately at half of the key. However, this theory has restrictions in the real-world that array processors to need a big amount of energy and unremitting operation for an extended period.
- **Known Plain Text Attack (KPA):** KPA is an attacking mechanism where the challenger has bites of plaintext (for example, detecting data in-network) and uses them to disclose secret keys. As an outcome, a challenger could interpret all the encrypted messages and also communicate untrue messages to the network.

- **Replay Attack (RPA):** RPA is an attack in contrast to the message which is recurrent or deferred. It could be using replicated authentication or malevolent data. In a network communication system, RPA can use for producing a novel session or avoid authentication.
- **Man-in-the-middle Attack (MITM):** MITM cryptography system is generally used for the intention to read, add, and change messages among two parties. It involves cut off messages among two parties.
- **Denial of Service Attack (DoS):** DoS is an attack to upset computer resources or hinder transmission among users and services. DoS in-network can be available in the form of an inundating network that disrupts interaction among nodes, or continuous communication burns out the gear's battery.

Subsequently, BFA is attentive on the key, quantum cryptography is cumulative of the calculation time that BFA essentially needs to decrypt a complete suite of information because of the occurrence of key variations in quantum cryptography.

Known Plain Text Attack (KPA) is more complicated in quantum cryptography since a quick change in the key all through the session is growing more varieties of encrypted relationships or patterns. Hence, an attack in quantum cryptography essentially needs greater dispensation or processing time. This is significantly greater than the general cryptography.

However, MITM and DoS are yet to happen in this quantum cryptography since these two attacks are not engrossed in cryptography technique but bank on trust and network protocol mechanisms.

Cryptography and communication are a few of the most important aspects of quantum information theory and quantum engineering. We discussed cryptography; now have a look at the communication part, even more, pinpointed on quantum internet.

[Quantum Internet](#)

Data confidentiality is one problem that requires to be addressed, whereas data security is one more chief challenge that we need to resolve. The world is becoming more and more linked through the **Internet of Things (IoT)** devices. Data is shared across these devices, among clients and industries,

like never before. Self-driven cars could be a usual thing in a few years, and with all this development in technology, there are requirements to hugely focus on cybersecurity.

If we have to do better to live a harmless and safe life when machines get more linked, the Quantum network is here to help. In a quantum network, information is communicated as quantum information, utilizing entanglement. This property as we know was defined as spooky action at a distance by Einstein. The particles are called entangled if they exist in the same quantum state, and consequently, detecting the state of one of them will mark the state of the other particle. These entangled particles perform like a solitary quantum object and can be demarcated by one wave function.

To communicate quantum information among particles, they want to be entangled. Henceforth, the process starts with producing entangled particles and communicating the particles securely to a distant site. Once this is efficaciously accomplished, we have a communication channel for quantum information. When one of the two entangled particles are allocated a state, it inevitably distresses the state of the further particle, which could be numerous miles away. When we do an operation on one of the particles, that is promptly imitated on the other. This procedure is called quantum teleportation.

Numerous pilots are being run across the world to produce a quantum internet that works on the code of quantum teleportation. A quantum network is anywhere possible when entangled quantum particles are conveyed among a network of nodes. Quantum information is transported across these nodes through these entangled quantum particles. China has done wonderful development with its efforts to attach different communication systems using photons.

The University of Science and Technology of China has accomplished the distribution of entangled particles to dual earth stations, one founded in South China and the other in the Tibetan Plateau. They used a satellite infrastructure for experimentation.

This resolution may not be feasible in normal applications yet, owing to the prices tangled. However, it was landmark experimentation. Further, some trials use drones as a fragment of the communication network to brand the procedure more commercially practical. The best benefit which we get from

these types of networks is in theory at least, when two entangled particles are established, quantum information can be communicated, utilizing no likelihood of anybody hacking into it.

But then again, one of the key challenges in applying quantum teleportation is producing the broadcast channel using entangled particles. The present state of quantum networks in many ways mirrors the state of the initial internet. Though, things are emerging fairly rapidly.

As we know, general-purpose QC is proficient and incompetently resolving tough problems that will be physically huge, encompassing millions or perhaps billions of quantum bits in distributed systems. Whereas, quantum computer architecture matches available physical qubit technologies to applications.

The driving force of architecture is quantum error correction, guarding against the loss of fragile quantum data. However, even with quantum computers, constant factors matter as much as an asymptotic computational class; researchers worldwide are working to bring error-corrected clock speeds up to operationally attractive levels. As the working of quantum computers is different, there is a need for different communication infrastructure. This is the core reason for something called the quantum network and its extended form of a quantum Internet.

Quantum networks are presenting an amalgamating set of occasions in terms of opportunities and challenges across thrilling intellectual, social, and technical fronts; this includes quantum collaboration, networking, communication, and security. The realization of quantum networks self-possessed of several nodes and channels needs new scientific competences for the generation and depiction of quantum coherence, superposition, and entanglement.

Vital to this endeavor is the quantum physics capabilities of quantum interconnects that renew quantum states from one physical system to those of one more in a reversible fashion. This type of quantum connectivity for networks, however, can be attained by optical connections of solitary photons and atoms, thus permitting entanglement distribution and quantum teleportation between nodes.

Within the broad context of the quantum information system, quantum networks play a vital role, together for the formal analysis and the ground-level physical implementation of quantum computation, communication, and metrology.

How to create a quantum network

A quantum internet will transform communication know-how by taking the benefit of marvels from quantum physics, such as entanglement. Investigators are working on skills that allow the communication of quantum bits among any two points on earth. This carries two topographies that the Internet cannot do, at least in its current shape. Firstly, in a quantum network, the entanglement settles to enhanced synchronization among distant sites.

This makes it tremendously suitable for tasks such as clock harmonization or the connecting of aloof telescopes to get improved pictures. The second is that entanglement is integrally protected. If two quantum bits are excellently entangled, then nothing else in the cosmos can have any part in that entanglement. This feature makes entanglement exclusively appropriate for applications that need security and privacy.

Numerous other applications of a quantum internet are already identified, and more are likely to be exposed as the primary networks online. Investigators at QuTech, a collaboration among the Delft University of Technology and the Netherlands organization for practical technical research (TNO) have now set forth phases of quantum internet development illustrious by technological competencies and consistent applications.

The quantum networks, letting the end-to-end transmission of quantum bits, are predictable to be appreciated in the coming years, signaling the beginning of a large-scale quantum Internet.

In the process of constructing a quantum network, quantum information is being produced, processed and stored locally in quantum nodes. These nodes are connected by quantum channels that transport quantum states from place to place with high fidelity and that dispense entanglement across the whole network. With comparatively diffident processing capabilities, one could even imagine a `quantum internet' that could achieve jobs that are otherwise

impossible within the kingdom of classical physics, together with the distribution of quantum software.

Apart from any specific algorithm, there is a significant benefit to be grown from quantum as opposed to classical connectivity between nodes. A network of quantum nodes related by classical channels encompassed of k nodes each with n qubits has a state space of dimension k^{2^n} , whereas a fully quantum network linked by quantum channels has an exponentially greater dimension 2^{kn} . Quantum connectivity also offers a possibly influential means to stunned size-scaling and error-correlation problems that would otherwise border the size of machines for quantum processing. At any phase in the growth of quantum technologies, there will be the largest size achievable for separate quantum processing units, which can be outdated by connecting such units collected into a completely quantum network.

Quantum internet consists of three essential quantum hardware elements

To construct a quantum network, the most important thing which we need is a quantum channel that would rope the transmission of qubits. To visualize it, take an example of standard telecom fibers since they are currently used to communicate light from one place to another. Thereafter, we essentially require the means to outspread these short distances. Quantum channels are characteristically lossy. For instance, the transmissivity of fiber optical channels scales exponentially with distance.

This scaling has robust insinuations for applications since, for both entanglement and key distribution, the attainable rates can at utmost be proportional to the transmissivity. Henceforth, to spread longer distances, in-between nodes named quantum repeaters are necessary. Such a repeater is located at sure intermissions along with the optical fiber joining. This concept permitting qubits to be transmitted randomly over long distances. In the coming times, influential repeaters may also twofold as long-distance routers in a quantum network.

Finally, we need the end nodes; what it means is that the quantum processors are associated with the quantum internet. These may range from enormously modest nodes that can only make and measure single qubits to large-scale

quantum computers. Additionally, end nodes may themselves act as quantum repeaters, though this is not a prerequisite. Quantum internet is not destined to substitute classical communication but somewhat to supplement or extend it with quantum communication.

In future, quantum repeaters could be used to transmit quantum information over extended remoteness (they are still in use in pockets). In its modest form, a quantum repeater works by first producing entanglement among the repeater and each of the end nodes separately.

Instinctively, this can be done since the distance of each endpoint to the repeater is still adequately small to permit the direct entanglement cohort by transmitting photons over telecom fiber. Subsequently, the repeater is fundamentally used to teleport one of the qubits which are entangled with node 1 onto node 2. In technical terms, this process is expressed as entanglement swapping.

The entanglement swapping permits the formation of entanglement over distances at which direct communication or the transmission of information is infeasible. After launching long-distance entanglement, information or rather a data qubit may here and now be directed or sent utilizing quantum teleportation.

The exact employment of a quantum internet may, like for a traditional network, be enhanced for distance, functionality, or both. The term network usually mentions circumstances that drive outside point-to-point communication.

The objective of a network is to offer any end nodes which are linked to the network to exchange data, essentially producing three end nodes. This is the instance of the slightest of the right network. External to the research laboratory, only righthand first stage repeater networks have been realized, that too in bigger cities of Europe and the US. Two solitary far-away end nodes have also been linked via satellite.

Quantum communications, Quantum Key Distribution (QKD) and Quantum Internet

Developing quantum communications is a prodigious development in securing data transfer, but if you don't shape a network to enable the

exchange of information, the efficiency and usefulness of that communication are imperfect.

Comparable to the method, classical networks exchange communications and data among different inter-connected entities, and a quantum network allows the secure transmission and exchange of quantum communications (quantum cryptographic keys) over fiber optic cable among different, physically-separated quantum processors, or endpoints. [Figure 8.15](#) showcases the quantum communication process:

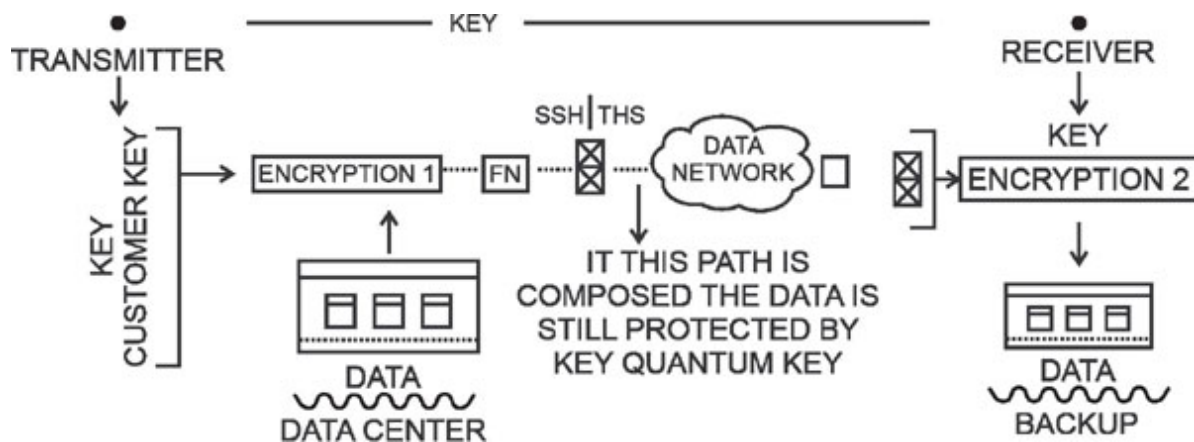


Figure 8.15: Quantum communication process

The QKD network works by conveying an encoded key in the arrangement of quantum bits (qubits) among endpoints over a fiber optic cable. The qubits are characteristically polarized photons, which can travel effortlessly lengthways with fiber-optic cables. Any effort to interrupt the quantum key abolishes the qubit's subtle quantum state and the information it holds, forewarning the endpoints that an intrusion happened. The detectability of the intrusions is what safeguards the security of the transmission.

The trick, or rather the challenge, of a quantum network, is that quantum communications are restricted by the distance photons can travel over fiber-optic cable, usually about 100km. To resolve this problem, Quantum Xchange has advanced trusted nodes, which function in a similar way a repeater does in a conventional or classical communications network. At each node, the key is decrypted into classical bits and then placed back into its quantum state for further transmission to the next node.

[Quick Facts](#)

The Quantum Xchange Trusted Node and QKD handling

The Quantum Xchange Trusted Node is constructed of more than two standard QKD system's endpoints located inside a shared secure housing, until continuous path of Trusted Nodes associates two inattentively parted endpoints. The endpoints can share quantum-derived key information over this much longer path. Among consecutive nodes, key information is endangered by the same quantum mechanisms as in a singular QKD system. While within a node, key information is protected by encryption with locally-generated keys and a secure borderline that averts tampering. And since a Trusted Node can include several QKD endpoints, the network topologies that can be realized by the Trusted Node are essentially limitless.

In the era when the global race is on to grow quantum capabilities ranging from quantum computers, quantum cryptography, quantum simulation, quantum sensing, quantum-safe cryptography, and quantum communication, building quantum networks is yet another milestone and endeavor.

A few countries in Europe have already made some treads toward this goal, though for the briefer distances. The US is also not behind in the race; they are showing progress with something called Quantum Xchange's Phio network and associated trusted node technology.

Fundamentally, the following technologies are used to achieve the same. Below is the flow mentioned. There may be a change in the sequential order between the points, sequence, and a bit of customization may happen based on vendor-specific implementations:

- QKD uses the quantum properties of single photons to exchange bits between two locations.
- QKD keys are derived from the exchanged quantum information.
- QKD security is proven. Security is based on the laws of physics, not on difficult math problems.
- If any effort to intercept the key is noticed because of a change in the quantum state, corrupted keys are rejected.
- Keys can be constructed as truly arbitrary, dissimilar conventional PKI keys, and are not susceptible to future computational hacking.

Pros and cons of expanding cryptographic infrastructure

Cryptographic infrastructure is complex as it involves software, crypto processors, hardware customization, and communications infrastructure. Any changes to this infrastructure must be carefully planned and tested to ensure compatibility with existing components.

Examples of these infrastructures are crypto-processors, which are hardware chips carrying several crypto functions such as encryption, signature generation, and hashing. They are embedded in many devices such as point-of-sale systems, automated teller machines, TV set-top boxes, smart identity cards, and smartphones (SIM cards).

These cryptosystems rely on the current cryptography standards depending on the applications. For lightweight applications with lower amounts of data, these crypto-processors mostly rely on small crypto algorithms such as AES, LED, and PRESENT.

Moving to quantum-resistant crypto primitives will affect the performance of these crypto processors since they involve more computations and even render some processors obsolete. Organizations may be required to buy new hardware with upgraded crypto-processors to handle the increased workload.

Teleportation

Teleportation is the theoretical idea of repositioning any matter from one site to another without moving through physical space. Quantum teleportation transmissions quantum states rather than objects. Let's try to understand it through an analogue of Patanjali, Madhav, Alice, and Bob.

Patanjali's latest task is to teleport a secret qubit state to Madhav. To attain this task, Alice and Bob need an entangled qubit pair and a classical information channel, like a phone or email, as shown in [*Figure 8.16*](#):

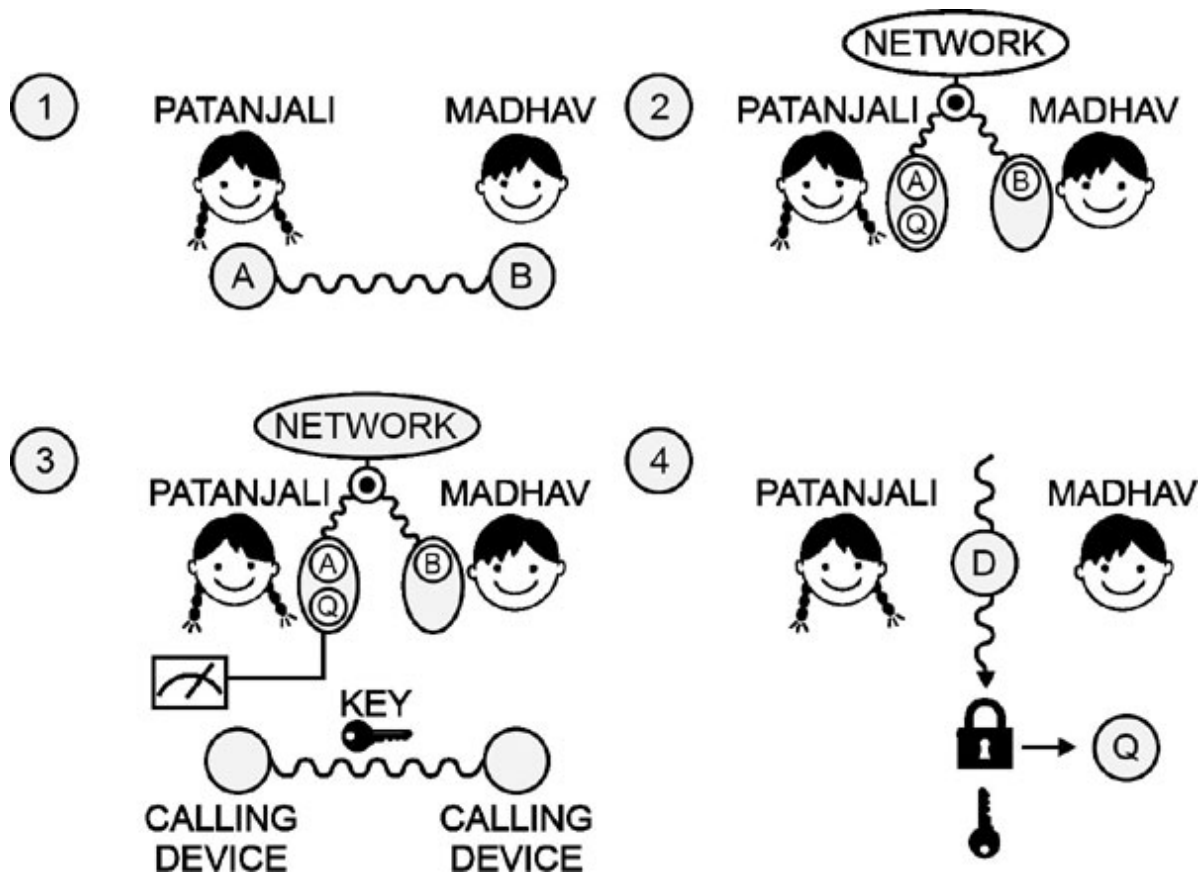


Figure 8.16: Quantum teleportation-- analogue of Patanjali, Madhav, Alice and Bob

Patanjali produces an entangled pair of qubits AB. Patanjali keeps qubit A and Madhav takes qubit B with him, itinerant to a new terminus. These qubits are used as a teleportation channel. Any information imprinted on one of the qubits instantly affects the other, regardless of distance.

The task needs Patanjali to teleport the quantum state of assignment qubit Q to Madhav. In groundwork, Patanjali syndicates qubit Q with her entangled qubit A. This act concurrently changes the state of Madhav's qubit B.

Patanjali does a Bell basis dimension of qubits Q and A on her own and becomes one of the four likely outcomes: 00, 01, 10, or 11. At a distance, this dimension also affects the state of Madhav's entangled qubit. Patanjali then makes a phone call to Madhav notifying him of her 2-bit measurement consequence.

Madhav has a code-table that maps 2-bit measurement results to a consistent quantum gate(s). The message Madhav gets from Patanjali serves as a key to select the precise pre-defined quantum gate(s) he must apply to his qubit Q.

By putting on the suitable gate(s), Madhav can re-form the original state of mission qubit Q without having possession of it.

Blockchain

Decentralization frequently appears to be a problem in attaining a commercial scale. Where scalability has been attained by Blockchain, decentralization has often been traded off, making it an indefinable perfect idea that we all want the world to move to. But Decentralization has been the backbone technology attached in popularity over the years for blockchain.

Decentralization fundamentally defends contrary to data loss; it can also safeguard that data possession is not central. It is starting to turn out to be progressively significant for each customer to achieve their data, economic, and social identity.

The mechanism of handling identities within a decentralized network is named self-sovereign identity management. Cheers to the Blockchain framework, the clue of self-sovereign identity is starting to appear less subtle than it once was. In a day and age where data is the well-thought-out novel oil, and attention (in the media and social media) is considered the new gold, identity must be possessed and managed by each individual.

Blockchain is a kind of database known as a dispersed ledger that functions on a consensus basis. When a user submits a new data block to the blockchain, the mainstream of other users must settle that it is valid.

The database does not have a central administrator. Each data block in the ledger is related to the preceding block by a cryptographic algorithm named a hash, with the related blocks starting a chain.

This is the main reason why it is named Blockchain. As per the standard of blockchain, each user holds a copy of the distributed ledger on their individual computer and the data is replicated and harmonized across all duplicates of the ledger in real-time. If one of the computers land a copy of the distributed ledger flops or comes below bout, the further computers continue to uphold the database.

Blockchain may be best recognized as the distributed ledger know-how that reinforces the digital currency Bitcoin, but it could also be used for a host of other determinations that include transmitting data securely.

These comprise payment processing, online voting, performing contracts, signing documents digitally, creating demonstrable audit trails, and recording digital assets such as stocks, bonds, and land titles. Its potential for application within the transaction-based financial services industry is principally huge, but it is pertinent to organizations in every sector.

Blockchain could potentially upturn the whole business model in certain sectors by authorizing the development of virtual organizations, also recognized as **Decentralized Autonomous Organizations** (DAOs). DAOs function through computer programs known as smart contracts that transmit the wishes of human shareholders by automatically performing the terms of a contract. For instance, transferring cash or other resources.

In theory, blockchain is very safe. Since each transaction on the blockchain is digitally contracted, it provides a guarantee that only a certain party could have recorded it. Besides, the data is authenticated by a widely held number of other users on the system.

But if the widely held users on the dispersed ledger turn out to be corrupt, it is possible to break the chain. Blockchain has other inadequacies like slowness that need to be fixed. Additionally, issues like weighty ingesting, confidentiality issues (other users on the ledger prerequisite to be able to see data if they are to authenticate it), an absence of standards governing the business, and its incomplete scalability to date. However, as time progressed, these issues are getting resolved.

Specified the chances and risks related to blockchain, it is not astonishing that the technology has involved the consideration of governments, central banks, and regulators. In June 2016, the European Securities and Markets Authority proclaimed that it was referring to whether dispersed ledger technologies should be used in the securities markets.

Bitcoin, Blockchain and their working

Bitcoin is one of the currencies of blockchain. The imprint of the blockchain explains how Bitcoin works since it is essentially related to the Bitcoin. However, blockchain technology is appropriate for any digital asset transaction where exchange happens online.

Internet commerce is wholly knotted to the financial institutions helping as the trusted third party who processes and facilitates any electronic transaction. The part of the trusted third party is to authenticate, protect and reserve the transactions. A certain fraction of deception is inevitable in online transactions and require arbitration by financial transactions. These consequences in high transaction costs.

Bitcoin uses cryptographic resistance in its place of trust in the third party for two eager parties to perform an online transaction over the Internet. Each transaction is safeguarded through a digital initial. Each transaction is driven to the public key of the overseer and digitally signed using the private key of the informant.

To spend money, the proprietor of the cryptocurrency wants to show possession of the private key. The object getting the digital currency confirms the digital signature, thus the possession of a conforming private key on the transaction using the public key of the dispatcher.

Each transaction is transmitted to all nodes in the Bitcoin network and is then logged in a public ledger after confirmation. Each solitary transaction desires to be confirmed for validity before it is logged in the public ledger. Confirming node requires to safeguard two things before recording any transaction, which is as follows:

- Spender owns the cryptocurrency.
- Spender has sufficient cryptocurrency in his/her account.

Though, there is a question of upholding the order of these transactions that are transmitted to each other node in the Bitcoin peer-to-peer network. The transactions do not derive in order in which they are made and henceforth there is a want for a system to make sure that double-spending of the cryptocurrency must not happen. Seeing that the transactions are approved node by node through the Bitcoin network, there is no assurance that instructions in which they are obtained at a node are the similar order in which these transactions were produced.

This fundamentally means that there is a requirement to grow a method so that the whole Bitcoin network can agree concerning the sequence of transactions, which is an intimidating task in a dispersed system.

Blockchain: the perfect partner for AI data

The supremacy of AI is depending on the admission to, degree, and fineness of the data it can process. So, while AI redefines the systems of business rendezvous, blockchain is recalibrating the schemes of records. Together, they will remap structural limits, moving them from siloed verticals to multifaceted processes to function professionally and in the process, releasing large amounts of currently trapped value.

Meeting the challenge--capturing the opportunity

AI systems are exceedingly altering the nature of services and experiences for customers. The present phase of the climbed use of AI has engrossed on the value and services separate organizations can carry to people. This internal application focus results from the natural marketable importance of individual businesses trying to get-up-and-go for profit and development.

Corporations typically start with what they can do and control along with the data they store in their systems. Infrequently, a single company widely meets a user's all-inclusive requirements, and utmost contemporary AI enactment efforts imitate that limitation. For instance, take the instance of a family shifting from one city to another, one of the very common cases in day-to-day life. Very common, yes, but is it handled with ease? Probably not.

Most organizations emphasize solitary objectives, including banks, insurance corporations, realtors, examiners, agents, retail, education systems, marketing companies, mail systems, the division of motor automobiles, tax establishments, and more.

To bring holistic integration in the chain (or to the supply chain) and sort things out, companies have started implementing AI along with data sharing. Now again go back to the example of shifting a family from one city to another, if the requirement of the family in the form of data is inputted in the AI system, life becomes easier and easier. There may be multiple systems and organizations involved in making this possible.

However, it should not be a problem until this could be done in a secure, transparent, and trusted way. This is exactly what AI along with blockchain is doing. Imagine a world where, through a blockchain-based system, where each party tangled in the move could see the data and information about the

end-to-end relocation procedure with suitable permissions approved by the stakeholders.

The corporations could access just the data they want, and since the information echoes blockchain's improved trust levels, parties no longer need to involve in messaging and reconciliation. As organizations extensively instrument these types of models, the likelihood of AI systems will raise expressively.

In the future, broader admittance to data across an ecosystem, and the developments in automatic business logic via keen contracts could permit new-fangled and better access for AI machines to cross-business ecosystems and bring more complete solutions to clients.

Quantum Blockchain

Blockchains, along with bitcoin, use tremendously progressive encryption which results in enduring, immutable records. The encryption is so robust that there is just no method anyone utilizing a classical computer can break down.

A quantum computer though could hypothetically crash right through all contemporary forms of encryption, including that used by bitcoin. So, for instance, an unlawful activity can be done with a quantum device by breaking private keys, snipping cryptocurrency and causing all types of chaos.

This knowledge enables us to understand how quantum computers pose an existential threat to the currently used asymmetric cryptography. Further, in asymmetric cryptography, a private-public key duo is produced in such a way that the dual keys have a mathematical relationship among them. As the term proposes, the private key is reserved as clandestine, while the public key is made publicly obtainable. This lets individuals harvest a digital signature by the use of private keys that can be confirmed by anyone who has the consistent public key. This arrangement is very common in the financial industry to demonstrate the authenticity and veracity of transactions.

The sanctuary of asymmetric cryptography is founded on scientific theory or rather a principle named a one-way function. This principle prescribes that

the public key can effortlessly result from the private key but not the other way round.

All recognized standard algorithms to initiate the private key from the public key need an astronomical amount of time to do such computation and are consequently not applied. Though, as we know, in 1994, the statistician Peter Shor's idea of the quantum algorithm fundamentally can break the security supposition of asymmetric cryptography. This means that anybody with an adequately big quantum computer could use this algorithm to stem a private key from its conforming public key, and thus, fabricate any digital signature.

In a mainly data-driven world, we essentially need an infrastructure that can bid intelligence and do simulation competencies that can blow into the fruitfulness of the data. Though, deprived of a framework to safeguard that the data being stowed and used by these machines are of high data excellence and integrity, all the intelligence would be of slight use.

Deprived of the intelligence, high-integrity data can take the weight off your feet on a computer starved of being of much use too. So, we want both know-hows to go mainstream to pact with the data challenges in the forthcoming time. Though, the two technologies are on a smash progression. Cryptography is an essential part of equal importance in quantum computing and Blockchain. There is a possible situation where the Blockchain solutions we have nowadays will be superseded as soon as quantum computers scale and develop mainstream.

Consequently, notwithstanding the commercial case for these technologies, we may only see one of the dual technologies being pertinent in the future. That is a situation that we could overwhelm though. Blockchain can still be pertinent if it can overwhelm another technology fence.

How likely is a quantum attack on bitcoin?

Obviously, a quantum attack on bitcoin to hack and destruct is not possible soon. Quantum computers are remarkable; they are also tremendously expensive and problematic to shape due to the error process, decoherence, and so on.

They are also actually not applied for whatever above and beyond progressive calculations, and they can only originate in R&D centres of

companies, academics, and government agencies. It is projected that we have about 10-15 years left from a quantum computer which is influential and sufficient to smash bitcoin's level of encryption.

At the moment, the threat from quantum computers may not be looming, but when a full-fledged version of it comes (it will come), can it make blockchains outdated, and bitcoin fated? Probably not, but the blockchain version of quant will emerge (actually... it is already emerging).

The progression of quantum computing means that encryption over-all will have to change, and that comprises all the know-hows that make use of encryption, like blockchains. Numerous projects such as **Quantum Resistant Ledger (QRL)** are being designed by teams working on preventive answers to quantum attacks. Bitcoin and other cryptocurrencies will likely implement additional progressive, quantum-proof blockchains in the future. There is a lot of research already happening in the area.

We will discuss one of the progress of import which happened recently. But before going into the details, just imagine a scenario.

A felonious person who is armed with a practical quantum device would be able to do opposite calculations much faster, which would allow them to falsify signatures, impersonate other users and get access to their digital possessions. In the situation of mining, such a malevolent actor could take over the process of updating the ledger, change transaction history, and double-spend coins.

To encounter such a situation, the Russian investigators recommended that the architects of encrypted systems flinch captivating defenses against this danger immediately. One resolution could be substituting conservative digital signatures with quantum-resistant cryptography. The quantum-resistant cryptography is the type of safety algorithm explicitly intended to endure an attack from an adequately influential quantum computer. Another preparation, the Russian physicists anticipated, will only be obtainable with the arrival of a quantum internet. This potential wireless communication architecture, founded on the joining among distant entangled quantum particles, will solve a wealth of novel blockchain models and designs.

Quantum networks and classical digital networks: impact and advantages in brief

Today's classical digital network on which the Internet is based is a play area for hackers. From unconfident communication links to incompetently protected data in the cloud, susceptibilities are everywhere. But if quantum physicists are working on these limitations of the network, it gives rise to the belief that such faintness will soon go by the means of the dodo.

The physicist wants to shape quantum networks sporting full-blown quantumness, where data is shaped, kept, and stirred around in ways that mirror the bizarre behavior of the quantum flora and fauna. Go back to our well-known symbolic cat (recall ... Schrödinger's cat paradox) that can be together lifeless and thriving or atoms that can use spooky action at a distance.

Unfettered from numerous limitations of classical networks, these systems could deliver a level of confidentiality, safety, and computational wallop that is unthinkable to attain with today's network. Though a completely understood quantum network is still a far-off dream, new advances in conveying, storage, and operating quantum information have persuaded some physicists that a humble proof of principle is looming.

The progress in quantum network endeavour is happening quickly. For instance, the first phase would be a quantum network using normal optical fiber to attach at least three small quantum devices about 50–100 km apart. When it comes to advantages one thing which we are sure of is that, quantum networks will do more than QKD. The next step would be to the transmission of quantum states unswervingly among nodes. While qubits encoded utilizing a photon's polarization can be directed over optical fibers, using such qubits to transmission great volumes of quantum information is problematic.

Future-ready: quantum cryptography and its influence in coming time

As there are two faces of every coin, Quantum cryptography also has pros and cons. The pros are that it would give a new dimension to cryptography, whereas the con is when utilizing in the incorrect way, it will be dangerous

for society. Therefore, exploration is also being conducted into what is ever so often named post-quantum cryptography. It might be thought of or termed as quantum-resistant cryptography. These are systems working on normal computers but originated on problems that are not in the concealed subgroup class.

These problems comprise resolving systems of multivariable polynomials, discovering the quickest remoteness from a point to an n-dimensional twisted network (or in other words, n-dimensional skewed grid) of other points, and its outcome would be the nearby bit string to a group of other bit strings.

Four to six years is also not an irrational amount of time to wait for a novel cryptographic standard on which post cryptography techniques would rely on. But the private and government agencies have started working on it. Because, government agencies are worried about shielding data that might have to endure protected for decades into the future, so they are preapering now for computers that could still be ten or twenty years into the future.

If you are concerned about quantum offenders receiving your credit card number off of the Internet. The cryptographers were supposed to be prepared for this. And you will be able to keep spending securely without buying your quantum computer.

Story 1: Quantum computing-based recommendation systems

Many auspicious quantum-algorithms have been projected for an extensive range of characteristic ML and AI tasks together with grouping, dimension decrease, unremitting estimation, classification, anomaly detection, recommendation systems, data creation, and all-purpose optimization problems.

The behavior of quantum-computing devices is best designated by linear algebra in very high-dimensional vector spaces, which is a capable starting-point for chasing improvement of traditional ML/AI. Likewise, quantum processing units (CPUs) can produce quantum states with likelihood distributions that are firm to sample from classically. This feature is supposed to permit QPUs to know data-patterns that are hard to identify

classically. Numerous additional classically stubborn ML/AI problems could feasibly advantage from quantum enhancements, with, for example, belief spread, tensor factorization, or submodular problems inside the quantum AI domain.

We can take one example of the research done by Iordanis Kerenidis and Anupam Prakash in 2016. The computer scientists Iordanis Kerenidis and Anupam Prakash published a quantum algorithm that resolves the endorsement problem exponentially quicker than any recognized classical algorithm.

They attained this quantum speedup in part by making the problem simpler; in its place of filling out the complete matrix and classifying the solitary finest product to indorse, they advanced a method of sorting operators into a minor number of groups. When Kerenidis and Prakash worked, there were a minor number of cases of problems that quantum computers appeared to be bright to resolve exponentially quicker than classical computers.

Most of those examples were focused and there were small problems designed to play to the fortes of quantum computers. Kerenidis and Prakash's outcome was thrilling since it provided a real-world problem that people were concerned about where quantum computers outstripped traditional ones.

Story 2: Disease spread, bacterial growth, and QC

Numerous quantum start-ups are now constructing software that they will use the gigantic computational power of quantum computing to produce medicines for the variability of situations. It's not too distant of a bound to trust that the power and the rapidity of quantum computers could be leveraged to contest a fast-developing disease, like Ebola, coronavirus, and multiple others. Preferably, epidemiologists would give the drug inventors an instant cautioning of a disease eruption and as much data on the condition as likely.

The creators would take over from there. Investigators recommend that present hybrid quantum-classical methods mean we're not too distant from this right now. As the technology grows, we might guess that instead of months or years to construct and produce a vaccination or medicine for an

epidemic, it may only take weeks or even days. For instance, take the example of the Corona Virus; it spread quickly and harms humanity on a wider scale globally.

Therefore, there is a need for a quick vaccine, and a drug is required to stop it, which needs a lot of research and consequently data crunching. Unfortunately, this is not happening fast enough through conventional systems. Hence there is a delay in the overall process. But if the QC is functional and put into these types of scenarios, the overall situation would be different.

In an upcoming epidemic, a quantum computer could permit out vaccines to healthcare contributors in the greatest effectual way imaginable, something that would save valued time and a method to regulate the blowout of the infection. This kind of harmonization would be categorically vital for a country such as the US, India, and actually to the whole world, which is mainly hard-hit by a coronavirus. Also, the medication in the hands of the correct individuals at a precise time would save an indescribable amount of lives.

Quantum computers could also be used to place the correct resources and the accurate alternative workers to help prevent the binge of disease. Several areas under quarantine, for instance, has a requirement of special considerations. We can count some as instant food, water, and fuel. Juggling these stream worries is logistically terrifying. For these types of instances, a quantum computer is capable to be well-matched to categorizing those provisions and transporting (through virtual or physical optimized channels) them to precise places.

The best defense against an epidemic or mass bacterial or viral disease is to halt them before they jolt; quantum computers could continually monitor data to reduce any flare-ups. From analyzing well-being figures to using natural language processing to analyze social media communications for signs of disease eruptions, quantum computers may brand a watchful guard contrary to the next destroyer epidemic.

Story 3: How quantum computers will transform travel

While marketable quantum computing has a few years left in the future, corporations are consecutively trying and producing branded algorithms today. For instance, since 2016, a global community of users has run more than 100 billion implementations on IBM quantum computers, accessible via the cloud.

Founded on these experiments, more than 200 technical papers have already been published. Quantum computing finds its best use cases in the travel industry. The quantum algorithms are very much well-matched to examine the travelling salesperson problem. This is a shared thought trial for quantum investigators that discovers the idea of a salesperson who is eager to find out how to style sales stops in numerous towns as proficiently as conceivable. And each time a new-fangled city is added to the salesperson's list, that difficulty surges exponentially. It turns out that quantum computers are hypothetically well-matched for resolving that task, perhaps more so than traditional computers.

Enhancing contextual personalized facilities for airlines, it's important to distinguish services offered by multiple vendors in any specific domain. For instance, in the airline industry, advanced customer experience would potentially provide get-up-and-go and handsome incremental income to the specific company. Broadly, good tailored customer engagement and services need the following four specific steps:

- Gathering and mining data, together with customer data and transactional data.
- Execution of data engineering to shape customer data landscapes.
- Training customer breakdown models based on client and trip context features.
- Scoring and identifying the finest suggestions dependent on customers' separate travel contexts.

Today's tailored offering systems repeatedly fall short of living up to their ability, mostly because of limits in the customer segmentation phase. Current segmentation approaches frequently trust on fundamental customer features such as demographic and sales data, but do not comprise contextual data, dipping the relevance of the optional offer.

The current systems also have a deficiency in providing multi-dimensional segmentation to efficiently imprison contextual differences in favorites, intent, and behavior of travelers. One of the whys and wherefores for the nonappearance of contextual features is inadequate computing measurements and scale to hold the high number of data elements obligatory to shape complex segmentation models.

Story 4: Quantum computer and personalized customer experience

As complexity in digital marketing increases, establishments are likely to see upsurges in the number of users for whom they need to generate custom-made experiences. It is one thing to monogram a landing page for one client part, but it's a different encounter when you have hundreds of personalities, numerous geographies, a dozen places, and thousands of places where personalization is desirable. Therefore, personalization approaches essentially to scale to be reasonable.

Quantum computing may resolve these problems, pleasing to the eye the individualization process by auxiliary better-off customer segmentation, including more complex client landscapes for multi-dimensional curbside segmentation, and permitting for advanced specificity in relative profiling to advance personalized aids. Also, for refining the correctness of machine learning models that bring insights and interpretability of outcomes to help marketers or customer service agents and help them to improve their understanding related to customer behaviour, quantum computation along with machine learning would play a major role. The interconnection between customer data, algorithm, artificial intelligence, quantum computers and delighted passenger's is something which is worth to keep watch on in near future. Further, this leads to enabling the identification of a Exceedingly better number of finely-tuned customer segments that are incontrollable for classical computers in some situations, whereas it is very easy for QC in almost all the scenarios to improve their understanding.

Use case 1: Quantum prediction

The learning of the financial market is extremely subtle, indeterminate, and changeable and is contingent upon many exterior factors. There is not at all a well-defined analytical process to forecast the financial market.

Subsequently, there is no distinct mathematical or analytical answer for financial market forecasts; people usually put on mathematical approaches and computer simulation to get improved outcomes.

The introduction of computer science to resolve the financial glitches is known as computational finance, where computer simulation stabs to discover out the greatest approximate consequence of a given problem. But the old-style traditional computers have restricted computational capacity with reverence to the difficulty of the problem. Henceforth, the dispensation time is contingent upon the rapidity of the processor and the swiftness of the processor hinge on the number of commands executed in each clock retro.

Therefore, if we want to upsurge the proceeding volume of the traditional computer, we have to enhance more and more computational gears to the physical system which eventually upsurses the hardware mechanisms of a system.

Quantum computation having the characteristic parallelism owing to the principle of superposition of quantum states and henceforth it is likely to upsurge the solution speed for many computational problems. In recent years, in some cases, the quantum algorithms, similar to Grover's algorithm for unstructured database search, Shor's factorization algorithm, and Deutsch's algorithm resolve the problems exponentially faster than their classical counterparts. Hence it is worth introducing quantum algorithms in the background of computational finance.

So, acquaintance with quantum ideas in the financial area may lead to a better estimate to make a decent choice in a specified period. For instance, now few research papers are available which discussed the financial health of the organization or predicted the market price of the stocks by using the algorithms like quantum search.

Precisely forecasting the weather and even the stock market could be on the cards as experts started using quantum computing to predict the future. Investigators from Queensland's Griffith University and Singapore's Nanyang Technological University have worked together to grow a quantum

device that can precisely measure likely futures for a set system. It means following the movement of photons, solitary elements of light - through the system, and calculating the consequences.

A few of those further multifaceted processes comprise forecasting the weather beyond a few days, which has possibly millions of variables, as well as effects like stock markets and traffic patterns. The Griffith University team accomplished to also show that using quantum computing took up less memory space than current simulation approaches using old-style computers.

The Griffith University trick functions and theory was defined by physicist Richard Feynman, whose experiential result explained that a particle itinerant among two points does not automatically trail a solitary track. In its place, it follows all paths due to superposition.

Use case 2: Caixa Bank journey to grow risk classification model through quantum computing

Caixa Bank has taken an additional footstep and advanced the first machine learning algorithm to classify risks in Spanish banking by the effective use of quantum computing. The bank turns out to be the first Spanish financial organization to put on a hybrid computing framework. To classify credit risk profiles, this framework combines quantum computing and traditional computing in different stages of the calculation process. Fundamentally, Caixa Bank used a public data set of around two thousand artificial users, with a similar profile to current customers, but with information configured explicitly for the test.

The bank is making developments in risk and regulation scenario-based simulations. At the same time, it is increasing its machine learning capabilities, behind progressively multifaceted algorithms by fulfilling the system's need for large quantities of data to learn. For analysis, it is using quantum computing-based applications. The outcomes of this test proves that hybrid computing can attain outcomes comparable to those presented by the conventional solution in a smaller amount of time.

Hybrid computing uses QC's exponential computing gain to do multifaceted calculations of parameters enhancing machine learning algorithms and

syndicates them with traditional computing approaches to make the most out of both systems. With the application of hybrid algorithms which comprises quantum and classical in risk analysis, the bank can spread similar conclusions as to the classical method in a much smaller amount of time.

Before this solution, Caixa Bank industrialized a project to bring out risk assessment simulations for financial possessions using quantum computing. The bank applied a quantum algorithm talented in determining the financial risk of two groups shaped explicitly for the project based on real data, one containing mortgage and the other, treasury bills.

Use case 3: Quantum computing and the automobile industry

Big car and tech corporations are scrambled in the investigation and growth of Quantum computing either by partnerships or connotations, as no single object has the competence to do what is essential, alone.

The automotive industry has made it in the newscast for being one of the few essential industries to tool quantum computing to resolve real-life problems linking to traffic management or autonomous vehicles. Apart from the logistics and production line linking problems, quantum computing will be furthestmost accommodating to autonomous vehicles from testing and prototyping phases to the sealable phase.

Autonomous vehicles need marvelous computing ability obviously for processing and wireless communication. This is where quantum computing would come into the picture and help. For instance, it could be used in optimizing direction and forecasting for autonomous vehicles.

Further, in a big country with massive roads spread across several miles, incredible data processing will be essential and henceforth the requirement to go quantum. Quantum computing will not only aid in the development of autonomous vehicles but also transform the complete mobility sector, where cars will be smart enough to interconnect with themselves, thus removing the need for traffic signals and simplifying package deliveries without much hassle.

Contemporary problems requiring immediate solutions are meticulous and the furthestmost relevant statement when keeping in mind that quantum

computing is futuristic a solution to current and future automotive problems. All the main automotive builders are viewing vehicles that run on batteries rather than the old-style fossile (diesel/ petrol) energies and transport automation in the procedure of direction-finding and power the vehicle itself. This radical change from remnant energies and driver-centric vehicles to battery and drive less or autonomous vehicles is as multifaceted as it echoes.

Numerous autonomous vehicles have been factory-made and are being frequently tested across many parameters, only to bomb in certain areas. The unified autonomous driving involvement is a little bit missing from being completely commercial and safe. The level at which these automobiles are technologically advanced is pretty high, and going forward, it is going to be even further multifaceted. This is where Quantum Computing comes in and aids automotive corporations with complex controls and approximations by which a dependable autonomous experience can be attained. Let's have a look at the progress and directions of automotive companies, and what they are doing to get the maximum benefit of QC.

Corporations like Volkswagen have started captivating minor steps to integrate QC into real-life applications. In association with tech giant Google and D-wave, VW plans to instrument quantum computing to decrease traffic overcrowding. Traffic data is taken with the help from smartphones and whisperers inside vehicles.

This data is treated in a non-quantum algorithm to decide the traffic compactness in a given area as well as the number of customers needing a carrier. This result can then be effortlessly treated by a quantum algorithm to enhance the number of carriers obtainable versus the number of customers requiring a carrier. Optimization can be completed by guiding cabs or a public transport service carrier suitably to the emphasis area.

This feature of quantum computing can very well be valid to autonomous vehicles of the near forthcoming time where these vehicles can be directed to a specific area of emphasis or can be furtive to evade that area all together mainly in heavy traffic circumstances.

Ford is one more car creator who is looking at Quantum computing as a likely solution to numerous complex problems. The corporation has borrowed skilled team members and is equipped to work on the theme with

other expert personnel. Ford has teamed up with NASA due to which the corporation will be talented to express some complex problems in a practical context, of the automotive industry or precisely to Ford specific problems. These initiatives are supported by NASA's Quantum Artificial Intelligence Laboratory. Teaming up with NASA will deliver Ford a chance to make use of NASA's quantum annealer, which is shared amongst NASA, Google, and the Universities Space Research Association.

Among other such companies, Denso and Toyota Tsusho started their hard work in the arena of quantum computing in 2017, where they projected to measure traffic data concerning time travel for this they uses, vehicle location for about 130000 profitable vehicles in Thailand. This combined exertion from the two big products used cloud-based quantum computers from D-Wave Systems Inc.

The primary focus area will be to help fleet owners optimize the resources and energy utilization of a large number of vehicles. A criterion based on vehicle to route mapping is required to be designed where the vehicles in focus happen to be diesel delivery vehicles. Diesel engines consist of filters that need to be tracked for optimal vehicle performance as well as compliance with the environment.

The filter element can be managed in the best way when the vehicle is having proper driving cycles. When delivery vehicles have multiple stops to make for drop off or face a huge traffic situation, this driving cycle is disrupted. Ford feels that an optimal route for the vehicle to follow while making stops can be found and then be applied across several other vehicles.

Naturally, when multiple parameters are stacked across multiple vehicles, a complex matrix of possibilities like congestion, breakdowns, number of drop off and pick up times, locations, customer feedback, and experiences come into play. A classical computer might be useful in terms of a limited number of vehicles, as well as other parallel parameters. As these parameters multiply, classical computing lacks the necessary firepower to prove adequate, hence, Quantum computing can be utilized in such scenarios. Ford intends to encode these problems into qubits and solve those using quantum computers in lesser time as compared to classic computers. Apart from the fact that quantum computing can provide tremendous help in logistical

calculation, the company thinks that it can be applied in other fields like new material development for batteries as well as chemistry optimization.

Latest research: path to quantum computing at room temperature

Military investigators at U.S. Army Combat Capabilities Research Laboratory forecasted that quantum computer circuits will no longer essentially need tremendously cold temperatures to do the job. It could turn out to be true after around a decade.

One of the main disadvantages/roadblocks of the development of quantum systems is the flimsiness of the odd states of the qubits. Most potential hardware for quantum technology must be set aside at tremendously cold temperatures (close to zero kelvins). As we know, it is required to stop the special states from being demolished by interrelating with the computer's environment. Any communication that a qubit has with whatever else in its environment will begin to misrepresent its quantum state.

Solid-state quantum technology that functions at room temperature appeared distant. While the application of see-through crystals with optical nonlinearities had arisen as the greatest probable way to this milestone, the credibility of such a system at all times continued in question.

But it will change soon; Army scientists at Army Research Laboratory formally confirmed the viability of a quantum logic gate, including photonic circuits and optical crystals, which could work at room temperature. This is the way forward towards building a cheap quantum computer. In the absence of these types of technologies, the future devices that use quantum technologies will need refrigeration to very cold temperatures, hence the overall devices would be expensive, bulky, and power-hungry. However, this research is aimed at developing forthcoming photonic circuits that will be able to operate the entanglement needed for quantum devices at room temperature.

Investigators have engaged in numerous labours to resolve this issue, but a sure answer is yet to be found. At the instant, photonic circuits that include nonlinear optical crystals have currently arisen as the sole possible way to quantum computing with solid-state quantum computing structures at room

temperatures. Photonic circuits are somewhat thought of as electrical circuits. However, the fundamental difference is that they operate on the light in its place of electrical signals (conventional computing systems). For instance, we can make networks in a translucent material that photons will travel down, a bit like electrical signals roving along chains.

Unlike quantum systems that use ions or atoms to stock information, quantum systems that use photons can avoid the cold temperature restriction. Though, the photons must still interrelate with additional photons to do logical actions. This is where the nonlinear optical crystals come into production.

Mind Map

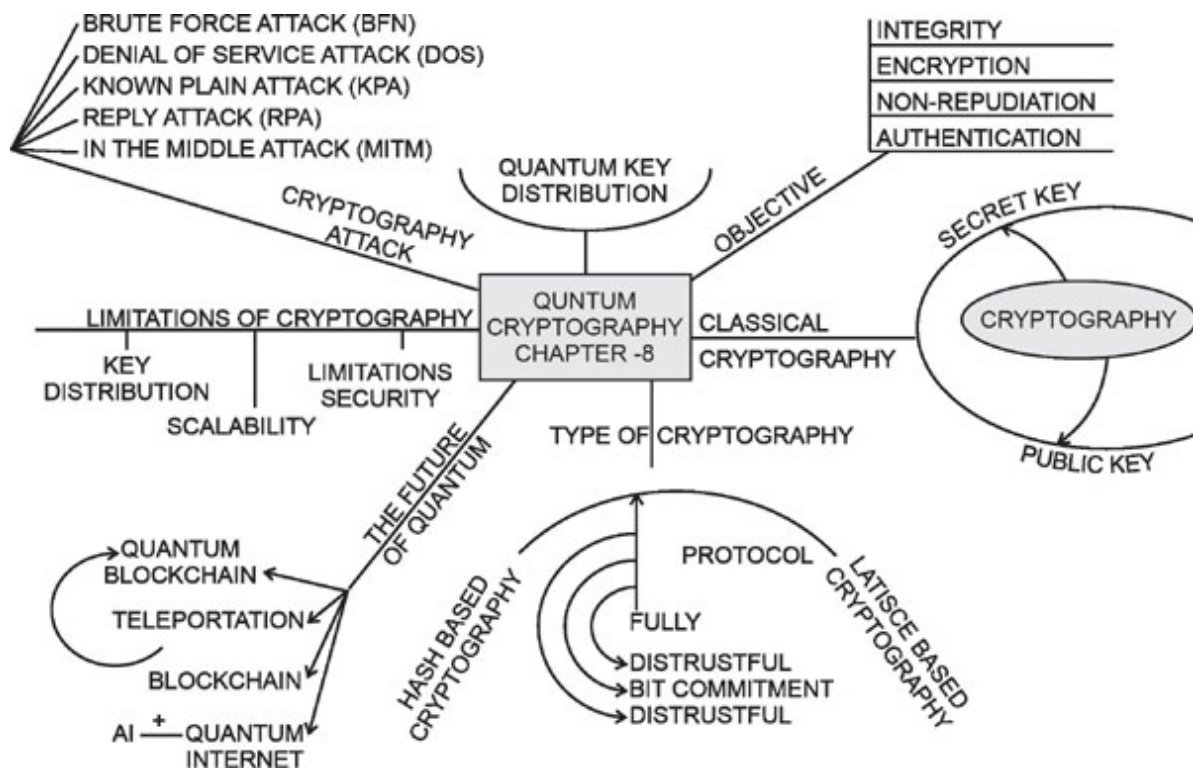


Figure 8.17: Mind map for [chapter 8](#)

Conclusion

It is vibrant that if a functioning quantum computer was to be built, it would directly challenge the security provided by symmetric-key algorithms and

public-key algorithms. Quantum computing would be able to breakdown public-key cryptosystems in insignificant amounts of time.

It is projected that a 1024-bit RSA key could be wrecked unevenly with 3,000 qubits. Given that present quantum computers have fewer than 100 qubits, public-key cryptography is harmless for the predictable future, but this is not a complete assurance.

Quantum cryptography, though still in a budding stage, is sure to have an enormous and radical effect on the world of cryptography and safe transportations. A few years back, quantum cryptography was not in substantial usage in th Fortune 1000 community, but it will probably discover better application in the future years as it develops and the price drops.

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CHAPTER 9

The Architecture of a Quantum Computer

A novel ‘world record’ has been set in the universe of Quantum computing after specialists were able to produce a quantum memory and then uphold it in that state for close to forty minutes at room temperature. Presently, the only quantum computers obtainable for procurement trust on tremendously cold temperatures—just above the toasty side of absolute zero, or about 450 degrees Fahrenheit—and they sell for about \$10 million each. Google and NASA share one made by D-Wave. Though quantum computing is still in its beginning stages, research has been ongoing in which the overall quantum computational processes were performed on a very small number of qubits (quantum bits). Both applied and theoretic research continues, and numerous national governments and military funding agencies provision quantum computing studies to develop quantum computers for both non-combatant and national security purposes, such as cryptanalysis.

The big companies, the Googles, the IBMs, Microsoft, and also some start-ups like Rigetti structure the quantum computing hardware. There are a lot of new start-up ecosystems happening around QC. In the last few years, a lot of money was being pumped by several companies to develop the Quantum ecosystem which is seen either as application development, use case identification, frameworks conceptualization, and realization or at the hardware level, as a development fault-tolerant practical system. There are start-ups constructing code and defining systems to handle 50 qubits or 72 qubits or more.

Therefore, it is important to understand the architecture of quantum computers. Let's deep drive in it.

Structure

In this chapter, we will cover the following topics:

- Introduction

- Different views of QC architecture
- A review of concepts
- Quantum computation
- Journey of Qubit
- What is quantum
 - Bloch's Sphere
 - Comparison of QuBits with classical bits
 - Quantum registers vs classical registers
 - Facts about quantum gate
- Quantum annealing
- Need for looking beyond classical computes
 - Logistics and traffic route optimization
 - Disaster recovery
 - Chemical and material science
 - Finance and Economic
- A review of quantum hardware
- General discussion on Core Qubit technologies development: in the background of history
- The desire of building a quantum computer
- Architecture of QC
- General discussion on QC
- Quantum annealing and computers
- Quantum simulations
- Universal quantum computing
- Multiple types of Qubit technologies
 - Topological QC
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 - Trapped ion QC

- Quantum dot QC
- Nuclear magnetic resonance QC
- Diamond-based QC
- Photon-based QC
- Few additional important technologies for quantum computers and Qubits
- Let's build a quantum computer: elements of a quantum processor
- Practicalities of quantum computers
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- Quantum processor: architectural thoughts, challenges, and steps
- Baby steps for quantum architecture
- Start-up stories: D-Wave Systems
- Story 1: quantum chemistry
- Story 2: quantum compilers
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- Use case 1: material design and quantum computer
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Objective

The objective of this chapter is to give an overview of architecture of the quantum computer. The chapter provide the details of:

- Why a quantum computer has both conventional and quantum fragments?
- Why is a quantum algorithm's usage conventional and quantum data?
- What is the architecture of quantum computers and what are different ways to achieve it
- Building blocks of quantum computers
- What are the different types of Qubit technologies?

This chapter covers different types of proposed and working quantum hardware and their potential along with their applications.

Introduction

The key challenge in implementing the quantum computing system is to recognize and isolate quantum mechanical systems that follow the dream of extended coherence time so that it could be operated or manipulated and further united in a scalable fashion. Multiple technologies exist for this purpose, and in this chapter, we will touch base on some of them. We will have a look at superconducting qubits, quantum dots qubits, trapped ion qubits photonic qubits defeat base qubits, topological qubits, NMR-based qubits, and a few more. As we know, long coherence time, error correction technologies, and scalability along with the cold environment play a significant role in the development of quantum computers.

In whatever form, we develop qubits, but these factors would be part of the overall drama. Now superconducting, iron trap, photonic crystals, and topological techniques are the leading platform for quantum computing hardware. Recently we saw very fast progress in the field of the development of QC and one of the reasons for this is the progress in the field of material science which fuelled nano-superconducting hybrids, and its progress is remarkable. Experts were now able to grow extremely clean versatile devices featuring various control and manipulations. As an outcome, the hybrid nature of qubits is gaining momentum to fulfil the requirements of stable QC.

Different views of QC architecture

Exploration at Intel® Labs has shed light to the development of something called Tangle Lake, a superconducting quantum processor that comprises 49 qubits in a bundle. This device exemplifies the third-generation of quantum processors created by Intel, rising from 17 qubits in its precursor. Tangle Lake is a footfall forward in the ongoing growth of bursting quantum computing systems. A superconducting qubit, could hypothetically and appropriately fit in one square millimetre of space.

In amalgamation with their physical resemblance to conservative transistors, this stuff of spin qubits could be contributory to climbing quantum computing systems rising to the projected millions of qubits that will ultimately be required in production systems. Recently, researchers have also advanced a spin qubit fabrication stream utilizing Intel's 300-millimeter process technology that is allowing the construction of small spin-qubit arrays in silicon. QuTech has now started testing moderate spin qubit grounded quantum computer systems.

It's problematic to shape perfect qubits. But that's nothing similar to making a quantum computing system proficient to do execution of operations on many qubits, which essentially work together with all system apparatuses and traditional computing organization. IBM is the company offering the complete quantum technology stack. For example, they provide the cutting-edge hardware, cohesive systems, cloud services, and extensively used software platform to deliver real-world resolutions to complicated problems.

A review of concepts: in the context of quantum architecture

In this chapter, as we discuss the fundamental architecture of quantum computers, it is good to recap a few important concepts in a nutshell. We will not relay here on textual content but would focus on the graphical representation of facts because of the simple fact. One picture has a worth of thousand lines of text. [*Figure 9.1*](#) summarizes facts about qubits and bits, Qubytes and Botch notation, and bar/ ket representation:

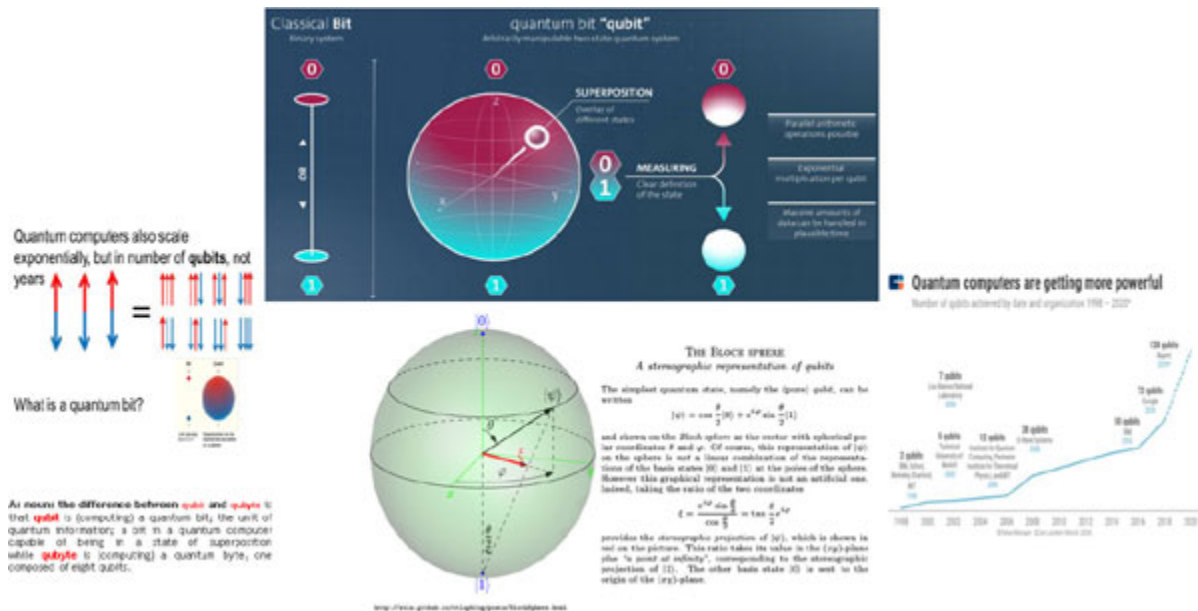


Figure 9.1: Facts about qubits and bits (source:google image)

A brief of quantum computation

Quantum computation includes manipulating probability amplitudes of several qubits and gaining the outcome in a procedure called measurement. When a qubit in-state $|S\rangle = a|0\rangle + b|1\rangle$ is measured, the likelihood of receiving the result $|0\rangle$ is given as $|a|^2$ and the likelihood of receiving the result $|1\rangle$ is given as $|b|^2$. Measurement delivers only one traditional bit of information: 0 or 1 and reduces the qubit in a state consistent with the measurement result. The measurement cannot recover the possibility amplitudes, so the information about the precise quantum state of a qubit earlier the measurement is lost.

In 1926, physicist Max Born (1882–1970) formulated a law known as the Born rule, which is of the utmost importance in quantum computing. Applied to a qubit, the Born rule says that the probability of measuring a particular basis state, $|0\rangle$ or $|1\rangle$, is equal to the absolute square of the probability amplitude of that state, $|a|^2$ or $|b|^2$.

In practical terms, the Born rule gives a link between the probability amplitudes of a qubit state and the number displayed on a measuring instrument after measuring the qubit. Measurement provides only one classical bit of information, 0 or 1, leaving the qubit in a state consistent

with the measurement result. Any additional measurement will yield the same bit value. The measurement cannot retrieve the probability amplitudes, so the information about the exact quantum state of the qubit before the measurement is lost.

Journey of Qubit

To perform useful computation, a quantum computer needs more than one qubit.

A single qubit in superposition is described as $|s1\rangle = a|0\rangle + b|1\rangle$ which is a linear combination (weighted sum) of two basis states $|0\rangle$ and $|1\rangle$. Refer to [Figure 9.2](#) that depicts working of the superposition in mathematical terms.

Measurement causes the state of each qubit to become, or collapse to, $|0\rangle$ or $|1\rangle$.

For more than one qubit (joined state) the combined state can be represented as below:

$a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle + a_{11}|11\rangle$ where,

$|a_{00}|^2 + |a_{01}|^2 + |a_{10}|^2 + |a_{11}|^2 = 1$.

Through measurement, the qubits are forced to collapse irreversibly through projection to $|00\rangle$, $|01\rangle$, $|10\rangle$, or $|11\rangle$.

The probability of their doing so is $|a_{00}|^2 + |a_{01}|^2 + |a_{10}|^2 + |a_{11}|^2$, respectively.

The a_{00} , a_{01} , a_{10} , and a_{11} are known as probability amplitudes.

If necessary, we can convert, $|00\rangle$, $|01\rangle$, $|10\rangle$, or $|11\rangle$ to classical bit string values of 00, 01, 10, and 11.

If qubits 1 and qubits 2 are in the combined state, it can be represented as

$$0|00\rangle + \frac{\sqrt{2}}{2}|01\rangle + \frac{\sqrt{2}}{2}|10\rangle + 0|11\rangle = \frac{\sqrt{2}}{2}|01\rangle + \frac{\sqrt{2}}{2}|10\rangle$$

- If we move from a one or two-qubit system to a multi-qubit system here are a few observations:
- By adding 3 qubits to the quantum system, we find that the number of states rose by 8.
- It reaches 16 in case we add 4. Hence the conclusion is in general, the number of states that “n” qubits hold in superposition would be 2^n .
- This property/capability of quantum physics enables a quantum computer to process vast information in “one go”.
- As an analogy potentially 273 qubits would be able to hold more than the number of atoms that exist in the visible universe.

Figure 9.2: Superposition in action

If we extend the preceding described concept for multi-qubits, then we get a clearer picture. In decimal form, we can generalize the overall process as follows:

$$|s\rangle a_0|0\rangle + a_1|1\rangle + \dots + a_{2^n-1}|2^n-1\rangle = \sum_{k=0}^{2^n-1} a_k|k\rangle$$

Here, $\sum$ stands for summation, $|a_0|^2 + |a_1|^2 + \dots + |a_{2^n-1}|^2 = \sum_{k=0}^{2^n-1} |a_k|^2 = 1$, k takes values from 0 to 2^n-1 . Qubit in a multi-qubit system in superposition can be measured by applying a measuring device to each qubit, thereby performing simultaneous measurements of all n qubits gives as:

$$n \text{ bit} \rightarrow \text{value} \rightarrow k \rightarrow \text{probability with } |a_k|^2$$

Now take an example to further understand it,

$k = 00\dots 0$ with probability of $|a_0|^2$

$k = 00\dots 1$ with probability of $|a_1|^2$

•
•
•
•
•
•
•

$k = 11\dots 1$ with probability of $|a_{2^n-1}|^2$

After the measurement, the information about all the probability amplitudes is lost and only n bits of information can be retrieved from n qubits.

The scale of operation

It is very important to understand the difference between multiple closely related and used terms. Quantum physics is known for strange things like objects have characteristics of both particle as well as wave and limits to the precision with which quantities can be measured, like, position and moment can't be measured at the same time. [Figure 9.3](#) gives a snapshot to the user about the concepts of quantum physics, and the difference between classical, relativistic, quantum and quantum field theory; this understanding

will help the reader to understand the architecture of quantum computers easily:

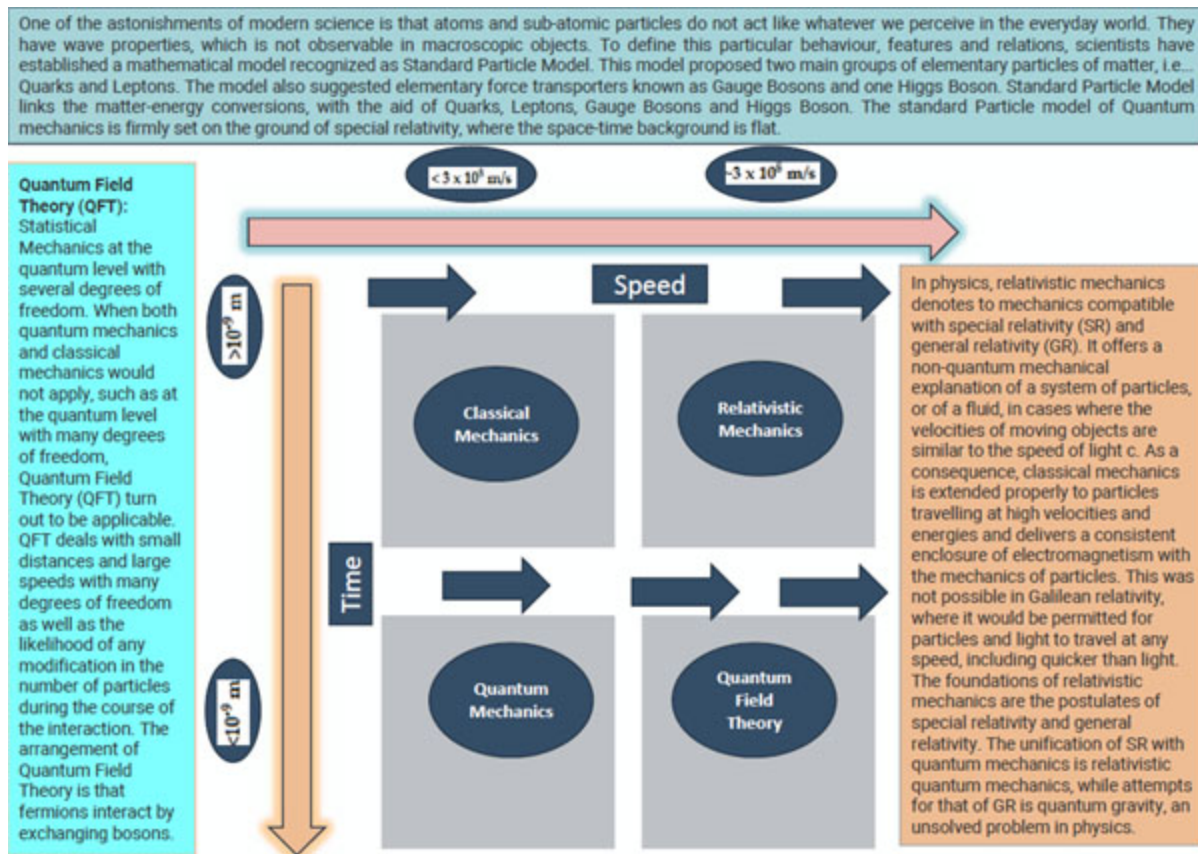


Figure 9.3: Difference between classical, relativistic, quantum and quantum field theory

Let's look at some other elementary concepts before moving towards quantum architecture (most of them are already discussed; it is just a recap).

Bloch's Sphere: a mathematical representation of the Qubit

Although we already discussed Bloch sphere in the previous chapters, it is good to recap the concepts at this juncture. Fundamentally, this model is linked to the representation of the state of a Qubit or of any quantum with two states by a two-dimensional vector whose so-called norm length is always 1. This vector has the particularity of having two elements: a real number α and a complex number β .

Comparison of Qubits with classical Bits: A Qubit is the fundamental element of quantum information; the quantum form of the traditional two-

bit is substantially realized with a two-state expedient. In a nutshell, a QuBit is fundamentally a two-level quantum-mechanical system.

Quantum registers versus classical registers: One Qubit can do nothing alone; we need more to be able to represent more data. Here is an example using a register of 4 QuBits. [Figure 9.4](#) showcases the storage of bits and qubits in the classical and quantum register in a comparative way.

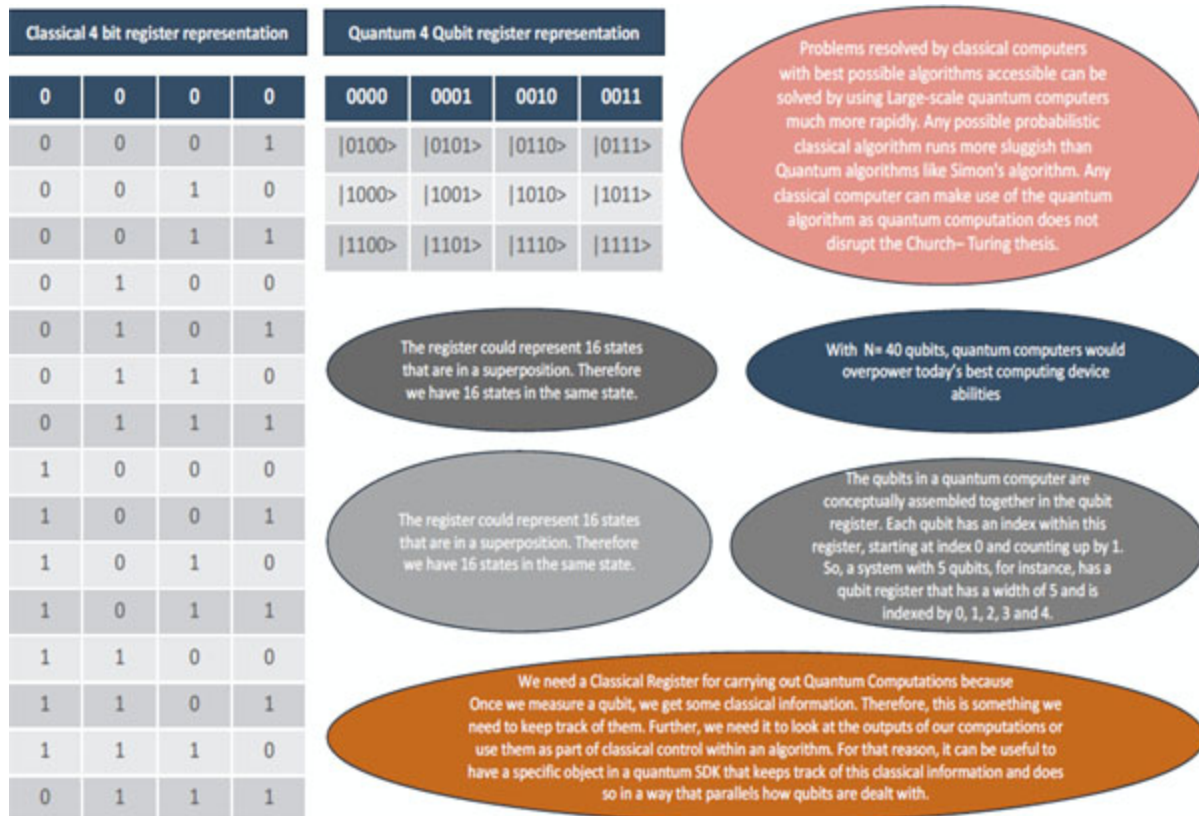


Figure 9.4: Quantum and classical register

Additional facts about quantum gate

Quantum gate plays an important role in quantum computer architecture. The quantum circuit model computation can be done through a sequence series of quantum gate. Fundamentally, each qubit is assigned 0 with which it represents a timeline along which the qubit moves to encounter quantum gates. There are multiple quantum gates; some of them are mentioned as follows:

- Measurement gate measures the qubit state. It provides output in the term of classical bit.
- The gate creates superposition. When a qubit $|0\rangle$ becomes $(|0\rangle + |1\rangle)/\sqrt{2}$ and when qubit is $|0\rangle$, it becomes $(|0\rangle - |1\rangle)/\sqrt{2}$.
- SWAP gate changes the state of two qubits.
- X gate flips a qubit state.
- CNOT gate is a 2-qubit gate that implements a conditional statement. It flips a target qubit if the control qubit is $|1\rangle$.
- CCNOT gate is a 3-qubit gate with one target and two control qubits. It flips the target qubit if both control qubits are $|1\rangle$.

A review of quantum hardware

Multiple types of hardware are associated with quantum computers specially in context of generation of Qubit. However, superconducting, trapped ions, photonic, topological photonics, and light are the front runner. But all these uses different technology for achieving the same. For instance, the trapped ion target an atom with one electron for building Qubit along with magnetic field to do that. Whereas photonic Qubit is use light and lasers . Further, in superconducting quantum hardware, no resistance is experienced whatsoever.

If the quantum particles are cooled down, they form Cooper pairs which are resistance-free. As we know, in the world of quantum, the particles use entanglement, and they're using this property to link in a quantum mechanical way. If we can shape that, they're kind of error-free, since to measure one part of that system, entangled system, and to interrupt any particle in that entangled system, the whole system is disturbed. However, it is much harder than just perturbing one single trapped ion or one single photon, or one single particle in a superconducting computer. We will go through the details of all these in the latter part of the chapter. However, it is good to have to see the picture of real quantum computer. [Figure 9.5](#) provides the pictures of quantum computers provided by multiple vendors.

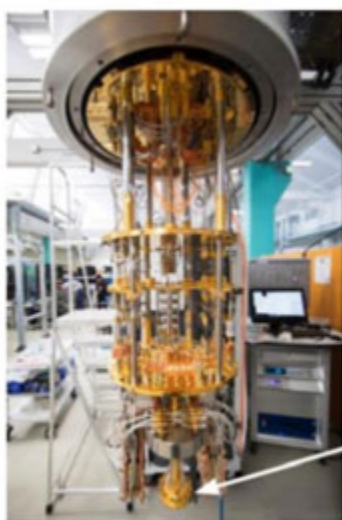
As we discussed earlier as well, Microsoft is betting big on trapped ion and doing a lot of research in this area. If Microsoft is successful in its efforts,

we will fundamentally be able to fulfil our dream of error-free quantum computing. This would be a big deal because it's the error and the noise in the so-called superconducting to photonic to trapped ion quantum computers that essentially spoil and at level destroy the quantum mechanicalness of a provided system. So, it means theoretically, there is a requirement of a lot of qubits for performing efficient error-correction.

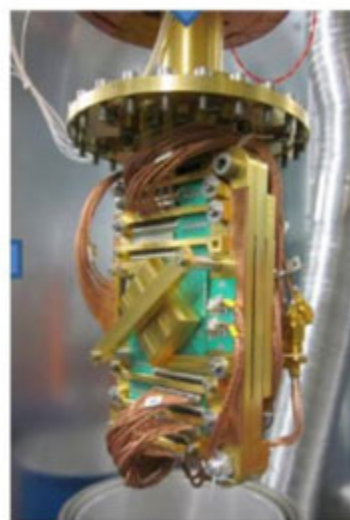
Further, if the sort of millisecond quantum-ness is achieved, then the meaningful calculations would be performed. Now if quantum gates, capable to perform 1,000, 10,000 quantum operations. Then this leads to the happening of significant quantum computation a reality. Let's accept the fact that with a smaller number of gates, the performance of meaningful quantum computation is not possible. This is a universal principle that remains relevant is applicable in the case of quantum as well as the classical world. Therefore, the number of gates is a significant fragment of having a good system.



IBM



rigetti



D::WAVE

Figure 9.5: Pictures of quantum computers provided by multiple vendors. (Source: Google image)

Some companies are working hard to realize QC hardware. Google's Bristlecone claimed that they created 72-qubits QC and this is the maximum number qubit QC at the moment, followed by IBM's 49. IonQ, are using trapped ions techniques and they claim they are creating a 60 qubits system. This is the first trapped ion device in the world. One of the

biggest processor maker Intel is also in the game. They have made a quantum processor of 49 qubits. They're experimenting with both superconducting and trapped ion. Whereas Xanadu based in Toronto uses light, so they're doing photonic quantum computing. One of the oldest players in the game of QC is D-Wave; they are using quantum annealing technology, which is not the universal quantum computing specifically.

Due to the different nature of the qubit technology which D-wave is using the debate is on whether they fall under quantum computers or not. But most of the experts agree that it is quantum. Recently they announced 5,000 qubits, quantum annealing based processor. It's more like an ASIC and they're decent at resolving optimization complications. They're not upright at chemistry or anything else. That is the main reason that experts doubt that they will ever be talented to out-compute a traditional computer. However, even with doubts, they provide full flagged so-called quantum computers for commercial uses and people are using them. Their clients include banks, oil companies, and so on. Fundamentally they are trying them out. Just because D-wave got a high number of qubits, it doesn't mean that they will outdo a traditional computer. Although it's verifiable that gate-based computers, where we have 50 or 72 qubits, can and will outdo traditional computing.

One of the biggest challenges of developing quantum hardware and specifically QC is its noise and fidelity state. If a 128-qubit system is constricted, it is noisier than 50 qubits. Then 50 qubit QC is more useful because in the case of a noisy system, it will decohere in a nanosecond. Fundamentally, it is an engineering nightmare.

General discussion on Core Qubit technologies development: in the background of history

Quantum inspired computer hardware is the type of hardware that readily solves combinatorial optimization problems using existing semiconductor technology. It is one big step in the quest for quantum computers and further to brain-like computers, which is a type of computer that exactly works like the human brain. It is possible with the help of Quantum computers along with deep and machine learning in the time scale of 2040

to 2050. However, we will confine our discussion on the types of quantum computer which are evolving and steps towards achieving bigger goals.

In 2005, investigators at the University of Michigan built a semiconductor chip that worked as an ion trap. Such devices are shaped by standard lithography techniques. A better-quality version was completed in 2006. In 2009, investigators at Yale University shaped the first basic solid-state founded workable quantum processor. The two-qubit superconducting chip was able to run simple algorithms. Another team, employed at the University of Bristol, also fashioned a silicon-based quantum computing chip, founded on quantum optics.

The team was talented to run Shor's algorithm on the chip. Additional developments were made in 2010. Further in April 2011, a team of experts from Australia and Japan achieved an advance in quantum teleportation. They effectively transported a complex set of quantum data with jam-packed communication integrity attained. Also, the qubits were demolished in one place but immediately resuscitated in another, without upsetting their superpositions.

In 2011, D-Wave Systems proclaimed the first marketable quantum annealer on the market by the tag D-Wave One. The corporation entitlements that this system uses is a 128-qubit processor chipset. In 2011, D-Wave publicized that Lockheed Martin Corporation arrived into a contract to buy a D-Wave One system.

D-Wave's engineers use an experiential method when scheming their quantum chips, concentrating on whether the chips can resolve specific problems rather than scheming founded on a thorough understanding of the quantum principles tangled. This method was adored by investors more than by some theoretical detractors, who said that D-Wave had not yet adequately established that they had a quantum computer. Such criticism unstiffened once D-Wave published a paper in Nature giving facts, which critics said showed that the corporation's chips showed few quantum mechanical properties which is very much needed for quantum computing.

In the same year, investigators working at the University of Bristol created an all-bulk optics scheme which was able to run a reiterative version of Shor's algorithm. They effectively achieved factorizing number 21. In

November 2011, investigators factorized 143 using 4 qubits. Further in February 2012, IBM experts said that they had created numerous breakthroughs in QC with something called superconducting integrated circuits. As proof, the IBM scientists came up with an initial version of IBMQ systems.

The year 2012 was a great year for QC development, especially in the context of Qubits, as multiple breakthroughs were achieved in that year. In April 2012, a cosmopolitan team of investigators from the different universities and Technological institutes in California, Santa Barbara, and a few others built a two-qubit quantum computer through a crystal of diamond doped with means of contamination. The system can effortlessly be scaled up in size and function at room temperature. Two logical qubit tips of electron spin and nitrogen grains spin were used for this purpose. As a result, the system shaped a surge of microwave radiation of sure duration and form was advanced for the upkeep of defence against de-coherence. Utilizing Grover's algorithm for four alternatives of search has further produced the correct response from the primary trial in 95% of the cases.

Around May 2012, a research team guided by Australian engineers shaped the primary working "quantum bit" founded on a solitary atom in silicon, summoning the same technological platform that forms the structural building blocks of present day computing devices including desktop computers, laptops, tablets, and phones. One of the seminal works performed by David J. Winelands and Serge Haroche in 2012, also got Nobel Prizes for this fundamental work; fundamentally they tried to understand the quantum world. Their work eventually helped to make quantum computing more robust, effective and efficient. Further, in November 2012, the first quantum teleportation from one macroscopic object (photon which exits in the form of energy quanta) to another was reported.

In February 2013, a novel method called Boson Sampling was testified by two groups utilizing photons in an optical lattice to produce quantum computer. Further, in May 2013, Google Inc proclaimed that it was beginning the Quantum Artificial Intelligence Laboratory, to be accommodated by NASA's Ames Research Center. The Laboratory will house a 512-qubit quantum computer from D-Wave Systems to study how

quantum computing and computer along with quantum information processing might aid in the growth of machine learning.

The desire of building a quantum computer

Minor quantum computing devices, tools, and gears are constructed on a variety of underlying physical applications. It may be created in the laboratory, where they have been developing for over a period, and have established the central characteristics essential for construction systems. The challenge lies in spreading these systems to be big enough, fast enough, and precise enough to resolve problems that are stubborn for traditional systems, such as the factoring of large numbers and the careful imitation or in technical teams' simulation of further quantum mechanical systems.

The construction or architecture of such a computer will be important to its recital. Organizationally, when saying that it was constructed, a quantum computer implies that there are building a hybrid device, with essentially contains a processing unit with the capabilities of doing quantum computations which, in general term, is known as quantum computing units. It is fundamentally a coprocessor portion of classical systems. The classical side is responsible for taking care of the program, control circuitry, and sizable amount of pre- and postprocessing functions. The fundamental architecture of the quantum system whose prime responsibility contains the algorithmic workloads for its alacrity and proficiencies in meeting these goals, its interfaces to the classical control logic, and the design of the traditional or classical control systems are overall the accountability of quantum processor engineers. The essential design of the commonly known classical control systems is accountable for the quantum computer architecture part.

A principal experience understood so far in investigating quantum computer architectures is that systems talented of resolving characteristically obstinate problems will be hefty, although the search for the minimum commercially striking machine endures. Device sizes will draw a boundary at integration levels, affecting architecture, and shaping logical clock speed involves the creation of numerous design decisions but theatrically affects what can and cannot be efficiently computed. By nature, any architectural

problem involves a bigger range but have acknowledged only modest amounts of consideration related to near-term experimental footprints. This leads to keeping much room for high-impact research, investigation, and study.

To improve the municipal emphasis on structure systems, it appears to be time to start demanding Moore's Law-like developments in scheme capacity. Reviewers of research papers and subsidy proposals must appear for truthful approximations of incorporating error correction in the quantum gated environment, with some target logical fidelity. Even more aspiringly, the recommendation is too needful of truthful approximations of application act. Making a communal understanding as well as language, ideas, and significant problems among the physics, material science, chemistry, and CS theory, algorithm design, physics experiment, engineering, and architecture groups have established to become tough, and limited journals or conferences now deliver decent sites for such interdisciplinary endeavours, but we imagine the number will raise.

Today, it is tough to imagine day-to-day living without IT; we are dependent on almost everything of information technology. It bids immediate communication and admittance to data that can then be worked and used as and when required. But if we ask a very fundamental question, like what is data, then the answer is not surprisingly in the realm of classical and quantum physics. Data is essentially the building blocks of information or rather digital information. The digital information is determined as visibly distinct values of 1 or 0, for instance, in a two-state system such as a voltage (on or off state) in an electronic device, the magnetic orientation of particles (up or down) in a memory, or a light pulse in an optical fibre. These binary "bits" are the foundations of today's complex computers, internet, collaboration, and communication systems. Note that these bits are nothing but the flow of current at its core, which is nothing but the complex interplay of electrons.

Initially, it was believed that such systems follow the acquainted laws of classical mechanics. Though, in the initial 1980's, the US physicist Richard Feynman recommended that the weird, "non-classical" properties of systems following the rules of quantum mechanics could bid a novel method of processing information; which is a full-scale quantum computer.

As per his proposal, the QC would be extraordinarily powerful. As we know, this was the eureka moment for something called the **Quantum Information Processing (QIP)** and the birth of a new field of study took place.

Architecture of QC

In a typical QC, we have the processor down there at 10 millikelvins. We have to govern these qubits somehow. Typically, it can be done with microwave in case of superconducting qubits, and with electromagnetic fields in case of trapped ion, and with lasers in case of photonics. We require to intensify the signal so we have input. Input is given to the system as per the need of calculation. It is provided to the system through the interface electronics, after quantum computation in a millisecond is performed to minimize the possibility of decoherence. So, that's the state-of-the-art quantum computers today. That's a high-level architecture.

The initial important effort to describe the technologies required to shape a computer came in the mid-1990s, when DiVincenzo enumerated standards that a feasible quantum computing technology must-have. The recommendations go as follows:

- A two-level physical system to perform as a qubit.
- A method to initialize the qubits into a recognized state.
- A universal congealed of gates among qubits.
- Measurement.
- Long memory lifetime.

These measures were later amplified with two collaboration criteria, as follows:

- The aptitude to adapt among motionless and hovering qubits.
- The capability to convey the latter among two places.

In some qubit technology, the first principle is utmost vigorous: What is the state variable? Corresponding to the electrical charge in a classical Von-Neuman architecture-based computer, what feature of the physical system encrypts the elementary 0 or 1 state? The initialization, gates, and

measurement procedure then follow from this indispensable stride. Numerous groups worldwide are presently employed with a variety of state variables, from the way of the quantum spin of an electron, atom, nucleus, or quantum dot, beyond the magnetic flux of a micron-scale current loop, further to the state of a photon or photon group. [Figure 9.6](#) provide high-level architecture and chipset-level information of quantum computers; this is a generic information and not specific to any vendor:

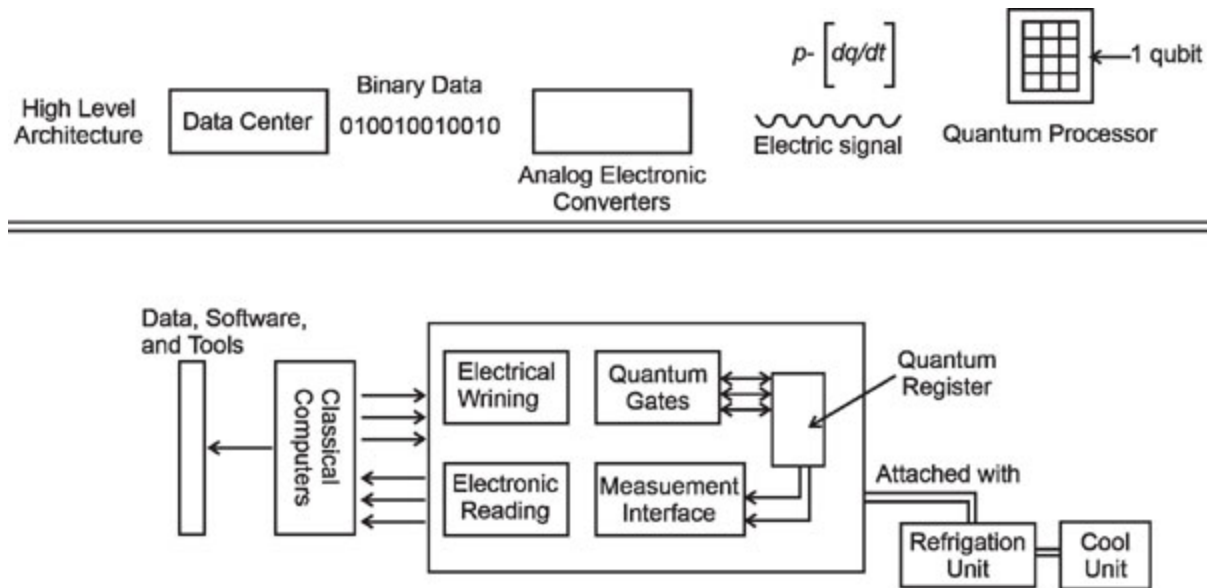


Figure 9.6: High level architecture of quantum computer

Controlling any physical system at the quantum level is tough. And it is true irrespective of technology we use. As we discussed, QC can be made through multiple techniques; Ion trap is one of them. But the fundamental working of devices would remain the same. That is, they would work on Qubit based technologies. Also, the Ion trap devices and optical systems currently would be able to hold the number of qubits that can be held in a device. Further, it can be controlled and entangled. Fundamentally in this method, ions are entombed in a vacuum by electromagnetic potentials, and qubit entanglement has been established.

Complex bench-top linear optical circuits are proficient in entangling eight-photon qubit states and have been exposed to do nontrivial computation over brief timescales. Both of these methods do not scale in those procedures, but scalable methods are also underneath expansion. Integrated nanophotonic (using photons as qubits) fabricated on silicon chips delivers

the way to getting optics off of the lab seat and into effortlessly scalable systems, and is making considerable development in groups.

In solid-state technologies, the new emphasis has been on refining the control of individual qubits rather than increasing their numbers, but that emphasis has started to shift. Because superconducting qubit memory lifetime has more than doubled every year since 1999 and has now touched the point where quantum error correction turns out to be operative.

In general, the scenarios are good for systems containing qubits to seem in manifold technologies over the following few years, permitting investigational validation of the lower spreads of the scaling behaviour of quantum algorithms and the efficiency of quantum error correction. Though, one feature that is often unacknowledged when observing at these qubit technologies is the physical scale of the devices, chiefly in contrast with classical digital technologies.

Numerous people associate quantum properties with minute objects, but most of these technologies use devices that are hugely related to the transistors in modern computer chips. Transistors are reaching down to atomic scales, with vendors having transported chips invented with a 22-nanometer process and less. In these chips, the minimum features will be only around 40 times the silicon crystal lattice cell size. In disparity, although the atoms themselves are minute, ion traps are restricted to inter-atom typography of conceivably tens of microns for RF and optical control.

Nanophotonic systems will necessitate the machinery of tens of microns to put up the 1.5 μm wavelength light that is needed for communications and silicon optics. Superconducting flux qubits need an existing ring micron across. All of this know-how consequences in macroscopic qubit devices, with spatial thicknesses of a million times less than computer chips. This detail will have a huge influence on important architectures.

With this background, now we move on to the different types of quantum computers, and in the following few sections, we will discuss the same.

General discussion on QC

There are multiple types of quantum computer but broadly three main kinds of quantum computers are popular. Each type varies and has different

characteristics. One of them is the quantity of processing power (qubits) required and the number of conceivable applications, as well as the time essential to turn out to be commercially feasible. These types of quantum systems fall under the category of quantum annealing.

Let's briefly look on the types of quantum computer. We will confine our discussion on quantum annealing, Quantum simulations/ simulator and general quantum computers.

Quantum annealing and computers

Quantum annealing is the best for resolving optimization problems. In other words, investigators are trying to discover the best (most well-organized) likely shape among many likely mixtures of variables. [Figure 9.7](#) showcases the placing of quantum annealing in the world of universal quantum computers.

For instance, Volkswagen (VW) of late showed a quantum trial to optimize traffic movements in the congested town of China (Beijing). The trial was run in cooperation with Alphabet and D-Wave Systems.

The algorithm could positively decrease traffic by selecting the perfect path for the respective automobile. The same concept could be applied on a global scale in the multiple fields including but not limited to optimizing airline route, railway timetable, climate data, petroleum costs, and traveller information, and therefore help in providing the most cost effectual travel and logistics resolutions. Classical computers would also do this but it will take a lot more time than quantum computers; theoretically, it can do it in just a minute, hours or less, as the number of qubits per quantum computer upsurges. Refer to [Figure 9.7](#):

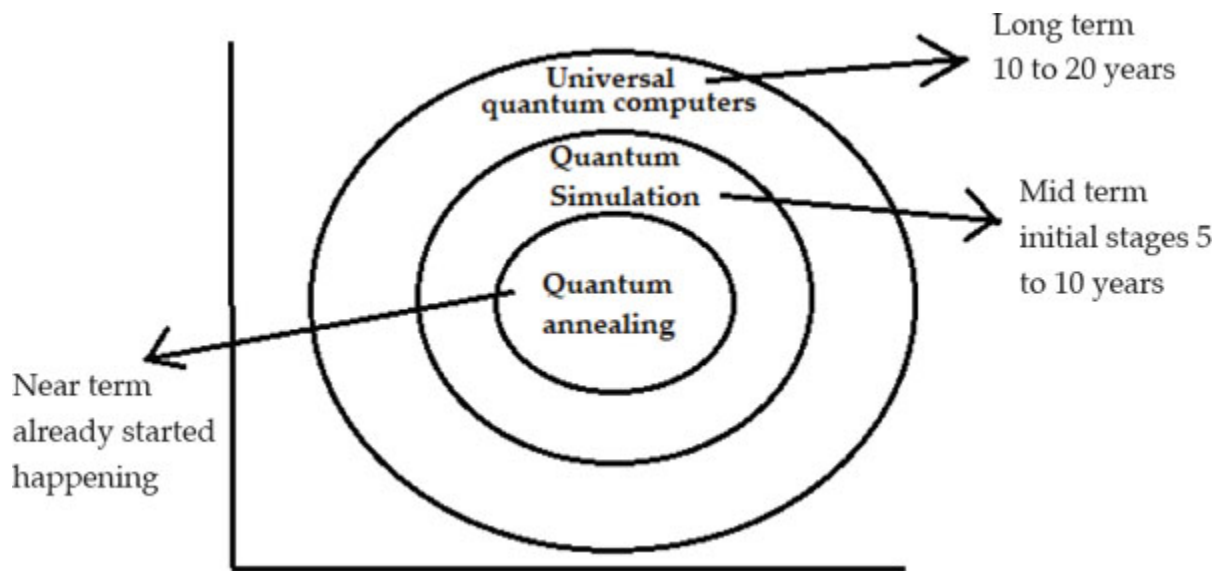


Figure 9.7: The placing of quantum annealing

Quantum simulations

The other type of quantum computer which takes researchers' attention is a quantum system which is specialized in simulation type of problems.

Quantum simulation-based computer discovers precise problems in quantum physics that are outside the capabilities of traditional systems. Simulating multifaceted quantum marvels could be one of the greatest significant applications of quantum computing. One part that is chiefly capable comprises modelling the consequence of a chemical stimulation on a huge quantity of subatomic particles. Quantum simulators could be used to simulate multiple things, one of them is protein folding. Misfolded proteins can be the source of illnesses like Alzheimer's and Parkinson's, and investigators may test novel dealings to learn which drugs were the basis of reactions for respective protein through the usage of haphazard computer modelling.

Scientifically, if a protein were to accomplish its properly folded configuration by a consecutive selection of all the conceivable drug-induced belongings, it would necessitate a time longer than the time of life of the cosmos to reach its precise natural state. A truthful charting of the protein folding arrangement would be a foremost technical and healthcare breakthrough that could save lives. This type of Quantum computers can help compute the vast number of possible proteins folding sequences for

making more real medicines. In the coming time, quantum simulations will allow quick designer drug testing by taking care of all possible protein-to-drug amalgamation.

Universal quantum computing

Universal quantum computers will be very powerful. A truly universal quantum computer would probably use over 100,000 qubits. Whereas a few other estimations put it close to one million qubits. The ground rule behind the universal quantum computer is that you could unswerve the apparatus at any enormously multifaceted computation and develop a rapid resolution. This comprises resolving the aforementioned annealing equation problems, feigning quantum phenomena, and other types of problems as well.

Investigators have been conniving algorithms for years that are only conceivable on a universal quantum computer. The furthestmost well-known algorithms are Shor's algorithm for factoring figures and Grover's algorithm for rapidly searching formless and enormous sets of data. Broadly, fifty exclusive algorithms have been advanced to run on a universal quantum computer.

The soul of QC is qubit. Whatever architecture it follows without qubit it is of now use. In the previous chapters we covered few facts about it. Now let's cover them in more organized ways, in next few sections.

Multiple types of Qubit technologies

The realization of qubit can happen in multiple ways with the help of different technologies. There is no fixed technology by which we can derive the conclusion that this is the best suited technique for qubit. All the technologies have their upside and down side. There are different technologies that exist for qubit realization. However, superconducting, trapped ion, silicon based, photonics based, topological and diamond vacancies are the most promising (at least at the moment).

- Number of qubits, coherence time, and fidelity are the most important parameters to evaluate a qubit technology.

- Scalability is also one of the important parameters for manufacturing of Qc. Companies approach towards different technology.

[Table 9.1](#) summarizes the few most popular qubit technologies and companies that are working on them:

Trapped ion	Silicon quantum dot	Topological qubit	Superconducting	Diamond vacancies
Logic success rate, 99.4%	Logic success rate, 99%	Logic success rate, NA	Logic success rate, 99.4%	Logic success rate, 99.2%
Honeywell AQT	Intel	Microsoft	Alphabet D-Wave IBM Intel Rigetti	Multiple companies

Table 9.1: Most popular qubit technologies and companies that are working on them

[Topological QC](#)

The Quantum Turing machine is hypothetically significant but the direct application of this model is not chased. Whereas, for physically realizing a quantum computer, several different candidates are being followed. The topological quantum computer is one of them.

Topology is a branch of mathematics that describes structures that contains some properties so that even the physical changes such as bent, twisted, compacted, or stretched are applied, yet they would uphold the possessions of the unique form. When functional to quantum computing topological properties produce a level of defence, it aids a qubit to hold information regardless of what's happening in the environment.

The topological qubit attains this additional defence in two different ways. Those methods are known as electron fractionalization and ground state degeneracy. Further due to the advanced reliability of topological qubits, fewer qubits are required for error correction; this intensely dips the number of qubits the system needs overall.

[Superconductor-based QC](#)

A qubit can also be realized by the state of minor superconducting circuits. One way of achieving this is through Josephson junctions. Fundamentally, superconductors are the type of material that conducts electricity (number of charges flows in the instance time) with no resistance. Therefore, superconductors can in principle hold a big volume of current and do their transmission from one position to another deprived of abundant loss. Further, electrical current can flow forever in the closed-loop with the simple reason that as there is no resistance, no friction will lead to almost zero loss. This makes the superconductor the closest thing to a continuous motion machine.

Multiple types of superconducting circuits have been used to implement qubits and quantum logic gates. However, three of them are most important. They are charge qubits, phase qubits, and flux qubits. In superconducting QC, a charge qubit is formed by a minute superconducting island recognized as a Cooper-pair box which is attached by Josephson Junction to a superconducting artificial lake. The state of the qubit is resolved by the number of Cooper pairs which have tunneled across the junction.

Artificial atom can be made from a minute superconducting circuit in which connected pairs of electrons experience quantized fluctuations. Superconducting qubits have been entangled to act as a quantum processor. They can be caused to be well-matched with normal chip technology and joined with photons. Two disadvantages have been their short coherence times and the truncated temperatures obligatory for superconductivity to happen. Though campaigns using high-temperature superconductors have been made and coherence times enlarged, quite a few groups are now starting research programmes.

Like present silicon integrated circuits, superconducting qubits are lithographically well-defined electronic circuits. When chilled to milli-Kelvin temperatures, they display quantized energy levels, and are thus occasionally named “artificial atoms”. Their compatibility with microwave control electronics, aptitude to function at nanosecond time scales, repeatedly refining coherence times, and possible to influence lithographic scaling, all meet to place superconducting qubits among the vanguard of the qubit modalities being careful for both digital quantum computation and quantum annealing.

Trapped ion QC

The first quantum logic gate was established in 1995 utilizing trapped atomic ions, following a theoretic proposal earlier in the same year. Subsequently, the original demonstration and technical developments in qubit control have permitted investigational demo of fully functional processors at a small scale and employment of an extensive range of modest quantum algorithms. Despite success in small-scale demos, the task of building scalable quantum computers by using current technologies and present computing industry standards is a big challenge. Especially when we are thinking in the terms of trapped ion technologies.

In Trapped ion an optical lattice is used which included interfering laser beams to make a crystalline array of ultracold atoms. It can control the quantum state of each atom using a strongly attentive laser and microwaves and is eager to create a scalable quantum processor of several hundred atoms.

Ions are atoms that have lost or gained one or more electrons. Therefore, it contains an electric charge. The ion traps are used for building the quantum registers. The number of qubits in the register is equal to the number of ions trapped. For instance, in Calcium ion, if one electron is removed, then they are positively charged. The ions pick up and drop something called kinetic energy as soon as they engross or release single photons (packets of energy). Ion traps can store quantum information with long coherence times and support universal quantum computations. Trapped ions possess a high degree of stability when it is in a superposition state. They have the advantage of being able to be entangled with each other.

A trapped-ion quantum data plane includes the ions that aid as qubits and a trap that holds them in exact locations. The control and measurement plane comprises a very exact laser source that can be rapt at an exact ion to distress its quantum state, additional laser to cool and allow measurement of the ions, and a set of photon detectors to measure the state of the ions by noticing the photons that they scatter.

A further capable method pays arrays of ions, cooled with lasers, and trapped by electric and radio-frequency turfs. A distinct laser is then used to stimulate the ions from one quantum state to another. The cold ions are

cajoled into entanglement with the laser, which reasons them to push back and forth in recital so that the couple work in synchronization. This type of coupling has allowed investigators to shape CNOT gates utilizing a couple of ions, and entangle eight or nine ions in a linear-shaped trap.

Constructing a computer would typically need big numbers of correlated ions. This might be best attained using arrays of micro-traps made via typical microfabrication methods. At Imperial College London, one of the types of research showcases with a type of trap based on static electric and magnetic fields that could be suitable. Also, at the **National Physical Laboratory (NPL)**, a technique is developed which uses linear traps on a standard silicon wafer utilizing gold electrodes.

In another research at the University of Sussex, a team of scientists has advanced a chip design that could be scaled to house a million ions, each in distinct micrometre-sized traps containing microfabricated cantilever electrodes. The idea is to transmit out quantum computing by travelling the entangled ions from place to place from one trapping zone to another.

Further, information is read out, utilizing optical fibres, via the photons released by the ions. Another important goal is to transfer entanglement between ions and photons, which would be essential for a quantum communication network. One of the standard ways is to capture the photons emitted by the ions between two mirrors forming an optical cavity. The photon rebounds back and forth in the hollow, which upsurges the communication with the ion, moving the quantum states among the two qubits with high reliability.

[Quantum dot QC](#)

The quantum dot is a technique where a nanometre-sized semiconductor structure confines single electrons in a host of the semiconductor substrate. This “artificial atom” can be controlled with a voltage and a laser. This is configured in such a way so that it emits single photons or entangled photon pairs.

A quantum dot is also thought of as a nanoparticle made of a semiconductor material such as Silicon. A free-electron which is not bound within an atom has resided inside. The surrounding solid is built up in layers which are

constructed with two semiconducting materials (Silicon) and called to a very low temperature. Characteristically, the temperature is one-tenth of a degree above absolute zero. The free-electron would further place using an electrical field. In this configuration, the electron spin can be swathed up and down electrically and can, therefore, be used to store the information.

Solitary electron spins can be controlled at ultrafast rapidity using optical beats. This is one of the necessities to attain a high number of quantum operations before a loss of coherence.

Determining the state of the qubit kept in an ion is fairly modest. Characteristically, a laser ray is functional to the ion that pair one of the qubit states. Once the ion breaks down into this state, during the process of measuring, the laser will stimulate it. This results in a photon release when the ion declines from the excited state. Subsequently, after deterioration, the ion is repeatedly excited by the laser and recurrently produces and emits photons. These photons can be further poised by a so-called **Photomultiplier Tube (PMT)** or a charge-coupled device (CCD) camera device. If the ion breaks down into the further qubit state, then it does not intermingle with the laser, and no photon is released. By including the number of poised photons, the state of the ion may be determined with more correctness.

[Nuclear magnetic resonance QC](#)

Nuclear magnetic resonance on molecules in solution is also known as liquid-state NMR. The first NMR quantum computer was constructed for the demo purpose in 1998 in Oxford. It was based on the principles of NMR. The qubits were linked spin states of hydrogen nuclei in carbon-based molecules which has the capabilities of being dissolved in solution. Though such quantum computers have been operative in testing quantum algorithms, and a seven-qubit NMR system has been prepared, it is not likely they could be scaled up any further.

Molecules are made of atoms and matters are made of molecules. Within every atom, there is a nucleus. The nucleus contains protons and neutrons. The chemical nature of an atom is defined by the number of protons in its nucleus. Take any element, for example, consider Carbon – all atoms of

carbon have six protons in the nucleus and all atoms of hydrogen have a single proton in the nucleus. Some nuclei are magnetic and therefore have a magnetic moment which means that they interact with the applied magnetic fields. **Nuclear Magnetic Resonance (NMR)** detects the bulk magnetism of the nuclei, meaning the sum of all the microscopic magnetic moments in the sample. NMR is the fundamental experimental observation of the resonant absorption of the energy by the nuclei from radio frequency sources.

When a nuclear magnetic moment is linked with a nuclear spin, it is put in an exterior magnetic field. This would lead to different spin states which are certain diverse magnetic potential energies. In attendance of a still magnetic field which shaped a minor quantity of spin polarization, a radio frequency signal of the power recurrently can induce a shift among spin states. This flip of spin places some of the spins in their advanced energy state.

Under NMR two approaches are generally used. The first one is all about using the spin property of atoms of specific molecules in a fluid sample as qubits. By its nature, this is known as liquid-state NMR. Whereas the second method uses a solid-state NMR sample, for instance, a nitrogen-vacancy diamond lattice than a liquid sample.

Diamond-based QC

In this arrangement, the qubit is appreciated by the electronic or nuclear spin of Nitrogen-vacancy hubs in a diamond. In this approach, an impurity in diamond (a rigid lattice of carbon atoms) is investigated.

Especially in situations when a single nitrogen atom sits next to a vacancy in the diamond lattice where a carbon atom would normally be, which traps a single electron (called an NV⁻ centre). The system's electron spin state is stable for milliseconds at room temperature and can be manipulated with radio-frequency waves and optical lasers. The NV⁻ qubit emits single photons that can then be entangled via interference. Similar defect-based systems such as chromium in aluminium oxide are also being investigated in order to produce QC.

The challenges are evident as we understand that, the system's electron spin state is stable only for milliseconds at room temperature and at that instance

only they could be manipulated with radio-frequency waves and optical lasers.

Photon-based QC

Photons exhibit neither mass nor charge (on a fundamental level, it is a packet of energy). As a consequence, they do not interact with each other. Photons are ideal candidates for many quantum information processing tasks and communication. As have a long coherent time and weak interaction with the environment. Further, they travel at the speed of light and allow multiple degrees of freedom encoding. Photonic qubits function works well at cryogenic and room temperatures.

Entangled photons with differing polarizations will be significant as “flying qubits” for moving quantum information. They are vigorous and processes can be carried out on them with high fidelity. The normal way of producing an entangled pair of photons has been via a so-called nonlinear procedure, whereby a photon is divided into two entangled photons of lesser energy when it permits through a certain type of crystal. Though, single photons from distinct sources do not interrelate, as would be desirable for a logic gate. Yet, photons can be made to “amalgamate”, by transferring them through a beam splitter, which reasons them to interfere in a means that outcomes in the generation of two equal photons – a linear process.

Intermingling photon states unlocked the likelihood of construction of a scalable, all-optical quantum computer from conservative optical elements such as phase-shifters, beam splitters, and polarizers – linear optical computing. At the University of Bristol, this breakthrough was achieved. Using silicon oxynitride as a substrate, the team has joined big statistics of photons in multifaceted interference outlines possibly apposite for a novel kind of QIP called cluster-state computing. The coupling between the molecule’s quantum states and photons can be fairly robust, permitting each molecule, through its optical nonlinearity, to persuade connections among photons that deliver a foundation for gate operations.

Few additional important technologies for quantum computers and Qubits

Apart from the preceding technologies, there are a few additional technologies as described as follows:

- **Electrons-on-helium quantum computers:** In this arrangement, qubit is founded on electron spin and all the manipulation is done on this basis.
- **Cavity Quantum Electrodynamics (CQED):** In this arrangement, qubit provided by the internal state of atoms are fundamentally trapped in and further joined to very high-finesse hollows.
- **Fullerene-based ESR quantum computer:** In this arrangement, qubits are founded on the electronic spin of atoms or molecules enclosed in fullerene constructions.
- **Optics-based quantum computer:** In this arrangement, QC is appreciated by suitable states of different styles of the electromagnetic field.
- **Transistor-based quantum computer:** In this arrangement, string quantum computers with entrainment of positive holes utilize an electrostatic trap.
- **Phosphorus in silicon:** It is a novel technique for QC. In this arrangement, one of the first solid-state qubits suggested was the nuclear spin of a phosphorus atom sitting in a silicon matrix. Each atom possesses a nuclear spin and an electron spin that can be controlled with radio waves and microwaves, while the qubits are coupled and measured using nearby electrodes. John Morton and colleagues at Oxford have worked on this system, recently demonstrating that large ensembles of phosphorus spin pairs can become entangled, as required for quantum computing.
- **Spin Qubits in the solid-state:** Several research groups are working on related systems that involve qubits of single atoms, ions, or molecules embedded in a solid matrix, in which the quantum states are the spins of atomic electrons or nuclei. Again, such systems should lend themselves to microfabrication and be scalable. However, the readout of the state of single-spin qubits has been achieved only recently.

Let's build a quantum computer: elements of a quantum processor

Just about any system that demonstrates quantum behaviour – from photons to molecules and superconducting devices – is being considered for QIP, and is generating a better understanding of quantum behaviour and nanoscale technologies.

Quantum computer architecture as a field is still not that mature, however, it contains much potential for creating machines that massively surpass present classical capabilities, for reliable systems intended to resolve particular problems. As we start to design greater quantum computers, it must be known that large systems are not just greater versions of minor systems. The theoretical stack of subfields necessarily provides an accessible, real-world machine. It is fundamentally divided into a set of layers. The fundamentals of this construction, in turn, is all steering to the predominant core of quantum computer architecture.

At the lowest of the stack, we have the technologies for storage of discrete qubits, and processing or conveying them to take a share in a greater multiplication. Given the fragility of quantum data, how can several qubits be earmarked in a coherence state for extended enough to wide-ranging complex computation become important? The answer to this problem would be found in the field of **Quantum Error Correction (QEC)**. The studies indicate that randomly accurate computation is theoretically conceivable even with flawed systems, but apprehension here is the design of subsystems for performing QEC, which can be named the quantum computer microarchitecture.

Essential parts of any computer are random access memory and information storage. While photons are excellent for communication, matter qubits such as an electron or nuclear spins are more suitable for holding information. Nuclear spins can have coherence lifetimes of more than a second, so are suitable for memory qubits. Their quantum states can be transferred via an electronic coupling. It is investigated that a solid-state quantum memory is stored in phosphorus nuclei in pure silicon. The coherence lifetime is almost two seconds and the overall store-to-readout fidelity is about 90%.

On the very top of the quantum stack, machines will be intended for precise workloads, to run firm algorithms such as factoring or simulation. These types of algorithms occur in computational complexity classes supposed to be unapproachable to classical computers. Without these algorithms, there will be no financial inducement to shape and organize machines. At both the top and bottom of the stack, the development that has been made in the direction of integrated architectures, and finished with a comprehensive instance of the huge scale-up in size. This further slows down in rapidity rising from the error correction essentials of a full-blown, digital quantum computer.

The furthestmost well-known advance to date in quantum algorithms is Shor's algorithm which was intended to factoring large numbers in polynomial time. And it was very successful in that regard. While the dialect presses often discuss factoring large numbers in seconds utilizing a quantum computer, in truth it is not even conceivable to debate the potential performance of a system deprived of knowing the physical and logical clock speed, followed by the topology of the interrelation between the elements. Further, the number of logical quantum bits (Qubits) accessible in the system, and the particulars of the algorithmic implementation is also very important. Therefore, without specifying the architecture and running the algorithms on actual quantum systems, it is very difficult to predict anything; this includes the performance aspect as well.

In the process of development of a quantum computer architecture, we will have much to input from multiple sources, and even apply our learning from classical computer architecture to its evolutionary design. However, a few things are different like the faint nature of quantum information. This strains that memory elements can be very vigorous. Second, long wires or long-distance networks within a quantum computer are either non-existent, needful adjacent neighbour, the cellular automaton-like transmission of data, or are of poor quality, and need much determination to transfer even a solitary qubit from place to place using quantum teleportation and error management techniques. Hence, the principles of classical computer architecture can be functional, but the responses inwards will likely fluctuate considerably from classical architectures.

The elementary structural block of any processor is a logic gate, which transmits out a logical operation on a set of inputs to give an output. In QIP, a standard gate would be the **Controlled-NoT (CNoT)** gate. These performances are on two connected qubits, a control qubit, and a target qubit. The quantum state of the control qubit controls the operation to be carried out on the target qubit. When the control bit has a value of 0, the value of the target qubit stays the same, but when it has a value of 1, the value of the target qubit is flipped.

The first CNOT gates were built in the 1990's, in the US and Europe including the UK, using ions, which were held in traps created by electric and magnetic fields, and excited into a superposition of electronic states using laser pulses. Another model system experimented with at the University of Oxford, as well as in the US, involved controlling the spin states of atomic nuclei in organic molecules, with strong magnetic and radio-frequency fields (nuclear magnetic resonance, NMR). It was used to execute Grover's search algorithm for the first time.

Practicalities of quantum computers

At first, it was not clear that QIP was a practical proposition – a robust and stable quantum computer is still a long way off. Qubits are fragile entities that collapse when they interact with their environment.

Other important aspects include scalability and communication. A practical quantum computer requires many thousands of interacting qubits, which need to be controlled. Data may also need to be passed from one processing system to another, which may involve preserving quantum coherences transferred between different kinds of qubits, and over long distances.

The simplest way to build a quantum computer

We can build quantum computers in multiple ways. However, one of the simplest ways it can be built is described in [Figure 9.8](#). The simplest way to build a modular quantum is to use single atoms as qubits. Each atom can represent the binary code value of 0 or 1 (or a superposition of the two) via different electronic orbits. At the bottom is a schematic three models: mini quantum computers made of five atomic ions, each connected in a way that

presents each model's quantum properties. Within each module, all five ions are elongated with one another. The two end ions are special and can emit photons to communicate with other modules. Refer to [Figure 9.8](#):

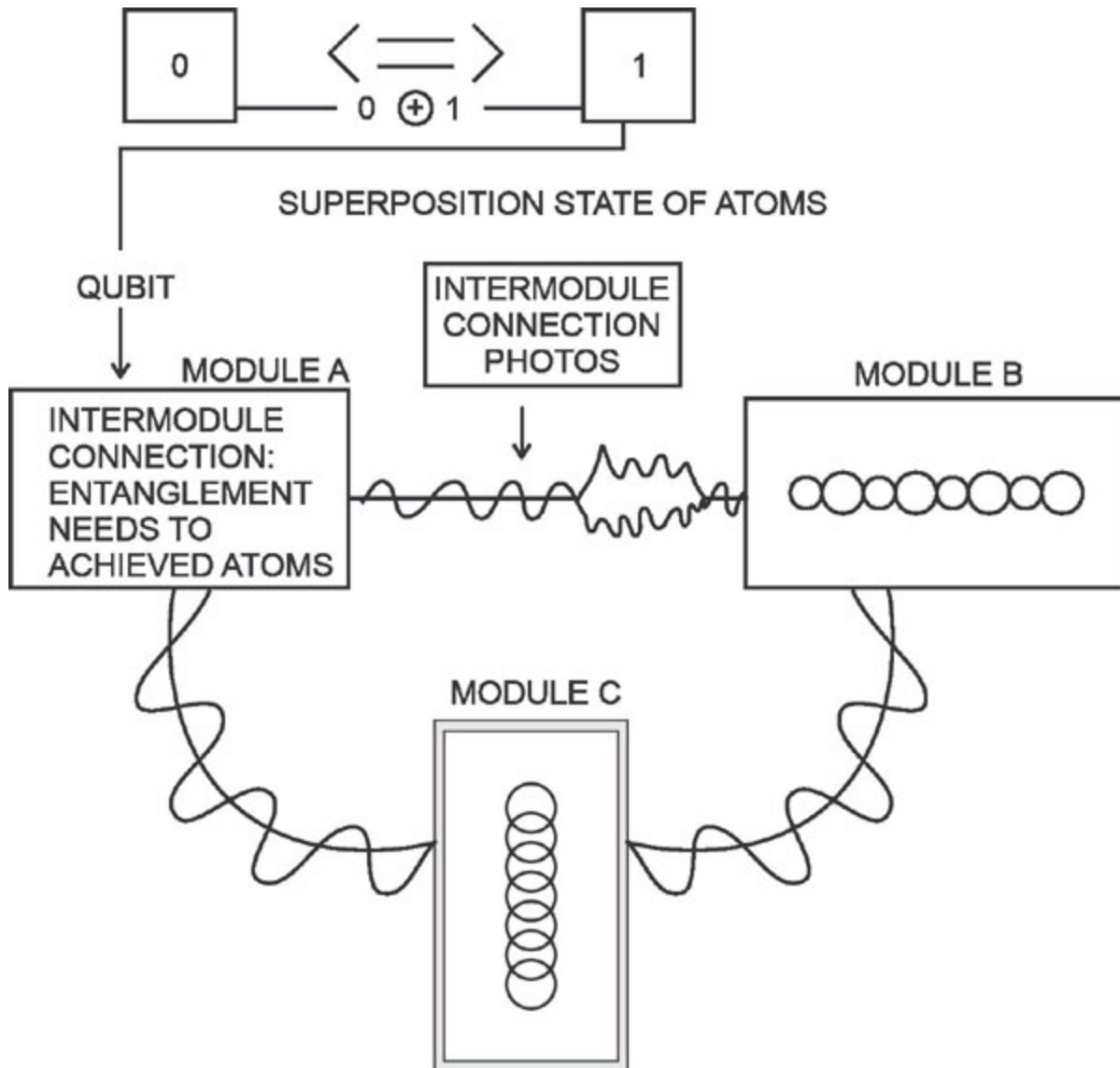


Figure 9.8: How to build a QC in a simple way

Entangled photons detection: Challenges and resolution

A key challenge for both quantum cryptography and computing is preparing the required single or entangled photons and then detecting them. The generation of more than one photon at a time could result in an eavesdropper collecting information without being detected. Developing

such photon sources is a major activity. For instance, the University of Bristol has built a single-photon source from a microstructure optical fibre.

Whereas a group at the University of Oxford has developed various single-photon sources, using both nonlinear crystals and optical fibres, holding the world record for the highest-purity single photon. The quest goes on at the University of Sussex which has generated single photons from single ions, which is important for transferring quantum information between different carriers. Further, Alastair Sinclair and his team at NPL are researching single and entangled-photon sources to determine standards for performance. Ongoing research at the University of Cambridge and Imperial College London, along with Toshiba Research Europe has created a scalable semiconductor device based on quantum dots grown in an optical cavity that emits single photons at a wavelength suitable for optical-fiber transmission.

Out of multiple technology present for quantum computers, one advantage of quantum dots techniques is that they emit single or entangled photons, in response to an electrical pulse, on-demand. It is unlike nonlinear crystals largely used to create entangled photon pairs, which emit photons probabilistically. Inexpensive chips hosting millions of such devices, each addressed by an electrical control signal, could be used in linear optical computing schemes.

Photon-counting detectors are also essential. Few groups including those at Oxford, Heriot-Watt, and Toshiba, have pioneered approaches to such detectors, which are particularly robust. Toshiba has developed a semiconductor photodiode that rapidly amplifies the faint signal from a photon and can detect up to 500 million photons a second. Whereas researchers are working on novel detectors based on various semiconductor materials, superconducting nanowires, particularly for single-photon detection in the near-infrared and infrared spectral regions.

Baby steps for quantum architecture

At the ground level, the work of quantum architecture is to govern how to separate qubits or qubit blocks that interrelate and communicate in instruction to process data. There are few chief areas where investigational

groups have started to consider architectural insinuations in designing their quantum system, let's have a look on them.

Heterogeneity: A few investigators have been examining technological heterogeneity by using amalgamations of electron spin, nuclear spin, magnetic flux, and photon polarization in a solitary organization. It is, though, likewise significant to deliberate structural heterogeneity, both in working competence and in interrelating speed or reliability. This can be achieved by cumulating the number of on-chip devices. Controlling nanoscale systems requires tens to hundreds of physical qubits. This further will need considerable venture in systems engineering. But with the challenges and opportunities, this is one of the major areas of research at the moment.

Interconnects: Inside separate chips, we have now sighted the start of designs with multiple kinds of intersects. Amalgamation levels will probably spread hundreds to thousands of qubits in a solitary chip or subsystem, but the accomplishment of millions to billions of devices in a system will need interrelates that continue only on the portrayal board. The inter subsystem connections are physically similar to intra-subsystem connections, while in others they can be radically different.

Error correction: The daring hard work of experimentalists has transported us to the point where a few qubits can be controlled and entangled. Receiving to that phase has been a colossal task as the delicate quantum system must be isolated from the atmosphere and its state-protected from wandering. What did you say will occur, then, when we thrust to hundreds, thousands, or millions of qubits? This transports us to the microarchitecture for quantum computers.

For a quantum architecture to being effective, both the correctness of the gate processes and the grade of isolation from the environment mandatory to perform the vigorous calculation of any important size lie far outside the domain of feasibility. To make matters worse, quantum data is subject to a restriction known as the no-cloning theorem which means normal, traditional, approaches of controlling error cannot be applied. Quantum data cannot be backed-up or imitated for humble recurrence code dispensation to detect and check for errors.

As we know, the accomplishment of error correction is not a humble task. In reality, the massive mainstream processing power will be used to correct errors in the quantum state. Application algorithms and data processing make their presence only at a level well above the real-time, (physical) qubit-by-qubit work of error correction. A structure for a universal quantum computer will consequently have as its main goal the implementation of exact types of error correction.

The employment of error correction is probably an important near-term investigational goal. As untried competencies have fully grown, assemblages have begun challenging to validate quantum error alteration in progressively comprehensive procedures. As in traditional computer architecture, microarchitecture is the bond between physical device competencies and architecture. Microarchitecture in the quantum case can be unstated to be the level devoted to the well-organized implementation of quantum error management, while system building is the organization of microarchitecture blocks into a whole system. Numerous exact quantum elements must be well-thought-out for this microarchitecture.

Clock speed: The adaptation factor from the physical gate cycle to the logical gate cycle has a strong, underappreciated influence on the act of an algorithm. It is contingent on a number of architectural topographies, as well as the error correction code itself. For instance, in the ion trap-based system, the code will be pragmatic in a recursive way, a follow-on in the development of the physical system by an order of scale and an upsurge in the logical clock cycle to 260msec. This melodramatic upsurge exemplifies the result of the base physical error rate on the architecture and act.

Most numeric quantum algorithms hinge profoundly on a three-qubit gate named a controlled-controlled NOT gate, or Toffoli gate (which we discussed in the previous chapters). In most quantum error correction examples, direct implementation of a Toffoli gate on programmed logical qubits is not probable. In its place, the Toffoli gate is done using numerous processes, together with one that devours a particularly ready ancilla state. The ancilla is shaped by utilizing distillation which is a quantum error detection code. This further takes noisy physical states and shapes more precise logical states. The formation of these states might control the workload of the machine. In conclusion, we can say that a functioning

quantum computer nearly undoubtedly will not trust on a solitary kind of error correction but integrate different forms of error correction/detection/suppression at different levels. Diverse error management methods have different fortes and faintness, and the combinatorial space for assimilating numerous sorts is huge.

Defects: A quantum architecture essentially considers that fabrication will unavoidably be a flawed procedure. Qubits may be professed faulty since they fail to properly hold the precise state mutable carrier (to trap a single electron), for the reason that memory lifetime is brief or gate control vague, or since they fail to couple correctly to other qubits. For gate-based, error-corrected systems, scheming to demonstrate that a severe meaning of declaring a device to be useful pays for itself in condensed error correction demands. A system's pliability to short harvest is very microarchitecture dependent. Otherwise, the digital quantum error correction itself can be revised to endure loss.

Now, with this, we conclude our discussion on the quantum computer and computing architecture. Further, we will witness of some stories and use cases, that would provide some practical insights into the topics covered in the chapter.

Startup stories: D-Wave revisited

D-Wave Systems, Inc. is a very famous first of its quantum computing company. It is founded in Burnaby, British Columbia. In 2011, D-Wave System declared D-Wave One, branded "the world's first commercially existing quantum computer," working on a 128 qubit chip-set using quantum annealing to resolve optimization problems. In 2013, it was proclaimed that there will be teamwork among NASA, Google and the Universities Space Research Association (USRA). It flung a Quantum Artificial Intelligence Lab using a 512 qubit D-Wave Two that would be used for investigating into machine learning, amongst further areas of study. The D-Wave One was constructed on initial prototypes such as D-Wave's Orion Quantum Computer. The prototype was a 16-qubit quantum annealing processor, established in 2007, at the Computer History Museum in Mountain View, California. D-Wave confirmed what they claimed to be a

28-qubit quantum annealing processor on November 12, 2007. The chip was made-up at NASA's Jet Propulsion Lab's microdevices lab in Pasadena, California. D-Wave was created by Haig Farris, Geordie Rose, Bob Wiens, and Alexandre Zagoskin.

Story 1: Quantum chemistry

For the chemical industry, the new quantum-computing capabilities open up the possibility of modelling quantum-mechanical systems, such as molecules, polymers, and solids, at a different level of precision. It would thus be possible to identify the most effective molecular designs or structures to accomplish specific tasks and achieve the required effects—before synthesizing a single molecule in the lab.

Access to this kind of computation resource could dramatically boost the effectiveness of R&D departments and change the way new products are developed, with ramifications across the entire chemical industry. Let's look at this in more detail in [Table 9.2](#):

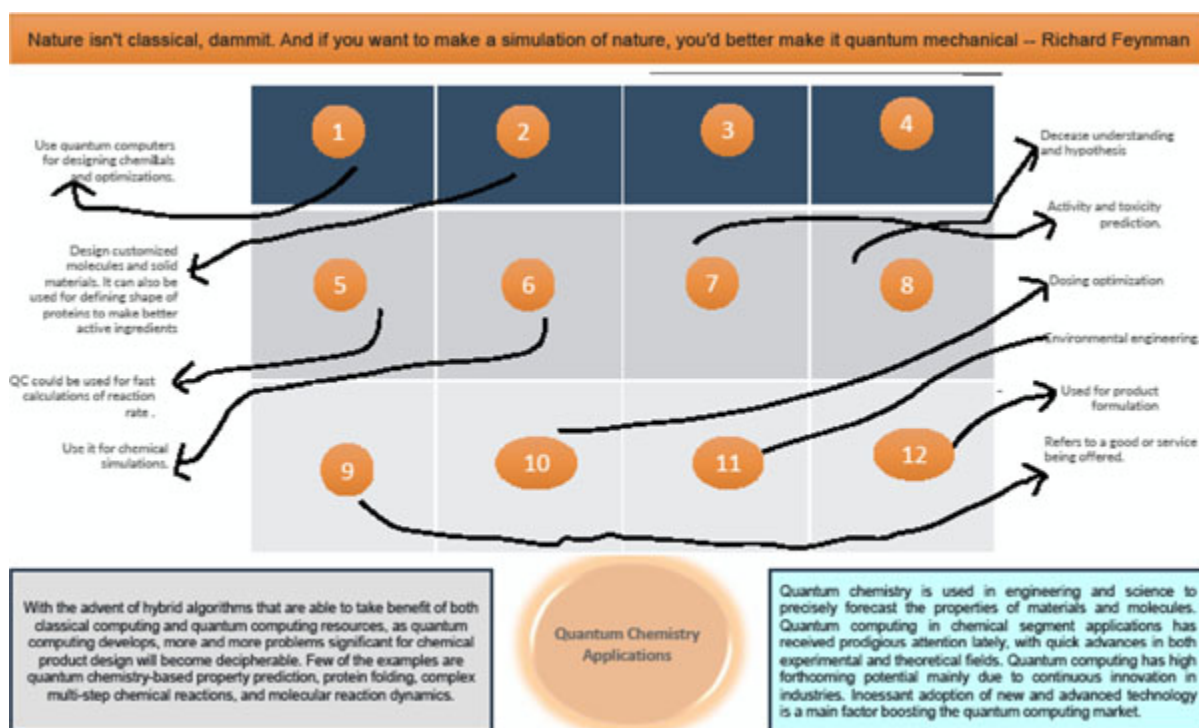


Table 9.2: Potential contribution of QC in the field of chemistry in coming times

Story 2: Quantum compilers

Quantum computing is seated at a significant modulation point. High-level algorithms for quantum computers have exposed significant capacity and new developments in quantum device construction bid confidence of utility. A gap still happens, though, among the hardware size and dependability, necessities of quantum computing algorithms and the physical machinery will be predicted within the next ten years. To bridge this gap, quantum computers need suitable software to interpret and enhance applications and abstraction layers. Given the severe resource restraints in quantum computing, information accepted among layers of software and implementations will fluctuate evidently in classical computing whereas in quantum computers/tools it is more physical in nature

The work of the compiler generally appears in two dimensions. First, on a given specific quantum circuit, it enhances it in some way. For instance, minimize the number of CNOT-gates used. But, even further, it changes the circuit in a way, that can be implemented by a quantum computer with a specific topology. To do this, SWAP-gates are performed to swap qubits from place to place, till the two qubits that the CNOT should be done on, are linked in the topology of the device.

Story 3: Quantum computer a developmental journey: adiabatic quantum computer

Adiabatic Quantum Computing (AQC) is a model of computation that practices quantum mechanical procedures working under adiabatic conditions. As a method of universal quantum computation, AQC services the values of superposition, tunnelling, and entanglement that is evident in quantum physical systems. The AQC model of quantum computing is illustrious using dynamical development that is sluggish, concerning the time and energy gages of the underlying physical systems. This adiabatic state applies the ability that the quantum computational state will continue in a well-defined and manageable way, therefore allowing the growth of new algorithmic principles.

Quite a few distinguished algorithms advanced inside the AQC model comprise approaches for resolving unstructured search and combinatorial

optimization problems. In a perfect setting, the asymptotic complexity analysis of these algorithms designate that computational rapidity ups may be relative to the state-of-the-art conventional approaches. Though, the attendance of non-ideal circumstances, together with non-adiabatic dynamics, non-zero temperature, and physical noise obscure the valuation of the potential computational performance.

A reduction of the adiabatic condition is taken in the balancing computational heuristic of quantum annealing, which lodges physical systems working at finite-temperature and in open environments. While **Quantum Annealing (QA)** provides a more exact model for the behaviour of real quantum physical systems, the likelihood of non-adiabatic belongings confuses a clear parting with conventional computing complexity.

A sequence of scientific developments in the control of quantum physical systems has allowed new realizations of AQC and QA. Protuberant instances comprise protests using superconducting electronics, which encrypt quantum information in the magnetic flux persuaded by actual weak current operating at cryogenic temperatures. A lot of devices particular for unimpeded optimization problems have been functional to resolving a variety of detailed domains counting logistics, finance, science, and numerical analysis. An associated infrastructure has also advanced to provision these investigational protests and to enable access to a wider community of users. The important development in our considerate of AQC has exposed numerous open themes that endure inspiring research into this model of quantum computation. Leading is the expansion of procedures for a fault-tolerant operation that will safeguard the scalability of AQC for solving large-scale problems. Besides, unambiguous experimental demos are desirable that distinguish the computational power of AQC and its variations from conventional computing approaches. This will also need developments in the construction and control of quantum physical systems under the adiabatic limitations.

[Use case 1: Material design and quantum computer](#)

Architecture is characteristically understood by choosing present materials for use in construction. As material science develops further urbane, though, architects will have augmented capabilities to design the architecture of materials themselves. To understand this impartiality, the aptitude to process big quantities of information is paramount.

Evolving novel materials for exact applications is a lively area of investigations for both material science, and Architecture groups. The quantity of atomic arrangements of molecular structures scales combinatorically with the extent of the molecules, restricting the competence of traditional algorithms. Quantum computers can deliver a well-organized solution to the specimen of the chemical compound space. A quantum algorithm with favourable scaling in resource requirements, permitting for the explanation of the material design problem is presently obtaining noisy quantum processors.

The anticipated arrangement rifts the problem into a classical optimization and a quantum search problem within a hybrid quantum-classical algorithm. Through their forward-thinking competences, quantum computers hold the potential of resolving previously tough problems in material design. Dreams like building a molecule or material on a computer and then totally predicting its behavior is not the things of science fiction by the use of QC.

These growths are not only stimulating from the belvedere of being talented to apply unparalleled control over material properties but it also increases thought-provoking queries about the line among materials and information. At the quantum scale, where information is militarized at the atomic level to straighten the preparation of other atomic building blocks, the unbending border among substance and data slackens.

Use case 2: Let's do the optimization: The QC way

Portfolio optimization in a quantum computing paradigm is one of the evolving fields of interest. The D-Wave adiabatic quantum computation optimization system is used to govern an optimal portfolio of stocks utilizing binary assortment. The stock revenues, alterations, and covariances are modelled in the graph-theoretic maximum self-governing set (MIS) and biased maximum independent set (WMIS) structures. These constructions

are charted into the model representation of the underlying D-Wave optimizer.

Optimization problems are at the core of numerous economic problems. In portfolio optimization, for example, the problem contains in choosing the finest possessions to capitalize on to equals the risk with the predictable returns. This is an NP-Hard problem, which implies an extremely difficult class of problem in the context of classical computers because it is difficult to solve efficiently.

Quantum annealing, provides approaches to classify improved solutions for these kinds of glitches. In an annealing process, we encrypt the problem we want to resolve into an equal physical problem. Precisely, we express the physical problem such that its lowermost energy state is scientifically corresponding to answer the problem we want to resolve. We then hypothesized this system in the annealer and permit the system to naturally grow to its lowermost energy state. Characteristically, the annealer meets to its inferior energy state through cooling. In quantum annealing, we use quantum tunneling events for the determination of the system to its global minimum in a quicker and more dependable style. Some talented methods have now been advanced for putting on quantum annealing to money, as follows:

- **Optimal feature selection in credit scoring:** Before yielding a loan, banks deliberate the debtor's revenue, age, monetary history, security, and so on, to recognize them as a high risk or little risk customer. The core problem is to control which client physiognomies are the most applicable to risk valuation. This is vital when some of the data might be immaterial or feebly connected to the output, when we do not have admission to all the data, or when using all the data is too computationally luxurious. This problem was similar on the 1 QBit SDK toolkit, in a form which can be run on a quantum annealer, displaying that these problems are also agreeable to quantum annealers.
- **Optimal arbitrage opportunities:** The impression of arbitrage is the impression of the creation of profit from contradictory prices of the same advantage in different markets, while manner no risk. For

example, we could alter euros to dollars, dollars to yens, and yens back to euros, and make a minor profit in the procedure. This is cross-currency arbitrage. While the problem of deciding the best arbitrage, occasion is NP-Hard, these can be noticed professionally using quantum annealers. Precisely, solving this problem is corresponding to discovering the route with the lowermost cost in a directed graph. This was resolved for minor sample sizes on the D-Wave quantum annealer, creating similar optimal solutions as a thorough traditional solver.

Use case 3: Development of new molecules and materials

The design of novel molecules or polymers trusts on precise forecasts of molecular possessions. Though chemical investigators have made a lot of progress with computational-chemistry gears to confront subjects that are eventually ruled by quantum mechanics, today's gears can deliver only uneven estimates. For instance, gears such as **Density Functional Theory (DFT)** deliver estimates of molecular systems and are rather real for investigating minor molecules but harshly incomplete for parts such as objects, molecules with heavy atoms, or big molecules.

The enhanced extrapolative power of quantum computing applied to molecular design effort could have significant applications in the growth of crop-protection chemicals and numerous other sections of the specialty-chemicals industry, where precise forethought into the possessions of novel molecules will haste growth. Take the example of novel solid-state materials; the design possibility unlocked by quantum computing could help new-materials' growth for a number of leading-edge sections, such as battery-operated materials, semiconductors, magnets, and superconductors.

Novel potentials would also expose in classifying and scheming molecular boards that interrelate best with proteins, custom-made to the cutting energy of proteins and small substrates. Today's computers are imperfect in their ability to predict the building of proteins. Utilizing the power of quantum computing, it may be likely to make progress in this kind of board credentials, again with significant applications in the growth of crop-protection chemicals, biocides, and certain other sections of the specialty-chemicals industry, as well as the pharmaceuticals industry. Likewise,

developing a novel cleaning-product preparation, for instance, is today founded on the triangulation of the involvement of a technician, fundamentally by trial and errors, and theoretic models.

Quantum computing could help this process with optimization intentions to comprehend precisely how, for instance, cleaner molecules interrelate with a wine stain on fibre and to recognize the best active components and preparations to eliminate it. A team using a quantum computer and a suitable algorithm could decrease the obligatory calculation time to seconds. Quantum computing could also aid growth work in the connected area of complex assemblies, such as amalgams, effect stains, and optoelectronic plans for displays.

These need hybrid approaches uniting simulations, optimization, and artificial intelligence to classify the finest materials to be used. All of which maps straight onto quantum computing's potential. Yoking quantum-computing competences here can hasten the investigation to process considerably. Three steps to engage with quantum computing are worth highlighting at this point, which are as follows:

- First, industries necessary to comprehend the precise probabilities that quantum computing could bring.
- Second, to thrive in quantum computing, one needs a make-partner-buy approach. Being open to collaboration will help chemical businesses to hit the ground running to pilot-test the first use cases and, at the same time, will help them to shape up the correct quantity of internal competences. Chemical businesses are improbable to capitalize on their hardware in the following few years, but they should contemplate how to safely access to a quantum cloud provider.
- Third, devoted subdivisions need to be built up in-house to transport the budding. At quantum computing's present level of technology willingness, this is likely to need founding central groups with the assignment to classify, test, and scale-up first application use cases through the organization.

Mind Map

The following is the mind map which summarizes the concepts explained in the chapter:

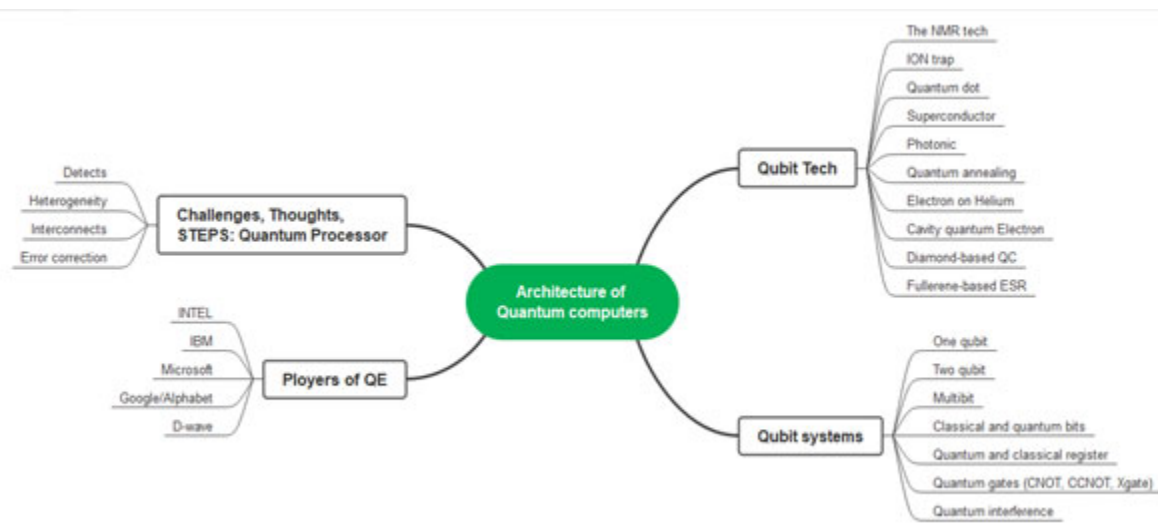


Figure 9.9: mind map for chapter 9

Question and answers

Question: What is the roadmap of a quantum computer?

The roadmap of quantum computers and computing (in terms of qubits) is explained in [Figure 9.10](#); it shows that how companies like Google, Altos, IBM and many more are working towards the dream of achieving a stable, efficient, and functional quantum computer at the moment and further what they are targeting in the future:

PHYSICAL QUBITS ROADMAP FOR QUANTUM COMPUTER

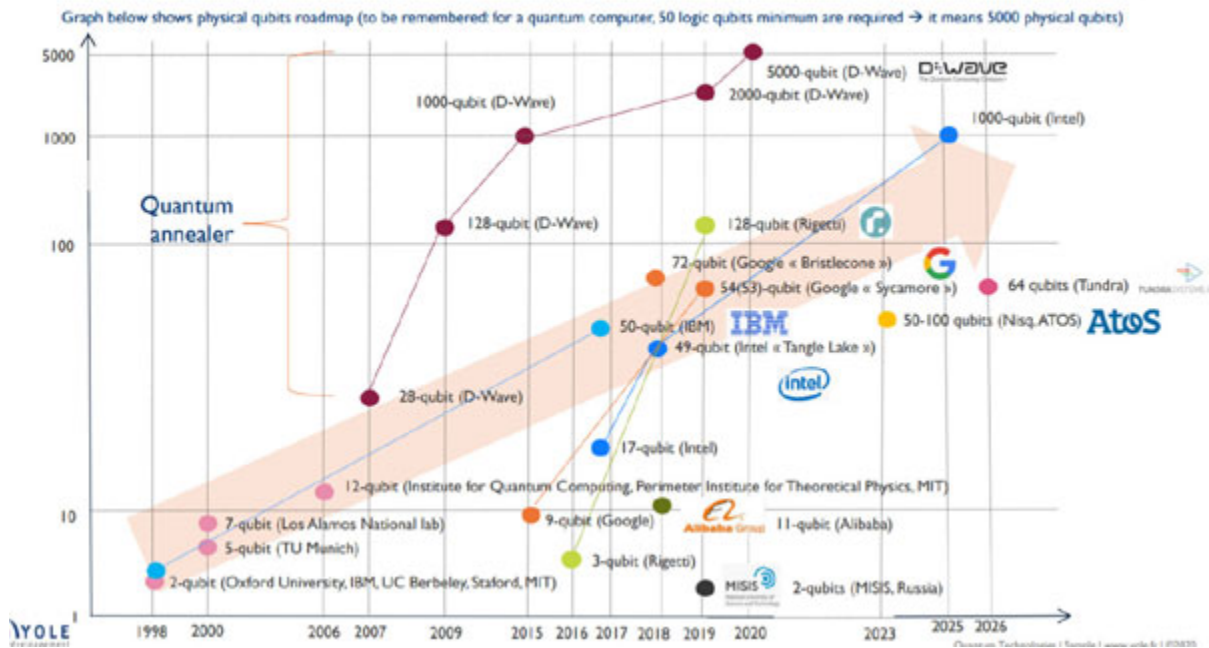


Figure 9.10: Roadmap of quantum computers (source: google image)

Conclusion

A quantum computer has both conventional and quantum fragments. However, quantum algorithms use conventional and quantum data. One method to visualize the overall system is to offload the quantum part as a coprocessor to a conventional system. It is very similar to the graphics processing unit in desktop or laptop computer and a specific task given to a quantum computer in the context of a larger program.

This chapter covered different types of proposed and working quantum hardware and their potential along with their applications. In the next chapter, we will cover DNA and photonic computers in detail.

CHAPTER 10

DNA, Quantum and Photonic Computers

Einstein first articulated the idea of conservation of energy, commonly recognized as the mass-energy equivalence. As per this concept, the mass of an object is the quantity of its energy content. This equation is one of the greatest discoveries because, before this theory, scientist was of the view that anything was made up of matter composed of the elements registered in the periodic table. Further after years of struggle in the workrooms, litres of sweat, and packages of dollars, with the trials performed 100 meters subversive in CERN, Switzerland, the scientific society finally determined, that certain matter is also a form of energy. Now utilizing this energy and minute elementary particles a new type of revolution is taking place in the area of machine development. Which is based on atoms, DNA, nano-particles, DNA and more.

Biological computers use DNA, RNA, or other carbon-based molecules for calculation. Some expert has constructed logic gates based on genes. Others have exposed how the enormous parallelism provided by DNA can be used to resolve some sorts of optimization problems.

In this chapter, we will discuss the architecture of some new evolutionary forms of computers. However. Mostly we will focus on quantum computers.

Structure

In this chapter, the following topics will be covered:

- Introduction
- Some new types of computing
- A Journey to the world of quantum, DNA and optical computing through AI
 - Quantum machines, computers, and AI

- Optical computing and AI
- DNA computers and AI
- The drive of DNA computing
- Explain to me in detail about DNA and DNA computing
- What is the Traveling Salesman Problem (TSP)?
- Let's solve it through DNA computers
- The advantages of DNA computing
- How long will it take for DNA computers to arrive?
- Optical Computing: A Journey
- Let's understand optical computer
- Benefits of optical computing
- Details of optical computing
- Laser transistor is playing a significant role in optical computer
- Present and future of photonics in the context of optical computing
- Optical processor architecture
- Microsoft and programming Biology
- Automation of DNA and Microsoft
- Can we have a DNA-based computer act as a Quantum Computer too? If yes, then how? If no, then why not?
- How Quantum, DNA and Optical computers would work together

Objective

After studying the chapter, you will be able to do the following:

- Deal with the fundamentals of DNA and optical computers.
- Develop the broader understanding of the architecture of DNA and optical computers.
- Understand the details of DNA computers along with the foundation of DNA concepts.

- Showcase how DNA, optical and the quantum computer would work together.

Introduction

There are some issues that exist on the micro as well as on the macroscopic level which traditional artificial intelligence finds difficult to solve. The specific problems range from the design of new materials to pharmaceuticals, and to the creation of clean energy devices. However, the good news is that if we could create a computing device which is based on the fundamentals of microscopic properties of matter, optics and life forces, we can go to the deeper level and resolve problems.

Quantum computers are potentially capable of handling complex real-life problems along with artificial intelligence, which may exist in genomics, science, humanities finance, or engineering. It can do this by building novel materials and by other means. Also, when it combines itself with optical computing infrastructures its power rise exponentially. Hence, the integrative field of optical quantum computing is emerging at a lightning speed and we started finding solutions to old age problems even faster. The arena of optical quantum computing (hybrid model of quantum and optical computers) is charming and there are lots of imperturbable likely applications, which we can visualize in the near future as well as in the longer term.

When this technology fully grows and assimilates itself with artificial intelligence, its use cases are massive, particularly in the area of number interpretations, interplanetary examination, finance, and genomics. So, from massive data centres to mobile phones, the power of computers continues to transform our lives and there are high possibilities that you would start seeing the appearance of quantum optical computation in our lives.

Some new types of computing

A group of a new generation of computing kids are knocking our door. Some of them use concepts like quantum physics, a few others use optical concepts, whereas some use electronics and biology. As we know, QC is

based on the principles of quantum theory and uses atoms for doing all the processing-related manipulations.

Optical computers use light (photons) in the place of electrons for computation. Some quantum computers may be optical, but most optical computer prototypes have used standard binary logic (bits) as an alternative to quantum entanglement. Investigators have identified the aptitude of optical computers to function as switches on optical networks to evade optical-to-electrical conversion, and turn them in reality.

A journey to the world of quantum, DNA and optical computing through AI

In 1995, Peter Shor's paper titled *Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer* discussed a large prime factorization algorithm to run on a quantum computer. This jumps a competition to make a practical quantum computer. Because he established mathematically that the time complication of his algorithm is expressively quicker than the present title holder of classical computing, corporates smell money in that and took it further. One area where QC is stated to make its mark is AI.

Quantum machines, computers, and AI

The experts stated predicted that AI and quantum computer combinations will take us to a new level of achievement. But there's also, constantly the anxiety that artificial intelligence programs powered by quantum computers could take over the world and crush us. Popular movies like Terminator in Hollywood and Robot in Bollywood play their roles in popularizing this fear. However, in reality, machine learning and broadly AI could bring transformation to the way we do many things without any element of fear (at least for now).

Machine learning along with other subfields of AI is augmenting and bringing change in the face of psychology, sociology, the brain, and core sciences. Shortly, it could bring alteration to the way we visualize and perceive the world around us. On the practical ground, for the successful implementation of any machine learning and artificial intelligence initiative,

data plays an important role. It is data which is fulling the revolution in the area of AI in recent times.

However, crunching those data in real-time to get effective insights is still a challenge. For the “silicon” based chips, it is very difficult to process a huge set of data in the coming time due to the tremendous volume of data humans are adding every day. Quantum computers come to the rescue of these types of limitations.

Quantum computers are tremendously hard to program with today's resource availability simply because they need extremely focused knowledge, infrastructure, and unavailability of technology for general people. Hence, classic computing are not going away. Quantum computers will also be very expensive from an energy and product price perspective. It is something like the first computers and their growth over the years into the initial desktop computers.

They will derive a lot of possibilities and are expected to solve “impossible” problems, but not all the so-called problems would be “solved” on QC. Therefore, the most probable situation is that both classical and quantum computers will just co-exist, and the quantum system's existence would counterpart to the classic system when it arises to resolve exact problems that would otherwise take an extensive period to calculate.

Optical computing, deep learning and AI

The idea of constructing a computer that uses light rather than electricity is not new. Almost all the researchers who are working in the field of optical computing, especially for photonic science have a firm belief that optical computing has widely guaranteed quicker performance and consumes less energy than its counterpart electronic computers.

However, the view of a practical optical computer has weakened as scientists have thrashed to create complete light-based components which is one of the requirements to outshine existing computers.

But regardless of hindrances, optical computers in recent times gets a renewed start. Investigators are testing a new-fangled kind of photonic computer chip, which potentially overlays the means of optical devices capable enough for integration with artificially intelligent devices and to

power self-driving cars and other smart tools. They are small enough and able to fit appropriately in one's pocket. Investigators at the Massachusetts Institute of Technology recently proposed light-based computing devices that would be particularly cooperative to fuel deep learning, which is one of the branches of artificial intelligence.

Deep learning needs an enormous volume of computation so that it feeds massive data sets into big networks of replicated artificial neurons. Artificial neurons are artificially created network which is based on the neural structure of the human brain and mimic its behaviour. Each artificial neuron takes in an array of numbers as input, does a simple calculation on those inputs, and thereafter directs the outcome to the next layer of neurons. By tuning the calculation of individual neuron, an artificial neural network can acquire to do tasks as varied as identifying cats and driving a car.

Deep learning as technology has become so vital to artificial intelligence that corporations like Google and high-performance chipmaker Nvidia have consumed millions of dollars into developing dedicated chips for it. Their innovative chips would take benefit of the fact that most of an artificial neural network's time is consumed on matrix multiplications processes in which individual neuron additions act as its inputs by employing a different value on each one. For example, in a facial-recognition neural network, some neurons might be tracing the signs of eyes; those neurons would place a larger value on inputs conforming to the capability of seeing, regions like retina, eyebrows, and a slightly lower value on light patches for establishing possibilities of skin and very little on, say, the colour neon black which is highly unlikely to embellish someone's eyes.

A dedicated deep-learning chip also does numerous of these weighted sums concurrently by farming them out to the chip's hundreds of minors, self-determining processors, yielding an extensive speedup. That kind of job demands processing power equal to a mini-supercomputer. Mercedes and BMW are developing self-driving cars that have the luxury of adding an entire stand of chips in the cars for taking advantage of these technologies. However, the main objective of optical chips is not to substitute a conservative CPU at least soon.

Optical chips are capable of performing or doing high-performance computing tasks with ease; therefore, scientists are building ultra-modern devices that incorporate photonic chips. In a few initiatives, they are integrating it with quantum computers for getting the utmost benefit. By dealing with information on light, chips bring very high speed, low latency, and little power consumption.

This complete package, therefore, bring massive orders of scale development over old-style electronic architectures. When all this is combined with artificial intelligence, the change becomes more evident. For example, it was observed in laboratory experiments that artificial neural networks have intensely enhanced performance for many machines learning tasks, including language and image recognition which are computation-intensive activities.

If we analyse today's market availability of hardware vs their performance in doing the massive parallel computation, we can easily figure out that computing hardware is incompetent at executing neural networks. The reasons are simple and one of the major ones is a considerable part of their design is based on von Neumann computing arrangements. So, an architecture that is based fully on the optical neural network will, in principle, present an improvement in the computational speediness and power effectiveness over state-of-the-art electronics for conservative inference tasks. Optical chips can enhance particular calculations for quantum computing.

At MIT, a research team had integrated a photonic chip into a popular computer device and instructed the system to perform a deep-learning program to divest its matrix multiplications onto the optical hardware. In the process of accomplishing the task, when the computer needs to perform a matrix multiplication which can be explained as a bunch of weighted sums of some numbers in row and column format, it first changes the figures into optical signs, with larger figures signified as sunnier beams. The optical chip then breaks down the complete multiplication problem into numerous lesser multiplications. Then each smaller task is moved by a single cell of the chip. To comprehend the process of a cell, you may visualize two streams of water flowing into it (the input beams of light) and two streams flowing out. The cell performs like a frame of channels and

drives which split up the streams, speeding them up or slowing them down, and mingling them back again. By regulating the rapidity of the drives, the cell can guide different volumes of water to each of the output streams.

The optical correspondent of drives is the animated channels of silicon. When animated, silicon atoms will feast out a bit, and due to this reason, it is possible for light to be transportable at a dissimilar speed, leading the light waves to either improve or overpower each other much as sound waves do. The conservative computer bands the warmers, so the volume of light flowing out from each of the cell's output channels is a weighted sum of the inputs, with the warmers shaping the weights.

DNA computers and AI

Modern silicon chips-based computers have helped us excel in almost all walks of life; however, our requirements grew and we need more contributions from their side. Computers along with AI can potentially take us to the next level by providing us insights by analysing the data we generate on a day-in, day-out basis. However, the data which we generate is huge; therefore, shortly, we will face severe challenges to store and process them. So, it is natural that we would start looking for options for storage. Researchers have started looking for the next big thing, and one of the possible novel materials which they are considering is somewhat astonishing called deoxyribonucleic acid or DNA. Investigators and academicians have identified for a long time that DNA can be used for data storage.

While we were decoding the myths and technicalities of optical and quantum computers, our main focus was on processing complexities and speed. Whereas the same focus is required for identifying techniques and devices for storing the processed data. Let's have a look ... how far we have moved in that direction and what are the future possibilities.

Nobel literate and one of the funniest scientists of our time, Richard Feynman, was a firm believer in something called sub-microscopic computers (quantum computers) and he proposes the idea that we can do this type of computation as early as the decade of seventies.

However, when we talk about the microscopic level of computation in biological terms, the first concept that comes in the mind is DNA. Because it is the smallest unit of processing, storage, and heredity, the first proposal was made in 1994 for using DNA as computers by Leonard Adleman. The computers are based on the functioning of DNA, the building blocks of our life. If we see the broader specifications, structures, organizations, and working of our genetic code, we start appreciating that it is a seamless tie for a computer. Let's examine a few reasons for this thought.

First, it is capable of storing the design for the creation of all live creatures on earth. Second, it communicates from one generation to the next with unbelievable trustworthiness recurrently. This is the reason, why the numerous genes still exist almost unaffected or unchanged for innumerable generations. Third, DNA is extremely tough, enduring and if saved under the correct circumstances, it will stay undamaged for millions of years. Fourth, it is not delicate and mild as thought previously. Therefore, after seeing all the reasons mentioned here, it's natural that researchers are at work on turning DNA into computer storage.

DNA can be used as a normal storage device because the binary code comes from utilizing the bases thymine (T), guanine (G), adenine (A), and cytosine (C) to signify 1s (T and G) and 0's (A and C). Each cell that's ever existed has been the consequence of the four-letter genetic script: A, T, C, and G, the elementary units of DNA. Now scientists are also working on generating synthetic DNA; the first living organisms are shaped with six-letter DNA.

The four natural letters plus two new man-made ones, X and Y – this breakthrough discovery could challenge our basic understanding of nature's design. But at the same time, it will help create customized man-made DNA computers. The other recent advancement in the field of DNA understanding, which adds to DNA computer research, is a success in editing and programming of DNA, just like coders do with their programs. However, this code creates life, giving scientists the power to translate digital data into biological material, similar to proteins and vaccines.

The term biological transportation, which holds the potential of beaming novel medicines across the world over the internet is not alien. With this

technology interconnection of DNA, computers become reality. Be mindful here, the DNA that we are talking about is not human DNA but laboratory-grown artificial DNA.

The drive of DNA computing

Since the previous decade, computer experts started looking at core physics and biology in the chase of more influential computers, transistors, and the on-off switches that power the computer processor. They started looking beyond the comfort zone of electronics and observing outside the silicon chip.

As an instinctive alternate is now being advanced using DNA to do kinds of complex calculations that silicon transistors do now. The problem with transistors is that they today occur at the scale of a few nanometres in size; therefore, it is only a few silicon atoms thick. They can't almost be made any lesser than they are now. If they get any lesser, the electrical current flowing through the transistor simply escapes out into other components nearby or distorts the transistor due to warmth, making it useless. You essentially need the least number of atoms to make the transistor work and we've functionally reached that limit.

Multiple corporations and research institutes are trying to get rid of this problem. Hence their engineers have found some workarounds for this delinquent by using multicore and multiprocessing systems to upsurge computational supremacy without consuming to shrink the transistors further, but this too comes with trade-offs in footings of programming challenges and power necessities, so the additional solution is desirable. This is the driving factor behind the development of alternative computing paradigms like DNA, quantum, and optical computing. Therefore, we are likely to see some more influential computers in the future.

DNA and DNA computing: Details

While quantum computing is getting a lot of press lately, DNA computing can be just as—or even more—powerful than even quantum computing and it doesn't run into nearly as many of the stability constraints that quantum

computing has. Plus, we know it works; we are living examples of the data storage and computational power of DNA computing.

The challenge for DNA computing is that when it is equated to traditional computing, it is slow. Evolution has had hundreds of millions of years to grow the complex order of DNA that occurs exclusively in each one of our cells, so that DNA is used to employ rendering to physical timescales, not the manifold gigahertz of modern classical processors.

But at the same time, benefit with DNA computing is what we would use to resolve problems outside the possibility of what a traditional computer can solve, the same way quantum computing can break RSA encryption in instants while it might take traditional computer thousands of years to do the same.

DNA computing pays a DNA molecule as a chief resource to fulfil computing responsibilities. Though, the idea of the primary DNA computing unit saves incomprehensible. It was later understood that there are manifold systems of elementary DNA computing units, all with the significant paraphernalia of nano-scale DNA remains of Waston-Crick pairing. Mostly five non-exclusive kinds of the chief DNA computing units were abridged one by one, and they are as follows:

- Adleman unit
- Tiling unit
- Rothemund-Shapiro unit
- Ribozyme unit and
- Paun unit

Adleman unit is fundamentally neutral elementary Waston-Crick pairing or an elementary form of DNA get-together, Tiling unit is subsequently frequent DNA slates, with assembly as computing process. Another type of Rothemund-Shapiro unit is Turing machine-like autonomous DNA automaton.

While Ribozyme unit is an endeavour that can primarily lead to protein enzyme-free DNA automaton. There is a supplementary special DNA computing unit named here as Paun unit, which is the membrane computing unit. It is not easy to comment which unit is most capable in the future, but

revisions on all these units will assuredly endorse DNA operation technologies and consequently the development of novel nano-scale DNA computing devices.

As we know, DNA computing was first designated in 1994 by computer expert Leonard Adleman. After the interpretation of the construction of DNA, he was enthused to write a paper in the journal Science screening on how you could use DNA to a disreputable mathematical and computer science problem known as the directed Hamilton Path problem, commonly called the travelling salesman problem (though the Hamilton Path problem is a slightly different version of the travelling salesman problem). Let's first understand the travelling salesman problem and further resolve it.

About Traveling Salesman Problem (TSP)

As per TSP, a corporation has a salesman that must officially visit n numeral of city for the sales purpose and can only visit each city on one occasion. Then, what order of visited cities delivers the shortest, and consequently the inexpensive, path?

Let's try to solve this; when n equals 6, the problem can be functioned out by hand on a piece of paper and a traditional computer can examine every conceivable path comparatively faster. But what if n equals 25? Discovering the shortest path through 25 cities becomes computationally more problematic and it would take a traditional computer exponentially longer to find the response.

Further, when it is 600, then finding the shortest path between 600 cities becomes very difficult and it would take a traditional computer longer than the whole lifetime of the Cosmos to find the shortest path since the only method to confirm that we have found the shortest path is to check every solitary permutation of cities. Some algorithms occur using dynamic computing that can hypothetically decrease the number of checks compulsory, but that might cut a limited million years off the top. The problem will still be all but computationally unbearable on a traditional computer.

DNA computers: In context of traveling salesman problem

DNA can be accumulated in such a way that a test tube filled with DNA blocks could accumulate themselves to encode all of the likely paths in the travelling salesman problem at the same time. In DNA, genetic coding is signified by four different particles, called A, T, C, and G. These four “bits”, when bound together, can hold an unbelievable volume of data.

After all, the human genome is prearranged in such a way that it can be crowded into a single nucleus of a cell. By fraternization of these four molecules into a test tube, the molecules unsurprisingly gathered themselves into components of DNA. If some mixture of these molecules signifies a city and a flight path, each strand of DNA could characterize a different flight path for the salesman, all actuality calculated on one occasion in the mixture of the DNA strands accumulating themselves in parallel. Then, it would just be a matter of sieving out the longer paths until you have only the shortest path left. This was shown by the originator of DNA computing in his original work for seven cities.

The reason this generated excitement was that DNA structures are cheap, relatively easy to produce, and scalable. There is no boundary to the influence that DNA computing can hypothetically have; subsequently, its power increases the more molecules you complement to the equation and different silicon transistors which can do a single logical operation at a time, these DNA structures can hypothetically do numerous calculations at a time as required to resolve a problem and do it all at once.

The problem though is rapidity. Even though it took instants for Adleman’s solution to the travelling salesman problem to be prearranged into his DNA strands in the test tube, it took days of cleaning out depraved solutions to find the best solution he observed for (afterwards scrupulous groundwork for this sole computation). [*Figure 10.1*](#) showcases how the digital data is stored in the DNA storage system:

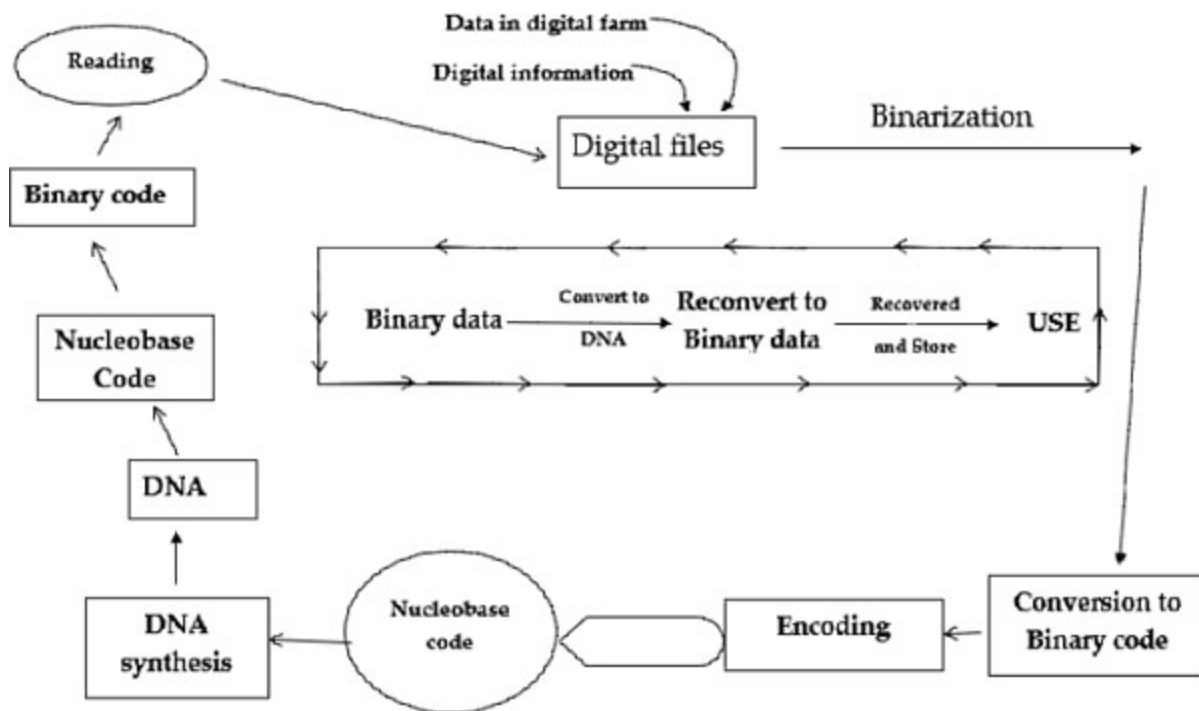


Figure 10.1: DNA computing storage process

Still, the idea was a complete one and the possibility for implausible gains in storage capacity and computational speeds was understandable. This jolted off two decades of investigation into how to produce applied DNA computing a genuineness. Storage of the digital data is one of the most important parts of the DNA computing infrastructure.

The advantages of DNA computing

As established with Adleman's paper, the main benefit of DNA computing over traditional computing and even quantum computing to a degree is that it can do limitless calculations in parallel. This impression of parallel computing can be categorized as novel and has been imitated in traditional computing for decades. When you run two applications on a computer at the same time, they aren't running concurrently; at any given time, only one training is being carried out.

So, if you are attending to music and shopping online using a browser, the computer is using something called context switching to give the attendance of concurrency. It runs an instruction for one program, protects the state of that program after the instruction is passed out, and eliminates the program

from vigorous memory. It then loads up the previously saved state of the additional program, runs its next instruction, saves its new state, and then unloads it from active memory. It then fills up the first program to carry out its following instruction and so on.

By construction of millions of incremental steps, a second across diverse programs, the arrival of concurrency is attained, but nonentity is only really being run in parallel. DNA computing can carry out these millions of operations at the same time. Over 10 trillion DNA molecules can be cuddled into a solitary cubic centimetre. This cubic centimetre could hypothetically do 10 trillion calculations at once and hold as much as 10 terabytes of data.

How long will it take for DNA computers to arrive?

We've come a long way since 1994. Shortly after Adleman published his paper, researchers were able to construct logic gates out of DNA. The portions of a circuit constructed from separate transistors can be made complex true-false logical equations out of the electrical current.

The clue behind DNA computing is to substitute chemical bonds for electrical signals and nucleic acids for silicon to produce biomolecular software. Since the last 20 years, several experiments have used molecular algorithms to do things like play tic-tac-toe or accumulate many shapes. In each of these cases, the DNA sequences had to be thoroughly intended to produce one precise algorithm that would create the DNA structure. What's different in this case is that the investigators are constructing a system where the same elementary bits of DNA can be well-ordered to position themselves to harvest completely different algorithms and hence, totally different end products.

The procedure starts with DNA origami, a method to collapse an extended part of DNA into the desired form. This crumpled part of DNA helps as the seed that jumpstarts the algorithmic assemblage line, analogous to the way a string dished in sugar water turns as a seed when mounting rock candy. The seed leftovers are mainly the same, irrespective of the algorithm, with vicissitudes made to only an insufficient minor sequence within it for each novel trial.

When the investigators have shaped the seed, it is added to a solution of around 100 additional DNA strands, known as DNA slates. These tiles, each of which is self-possessed of sole prearrangement of 42 nucleobases, are occupied from a greater gathering of 355 DNA tiles shaped by the investigators.

To generate a different algorithm, the investigators would select a different set of starting tiles. So, a molecular algorithm that implements a random walk needs a different group of DNA slates than an algorithm used for counting. As these DNA tiles link up throughout the assemblage procedure, they form a circuit that chains the selected molecular algorithm on the input bits provided by the seed.

All this is not in the air now and not even confined to the laboratory only. Microsoft even has a programming language (DNA Strand Displacement) for DNA computing that can help make DNA computing applied when the technology of bio-processors advances to the point that it can run more classy algorithms. Also, Microsoft is planning to have DNA computing to its cloud services in the next few years and vigorously emerging a DNA data storage to assimilate into its cloud services. These loans will likely be understood much earlier than developments in quantum computing.

Quantum computing needs erudite machinery, superconductors, and tremendously cold circumstances to retain qubits stable to do any valuable computational errands, unless we develop a material that can act as a superconductor at room temperature. However, these are not the mandatory criteria of DNA computing, as DNA computing uses DNA that we have developed as a practiced at operating to the point of substituting a solitary gene of a DNA strand through CRISPR.

The materials desirable to synthesize DNA molecules are inexpensive and readily obtainable and continue stable at room temperature and beyond. What DNA Computing is possibly able to attain the given DNA's resiliency and biological parallelism signifies a vital step in the direction of the upcoming time of computing.

[Optical computing: a journey](#)

In contrast to the old-style memories that were existing, optical memories had two outstanding topographies; first a possible high density and the option of parallel access. In 1963, van Heerden industrialized the idea of the optical storage in 3D materials. In 1968, the Bell Labs constructed the primary holographic memory with a holographic matrix of 32×32 pages of 64×64 bits each.

In 1974, d'Auria et al. from Thomson-CSF constructed the first complete 3D optical memory system storing the data into a photorefractive crystal with angular multiplexing and accomplishing the storing of 10 pages of 104 bits. Synthetic holography (imaging technique, able to retrieve the complex scattered field from a sample, by measuring a single hologram acquired by a CCD camera) has been practised for the recording and storing of digital data in the Human Read/Machine Read (HRMR) system industrialized by Harris Corporation in 1973.

Additionally, Optical computing was invented as an original method of analysing optical problems; surely, the ideas of communications and information theory found the basis of optical information processing. In 1952, Elias analysed the optical systems with the support of the communication theory.

Information optics is a known division of optics. However, truthfully, the knife-edge test by Foucault in 1859 was generated optical information processing. Additional funders can be renowned such as Abbe in 1873 who Popularized the concept of image formation in the microscope, or Zernike who presented in 1934 the phase contrast filter.

In 1946, Duffieux completes the main influence with the publication of a book on the use of the Fourier methods in optics. This book was printed in French and deciphered in English by Arsenault. The work by Maréchal is an additional main influence in 1953. Holography can be comprised of the area of information optics; meanwhile together these fields are thoroughly connected and advanced.

There was a lot of eagerness in the field; the coming time is very bright for optical computing, and there was a subsidy for the agendas on the theme and the investigative exertion was very concentrated internationally. Each

year, numerous international conferences were prearranged by different international societies on topics related to optical computing.

Recent research was productive in all the domains of optical information processing as well as academic work on algorithms, analogue, digital, linear and nonlinear computing. Optical correlators for real applications were unfluctuating commercialized. Though, around 2000, the interest in the topic happened to weaken. The reasons are many, but the development of digital computers in terms of their act, influence, and also plasticity can be thought of as important ones. They are correspondingly very laid-back to be used even for a non-specialist.

However, due to the limitation of digital computers and in search of new paradigms, the renewal of sentiment towards optical computers is observed. And few very promising start-ups like Lighton are taking this further with their continuous innovation talks about the optical quantum chip.

Understanding the optical computer

Optical computing alternatively recognized as optoelectronic computing and photonic computing is a computation model that uses photons (small packets of light energy) shaped by laser/diodes for digital computation. Photons have shown to give us an advanced bandwidth than the electrons.

The optical computers, would give us an advanced concert and henceforth be quicker than the electronic ones. Optical information processing is founded on the knowledge of using all the possessions of rapidity and parallelism of the light to the development of the information at a high data rate.

The information is in the procedure of an optical signal or image. The characteristic parallel processing was often tinted as one of the important benefits of optical processing associated with electronic processing using computers that are typically sequential. Consequently, optics has a significant potential for processing a large amount of data in real-time.

The Fourier transform property of a lens is the foundation of optical computing. When using coherent light, a lens accomplishes in its back focal plane the Fourier transform of a 2D transparency situated in its front focal

plane. The meticulous Fourier transforms with the amplitude and the phase is calculated in an analogue way by the lens.

The rapidity of computation is contingent on two factors, how debauched the information can be transported and how fast that info can be treated that is data computation. Photons essentially use wave propagation and the interference pattern of waves to control outputs. This permits prompt computation without persuading dormancy. Data is treated while it's spreading. There is no requirement to halt the data drive and movement for its processing. This rapidity factor would potentially alter the computer industry.

The building block of any conservative electronic computer is a transistor. For optical computing, we attain an equal optical transistor by making use of materials with non-linear refractive indices. Such materials can be used for the creation of optical logic gates, which go into the CPU. An optical logic gate is just a switch that controls one light beam. The systems signal ON in case the light is being transmitted and debated, whereas it appears OFF in the absence of the blocks of light.

Photons are almost massless, henceforth we essentially prerequisite very fewer quantity of energy to stimulate them. Likewise, in its place of operating in a sequential style like most of the traditional computers, optical computing functions in a parallel way, which aids it to challenge complex problems using light reflection, as well as have augmented bandwidth as associated to electron-based systems. Approaching security, as optical computing processes data though it in motion, very fewer data is unprotected. These lead to augmented safety than the conservative systems.

Benefits of optical computing

Some of the benefits of optical computing are described as follows:

- Less heating
- Able to handle complex computations rapidly
- Quickly scaled to bigger networks proficiently
- Augmented computation rapidity
- Advanced bandwidth with low data damage broadcast

- No problem of electrical short circuits

Not so good about optical computing

Some of the disadvantages of optical computing are described as follows:

- Components and parts of optical computers would be expensive
- Huge in size
- Integrating optical gates is multifaceted
- Interference can be produced by dirt or any inadequacies

Details of optical computing

The history shows that considerable exertions were keen on the structure of optical processors that could process in real-time a big volume of data. Today, we see that optics is effective in information systems such as transferences and memories related to its comparative disaster in computing.

This could have changed, if, in the seventies, the electronic computers were not lethargic and had an imperfect power. Today components such as well-organized spatial light modulators (SLM--is an object that imposes some form of spatially varying modulation on a beam of light) laser diodes, or high rapidity and high steadfastness detectors would have been obtainable. Though, all the investigations result in optical computing subsidizing strongly to the growth of new research topics such as bio photonics, nanophotonic, optofluidic, and femtosecond nonlinear optics.

The vision of an all-optical computer overwhelming the digital computer not ever became reality and optical correlators for pattern acknowledgment have nearly vanished. The explanations are numerous. The speed of the optical processor was continuously restricted by the rapidity of the contribution and output devices.

The digital computer has advanced very speedily; Moore's law is still effective, multi-core processors are more influential, and it is clear that digital computer is calmer to use and bids more elasticity. Digital computers have advanced quicker than optical processors. Optical computing is mostly analogue when electronic computing is digital. The digital optical

computers were not intelligent to contest with the electronic owing to the absence of suitable optical components.

It seems clear that the explanation is to subordinate optics and electronics and to use optics only when it can bring to some degree that electronics cannot do. Optical processing is valuable when the information is optical and that no electronics to optics transducers are desirable. The potential of optics for parallel real-time handing out leftovers and the future will tell if optical computing will be back, for instance, employing nanotechnologies (a branch of engineering that focuses on creating incredibly small and intricate electronic parts and devices, measured in nanometers).

At present, quantum computing holds self-assurance for an original technological bound, but there is an additional choice on which many are holding their hopes, optical computing, which swaps electronics (electrons) with light (photons).

The end of Moore's law is the usual importance of physics; to carton additional transistors into a similar space, they have to be contracted down, which increases their speed while concurrently plummeting their energy ingesting. The miniaturization of silicon transistors has prospered in contravention of the 7-nanometre barrier, which used to be considered the limit, but this reduction cannot last forever. And although more influential systems can always be gotten by increasing the number of transistors, in charge, so the dispensation speed will decrease and the warmth of the chips will rise.

Henceforth, the potential of optical computing, photons transfer at the rapidity of light, quicker than electrons in a wire. Optical technology is also not a stranger to our lives; the massive universal traffic on the information thoroughfares nowadays travels on fibre optic channels, and for years we have used optical readers to burn and read our CDs, DVDs, and Blu-ray discs. Though in the gluttons of our systems, the photons impending done on the fibre optic cable must be rehabilitated into electrons in the microprocessors, and in turn these electrons must be transformed into photons in the optical readers, decelerating down the process.

Therefore, our present technology is now a hybridization of electronics and optics. In the short-term, it is clear that hybrid optical-electronic systems

will rule. For instance, the massive widely held communications data is channelled through photons, though nearly all computation and logic is done by electrons. There are important explanations for this separation of labour because while less energy is desirable to communicate information in the form of photons, the waves related to the electrons are smaller.

Meaning the advanced rapidity of photonic devices has as its complement a greater size. This is the reason why some specialists realize restrictions in the diffusion of optics in computing. It is agreed that the utmost significant for short-term application for optics is communications, that is, handling the movement of optical data from fibres to electronics.

Laser transistor and optical computer

Though we are still distant from the full optical microchip (optical microchip is a practical system skilled of computing only employing photons). The progress is snowballing contribution of photonics in computers. In 2004, experts advanced the impression of the laser transistor, which changes one of the two electrical yields of normal transistors with a light signal in the arrangement of a laser, providing a progressive data rate.

For example, nowadays it is not probable to use light for internal communication between varied components of a computer, owing to the device that would be indispensable to alteration of the electrical signal to optical and vice versa, the laser transistor would make this believable.

Comparable to transistor integrated circuits, in this arrangement, we are sure that the transistor laser will be used for electro-optical integrated circuits for optical computing. The co-author of this loan is betting on optical over quantum computing, then it does not necessitate the cold temperatures at which quantum superconductors essentially function.

The buoyant of the attention in this type of system is the penetrating examination in this field, which includes novel materials talented in supporting photon-based computing. Between the tests still to be met to becoming optical chips, the integration density of the components in instruction to the reduction of the size.

Though, irrespective of these progressions, one shouldn't be unreasonably self-assured that a photonic laptop will one day feast on the hands of

customers. Since, as with quantum computers, the anticipation is vibrant that optical computing will displace electronic general-purpose computing in the short term but not substitute the digital one.

Present and future of photonics in the context of optical computing

Application-specific photonics is now available and working, predominantly, in data centres and more lately in machine learning. **Artificial Intelligence (AI)** neural networks are hyped as one of its prodigious applications, with the possibility to attain 10 million times better competence than electronic systems. Numerical workloads such as those working in AI algorithms are flawlessly suitable for optical computing.

Therefore, optical computing can resolve very complex network optimization glitches that would take periods for traditional computers. For instance, in Japan, the NTT corporation is constructing an enormous optical computer that encircles five kilometres of fibre in a container the magnitude of a part of the building, and will be practical to tough power or communications networks improvement tasks.

Scientists believe they influence the ecosystem shaped by optical telecommunications in the parts of integrated circuit project, assembly, and wrapping, and optimize for the explicit operating points mandatory by optical computing. Though, poignant from a prototype to full-scale engineering will be a hard challenge. In a nutshell, there are motives for optimism about the growth of optical computing but deprived of overrating its potentials.

Optical processor architecture

The architecture of an overall optical processor for information processing is given in [*Figure 10.2*](#):

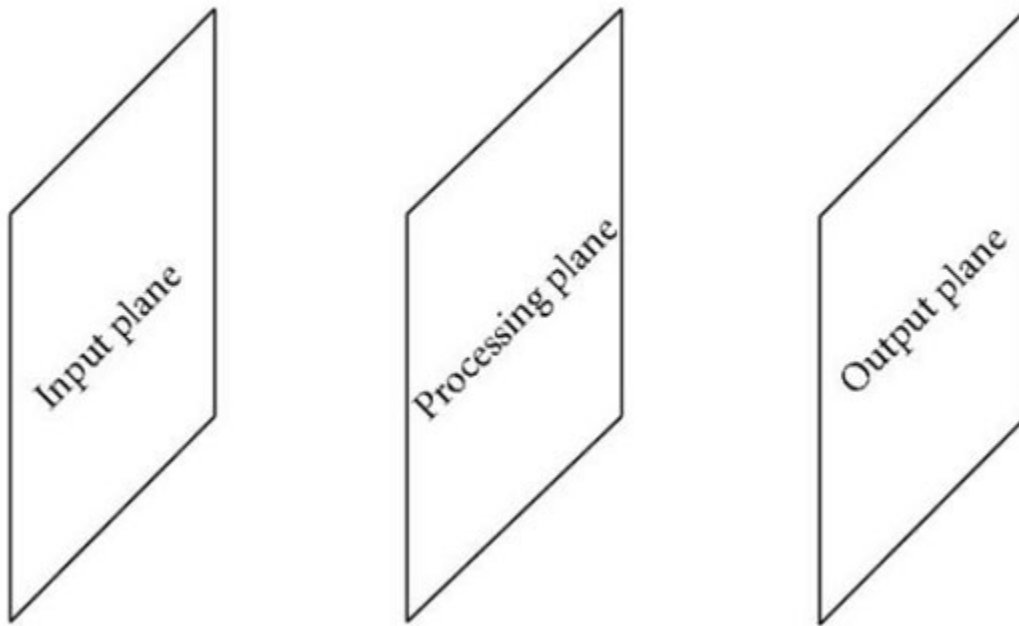


Figure 10.2: architecture of optical computer

The optical processor comprises three planes, mutually termed as, the input plane, the processing plane, and the output plane. The data to be treated are presented in the input plane; most of the time, this plane will use for an electrical to optical conversion.

A **Spatial Light Modulator (SLM)** does this conversion. The input signal can be 1D or 2D. An acousto-optic cell is frequently used in the case of a 1D input signal and 2D SLMs for 2D signals. The processing plant can be composed of lenses, holograms (optically recorded or computer-generated), or nonlinear components. This is the heart of the processing, and in most optical processors, this part can be performed at the speed of light. A photodetector, a photodetector array, or a camera composes the output plane where the results of the processing are detected.

The figure shows simply that the speed of the entire procedure is restricted by the speed of its sluggish component. The SLM is a key component for the growth of applied optical processors, but inappropriately also one of their feeblest components. Certainly, the poor performance and high cost of SLMs have overdue the construction of an optical processor for real-time applications.

After developing a strong background in quantum, DNA and optical / photonic computing in this chapter. Let's see how the corporates are reacting to these developments. This dimension of understanding is necessary because ultimately what matters is the practical uses of any classic or extraordinary concepts. Further, we will also see how these technologies would potentially work together. let's capture the Microsoft and DNA combo first.

Microsoft and programming Biology

The capability to code biology could permit important breakthroughs across a comprehensive range of industries, together with medicine, healthcare, storage solutions farming, nutrition, construction, fabrics, resources, and substances. It could also lay the groundwork for an upcoming bioeconomy based on maintainable technology. Regardless of this potential the concepts of programming biology at the moment is still done mainly by trial-and-error.

As confrontation of this challenge, the field of synthetic biology is working together to grow approaches and technology for coding biology more methodically. Station B is a portion of this wider effort from Microsoft, with an emphasis on evolving integrated platform for programming biology. The designation Station B is stirred by Station Q, which hurled Microsoft's efforts in quantum computing, where Station B stresses in its place on biological computing.

The Station B platform is being advanced at Microsoft Research mainly at the Cambridge centre of research, which houses Microsoft's primary molecular biology research laboratory. The platform purposes to advance all phases of the Design-Build-Test-Learn workflow characteristically used for programming biology. Station B is a cooperative effort being supported with designated technology, theoretical and profitable partners.

Microsoft is presently concentrating on two chief concepts: programming genomic circuits that purpose in living cells and programming nucleic acid circuits that do information processing.

On the other font, investigators from Microsoft have confirmed the first completely automated system to store and recover data in manufactured

DNA. This is an important stage in stirring the technology out of the research laboratory and into commercial datacentres. In a proof-of-concept trial, the team positively programmed the term hello in snippets of fabricated DNA and transformed it back to digital data through a completely automated end-to-end system.

DNA can stock digital data in a space that is orders of scale lesser than datacentres use today. It's one hopeful solution is storing the ignition amount of data the world produces each day, from business archives and attractive animal videos to medical scans and pictures from outside astronomical details. Microsoft is discovering methods to close an impending gap among the amount of data we are creating that desires to be conserved and the volume to accumulate it. This comprises emergent algorithms and molecular computing knowledge to encrypt and recover data in fabricated DNA, which could suit all the information presently stored in a warehouse-sized datacentre into space unevenly, to the extent of a few boards game cubes.

Microsoft's final goal is to place a system into production, that to the end-user. This would also work like cloud-based solutions. Quantum computers are already available in this fashion. That is bits are sent to a datacentre and stored there and then they are just available when the client wants them.

Information is stored in synthetic DNA molecules shaped in a laboratory, not DNA from humans or other living things, and can be encoded already as it is directed to the system. Though urbane machines such as synthesizers and sequencers now do important portions of the procedure, several of the middle stages until now have compulsory physical labour in the research lab.

However, for the method to make sense as a profitable storing solution, prices need to decrease for both synthesizing DNA. Which essentially custom construction strands with evocative sequences and the sequencing method that excerpts the kept information. Drifts are rapidly happening in that way.

Under the right conditions, DNA can last much longer than current archival storage technologies that worsen in a matter of decades. Some DNA has accomplished to persevere in fewer than ideal storage conditions for tens of

thousands of years in enormous tusks and bones of initial humans, and it should have relevancy as long as individuals are alive.

Automation of DNA and Microsoft

The automated DNA data storage system uses software advanced by the Microsoft and UW team that changes the ones and zeros of digital data into the As, Ts, Cs, and Gs that make up the edifice blocks of DNA. Then it uses cheap, mainly off-the-shelf laboratory apparatus to stream the essential liquids and chemicals into a synthesizer that shapes manufactured scraps of DNA and to thrust them into a storage container.

When the system desires to regain the information, it enhances other chemicals to correctly make the DNA and uses microfluidic drives to push the liquids into additional fragments of the system that read the DNA sequences and change it back to information that a computer can comprehend. The goal line of Microsoft was not to demonstrate how fast or reasonably the system could work but just to prove that automation is conceivable.

One instant advantage of having an automated DNA storage system is that it liberates investigators up to inquiry profound queries, in its place of expenditure time searching for flasks of reagents or repeatedly squeezing drops of liquids into test tubes. Having an automated system to do the monotonous work permits scientists and professionals occupied in the laboratory to take an advanced interpretation and start to accumulate novel plans. The team from the Molecular Information Systems Lab has now confirmed that it can stock cat photos, prodigious mythical works, pop videos and archival recordings in DNA, and recover those files without mistakes in a research setting. Microsoft has been capable of storing 1 gigabyte of data in DNA, overcoming their preceding world best of 200 MB.

Further, the investigators have also for advanced methods to do meaningful computation like searching for and recovering only pictures that comprise an apple or a yellow bicycle using the molecules themselves and deprived of consuming to change the files back into a digital format.

This leads to a novel kind of computer system being born where molecules are used to store data and electronics for control and dispensation. Tapping them in a composed manner holds some truly thought-provoking potentials for the future. Unlike silicon-based computing systems, DNA-based storing and calculating systems have to use liquids to transfer molecules everywhere.

Can we have a DNA-based computer act as a Quantum Computer too? If yes, then how? If no, then why not?

Reading and writing in DNA is sluggish than in other media, however, which makes it more matched for archival storing of huge volumes of data, rather than for rapid recovery or data processing. Google investigators said they anticipate the foremost profitable quantum computers in five years to 10 years. Further, some specialists say that a 50-qubit computer could outdo any conservative computer.

But there's a huge problem. You can't save or copy information on a quantum computer. All that computing power is of little use if you can't back up your work. You can change quantum data and put it on an old-style storage device, but all that rehabilitated data takes up a ration of space. So, physicists are following for dependable, super-compact hard drives completed with novel materials, together with DNA.

Quantum computers are so influential, precisely because of their data density and computing power. Therefore, if you control a quantum particle in a superposition of two states, you can do errands in parallel, which speeds up convinced computational responsibilities exponentially. That speed won't advance your Hot star experience or make Microsoft Excel more manageable, but it will be much quicker at successively searching algorithms or simulating complex systems like organic materials or the human brain.

But the strangeness of quantum mechanics has its disadvantages. Its rules permit superposition, but they also prohibit everybody from copying a quantum particle, which as we know is termed as a no-cloning theorem. Understand in this way, that a quantum computer programs an atom to be in a precise quantum state that characterizes a set of numbers. It is materially

unbearable for the computer to program one more atom to be in the precisely similar quantum state. Therefore, it is essential to change it into binary data that is, interpreting the numbers that pronounce quantum superposition into humble 1's and 0's.

At that point, you stock that rehabilitated data in a traditional storage format to the hard drives. But the space required for this is huge; as a predicted capacity, the size of each quantum data file from a 49-qubit computer will be on the gage of 40,000 videos. To stock that abundant data, quantum computer creators want novel data storage technologies; commercial drives aren't dense enough right now. A solitary quantum file would inhabit a stamp-sized area on a solid-state hard drive.

Therefore, one substitute storage candidate is DNA. This is a reality now as scientists confirmed a technique that could stock 215 petabytes, or 215 million gigabytes, in a sole gram of DNA. At that density, all of humankind's data could fit in a twosome pickup van. Different from conservative hard drives, which only stockpile data on a two-dimensional surface, DNA is a three-dimensional molecule. That additional perpendicular dimension lets DNA hoard much more data per unit area. Additionally, it continues a long time.

How quantum, DNA and optical computers would work together?

Here is the deal; imagine a time when the power of quantum, optical, and DNA computer combined is not a dream, but is already happening. Technically, a single quantum file would occupy a coin-sized area on a solid-state hard drive which makes it difficult to store lots of information on the limited memory, and this is where DNA computation techniques come in. Scientists managed to demonstrate a new DNA storage method that could store 215 petabytes, or 215 million gigabytes, on a single gram of DNA, and at that density, you could fit all the information on the internet onto just one gram.

Furthermore, as the University of Manchester scientists developed a DNA storage system that could self-replicate, scaling almost instantly to meet

whatever demands a quantum - optical computer can throw at it. Plus, it lasts a long time, a very, very long time without degrading.

Ok, so we discussed how processing happens at a fundamental level, how we can store the mammoth data, how we can make technologies and science to coordinate and collaborate to provide us the maximum benefit. We also learned what is the future of computing technologies along with traditional and forthcoming inventions. However, only the coordination of science and technology is not enough. In a complex ecosystem of humans, their emotions, science, technology, and different emergent fields to understand humanity is required. The [chapter 13](#) will provide you with this understanding.

Question and answers

Question: What is neuromorphic technology?

The neuromorphic computing is about emulating the biological neural structure and process of the human brain, as well as probabilistic computing, which produces algorithmic methods to deal with the indecision, vagueness, and illogicality in the natural world.

The important challenges in the neuromorphic investigation are matching a human's elasticity, and aptitude to study from unstructured incentives with the energy competence of the human brain. The computational basic structural units within neuromorphic computing systems are rationally similar to neurons. Thwarting neural networks (SNNs) are a new model for positioning those elements to emulate natural neural networks that exist in organic brains.

Mind Map

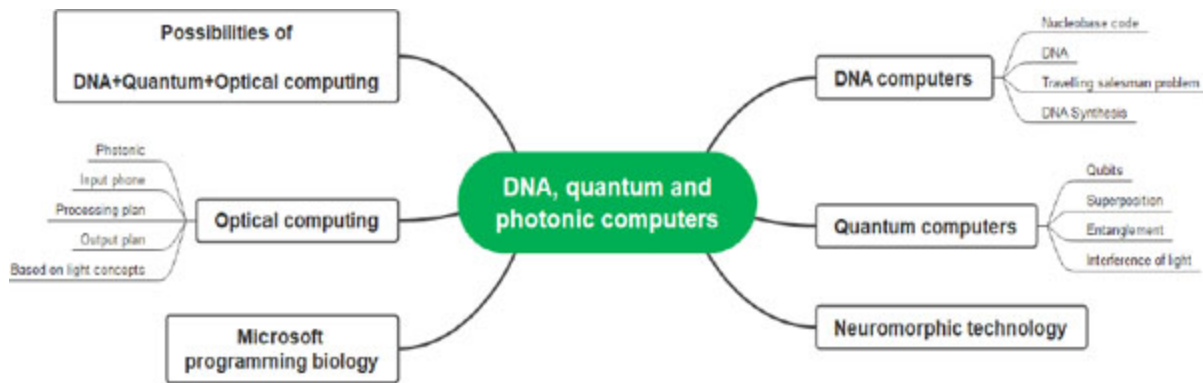


Figure 10.3: Mind map of the [chapter 10](#)

Conclusion

This chapter discussed about the integration of DNA, Optical and quantum computers. Also, it explained in details about the DNA and Optical computers and showcased how the coming time would witness the adventures of these technologies.

CHAPTER 11

Let's Realize It: Quantum Start-Ups and Giants in Action

Quantum benefit comes from producing a solution that is faster, inexpensive, or of improved quality. It's an open interrogation as to which industry will attain the first commercially beneficial applications. But even a minor recital development over traditional machines can add marvellous value for investigators and businesses around the world who run, test and optimize. For these start-ups, big companies, research laboratories, and educational institutes are here to play their role. They are running trials that educate us a lot about how these machines work and can best be put to use. Neither sustained hardware development nor applications alone will drive value. But only the thoughtful strategies around optimal uses of component and their tight integration with the bigger ecosystem is the key to success. They will help solve real-world problems. They realized that joining with researchers and enterprises to recurrently advance these competencies and adapting them to long-term business and research challenges will solve the power of quantum for everybody.

In recent years, the field of QC has seen marvellous growth and has moved out of the hypothetical realm to the real technological world. IBM played a pivotal role by converting research to commercial products in terms of QC. They showered to the world that practical quantum computers are possible and can be used for commercial purposes. In the upcoming years, we will see supplementary corporations displaying a sustained inquisitiveness in this field and would grow new-fangled innovations connected to the same. Because, enterprises like Google, AT&T, Microsoft, Raytheon, Honeywell, and Intel, are creating a robust quantum portfolio.

Apart from all these corporates, two of the world's top banks, Goldman Sachs and JP Morgan Chase are also consuming quantum motivations. Global tech companies are not alone. Multiple start-ups have nowadays

augmented fame in this area. For instance, Canadian quantum computing player D-Wave has developed themselves as an innovator in this field and claiming that they are the first quantum company in the world. One more start-up that has an aggressive development target is IonQ; they lately signed long-term partnerships with Microsoft. They are working on trapped-ion technology and want this will create novel developments in physical science, drug discovery, and progressions in AI along with quantum computers. These start-ups are not alone; they are escorted by Atom Computing, Artiste-qb.net, Aurora Quantum Technologies, Xanadu AppliedQubit, and so on. This chapter will cover some stories and technologies that technology companies, start-ups, and big players are offering.

Structure

In this chapter, we will cover the following topics:

- Introduction
- A story of startups
- The generation of start-up
- Start-up stories
 - IONQ
 - Cambridge quantum computing
 - GTN
 - Entropica labs
 - HQS quantum simulations
 - ProteinQure
 - QCWare
 - Strangeworks
- Giants stories
 - Microsoft
 - IBM

- Intel
 - Raggett
 - Alphabet
- Patents
 - Quantum patents 1
 - Quantum patents 2
 - Quantum patents 3
- Point of view 1
- Latest research

Objective

This chapter will discuss the role of start-ups, tech giants, and patents in the field of quantum innovation. Therefore, after reading this chapter, you will be able to describe the strategies of quantum computing startups and big corporations like Microsoft and IBM who are active in the field. In a nutshell, you would learn

- Working of startups and big corporations
- You will learn broader strategies of quantum companies
- What have the innovative and important patenting happened in the field
- And much more....

Introduction

A lot of companies have been competing to determine the power of quantum computing in multiple domains. For example, in the automotive industry, the Volkswagen (VW) Group vanguards quantum computing technology. Their authorities design intelligent, quantum-assisted responses for future mobility. The patent scenery delivers ideas on how the businesses are embattled to attain and where they are heading. For instance, Accenture has applied and allocated a US patent for a quantum computing machine learning module that recommends methods to train **Artificial Intelligence**

(AI) models to govern at what time and in what way, computational responsibilities would be best done by quantum computing against classical computing approaches, and direct them to the suitable possibility. Therefore, it could aid organizations to comprehend where quantum computing can have the greatest influence on their businesses, and when traditional computing may still be the best choice for a certain job.

A story of startups

There are a lot of companies, including a few small companies and start-ups ([*Figure 11.1*](#)) that are active in the field of quantum computing. Apart from them, a few are established corporates, also working in the field of quantum computing and computers. They are building hardware, and software, and provide service. Therefore, it is a good idea to have a look at them. This is exactly what this chapter is meant for. It will discuss the details of companies' offerings and solutions, all in the context of quantum computers. Refer to [*Figure 11.1*](#) for a detailed list of quantum computing startups:

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| <ul style="list-style-type: none"> • 1QBIT • Accenture • AegiQ • Aliro Technologies • Alpine Quantum Technologies • Alter Technology • Amazon Web Services • Archer • Artiste QB Net • Atom Computing • ATOS SE • AUREA Technology • BCG • Bleximo • Bluefors • Cambridge Quantum Computing • Chicago Quantum • Cogisen • ColdQuanta • D Wave • Data Vortex Technologies • Delft Circuits • EeroQ • Element Six • Elyah • Entropica Labs • Entropy • Fujitsu • Google • HQS Quantum Simulations • Honeywell Quantum Solutions • IBM • ICE Oxford | <ul style="list-style-type: none"> • ID Quantique • Intel • IonQ • IQM • ISARA • JOS Quantum • Keysight Technologies • Kiutra • LIGENTEC SA • Low Noise Factory • Microsoft • M Squared • MUQUANS • NU Quantum • NVIDIA • Oxford Instruments • Oxford Quantum Circuits • Parity Quantum Computing • PsiQuantum • Qbit Logic • QC Ware • Q-Ctrl • QDevil • Qilimanjaro • Q-LION • QLM Technology • Qnami • Qsimulate • QuanSys • Quantaneo • Quantropi • Quantum Benchmark | <ul style="list-style-type: none"> • Quantum Circuits • Quantum Computing Report • Quantum Microwave Components • Quantum-South • Qubitekk • Quintessence Labs • QULab • QURECA • QuSoft • QuTech • QxBranch • Qzabre • Rigetti Computing • Rohde & Schwarz • Silicon Quantum Computing • Sparrow Quantum • Spectratime • Strangeworks • SwissLitho • Teledyne e2v • TMD Technologies Ltd • Toptica • Toridion Quantum Computing • Toshiba • UnikLasers • Xanadu • Zapata Computing • Zurich Instruments |
|---|---|--|

Figure 11.1: A list of some innovative startups for quantum computers and computing (Source: a compilation from multiple places)

The generation of start-up and giants collaboration

In the web of emerging technologies, be it AI, blockchain, **Internet of Things (IoT)**, **Machine Learning (ML)**, and so forth, quantum computing is considered one of the most promising areas which could potentially fight them. It will turn out to be a technological game-changer. This is the technology that will exponentially make the improvement and quicken the growth and arrangement of artificial intelligence and machine learning algorithms. In the era of cloud computing and visualizing the industry trend of the future, efforts are on to merge these technologies. So, there is the deal; imagine a future where **Quantum Computing (QC)**, AI, blockchain,

IoT, ML would merge along with other evolving technologies. Who will provide these solutions and integrative technologies? Start-ups and giants are here to make our life easier. For instance, things like qSaaS and qPaaS are already taking off. Companies like BMI have already started doing this. Refer to [Figure 11.2](#) for the visualization of companies who are active in the field of QC; Also, there is a lot of collaboration happening between companies in the field of QC. Because of the holistic need, as big corporations require innovation quickly and start-ups need infrastructure within no time. Due to the higher cost of the infrastructure, it is sometimes difficult for them to arrange the fund. Therefore, it is a winning-situation for both types of corporations. Now let's see some recent collaborations, it would provide you with a view of the playground. Microsoft is collaborating with 1Qbit, and IBM and Honeywell have the same types of collaborations.

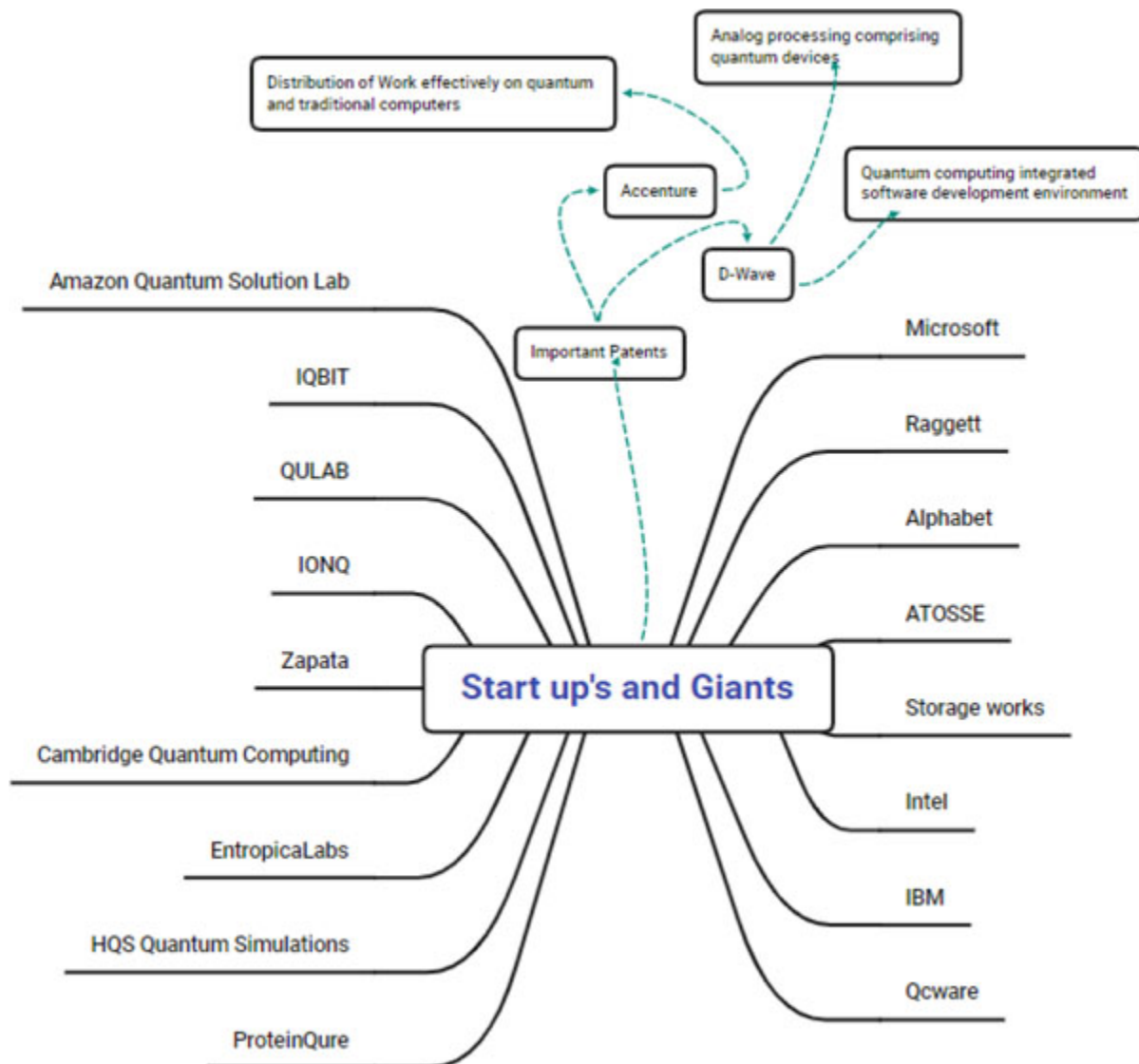


Figure 11.2: Mind map for some established quantum computers and computing companies

With qSaaS, the user will gather prompt access for ready-to-use mobile, tablet, and other smart device-based access, along with web-based applications for precise services. The quantum software and user data are delivered online in the cloud environment. The utmost suitable expense option for **Quantum Software as a Service (qSaaS)** might be a customized subscription with expense plans that are founded on the number of available services that would be presented to the user. For instance, in the case of automobile management and taxi services scenario, the trio expense plans might be thought of as a platinum tier that would give the full-blown version with cutting-edge control facilities and abilities, including hotspot

demand forecast as well as vehicle, route, and speed limit job. The two inferior plans, viz. a bronze and silver base tier, might comprise a restricted set of facilities and topographies. Additionally, pay-per-update choices can be obtainable.

The **Platform as a Service (PaaS)** contribution will deliver a cloud-based platform with scalable, extremely obtainable applications, or middleware (APIs), that are constructed into the platform with special software mechanisms. This platform would comprise an operating system, a software development language, an implementation environment, a data store, web servers, and so on. Creators can modify the pre-coded quantum algorithms and quantum computing models. Thereafter, they test, check maintainability, including reliability and management, and deploy them on the customized platform which is a specific target for the quantum computing infrastructure. All this generally happens over the cloud. The qPaaS contribution might be valued based upon data ingesting either in the form of a tiered payment with pre-defined tiers for computing examples, data storage or just as a pay-per-use option.

As we discussed at the beginning of the chapter. There are a lot of success stories are built around quantum computers and computing. Few of them are specific to qPAAS and qSAAS. Therefore, it is a good idea to cover them here in order to get a holistic understanding of quantum services, progress, research, strategies and current state. Here we go...

[Zapata](#)

Zapata Computing is the leading software corporation for NISQ-based quantum applications. It's one of the products called Orquestra™, which is the end-to-end platform for quantum-enabled workflows™. The platform is providing the utmost software- and hardware interoperability and quantum toolset. The Orquestra empowers cutting-edge technology, **Research and Development (R&D)**, and academic teams to quicken quantum solutions for complex computational glitches in optimization, Machine Learning, and simulation across diverse industries. With wide backing and direction from Zapata specialists, Orquestra will permit teams to quicken their work in quantum. The Orquestra is an amalgamated Quantum Operating

Environment for emergent and running quantum and quantum-inspired workflows.

Zapata's Orquestra doesn't only run on all the quantum hardware, but we can use Orquestra to crush the top act from quantum devices. Zapata does this with deep hardware corporation partnerships and years of R&D. They have algorithms and methods to exclusively control the fortes of each device across the full variety of quantum technologies. A few innovations they deal with are described as follows:

- Zapata used their insights on leveraging classical computing techniques in novel ways. Their platform, Orquestra, is constructed to run enterprise workloads that utilize these technologies.
- Several Zapata scientists have worked with quantum annealers since their beginning. Hence, it is good for optimization problems.
- Hardware companies are exploring a range of methods. Zapata has deep proficiency in gate model algorithms and physics that help the organization to build gate model-based quantum computers. Through Orquestra, Zapata aims to provide a quantum advantage to its users.

The best part of Orquestra is that it is agnostic across the complete software and hardware stack. It bids the utmost extensible library of open source. As a corporation, Zapata-created components for writing, operating, and enhancing quantum circuits and running them across quantum computers, simulators, and classical computing resources. It derives armed with a multipurpose workflow system and APIs to attach all modes of quantum devices. The main capability of Zapata's frameworks (Orquestra) and its applications are shown in [Figure 11.3](#). Precisely, Orquestra benefits as follows:

- **Widespread admittance to quantum libraries for flawlessly combining workflows:** With the utmost vigorous competence for quantum workflow strategy, implementation and examination, Orquestra users can start with sample workflows, open-source algorithms, and an increasing variety of branded Zapata algorithms. Further, Orquestra permits users to shape their algorithms and IP. They can fuse and contest modules inscribed with general quantum libraries

and tongues of their high quality, together with Cirq, Qiskit, PennyLane, and PyQuil, leveraging a rising library of backends with no boundaries on software or device types of machinery.

- **Multi-cloud and hybrid cloud classical backends:** Users can give in to workflows to the **Orchestra Quantum Engine (OQE)** which is a command-line tool. The Orchestrate workflow tool has a variety of backends that include gate-based model devices, quantum annealers, quantum emulators, and HPC resources. Users can benefit from OQE servers and run work on their cloud of choice. It may be Zapata's cloud or on-premise, linking across networks to quantum devices and simulants.
- **Shifting outside hardware-agnostic into hardware-smart expansion:** Orchestra is the only quantum platform that drives outside hardware independent or agnostic capabilities, lets users watchful about devices' strong points for specific problems and measure how workflows perform along devices. More than that, they can quicken their quantum trials by using Zapata's branded algorithms for quantum hardware resource optimization.
- **Scalable orchestration:** From handling multifaceted data histories to the automatic realization of parallelization through ample orchestration, quantum teams advantage from the iterative aspect, flexible trials at scale.
- **Multi-user data organization and interpretability:** Quantum teams can quickly record and liken consequences, then review and repeat as a multi-user team, joining benchmarking across quantum hardware and algorithms.

Refer to [Figure 11.3](#) for the service offering of Zapata:

Selected Quantum Use Cases Available on Orquestra™		
Proprietary Zapata techniques in green		
FINANCE		
Optimization - Sampling from risk-neutral probability measure for asset pricing - Portfolio optimization by quantum annealing or gate-model quantum heuristics - Improved asset allocation by quantum-inspired optimization	Simulation & Modeling - Quantum-assisted Monte Carlo for derivative pricing, credit valuation adjustment - Imaginary-time propagation for multi-asset Black-Scholes equation - Accelerated sampling from stochastic processes for risk analysis	Machine Learning - Improved graph clustering analysis for anomaly and fraud detection - Time series analysis of financial data using quantum Boltzmann machines - Dimensionality reduction for financial data using quantum-classical hybrid optimization
HEALTHCARE & PHARMACEUTICALS		
Optimization - Optimal healthcare resource allocation e.g. patient-treatment matching and priority scheduling of doctors and therapies - Molecular structure analysis by combinatorial optimization - Protein design using unconstrained discrete optimization	Simulation & Modeling - QM/MM method for molecular binding affinity prediction in drug discovery - Ab initio transition state analysis for catalytic reaction simulation - Ab initio determination of the crystalline structure of organic molecules	Machine Learning - Quantitative structure-activity relationship (QSAR) using quantum enhanced supervised learning - Molecular screening in drug discovery using continuous latent space - Natural language processing for automated literature analysis using quantum Boltzmann machine
TRANSPORTATION & LOGISTICS		
Optimization - Generating efficient transportation routes for vehicle fleets with improved qubit cost - Improving timing in job scheduling	Disruption management by optimized traffic re-routing	Machine Learning Irregular operation (IROP) prediction using quantum-enhanced supervised learning
MATERIALS		
Optimization Prediction of alloy ground state orderings from cluster expansions	Simulation & Modeling Determining thermodynamics of alloys by applying quantum amplitude amplification to cluster expansion models	Ab initio simulations to accelerate materials discovery (e.g., Li-ion electrolytes/electrodes, photovoltaics, catalysts, alloys)
ENERGY & UTILITIES		
Optimization Finite time-window scheduling in energy allocation for smart grids	Simulation & Modeling Turbulent mixing simulation in predictive environmental modeling for climate change	Quantum accelerated stochastic simulation of Reynolds-Averaged Navier-Stokes (RANS) systems
AEROSPACE & AUTOMOTIVE		
Optimization Identifying failure modes through fault tree analysis as combinatorial optimization	Simulation & Modeling - Variational methods for general linear PDEs arising in engineering design - Heat transfer related PDE-constrained optimization for engine design	Wave equation simulation using variational linear systems solver
CHEMISTRY		
Optimization Process optimization in chemical reaction networks for maximizing yield	Simulation & Modeling - Homogeneous and heterogeneous catalysis modeling using electronic structure calculations - Singlet-triplet transition energy prediction for OLED molecules	Quantum-enhanced force-field methods for chemical dynamics simulation
DIGITAL CONTENT		
Optimization Display advertising optimization for maximizing click-through rate (CTR)	Machine Learning Hybrid quantum-classical generative modeling for content creation	
BLOCKCHAIN		
Optimization Hash collision for proof-of-work cryptocurrency mining	Simulation & Modeling Certified randomness from quantum supremacy for proof-of-stake blockchains	
DEFENSE		
Machine Learning Predictive maintenance using variational quantum classifier	Anomaly detection by time series analysis using quantum Boltzmann machine	Image and audio generation and classification using quantum-enhanced supervised learning

Figure 11.3: The main capability of Zapata's frameworks (Orquestra) and its applications in diverse areas (Source: Zapata website)

Cambridge quantum computing

CQC is a world-leading self-governing quantum computing corporation that grows architecture-agnostic, enterprise quantum solutions to challenge some of the industry's most fascinating tasks. They are a globally acknowledged player in the fields of quantum chemistry, quantum machine learning, quantum cybersecurity, and quantum software. Their know-hows are permitting some of the world's major chemical, energy, financial and material science corporations to yoke the transformative influence of quantum computing. They are giving solutions in the following areas:

- Quantum algorithm
- Quantum chemistry
- Quantum simulation
- Quantum compiler
- Quantum machine learning
- Quantum natural language

Generative Tensorial Networks (GTN)

GTN is a London-based computational start-up. GTN industrialized patented technology and software, Reproductive Tensorial Networks, which can analyze big chemical spaces of minor molecules to recognize talented drug-like preliminary points for detection. The software is beached on methods of machine learning, and quantum physics, permitting to attain unresolved outcomes, associated with outdated computational approaches.

Bringing a solitary novel drug to the market costs very high and takes 15 years, with a chance of disappointment and frequently bringing an interference that hardly fluctuates from those previously in the market. Forecasts into the future are not capable, with a predictable 50% drop in R&D output every nine years.

GTN, a multidisciplinary, innovation-driven business, are penetrating the exorbitantly big space of drug-like molecules with sole patented technology, Generative Tensorial Networks. The software would syndicate and shape upon methods from machine learning and quantum physics to put on, sieve, and hunt for molecules. This will produce considerable competences in the whole drug growth cycle.

Entropic labs

This company designs its software to work on the leading cloud quantum computing platforms. Their products guarantee a smooth amalgamation, testing and benchmarking across models, computing systems, and workflows. Fundamentally, the company provides experience so that one can feel that he/she is running the code on the physical platform. It proved from experience that one code can run on real physical quantum computers. They indulge in the construction of an all-inclusive framework that would provide the organization the skill to trial with benchmarking protocols, machine learning models, and optimization workflows on physical quantum hardware.

Before quantum computing can spread large-scale acceptance, users and scientists will need a wide-ranging software layer equivalent to the one that happens to be for conservative computing.

Entropica is here to help. They are constructing the algorithms, software tools, approaches, and models to make quantum computers valuable. This alteration shadows a straightforward day-to-day realism. Coding quantum computers at the assemblage level while the open-handed response to quantum hardware builders and subsidiary end-users along with developers are something where Entropica labs are involved in. Entropica's software roadmap is revolved around their Entropy Development Framework. The Entropy is intended to empower both skilled designers and infrequent users to shape and trial with quantum computing workflows. Entropy comprises precise quantum optimization and machine learning methods that run on near-term processors. Outside the short-term era, Entropy's modular design ropes the upcoming integration of error-corrected quantum computing architectures. Just to highlight, Entropy fully integrates with the major cloud quantum computing platforms.

HQS quantum simulations

HQS Quantum Simulations is providing emerging quantum algorithms to forecast molecular possessions for presentation resources, shaping chemicals and pharmaceutical companies. The growth of field chemicals, performance materials, agrochemicals, and medications is an iterative, time-

consuming, and often very expensive process. Further, designing products are founded on purposeful molecules, formulations or solid-state compound needs, often of hundreds succeeding synthesis, preparation, and testing steps to harvest the anticipated stipulations. Simulations are an influential tool to diminution the time-to-market and lessening costs for R&D and product development cycles. However, they can't be used in a competent method in numerous cases or is just not progressive enough to tackle certain problems. Further, HQS algorithms are the key to quick simulations and solving the previously unsolvable problems.

HQS branded software covers the method of commercial quantum computing. Further, the HQS algorithms work together on conservative state-of-the-art computers and quantum computers, permitting their customers to now tap into the potential of this novel know-how. They provide an upsurge in the speed and competence of product growth processes for their clients. This increase broadens the considerate of chemical and physical communications which leads to improved products and procedures and quicker development cycles. HQS is trying to use the potential of quantum computers.

Quantum computers will disrupt the method in which specialty compounds, ingredients, agrochemicals, and medications are being advanced. HQS bid their customers direction to advantage from the quantum revolution at an initial stage. New-fangled chances can be detained and the characteristic menace of change can be alleviated via meeting sessions, referring services, experimental and joint growth projects with HQS algorithms, as well as quantum simulation of molecules and procedures.

ProteinQure

PQ, a Canadian start-up, has its office in Toronto. It provides a computational platform for protein-based drug discovery. ProteinQure syndicate molecular simulations, AI, and quantum computing to do the structure-based approach of drugs. These physics-based approaches make companies less reliant on big data sets and that lets interested parties discover novel protein structures and connections. They use their computational platform to resolve partners' drug design tests and deliver

novel beneficial applicants. As they partner with pharma to bring experimentally authenticated novel chemical substances, this initiative enables them to syndicate molecular imitations or simulation, using ML and the addition of high-performance algorithms to do something called structure-based drug design. This brings them to the different league as they fundamentally use their physics-based methods to make different interested parties less reliant on large data sets.

ProteinQure is working hard uninterruptedly to assert the limitations when it comes to protein foldaway on quantum computers. One of their achievements is to fold the principal proteins on binary dissimilar quantum computing architectures, quantum annealers as well as universal quantum computers.

ProteinQure brings quantum-classical hybrid approach to protein design

The high-level summary of ProteinQure hybrid classical/quantum approach to protein design is mentioned as follows:

- They utilize state-of-the-art classical algorithms for producing protein folds and scoring them for a specified protein sequence.
- These simulations are computationally concentrated, even on contemporary **Graphical Processing Units (GPUs)**, and quickly becoming computationally stubborn for big proteins.
- They do comprehensive calculations of protein folds using abridged models with quantum computers which speed up their classical algorithms.
- By mixing machine learning techniques with chemical perception from biologists and chemists, they can produce proteins for therapeutic applications.

A collaborative effort of ProteinQure: Rigetti Computing and ProteinQure

There is a unique scenario where ProteinQure and one of the leading quantum start-ups called Rigetti Computing are working together. Further,

Rigetti Computing is constructing universal quantum computers and is photocopying down with its separate semiconductor fabrication plant. They have lately proclaimed a determined one-year goal to shape a quantum chip with 128 qubits. Composed with its hardware developments, Rigetti has completed its stand-up with its over-all and well supported full-stack quantum software library Forest. Forest is an open-source project obtainable on GitHub. From the last few months, ProteinQure energetically underwrote to the development of this software. By the use of the novel Cloud Service which is targeted to Quantum computing, fundamentally Rigetti can democratize the quantum algorithms. On the broader level, this is very useful for application-based quantum computing. This implies that quantum software start-ups like ProteinQure will be bright to inscribe apps that can be retrieved through the new QCS virtual environment. This will allow big multinational corporations to access the aids of quantum computing algorithms without demanding to be specialists themselves.

ProteinQure as part of this exertion, built the world's first quantum app for protein folding on top of Forest. This app can then be used as a plug-and-play research tool by pharma corporations and self-governing researchers in the biotech space.

[QC Ware](#)

QC Ware is a quantum computing software corporation that is constructing innovative solutions that run on quantum hardware. The corporation's stated objective is to make quantum computing effortlessly available to classically-trained data experts and to bid act speed-ups on short-term hardware. QC Ware is employed to that goal with one of the world's sturdiest teams of quantum algorithms experts. The company is based in Palo Alto. It newly opened an office in Paris as well; further, it plans to launch multiple office across geographies. There are five important areas that QC Ware is employed on that will have an important influence on enterprise business procedures. Outside of its government corporations, QC Ware's list of customers at present comprises a number of Fortune 500 corporations, but its medium-term goal is to enlarge its client base to characteristically skilled data scientists, optimization, chemistry specialists, and more.

One of the drivers of QC Ware, which also leads to its product strategy, is to make quantum computing effortlessly available for as many users as conceivable. User-friendliness, together in footings of hardware admittance to different hardware platforms, and in relationships of conceptualizing away difficulty. So that more traditional specialists can use quantum computing algorithms.

QC Wire target user is everybody who has a GitHub account; this means every programmer, each data scientist, and anybody interested in Quantum. So, it means there are about 25 or 30 million people as their customers/users. These people are not trained in quantum. However, they're trained to write programs that function on CPUs and GPU's traditional processors. The fundamental goal of QC Wire then is to offer to those individuals and parties a way to stay in their high-level language like Python, Java, and C++, but at the same time, be able to run their job on any of the quantum computers that are presented. It is a black-box approach. This means the user doesn't essentially need to know that, but the user has the aptitude to board or direct his or her job onto any of those hardware platforms. Areas on which they work are as follows:

- Chemistry simulation
- Optimization
- Machine learning
- Differential equations
- Monte Carlo simulations

[Strangeworks](#)

Strangeworks is an avantgarde computing software corporation based in Austin, initiated by whurley. Strangeworks is now making quantum computing available by constructing and bringing tools for software developers, investigators, and systems management for IT Managers and CIOs. Strangeworks' task is to transform technology by superintendent corporations and making it effortlessly available to all.

trangeworks transport the power of many quantum computers into a solitary platform, available to everyone. The Strangeworks Quantum Computing

Platform is a hardware-agnostic, software comprehensive, cooperative development environment that transports the newest progressions, frameworks, open-source, and tools into a sole user interface. Strangeworks trust in the power of public collaboration, teamwork, and influence that make groundbreaking detections and breakthroughs.

The Strangeworks **Quantum Computing Platform (QCP)** delivers gears to use and install short-term quantum computing devices which are supportive of both gate and annealing models. Strageworks's core functionality is to assimilate all hardware, frameworks, and supporting languages to deliver users with the aptitude to turn their concepts into experimentations. To ease this goal, Strangeworks QCP accomplishes assembly, integrations, and product updates so the user can focus on their work. In addition to developing experiments, users may collaborate, publish, and share their work directly from the platform.

As the technology world is attentive towards the cloud and wants to run almost everything on the cloud including the software, so is Strangeworks. Presently, the Quantum Hardware wage-earners need distinct accounts (IBM, D-wave). What it means is if you invest in one platform, porting over to additional is tougher, costlier, and sometimes troublesome. Strangeworks aims to be that hardware agnostic middle-ware that permits users to emphasize on resolving the problems rather than converging on a detailed hardware solution structure. It provides the tooling that a team desires to work collaboratively on Quantum algorithms. Even if the computer hardware is not there, such as in the case of Microsoft, you can still simulate in the Q# language, for instance in the cloud-based environment. [Figure 11.4](#) shows the highlights of Strangeworks ecosystem:

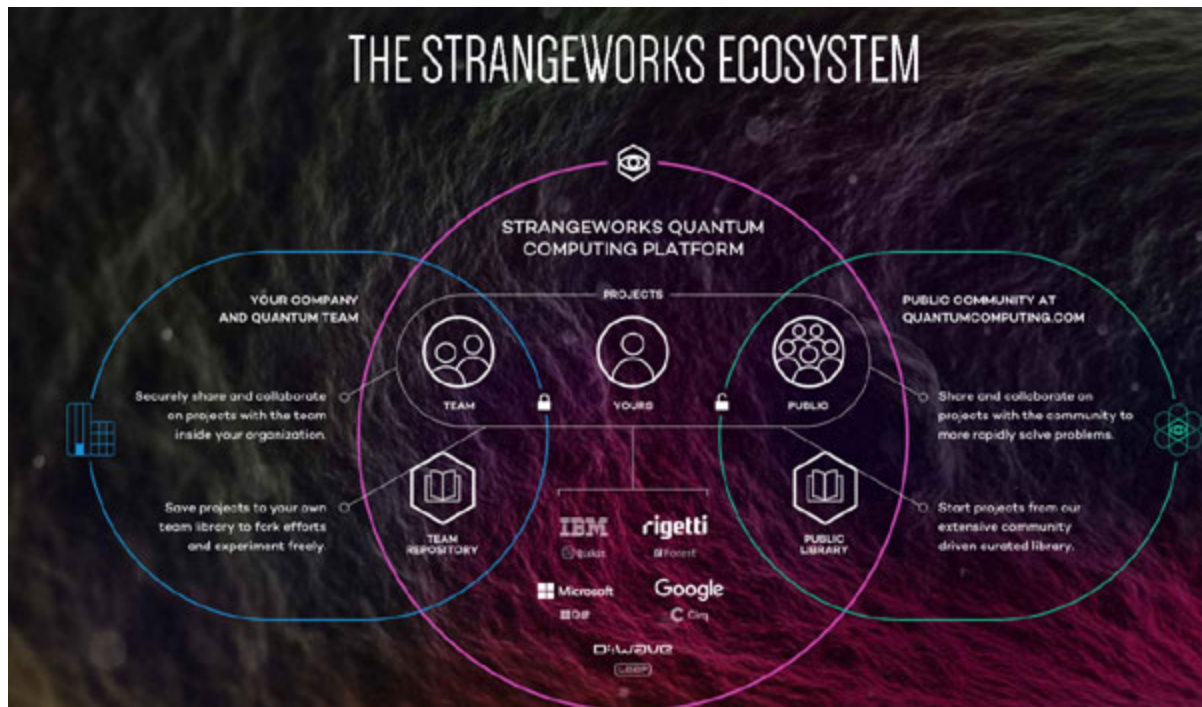


Figure 11.4: The highlights of Strangeworks ecosystem (Source: Strangeworks website)

[Amazon quantum solutions lab](#)

Google and IBM may be fighting for quantum supremacy, but Amazon is currently content to be a middleman. Recently, they announced and launched something called Amazon Braket. It is an Amazon effort to turn the budding field of quantum computing into a service which one can access over the internet. Amazon Braket is a completely managed AWS service, with safety & encryption forged in at each level. As it is a fully managed service, it permits experts, investigators, and developers to begin testing with computers from multiple quantum hardware providers in a single place. Bra-ket notation is usually used to represent quantum mechanical states and inspire the name of the service.

It may sound like a pretty restricted matter, where you will mean Amazon's business clients, and where "service" means the aptitude for trial by consecutive simulations on a set of prevailing quantum computers from associates D-Wave, IonQ, and Rigetti. This novel service is intended to let you get some hands-on involvement with qubits and quantum circuits. One can shape and test their circuits in a virtual computing environment and

thereafter run to enable them on a real quantum computer (ideally it is machine agnostic).

Amazon established an AWS Centre for Quantum Computing. It did it to bring together the world's foremost quantum computing investigators and engineers to quicken the growth of quantum computing hardware and software. Further Amazon Quantum Solutions Lab engagements are cooperative investigating programs that permit to effort with leading specialists in quantum computing, machine learning, deep learning, Natural language processing, and artificial intelligence even to **High-Performance Computing (HPC)**. The programs help explore and classify the greatest efficient applications of quantum computing for business and make them quantum ready. Amazon Quantum Solution Lab specialists are working backwards and studying the science of quantum computing, its growth and measure of novel algorithms and solutions. Further, based on this idea, they shape the internal know-how and approaches obligatory to make the organization ready for the future of quantum computing.

The Amazon Quantum Solutions Lab has joined businesses with different specializations, together with computational chemistry, machine learning, and optimization. In accumulation to quantum computing, the experts at Amazon have developed the know-how in hybrid quantum/classical systems, quantum-inspired classical approaches, and high-performance computing which are further based to provide assistance to create a plan for the organization. The Amazon Quantum Solutions Lab will help you classify and cooperate with the right partners for your business. It cooperates with businesses like 1QBIT, rakho, rigetti, QCWARE, QSMULATE, ZAPATA.

[A few highlights for Amazon quantum initiatives](#)

Let's focus on Amazon quantum initiatives, as follows:

Amazon Quantum Solutions Lab makes you quantum ready: The Amazon Quantum Solutions Lab will aid to comprehend organizations if and when quantum computing assists businesses to resolve problems. Through exploration and teamwork with the Amazon team, one will become talented to classify the greatest chances of quantum computing for

business and shape a roadmap to make an organization for the arrival of this technology. Amazon's Quantum Solutions Lab specialists will communicate knowledge to the wider population of interested people, so that organization can shape quantum skills, knowledge, and knowhow. Amazon's specialists will work side-by-side with your team to dive deep into the greatest real ways of using quantum computers, cooperate on trials, and deliver insights about current quantum technology so that one is prepared to organize quantum solutions when the time is right.

Amazon Quantum Solutions Lab would aid to Determine Solutions:

Quantum computing specialists will partner with the organization to explore and determine the utmost capable near-term quantum computing solutions for commercial/business. At the same time, they will aid organizations to appraise and enhance the present method to target business problems by structure and organize traditional and quantum-inspired solutions using ML and HPC. This will permit one to measure the recital and feasibility of initial stage quantum computing solutions as well as determine additional real classical solutions that meet the organization's needs and transport business value today.

Microsoft

While we might not realize applications consuming quantum computing in everyday life, corporations employed on these technologies are presenting numerous initiatives so developers can get a head start on it. For instance, Google launched an open-sourced version of TensorFlow for quantum computers. Prior to that, Amazon also was on track to present quantum computers as-a-service with Amazon Braket. However, the service is obtainable in a restricted phase and lets individuals and corporations experiment with quantum circuits. One of the underlying Qubit technologies used by Microsoft is a different kind of topological qubit. And as we know, it is offering like Microsoft Azure Quantum, which is a platform for quantum computing for developers and corporations.

Microsoft is structuring a full-stack quantum ecosystem, distributed through the well-established power and scale of Azure's worldwide cloud services platform. Azure Quantum is a full-stack, open cloud ecosystem. Once its

ecosystem is matured, the individuals and organizations have access to a varied set of quantum resources. The probable list of resources would be endless, including but not limited to, pre-built solutions and, quantum development tools. Further, resource approximation tools would also be available in collaboration with quicker traditional hardware and diversity of quantum hardware. The Microsoft Quantum Network is a group of futuristic, connected and informed innovators who are cooperating with Microsoft to enhance their knowledge and learn about, investigations, research, and launch quantum computing applications, tools, technologies (including hardware). Further, the Network partners or collaborates to obtain access to quantum development properties, high-class Azure services (in case of Microsoft), and workshops on quantum programming and algorithm growth. In a nutshell, Microsoft is providing the following:

- An open ecosystem, allowing you to the entry of varied quantum software, hardware, and solutions.
- A righthand, scalable and safe platform that will endure acclimatizing to our quickly developing quantum future.
- An aptitude to have quantum influence today with pre-built answers that run on traditional computers.
- An acquainted setting with Azure to help you rapidly ramp up and have influence nowadays while building for the future.

Quantum computing at IBM

IBM is employed to brand its quantum computers available to all individuals who are interested in it, and not only to just those who comprehend the strange physics ideas like superposition and entanglement that brand quantum computers tick. It calls circuits that smear an arrangement of alterations to qubits, the quantum data store elements that are far more flexible than conservative computers' bits. They also found multiple use cases and did collaborations with different corporations including financial institutes.

IBM's Quantum computing devision is an industry primary innovation to shape universal quantum computers for professional, engineering and scientific purposes. They are dertermin to make the difference in the field.

This determination comprises proceeding the complete quantum computing technology stack and discovering applications to make quantum broadly practical and reachable. With a worldwide network of Fortune 500 businesses, academic institutions, investigators, educationalists, and aficionados, IBM is dedicated to driving innovation for clients in the IBM Q Network and the protracted IBM Quantum Community.

At the core of IBM quantum computing systems is the transmon qubit. Consecutive compeers of IBM Quantum processors have established the likelihood of superconducting transmon qubits as the foundation for electrically controlled solid-state quantum computers. IBM quantum computers are based on a scalable method of chip architecture and research related to error correction and mitigation. IBM Quantum is at the vanguard of emerging systems with satisfactory quantum volume to determine benefits in real-world applications.

Numerous IBM quantum devices are obtainable to them by their quantum services available on cloud infrastructure. Users can access devices free of cost through Qiskit. Further, more progressive quantum systems are obtainable to IBM customers in the IBM Q Network. Qiskit permits investigators, educationalists, and coders to get started with quantum programming. Qiskit could be used to distinguish errors, autocomplete code, and produce visualizations or complex quantum states. Qiskit also enables flawless integration of quantum development into normal workflows using general-purpose programming languages and tools.

IBM is on a trail to at least double the performance of its quantum computers each year. At least from the last four years, they are achieving this consistently. It notches its machines on a measurement. IBM termed this invention as quantum volume, which leads both the number of qubits in the machine and the decrease in error rates that limp quantum calculations.

IBM's quantum computing competitors haven't usually signed up to use a similar standard, but the greatest distinguished one that has is Honeywell. It's constructing a different assortment of quantum computer which is based on an ion trap machine that can run, to some extent, at fewer frosty temperatures than IBM's superconducting designs. Players like Honeywell further promises that its machines will fundamentally upsurge the so-called

IBM invented the concept of quantum volume by a factor of 10 each year. This is a faster improvement than IBM.

IBM today has 18 quantum computers. Eighteen quantum computers might not look like an impressive figure at a first glance. However, given that each one is a bulky device cooled inside a portion of a degree above absolute zero and operated by experienced scientists and researchers, it's a pretty big fleet. In contrast, Google's quantum computer laboratory close to Santa Barbara has only five machines, and Honeywell only has six quantum computers.

IBM Quantum is hurrying the application of quantum computing to resolve the furthestmost stimulating glitches in chemistry, optimization, and machine learning. The following are a few of its capabilities:

- **Machine learning:** By the use of quantum systems, IBM is trying to train and run machine learning algorithms that could allow their customer to resolve complex problems further rapidly, perhaps refining applications like illness verdict, fraud discovery, and well-organized energy management.
- **Materials:** Simulating quantum mechanical schemes is a talented initial application of quantum computing. This method can be functional to fields such as physical and chemical sciences.
- **Optimization:** Quantum computers may discover the finest resolution amongst variable weighted choices more professionally than traditional computers, and could deliver benefits in multiple areas.
- **Finance:** In the financial service sector, numerous computationally concentrated problems occur. A couple of examples of this are financial optimization and risk analysis of big portfolios. For a few of these glitches, quantum computing may have the probability to attain an important benefit associated with traditional computing.

Intel

Research at Intel® Labs has controlled straight to the growth of Tangle Lake. Fundamentally Tangle Lake is a superconducting quantum processor developed by Intel. It joins 49 qubits in a set that is factory-made at Intel's

300-millimetre construction ability in Hillsboro, Oregon. This device signifies the third generation of quantum processors shaped by Intel, scaling rising from 17 qubits in its predecessor.

Tangle Lake is a stage onward in the continuing growth of full quantum computing systems, which is still possibly ten years in the future. Such systems have potentially unparalleled aptitude to simulate and analyze natural marvels, leading to quick responses to phenomenological queries that would necessitate high-priced amounts of time on today's supercomputers.

Tangle Lake must operate at deeply cold temperatures, inside a minor fraction of one kelvin from absolute zero. Furthermore, there are important subjects of scale, with real-world applications at a marketable scale needing at least one million qubits. Given that reality, the comparatively big size of quantum processors is an important limitation in its right. Intel is vigorously developing strategy, modeling, wrapping, and construction techniques to allow the formation of more complex quantum processors.

Intel started work together with QuTech, a quantum computing startup, in the Netherlands and invested US\$50M to increase its footprint and capabilities. QuTech delivers continuing engineering resources that will help quicken growths in the field. QuTech was shaped as a forward-thinking research and education center. QuTech joint hands with Intel's skill in fabrication, control electronics, and architecture; this corporation is exceptionally suitable to the challenges of developing the foremost feasible quantum computing systems.

Presently, Tangle Lake chips shaped in Oregon are being transported to QuTech in the Netherlands for analysis. QuTech has advanced vigorous techniques for simulating quantum assignments as a means to address subjects such as joining, controlling, and measuring many, entangled qubits. In accumulation to its partnership with QuTech, Intel® Labs is also employed with other ecosystem members both on essential and system-level challenges on the whole quantum computing stack.

Rigetti computing

It is building the world's greatest influential computers to help crack humanity's most persistent and significant problems. These systems will do computations that today's wildest supercomputers are unable of solving, i.e., completely novel classes of problems and contributing a straight path to solutions. Rigetti is a group of specialists, engineers, manufacturers, and thinkers. They trust quantum computing is going to suggestively affect health care, how we treat illness, how we produce energy, and how we feed humanity. Rigetti is the only corporation organizing full-stack solutions for hybrid classical/quantum computing. Their 19-qubit quantum computer is made available on their cloud infrastructure through the Forest platform. They were started in 2013 by Chad Rigetti, and are headquartered at Berkeley and Fremont, California.

Novel advances in hybrid algorithms have made the applied use of near-term quantum computers ascend preferably than countless of us thought conceivable. That's why we're absorbed on consolidating each layer of the quantum computing stack, from the hardware to the applications, to transport the applied use of quantum computing to life.

The following phase of quantum computing will need more power at the hardware level to get-up-and-go for improved outcomes. Over the past year, they have constructed 8-qubit and 19-qubit superconducting quantum processors, which have been available to users over the cloud through their open-source software platform Forest. To drive applied use of quantum computing today, we must be intelligent to scale and progress the recital of the chips and connect them to the electronics on which they run which has been established to be one of the greatest challenging features of quantum computing.

Rigetti is in the only place to resolve these problems which are most suitable to shape quantum systems that scale. Their 128-qubit chip is advanced on a novel form influence that lends itself to speedy scaling. Since their in-house design, fab, software, and applications teams work thoroughly composed, they are talented to repeat and organize new-fangled systems speedily. Rigetti custom control electronics are intended exactly for hybrid quantum-classical computers, and we have started integrating a 3D signaling architecture that will permit for accurately scalable quantum

chips. Over the next year, we'll put these pieces together to bring more power to investigators and developers.

The patent journey of quantum computing companies

As we understand, there are a lot of corporates that are working hard in the field of quantum computers and computing. This is one of the reasons why there is a significant rise in the filling of quantum computing patents in recent times. The following image would give a glimpse of the overall gameplay. [Figure 11.5](#) shows different patents filled by a few leading companies:

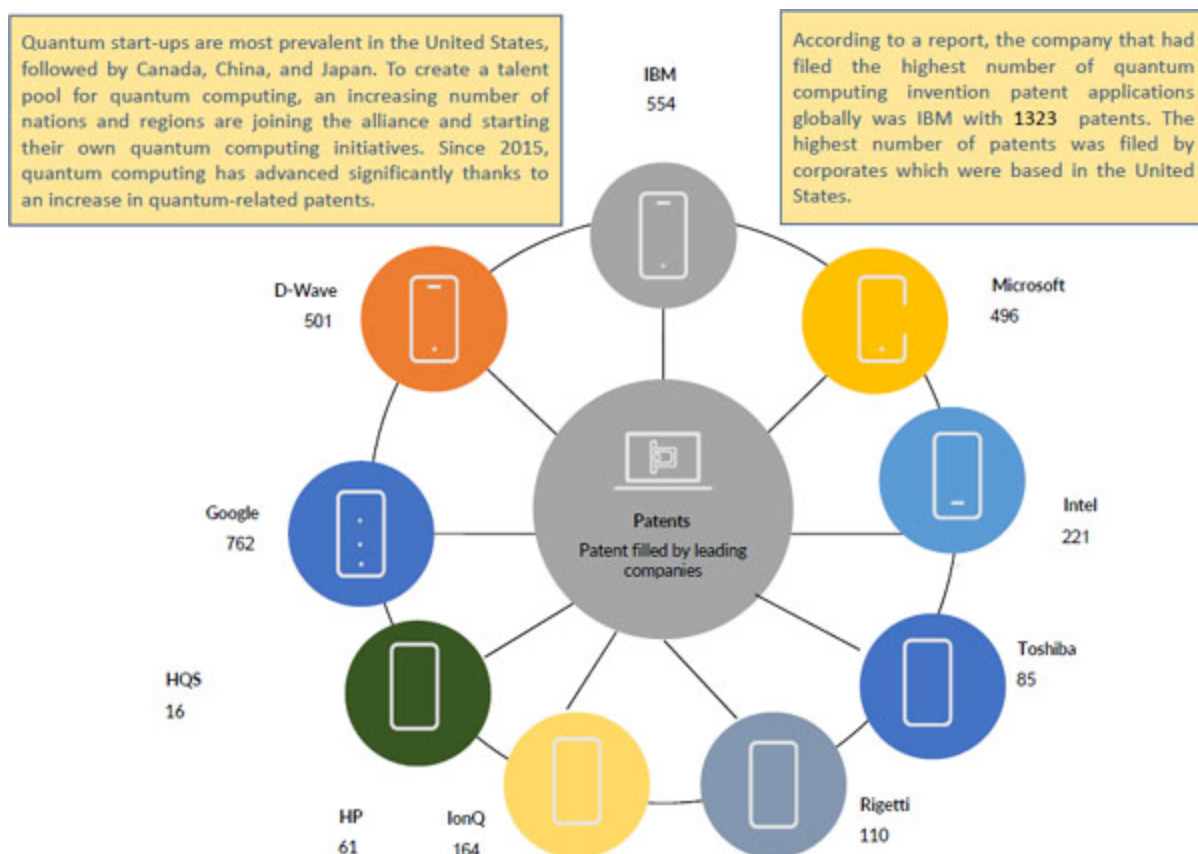


Figure 11.5: Patents filled by leading companies

The data presented in the picture says that Microsoft is leading the way with 425 applications, their keen quantum team is employed to shape an accessible quantum system to resolve the world's utmost complex challenges.

Patent 1: Accenture US patent for quantum computing – distribution of work effectively on classical/traditional versus quantum computer

Quantum computing has huge potential and capabilities, offering truly ground-breaking competences to quickly unravel corporate's utmost hard computational encounters. This would determine at what time to employ quantum vs classical computing. Recently Accenture has been granted a US patent. The patent is exploring the dimensions of quantum computing and machine learning. The patent is for a machine learning module that chooses quantum or traditional computing for the best commercial results. For a quantum computing machine learning module that trains AI models to govern at what time and in what way, computational tasks would be superlatively handled by quantum computing against traditional computing approaches, and be routed to the suitable opportunity. This is a big deal because it could assist cooperates and governments to comprehend where quantum computing can have the utmost influence inside their businesses, and when traditional computing may still be the finest fit for the provided task.

The US Patent No. 10,275,721 fundamentally calls a resolution to discourse the trade-off among the reimbursements of using quantum computing as opposed to traditional computing resources. It achieves this by shaping when and how to exploit the influence of quantum computing; such a system can help do computational errands in the utmost well-organized and cost-effective way conceivable. Furthermore, the module can learn to prioritize certain responsibilities. As more advanced and effectual quantum and traditional systems are familiarized over time, the quantum computing machine learning module can acclimatize consequently.

Patent 2: Analog processor comprising quantum devices (10691633)

Methods and systems for resolving numerous computational problems with quantum CPUs are provided by a patent filed by D-Wave systems in a patent titled Analog processor comprising quantum devices (10691633).

This type of quantum processor includes many quantum devices composed of a variety of coupling devices. The quantum processor is set by setting states of the quantum devices and coupling devices and permit to grow to a concluding state which moves toward a natural ground state of the computational issues.

The last state can take in values of nodes organized in a lattice in the quantum processor and can signify a resolution to the computational processor. The computational problem can have difficulty or better complexity P, NP, NP-Hard, or NP-Complete and may be charted to a quantum processor with nearest-neighbor and next-nearest-neighbor couplings. The resolution to the computational problem can be read out from the quantum processor and communicated as a data signal personified in a carrier wave.

Patent 3: Quantum computing integrated development environment (US20030121028)

D-wave system filled a patent for a quantum integrated development environment which provides facility for designing quantum logic that uses N qubits, accumulating the quantum logic into quantum machine language commands, and running the machine language commands on a QC system. Further, the outcomes of the implementation are provided as a final result.

Mind Map

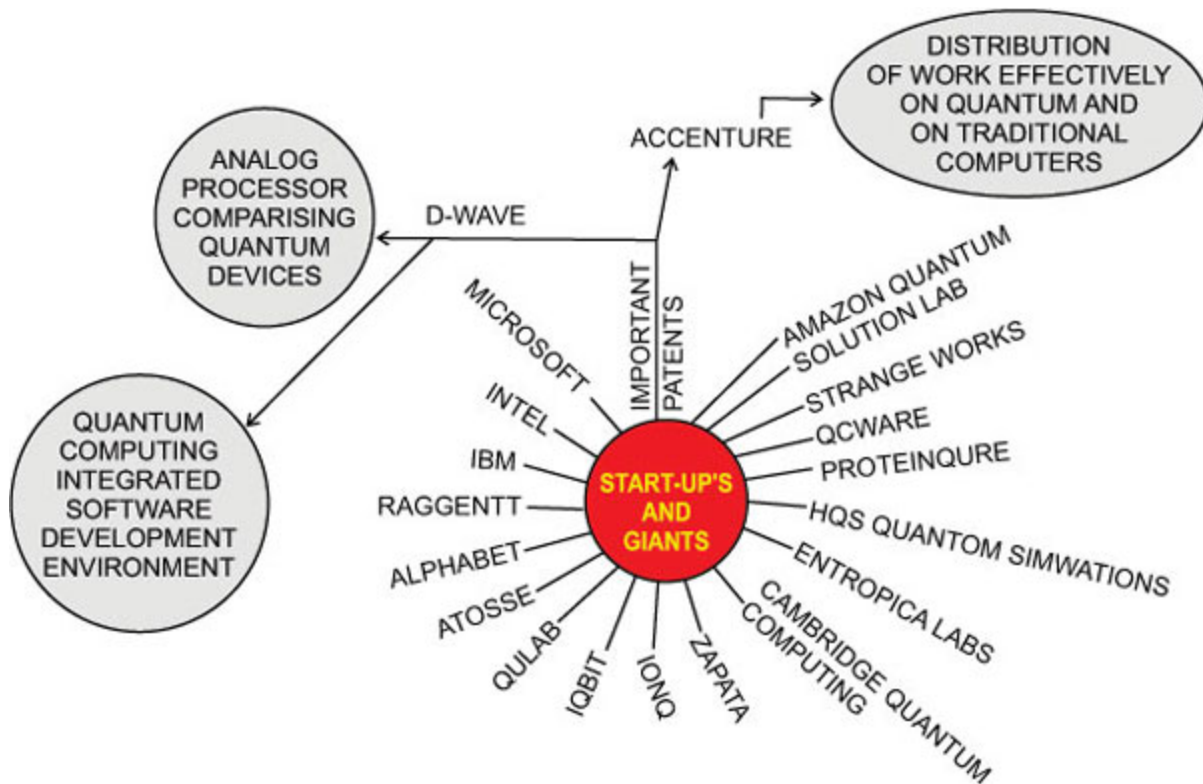


Figure 11.6: Mind map for the vchapter

Conclusion

This chapter discussed the details of multiple start-ups that are active in the field of quantum computer and computing, and the working and plan for the present and future. It also provided a glimpse of some details of established giants who are working in the same field. Apart from this, the chapter discussed a few important patents.

In the next chapter, we will discuss strategies of the new generation of organizations, especially which are targeted towards quantum incentives.

CHAPTER 12

The Quantum Strategies

This chapter will discuss the business implication of quantum computers and computing. Quantum computers wedged business and economics in an inimitable way. It altered the methods by which we plan, act, and do things. In the current time, when the data cantered economics along with analytics mindset is thriving, there are few terms are coined by various experts and professionals to present zest and impact of data on changing economy. I coined the term called “Datanomcis” which fundamentally means data-based economy. I am not an economist or a data expert but I am a person who has an overall understanding of economy and technology like AI, ML, QC and big data. “Datanomics” fundamentally compromise all these together. And after my book -- Machine learning for decision-makers, it is not the alien term. You may rename it with any other term you like to represent the concept but the idea remains the same, that is, “economics is dependent on data, and not on bricks and other physical infrastructure”. Quantum computing along with datanomics will potentially change the complete landscape of economic and commercial vision. We will discuss these things in the forthcoming sections.

Structure

In this chapter, the following topics will be covered:

- Introduction
- A new way of doing business
- A new start called quantum computing
- The risk of ignoring quantum computing
- race for quantum supremacy
- Datanomics, AI and QC
- The realities of quantum computing

- Why your organization may need to act now
- Quantum software strategies
- Seizing quantum advantage for your business
- CIOs, business value and quantum computing
- Business barriers
- Strategic alliance for QC
- Quantum computing market structure
- Market forecasts
- Opportunities to apply quantum computation on industry
- Innovating with quantum computing now
- Let's be agile ... in context of quantum computing incentives and machine learning (applicable for QML as well)
 - Why you need agile?
 - Difficulties in implementing an agile approach
 - Waterfall methodology
 - Pit bits
 - Agile disadvantages
 - Agile usage
 - Growth values
 - Personal values
- Short- and long-term goals for corporates and additional thoughts on quantum strategies
- Mind Map

Objective

This chapter will discuss how and what strategies are needed to take maximum advantage of quantum technologies, and what a corporation would do to make it future-ready. It will also discuss what strategies will enable them to train their workforce to lead the quantum revolution.

Introduction

There is a clear shift visible in most of the world's governance and business ecosystems. Data-based decision-making is not only the elite and unique capability, but it is also the reality and it is here to stay. Also, states, subcontinents, and governments are started using machine learning analytics to raise the lifestyle of their population and the welfare of people. Business houses, enterprises, and companies are using this for acquiring the market and improving their profit margins and revenue.

Data is the new blood in the economic systems and it's changing its DNA and RNA. As the fundamentals are changing, the way of reaction and working style of the market is making a paradigm shift. Machine learning tools, technologies, frameworks, and platforms are enabling consumers, governments, and businesses to become more productive and innovative. For example, the personal computer provided the power of doing our calculations by ourselves and creating documents without any hassle through Excel and Word.

A new way of doing business

Mobile phones changed the way we communicate, and social media like Facebook changed the meaning of our social interaction. Similarly, with the wide availability of machine learning-enabled platforms, frameworks would provide us with the power of intelligence and prediction capabilities.

Also, companies like IBM and Microsoft are providing machine learning algorithms for general use in the cloud environment. Therefore, it is readily available to everyone. Whether it is a desire or needs to recognize the voice, face, emotion, picture/image, or anything else, an algorithm is readily available for use. You can train them and use them for your personal or commercial purposes. For training, data is needed, which is now not only the monopoly of big players like Facebook, LinkedIn, or Microsoft but websites like Kaggle provides the enormous datasets to train and use algorithms.

Quantum computing for getting business edge

If the real value of quantum computing-based transformation is five to ten years away, then why should organizations and governments be considering

investing in it at the moment? Does that make any sense? The modest response of this question is that this is an essential technology that is offering daunting ramp-up challenges, even for corporations with cutting-edge supercomputing competencies. Quantum programming and the quantum tech stack allow diminutive similarity to their conventional counterparts. Early starters are in a situation to take advantage, perceptibility into knowledge and know-how slits, and even intellectual property that will put them at an operational benefit as quantum computing advances marketable adhesive friction.

Numerous experts believe that growth on the way to maturity in quantum computing will not track a flat, constant curve. In its place, quantum computing is a runner for a rushed innovation that may come at any time. Few companies that have taken advantage of integrating quantum computing into the workflow are far more probable to be in a place to make the most of it. This will caution substantial advantage in industries in which characteristically stubborn computational difficulties lead to blockages and wasted revenue prospects.

Quantum economics is a developing scientific investigative field that put-on approaches and thoughts from quantum physics to the arena of economics. It is driven by the trust that economic processes such as financial transactions have much in common with quantum procedures, and can be suitably modelled using quantum formalism. With the advent of full-fledged quantum computers and computing infrastructure, this field would take new heights.

Quantum computers would provide an impact on industries, decision-making, and government in the coming time. However, it will not only be the QC but it would be complemented by Datanomics and they will holistically lead the future.

[The risk of ignoring quantum computing](#)

Quantum computing will enhance the information systems we are contingent upon to a level not ever experienced by the human race before. It would become the perilous enabler of the subsequent phase of economic, social, industrial, and ecological renovation. Fundamentally, the full realization of quantum computing, will, in the end, transform each form of life, both at an

individualistic and external level and therefore in every form of human exertion. If we can try to visualize, in the future something strange and unparalleled is going to happen. That is a solid change from the normal change we've been feeling for the previous few decades, to somewhat more disruptive.

The Internet uprising was caused by a mixture of two factors – computing and communications – but in the forthcoming time, it expands its dimensions. It won't be an impartial additional session of foreseeable and incessant alteration that we can comprehend and make for, but somewhat unexpected, intermittent change.

A huge phase of changeover for humanity is coming. This is because quantum computing technology is different from today's transistor-based computers. Quantum computers can attack complex problems that are presently impossible, or that would take an excessively long time to solve. In one trial, a solitary quantum computer resolving an optimization problem has beaten 1 million computers at Google by a factor of 100. That's 100,000,000 times quicker, which is an entire lot more than doubling. Hence, if we ignore it, we would lose out on the future which is full of opportunities. However, having said that, it is a transistor-based system which brings digital revolution to the life of individuals, corporates, and government. [Figure 12.1](#) represents the revolution around digital technologies and also its drivers:

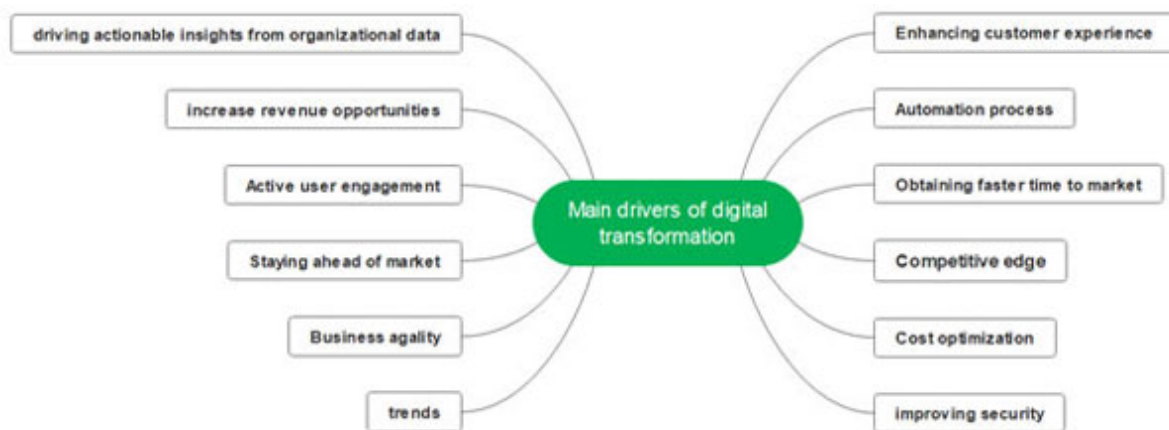


Figure 12.1: Digital drivers

Quantum Computing (QC) is seen by many as a likely upcoming choice for undertaking high complexity or stubborn problems by completing

traditional computing with a different compute paradigm. As we know, quantum computers have a few essential competencies that distinguish them from today's classical computers.

One of the important ones is a quantum simulation, which is fundamentally a quantum computer model and is used for complex molecules, optimization that is, at the moment though as a good candidate for resolving multivariable problems with unparalleled rapidity. The other one is quantum **Artificial Intelligence (AI)** and thought as the improved algorithms which could alter business models by enhancing improved intelligence across industries as assorted as pharma and automotive finally the **area of computational efficiencies** like prime factorization, which could potentially transform encryption.

It is predicted that by 2023, around 90% of corporate quantum computing investments will engross in quantum consulting establishments to help from problems that can influence quantum algorithms. Knowing how to identify and extract business value from a quantum computing initiative is a key skill to develop. Zapata, Altos, Honeywell, IBM, Microsoft, and others have client rendezvous services for organizations attracted in recognizing conceivable commercial prospects that quantum computing could one day address.

The best method to comprehend the commercial potential of quantum computing is to understand how those competencies could confront a variety of use cases. Convinced industries have precise problems that are chiefly well suitable to quantum computing, including medicines, cybersecurity, economics, materials science, and broadcastings. The following areas are (at the moment) thought as highly influenced by QC in the coming time:

- The reduced development time for chemicals and drugs by the use of simulations.
- Resolve optimization problems with unparalleled rapidity.
- Quicken autonomous vehicles with quantum AI.
- Transmute cybersecurity.

The physics, materials, and control systems of quantum computers continue to be evolving, as big organizations like IBM, Google, Intel, and Microsoft

to deeply invest in quantum hardware and software. Start-ups in multiple industries are emerging, together with novel skillsets — from quantum algorithm specialists and creators to quantum circuit engineers and practical physicists. CIOs should interpret quantum computing as a modest advantage, as novel quantum-inspired algorithms could fetch ground-breaking solutions and methods for product growth. It could also decrease the time to market and enhance customer distribution.

Moreover, waiting or overlooking quantum computing valour places **Intellectual Property (IP)** and patent collections at risk. Initial organizations will have the benefit of patenting quantum enthused innovations within their exact domains. For instance, a competing business could grow a quantum enthused innovation that progresses Monte Carlo imitations by 1000% or a pharmaceutical company could cut the time to market for novel drugs. [Figure 12.2](#) depicts those multiple areas would be affected if quantum technologies are not taken seriously:

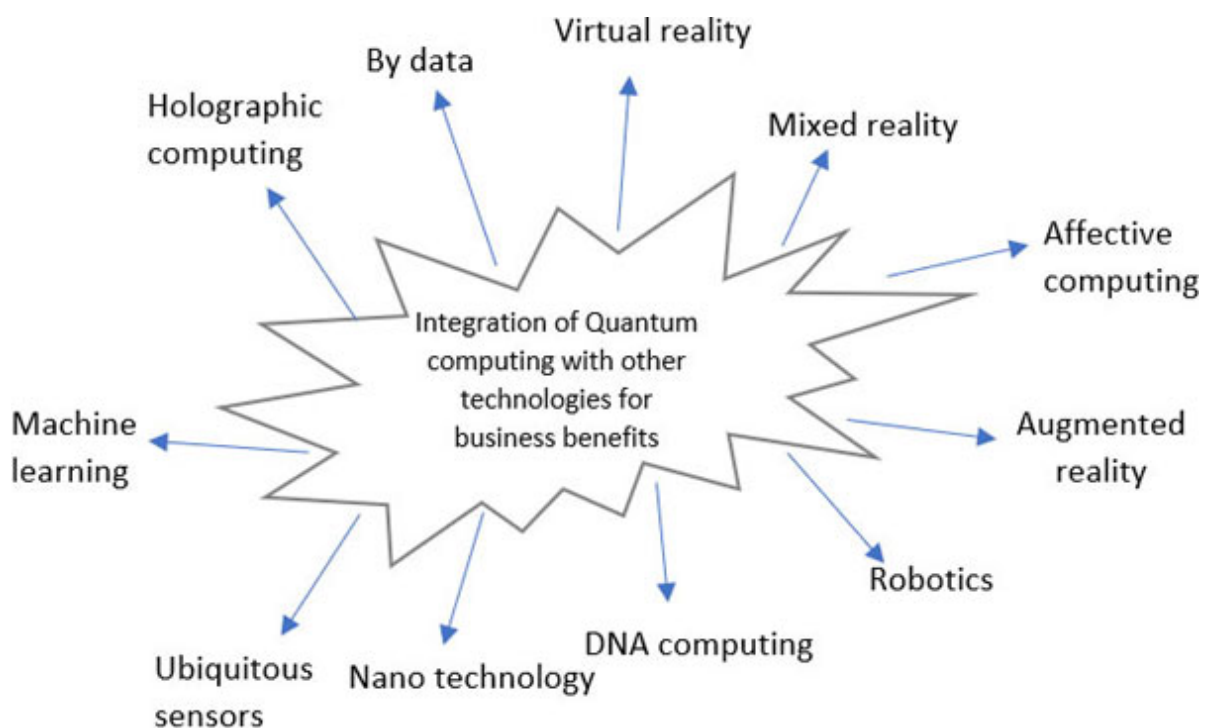


Figure 12.2: Areas affected by quantum technologies

Apart from the fields and areas which are mentioned in the preceding image, success of different future trends depends on the quantum. Therefore, it is necessary to make appropriate strategies for the same. First, visualize the

emerging trends and we will see the impact of QC on them and the side effect, as it is not cultivated and incorporated in the overall vision.

Democratization of technology and race for quantum supremacy

Democratization would allow developers to produce data models without containing the skills of a data scientist. They would in its place depend on AI-driven expansion to produce code and automate testing. The human increase is the use of technology to improve a person's reasoning and bodily understanding. Bodily increase changes a characteristic physical ability by embedding or accommodating a technology outside or inside on the physique. For instance, the automotive or mining industries use wearables to advance worker security. In businesses, such as trade and travel, wearables are used to upsurge worker efficiency. Bodily augmentation cascades into the following four key groups:

- Sensory augmentation
- Biological function augmentation
- Brain augmentation
- Genetic augmentation

Reasoning augmentation improves a human's aptitude to reason and brand improved choices, for instance, take advantage of information and bids to improve learning or novel experiences. Cognitive augmentation also comprises some technology in the brain expansion group as they are bodily inserts that deal with reasoning perceptive. Though science fiction may portray AI robots as the bad guys, a few technology companies now employ them for security; mind here, robots need not be made of iron and steel, it can be a piece of software as well. Corporations like Microsoft, Google, and Amazon use robots for multiple purposes like car parking lots and bulky outside areas to forecast and avoid crime.

The robots can read authorization plates, report apprehensive action, and gather data to report to their proprietors. These AI-driven robots are just one instance of autonomous things; there are multiple other categories and use cases that exist.

But why are we discussing these things here, in a book on quantum computers? Because, the future of these developments including visions, strategies, and technologies lies in QC. Quantum computers and computing are here to take them to the next level. Just imagine a scenario when these are integrated with the power of QC; how will the future take shape?

The fact is that quantum solutions could transform the complete IT industry with the foremost financial, engineering, academic, and social influences. However, at least at the moment, they won't function quicker than light travels or substitute current computing systems, and though they'll challenge a few safety encryptions, they won't render them all in the old-fashioned instantaneous way. But definitely, they would be integrated with Blockchain in the future and change the complete dynamics of today's security landscape. As we know, Blockchain is a kind of distributed ledger, an increasing chronologically well-ordered list of cryptographically contracted, irreversible transactional records, common for all members in a network. Blockchain also permits parties to hint resources back to their source, which is helpful for old-style resources but also overlays the method for other uses such as outlining food-borne diseases back to the unique dealer. It too permits two or more parties who don't know each other to securely interrelate in a digital setting and exchange value deprived of the necessity for a central consultant.

[Datanomics, AI, and QC](#)

We touched base datanomics in the previous sections including introduction in this chapter. Now let's go a bit deeper in the notion of datanomics. The fundamental point which I want to highlight is, modern economy is different from the traditional economy which was based on land, labour, machine, and industries. It is also different from the knowledge-based economy. Essentially, it is the economy that is based on data and its intelligent use. Knowledge could further be derived out of this. It affects all walks of life of individuals, groups, society, and countries, and ultimately, the whole world. Either it is politics, fashion, health care, manufacturing, or retail, its presence and effects are all-round, just like the way the traditional economy had a few years back. Refer to [Figure 12.3](#):

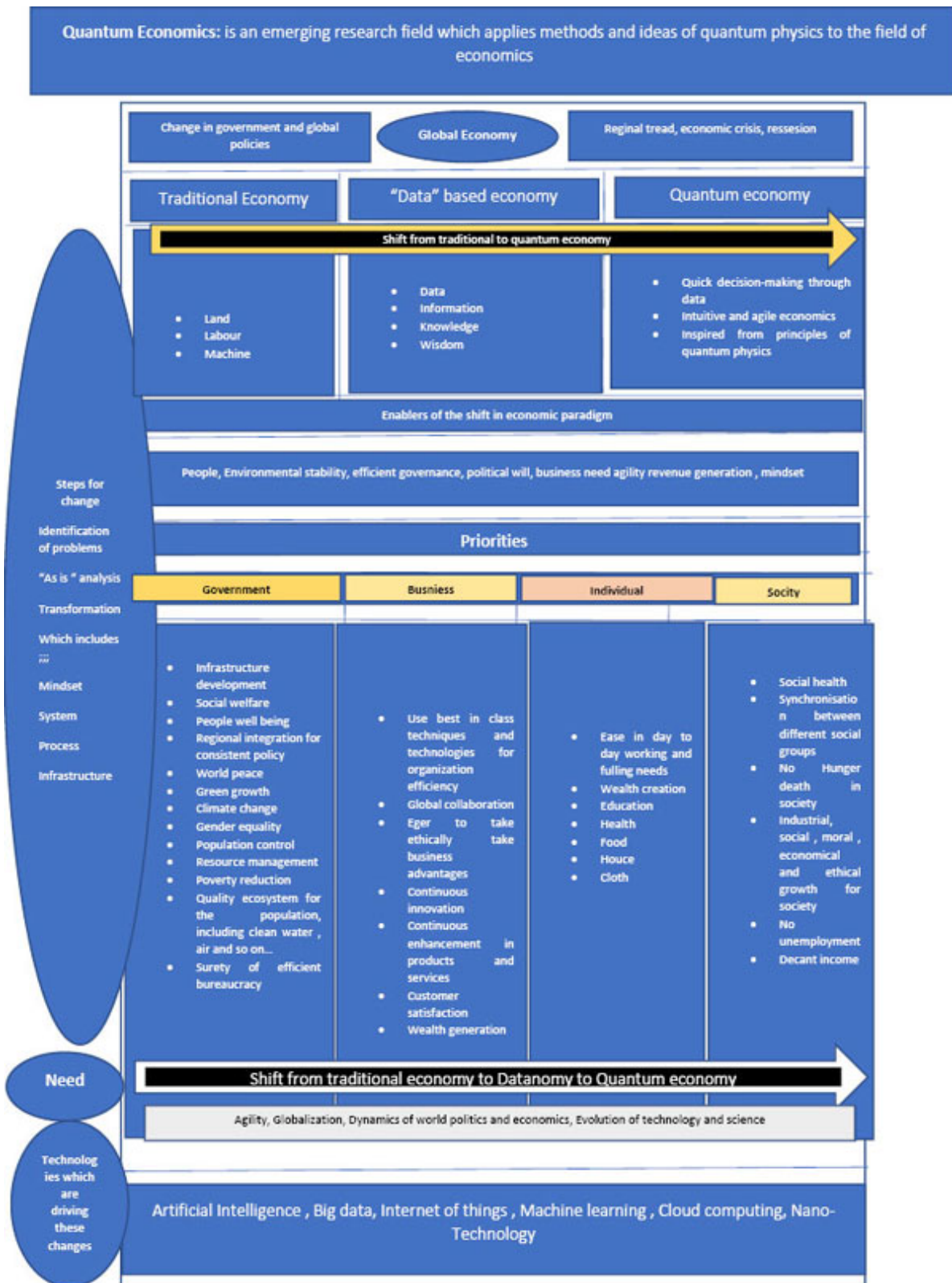


Figure 12.3: Quantum economics and its challenges, evolution and benefits

There is an exponential advancement in data generation and exploration happened in recent times. Hence it is fuelled to analytics and machine learning-based industries, technologies, and services (10 to 15 years of timeframe). Old-style businesses like manufacturing, healthcare, and retail are ailing and adopting the finest portion of artificial intelligence and analytics at the super-fast speed. At the same time, relatively new industries like fashion and cloud-based customer care are fundamentally using AI or in other words data-driven mindset, strategy, application, and culture as the base of their business model. However, analytics and machine learning did not confine itself to business, industries, and enterprises.

On the broader level, government policies and world politics have also started being impacted by it in a great deal. For example, before establishing a smart city, a lot of data crunching and data-based technological assessment are some of the mandatory criteria for any government. For making people-friendly policies or to prevent crime, data analytics and insight generation slowly become mandatory criteria for government agencies. Shortly, all the policy, social welfare activities, and Infrastructure developments must be guided by the data and its processing.

In the coming time, we will move from just processing of data and generating insights. But we want to go to the era where data processing (of whatever volume) is just a mandate and enabler for decision making. And this is again possible with the help of QC. This will take us to the depth of quantum economics.

Quantum economics is the area of study which is based on the principles of quantum physics. In my view, Quantum computers and computing would sit at the centre of this type of economics. Potentially, quantum economics would transform the world and more specifically it would change business. This would impact almost all the entities starting from individuals, to social groups to the government.

This is totally my view and you have all the right to agree or disagree with the opinion.

To make this point more clear, [Figure 12.3](#) provides a snapshot of the transition between traditional and quantum economics, including triggers, challenges, priorities and advantages.

The realities of quantum computing

There is an enormous gap between the glitches for which a quantum computer valour is beneficial and what we can presently shape, package, and run. Many well-known QC algorithms have qubit resource necessities that far surpass the present scale at which QCs can be built. The aim of the QC study community is to close the gap such that advantageous algorithms can be run in applied amounts of time on dependable real-world QC hardware. Though the present pace of growth is high, the predictable time to close this algorithms-to-machines gap is still frequently predicted as ten years or more in the future.

Present labour in QC are attentive on hurrying exploration and growth to close the gap more readily. CS researchers bring priceless know-how in the design of software design, languages, in techniques for systems structure, scalability and confirmation, and in architectural methods that can bring applied QC from the future to the present.

Between 2024 and 2028, we imagine numerous businesses with optimization issues to accept hybrid methods, in which, portions of the problem would be handled by traditional computing and fragments by quantum. In that similar time frame, quantum computers are probable to become influential enough to flinch handling expressive simulations. This simulation will ride on the exponential processing capability of a quantum computer. The ultimate quantum computer which is “universal” in nature is probably far away from now, but we already made significant progress in the short version of it like quantum computers which is based on quantum annealing. The tentative timeline of this is shown in [Figure 12.4](#).

The influx of quantum AI is more, and we don’t imagine quantum computers to be influential enough for prime factorization, further, refer [figure 12.4](#) for the tentative timelines for the universal quantum computers and other in between. Refer to [Figure 12.4](#):

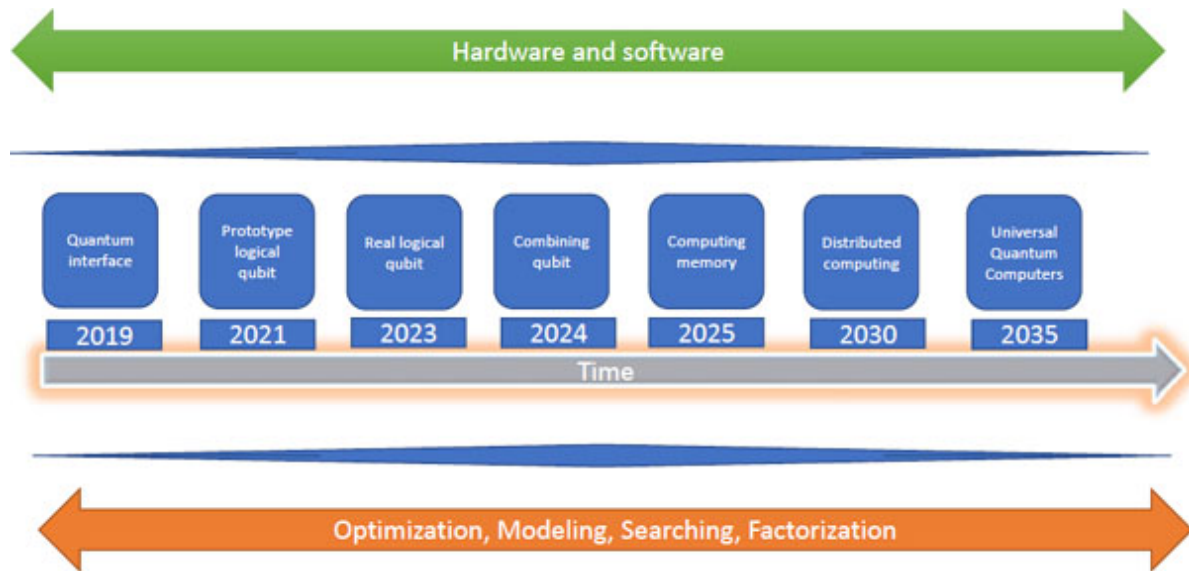


Figure 12.4: Tentative timeline for universal quantum computers

This timeline for the growth of the expertise notifies our approximations of when different industries are probable to advantage most from quantum computing. The experts who are forerunners in progressive industries, global energy, and materials, aggress that the significant value will start coming from quantum by 2025. The large reimbursement for pharmaceuticals may not come till the next decade, given that resolving the greatest multifaceted medical problems includes imitating intensely complex molecules. By the mid-2030's a wide range of industries will have the latent to produce important values from quantum computing.

To contest with the prospects of today's challenging customers, data plays an imperative role and sits at the heart of organizations. Business wants far-sightedness and more customized method to serve their customer in an improved way; hence fast processing of data for prediction and classification plays an important role in making strategies.

Quantum Machine learning and analytics are good in providing classification, prediction, and generalization. Thus, a lot of companies are watching for quantum data analytics and machine learning for the resolutions of their problems. However, sightlessly and thoughtlessly going for a quantum machine learning-based approach and changing the existing infrastructure of the business would not solve many purposes.

It must be applied based on the requirements and competence of applying the necessities in arrangement with the business vision. So, before going for creating a solution(s) or adopting QC oriented approach, it is significant to recognize and record the efficiency and readiness of the organization on the preceding-mentioned parameters.

There are many conducts to test and access it. Though, the utmost effective method is to produce proof of concepts or several connected proof concepts. The Proof of concept(s) would help recognize the strong areas and zones where the enhancements are vital before applying the solution. Correspondingly, business and leadership across organizations would get acumens on the real-world ground and get the knowledge on how product/solution would look like if it turns out to be a certainty.

It would aid leadership to discover the business parts which would take the most advantage with QC and QML employment in the organization. It would also help produce strong business cases for investment. The proof of concept would also be capable of delivering insight on benchmarking and monitoring the data set and the imagining of tentative results. The integration challenges would also pop up if it exists and proof of concept is realized properly. However, to get the maximum benefit from the PoC method is to make it strong as near as to the real problem. In a nutshell, it must not be just a mere hollow piece of code.

Innovation, disruptions, business expectations and quantum computing

The world of business is quickly developing; clients are asking for more robust, intelligent products, and improved services. The conjunction of communication and computing is the start of novel business models and revenue streams. The IT systems must provision new-fangled ways of doing business collaboratively with partners and customers. The outcome is a smart, fast, and intelligent ecosystem that allows interaction at more touchpoints and penetrates not before being endeavoured.

Today's business challenges are not successful to be encountered by just adding a few systems here and there, nor by onslaught up a limited platform in the enterprise IT landscape. The only solution may be to rearrange the

innovativeness's operative and legislative rules in the novel environment. It will be a transformational drive that has to start with a new thought process and an agile mindset. The strategies, vision and mission would act as a director, maybe a trailblazer, who takes the enterprise on a transformational trip from a disjointed and multifaceted world with line-of-business parting, product-specific stovepipes, legacy systems domain, and costly process to a more reasonably prearranged and valuable state with multiservice, revenue-generating platforms and a well-organized operational regime. On the way, fundamental operations may be obligatory to remove duplication, reduce costs, improve dependability, and upsurge agility in the business.

Along with the vision, mission, and goal, technology is the centre place for achieving this. In the next three to four years, the demand and agility of customers are going to be increased. Hence the competition in the market to grab the customers are bound to rise. Then how to bring differentiation and prepare for the new set of challenges is the million dollar question for the organization.

Along with the business strategy, technical inquisitiveness is going to play a major role. To make themselves comparative, relevant, and effective, adopting new technology would play a major role. Quantum computer is one such technology that would help companies in doing that. The transformational journey that begins with EA is long and hard. The effort may be cost-justified by identifying IT-centric benefits, but the real value of enterprise architecture can only be appreciated when it delivers tangible business benefits.

The enterprise decision-makers must imagine how the organization will change the way it operates and how big it produced business benefits against the spent. The benefits must be understood in the interpretation of both long-term and short-term business priorities. For instance, when embarking on a multiyear plan for the placement of a united, strategic **Service-Oriented Architecture (SOA)** platform for the enterprise, the corporate should be intelligent to understand the benefits of such a platform early enough (say, within the first six months) and then intermittently thereafter (for example, every couple of months).

[Quantum software strategies](#)

A primary wave of programming and toolchain growth happened in the past ten years, but much of that effort engrossed courtesy on high-resource requests and algorithms far outside NISQ competences. Present work must emphasise on methods that give programmers the communicative power desired to communicate the wishes of QC algorithms comparative to close-fitting resource restrictions on current employments.

To an extent, the present state of QC programming and system layers look like that of traditional computing roughly like what existed in the decade of fifties. In specific, while research consideration has been located on algorithms and devices, more consideration is required to be dedicated to the functionality in between, to regulate how best to instrument and enhance programming, mapping, and resource management for QC systems. This is similar to the software stack contemporary in today's traditional computers, though it may well take a different form in QC due to the exceptional resource restraints at play. [Figure 12.5](#) shows evolution of hardware from vacuum tube to transistor to quantum processors and placing of algorithm:

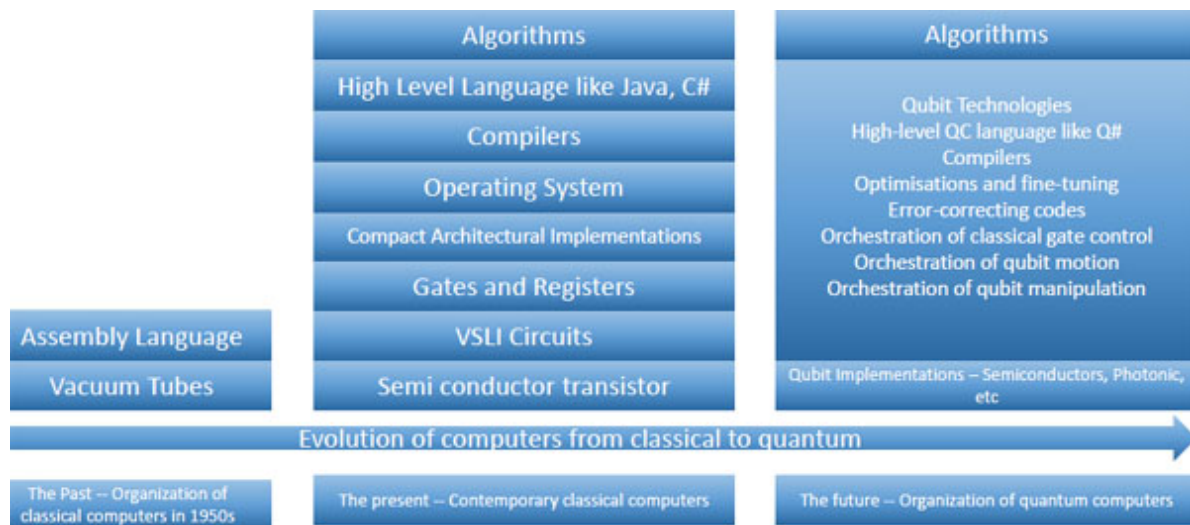


Figure 12.5: Evolution of hardware from vacuum tube to transistor to quantum processors and placing of algorithm

QC systems want the hefty full-stack distribution of information and data from applications and further programs down to target specific devices. The languages and software systems will profit from approaches to cumulate and share such information. Similarly, and in more human footings, it will also be significant for discussions and groups to be established to aid the persons

working on different portions of the toolchain to segment methods and to cooperate with an extensive range of professional applications and devices. In the extended term, typical interface layers might permit toolchains and methods from different academic or industry objects to interoperate with each other.

There is an overall necessity for research concerning how best to instrument and enhance programming, charting, and resource management for QC systems through the functionality among algorithms and devices. For near-term NISQ machines, we will have a requirement to produce and improve languages and compilation methods that give programmers the easy-to-read power required to communicate the wants of QC algorithms comparative to close-fitting resource restraints on present operations. In the long term, the use of thoughts to improve output will be required, once quantum resources are additional abundant. Organizations must find the kinds of modularity and layering usually desirable for scalable systems. These comprise libraries for commonly-used functions, as well as APIs and instruction set to encourage growth and optimization.

Quantum debugging is an essential encounter, in cooperation. Because, while computing a qubit finding the reasons of the failure of its state is important. As it helps in detecting operative precision errors and harvesting probabilistic consequences on applications. Consequently, research is desirable to establish how we can path the errors that accrue through a quantum program, and how we can confirm that the calculation functions within an error, which can enable the procedure of QC software debugging.

Computer architectures for QC systems is required to be investigated. Real-world quantum schemes will be hybrids of traditional and quantum units. Research is desirable on how to program and map professionally to both sides of such machines. Thoughts differ on the degree of architectural complexity necessary on each side, as well as on subjects of message between them. As with other systems and architecture research, the growth of computer hardware and package methods is essential be paralleled by the simultaneous growth of metrics to describe the act and depend, and assessment infrastructure to guesstimate, pretend, or measure them.

[Seizing quantum advantage for your business](#)

QC is the consumption of the philosophies of quantum mechanics to do calculations and solve problems, utilizing the sole marvels of separate subatomic particles as compute elements. The quantum computers are likely to rapidly resolve problems that are incredible to calculate in time frames that are valuable to humans.

A quantum system is capable of looking at every possible answer to a problem concurrently and not just the single best solution; it can also provide thousands of near replacements as well, all within a second. This grants noteworthy compensations over an old-style computer, which must successively discover the possible solutions to a mathematical problem, taking more time. The actual influence of quantum computers is the aptitude to upsurge compute influence by adding additional qubits to the system, which delivers an exponential haste upsurge every time a novel qubit is added.

CIOs, business value, and quantum computing

The market for QC is being determined mainly by the craving to solve problems that just would take too long time remove, or too much compute power, to be resolved by traditional computers. Numerous issues are driving this desire for more power and speed. A few drivers are mentioned as follows:

The arrival of huge volumes of data: The mammoth growth of data in both the areas that are consumer-produced and business-to-business oriented data have blasted in the resent times. The growth is multidimensional which includes size and rate of recurrence. Today's computers are just incapable of processing this information in a sensible amount of time. Furthermore, enterprises are emerging systems that are dependent on more so-called complex data, which necessitate additional powerful analytical tools and processing than is presently available.

Security anxieties: There is a robust craving to invent novel skills that can save systems from hackers, most of which are predicted to use QC technologies once it turns out to be available.

The growing use of simulation and modelling: The usage and request to digital simulations of real-world procedures have augmented meaningfully,

mainly as a method to decrease design, development, and testing prices and time scales. Simulations are data- and processor-intensive and can significantly advantage from the exponential power of QC. These models and simulations are even being protracted to modelling real-world processes, such as crowd forecast and control, fleet optimization, and logistics.

Business barriers

Regardless of the potential of quantum computers, there are important barricades to overcome, both technical and non-technological. Though these are not insuperable challenges, they will retain the market from producing substantial revenue throughout much of the prediction period. Currently, there are quite a few definite technological challenges that must be addressed over the impending years as QC develops. The primary one is that quantum computers have not been capable to determine quantum supremacy on a consistent and flawless basis. This is the main fence to acceptance; quantum computers are hard to shape and govern, and if they cannot deliver a benefit over orthodox machines, companies are not going to adopt them. However, companies like Alphabet are claiming that they achieved quantum supremacy. But the error correction is still a bigger problem to encounter.

As mentioned earlier, Quantum computers are also likely to produce errors and instability, as qubits used to compute the outcome of a problem can be prejudiced by magnetic interference, noise from the nearby qubits, or they can just fall out of a state of entanglement, which signifies the qubit can no longer develop, hence a calculation is not possible as a qubit. Modifying these errors is challenging, since present solutions need that a huge number of qubits be grouped to form the so-called logical bits, to handle error correction, and that distracts power away from the authentic qubits that are essential to do the work. As physical qubits are so unbalanced, there is a natural requirement of the extra qubits to safeguard error correction and fault tolerance. For semiconducting quantum machines, the number of physical qubits essential to generate a logical qubit can be as high as 3,000 to 1. Other researchers have estimated that this proportion is far too low; in its place, they refer to approximations of 10,000 qubits to 1 logical qubit, or higher. The proportion of physical qubits to logical qubits is tremendously significant in evolving a quantum machine that can scale to address

multifaceted, real-world problems, and might compel growth pending experts to decrease this proportion or grow another form of error correction.

Further, to attain quantum supremacy, machines will probably need to be tailored anywhere from 75 to 150 physical qubits, as the computational power raises exponentially with the accumulation of each separate qubit. While it is likely that investigators may grow a machine that can determine quantum supremacy, emerging a quantum machine that can address real-world, enterprise-grade problems will probably require qubit counts of more than 600 to address basic problems, and qubit totals in the approximate thousands will be essential to handle multifaceted problems. But it is not just technical challenges that are antagonizing the QC market. Comparatively, few quantum physicists comprehend how to write quantum algorithms that will be valuable to enterprises, and several of the top minds have now been employed to work for the principal vendors and enterprises.

Further, quantum algorithms can be run on quantum simulators, but these simulators are only effective at mimicking quantum machines with about 50 qubits. Complex, real-world work (such as developing composite material simulations, chemistry simulations, or large-scale optimization tasks) simply cannot be simulated and must be conducted on an actual quantum machine with hundreds or thousands of qubits (which does not exist). That is why, it is extremely challenging for companies to devise future quantum algorithms that will yield meaningful results.

A key issue not often discussed is separate from potential regulatory activity. While quantum computers are essentially a catalyst—specific algorithms must be written to solve specific problems—there is always the likelihood that privacy and data-protection advocates will want to ensure that specific policies and procedures are in place that extend to data that is used in quantum calculations or systems.

Further, because quantum computers will have the capability to ingest and analyse a wide variety of public and private data points very quickly, there is the distinct possibility that data that may not be tracked or used today could get swept up in a quantum-enabled system. Because of the exponential increase in speed and capacity to handle massive amounts of data, QC-enhanced systems are likely to attract additional scrutiny from privacy

experts. The following are the few advancements in quantum problem solving by different organizations:

- Accenture Labs and Biogen are utilizing quantum computing molecular evaluations to help quicken drug discovery.
- Experts are not discovering how to substitute the century-old Haber process to harvest ammonia for use in composts.
- It may help to recognize new-fangled materials that can hold a room-temperature superconductor.
- Quantum methods may help in the discovery of a catalyst that can advance the competence of carbon confiscation, a possible critical technology in the bout in contradiction of global weather change.
- It can aid to grow new-fangled battery chemistry that can meaningfully advance.

Quantum computing market structure

The enterprise quantum computing market is mainly attentive to research and development and classifying which use cases that may be appropriate for quantum computers. Vendors providing professional services, quantum algorithm development knowhow, and amalgamation services are controlling the marketable action in the market and are likely to do so over the following years.

The vendors developing quantum computers have some chief goals, which comprise growing the computing power of quantum machines by rising the number of qubits in use, plummeting noise (interference), and error rates to produce a machine that can be used to address problems with additional correctness and dependability, emerging an ecosystem of tools and platforms to permit enterprises to yoke the power of quantum computers in the close term, and generating industry, use-case and process-specific quantum algorithms that can be original and dispersed, thereby making a feasible business model for QC applications.

Market forecasts

The QC market is predictable to rise strongly over the subsequent ten years, with the majority of revenue coming from the areas of North America, Europe, and the Asia Pacific. These regions are home to the vendors that are contributing to the hardware, software, and services essential to plug out the QC ecosystem. The big enterprises are trying to discover appropriate use cases where QC will be helpful. Fundamentally, these two sections of the market are the academic and other research institutes, which are foundations of both know-how and future workers to the profitable companies operating in the market. [Figure 12.6](#) highlights how market and business is perceiving quantum computing applications:

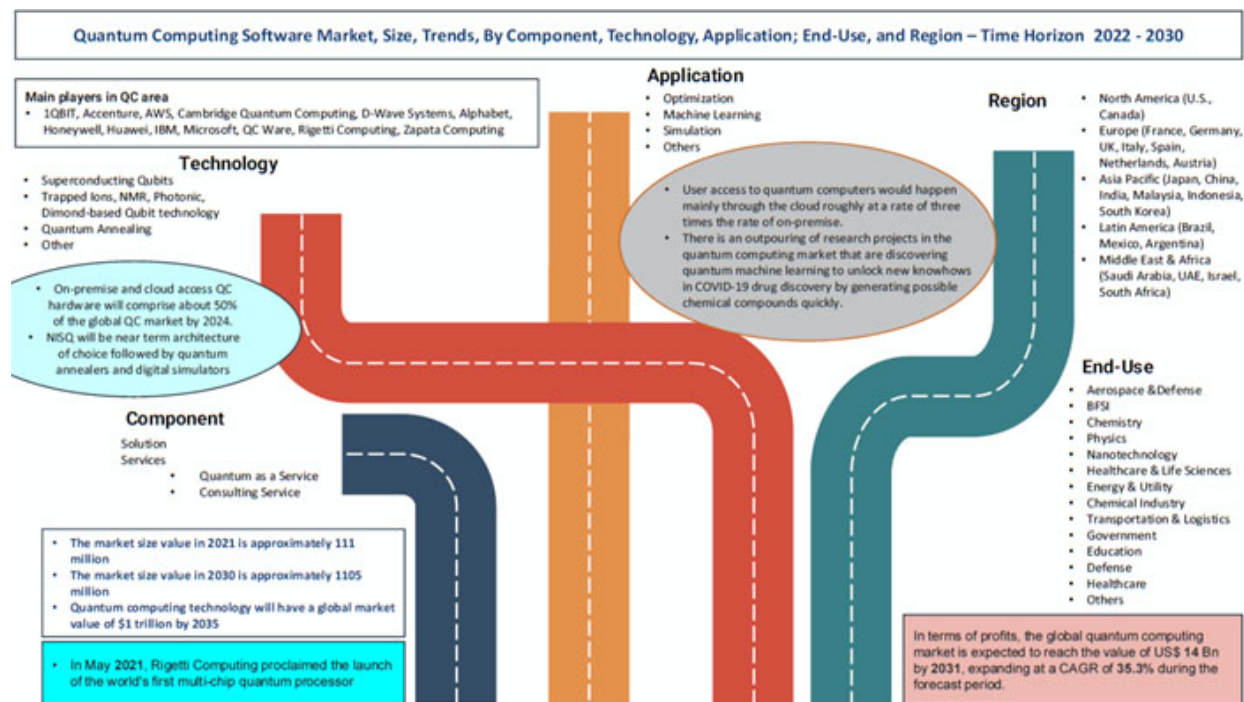


Figure 12.6: Highlights how market and business is perceiving quantum computing application

From an industry viewpoint, in the coming time, a moderately minor numeral of industries will be vigorously spending and making revenue from QC within a period of five to seven years. Normally, the industries that will be capitalizing in quantum technology will feature problems that fit together nicely with the kinds of solutions that quantum computers can deliver, mainly focused around optimization, simulation, design, and chemistry.

At present-day, there is a very tiny real product that can be bought. It is highly improbable that the utmost Fortune 500 businesses will be lining up

to buy a quantum computer. Much of the income in the market will be made by services, together with over-all consulting services that are provided by vendors, specialized services companies, and third-party industry advisors. Cloud-based admittance to quantum computers will be the leading way that enterprises will use quantum machines in the adjacent term, that will also underwrite to services revenue.

The market for QC is obviously on the increase. Though, several of the real applications and use cases are probable to be organized in a test phase. Though QC may be in its beginning stage today, a decade will pass very fast, and no enterprise or vendor will be required to be wedged deprived of having a vibrant map to guide them. As per one prediction of Gartner, by 2023, 20% of establishments will be accounting for quantum computing projects, compared to fewer than 1% today. CIOs must look for possible changes from quantum computing and be prepared to help the business influence them.

Quantum computing is a multifaceted technology. It's not like the usual software that's going to give the impression one day and be accepted by millions of individuals the next. The quickly rising quantum ecosystem would advance a clear approximation of how the technology will go on the track of improvement over the next couple of decades. These prospects will essentially be completely combined with old-style IT and will need novel cross-collaboration from research experts, computational data researchers, and quantum data researchers. This novel growth model is perilous to the achievement of any quantum program.

Human augmentation raises apparitions of revolutionary cyborgs, but humans have been supplementing portions of the physique for hundreds of years. Glasses, hearing assistance, and prosthetics grew into cochlear transplants and wearables. Even laser eye operation has become ordinary. But what if experts could supplement the brain to upsurge memory storage, or transplant a chip to decode neural patterns? What if exoskeletons turn out to be a normal uniform for autoworkers, allowing them to lift superhuman masses? What if medics could insert sensors to map how drugs travel inside a body? These things will get a kick by the combination of technologies like nano-technology, artificial intelligence, deep and machine learning, and QC will play a crucial role here.

Quantum computers will be luxurious machines industrialized and functioned by a few key players. Businesses such as Alphabet, Microsoft, and IBM hope to double the competences of quantum computers, in Moore's law-like style, every year. Along with a minor but noteworthy unit of talented start-ups, they will progressively drive up the number of qubits that can be identified by their computers. Subsequently, the technology is blossoming, though their development may be sluggish. By the early decade of the 2040's, thousands of quantum computers would be available and operational. Maturing of robust hardware and software capable to handle the utmost complex problems takes time and optimistically available by the late 2040's.

Having said that, quantum will start bringing value to a few businesses well beforehand. As far as access of QC is a concern, mostly in the near term and in the long-term, businesses and individuals will obtain quantum services via the cloud from almost in the similar way as it is available now. However, it is more democratized, robust, and easy to use. Amazon, Microsoft, IBM, Zapata, Honeywell, and a few other start-ups have already started taking steps to build the quantum hardware, therefore possibly, they would exist in the future of QC as well, because the first step is a major requirement for success in the field of technologies like Quantum computers. This is one of the reasons why a few other vendors have already proclaimed quantum offerings. These cloud contributions could rapidly enlarge acceptance and demand.

Opportunities to apply quantum computation on industry

Whether it's carriages, robots, or farming, autonomous things use AI to do tasks that are usually finished by humans. The sophistication of the intellect differs, but all autonomous things use AI to cooperate more obviously with their surroundings. Autonomous things are across five kinds, as follows:

- Robotics
- Vehicles
- Drones
- Appliances
- Agents

Those five kinds dwell in four types of surroundings: aquatic, land-living, air, and digital. They all function with fluctuating degrees of competence, harmonization, and intellect. For instance, they can span a drone functioned in the air with human-assistance to a farming robot functioning autonomously in a field.

This tints a comprehensive depiction of possible applications, and virtually every application, service, and IoT object will join some form of AI to systematize or supplement procedures or human movements. Cooperative autonomous things such as drone swarms will progressively drive the upcoming of AI systems. At the moment, we can see the sights of the potentials of AI-driven autonomous competences in any physical object in corporate, society, government, individual, or customer environment, but these technologies are best used for barely demarcated purposes and for the confined purpose. They do not have a similar competence as a human brain for decision making, intellect, or general-purpose learning. However, with the integration of quantum computation, the complete landscape of automation itself would be changed. Because all for devices, software, or wearables, automation uses data crunching to make human-like decisions which “only” QC would be able to give at least at the rate of desired requirements.

Further to normal automation, Hyper automation deals with the application of progressive know-how, including AI and **Machine Learning (ML)**, to gradually automate procedures and supplement humans. Hyper automation spreads across a range of tools that can be automated, but likewise denotes to the sophistication of the automation (that is, discover, analyze, design, automate, measure, monitor, reassess). Intelligent Hyper automation is the future and it is going to be driven by again a mix of IoT, AI, virtual reality, quantum computers. The following are the areas which are broadly going to be affected by this:

- Customer-related values
- Team-related values
- Product-related values
- Sustainability and growth values
- Personal values and qualities

- Client Satisfaction
- Social responsibility
- Economic sustainability
- Environmental sustainability

Innovating with quantum computing now

While making quantum centric policies or adopting it for profitable or transformative purposes, a few fundamental but important things are to be considered. It is not out of the moon ingredients but any business needs to follow them and quantum strategies. Adopting or innovating with quantum computers also need some strategies to be adopted. The following are the strategical guidelines which need to be followed while going for quantum:

- Understand your business challenge with the context of your vision, mission, and goal.
- Make your strategies in the alignment of the preceding point.
- Look at the issues with different lenses.
- Integrate business, process, people and technology expertise to develop outline concepts.
- Develop joint (with partners, vendors, and other stakeholders) for immediate experimentation.
- Collaborate with academia, researchers and corporates.
- Create a dynamic agile ecosystem of innovation and change.
- Hire and develop talent.
- Keep track of market trends.
- Put agility at the heart of whatever you do.

On top of these, apply the engagement model which comprise the following:

- **People:** Experts from business and technology.
- **Process:** Focus on a business imperative.
- **Design:** Develop solutions quickly for experimentation. Adopt proof of concept way.

- **Outcome:** Deliver value by implementing at scale.

Based on mindsets, further, extend it for validation and verification, based on the following dimensions of strategies:

- **Consulting:** Address an imperative through collaborative engagement across business.
- **Co-creation:** Coming up with the ideas to resolve a challenge.
- **Iterative implementation:** Exploring a range of opportunities.
- **Agile based concept development:** Develop, test, and deploy quickly and iteratively.

Let's be agile ... in context of quantum computing incentives and machine learning (applicable for QML as well)

Agile means dynamic in plain English. Agile methodology is all about adopting a dynamic way of implementing anything in small chunks, instead of doing in the big bang fashion and in a monolithic way. To create a quantum organization that would revolve around quantum computers, this approach becomes more relevant. Because quantum computers are a big investment to make, there is less bandwidth of failure. Taking baby steps that are tested and verified would help get what you need.

If you apply agile analytics to machine learning-based projects, you come to realize that this style is not a rigid type of methodology which “provides you list of steps like what to do, and how to do it; you have to accomplish this before moving to the next phase. No, it would definitely not tell you to do all this. What it does for you is that, it exactly outlines what you must do and how you must do it. Agile methodology is a repeat / iterative, incremental, and evolutionary way of developing a project.

It works in short iterations that are typically one to three weeks long. The goal of every iteration is to provide user-valued features (what matters for users). Each developed feature must be fully tested during the development of specific iteration. The agile development methodology along with Devops (a culture cohesive thinking in terms of development and operation) is not about creating dull examples, which has no value in the real world. Also, one

of the expectation from the implementation of agile methodology is to automate as routine processes wherever possible. Test automation is possibly the most important among them.

Agile exists in multiple variants and types, like extreme programming, SCRUM, LeanUX, Kanban, ScrumBan, and so on. All the approaches/methodology has their own benefits, suitability, and drawback. Scrum, Kanban, and XP are the most famous among the chunk. [Figure 12.7](#) highlights key ingredients of agile methodology in a nutshell:

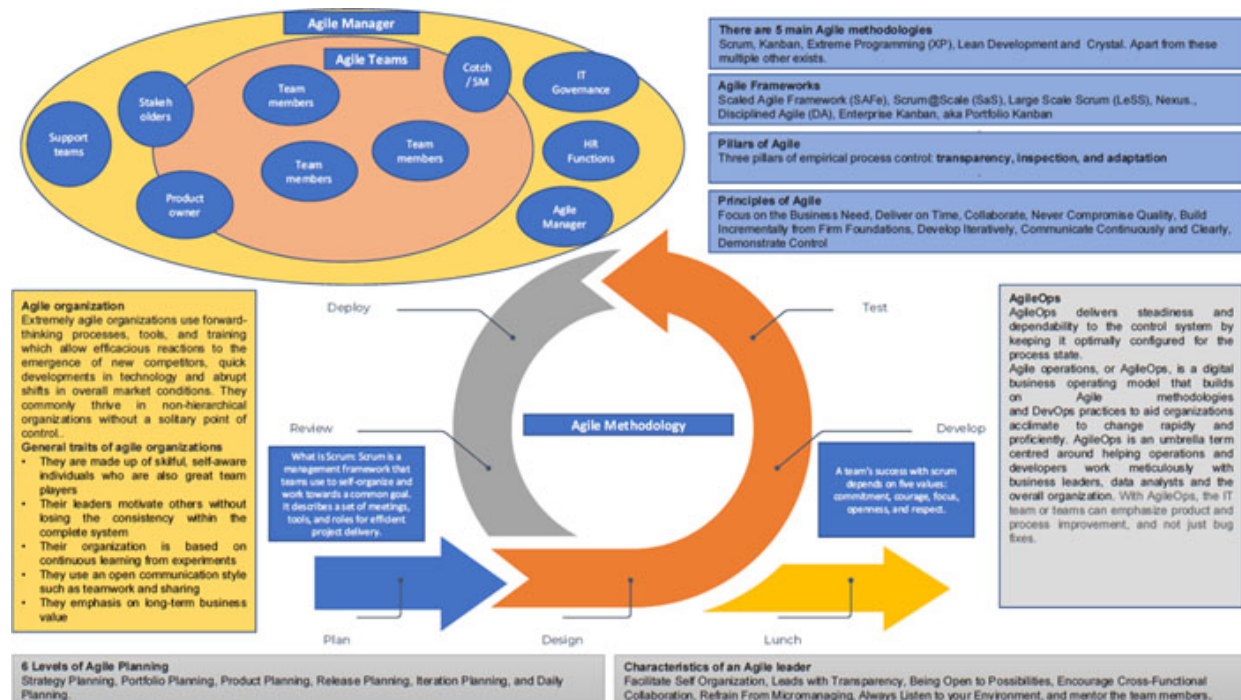


Figure 12.7: Agile in nutshell

Why do you need agile?

Analytics and quantum software development projects are complex; if you track it well, you could potentially develop it better. If not, then you fall under the following trap:

- The project is over budget.
- The schedule has slipped.
- Some expected functionality was not implemented.
- Users are unhappy.

- Performance is unacceptable.
- There is no ability to expand.
- The data and/or reports are poor.
- The project is not cost-justified.
- Management does not recognize the benefits of the project.

These happen, because of the following:

- Lack of experience.
- Ambitious and unrealistic goals.
- Lack of domain knowledge and subject matter expertise.
- Unrealistic expectations from the project.
- Only focus on technology, not as much on business.
- Lack of collaboration.
- Lack of Specialized skills.
- Lack of Multiple other skills like people management, open to adopt changes in the project/organization.

The difficulties in implementing an agile approach are as follows:

- **Tool support:** There aren't many tools that support technical practices such as test-driven quantum machine learning development.
- **Data volume:** It takes creative thinking to use lightweight development practices to build high-volume data warehouses, quantum - legacy system synchronization and BI systems. We need to use small, representative data samples to quickly build and test our work.
- **Lack of knowledge:** Not enough knowledge of end-to-end quantum technology based agile development practice.

Agile is needed because it brings rapidity, quick learning and predictability in the project success. We could use other methodologies including waterfall methods. But it is not advisable, as it is rigid in nature. For instance, suppose you procure all the infrastructural component before the start of the quantum project. Then after some time realized it is not feasible to go with that.... outcome of it? you failed miserably and suffer loss. So, the efficient

approach is go for bare minimum requirements and do the development and test and then decide. Let's have a look on waterfall methodology. The purpose of this discussion is to provide you clarity about why agile the better choice than the other.

Waterfall methodology

The waterfall software development life cycle or SDLC is a traditional methodology. In waterfall development methodology, every stage of a product's life cycle sequentially takes place. It is known as the waterfall because in this, the progress flows steadily downwards through these phases like a waterfall. In Waterfall, the software development model operates in a phased manner and is often rigid. Typically, it contains phases like, Define, Develop, Test, and implement. At this point, it is good to understand the difference between agile and waterfall methodology in a clear way. [Figure 12.8](#) will help in doing that:

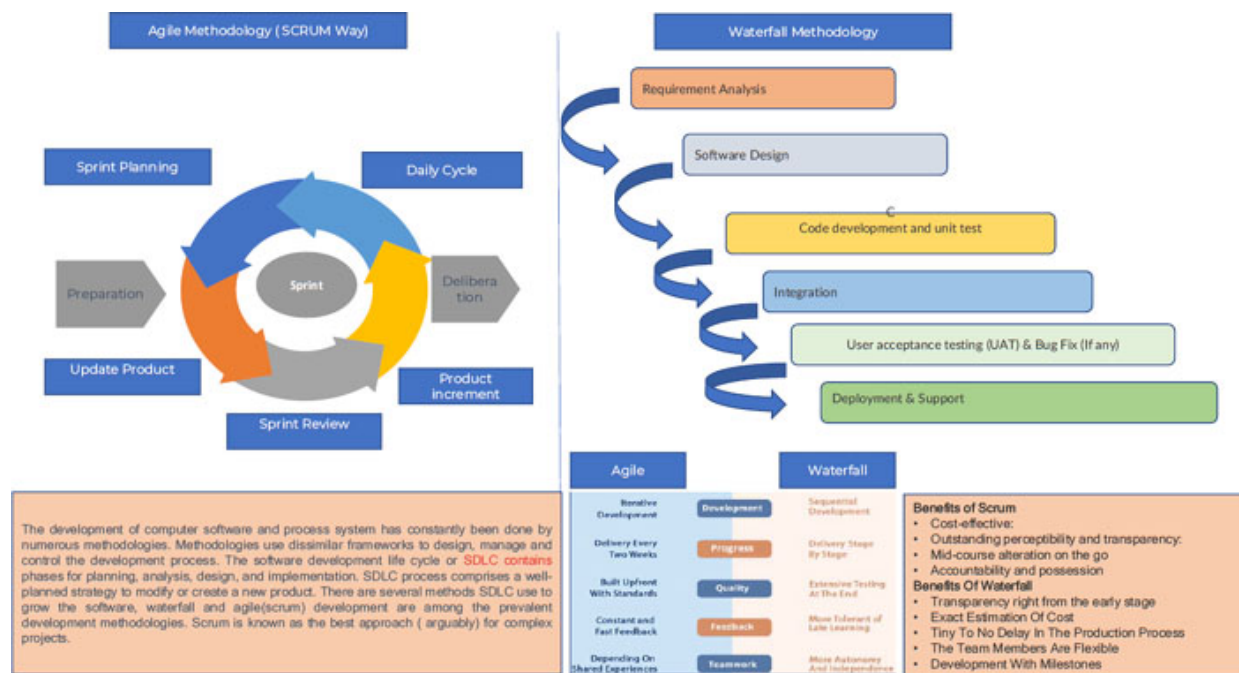


Figure 12.8: Difference between agile and waterfall methodology

Pit bits

Making changes in Waterfall projects is comparatively costly. Whereas in Agile, the changes to requirements can be generally incorporated at any

point of the process. Even at the advanced stage of development, agile is willing to accept and accommodate the changes. Therefore, it is less costly. And hence, Most suitable for quantum projects.

Agile disadvantages

Following are some disadvantages of agile.

- It desires stakeholder's availability, involvement, and cooperation continuously.
- It needs customers with clear priorities and vision, which sometimes is not possible. Noncompliance of the same sometimes leads to off tracking and failure of the project.
- Senior and experienced programmers with decision-making capabilities are required during the implementation of the process. Hence, sometimes unexperienced programmers may feel uncomfortable if experienced resources are not available to help. This parameter again becomes very important while we planning and executing quantum projects.
- It is hard to quantify the total effort, estimation, and cost due to the qualitative nature of the process.
- Difficult to execute in a distributed team.

Agile usage

- The bust is suited if the deadlines are aggressive.
- Good for the project with ambiguous requirements, as it is excellent in accommodating changes in the later part of the project.
- Good for complex projects.
- Good for innovative projects.

Growth values

- Curiosity, creativity
- Self-improvement

- Self-discipline
- Continuous improvement
- Open-mindedness, questioning old, provide risk-taking capabilities
- Enable innovations

Personal values

- Integrity
- Trust
- Respect

Short- and long-term goals for corporates and additional thoughts on quantum strategies

With untried demonstrations spread more than two decades, the quantum computing community has attained a main milestone over the previous few years. The capability to shape systems that are extending the boundaries of what can be characteristically simulated, and which allow cloud-based research for a wide variety of scientists, therefore cumulate the pool of talent discovering early quantum systems. Though experimental research in quantum computing has been vigorous for over two decades, it was not until the mid-2010's that it developed to physically attach a minority of superconducting qubits collected to instrument minor multiqubit tests with adequate reliability for evocative results. There is already demand from physicists, scientists, developers, and many others to access and test various aspects of quantum computing, even if the systems are still small, comprise only a few qubits, and suffer from noise levels worse than the fault-tolerance threshold. Over time, other groups in the industry and academia have also started to offer quantum cloud services to varying degrees, either in the form of real system access or simulators. In the shorter term, the improvement in the quantum error correction would get the most focus. It will help to come up with a robust quantum processor.

Any organization which wants to be quantum ready has to give attention to the following initiatives:

- Ready to invest in the future.
- Ready to finetune existing stature, and if required come up with the new one.
- Ready to invest in talent.
- Build an ecosystem of researchers, experts, scientists, academia, corporates.
 - Attractive, lively demos that elucidate quantum computing applications, and present their programming model.
 - Line up cooperative education materials with live code, equations, visualizations, and story text to jump-start application development.
 - Adopt open source Ocean SDK, which aids to use the system and grow code within the acquainted language of Python or Q# or any other quantum-enabled programming language.
 - Admittance to the rising community of developers who need to shape the quantum apps.
 - Provide comprehensive documents containing an explanation of the whole thing from the physics of quantum annealing to learning how to use the Ocean software.
- Enable innovative mindset within the organization.
- Willingness to be agile.

The quantum physicist owns a profound considerate of the fundamental device and would like to discover additional applied technical particulars, such as optimal control methods, original pulse-shaping methods, procedures to enumerate the fundamental system Hamiltonian, and error mitigation approaches. These operators want more of the nitty-gritty particulars and the aptitude to inspect device-level possessions of the system, for example, control over the occurrence, effectiveness, pulse shapes, and dimension addition kernels that are sent to the trial.

Benchmarking quantum systems will be a need to measure development. Assuming quite a lot of physical understandings of quantum computers will arise over time, and there must be a way to enumerate their act much like

conventional benchmarks. While benchmarking seems to be an understandable necessity, it is far less noticeable than how to develop a hard set of metrics appropriate to quantum computers. Significant factors for expressing a suitable quantum benchmark include the following:

- **The numeral of Qubits:** More qubits are essential to resolve progressively hard problems, and therefore, the additional qubits a quantum computer has, it will get the supplementary computational power. Systems with more than a few tens of qubits can be replicated on a traditional computer, and consequently, consuming a few qubits will not be helpful in the long run. However, it is important to mention that just having an additional qubit is not enough for more computational power. It depends on some other additional factors as well,
- **Connectivity:** In what way qubits are linked to one another matters. At one extreme, qubits linked on a line would need a noteworthy overhead for any arbitrarily designated gate among any random pair of qubits. At the other extreme, if all qubits are associated with each other, there is no extra overhead for an arbitrarily designated gate between any haphazard qubit pair. Though at the hardware level, connectivity staples a lot, and it importantly affects metrics such as crosstalk and trustworthiness. It is significant to foray an equilibrium among connectivity and overhead for a given application.
- **Gate Set:** The choice and act of the fundamental gate set are significant. A huge set of gates decreases the chances to produce random gates or move quantum information but it also needs far more difficulty from a standardization and constancy position.
- **Compilers and Software Stack Concert:** Compilers are critical for the best translation of circuits to the basic hardware. A compiler wishes to deliberate and enhance the device connectivity and possibly even differences of gate reliabilities and spectator errors across the device.

Mind Map

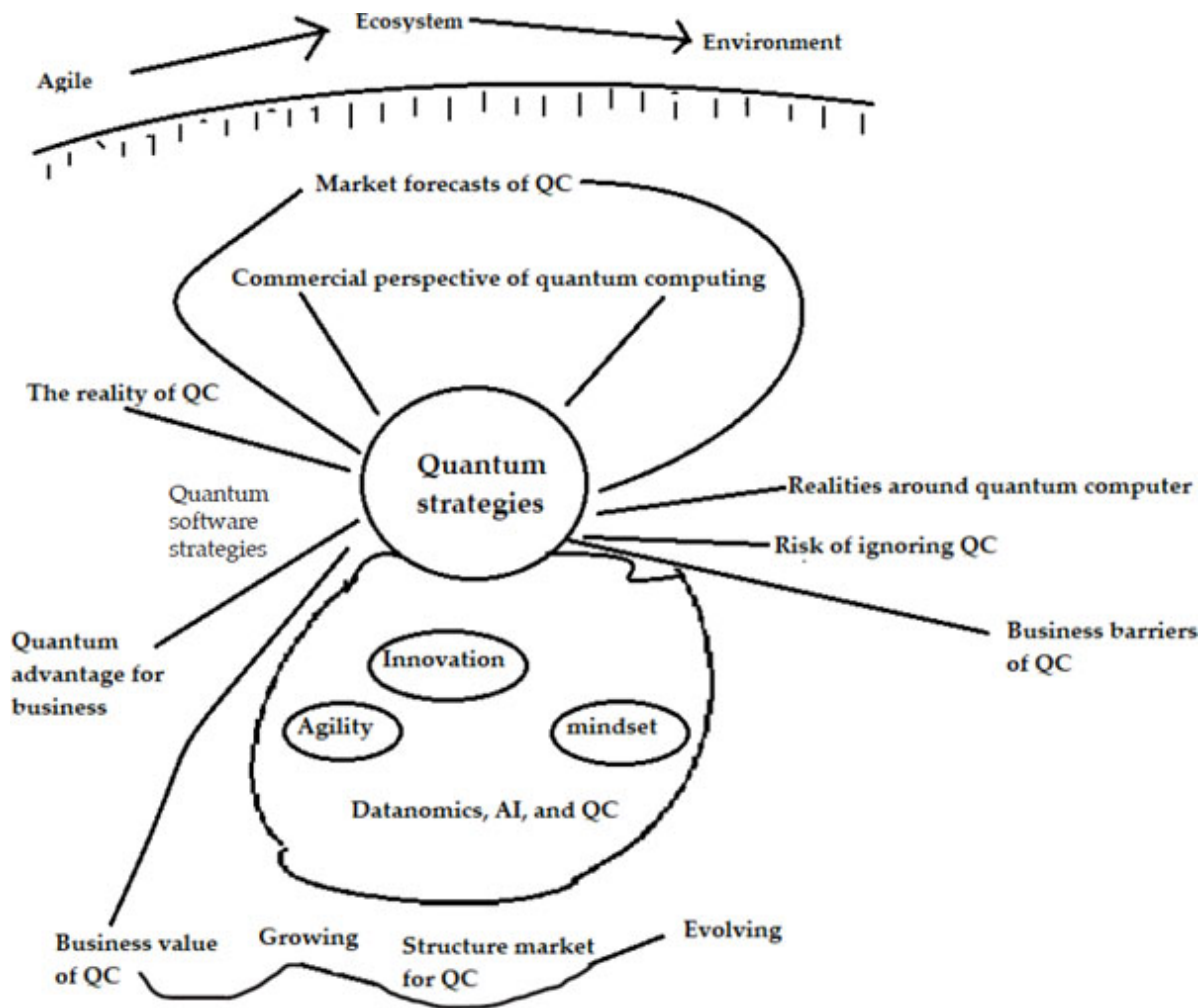


Figure 12.9: mind map for [chapter 12](#)

Conclusion

This chapter is all about strategies. It discussed strategies related to different parts of quantum computers and computing, especially when an organization is thinking to implement it. It tried to answer questions like how to approach quantum software, application, what are the main ingredients to become a successful quantum computing-based enterprise, how data, economics along artificial intelligence is playing a critical role in the overall dynamics of building next-generation organization, why quantum computer is now the need of the hour, and what would be the outcome of being reluctant to join the wave of quantum.

In the next chapter, we will focus on the people aspects of the quantum organization. For instance, the upcoming chapter will provide answers to

questions like, what are the skills needed in the coder, leader, and manager for being successful in the world of quantum computers.

CHAPTER 13

The Human Side of Quantum Computer

This chapter will discuss the people, culture and team-building aspects of an innovative or quantum organization (fundamentally, quantum or innovative organization does not follow any rigid rule; broadly they have a flexible structure, rule, thoughts, and hierarchies). Therefore, it is a very important chapter because it is about people who lead, build, and realize any technology within an organization. It is ultimately people who, on the ground level, bring the vision to reality. Hence, understanding their behaviour and reaction to typical situations is important. This chapter would help us in understanding these aspects. Also, the chapter will turn out to be significant since it discourses on how to accomplish quantum machine learning projects and effectively use the human resources who are involved in the overall process of technology realization.

This chapter will provide a view of how to build a team for quantum computer-centric initiatives within an organization. Broadly, it will discuss what are the skills, technologies, and tools required for a quantum computer team. It will throw light on the behavioural and technical competencies and requirements of a team member, manager, and leader. Also, it will discuss what are the new job role and mindsets that are emerging in the arena.

To be successful in the journey of quantum initiatives and for building a good development team, planning, producing and executing a strategy (quantum focused), and putting the right people in the right place in the project, we need a good leader. However, finding a good leader is not an easy task. But we can build a good leader and deploy him/her appropriately.

Structure

This chapter will be organized in the following fashion. It will start by giving some knowledge on innovation, agility, and a new generation of organizations. Then it will move towards the aspects of building a team and

the skill requirements of a quantum organization and team. Finally, it will discuss about the details of EQ, SQ, and MQ. A broader outline of the chapter is given in the following section.

This chapter will cover the following topics:

- Why are social, emotional and other types of intelligence like ecological and moral are important apart from intelligent quotient (IQ) for quantum teams?
- What are the fundamental potentials that the team members process for the planning and implementation of the projects?
- How to attain the best performance and promise from the team members in a QC project?
- What are the ingredients of thought and ground-breaking leadership? And why are they compulsory in QC projects?

In addition to the preceding points highlighted, the following are the few major topics which will be discussed throughout the chapter:

- Some thoughts on leader, leadership, and intelligence
- Interrelation of creativity and quantum computing
- Agility and modern organization
- Meaning of organizational agility
- Emotional, social, and moral intelligence
- Some of the benefits of equipping technical teams with these competencies
- What are innovations and innovative mindset
 - Flexibility
 - Creativity
 - Holistic and bigger thinking
 - Courage
 - Quick action
- Skills requirements for QC team
- Skills requirements of QC for an individual

- Ideal skills set for quantum computation
- How to build a quantum team
- Qualities of great quantum team members
- Difference between leader and manager
- How to build data culture and train resource on innovative technologies
- Dimensions of team
 - Team leader
 - Team members
 - Technology manager
 - Organizational leader
- Questions for transparency
- POV: How to be successful in quantum initiatives from the team, organization, managerial and leadership perspective
- Mind Map

Objective

The objective of this chapter is to give an idea about the human aspects of quantum computers. What it means is that this chapter will answer the questions like, how to build an effective quantum computing team, how to develop talents who would be able to handle requirements related to quantum computers which include software and hardware, and multiple others.

Introduction

We can find a general pattern in successful leaders; the leaders must be purposely focused on building up their self. Self-development is an extremely huge area like the sea and it takes a lifetime to refine and mature. Things are continually changing, and individuals are acclimatizing quickly to endure the new-fangled conditions. It is a lifelong process. The persons

who have a remarkably healthy self could be talented to steer through numerous complex conditions.

The leaders require to continuously upskill themselves and know what it takes to loan anyone. And, allow others to develop their self. This is again another vast knowledge area like an ocean and will take a lifetime to master.

When a leader gives his / her best to others, others will follow them. Others will make the leader a role model. Mind it, leadership is not a position or a title, it is an action and an example. Leadership development is self-development. Employees in the Knowledge and innovation-based industry encompassed humans and their multifaceted social, moral, emotional, and behavioural connections. High practical skills do not automatically produce high performance without the team members' commitment and motivation. It is also one of the fundamental criteria that one must like their work.

Further, it is also vital how they knob pressure and conflicts and interconnect and cooperate at the workplace. Orientation of worker morals, skills, capabilities and goals with organization, clients/vendors and the viewpoint of technical practices are required at the office for attaining brilliance.

In fact, numerous theoretical and specialized viewpoints have recognized the important influence of the aptitude to be aware of and accomplish emotions (emotional intelligence, EI) in the workplace (in personal life as well). The aptitude to circumnavigate and ease social relations (social intelligence, SI) and the skill to put on universal principles to one's values and actions (moral intelligence, MI) would usually lead in the direction of triumph in the workplace. Consequently, organizational competence and efficiency would be attained. As quantum computing team is the extremely technical application of this innovative motivational practice (EQ, MQ) become very significant.

Interrelation of creativity and quantum computing

QC and related technologies are extremely technical. Therefore, it requires creativity and innovative approach towards problem statements. Hence, a knowledge worker who is intelligent to use his logical/ rational as well as

psychological competence in a collected way would high probabilities to succeed, than one having only good scientific/ rational/ logical skills. For example, a quantum engineer who is outstanding in problem solving but not decent in articulating, presenting and imaginatively linking the examination to the broader picture is perhaps not a good choice for the a project where these competencies are required.

Thus, organizations and in turn projects require a group of people who are good at mathematics, logic, and problem-solving and at the same time they must possess good communication and social skills. Also, an archetypal organization deals with technologies which are sometimes not supposed to be exposed to the wider world, and hence it needs confidentiality which needs to be maintained. For example, a quantum organization deals with some patented methods.

In this typical case, exposing the method to the outside world may give advantages to the competitor. Hence, the moral values and ethics of an employee come into the picture. Until an organization does not hire morally strong people, their Know-how is at stake. Technologies and processes are here to help but to an extent. Finally, the person who is dealing with this is the most important. However, if the employee is morally strong, all this would be taken care of automatically. And the good news is these competencies can be developed, groomed and cultivated.

Agility and modern organization

The proverb change is the only constant in the commercial world, is very factual. However, it had been true from the start of free market capitalism. But with ground-breaking technology, the stride of change has only become faster in the 21st century.

An organization's aptitude to encounter those changes and rapidly acclimatize is essential for accomplishment in the contemporary business world. The greatest significant feature swaying project success rates is an organization's agility level. Agility is even further significant than the project approach – the better the agility, the improved the outcomes: financial results, customer gratification, project metrics, strategy application and benefits understanding.

Organizational agility includes features like responding fast on the change. It mentions an organization's aptitude to familiarize and reply to changes in external market circumstances. This comprises not just rapidity but residual in trace with the requirements of customers.

Agility is about elasticity also. Preferably, organizational leaders keep in touch with customer needs, learning and then adapting to changes, with a focus on "iteration, interaction and collaboration." It also involves the ability to effortlessly switch priorities and face market changes ahead rather than clinging to a business scheme five years ago that is no longer viable.

Using Agile approaches has also provided the organization with a stripper method of working, which is likely to have follow-on aids for the growth, construction, and formation of their other products. The power of fixed limits is why the Agile approach's structure work to be begun and transported in fixed time iterations with secure review sessions at the end of each repetition.

Meaning of organizational agility

Organizational agility jolts at the top. Leadership behaviour is one of the most vital features in producing culture. For instance, consistently recognizing and rewarding behaviour by employees that foster an agile approach to organizational goals.

Another critical component to success includes getting everyone involved with a specific business value change by working together on agile initiatives. Otherwise, there might be pockets within an organization working on agility, but they go nowhere because they are isolated.

According to the PMI report, organizations should look into the following:

- Communicate messages from the C-suites in a consistent way.
- Enable collaboration across diverse enterprise groups – agile is about people over processes.
- Address a whole business value chain at once, allowing end-to-end agility.
- Accept multiple approaches that are tried during the transformation phase (in other words, some failure must be expected and accepted).

- Commit to transforming and modernizing current approaches before introducing new ones.
- Minimize friction between traditional phase-based delivery and more modern approaches.

One way to approach handling all these issues is to run agility initiatives through an organization's **Project Management Office (PMO)**. Because of their experience on working across all areas of an organization to implement successful projects, project managers in a PMO have exactly the experience and relationships needed to foster agile transformation.

To start and build a quantum initiative in the organization, a few steps must be taken. This can be spread over the building culture of developing agility in the organization to developing quantum leadership to creating a team. Here, let's see quickly for forming a good quantum team, what a leader along with a manager and other support functions like HR and administrative could do in the organization in stepwise fashion.

In [*Figure 13.1*](#), there are the steps to make an organization quantum ready in an agile way:



Figure 13.1: The agile ways or steps to make an organization quantum ready

The steps are explained as follows:

- Build a quantum innovation team, with defined roles and responsibilities.
- The team's role is to orchestrate internal and external quantum computing resources to drive exploration and adoption.
- Track the evolution of quantum computing, development of QC ecosystem and bear activity in this space.
- Develop your talent roadmap and initiate investment in training. Identify potential use case, build business case for QC prioritization and formulate a roadmap for further exploration.
- Develop algorithms to stimulate use cases on classical computers, and improve these algorithms' efficiency by iteratively simulating on a classical computer.

- Start coding in quantum assembly Languages and deploy the program using commercially available computers (IBM, Microsoft) and start-ups.

Building a good team is not enough. Keeping team members motivated so that they would give their best is far more important. An innovative and technical team by nature is "restless". They always look for a new set of challenges. And guided by their surroundings including social and emotional set-up. Therefore, understanding their social, emotional and intellectual need is necessary. By understanding them an organization with the help of leaders, managers and other associated parties would be able to cater for these challenges. In the next section, we would understand the challenges that a typical innovative organization face from the perspective of talent, client and project and will see how Emotional, social, and moral intelligence would help in resolving them. In this journey first, we define these competencies and then go deeper.

Emotional, social, and moral intelligence

The following are the utmost challenges their organizations face from people, talents and leadership context:

- Individuals need to deal with enormous, fast change.
- Individuals need to be additionally imaginative in directive to get-up-and-go for innovation.
- Individuals need to achieve enormous volumes of information.
- The organization needs to upsurge client faithfulness.
- Persons need to be additionally inspired and dedicated.
- Individuals need to work in a composed manner.
- The organization needs to brand improved use of the unusual talents obtainable in a varied workforce.
- The organization needs to classify possible leaders in its positions and make them move up.
- The organization desires to recognize and recruit top talent.

- The organization needs to brand good choices about novel markets, products, and strategic associations.
- The organization needs to skill people for foreign projects.

These are the strong challenges for all organizations today from people and talent perspective that includes both public sector and private. And in almost every case, emotional intelligence must play a significant role in sustaining the need. For example, managing huge change involves, among other things, the aptitude to observe and comprehend the emotional influence of change on ourselves and others.

Apart from people and talent-specific challenges, additionally, organizations face multiple other issues as well which range from staff holding to improving efficiency to getting desired outcomes from clients. And in all these issues emotional, social and other little unknown mentioned intelligence is effective. Further, to overcome these challenges leader's role is also very important. For instance, to be actual in serving their organizations by managing change, leaders first need to be conscious of and manage their individual emotional states of nervousness and doubt. Then they need to be conscious of the emotional responses of other organizational associates and turn to help people cope with those responses. At the same time in this procedure of managing efficiently with huge change, other associates of the organization need to be vigorously tangled in monitoring and managing their emotional reactions and those of others. Emotional intelligence affects organizational efficiency in a number of areas, as follows:

- Worker staffing and holding
- Growth of aptitude
- Cooperation
- Worker commitment, confidence, and fitness
- Novelty
- Efficiency
- Competence
- Quality of service

- Client loyalty
- Client outcomes

Organizations along with leaders we can take these benefits to next level with social and moral intelligence. Now, formally look on the definition of these competencies. Short definitions of these competencies are as follows:

Emotional Intelligence (EQ) is the capability to find, practice, appreciate, and accomplish one's own emotions in constructive ways to release stress, communicate effectively, empathize with others, overcome challenges and resolve conflicts. This skill also enables to identify and appreciate what others are feeling emotional. This is mostly a nonverbal process that notifies thinking and influences how well one connects with others.

Social Intelligence (SI) is the aptitude to connect well with others and to grow them to collaborate with you positively. It is also denoted in some literature as to people skills; however, it is more than just that. Social intelligence is about mindfulness of the situations and awareness to the social forces at work which lead them. This competency helps one to equip with the knowledge of communication styles, strategies, policies and practices that would help attain desired objectives in the context of dealing with others and society. Social intelligence helps one to get self-insight and awareness.

Moral intelligence (MQ) is the ability to appreciate right from wrong in the rational and empathic way. It is the ability to have solid ethical beliefs and to act on them positively, to enable one to behave in the correct and honourable way.

After having an understanding of EQ, MQ, and Social Q, let's discuss how these are affecting different functions of organization. To understand it in a better way, let's refer to [Figure 13.2](#) which depicts organization efficiency in terms of EI, SI and MI.

Let's understand this with the help of a model the left-hand side of the model ([Figure 13.2](#)), exemplifies three structural factors that are interconnected. Each of these features influences emotional intelligence through its impression on relations, and each aspect influences the other two. For instance, the data showing how the emotional intelligence of

organizational leadership affects organizational efficiency through its effect on organizational climate.

At the same time, the HR functions of staffing and assortment, training and growth, and management act have a robust influence on leadership. Though, leadership will affect the extent to which HR functions are real in serving organizational members upsurge their EI. The leaders who have deficiency of EI will deliver deprived models for growth and are not able to help others to develop EQ. Because they are improbable to deliver the kind of backing and reassurance essential for real EI promotion efforts. Refer to [Figure 13.2](#) for details.

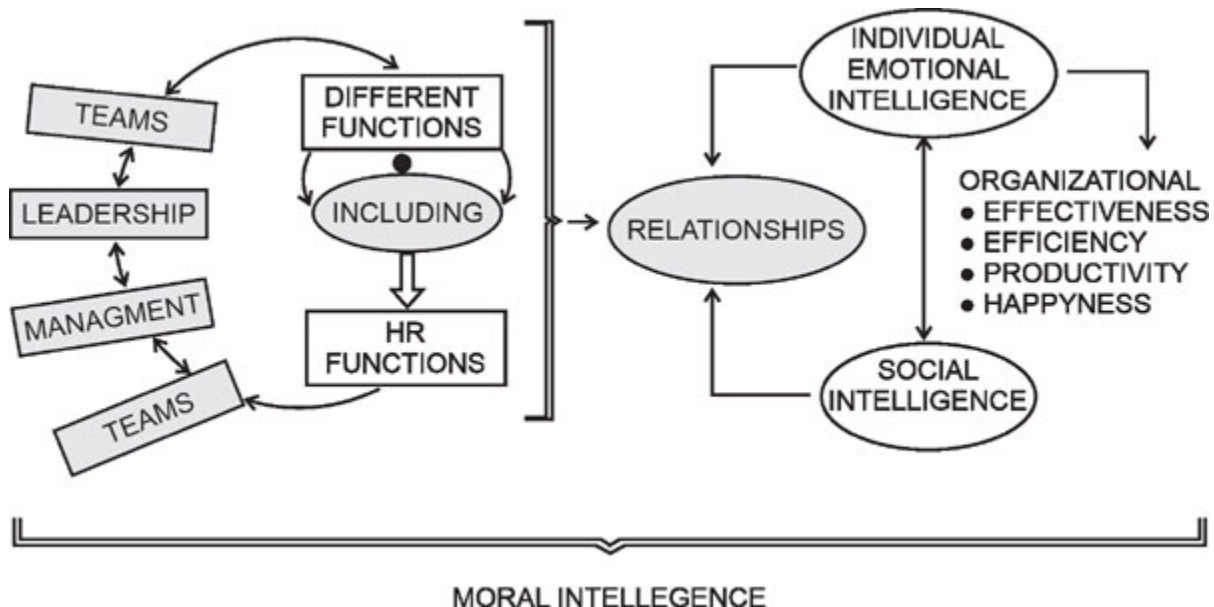


Figure 13.2: Organization efficiency in terms of EI, SI and MI

The model proposes two significant insinuations. First, any exertion to recover the EI of organizational members will eventually nosedive, except it touches naturally happening relations amid those associates. Formal, off-site training programs can have worth, for instance, but only if they lead to continued vicissitudes in relational and intergroup relations back in the organization. The second significant consequence is that interferences that emphasis on only one part of the model is not probable to be very real.

So, for instance, a training program intended to aid organizational members that become more expressively intelligent will be of incomplete value by itself since it marks only one part of the model—HR functions. Such

training exertions will prosper only if the executive leadership and culture provision them.

Benefits of equipping technical teams with EQ and other behavioral competencies

To assess an individual, a measure of his/ her holistic intelligence is required. **Holistic Intelligence (HQ)** is nothing but combined or integrated intelligence which combines all the intelligence. It can be represented in the form of $HQ = IQ + EQ + MQ + \text{other intelligence (like Social Q, multiple intelligence, and ecological intelligence)}$. Now let's look at some of the benefits of the different competencies which we discussed:

- EI, Social Q, and MQ have a meaningful impact on the performance of human resources.
- Increased EI, Social Q, and MQ correlates with profitability via customer satisfaction.
- EI, Social Q, MQ can inculcate service quality that improves customer satisfaction.
- These competencies can be used for improving team performance and human behaviour in totality in all dimensions including intellectual capabilities.
- Theory and practice of management science, human resource development, organizational behaviour and industrial psychology can be integrated with these competencies and could be applied to the team through the proper use of appropriate methodology and frameworks.
- These competencies can be integrated with gamification, storytelling (automated and manual) and concept comics. In turn, these can be used to improve the efficiency and effectiveness of software development and support teams.

Innovations and innovative mindset

Innovation is an enquiry of mindset, and producing that mindset leads the whole thing elsewhere. It's the innovative mindset that supersedes the

features of human nature that are frequently brings innovation in great organizations. However, the fundamental question is how do we create an innovative mindset? Well, there is no formula for this but there are some guidelines which need to be followed.

Flexibility

Organizations need to be open to change. It implies that they have to admit and absorb the idea that the world is in continuous transformation and all parts of society are challenged by this alteration. It also means to be conscious of where this altering world is heading and to inquisitively retain the path of change and novel marvels. Lastly, it also means that you have to continually possess and analyse what the transformation means and what the conceivable significance of the transformation will be for your business. Change is a complicated thing. Organizations have to deal with it. Accepting the circumstance that technological transformation is about to influence business is typically very tough for recognized establishments. However, if they accept and adopt this fact, they will give the impact everywhere.

Creativity

Another feature of an innovative mindset is to truthfully accept creativity. An innovator's boldness is that creativity is the answer to problems, rather than an old-style scientific method. This argument is principal among numerous of those who have positively experienced innovation in the arena of daily business actions.

Nonetheless, to associate innovation with art doesn't rule out the requirement of structure, procedures and practice for innovation. All these are compulsory also when working on the art of innovation. Though, the innovation-as-art viewpoint pressures that the preliminary point for innovation is creativity, rather than application of organization processes and structural structures for innovation. Which, in turn, needs a convinced the kind of culture and organization that allows creativity. In this part, we understand numerous digital businesses position themselves.

Holistic and bigger thinking

Most academic investigators and specialists on innovation settle on the idea that innovation is about more than just incremental developments to current products or product postponements. This leads to the point that innovation needs an aptitude and the bravery to contemplate better and outside the present standards and facts in the market.

Innovation is about widening one's opinions out of commonplace regular thinking and analysis. Big thinking and innovation are a mixture of logical skills, business spirit and the aptitude to visualize. Few persons are consecrated with all these competencies, however, not all are equipped with this capability. Hence surely organizations must prepare themselves to accommodate these competencies under the same roof.

This is also one of the motives why a varied organizational culture has arisen as the main precondition for novelty. Meaning miscellaneous groups that syndicate skills and competencies can achieve big rational more effortlessly than consistent groups that are probable to replicate forms of similar rational over and over again.

Courage

Innovation doesn't occur without establishments and the go-getters inside them have the courage to continually reconsider how effects can be done. It takes courage not to imitate to extensive politics and general realities in large organizations. It takes courage to challenge established plans and positive products and services before they go into weakening.

It takes courage to query organization and colleagues for doing things the way they have continuously done. It takes courage to continually problematize and be that one individual who continually goes in contradiction of the grain and tries to think about things from a different viewpoint. It takes courage to be susceptible rather than in concert of harmless rendering to recognized commercial repetition.

It takes courage to undertake the novel and indeterminate, endangering disappointment. But all of the preceding is essential to thrust innovation onward and to produce an innovative climate in establishments. Since when has whatever new ever occurred except when someone ventured to take that

stage into the doubt of the unidentified – the courage needed to take that first step?

Quick action

Innovation inside an organization essentially is a fast-moving procedure to keep up with the alteration going on outside of the organization. Twentieth century innovation was frequently a sluggish process, with extended lead times from impression to idea, and idea to market.

A lot of time typically went into wide R&D. In the automotive industry, for instance, the timeframe to create, project and inaugurate a novel car model has been about seven to nine years. However, in current time, seven to nine years is an eternity in an automotive market that is altering year by year.

Possibly, disruptive business models for auto manufacturing like Local Motors have showed that a novel car can be Imagined and hurled in the market in 11 to 19 months, and with progressive 3D printing technologies and VR-aided design and manufacturing, that timeframe can perhaps be cut even shorter in the years to come.

In a nutshell, there are five chief elements to an innovative mindset. We need to be exposed to change, have a prejudice towards creativity, an aptitude to think big, inexorable courage to challenge the standard, and be branded by the speed of thought and action.

An organization with the wish to be innovative must reason reckless and smear a fast-paced innovation procedure with a well-organized go-to-market roadmap. In this setting, it's also important to follow the idea of deteriorating fast, as novel ideas and thoughts have to be verified rapidly and be closed down just as fast if they don't hover. In this way, the organization can move resources to the following idea instead of receiving a wedge in a dead-end innovation project; after all, the world's subsequent big idea is just around the corner.

Skills requirements for QC team

Quantum Mechanics deals with the most basic structure of the universe. The rules of computation of properties of the fundamental particles are

highly counterintuitive; they are completely different from our well-known classical world.

The properties of spacetime in quantum regimes highly fluctuate, leading to abnormal experimental outcomes. The basic idea behind the construction of quantum computers is that they can be extensively utilized to unriddle most of the present-day problems of prime factorization and tedious optimization problems, which cannot be resolved by the classical computers that we use today (even though they helped us immensely). But quantum computers are faster than our present-day classical ones and can resolve these problems by using quantum phenomena like superposition, entanglement, tunnelling, and so on. There are some specific skills required for building a QC team, which would spread across software and hardware. There are multiple skills desired however, few must-have skills are mentioned below.

- **Physics:** Quantum computing appears to be the connection of mathematics and physics with admiration to practical math/computer science. It's fairly stimulating. You can really influence the mathematics leading the physical process to abridge the model's mathematics. Understanding Physics is extremely important in the context of quantum mechanics is desired to perform the expert level of activities in the areas of QC. This includes learning about the quantum information overlaid with condensed matter and high-energy physics concepts. So, if someone wants to build a high-end career in QC, Physics would be the healthier choice as a major subject. Followed by a minor in math--Calculus, linear algebra, differential equations, maybe PDEs or differential geometry.
- **Maths:** is another skill along with linear algebra and complexity theory which are required for building a career in QC. One should also learn collection and symbol theory, random matrix theory and functional analysis. Elementary linear algebra is a pretty overall requirement. With either a math or a physics major, one can necessarily learn that. From there, the response is contingent on specific curiosity and will follow that interest. Meaning that if one needs to do algorithms, then he/ she needs the mathematical background essential for complexity theory as well as for the applications of the specific algorithms. If you want to emphasise

crypto, then one needs mathematics for that. If someone is attention to quantum error correction, there is the mathematics behind the numerous codes.

- **Computer science:** In computer science, themes such as data theory, machine learning, artificial intelligence, big data, error-correcting codes, optimisation and complexity are vital.

Additional and specific skills requirements of QC for an individual

The following is a list of prerequisites of topics and concepts needed before diving into quantum computing:

- Elementary quantum mechanics
- Linear algebra
- Basic collection theory
- Elementary abstract algebra
- Elementary probability
- Stochastic procedures
- Fourier transforms
- Elementary algorithms and capability of examination of algorithms
- Combinatorics
- Discrete algebra
- Number theory
- Topology
- Laplace transforms
- Matrices (basic, intermediate and advanced)
- Functional analysis
- Theory of computational complexity
- Solid state physics
- Holography (advanced)

People who will work on the field of quantum computation must possess the understanding of quantum information theory. It is a subtopic, where previous acquaintance of probability theory and traditional information theory is a must. Also, the idea about quantum noise modelling and quantum error correcting codes is required. There are three different specializations which can lead to an occupation in quantum computing. For instance, a Physics major can be chiefly helpful if he/she is attentive in construction of a quantum computer.

A Physics major with theoretic Computer Science emphasis can help one in scheming algorithms for a quantum computer. If one is involved in Quantum Mechanics, then a major in computer science and a minor in Math's with an emphasis on abstract linear algebra is obligatory to shape a basis in quantum computing. From these, we can determine that knowledge of computational physics/science and quantum mechanics is critical for this arena.

How to build a quantum team

World class technical team needs courage, perseverance, and aptitude to pact with vagueness. Also, giving unrestricted access to the team members is one of the required standards for an actual and well-organized team. If you limit the team in the limitations of valued pecking order and expert level, the team would not prosper. There are some vital critical accomplishment influences for the real technical team.

The following are some of the general criteria for a great team from quantum computing team perspective:

- Senior decision-making participation in setting up the team.
- Incessant provision from the senior management.
- Arrangement of team's goals with dream and assignment of the organization.
- Quantum team must be a combination of business and technology authorities. Since requirements and commercial need to be tacit before going for technical application.

- Team must be talented to work in cross functional setting and situations.
- Provide independence for the team to grow this own working style and creating internal roles.
- Robust rules need to be defined about how to measure the attainment of team and team members.
- Team members and leaders must be designated on the foundation of diversity, knowledge and capabilities.
- Team must be a combination of individuals who are freshly hired from outside the organization and who are already employed in the team. So that the team has different viewpoints.
- The culture of continuous learning, real knowledge management and growth environment would be cultivated.
- Appropriate communication process and infrastructure availability would be established.
- Good recompense and credit mechanism for the team, leader, manager and separate team member. Value addition must be appreciated at all the levels.
- Repetitive, timely and operative feedback and performance assessment mechanism needs to be implemented in the team.
- Good training for developing technologies, team skills, problem solving and decision making.
- Happiness and fitness frameworks arrangements with the team members.
- Clear and well-defined RACI matrix need to be in place.
- Team must know how to interconnect visually. What it means is that collaborating the vision received from data to the stakeholders in an appropriate way through graphs, infographics, animation and reports. Mere raw truths is of no use in business.
- Team members must have an inquisitive mind, they must have boldness to know how, what and why of a problem.

In a nutshell, the team and consequently the team members must process mathematical, statistical, computational, programming, data management, positive psychological, communication and change management skills.

Qualities of great quantum team members

Building a good team depends on a lot of other things. But obviously the most important aspect is the quality of an individual team member. [*Figure 13.3*](#) describes the qualities of a good team member; these qualities need to be evaluated, identified, and developed in the team members through continuous training and other means of grooming. Also, some qualities of great team members are described in the following section. These qualities have become more important because the field of quantum computers and computing is by nature a very innovative, technical and creative field.

1.Organization 2. Working tough 3. Consuming fun 4. Faith 5. Value addition and Interest 6. Balancing skills 7. Shared goals 8. Output 9. Creativity 10. Collaboration 11. Optimization skills 12. Clear goals 13. Mutual development of mission, value and vision 14. Agreed on priorities 15. Each person capable of doing tasks 16. Measurable results 17. Well trained people 18. Positive attitude 19. Knowledge sharing 20. Focused discipline 21. Commitment 22. Positive stress 23. Good planning 24. Continuous improvement and delivery 25. Striving for excellence 26. Being open minded 27. Demonstrating empathy and support 28. Setting clear expectation 29. Role clarity 30. Balanced task and process focus 31. Stable workload among members 32. Broad-mindedness of diversity 31. Elasticity 32. Open and candid feedback mechanism 33. Need based on demand coaching 34. Admiration to each other 35. Accountability 36. Trustworthiness 37. Honesty 38. Team rewards 39. Impartiality 40. Courage 41. Risk management capability 42. Self-awareness, and a few more, that are described in [*Figure 13.3*](#):



Figure 13.3: The qualities of a good team member

Further, the role of a leader and manager is very important in leading quantum initiative. However, to assign the proper role to the specific person, it is a must to understand the difference between the two roles.

Difference between leader and manager

The following are some of the important difference between a leader and a manager:

- The manager directs, the leader revolutionizes.
- The manager maintains, the leader grows.
- The manager emphasises on systems and structure, the leader efforts on people.

- The manager trusts on control, the leader stimulates faith.
- The manager has a short-range sight, the leader has a long-range viewpoint.
- The manager has his eye continuously on the bottommost line, the leader's eye is on the prospect.
- The manager reproduces, the leader creates.
- The manager accepts the status quo, the leader challenges it.
- The manager is the definitive good combatant, the leader is his own person.
- The manager does things right, the leader does the right thing.

How to build data culture and train resource on innovative technologies

As we know, now the data and its processing took the central stage in the modern or even legacy business (Asian paints are using supercomputers for processing their customer's data, and it is one of the open secrets of their successes). Even after the entry of quantum computers data plays a very important role in the success of the organization. Therefore, some of the steps and measures one can take to build data culture and train resources on innovative technologies like machine learning and QC are as follows:

- Train your team (almost everyone) on data dynamics, Datanomics, statistics. Provide a refresher course on it to the stakeholders.
- Provide knowledge of domain, information flow, data flow, storage capabilities, external and internal data source to the stakeholders.
- Establish and educate stakeholders about the importance of data within and outside of the organization.
- Explain technology to relevant people in deep and to loosely coupled people at 200 feet level. This must include the architecture.
- Leader must be involved in the project.
- Proper reward and recognition must be provided to the people.

- Encourage research, innovation and thought leadership at all the levels.
- Knowledge sharing sessions must be incorporated in the DNA of the organization.
- Create a culture of shared success celebration; this would bring people together. Team must not be bothered about the organizational policy and changes. Essentially, they would be free from the organizational bureaucracy and hierarchy.
- Equip the team with the latest technologies and tools. Give them challenging work and encourage them to solve them. Encourage teams to publish their work on internal and external forums like, conferences, filing patents and writing white papers.

Important roles and dimensions of a quantum team

In an organization, a team has multiple dimensions and it includes multiple roles, including team members, team leaders, managers and leaders at multiple levels. Going into details of all the levels are out of the scope of this chapter. However, it is good to discuss a few important roles and their qualities which are mentioned in the following section.

Team leader

The characteristics, qualities, and skills of team leader are described as follows:

- Must not be alien with business perspective and issues. Having strategic view on the same.
- Aware about the organizational level frameworks, process and the goal of the company.
- Aligned with organization vision and mission updated with the technical, process and people issue which the team is facing and be able to solve them.
- Must have good leadership and intermediate managerial skills.
- Good project management skills.

- Good problem solving, conflict resolution and team management skills.
- Good collaboration and communication skills (including inter and intra personal).
- Passionate, creative and innovative
- Self-motivated, good in time management and administrative work.
- Must know business
- Be able to lead and guide a multi-disciplinary team of technical experts.
- Good at statistical analytics techniques and in mathematical concepts.
- Experienced in the agile software development methodology like SCRUM and extreme programming.

Team members

The characteristics, qualities, and skills of team members are described as follows:

- Capabilities of finding problems smartly on their own by analysis and then completing the tasks by minimum intervention of the others within the timeline.
- Technically excellent in their work.
- Multi-task management skills.
- Capable of individual as well as group decision making and good in conflict management.
- Open to taking feedback.
- Good at planning and organizing.
- Good problem solving, analytical and mathematical skills.
- Good collaboration and communication skills.
- Excellent skills of taking Initiative. Must be flexible.
- Passionate about data crunching.
- Good at adopting the concepts.

- Good at handling unstructured data and be able to do analysis.
- Knowledge of high computing and multithread programming model.
- Knowledge of agile software development methodology.

Technology / Engineering manager

The characteristics, qualities, and skills of Technology / Engineering Manager are described as follows:

- Excellent communication skills.
- Leading with transparency and honesty.
- Supporting your employees with clear direction and removing their roadblocks for them.
- Embracing technology.
- Motivating with positive feedback and recognition.
- An expert in the field.
- Mediating with productivity and calmness.
- Promoting cross-level and cross-functional collaboration.
- Creating a productive and lively work environment.
- Trusting your employees.
- Must be experienced in machine learning projects and be able to distinguish between normal technical projects and machine learning projects.
- Must know business in detail.

Organizational Leaders

These are the people who drive the organization by making policies, and providing directions. The characteristics, qualities, and skills of Organisational Leader are described as follows:

- Good social, moral, ethical, and emotional intelligence skills.
- Visionary, committed and ability to inspire others.
- Honest and self-aware.

- Good at making decision.
- Take actions with responsibilities.
- Must be productive, innovative and thoughtful at work.
- Ability to simplify the things and communicate / present information in the understandable way.
- Must have a positive outlook towards, work, life and self.
- Realistic at work. Ability to realize imagination on the tangible entity.
- Clear thinking and good judgment skills.
- Being aware and considerate and - showing sensitivity.
- Ability to lead by example.
- Desire for excellence and for getting things done.
- Excellent coaching and motivating skills.
- Good listener and strong executor.
- An excellent understanding of business, process and people skills.

So, in summary, the preceding discussion tells us what are the must have qualities of an extraordinary teams, team members, managers and leaders in general. Now we will discover some specific qualities of QC teams and the people around them. The following is the compiled list of specific quantum machine learning and quantum artificial intelligence team:

- They must process data science skills, and be good at statistical, numerical, problem solving, analysis, data crunching and modelling techniques. Be able to process research and have an innovative mind set.
- Exposed to handling large volumes of data, knowledge of diverse source of data and gathering techniques.
- Knowledge of tools, framework and libraries related to quantum machine learning and quantum artificial intelligence team.
- Knowledge of business process, company specific policies and business jargon.
- Excellent capability to present complex things in an easy way to laymen with the help of data visualization tools.

- Flexible, adaptable, learning, innovative and self-motivating skills.

Questions for transparency

A team is a complex entity, as majorly they revolve around humans. Therefore, managing them is always a challenge and satisfying their need is the organizational responsibility. Every team member want to excel in their life and in the organization. They always want to perform better if the surrounding environment supports. They have some questions for that; they want answers. If proper answers are provided, it motivates them to work better. The following are the list of questions which every team member wants answered:

- What business challenge is the team supposed to solve?
- Who would select the team leader?
- What are the role and responsibilities of the team members and team leader?
- What are the role and job description of the team member and team leader?
- What are the selection criteria of team member and leader?
- What are the technologies, tools and programming language that the team member and leader are supposed to work on?
- How will individuals allocate their time to the team verses their other functional responsibilities?
- Who measures the overall team member performance and what are the metrics for that?
- What is the appraisal cycle for the team member and leader?
- What would be the optimum size of the team? Who establishes the team priorities?
- Who manages the team borders with the other areas in the organization? What authority and responsibilities are implied by the dotted line reporting structure?
- How will the cross-functional and cross-cultural conflicts and priorities be resolved?

- What process helps teams to improve productivity, efficiency and effectiveness?
- How can the team improve internal and external communications?
- How often should the team meet and in which format should they share the minutes of the meetings?
- What are the location teams are residing on?
- How can they meet if face to face is not an option; what are the technical means of communication and collaboration (video conference)?
- What are the reports that the team needs to produce and who are the audience?
- What is the process that can be used to help team members to evolve and improve continuously?
- What are the team values?
- How would the team be motivated? How could the team loyalty be increased?
- How should team results and individual's performance be measured?
- Who will evaluate team members and what are the criteria for the same?
- How is an individual and team performance linked with company's performance and therefore to their salary?
- What is the training program that exists to develop the skills?

POV

How to be successful in quantum initiatives from team, organization, managerial and leadership perspective?

For any commitment, rebelliousness matters the greatest. And certainly attitude plays a vital role. One significant portion of correct attitude is competences to understand things in a different way. Since it gives you different viewpoints. Quantum projects need different standpoints all the times. It endures learning, trying novel methods and aptitude of being stretchy and adoptable. Hence, attitude counts here.

Further, to be efficacious in Quantum projects, willingness to learn, competence of attaining novel opinions and aptitude to alteration are critical. In a technology space like quantum computing and related ecosystem, transformation is the rule. So, no one can be continuously proficient at everything. Hence, readiness to familiarize whatsoever comes to the rescue. Intelligent and behavioural meekness are the structural blocks of the learning attitude.

In quantum initiatives and executing project, creating suitable culture need not be the obligation for the businesses who pacts with technology. There is an archetype change required to produce a technical culture if it is not triumphed inside the organization. Though, it is a two-way trip, people also have to accept it and fit in to that framework of the culture of the organization.

One of the mantras to be successful in a quantum team is to looking for 'deucedly great' innovations or at least keep trying for the same. Also, high on energy and life-centric are the other necessities for quantum computing professionals. Businesses and, in turn, managers have the poise on their persons that they want to attain somewhat big and are hopeful in winning. For instance, a corporation known as Chumbak, has a team of people who are explicitly skilled to manage a culture test.

They demeanour this test once the specific resource clears all the rounds of technical and managerial interviews. One of the common questions they ask all through their test is, if money were not an object, what would you be doing in its place? This board of interviewers have veto authorities. What it means is that if they do not fit with the culture of the resource in the organization, they have all the right to used the applicant, even if the contender is outstanding from the technical viewpoint.

Failure is the reality in the quantum projects and this is here to stay in everyone's life. Failure is common but how somebody is to deal with that is very significant. The way of coping with failure separates someone from others. In quantum projects, these coping skills are compulsory.

Since, there are times where you are trying numerous things and waiting for results, easily getting upset would not help. For instance, trying multiple algorithms to come with the best result, needs endurance and hunger to

leave something which is not working and move on. Also, while accomplishment at preceding roles is good, answers to questions like when were the last time you failed, make sense. Discovering a contender who does not repeat errors is outstanding to the business since they know where they've gone wrong and thus they work on safeguarding that they don't repeat those errors.

Mind Map

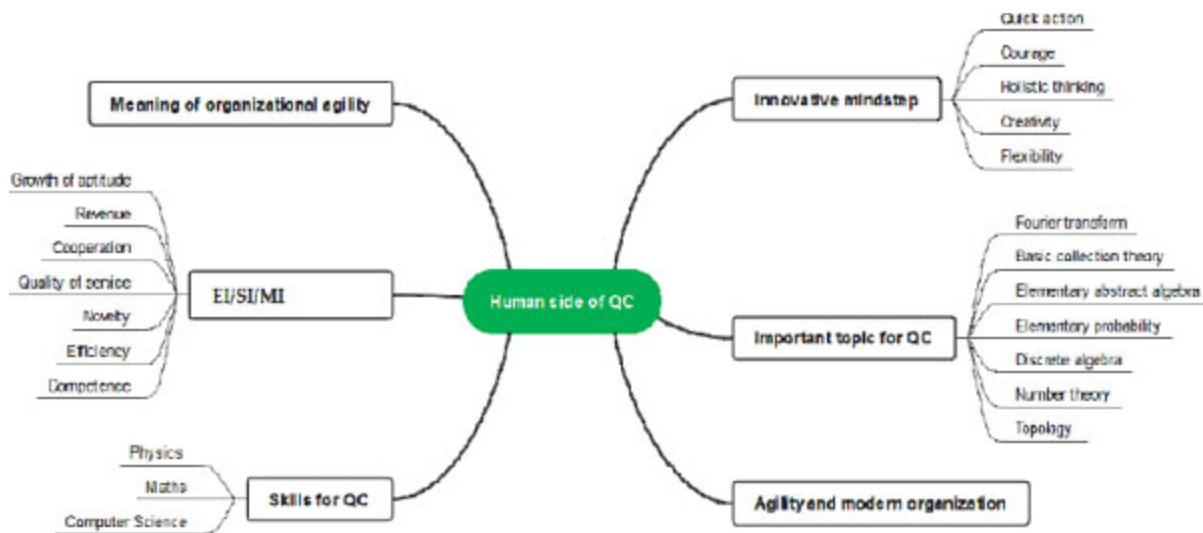


Figure 13.4: Mind map for chapter 13

Conclusion

This chapter discussed about the human aspects of quantum computer and computing. Further, the chapter discussed about people, culture and team building aspects of an innovative organization. It was an important chapter because it focused on the human's behaviour and their role.

As we know that it is ultimately people who lead, build and realize any technology with an organization. It is ultimately people who on ground level bring vision to reality. Also, the chapter turned out to be significant since it discussed about how to accomplish the quantum machine learning and artificial intelligence projects.

ANNEXURE 1

Q# for quantum computation

Q# is Microsoft cooperation provided an open-source programming language. It is used for developing, programming and executing quantum algorithms. It's part of the Quantum Development Kit (QDK), which comprises Q# libraries for the quantum program development, quantum simulators for the visual realization of algorithms, extensions for another programming ecosystem like C#, and API documentation. In accumulation to the Standard Q# library, the QDK comprises Chemistry, Machine Learning, and Numeric libraries. As per Microsoft, a lot will be included in it in the coming time.

As a programming language, Q# attractions acquainted elements from Python, C#, and F# and ropes a basic technical model for script programs that may contain loops, conditional statements like if, then, else and most common data types similar to integer, variable and float. It also presents novel quantum computing-specific data structures for doing multiple types of operations on them.

If one analyses Microsoft QDK, it becomes evident that QDK is a full-featured development kit coupled with Q#. That can be further used along with multiple shared tools and languages to write, develop, test and execute quantum applications by using the coding framework. Additionally, you can run it in numerous environments. It is important to note that Q# programs can run as a console application. For running the programme Jupyter Notebooks could be used, further, Python or .NET would be used for hosting programs.

Q# features

- Q# File Structure: is very important in the world of Q# because it defines the fundamental structure and associated syntax of a *.qs (extension) Q# file.

- Operations and Functions: are the two callable kinds of basic Q# language operations constructs. This is based on the fundamental logic of acting on qubit registers and functions that work with traditional information.
- Variables: just as in any other programming language in Q# also variables are defined and used for storing and manipulating values. For instance, it could be used for providing information about compulsory possibilities, as well as for the change, modification and shift between immutable and mutable variables. Further through variable, it could be explained how to allocate or re-assign variables.
- Control Flow: Again, like any other programming language in Q# also control flow is not different. Meaning here also they specifics the programming control flow, that decorations obtainable in Q#. It comprises many standard techniques which are common in popular programming languages and in the mainstream for years like conditional processing (if, then, else) looping construct (for, *while* loops). However, it has some additional constructs as well which are specific to a quantum programming language like the *Repeat-Until-Success* pattern.
- Testing and debugging: these techniques are very important for any programming framework because they would give techniques for making sure that written code is fulfilling its responsibilities.

Due to the overall impenetrability of quantum information, debugging, testing and pinpointing errors in a quantum program can necessitate the need for dedicated techniques. Advantageously, Q# supports many of the traditional debugging techniques programmers are familiar with, as well as those that are quantum-specific. These comprise planning, creating and running unit tests in Q#, embedding *assertions* on values and likelihoods in your code, and framework.

[Q# Libraries](#)

The Quantum Development Kit is providing numerous libraries to brand it calmer to grow quantum applications in Q#. below are some of the libraries are describe which can be used in the quantum program.

- **Standard libraries:** describes the interface among Q# programs and mark machines, and the standard, a Q# library that delivers general-purpose processes and functions for use in script Q# programs.
- **Quantum chemistry library:** the quantum chemistry library, delivers a data model for loading depictions of fermionic Hamiltonians and quantum simulation processes and functions which perform on these depictions.
- **Quantum numeric library:** this library defines the quantum numeric library, which delivers applications for a crowd of mathematical functions. It ropes integer and fixed-point depictions.
- **Quantum machine learning library:** This library delivers employment of the consecutive classifiers that take the benefit of quantum computing to comprehend data.

Foundations of the libraries and code samples can be found on GitHub. Note that package references (“binaries”) are obtainable also for the libraries and bid additional method of together with the libraries in projects. A suitable way of obtaining them is via NuGet.

ANNEXURE 2

Python for Quantum computing

Python is perhaps one of the utmost prevalent programming languages. This is due to the fact that Python is a very adaptable, high-level programming language with a very lively and strong developer and open-source community. Python's usefulness and ease make it the language of choice for varied applications such as embedded systems, artificial intelligence, gaming, website development, graphics design, data science, machine learning, virtual reality and complex scientific and numerical computing.

For quantum computing and quantum information processing (QIP) prominent corporations such as Google and IBM, together with some QIP start-up companies such as Xanadu in Canada are using it to fulfil their quantum programming needs.

For using Python for your quantum programming needs one needs to install Python packages. It can be installed using installation tools such as pip. Usually, in order to accomplish Python packages and dependencies, diverse Python libraries are used. For example, pip is used for package installation, update, and uninstallation, while another tool such as venv is used to accomplish virtual environments. These packages are characteristically used distinctly. This in practice makes the management of a Python project very bulky. The solution to this is the use of a tool that can do the tasks of package management which includes installation, modifications, deletions, updating, and uninstallation, lengthways with virtual environment creation and support. The suitable tool for this is pipenv. Fundamentally pipenv is a Python packaging tool. This tool is used to manage and streamline Python package dependencies. It can be installed using the command: `pip install pipenv`. Once pipenv is installed, it substitutes pip as a package management tool. Additionally, it becomes accountable for all the package dependency jobs.

Vectors, matrices, and tensors manipulations

Vectors, matrices, and tensors can be created and manipulated in Python. One of the most prominent Python packages used for linear algebra is NumPy. The following code snippet shows how vectors, matrices, and tensors can be created using NumPy

```
import numpy as NumPyForDemo
x = NumPyForDemo.array([10,20,30]) # Three components vector.
y = NumPyForDemo.array([40,50,60]) # One more vector.
A = NumPyForDemo.array([[10,20,30], [40,50,60], [70,50,60]]) # a 3x3
matrix.
B = NumPyForDemo.array([[20,30,40], [10,20,00], [30,30,30]]) # another
3x3 matrix.
v = NumPyForDemo.kron(x,A) # Representation of a tensor product of
vector x and matrix A.
w = NumPyForDemo.kron(y,B) # Representation a tensor product of vector
y and matrix B.
```

Here, array (), Kron () are the functions from NumPy. Whereas NumPyForDemo is used for importing NumPy.

Arithmetic operations in Python

Arithmetic operations, such as total, deduction, and scalar multiplication, can be performed on the vectors. The same is also true of matrices and tensors. Some examples of arithmetic operations on vectors, matrices, and tensors are given below

```
# Arithmetic operations
# Vectors
z = x + y # Vector addition
z = x - y # Vector subtraction
z = 2 * z # Scalar multiplication
# Matrices
C = A - B # Matrix multiplication
C = x * A # a product of a vector and a matrix
# Tensors
```

```
u = v + w # Tensor addition
```

```
u = 5 * u # Multiplication of a tensor by a scalar
```

The previous code piece demonstrates basic operations that can be executed on vectors, matrices, and tensors. These operations comprise addition, multiplication, and scalar multiplication.

Vector and matrix properties and python

The Python code for computing an inner product for two vectors using numpy is described below

Inner product

```
import numpy as NumPyForDemo
```

```
x = NumPyForDemo.array([1,2,3]) # a vector with three components
```

```
y = NumPyForDemo.array([4,5,6]) # another vector.
```

```
# The output of an inner product of vectors is a scalar
```

```
a = NumPyForDemo.inner(x,y)
```

```
print(a)
```

Now let's see determinant

Determinant

The Python code for computing the Determinant of a matrix is given below. A Determinants of matrices are only defined for square matrices, where the number of rows is the same as the number of columns. Generally, the determinant of a matrix A, denoted by $\det(A)$ or $|A|$, is a scalar.

The Python code snippet for computing the determinant of a matrix is given below:

```
import numpy as NumPyForDemo
```

```
A = NumPyForDemo.array([[1,2,3], [4,5,6], [1,5,1]]) # a 3x3 matrix.
```

```
det_A = NumPyForDemo.linalg.det(A) print(det_A)
```

Similarly, all types of quantum computing-related manipulations can be done in python. It is as vast as even a book is not enough to cover all its features of it.

ANNEXURE 3

Miscellaneous topics: reduction in emissions, global warming, fearless leadership and important facts

In this annexure, we would discuss some evolving themes which impact quantum computing holistically in the realm of technology, corporate social responsibilities and more

Theme 1: Reduction in emissions and controlling global warming

In some of the most difficult or emission-intensive fields, including farming or direct air capture, quantum computing could enable reducing emissions. It could also hasten improvements in technologies that are essential on a large scale, like solar panels or batteries. Without significant advancements in climate technology that aren't possible now, nations and some corporations won't be able to meet the net-zero emissions that they have committed to. Some of these issues are beyond the capabilities of even the most potent supercomputers at this time. In those fields, quantum computing might be a game-changer. Overall, it is believed that quantum computing could contribute to the development of climate technologies that, by 2035, have the potential to reduce carbon emissions by around 7 gigatons per year.



Figure A3.1: Use cases for sustainability: McKinsey study (source: McKinsey website)

Theme 2: Main players in the field of quantum computing

The main players of quantum computing are described in [figure A3.2](#). We already covered most of them in the previous chapters. Here we would give a consolidated view.

Table presenting the main players



Figure A3.2: Main players: McKinsey study (source: PWC website)

Theme 3: Growth Road map of Quantum computers

Below [figure A3.3](#) gives a growth journey of QC

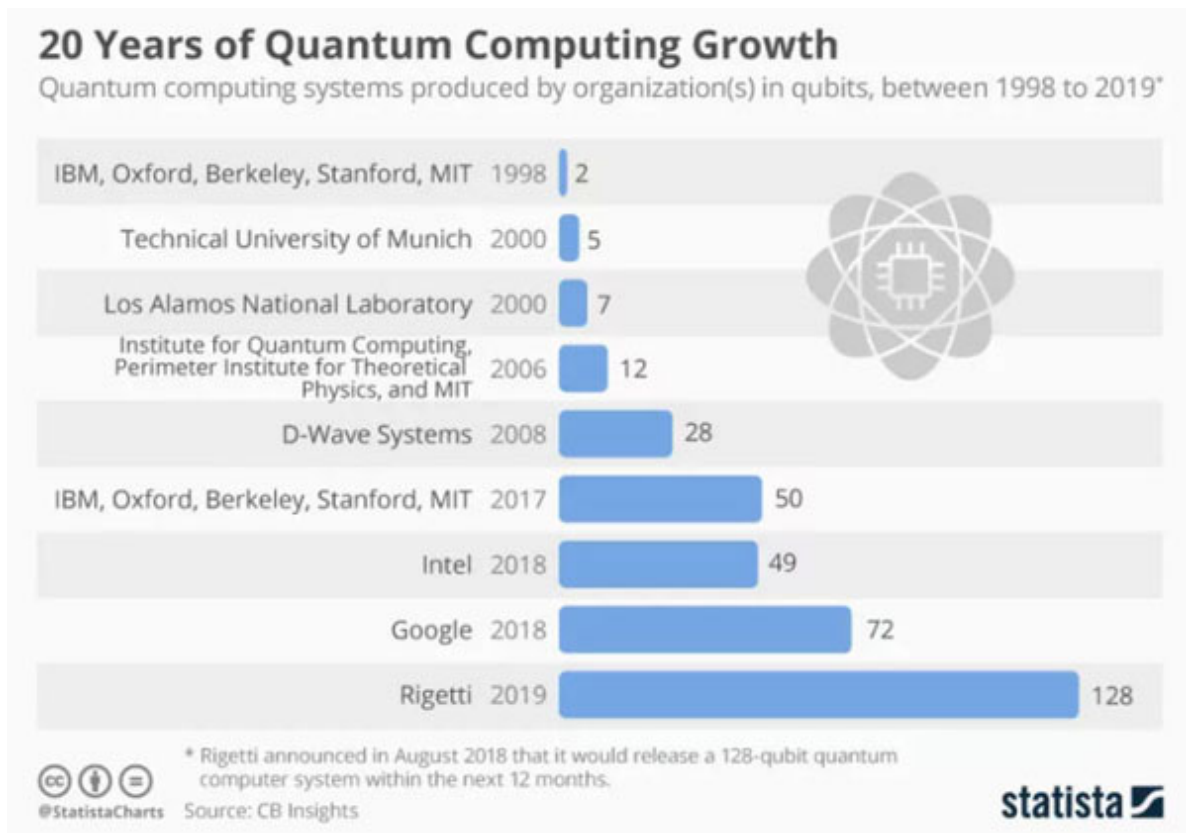


Figure A3.3: 20 years of growth journey (source: PWC website)

Theme 4: Interesting facts

Now let's look at a few interesting facts in the context of quantum computers and computing

- Richard Feynman, a scientist, proposed quantum computing as a substitute for classical computers in 1982. (INSEAD)
- Since particles must be in a stationary phase in order to be observed, quantum computing necessitates a chilly operating environment. The quantum computer chip operating temperature record is -272°C . (NewScientist)
- D-Wave created the world's first commercial quantum annealing machine in 2011, and it cost about US\$10,000,000. It operated on a 128-qubit system. (HandWiki)

- As of 2018, China leads with almost 3000 research articles on quantum computing. China is followed by the United States with about 2500 articles. (BCG)
- The most recent Google quantum computers can accomplish a task in 3 seconds, whereas Boolean logic computers need 10 years to do it. Source Scientific

Theme 5: Funding for start-ups

Let's look at the funding of a few important start-ups

- With \$216M in 19 rounds, D-Wave is the most well-funded private quantum computing company (Crunchbase)
- \$198.5M in 9 rounds for Rigetti Computing (Crunchbase)
- The organisation that has received the most public financing is the University of Oxford (UK): 62 grants worth \$117.59 million (PwC)
- IonQ: \$52M over three rounds (Crunchbase)
- Silicon Quantum Computing: \$66 million (PwC)
- ££1M in 5 rounds for Cambridge Quantum Computing (CQC) (Crunchbase)
- 1QBit: \$45.00 CA over 4 rounds (Crunchbase)
- IQM: €29M over three rounds (Crunchbase)

Theme 6: The fearless organization

The first step to enabling an organization to take the decision to incorporate technologies in its DNA. For taking these type of initiatives fearlessness is required. Quantum computing is an evolving and extraordinary technology to implement, the organizations need an altogether new mindset for the same. With the help of Amy C. Edmondson's model of fearless organization. The leaders of the corporate can do that. Hence, it is necessary to have a look at the model. Let's start with some issues of the modern organization.

As we know with the progressively unstable, questionable, complex and vague (VUCA) world we live in, it's indeed more critical to completely use the idle potential of the workforce (to discover arrangements for progressively challenging issues). We require cooperation where information is openly shared over the organization. In case the culture makes individuals

anxious to share ideas/thoughts/challenges/concerns/questions/mistakes/knowledge, at that point everybody loses. Our working environment is moving much more in collaboration (both interior an association, over geological boundaries as well as with outside associations – 50% more time is gone through collaborating than 20 a long time prior).

Each choice in an organization these days are made by looking for numerous viewpoints from others. The more noteworthy an issue is examined, at that point, it is more likely the organization will make a distant better, higher; stronger, improved choice. It's an ancient cliché that terrible news does not travel up the progression. The inconvenience is in today's boisterous work environment, there's small time or want to engage 'blocks or challenges' It takes time and vitality.

Due to a lack of openness with the supervisors and senior leaders, individuals are anxious of the results of talking up and out – indeed when they perceive what they needed to say would be of significance for the association (and this could in some cases lead to stretch and individual regret). When tested, individuals say they did not need to be seen in a terrible light or harm connections while others communicated the seen worthlessness of talking up (i.e., tall chance, moo remunerate). Commonplace zones they needed to comment on were destitute execution of a chief, badgering, or recommended enhancements for work processes. In one piece of inquiry, 85 % of respondents detailed that on at slightest one event at work, they did not feel they seem to rise a critical concern with their boss. Indeed, senior individuals can still feel the weight not to comment or challenge. The research highlighted a few of the 'taboo' transgressions you do not commit:

- Don't criticise the boss on something s/he was involved in creating
- Don't speak unless you have solid data
- Don't speak if your boss's boss is present
- Don't speak up in a group which could make your boss lose face

The reality is, keeping calm is as a rule continuously the more secure choice and thus is the default setting (no one gets let go for quiet) – whilst speaking up has tall seen hazard. The inconvenience is since individuals don't talk up; its effect is covered up from investigation. We don't know the genuine effect

of hush. Tragically, numerous directors still accept administering through fear is a successful way to boost execution. However, the investigation has clearly illustrated that fear restrains learning and cooperation. Fear occupies assets to oversee this seen danger, counting lessening working memory. This impedes expository considering, inventiveness and issue understanding abilities. ‘Workarounds’ could be a term characterized by Anita Tucker when she watched medical attendants. She noticed that rather than challenge the bosses, they found ways around the issue. Workarounds are frequently less productive and basically cover up the basic issue. Groups with moo mental security tend to enjoy more walkarounds. Inquire about has appeared ventures with more noteworthy mental security were more fruitful in general than ones with moo mental security. This in turn drove to more prominent money-related returns.

The fearless organization is one where interpersonal fear is limited, permitting a more liberated stream of information. It’s an environment where an individual feels able to specific their sees on something straightforwardly and truly without fear of recrimination, mishandling, putdown or humiliation. According to research mental security is the foremost critical component of the intrepid organization and of is the key figure that drives tall performing groups. This was bolstered by four other features:

- Clear goals
- Dependable colleagues
- Personally meaningful work
- The belief that the work has an impact

Psychological security isn’t around being decent. It’s around being able to talk the truth. And it’s affirmed to oppose this idea. Nor is mental security almost bringing down benchmarks. It is really the inverse. It’s around making an environment that permits individuals to be fairer: to challenge, make change proposals – that basically, get acted upon. Psychological security shows up to live at the group/team level. In any association, you may discover pockets of both tall and moo mental security (frequently connected to the pioneer in that area). Psychological security begins at the best (but everyone’s moreover capable of it).

Benefits which we get by developing fearless organization and team

The benefits of mental security are colossal and widespread. Research has appeared mental security makes a difference boost engagement/fulfilment/ more noteworthy meaning, and so makes a difference unleash the ability of the workforce. It increments individual development, group building, learning/new bits of knowledge, blunder announcing, information sharing, and imagination. It too makes a difference diminish work push, wiped out days, turnover, and makes a difference overcome interpersonal challenges (such as geographic dispersants, strife as well as grasping ethnic differing qualities). Eventually, it leads to expanded extended adequacy, development, in advertise competitiveness and monetary return.

For the summary, characteristics and important elements, refer to [figure A3.4](#)

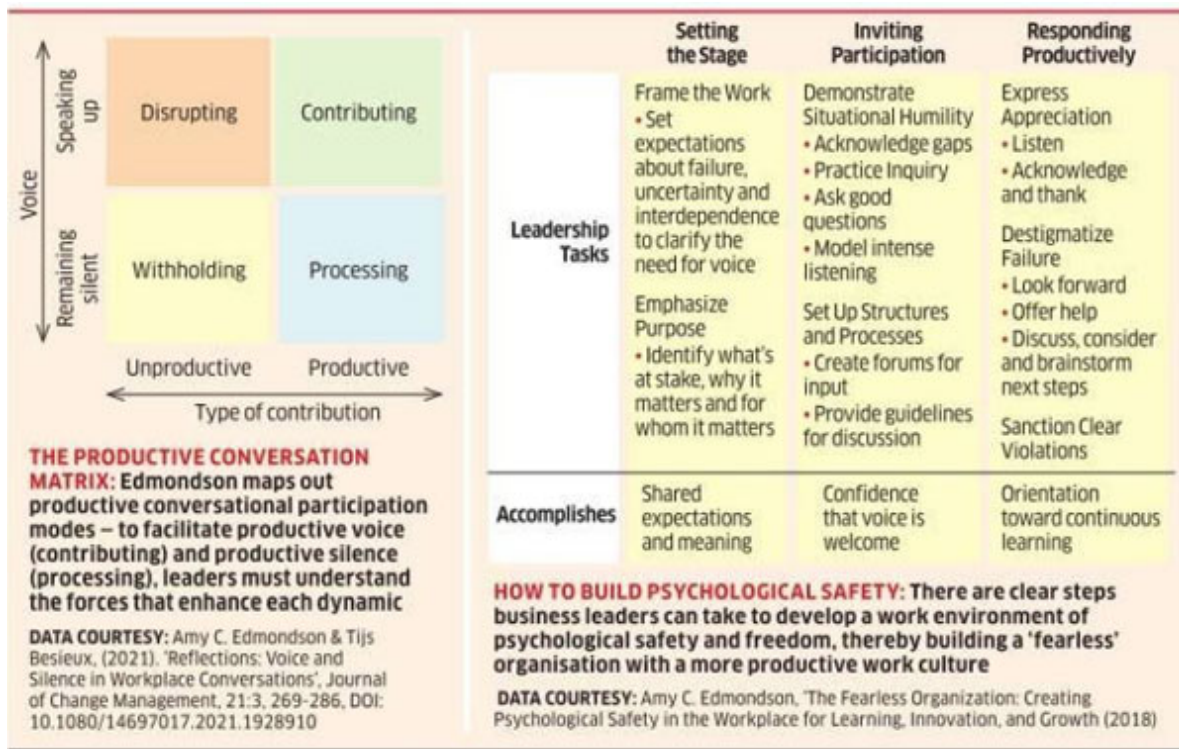


Figure A3.4: Fearless organization-characteristics (source: the fearless organization: Creating Psychological Safety in the Workplace for Learning, Innovation, and Growth, Amy C. Edmondson, HARVARD BUSINESS SCHOOL and the economics times, Friday, 19th August 2022, page 13)

Theam 7: Metaverse and quantum computer

Imagine the internet in three dimensions – you are being able to move inside it like you would in your physical world. That is, probably where the

metaverse is taking us. It makes use of advanced telecommunication infrastructure, artificial intelligence and extended reality (XR)- that is virtual reality, mixed reality, augmented reality and mixed reality and the likes- to create online platforms that are more immersive, interactive and interoperable than they are today. These online platforms or virtual worlds can be accessed through applications on your mobile phone, personal computer or any other digital device. You will be represented by your avatar in the metaverse. With the help of a VR device one can immerse oneself completely in the virtual world but considering its prohibitive cost many platforms just let users engage through touch screens. The metaverse also has a dynamic digital marketplace, driven by the forces of demand and supply.

As it becomes popular, the data and its processing start playing a bigger role in the complete ecosystem. And here is the entry point of quantum computers and optical computers combination for processing and storage purposes. They can potentially make the complete experience more immersive and interactive.

Microsoft metaverse technology stack

Microsoft Mesh & HoloLens

Enable anyone across the planet to collaborate in mixed reality environments.

Microsoft Power Platform

Empower anyone within your organization to interact with and act on the data flowing through your environment. Build applications, dashboards, and intelligent chatbots that modernize workflows.

Azure AI & Autonomous Systems

Azure AI delivers breakthrough insights into all your data. Microsoft Project Bonsai provides a low-code approach to machine teaching and creating intelligent autonomous systems that learn and improve over time.

Azure Synapse Analytics

A comprehensive set of data services that work together to provide predictive analytics, the ability to analyze data across systems, and the ability to track the historical state of your environment.

Azure Maps

The location of things. Indoor private maps enable you to apply location and routing services to people and things in your private environment while keeping data locked down to your enterprise.

Azure Digital Twins

Simplify the creation of detailed, comprehensive digital models. Support for the Digital Twin Definition Language enables modeling of complex relationships between things and systems in your environment.

Azure IoT

Connect physical assets and run cloud intelligence directly on premises. Sense and monitor anything in your physical environment seamlessly and securely.

The physical world

Figure A3.5: Microsoft version of Metaverse (Source: Microsoft website)

ANNEXURE 4

References, Notes and Bibliography

In this annexure, we would discuss some References, Notes and Bibliography related to quantum computing and associated technologies, topics, and subjects.

Web links

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